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Crazy Merged 3

Sections 9–12

One merged book: the condensed books and the Paper 1–4 answers combined, section by section, with each topic appearing once. Read each chapter top to bottom. Where a topic was answered in the papers, the full paper answer sits in its place (marked "Paper answer", with the papers it was asked in), followed by a dashed box of anything extra the book had. Paper answers not tied to a chapter are at the end of each section, then the section's Rapid Revision Sheet.

Section	Title
9	Andrology & Sexual Medicine
10	Testis, Penis & Urethra
11	Stones, Obstruction & Trauma
12	Bladder Cancer & Diversion

SECTION 9

Andrology & Sexual Medicine

66 Surgical, Radiographic and Endoscopic Anatomy of the Male Reproductive System

1 · Scrotum

Each layer is a surgical plane from the abdominal wall.

Layer (superficial to deep)	Continuous with	Note
Tunica dartos	Scarpa, Colles, penile dartos	Smooth muscle
External spermatic fascia	External oblique	—
Cremasteric fascia	**Internal oblique**	**Striated** cremaster; thermoregulation
Internal spermatic fascia	Transversalis fascia	—
Tunica vaginalis (parietal, visceral)	Peritoneum	Hydrocele collects between its layers; visceral layer fused to albuginea

- The anterior wall carries the external pudendal vessels and the ilioinguinal and genitofemoral nerves; the posterior wall carries the perineal vessels and nerve. Nothing crosses the raphe.

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

> **KEY POINT**
>
> Scrotal lymph goes to the groin, testicular lymph to the retroperitoneum. †So scrotal violation in testis cancer adds the inguinal nodes to the at-risk field.

2 · Testis

Structure

- Volume is **15–25 mL**, length 4.5–5.1 cm. The capsule is visceral tunica vaginalis, **tunica albuginea** (with smooth muscle) and tunica vasculosa.
- The **mediastinum testis** is the posterior echogenic band on ultrasound (US). The appendix testis is a **müllerian** remnant.
- About 250 lobules hold **600–1200 seminiferous tubules**, which drain to the rete testis.
- The interstitium is **20–30%** of volume and holds the **Leydig cells**. **Sertoli** tight junctions form the blood–testis barrier, aided by endothelium and peritubular myoid cells (see Ch 67).

Arterial supply — three sources

Artery	Origin	Point
Testicular (internal spermatic)	Aorta below renal artery	Main supply; larger than the other two combined

Artery	Origin	Point
Deferential	Superior (or inferior) vesical	Vas, cauda; key collateral, lost after vasectomy
Cremasteric	Inferior epigastric	Tunica vaginalis; anastomoses at mediastinum

- Collaterals may save the testis after testicular artery ligation, but atrophy or azoospermia can occur.

> **KEY POINT**
>
> **Surgical rules**: no traction suture through the lower pole. Biopsy the **medial or lateral upper pole**; the midportion has fewest vessels. After vasectomy the deferential collateral may be gone, so preserve the testicular artery in later cord surgery and counsel about devascularisation.

Veins — pampiniform plexus

The plexus cools the artery by **countercurrent heat exchange** to **2–4 °C below rectal** temperature. A varicocele disrupts this.

Group	Accompanies	Drains to
Anterior	Testicular artery	Inferior vena cava (IVC) on the right; left renal vein on the left
Middle	Vas	Pelvic veins
Posterior	Posterior cord	Epigastric and pudendal veins

*See the paper answer **Varicocele: pathophysiology, controversies and surgical management** (Section 10, Ch 87).*

Also, from the condensed book (not in the paper answer):

- **High (retroperitoneal) ligation** can take artery and vein, since collaterals suffice; **inguinal or subinguinal** ligation takes veins only, because arterial ligation there risks devascularisation.

Lymphatics and nerves

- Dividing lymphatics at varicocelectomy causes a **hydrocele**; microsurgery preserves them.
- Supply is autonomic, with **no somatic innervation**; the genital branch of the genitofemoral nerve supplies the tunica vaginalis.
- The **"trifecta" nerve complex** (perivasal, posterior periarterial/lipomatous, intracremasteric) is the target of microsurgical cord denervation for chronic orchialgia.

Imaging

Modality	Role
Ultrasonography (US), 7.5–10 MHz with Doppler	**First line**: pain, mass, swelling, epididymo-orchitis, recurrent varicocele
Computed tomography (CT)	Not primary; cord or inguinal disease; air in Fournier gangrene
Magnetic resonance imaging (MRI)	Second line: indeterminate US, intra- versus extratesticular; nonpalpable testis (MR angiography)

- If torsion is suspected, **explore**; Doppler must not delay surgery.

- Varicocele is a **clinical diagnosis**. Use US (standing, Valsalva) only if examination is equivocal.

3 · Epididymis

- Caput, corpus and cauda lie on the posterolateral testis. **Principal** cells form the blood–epididymis barrier, and **clear** cells acidify the lumen to keep stored sperm quiescent.
- The testicular artery supplies caput and corpus, and the deferential artery the cauda. Nerves and smooth muscle increase toward the cauda, which shifts from storage to propulsion.
- Wolffian (mesonephric) remnants include the **appendix epididymis**; müllerian (paramesonephric) remnants are the **appendix testis** and prostatic utricle.

4 · Vas deferens

- It runs **posterior** to the cord vessels through the inguinal canal, crosses **anterior to the ureter**, forms the ampulla and joins the seminal vesicle.
- It is **30–35 cm** long with a **0.2–0.7 mm** lumen and has **three muscle layers** (longitudinal, circular, longitudinal).
- About half the cord nerves surround the vas: sympathetic fibres drive emission, sensory fibres cause chronic scrotal pain.
- A **vasogram** can cause obstruction, so do it **only at reconstruction**.

5 · Seminal vesicles and ejaculatory ducts

- Behind the bladder base, they join the ampulla as ejaculatory ducts opening on the verumontanum at **5 and 7 o'clock**.
- They bud from the mesonephric duct alongside the ureteric bud, so **renal anomalies coexist**.
- The fluid is ~70% of semen and alkaline. It carries **fructose**, prostaglandins and **semenogelin**, which forms the coagulum later liquefied by prostate-specific antigen (PSA).
- On **transrectal ultrasound (TRUS)**, a post-ejaculation anteroposterior (AP) diameter **>16 mm** or volume **>6 mL** means obstruction. **MRI** is the modality of choice; CT is for stones.

> **KEY POINT**
>
> A normal vesicle empties at ejaculation; one that stays large is the imaging signature of ejaculatory duct obstruction.

6 · Prostate

*See the paper answer **Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate** (Section 16, Ch 148).*

Also, from the condensed book (not in the paper answer):

- TRUS volume is length × width × height × **0.52**, which underestimates. The normal gland is **~25 mL**.
- **Multiparametric MRI** is scored by PI-RADS v2.1; targeted biopsy finds less insignificant cancer.

7 · Urethra

LAQ Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias

Paper answer · P1

Asked as:

- Describe the embryology and anatomy of male Urethra. Describe different endoscopic management techniques of treatment of urethral strictures. — *Summer 2019 Paper I Q1 (LAQ)*
- Discuss anatomy of urethra. Classify its congenital anomalies and discuss management of epispadias. — *Practice Set 5 Paper I Q1 (LAQ)* (male urethral anatomy also answered in 2019-Summer Paper 1, Q1; isolated epispadias in 2022-Summer Paper 3, Q3f)

Type: LAQ (25 marks)

Embryology

- **Cloaca** is divided by the **urorectal septum** (weeks 4–7) into the anterior **UGS (urogenital sinus)** and the posterior anorectal canal.
- The UGS has three parts: **vesical** (bladder), **pelvic** (prostatic and membranous urethra) and **phallic** (urethral plate of the penis).
- **Posterior urethra:** from the pelvic UGS. Prostatic buds appear at 10 weeks; the **utricle** (Müllerian remnant) and ejaculatory ducts (Wolffian) open on the verumontanum.
- **Anterior urethra:** Leydig cells make testosterone from about 8 weeks; type 2 5α-reductase converts it to DHT (dihydrotestosterone), which masculinises the genital tubercle. Germ layers: ectoderm gives skin and prepuce, mesoderm the corpora, endoderm the **urethral plate**.
- **Penile urethra: "double zipper"** 1. **Opening zipper:** the urethral plate canalises proximal to distal into a urethral groove between the urethral folds. 2. **Closing zipper:** the folds fuse in the ventral midline, forming the tube and the median raphe. Folds fused by 16 weeks; urethra complete by about 18 weeks.

Glanular urethra	Ectodermal ingrowth theory (Glenister)	Endodermal theory (current)
Formed by	Surface ectoderm grows into the glans to meet the endodermal urethra	Direct canalisation of the urethral plate inside the glans
Epithelium	Mixed ectoderm and endoderm	Endoderm (UGS) along the whole urethra

- **Clinical link:** the shaft–glans junction (change of mechanism) is the commonest hypospadias site. **Also know:** urethral duplication; PUV (posterior urethral valves); utricular cyst.

Anatomy of the male urethra

- About 18–20 cm long in adults; normal diameter 8–9 mm (wide). **Posterior urethra** runs from the bladder neck to the **perineal membrane**; **anterior urethra** from there (perineal membrane) to the meatus, inside the corpus spongiosum.

Segment	Length / features	Lining
Prostatic	Length of prostate; urethral crest, **verumontanum** with utricle (Müllerian remnant) and ejaculatory ducts; angle divides preprostatic and prostatic parts	Urothelium
Membranous	2–2.5 cm; shortest and narrowest; striated sphincter	Pseudostratified columnar
Bulbar	Widens as the bulb; **Cowper ducts at 3 and 9 o'clock**; lies **eccentric, close to (nearer) the dorsum** of spongiosum	Pseudostratified columnar
Penile	About 15 cm; central, uniform calibre; glands of Littré	Pseudostratified columnar
Fossa navicularis	Glanular; lacuna magna on the roof	Stratified squamous

- **Sphincters:** (1) **internal (preprostatic) smooth muscle sphincter** at the bladder neck, autonomic, gives continence at rest; (2) **external striated rhabdosphincter**, horseshoe-shaped around the membranous urethra, supplied by the pudendal nerve (mainly S3, some S2; Onuf nucleus), plus its intrinsic smooth muscle.
- **Blood supply (arteries):** internal pudendal → common penile artery → artery to the bulb (proximal; bulbourethral branches) and **retrograde flow via the dorsal arteries through the glans**. This **dual supply** allows bulbar transection in anastomotic urethroplasty.
- **Venous** (veins) to pudendal and prostatic plexuses; **lymph** to internal and external iliac nodes.
- **Female urethra:** about 4 cm; smooth muscle plus striated sphincter over the distal two-thirds; **Skene glands** (prostate homologue) near the meatus; oestrogen-dependent mucosal seal.

Endoscopic management of stricture

Principle: healing after incision or dilation is by **secondary intention**, a race between epithelialisation and wound contraction. No endoscopic method removes **spongiofibrosis**, so cure is limited to short, superficial bulbar strictures.

Favourable	Unfavourable
Bulbar, **<1.5–2 cm**	Penile, fossa, longer
No dense spongiofibrosis	Straddle injury, LS (lichen sclerosus)
First procedure	Repeat; luminal narrowing >74%

1. Dilation

- Stretches the scar without tearing it (bleeding means tearing).
- **Safest: balloon dilation over a guidewire** under vision; blind filiforms condemned. Sounds or self-dilation for palliation.

2. DVIU (direct vision internal urethrotomy)

1. RUG (retrograde urethrogram) and VCUG (voiding cystourethrogram) define site and length.
2. Guidewire across; **normal saline** irrigation.
3. **Sachse cold knife** at 12 o'clock, through scar to healthy tissue; avoid deep cuts (corpora: ED (erectile dysfunction), fistula).

4. Catheter **3–7 days**; longer adds nothing. - **Laser urethrotomy** (holmium, others): **no advantage over cold knife**. - Results: **Steenkamp and Heyns** (randomised): dilation equals urethrotomy; a second attempt rarely and a third never succeeds. **Santucci (2010):** about 8% stricture-free long term.

3. Adjuncts against recurrence

- **CISC (clean intermittent self-catheterisation)** tapering over 3–6 months: delays recurrence, which returns on stopping.
- **Mitomycin C** injection reduces recurrence; steroid injection, colchicine: weak data.

4. Urethral stents

- **UroLume** (permanent): **withdrawn** (ingrowth). **Memokath**, removable stents: small series only.

5. Optilume (paclitaxel-coated balloon)

- FDA (Food and Drug Administration) 2023. **ROBUST III:** 74.6% vs 26.8% stricture-free at 1 year after failed endoscopy.

Complications: recurrence (commonest), bleeding with erections, irrigant extravasation, UTI (urinary tract infection) and sepsis, ED, spongio-cavernous fistula, incontinence.

Guideline (AUA (American Urological Association) 2016): urethrotomy or dilation for short (<2 cm) bulbar strictures; **urethroplasty, not repeat endoscopy**, after failure. The reconstructive ladder is abandoned.

Classification of congenital urethral anomalies

Group	Anomalies
Abnormal meatal position	**Hypospadias** (ventral); **epispadias** (dorsal)
Obstructive	**PUV (posterior urethral valves)**, Young types I–III; **anterior urethral valves** (with diverticulum); urethral atresia or hypoplasia (congenital stricture); congenital meatal stenosis
Dilatation	**Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); anterior urethral diverticulum; Cowper duct cyst (syringocele); enlarged prostatic utricle
Duplication	**Effmann classification**: type I blind incomplete; type II complete (IIA2 "Y-type"); type III with caudal duplication. The ventral channel is the functional one
Accessory openings	Congenital urethrocutaneous fistula; parameatal urethral cyst
Female	Epispadias; urethral prolapse; persistent urogenital sinus

Epispadias: definition and classification

- A **dorsal urethral wall defect**, replaced by a broad dorsal mucosal strip (urethral plate); part of the **exstrophy–epispadias complex**, mostly seen with exstrophy.
- Embryology: an overdeveloped cloacal membrane blocks mesenchymal ingrowth; genital tubercles fuse caudal to it.

	Male	Female
Incidence	About 1 in 117,000	About 1 in 484,000

	Male	Female
Types	Glanular, penile, **penopubic** (about 70%, incontinent)	**Davis:** vesicular (patulous meatus), subsymphyseal, retrosymphyseal (incontinent)
Genitalia	Short, upturned penis with **dorsal chordee**	**Bifid clitoris**, flat glabrous mons
VUR (vesicoureteric reflux)	30–40%	30–75%

- The more proximal the meatus, the worse the continence (glanular never incontinent). Pubic diastasis is less than in exstrophy.

Evaluation

- Renal ultrasound; VCUG (voiding cystourethrogram) for reflux; cystoscopy and **bladder capacity** under anaesthesia.
- **Key decision:** use the exstrophy pathway (osteotomy and closure) if pubic diastasis >3 cm, poor intrasymphyseal bar, absent sphincter tissue or tiny bladder.

Management

Aims: continence with safe upper tracts, and a straight penis of adequate length that works for intercourse.

- **Preoperative testosterone** (2 mg/kg intramuscular, 5 and 2 weeks before).

Stage 1: urethroplasty and penile reconstruction (6–12 months)

1. Release dorsal chordee: divide suspensory ligaments; free corpora from the inferior pubic rami.
2. **Modified Cantwell–Ransley:** an 18-mm plate strip is outlined; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; the plate is tubularised over an 8-Fr stent and moved **ventral to the corpora**, which are rotated medially over it (diamond corporotomies if curvature persists).
3. Glans wings closed; skin flaps resurface the dorsum; base Z-plasty; stent 10–12 days. - **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias. - Urethroplasty adds outlet resistance, so the bladder grows before stage 2.

Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **Young–Dees–Leadbetter** repair with Marshall–Marchetti–Krantz suspension and **ureteric reimplantation**. Osteotomy is rarely needed.
- **Good capacity at presentation:** single-stage repair is possible.
- **Female:** urethroplasty and genitoplasty (clitoral halves and labia joined, mons reconstructed) at 6–12 months; bladder neck repair at 4–5 years.

Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula about 1 in 6 (penile base); stricture rare; neourethra catheterisable; erections and paternity preserved.
- Better than exstrophy: the bladder was never exposed or scarred. Failures: injectables, augmentation.

Diagram to draw: sagittal male urethra with segments, sphincters and a 12 o'clock urethrotomy; Cantwell–Ransley with plate transposed under the corpora.

> **Key points to remember:**
> - Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
> - Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
> - DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
> - Optilume ROBUST III 74.6% vs 26.8%.
> - Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
> - About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

8 · Penis

Structure

- There are paired **corpora cavernosa** (crura on the pubic arch) and a **corpus spongiosum** (bulb, urethra, glans).
- The septum is incomplete distally, so the corpora act as **one unit** (†one-sided injection fills both).
- The **tunica albuginea** (2–3 mm) has an inner **circular** and an outer **longitudinal** layer; it is thin ventrally.
- **Buck fascia** encloses the corpora and splits around the spongiosum. From midline outward it holds the **deep dorsal vein, then arteries, then nerves**.
- Sinusoidal smooth muscle is **~45%** of volume. Endothelial nitric oxide synthase (NOS) sustains erection.

Arteries

The internal pudendal artery runs through Alcock canal to become the **common penile artery**:

Branch	Supplies
Bulbourethral	Spongiosum, urethra, glans
Cavernosal	Enters at hilum, runs centrally; helicine arteries fill the sinusoids (erection)
Dorsal	Under Buck fascia to the glans; circumflex branches

Veins, nerves

- The superficial dorsal vein drains skin; the **deep dorsal vein** (under Buck fascia) drains to the periprostatic plexus.
- In the distal two-thirds, sinusoids drain via the subtunical plexus and **emissary veins**; the proximal third drains via cavernous and crural veins.
- The **dorsal nerve** (pudendal, **S2–S4**) gives sensation; the **cavernous nerves** (pelvic plexus) give erection.

> **KEY POINT**
>
> Erection is veno-occlusive: filled sinusoids compress subtunical and emissary veins against the inelastic tunica. †Tunical or smooth-muscle damage defeats this and causes venous leak.

Penile duplex (after intracavernosal injection)

Parameter	Threshold	Meaning
Peak systolic velocity	**<30 cm/s**	Arterial insufficiency
End-diastolic velocity	**>5 cm/s**	Veno-occlusive dysfunction
Resistive index	**<0.80**	Veno-occlusive dysfunction

MRI shows fracture, prosthesis malposition and tumour.

67 Male Reproductive Physiology

1 · Hypothalamic–pituitary–gonadal axis

Every hormonal work-up reads this axis.

	Peptide hormones	Steroid hormones
Examples	Luteinising hormone (LH), follicle-stimulating hormone (FSH)	Testosterone, oestradiol
Storage	Secretory granules	Not stored; secretion equals production
Receptor	Cell surface, acting via cyclic AMP (cAMP)	Intracellular, altering transcription; carried bound in plasma

Hypothalamus

- Gonadotrophin-releasing hormone (GnRH) is a decapeptide whose only function is LH/FSH release. Half-life is **5–7 min**.
- Its rhythms are pulsatile (every **90–120 min**), circadian (testosterone peaks early morning) and seasonal.
- **Pulsatility is essential**: a continuous GnRH agonist (leuprolide) desensitises the pituitary, giving castrate testosterone.
- In **Kallmann syndrome** GnRH neurones fail to migrate, which causes hypogonadotrophic hypogonadism. †Anosmia coexists because the neurones migrate with the olfactory placode.

GnRH modulation	Factors
Negative	Opioids (β-endorphin), FSH, LH, testosterone
Positive	Prostaglandin E2, insulin, **kisspeptin** (puberty), **leptin**
Variable	Dopamine (catecholamines)

Anterior pituitary

- LH and FSH share an **α subunit** with TSH and hCG and have a specific **β subunit**.
- **Hyperprolactinaemia abolishes gonadotrophin pulsatility**, so testosterone falls. Low prolactin is not pathological.

	LH	FSH
Pulses	8–16/24 h; follow GnRH	Smaller; partly GnRH-independent (inhibin, activin)
Target	**Leydig cell**	**Sertoli cell** (and spermatogonia)
Action	Steroidogenesis: cholesterol to pregnenolone to testosterone	Tubule growth; **initiates spermatogenesis at puberty**; maintains quantitative output
Main feedback	Testosterone (hypothalamus)	Oestradiol (pituitary) and **inhibin B**

Testicular hormones and feedback

- Testosterone output is ~**5 mg/day**. 5α-reductase converts it to the more potent **dihydrotestosterone (DHT**), needed in most peripheral tissues; aromatase converts it to **oestradiol**, which feeds back especially on FSH.
- FSH makes Sertoli cells produce androgen-binding protein (ABP), transferrin, lactate and inhibin. **Inhibin** suppresses FSH; **activin** stimulates it.

> **KEY POINT**
>
> There are two loops: LH with Leydig cells (testosterone), and FSH with Sertoli cells (inhibin, oestradiol). So **LH reports Leydig function, while FSH and inhibin B report the tubules**. High FSH with low inhibin B means spermatogenic failure.

Development and ageing

- **SRY** (Yp) starts the testis. **DAX1** and **WNT4** are "antitestis" genes; the female pathway is not a passive default.

Fetal cell	Product	Effect
Leydig	Testosterone	Wolffian duct forms internal genitalia
Leydig	Insulin-like peptide 3 (INSL3)	Transabdominal testicular descent
Target tissue	DHT	External genitalia
Sertoli	Müllerian-inhibiting substance (MIS/AMH)	Müllerian regression; keeps germ cells out of meiosis

- Puberty (~12 years) is triggered by kisspeptin, leptin and melatonin.
- **Ageing**: testosterone is ~one-third lower by the 7th decade and FSH ~threefold higher. SHBG rises and feedback is blunted, so failure is both peripheral and central.

2 · Testis

Architecture and blood flow

- It has **600–1200 tubules** and ~**700 million Leydig cells** in young adults.
- The cord holds one artery in ~half of men, so at microsurgical varicocelectomy ligating a single artery can cause atrophy.
- **Countercurrent exchange** cools arterial blood **2–4 °C** below core and recycles testosterone. It is lost in varicocele and cryptorchidism.

Leydig cells and testosterone synthesis

- **Rate-limiting step**: LH raises cAMP, and **steroidogenic acute regulatory (StAR) protein** moves cholesterol into the inner mitochondrial membrane to become pregnenolone.
- Further enzymes (3β-HSD, 17α-hydroxylase/17,20-lyase, 17β-HSD) make testosterone; their mutations cause 46,XY disorders of androgen biosynthesis.
- †In blood, ~44% of testosterone is bound tightly to SHBG, ~54% loosely to albumin and ~2% is free. Free plus albumin-bound is "bioavailable".

Testosterone peak	Timing	Role
Fetal	12–18 weeks	Builds reproductive tract
Neonatal	~2 months	Imprints target tissues
Pubertal	12–17 years	Masculinisation
Adult	Peak in 2nd–3rd decade, then decline	Maintains androgen-dependent organs

LAQ Spermatogenesis, azoospermia and non-obstructive azoospermia

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P1 LAQ "Spermatogenesis, azoospermia, non-obstructive azoospermia and maturation arrest" · P2 SAQ "Azoospermia and non-obstructive azoospermia" · P3 SAQ "Spermatogenesis"

Maturation arrest is taken as spermatogenic maturation arrest on testicular histology.

Spermatogenesis: definition

- Production of haploid spermatozoa from diploid spermatogonia in the seminiferous tubules: about **64 days** (42–76); about 123 million sperm a day.

Hormonal control: HPG (hypothalamic–pituitary–gonadal) axis

- The hypothalamus releases **GnRH (gonadotropin-releasing hormone)** in pulses; pulsatility is essential (constant agonist exposure shuts the axis down). Kisspeptin triggers puberty.
- **LH (luteinising hormone)** acts on Leydig cells to make testosterone (about 5 mg/day). **FSH (follicle-stimulating hormone)** acts on Sertoli cells.
- Feedback: testosterone (and oestradiol) on LH; **inhibin B** from Sertoli cells on FSH. So FSH and inhibin B report the tubules, and LH reports the Leydig cells.
- **Intratesticular testosterone** is about 50-fold higher than serum (held by **ABP, androgen-binding protein**) and is essential for postmeiotic maturation. Exogenous testosterone suppresses LH and so stops spermatogenesis.
- The testis is kept 2–4 °C below core temperature by the pampiniform countercurrent exchange.

Sertoli cells and the blood–testis barrier

- Nurse cells: nourish germ cells; secrete ABP, inhibin and fluid; phagocytose residual bodies.
- **Sertoli–Sertoli tight junctions** divide the tubule into a **basal** compartment (spermatogonia) and an **adluminal** compartment (spermatocytes, spermatids). Post-meiotic cells are "foreign" to the immune system; the barrier and local tolerance protect them.

Phases of spermatogenesis

Phase	Events
1. Proliferation (spermatocytogenesis)	**Dark type A (Ad)** reserve stem cells self-renew (GDNF [glial cell line–derived neurotrophic factor] niche); **pale type A (Ap)** divide every 16 days into type B spermatogonia, then primary spermatocytes. Daughter cells stay joined by cytoplasmic bridges
2. Meiosis	Primary spermatocytes cross into the adluminal compartment. Meiosis I (reduction division; synapsis and crossing-over) gives secondary spermatocytes; meiosis II gives 4 haploid round spermatids
3. Spermiogenesis	No division: acrosome from Golgi, flagellum from centriole, nucleus condenses to about 10%, mitochondria form the midpiece, cytoplasm shed
4. Spermiation	Sertoli cell releases sperm into the lumen

- **Cycle, stage, wave:** six stages in humans; cycles are arranged as a helical wave, so production is continuous.
- Defective meiotic recombination causes maturation arrest and sperm aneuploidy.

Epididymal maturation and transport

- Testicular sperm are immotile. During 2–12 days of epididymal transit they gain **progressive motility** and fertilising ability; they are stored in the cauda.
- **Capacitation** and the acrosome reaction occur in the female tract.

Clinical relevance of spermatogenesis

- A semen sample reflects events about 3 months earlier; reassess 3 months after treatment.
- Disruptors: heat (varicocele, cryptorchidism), exogenous testosterone, chemotherapy, radiation.

Azoospermia: definition

- No sperm in the **centrifuged** ejaculate on **at least two** semen analyses; 10–15% of infertile men.
- **NOA (non-obstructive azoospermia):** azoospermia due to **failure of spermatogenesis** (testicular failure); about 60% of azoospermic men.

Classification

Type	Causes	Clues
Pre-testicular (hypogonadotropic)	Kallmann syndrome, pituitary tumour, hyperprolactinaemia, **exogenous testosterone or anabolic steroids**	Low FSH, LH, testosterone
Testicular (NOA, spermatogenic failure)	Klinefelter, Y microdeletion, cryptorchidism, mumps, chemotherapy or radiation, idiopathic	Small soft testes, **FSH >7.6 IU/L**, long axis ≤4.6 cm
Post-testicular (OA, obstructive azoospermia)	CBAVD (congenital bilateral absence of the vas deferens), epididymal obstruction, vasectomy, hernia repair, EDO (ejaculatory duct obstruction)	Normal testes, **FSH ≤7.6**, dilated epididymis
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Causes of NOA

- **Genetic: Klinefelter (47,XXY)**, the commonest; **Y microdeletion** (AZF [azoospermia factor] a, b, c); other karyotype anomalies.
- **Acquired:** cryptorchidism, mumps orchitis, torsion, **chemotherapy or radiation**, trauma, varicocele.
- **Idiopathic:** commonest overall.
- *Note:* hypogonadotropic hypogonadism is pre-testicular and treatable medically (hCG [human chorionic gonadotropin] ± FSH).

Evaluation

- History (mumps, surgery, chemotherapy, drugs, testosterone use); examination (testis size, **vasa**, epididymis, varicocele).
- **Hormones:** FSH (reflects germ-cell mass), LH, testosterone, oestradiol, prolactin.

- NOA: small soft testes, palpable vasa, normal ejaculate volume. **FSH >7.6 IU/L and testis long axis ≤4.6 cm** point to NOA; FSH ≤7.6 with normal testes (>4.6 cm) suggests obstruction.

```
Azoospermia (2 centrifuged samples) ├─ Vasa absent → CBAVD → CFTR testing, both partners; renal scan/US ├─ Volume <1 mL → post-ejaculate urine: sperm = retrograde; none → TRUS for EDO └─ Normal volume: FSH ≤7.6 IU/L + testis long axis >4.6 cm → obstructive FSH >7.6 + testis ≤4.6 cm (small) → spermatogenic failure
```

- **Genetics (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine): karyotype** and **Y microdeletion** in all azoospermic men with NOA, with counselling. **AZFa or AZFb complete deletion: no sperm, do not operate**; AZFc: sperm often found. **CFTR (cystic fibrosis transmembrane conductance regulator)** testing in CBAVD.
- **TRUS (transrectal ultrasound):** seminal vesicle >1.5 cm wide or a midline cyst suggests EDO.
- **Testis biopsy:** combined with sperm extraction and freezing.

Histology patterns in NOA

Pattern	Histology / meaning	Micro-TESE retrieval
Hypospermatogenesis	All stages present, fewer cells	Best
Maturation arrest	Halts at one germ-cell stage	About 40%
SCO (Sertoli cell–only)	No germ cells	About 30%

Maturation arrest

Definition

- A histological pattern of **spermatogenic failure** in which germ cells develop normally up to one stage and **the sequence halts** at a premeiotic, meiotic or postmeiotic step. No mature sperm, or very few, are made.

Classification

- **Early:** arrest at spermatogonia or **primary spermatocyte** (meiotic failure).
- **Late:** arrest at the **spermatid** stage.
- **Complete** (all tubules) versus **incomplete** (foci progress to sperm): incomplete gives better retrieval.

Causes

- **Genetic:** complete **AZFb deletion** causes arrest; meiotic recombination defects in about half of men with maturation arrest; karyotype anomalies.
- **Acquired:** heat (varicocele, cryptorchidism), chemotherapy, radiation, drugs, orchitis; often idiopathic.

Clinical features and diagnosis

- Azoospermia or severe oligospermia, normal semen volume, palpable vasa.
- **Testis size and FSH are often normal**, so it can mimic obstruction (FSH ≤7.6 IU/L, long axis >4.6 cm).
- **Diagnosis needs testicular biopsy** (Bouin's fixative, never formalin), usually done as sperm extraction.

- Karyotype and Y microdeletion testing in all azoospermic men (AUA/ASRM).

Management

- Optimise: treat varicocele, stop exogenous testosterone; off-label hormonal stimulation (clomiphene, hCG) unproven.
- **Micro-TESE** with ICSI: retrieval **about 40%** (versus 30% in SCO).
- **Complete AZFb deletion: retrieval highly unlikely; do not operate.**
- No sperm: donor sperm or adoption. Round spermatid injection is experimental.
- Genetic counselling: sons inherit Y deletions.

Management of azoospermia

1. Pre-testicular (hypogonadotropic)

- Stop exogenous testosterone. **hCG 1500–5000 IU 2–3 times weekly ± FSH**; pulsatile GnRH in hypothalamic failure; dopamine agonist for prolactinoma.

2. Obstructive

- Microsurgical **vasovasostomy** or **vasoepididymostomy** (clear watery fluid or sperm parts in vasal fluid: vasovasostomy; thick toothpaste-like fluid without sperm: vasoepididymostomy).
- **TURED (transurethral resection of the ejaculatory ducts)** for EDO.
- CBAVD or unreconstructable: sperm retrieval by **MESA (microsurgical epididymal sperm aspiration)**, PESA (percutaneous epididymal sperm aspiration) or TESA + **ICSI (intracytoplasmic sperm injection)**; freeze sperm at reconstruction.

3. Non-obstructive

- **Optimisation (before retrieval):** stop exogenous testosterone; repair a clinical varicocele (evidence weak; no definitive evidence in NOA); off-label hormonal therapy (clomiphene, anastrozole, hCG) for low testosterone.
- **Micro-TESE (microdissection testicular sperm extraction) + ICSI:**
- **Principle:** spermatogenesis in NOA is **focal**. Under ×20–25 magnification, **larger, dilated, opaque tubules** are chosen; the least tissue is removed and the blood supply is preserved.
- Sperm found in **about 50%**; better than conventional TESE (Klinefelter 68%, Sertoli cell–only 30%).

Predictor	Retrieval (%)
Post-orchidopexy	74
AZFc deletion	70
Klinefelter	68
Post-chemotherapy	53
AZFa or AZFb complete deletion	**0: do not operate**

- FSH and testis size **do not** reliably predict retrieval.
- **ICSI:** fresh or frozen testicular sperm. Live birth per cycle is modest; miscarriage is higher (sperm DNA damage). Male offspring of AZF deletion carriers inherit the deletion.

4. Ejaculatory

- Retrograde: sympathomimetics, or sperm from alkalinised urine. Anejaculation: vibratory or electroejaculation.

5. Failure (no sperm): donor sperm or adoption; genetic counselling.

Also know

- Spermatogonial stem cell renewal by GDNF; in-vitro spermatogenesis and stem-cell transplantation are experimental.

Diagrams to draw

- Cross-section of a seminiferous tubule showing basal and adluminal compartments with germ cells maturing towards the lumen.
- Hypothalamic–pituitary–testis axis with the three levels of failure.
- Opened testis under the microscope: plump tubules versus thin Sertoli cell–only tubules.
- Maturation arrest: seminiferous tubule showing germ cells up to spermatocytes with no spermatids or sperm.

> **Key points to remember:**
>
> - Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
> - Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
> - FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
> - Confirm azoospermia on two centrifuged samples.
> - FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
> - Karyotype and Y microdeletion in all.
> - AZFa/b complete deletion: do not operate (no retrieval).
> - NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
> - Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
> - FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
> - Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

SAQ Male reproductive immunology and immunobead test

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "Male reproductive immunology" · P3 SAQ "Immunobead test for semen"

Introduction

- Haploid germ cells appear at puberty, after self-tolerance is established, so their antigens are "foreign". The testis is therefore **immune-privileged**.

Immune privilege

Anatomical barrier: blood–testis barrier (BTB)

- **Tight junctions between Sertoli cells**, formed at puberty, divide a **basal compartment** (spermatogonia, plasma-like fluid) from an **adluminal compartment** (later spermatocytes, spermatids). If the barrier is breached, sperm antigens meet the immune system.

Functional barrier

- A small antigen leak at the **rete testis, efferent ductules and epididymis** induces suppressor T cells and tolerance; a large load provokes immunity.

- Sertoli and local mediators: galectin-1, TGF-β (transforming growth factor-β), interferon-γ, soluble Fc receptor, **androgens**.
- Insult before puberty does not induce antibodies; the barrier also limits chemotherapy (isolated testicular relapse).

Antisperm antibodies (ASA)

- **Causes:** obstruction (**vasectomy** commonest); torsion, trauma, biopsy; infection (orchitis, prostatitis, STI [sexually transmitted infection]); heat (varicocele, cryptorchidism); genetic (human leukocyte antigen HLA-B28).
- **Effects:** agglutination, poor motility, failure to cross (poor penetration of) cervical mucus, failure to bind the zona and impaired fertilisation; head-bound antibody matters most.
- **Indications to test:** sperm agglutination or isolated asthenozoospermia; unexplained infertility; after vasectomy reversal.

Tests for ASA: immunobead test (IBT) and MAR

- The IBT detects ASA bound to sperm (direct) or free in body fluids (indirect).
- **Principle:** polyacrylamide beads coated with rabbit anti-human **IgG, IgA or IgM**.

Test	Method	Use
Direct IBT	Patient's motile sperm washed free of seminal plasma; mixed with beads coated with anti-human IgG/IgA; motile sperm with attached beads counted	Sperm-bound antibody: % bound, class and site (head/tail)
Indirect IBT	Antibody-free donor sperm incubated with heat-inactivated patient serum, seminal plasma or female serum/cervical mucus; then direct IBT done	Free antibody; patient sperm too few or immotile; female partner
MAR (mixed antiglobulin reaction)	Fresh unwashed semen + immunoglobulin (Ig) G (or IgA)-coated latex beads + antiglobulin (anti-IgG); beads stick to antibody-coated motile sperm; mixed sperm–bead clumps counted	Quick, cheap screen; IgG/IgA only

- Report: **% of motile sperm with beads**, immunoglobulin class, and **site** (head, midpiece, tail). Tail-tip binding is ignored.
- **WHO (World Health Organization) 2010 (5th edition) threshold: ≥50% of motile sperm bound is clinically significant.**
- **Head binding** and **IgA** matter most (IgA is produced locally and blocks mucus penetration; IgM is too large to enter semen).
- Both direct tests need motile sperm.
- **Also know:** ELISA (enzyme-linked immunosorbent assay) and flow cytometry; serum tests correlate poorly with sperm-bound antibody.

Management of ASA

Treatment	Comment
Condoms	Lack good evidence
Corticosteroids	Two controlled trials conflict; risk of avascular necrosis; not recommended

Treatment	Comment
Sperm washing + intrauterine insemination (IUI)	Bypasses cervical mucus; for mild binding; modest benefit, lacks good evidence (washing removes unbound, not bound, antibody)
IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)	Preferred, usual treatment; fertilisation largely unaffected, bypassing antibody effects

Infection and inflammation

- **Leukocytospermia:** >1 × 10^6 peroxidase-positive leukocytes/mL. Reactive oxygen species and cytokines damage sperm.
- **Orchitis:** mumps (post-pubertal, atrophy); bacterial (*Chlamydia*/gonococcus if young, *E. coli* if older); tuberculous; **autoimmune**, primary (antibodies with azoospermia) or secondary to vasculitis.
- **Also know:** sperm DNA fragmentation testing; prostatitis harms sperm via oxidative stress and ASA; seminal plasma suppresses female tract immunity.

Diagrams to draw

- Seminiferous tubule with Sertoli tight junctions and compartments.
- Motile sperm with beads attached at head and tail.

> **Key points to remember:**
> - BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
> - ASA: vasectomy commonest; ≥50% bound significant; head-bound IgA matters most.
> - Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
> - Treatment of choice: IVF/ICSI.
> - Leukocytospermia >1 million/mL.

Also, from the condensed book (not in the paper answer):

- **Inhibin B** marks Sertoli function; Sertoli tight junctions are the body's strongest, yet open to let spermatocytes pass.

3 · Spermatogenesis

A semen test reflects events about two months earlier.

*See the paper answer **Spermatogenesis, azoospermia and non-obstructive azoospermia** (Section 9, Ch 67).*

Also, from the condensed book (not in the paper answer):

- Output is ~ **123 million/day**, about 1000 per heartbeat.
- †Spermatogenesis needs high intratesticular testosterone, so exogenous testosterone suppresses it.

Stem cells

- Failed migration of primordial germ cells causes extragonadal germ cell tumours or Sertoli cell–only testes.

- "Selfish spermatogonia" (FGFR2/FGFR3 mutations) cause paternal-age disorders and spermatocytic seminoma.

Meiosis

	Mitosis	Meiosis
Divisions / daughters	1 / 2 identical	2 / 4 varied, haploid
Synapsis, crossover	None	Prophase I (synaptonemal complex, chiasmata)
Centromere division	Anaphase	**Anaphase II**

- **Defective recombination** occurs in ~half of maturation arrest. Altered crossover causes aneuploidy, so IVF/ICSI offspring have more chromosomal abnormalities.

Genetics

Y microdeletions of the azoospermia factor (AZF) region occur in **15%** of azoospermic and 6–8% of severely oligospermic men, the commonest defined molecular cause.

Deletion	Histology	Sperm retrieval
Complete AZFa	Sertoli cell–only	Highly unlikely
Complete AZFb	Maturation arrest	Highly unlikely
AZFc	Hypospermatogenesis / focal spermatogenesis	**Often found**
Partial AZFa/b	—	Ejaculated sperm possible

Paternal age matters because spermatogonia divide ~23 times a year, raising de novo dominant mutations (achondroplasia, Apert, Marfan), autism, schizophrenia and **DNA fragmentation**.

4 · Epididymis

- The 3–4 m duct runs from caput through corpus to cauda. Testicular sperm are **immotile and cannot fertilise**, except by ICSI.
- **Transport** takes **2–12 days**, driven mainly by spontaneous rhythmic contraction.
- **Storage**: the cauda holds about half the sperm; prolonged storage increases DNA fragmentation.
- **Maturation** gives progressive motility and zona binding. **Fertility is reached mostly by the late corpus or early cauda**.

> **KEY POINT**
>
> Pregnancy occurs even after vasoepididymostomy at the efferent ductules, but **the more distal the anastomosis, the better the fertility**. After vasectomy for >15 years, motility is lower after reversal because of epididymal dysfunction.

- The epididymis is **androgen dependent**, mainly on DHT, which explains impaired semen on 5α-reductase inhibitors.
- **Heat** abolishes storage, and **denervation** (spinal cord injury, diabetes) impairs transport.

5 · Vas deferens, seminal vesicles, ejaculatory ducts

Semen chemistry shows whether these organs are present and open.

- With a **muscle:lumen ratio of ~10:1** the vas is the most muscular hollow viscus.

- The **seminal vesicle** gives ~**80%** of seminal volume in the late fractions. Its fluid is alkaline and carries **fructose** and prostaglandins; **semenogelin I** forms the coagulum.
- The **ejaculatory duct** has no sphincter; reflux is prevented by its **acute insertion angle**.
- Obstruction may be **physical** or **functional** (vesicle myopathy from diabetes or drugs), and static TRUS may not tell them apart.
- An **ejaculate with pH <7.2, low volume and no fructose** means absent or blocked seminal vesicles, as in congenital bilateral absence of the vas (CBAVD) or ejaculatory duct obstruction.

6 · Spermatozoon, capacitation and fertilisation

Part	Contents
Head (4.5 × 3 µm)	Compacted nucleus; acrosome (enzymes)
Neck	Proximal centriole; axoneme starts
Midpiece	~75 mitochondria in a helix, outer dense fibres; ends at annulus
Principal piece / endpiece	Axoneme with fibrous sheath

- The axoneme is **"9 + 2"**, and **dynein** slides the doublets. Inner dynein arm mutants have no motility; outer arm mutants have reduced motility.
- **Paternal mitochondria are degraded** in the egg, so mitochondrial inheritance is maternal.
- **Capacitation** in the female tract gives hyperactivated motility and readiness for the **acrosome reaction**.
- Binding to zona **ZP3** triggers the acrosome reaction. Sperm **phospholipase C zeta (PLCζ)** triggers calcium oscillations, which activate the oocyte.

Sperm defects

- Morphology is graded by strict (Kruger) criteria. Head defects rarely reflect DNA quality, except **pinhead** and **multiple-tailed** sperm.
- In **globozoospermia** (round head, no acrosome) fertilisation fails even with ICSI, because of PLCζ deficiency.

	Ciliary dyskinesia (immotile cilia)	MMAF (multiple morphological abnormalities of the flagella; fibrous sheath dysplasia)
Motility	<10%	Severely reduced
Morphology	Normal on light microscopy; axonemal defect on electron microscopy (EM)	Short, bent, coiled or absent flagella
Cause	Recessive; respiratory disease, dextrocardia (Kartagener)	Genetic; also varicocele, reactive oxygen species (ROS)
Treatment	IVF-ICSI	Correct cause; else IVF-ICSI

7 · Ejaculation physiology †

Each cause of ejaculatory failure hits a specific phase.

ALGORITHM

1. **Emission** (sympathetic, T10–L2 via the hypogastric nerve; α1): vas, seminal vesicle and prostate contract and deliver semen to the posterior urethra.
2. The **bladder neck closes**. If it fails to close, then ejaculation is retrograde.
3. **Expulsion** (somatic, S2–S4 pudendal): rhythmic bulbospongiosus and pelvic floor contractions, with a relaxed external sphincter, give an antegrade ejaculate.

SAQ Retrograde ejaculation

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3f

Definition

- RE (retrograde ejaculation): semen passes backwards into the bladder at orgasm because the **bladder neck fails to close**; orgasm is preserved, ejaculate absent or low volume.

Physiology

Phase	Event	Control
Emission	Vas, seminal vesicles, prostate deliver semen to posterior urethra	Sympathetic T10–L2
Ejection	**Bladder neck closes**, external sphincter relaxes, bulbospongiosus contracts	Somatic S2–S4 + sympathetic reflex
Orgasm	Cerebral perception	Cortical

Causes

- **Anatomical: TURP (transurethral resection of prostate)** 42–100%, TUIP (incision) 5–45%; most other resections and enucleations; previous bladder neck reconstruction. Ejaculation-sparing techniques, PUL (prostatic urethral lift) and Aquablation preserve ejaculation.
- **Neurogenic: diabetic autonomic neuropathy** (commonest medical), RPLND (retroperitoneal lymph node dissection; less with nerve sparing), colorectal surgery, sympathectomy, aortic surgery, spinal cord injury, multiple sclerosis.
- **Drugs:** α-blockers (silodosin, tamsulosin), antipsychotics, antidepressants, thiazides.
- **Congenital:** Müllerian duct cyst, Wolffian duct anomaly, prune belly syndrome.

Evaluation

- Low-volume or absent ejaculate with orgasm; infertility.
- **Post-ejaculatory urinalysis:** void, ejaculate, void again; centrifuge. **>5–10 sperm per HPF (high-power field)** confirms RE; significant if urinary sperm approach or exceed antegrade sperm.
- Sperm and fructose in urine separate RE from **failure of emission**; TRUS (transrectal ultrasound) if EDO (ejaculatory duct obstruction) suspected.

Treatment

- Reassure if fertility not desired; stop offending drug.

- **Sympathomimetics** (close bladder neck): **pseudoephedrine** 60–120 mg (120 mg 6-hourly the day before and twice on the day); midodrine, ephedrine, phenylpropanolamine. Side effects: dry mouth, hypertension.
- **Imipramine** 25 mg twice daily; antihistamines (brompheniramine) add to sympathomimetics.
- **Bladder neck reconstruction:** poor results; collagen injection experimental.
- **Sperm retrieval:** alkalinise urine (sodium bicarbonate), void or catheterise after orgasm, wash sperm in buffer → IUI (intrauterine insemination), IVF (in vitro fertilisation) or ICSI (intracytoplasmic sperm injection).
- **Prevention:** nerve-sparing RPLND, ejaculation-preserving procedures for BPH (benign prostatic hyperplasia).

Diagram to draw: bladder neck open at ejaculation with semen entering the bladder.

> **Key points to remember:**
> - TURP and diabetic neuropathy are the commonest causes.
> - Post-ejaculatory urine: sperm and fructose confirm RE.
> - Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

68 Integrated Men's Health: Androgen Deficiency, Cardiovascular Risk and Metabolic Syndrome

1 · The men's health centre

The urologist is often the only doctor a man sees.

- Men avoid preventive care but seek help for erectile dysfunction (ED), infertility and voiding, so the urologist is the **front door** to a multidisciplinary centre and triages the whole man.

Age band	Dominant need	Key partners
18–39	Sexual, reproductive, testicular ("rule out hypogonadism"); preventive	Sports medicine, physiotherapy, mental health
40–64	Prostate and colon screening; disease management	Gastroenterology (GI), cardiology, endocrinology, bariatrics, pulmonology
65+	Falls, neurodegeneration, hearing/vision, nutrition	Geriatrics, neurology, audiology, ophthalmology

2 · The longevity gap

- Male life expectancy was **76.1** versus 81.1 years for females (2016).
- Males have higher mortality in **12 of 15** leading causes; the largest gap is **suicide (>3-fold)**.
- Prostate cancer is the only US malignancy with rising stage IV at diagnosis, so discuss screening.

3 · Urologic screening

- Screen sexual and voiding complaints together, with PSA and urinalysis per guidelines.

Questionnaire	Screens
Sexual Health Inventory for Men (SHIM)	Erectile function
Androgen Deficiency in the Aging Male (ADAM)	Androgen-deficiency symptoms
International Prostate Symptom Score (IPSS)	Lower urinary tract symptoms (LUTS)

4 · Urologic disorders of focus

Testosterone deficiency (TD)

SAQ Male menopause (andropause, late-onset hypogonadism)

Paper answer · P4

Asked as:

- Male menopause. — *Summer 2017 Paper IV Q3e (SAQ)*
- Andropause — *Winter 2017 Paper IV Q3e (SAQ)*
- Male menopause — *Summer 2019 Paper IV Q3f (SAQ)*

Type: SAQ (10 marks)

Read as: late-onset hypogonadism (testosterone deficiency of ageing men, "andropause"). Same topic as 2017-Summer Paper 4, Q3e and 2017-Winter Paper 4, Q3e; full answer given here.

Definition

- **Symptoms plus consistently low serum testosterone** in an ageing man.
- "Menopause" (or "Andropause") is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

Pathophysiology

- Testosterone falls slowly from middle age.
- Testosterone in the 70s is about one-third lower than in young men. (Another source: mean testosterone in the seventh decade is about **one-third lower** than in young men.)
- **Testis:** fewer Leydig cells, lower sperm output; FSH (follicle-stimulating hormone) rises about threefold.
- **Centre:** blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises with age**, so free testosterone (bioavailable [free + albumin-bound] testosterone) falls more.
- Worse with obesity, diabetes and **opioids**.

Clinical features

- **Sexual (most specific):** low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood.
- **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

Diagnosis

- **Morning (7–11 am) total testosterone, twice (repeated).** AUA (American Urological Association): **<300 ng/dL** (≈10.4 nmol/L). EAU (European Association of Urology): about **12 nmol/L**; calculated free testosterone if borderline. Calculated bioavailable testosterone when SHBG is abnormal.
- **LH (luteinising hormone):** high in primary, low or normal in secondary hypogonadism. Check **prolactin** if LH is low (Low LH → **prolactin**); pituitary MRI (magnetic resonance imaging) if testosterone <150 ng/dL or prolactin high.
- Baseline: PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

Treatment

- Lifestyle first: weight loss, exercise, treat comorbidities.
- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at 3–6 months (another source: at about 3 months).
- **Contraindications:**
- Prostate or breast cancer.
- **Wish for fertility** (TRT stops sperm production [suppresses spermatogenesis]; use hCG [human chorionic gonadotrophin] or clomiphene).
- Haematocrit >50% (raised haematocrit).
- Untreated severe sleep apnoea; recent MI (myocardial infarction) or stroke.

Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (stop/hold if >54%), PSA.

Safety evidence

- **TRAVERSE (2023):** 5,246 men aged 45–80 at high cardiovascular risk. TRT was **non-inferior for major cardiac events**; more atrial fibrillation and pulmonary embolism.
- **Testosterone Trials**: modest sexual benefit, **no cognitive benefit**.

Diagram to draw: hypothalamic–pituitary–testis axis with ageing defects.

> **Key points to remember:**
> - Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
> - SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
> - TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

Also, from the condensed book (not in the paper answer):

- For haematocrit, the AUA withholds TRT at >50% and the Endocrine Society at >54%; then reduce the dose or phlebotomise.
- **Randomised trials show TRT benefit** for obesity, metabolic syndrome, diabetes, muscle and bone, but **TRT is not a treatment for depression**.
- For men on chronic opioids, the Endocrine Society advises screening for TD and considering TRT. Exclude secondary causes before starting TRT.

> **KEY POINT**
>
> † **Fertility caution**: exogenous T suppresses LH/FSH and intratesticular T, which causes oligo- or azoospermia. Men wanting children should avoid TRT and use clomiphene, human chorionic gonadotrophin (hCG) or aromatase inhibitors (see Ch 69).

† TRT	Points
Contraindications	Near-term fertility wish; breast cancer; untreated prostate cancer; haematocrit >50%; untreated severe obstructive sleep apnoea (OSA); uncontrolled heart failure; MI/stroke within 3–6 months
Target	Mid-normal T (≈450–600 ng/dL)
Formulations	Gels, injections, pellets, each with its own monitoring plan

Erectile dysfunction

- ED affects >50% of men aged 40–70.
- **Every man with ED, of any age, is at cardiovascular disease (CVD) risk**. †Penile arteries are small, so ED precedes coronary events by ~2–5 years.
- Treatment steps from lifestyle to PDE5 inhibitors (never with nitrates), then injection, intraurethral or vacuum therapy, then prosthesis, which has the highest satisfaction. Platelet-rich plasma and shock wave therapy remain in trials.

BPH and infertility

- Refer BPH when medication fails, is not tolerated or is not wanted.
- Infertility is no pregnancy after 1 year; a male factor is present in ~half (see Ch 69).

5 · Cardiovascular prevention

- Only **lowering low-density lipoprotein cholesterol (LDL-C)** benefits primary prevention; raising HDL or lowering triglycerides does not. Screen lipids from age 20.

Statin decision (ACC/AHA)	Action
LDL-C ≥190 mg/dL	High-intensity statin with or without a nonstatin
LDL-C 70–189, 10-year risk ≥**7.5%**	Moderate–high intensity statin
Risk 5–7.5%	Moderate intensity may be started
Statin-intolerant, risk >20%	Ezetimibe or PCSK9 inhibitor

- USPSTF, NICE and ESC start statins at ≥10% risk.
- A **coronary artery calcium (CAC) score** is most useful at **intermediate risk**; zero means <1% atherosclerosis.

Blood pressure (BP) (ACC/AHA 2017)	Systolic	Diastolic
Normal	<120	and <80
Elevated	120–129	and <80
Stage 1	130–139	or 80–89
Stage 2	≥140	or ≥90

Elevated or stage 1 BP gets lifestyle change. Drugs are for stage 2, or stage 1 with age ≥65, T2DM, CVD, chronic kidney disease or 10-year risk ≥10%.

6 · Obesity and metabolic syndrome

Fat lowers testosterone, and low testosterone adds fat.

- Body mass index (BMI): overweight 25–29.9; obesity I 30–34.9, II 35–39.9, III ≥40.
- Advise weight loss at BMI ≥30, or 25–29.9 with CVD risk, OSA or osteoarthritis; GLP-1 agonists are approved.
- †Metabolic syndrome (ATP III) needs ≥3 of: waist >102 cm, triglycerides ≥150 mg/dL, HDL <40 mg/dL, BP ≥130/85, glucose ≥100 mg/dL. Adipose aromatase raises oestradiol, which lowers LH, so low T and metabolic syndrome drive each other.

7 · GI, skin and mental health

- Evaluate **constipation** in every man with UTI or retention. Colorectal screening runs from **45–75** years.
- Iron deficiency anaemia carries a >30-fold GI malignancy risk and needs investigation.
- Psoriasis is linked to MI, and suicide is much higher in men.

69 Male Infertility

1 · Definition and overview

Half of infertility involves the man, so always examine him.

- Infertility is no pregnancy after **1 year** of unprotected intercourse. It affects ~15% of couples; male and female factors contribute equally.
- Most conceptions occur in the first 3–6 cycles, so evaluation may start after 6 months.
- Idiopathic infertility is the largest group. **Varicocele is the commonest identifiable cause**, then obstruction.

† Classification	Examples
Pre-testicular (endocrine)	Hypogonadotropic hypogonadism (HH), Kallmann, prolactinoma, exogenous androgens, opioids, obesity, congenital adrenal hyperplasia (CAH)
Testicular	Klinefelter, Y microdeletion, cryptorchidism, torsion, orchitis, varicocele, chemotherapy/radiation, Leydig failure
Post-testicular	Obstruction (epididymal, vasal, ejaculatory duct), congenital bilateral absence of the vas (CBAVD), ejaculatory dysfunction, antisperm antibodies, structural sperm defects

2 · History

LAQ Discuss the role of urologist in reproductive endocrinology and male infertility practice.

Paper answer · P2

Type: LAQ (25 marks)
Source: 2022-Summer Paper 2, Q1

Introduction

- **Infertility:** no conception after 12 months of regular unprotected intercourse; about **15% of couples** (WHO, World Health Organization). The male contributes about half, yet many men are never examined.
- The urologist (andrologist) has five roles: **diagnose** the male factor, **treat endocrine causes**, **correct surgical causes**, **retrieve sperm for ART (assisted reproductive technology)**, and protect **general health and future fertility**.
- Commonest diagnoses: idiopathic, **varicocele** (commonest correctable), obstruction.

1. Diagnostic role

- **History:** duration, coital timing (semen is best after 1–2 days' abstinence), partner's age (fecundity falls after 35), cryptorchidism, mesh herniorrhaphy, RPLND (retroperitoneal lymph node dissection), diabetes, spinal injury, smoking, cannabis, opioids, **exogenous testosterone or anabolic steroids**.
- **Examination:** androgenisation; **testis size** (reflects spermatogenic mass); epididymis; **both vasa present?**; varicocele (standing, Valsalva).
- **Semen analysis**, at least two. WHO 6th edition (2021) 5th centiles: volume 1.4 mL, concentration 16 million/mL, total count 39 million, progressive motility 30%, normal forms 4%. These describe fertile men, not fertility cut-offs.

- **Genetics:** karyotype and **Y-chromosome microdeletion** (AZF, azoospermia factor) in non-obstructive azoospermia or severe oligospermia; **CFTR (cystic fibrosis transmembrane conductance regulator)** testing of both partners in CBAVD (congenital bilateral absence of the vas deferens).
- **Imaging:** scrotal US (ultrasound); **TRUS (transrectal US)** if volume <1 mL (ejaculatory duct obstruction, EDO), after post-ejaculate urine excludes retrograde ejaculation; renal US in vasal agenesis.

2. Endocrine evaluation (the urologist's reproductive endocrinology)

- Spermatogenesis needs a **high intratesticular testosterone (T)**. Morning total T + SHBG (sex hormone-binding globulin) + albumin gives bioavailable T; low is below about **280–300 ng/dL**; repeat borderline values (T is pulsatile and circadian).

Test	What it tells
LH (luteinising hormone)	Localises low T: high → testicular (Leydig) failure; low/normal → pituitary or hypothalamus
FSH (follicle-stimulating hormone) / inhibin B	Germ-cell mass. Azoospermia with **FSH >7.6 IU/L and testis long axis ≤4.6 cm → spermatogenic failure**; FSH ≤7.6 and >4.6 cm → obstruction
Oestradiol (E2)	**T (ng/dL) : E2 (pg/mL) <10** → aromatase excess (obesity)
Prolactin	Only with low T, low libido/ED (erectile dysfunction), headache or field defect
TSH (thyroid-stimulating hormone), cortisol, 17-hydroxyprogesterone	Only if signs; CAH (congenital adrenal hyperplasia)

3. Medical (endocrine) treatment

Condition	Mechanism	Treatment
Hypogonadotropic hypogonadism (Kallmann, pituitary)	Testis normal, signal missing: most treatable cause	**hCG (human chorionic gonadotropin) 1500–5000 IU 2–3 times weekly**, add hMG (human menopausal gonadotropin) 75 IU or recombinant FSH; pulsatile GnRH (gonadotropin-releasing hormone) pump for hypothalamic failure. Sperm may take 1–2 years
Anabolic steroids / exogenous T	Negative feedback suppresses LH → intratesticular T falls → azoospermia	Stop; hCG ± clomiphene to speed recovery. **Never give T to a man wanting fertility** (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine)
Hyperprolactinaemia	Prolactin inhibits GnRH	Repeat level; pituitary MRI (magnetic resonance imaging) if high; **cabergoline** (fewer side effects than bromocriptine)
Partial hypogonadism, obesity	Aromatisation of T to E2 suppresses LH	Weight loss; **clomiphene 25 mg daily** (off-label), **anastrozole 1 mg daily** or letrozole if T:E2 <10
CAH	Adrenal androgens suppress gonadotropins; testicular adrenal rest tumours	Glucocorticoids

Condition	Mechanism	Treatment
Primary testicular failure (Klinefelter 47,XXY)	Germ cells absent or few	No drug restores sperm; exogenous T worsens it; **micro-TESE** (microdissection testicular sperm extraction)

4. Surgical role

Problem	Operation
Clinical varicocele + abnormal semen	**Microsurgical subinguinal varicocelectomy** or embolisation: improves semen, DNA fragmentation and pregnancy. Not for US-only varicocele
Vasal or epididymal obstruction	**Vasovasostomy / vasoepididymostomy** (microsurgical intussusception)
EDO	**TURED** (transurethral resection of ejaculatory ducts)
Non-obstructive azoospermia	**Micro-TESE**, best technique; no operation for **AZFa/AZFb** deletions (no sperm found)
Unreconstructable obstruction	PESA/MESA (percutaneous/microsurgical epididymal sperm aspiration), TESA (testicular sperm aspiration)
Retrograde ejaculation; anejaculation	Sympathomimetics, urine sperm retrieval; **penile vibratory stimulation**, electroejaculation

5. Partner of the ART team, counselling and men's health

- Choose the least invasive route: natural conception, IUI (intrauterine insemination), IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection). **Cryopreserve sperm** at retrieval.
- **Fertility preservation:** sperm banking before chemotherapy, radiotherapy or RPLND; **onco-TESE** if azoospermic.
- **Genetic counselling:** AZFc deletion passes to sons; CFTR risk to offspring; paternal age ≥40 years raises offspring risk.
- **Infertility is a marker of general health:** higher risk of **testis cancer** (scan every azoospermic man), later comorbidity and mortality.

Recent advances: sperm DNA fragmentation testing; microfluidic sperm selection; genomic panels; prepubertal testicular tissue cryopreservation (experimental).

Also know: antisperm antibodies; globozoospermia and ciliary dyskinesia (ICSI); androgen receptor defects (high T and LH).

Diagram to draw: hypothalamic–pituitary–testis axis with feedback loops (T and E2 on LH; inhibin B on FSH).

> **Key points to remember:**
> - FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
> - Exogenous testosterone is a contraceptive: never give it for infertility.
> - Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
> - Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

Also, from the condensed book (not in the paper answer):

- The **female partner's age is the most important determinant**; fecundity falls after 35.
- Paternal age ≥40 needs counselling about offspring bipolar disorder, schizophrenia, autism and ADHD (ASRM/AUA 2021).
- Semen quality peaks after **1–2 days' abstinence**, so advise daily intercourse around ovulation; most lubricants impair sperm.

Gonadotoxins and drugs

> **KEY POINT**
>
> **Exogenous testosterone (T) is a contraceptive**. It is aromatised to oestradiol (E2), which feeds back on the pituitary and lowers LH. Intratesticular T then falls, and spermatogenesis falls with it. Recovery usually takes ≥4 months after stopping (after ≤2 years' use); after longer use it is unknown.

- Other harmful drugs include antiandrogens, spironolactone, corticosteroids, anabolic steroids and some antiretrovirals. Cannabis, alcohol and smoking also impair semen.

Infection and childhood conditions

- Infection injures either **anatomically** (duct scarring) or **functionally** (leucocytes, reactive oxygen species [ROS]).
- After torsion, about half have later spermatogenic effects.

Cryptorchidism (after orchidopexy)	Unilateral	Bilateral
Paternity	Near normal	Markedly reduced
Inhibin B	~⅔ of controls	~⅓ of controls

- **Laterality is the key prognostic factor.**
- The American Urological Association (AUA) advises orchidopexy within **18 months**, because germ cells are lost from 15–18 months. Later surgery before 10 years still improves sperm retrieval.

3 · Physical examination

Testis size and the vasa sort most azoospermic men.

- Examine the man standing. An engorged epididymis suggests obstruction, and testis size reflects spermatogenic mass.

Testis size	Abnormal
Seager caliper (long axis)	**≤4.6 cm**
Prader orchidometer (volume)	≤20 mL
Scrotal ultrasound (US)	Most accurate; added value unclear

- Unilateral vasal absence needs renal US; bilateral absence needs CFTR testing and renal US.
- Palpable seminal vesicles suggest ejaculatory duct obstruction (EDO).

Varicocele grade	Definition
Subclinical	Seen only on Doppler; **do not treat**

Varicocele grade	Definition
I	Palpable only with Valsalva, standing
II	Palpable without Valsalva, not visible
III	Visible through scrotal skin

4 · Endocrine evaluation

ALGORITHM

1. Take a morning sample for total T, sex hormone-binding globulin (SHBG) and albumin, and calculate bioavailable T.
2. Measure LH and follicle-stimulating hormone (FSH), then E2.
3. Measure prolactin only if indicated.

- Adequate total T is ≥**280–300 ng/dL**. Low T is common even with density >20 million/mL, so assess every infertile man.
- Low SHBG (obesity, diabetes) makes total T under-read; high SHBG (ageing, cirrhosis) makes it over-read.
- A T (ng/dL) to E2 (pg/mL) ratio **<10:1** means aromatase excess.
- Prolactin of 20–50 µg/L needs no further work-up; a significant rise needs pituitary MRI.

5 · Semen analysis

SAQ Semen analysis nomenclature.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2018-Summer Paper 4, Q3e

Collection (WHO, World Health Organization, 2021, 6th edition)

- Masturbation into a clean wide-mouthed non-toxic container (special condom if needed); complete sample.
- Abstinence **2–7 days** (evidence: 1–2 days best). Keep at 20–37 °C; analyse within **1 hour**, after liquefaction (~30 min).
- At least **two samples, 2–3 weeks apart**, because of variability.

Lower reference limits (5th centile, fertile men)

Parameter	Lower limit
Volume	**1.4 mL**
Concentration	**16 million/mL**
Total count	**39 million/ejaculate**
Total motility	**42%**
Progressive motility	**30%**
Normal forms (strict)	**4%**
Vitality	**54%**
Peroxidase-positive leucocytes	<1 million/mL

- These are reference values, not fertility cut-offs. The 6th edition grades motility as progressive, non-progressive or immotile.

Nomenclature

Term	Meaning
Normozoospermia	All parameters at or above reference
Oligozoospermia	Total count (or concentration) below reference
Asthenozoospermia	Progressive motility below reference
Teratozoospermia	Normal forms below reference
OAT (oligoasthenoteratozoospermia)	All three abnormal
Azoospermia	No sperm, **including centrifuged pellet**
Cryptozoospermia	Sperm absent in fresh smear, present only after centrifugation
Aspermia	No ejaculate (dry orgasm)
Hypospermia	Volume below reference
Necrozoospermia	Low vitality (many dead sperm)
Leucocytospermia	Peroxidase-positive white cells above 1 million/mL
Haemospermia	Blood in ejaculate
Globozoospermia	Round-headed sperm without acrosome

Notes

- **Low volume or aspermia:** post-ejaculatory urine for retrograde ejaculation; TRUS (transrectal ultrasound) for EDO (ejaculatory duct obstruction).
- **Vitality:** eosin–nigrosin, or **HOS (hypo-osmotic swelling)** test, which keeps sperm usable for ICSI (intracytoplasmic sperm injection).
- Only azoospermia is definitive; values overlap between fertile and infertile men.
- **Also know:** Kruger strict morphology; DNA fragmentation; antisperm antibodies (MAR, mixed antiglobulin reaction); semen pH no longer recommended.

Diagram to draw: normal sperm (head, midpiece, tail) beside a WHO reference box.

> **Key points to remember:**
> - WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
> - Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
> - Two samples 2–3 weeks apart; abstinence 2–7 days.

Also, from the condensed book (not in the paper answer):

- **Low volume (<1.0 mL)** suggests EDO, CBAVD, hypoandrogenism or retrograde ejaculation. Do a **postejaculatory urinalysis**; if there are no sperm in urine, do transrectal ultrasound (TRUS).
- †Sperm DNA fragmentation testing is not routine; use it for recurrent pregnancy loss or failed assisted reproduction (ART).

6 · Genetic testing

ALGORITHM

1. If non-obstructive azoospermia (NOA) or density **<5 million/mL**, then do karyotype and Y-microdeletion testing (ASRM/AUA 2021).
2. If CBAVD (or idiopathic EDO), then test CFTR in the man **and his partner**.

SAQ Genetic causes of male infertility. Enumerate and classify.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2017-Summer Paper 3, Q3d

Introduction

- Genetic defects underlie many cases of **azoospermia** and **severe oligozoospermia**. They decide **sperm retrieval odds** and **transmission to offspring** through ICSI (intracytoplasmic sperm injection).

Classification

Group	Condition	Key features
1. Chromosomal, numerical	**Klinefelter syndrome (47,XXY)**	**Commonest genetic cause**; 1 in 500–1000; small testes, high FSH (follicle-stimulating hormone), low testosterone, usually azoospermia. Non-mosaic in 80–90%. Testicular sperm in about 50% (micro-TESE, microdissection testicular sperm extraction)
	47,XYY; **46,XX male** (SRY gene translocated to X)	XX male: azoospermia, no sperm retrievable
2. Chromosomal, structural	**Robertsonian and reciprocal translocations**, inversions, ring Y	Oligozoospermia; recurrent miscarriage
3. Y chromosome microdeletions (Yq, AZF (azoospermia factor))	**AZFa** (USP9Y, DDX3Y), **AZFb**, **AZFc** (DAZ genes)	Complete AZFa or AZFb deletion: **no sperm at retrieval**. AZFc: sperm often in ejaculate or testis; sons inherit the deletion
4. Single-gene: excurrent ducts	**CFTR (cystic fibrosis transmembrane conductance regulator) mutations**: CBAVD (congenital bilateral absence of vas deferens); idiopathic ejaculatory duct obstruction	Low-volume azoospermia, normal spermatogenesis; **test both partners** (ΔF508)
5. Single-gene: hypothalamic–pituitary	**Kallmann syndrome** (KAL1, anosmia); GnRH (gonadotropin-releasing hormone) receptor mutations	Low LH (luteinising hormone) and FSH; treat with hCG ± FSH
6. Androgen action	**Androgen receptor** mutation (long CAG repeat); 5α-reductase type 2 deficiency	High testosterone and LH, poor sperm

Group	Condition	Key features
7. Sperm structure and motility	**Globozoospermia** (SPATA16, DPY19L2: no acrosome); macrocephalic sperm (AURKC); **PCD (primary ciliary dyskinesia)**/Kartagener (DNAH5, DNAI1: immotile sperm)	ICSI; macrocephalic sperm: ICSI not advised (aneuploidy)

Genetic testing indications

- **Karyotype and Y microdeletion:** non-obstructive azoospermia and severe oligozoospermia (**<5 million/mL**, ASRM/AUA (American Society for Reproductive Medicine/American Urological Association) 2021; EAU (European Association of Urology) uses <10 million/mL for karyotype).
- **CFTR:** CBAVD, idiopathic obstructive azoospermia; screen the partner; counsel before ICSI.
- **Also know:** Noonan syndrome, myotonic dystrophy.

Diagram to draw: Y chromosome long arm showing AZFa, b, c.

> **Key points to remember:**
> - Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
> - AZFa or AZFb complete deletion: do not attempt retrieval.
> - CBAVD = genital form of cystic fibrosis; test both partners.

Also, from the condensed book (not in the paper answer):

- AZFc deletion with oligozoospermia declines progressively, so **cryopreserve sperm** early.

7 · Imaging

- Do not screen, because incidental microlithiasis causes harm. Varicocele is clinical.

SAQ Ejacultory duct obstruction

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 3, Q3b

Assumed meaning: ejaculatory duct obstruction (EDO).

Anatomy

- Intraprostatic continuations of the seminal vesicles, joined by the vasal ampullae; they lack muscle and open at the **verumontanum**. Seminal vesicles give about 60% of the ejaculate and its fructose.

Aetiology

- **Congenital (commonest):** midline (Müllerian/utricular) cyst, Wolffian duct cyst, atresia. Idiopathic EDO carries CFTR (cystic fibrosis transmembrane conductance regulator) mutations in about half.
- **Acquired:** infection, instrumentation, duct stones, compression.
- **Zinner syndrome:** seminal vesicle cyst, ipsilateral renal agenesis, EDO.

- Complete bilateral EDO causes under 5% of azoospermia. Partial EDO gives oligoasthenospermia.

Clinical features

- Infertility, painful ejaculation, haematospermia, perineal pain.

Diagnosis

Semen (complete bilateral EDO)

- Volume **<1.0 mL**, **acidic pH**, low or absent **fructose**, azoospermia.
- Palpable vas, normal FSH (follicle-stimulating hormone) and testes.
- Exclude retrograde ejaculation and CBAVD (congenital bilateral absence of the vas deferens).

Investigations

Test	Method / finding
TRUS (transrectal ultrasound)	Seminal vesicle AP diameter **>1.5 cm**, duct >2.3 mm, midline cyst. Low specificity
Seminal vesicle aspiration	**≥3 sperm per high-power field** confirms obstruction
Vesiculography	Contrast into vesicles; invasive
Chromotubation	Indigo carmine into vesicles; dye at the veru on cystoscopy

Management

TURED (transurethral resection of the ejaculatory ducts)

- For azoospermic or severely oligoasthenospermic men, after the couple declines IVF (in vitro fertilisation).
- **Steps:** TRUS-guided aspiration (cryopreserve sperm) → instil indigo carmine with contrast → 24-Fr resectoscope, finger in rectum → resect the veru until dye effluxes. Spare bladder neck, sphincter and rectum. Catheter overnight.
- **Results:** volume rises in most; sperm appear in about half. Cold-knife incision alone re-obstructs.

Complications

- **Urine reflux** into the ducts, **epididymitis**, retrograde ejaculation, rectal injury, re-obstruction.

Alternatives

- **Seminal vesiculoscopy** (4–6 Fr via utricle): avoids reflux; restenosis unknown. Sperm retrieval with ICSI (intracytoplasmic sperm injection).

Diagram to draw: sagittal prostate with midline cyst compressing the ducts at the veru.

> **Key points to remember:**
> - Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
> - TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
> - TURED: sperm return in about half; reflux and epididymitis are its price.

8 · Azoospermia: obstructive (OA) vs non-obstructive (NOA)

This split decides between reconstruction and sperm extraction.

	OA	NOA (spermatogenic dysfunction)
FSH	**≤7.6 IU/L**	>7.6 IU/L
Testis long axis	>4.6 cm	≤4.6 cm
Management	Reconstruction (see Ch 70) or sperm retrieval	Genetic tests; microdissection testicular sperm extraction (micro-TESE) with intracytoplasmic sperm injection (ICSI)

ALGORITHM

1. If the vasa are absent, then test CFTR in both partners.
2. If the vasa are present and volume is <1 mL, then do postejaculatory urinalysis. Sperm in urine means retrograde ejaculation; no sperm means TRUS for EDO.
3. If volume is normal, then use FSH and testis size to separate OA from NOA.

9 · Specific causes and treatment

Klinefelter syndrome (47,XXY)

SAQ Klinefelter's syndrome.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2018-Summer Paper 4, Q3b

Definition and genetics

- **47,XXY**: commonest sex chromosome aneuploidy and **commonest genetic cause of male infertility**; 1 in 500–1000 male births.
- Non-mosaic in **80–90%**; the rest mosaic **46,XY/47,XXY** (milder, more sperm) or 48,XXXY variants. Extra X from non-disjunction (maternal age).
- Underdiagnosed: most never diagnosed; increasingly detected on NIPT (non-invasive prenatal testing).

Pathophysiology

- Seminiferous tubules **hyalinise** progressively from puberty; germ cells are lost. Leydig cells fail, so testosterone falls. Puberty starts on time, then stalls.

Clinical features

Age	Features
Infant	Usually normal; micropenis, cryptorchidism, hypospadias
Adolescent/adult	**Small firm testes (<3.5 cm, 1–4 mL)**, gynaecomastia, tall stature with long legs (only ~30%), sparse hair, female fat distribution
Neurodevelopment	Language, learning and attention difficulties

Hormones and tests

- **High FSH (follicle-stimulating hormone) and LH (luteinising hormone), low or low-normal testosterone, high oestradiol** (raised oestradiol:testosterone ratio). Inhibin B

low.

- Semen: **azoospermia** in most (~8% have ejaculated sperm).
- **Karyotype** confirms (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine, guideline: karyotype all men with non-obstructive azoospermia or severe oligozoospermia).

Management

- **Fertility: micro-TESE** (microdissection testicular sperm extraction) + ICSI (intracytoplasmic sperm injection). Sperm found in **~40–50%** (68% in one series); offspring are usually chromosomally normal.
- Spermatogenesis declines with age, so retrieve early (late adolescence or young adulthood). Some give clomiphene, anastrozole or hCG (human chorionic gonadotropin) beforehand.
- **TRT (testosterone replacement therapy):** for hypogonadism (bone, muscle, mood, libido). It **suppresses spermatogenesis**, so offer sperm retrieval or banking **before** starting.
- Gynaecomastia: reduction mammoplasty if persistent. Speech, educational and psychological support.

Associated risks

- Osteoporosis, metabolic syndrome, diabetes, VTE (venous thromboembolism), autoimmune disease.
- **Mediastinal (extragonadal) germ cell tumours**; male breast cancer (~1% lifetime).

Diagram to draw: hypothalamic–pituitary–testis axis showing high FSH and LH, low testosterone.

> **Key points to remember:**
> - 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
> - Small firm testes, high FSH and LH, azoospermia.
> - Micro-TESE ~50% success; retrieve sperm before testosterone.
> - Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

Also, from the condensed book (not in the paper answer):

- **Trap**: exogenous T suppresses residual spermatogenesis, so extract sperm before starting T. If fertility is wanted, use aromatase inhibitors or human chorionic gonadotrophin (hCG).

Hypogonadotropic hypogonadism — most treatable cause

State	Hormones	Treatment
Kallmann (anosmia; KAL1, GNRHR)/complete HH	Low LH, low FSH	hCG with or without recombinant FSH (rFSH) or human menopausal gonadotrophin (hMG)
Isolated LH deficiency ("fertile eunuch")	Low LH, normal FSH	hCG
Isolated FSH deficiency	Low FSH	FSH
Hypothalamic GnRH deficiency	Low GnRH	Pulsatile subcutaneous GnRH every 2 h, for ≥6 months

State	Hormones	Treatment
Partial HH (common)	LH inappropriately low for T	Selective oestrogen receptor modulator (SERM) or aromatase inhibitor

- Give hCG 1500–5000 IU 2–3 times weekly (max 10,000/week), titrated to T; add hMG or rFSH 75 IU 2–3 times weekly. After androgen abuse, allow 1–2 years.
- Treat prolactinoma with cabergoline, and CAH with corticosteroids.

Empirical medical therapy

Drug	Dose	Notes
Clomiphene (SERM)	25 mg daily or 50 mg alternate days; max 100 mg	Blocks E2 feedback, so LH and FSH rise; off-label; no change in prostate-specific antigen (PSA) or haematocrit; target T 600–800 ng/dL
Anastrozole / letrozole	Anastrozole 1 mg daily	Raises T and lowers E2; use when T:E2 <10:1 (obesity); caution with bone density

Primary failure and androgen resistance

- High LH with low T has no accepted therapy; exogenous T is contraindicated, so offer sperm extraction.
- Androgen receptor defects give high T, LH and E2 and are treated with ICSI.

CBAVD

SAQ CBAVD

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 3, Q3f

Definition

- **CBAVD (congenital bilateral absence of the vas deferens):** both vasa absent, causing **obstructive azoospermia** with normal spermatogenesis.
- Often a **genital form of CF (cystic fibrosis)**: nearly all men with CF lack both vasa.

Genetics

- **CFTR (cystic fibrosis transmembrane conductance regulator)**: chloride channel gene; more than 1600 mutations.
- Severe mutation on both alleles: CF. **Mild mutation on one or both alleles: CBAVD.** Commonest severe mutation **ΔF508**; the **5T variant** (intron 8) is a common mild allele.
- A minority have other mesonephric duct gene defects instead, with more **renal agenesis**.

Clinical features

- **Vasa not palpable** (Meacham: a presumed vas slipping away three times is absent).
- Caput epididymis present and distended; body and tail often absent.
- Normal testis size, FSH (follicle-stimulating hormone) and testosterone.
- Seminal vesicles absent or hypoplastic: **semen volume <1.0 mL, acidic, fructose-negative**, azoospermic.
- Sinus or chest symptoms may reveal mild CF.

Evaluation

- Semen analysis; FSH; **TRUS (transrectal ultrasound)** for seminal vesicles.
- **Renal ultrasound** (renal agenesis in about 1 in 10).
- **CFTR testing of the man and his female partner** (carrier rate about 1 in 20 in northern Europeans). Panels cover 25–40 mutations, so a negative panel leaves residual risk.
- Testis biopsy unnecessary when FSH and testis size are normal.

Management

- **Genetic counselling**: if the partner carries a mutation, offspring may have CF; PGT (preimplantation genetic testing) or donor sperm can be offered.
- Reconstruction is impossible, so **surgical sperm retrieval + ICSI (intracytoplasmic sperm injection)**:

Technique	Points
MESA (microsurgical epididymal sperm aspiration)	Best: many motile sperm; cryopreserve several vials
PESA (percutaneous epididymal aspiration)	Local anaesthetic; fewer sperm
TESA/TESE (testicular aspiration/extraction)	If epididymal retrieval fails
- ICSI results are good; one study reported more miscarriage and stillbirth.	
- **Also know:** CUAVD (unilateral absence) is a Wolffian duct defect with ipsilateral renal agenesis.	

Diagram to draw: absent vas and seminal vesicle with intact testis and caput epididymis.

> **Key points to remember:**
> - CBAVD = mild CFTR mutation; test both partners.
> - Low-volume, acidic, fructose-negative azoospermia; normal FSH.
> - Renal ultrasound; MESA + ICSI.

Varicocele

*See the paper answer **Varicocele: pathophysiology, controversies and surgical management** (Section 10, Ch 87).*

Also, from the condensed book (not in the paper answer):

- Varicocele affects ~15–20% of all men versus 35–50% of infertile men (see Ch 70).

Antisperm antibodies and structural defects

- For antisperm antibodies, washing fails and steroid trials conflict, so use **ICSI**.
- Globozoospermia needs ICSI with or without oocyte activation, and ICSI is not recommended for macrocephalic multiflagellate sperm (AURKC).

10 · Lifestyle and obesity

- Obesity raises E2 and lowers LH and T, and worsens ART outcomes, so advise weight loss. Stop smoking, cannabis, excess alcohol and anabolic steroids, and avoid heat.

11 · Assisted reproduction

Sperm count decides the technique.

- †Choose by total motile sperm count (TMSC): IUI if processed TMSC is ≥5–10 million; IVF below that or after failed IUI; ICSI for severe oligozoospermia, retrieved sperm, antisperm antibodies or prior fertilisation failure.
- †Retrieval in OA is by percutaneous or microsurgical epididymal sperm aspiration (PESA/MESA) or testicular aspiration; in NOA it is micro-TESE.

70 Surgical Management of Male Infertility

1 · Principles

- In vitro fertilisation (IVF) with intracytoplasmic sperm injection (ICSI) treats even non-obstructive azoospermia (NOA), but it is costly, burdens the female partner and bypasses natural genetic barriers.
- Treating the man (reconstruction for obstructive azoospermia (OA), varicocelectomy) is safest and cheapest, and moves the couple **down the ladder**: donor sperm, then IVF/ICSI with testicular sperm, then IVF/ICSI with ejaculated sperm or intrauterine insemination (IUI), then natural conception.

2 · Surgical anatomy

Structure	Arteries (origin)
Testis	**Testicular** (aorta) — main; deferential (internal iliac/superior vesical); cremasteric (inferior epigastric; mainly tunica vaginalis)
Epididymis	Superior/medial epididymal (from testicular a.) supply the caput; inferior epididymal (from deferential a.) supplies the cauda; interconnected
Vas	Abdominal end: deferential only. Testicular end: deferential and inferior epididymal. **No supply from cremasteric or cord vessels**

- The testicular artery **exceeds deferential and cremasteric combined**; after vasectomy, preserve it at any later surgery.
- The epididymis can be dissected off the testis if the **superior epididymal artery is intact**; this gains **4–6 cm**.
- If the vas is blocked and devascularised at two sites, the segment between fibroses. So **never do two simultaneous vasovasostomies on one vas**; excise it and do one tension-free anastomosis.
- **Excurrent ducts**: 7–11 efferent ducts enter the caput, but a **single tubule** runs from the caput–corpus junction to the vas. So injury distal to the caput (even **one stitch**) obstructs the whole side.

3 · Diagnostic testis biopsy

LAQ Azoospermia: testicular biopsy and surgical sperm retrieval techniques

Paper answer · P1 · P2 · P3 · P4 · merged from 4

Asked as: P4 LAQ "Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival." · P2 SAQ "Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction)" (asked as "Sperm extraction in Obstructive azoospermia", "Testicular aspiration of sperms and assisted reproduction" and "Sperm Retrival techniques") · P1 SAQ "Testicular biopsy and TESA (testicular sperm aspiration)" (asked as "TESA" and "Testicular biopsy") · P3 SAQ "Sperm retrieval techniques"

Definitions

- **Azoospermia:** no sperm in the **centrifuged pellet** of the ejaculate on **at least two** semen analyses. Found in about 1% of men and 10–15% of infertile men.
- Not **cryptozoospermia** (rare sperm in pellet) or **aspermia** (no ejaculate).
- **OA (obstructive azoospermia):** no sperm in the centrifuged ejaculate despite **normal spermatogenesis**, from blockage of the ducts.
- **TESA (testicular sperm aspiration):** percutaneous needle aspiration of seminiferous tubules under local anaesthesia to retrieve sperm for **ICSI (intracytoplasmic sperm injection)**, which needs only one sperm per oocyte.

Classification

Type	Causes	Clues
Pre-testicular	Kallmann syndrome, pituitary disease, **exogenous testosterone**	Low FSH (follicle-stimulating hormone), LH (luteinising hormone)
Testicular: **NOA (non-obstructive azoospermia)**	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy, mumps	Small testes, high FSH
Post-testicular: **OA**	CBAVD (congenital bilateral absence of the vas deferens), vasectomy, epididymal or ejaculatory duct obstruction, hernia-repair injury	Normal testes and FSH
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Evaluation (brief)

- Absent vasa (CBAVD): CFTR (cystic fibrosis transmembrane conductance regulator) testing of both partners. Volume <1 mL: post-ejaculate urine, then TRUS (transrectal ultrasound).
- **FSH ≤7.6 IU/L and testis long axis >4.6 cm: obstruction likely;** FSH >7.6 and axis ≤4.6 cm: spermatogenic failure.
- **Karyotype and Y microdeletion** in NOA. **AZF (azoospermia factor) a or b complete deletion: no sperm, do not operate.**

Testicular biopsy

Indications

- Azoospermia with **normal testicular size, palpable vasa and normal FSH**: separates OA from NOA (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine).
- Since ICSI needs one sperm, biopsy is now **diagnostic and therapeutic**: plan it as TESE (testicular sperm extraction) and **cryopreserve** sperm.
- **Not needed:** CBAVD with normal FSH (spermatogenesis always present); high antisperm antibodies after iatrogenic vasal injury. Do not operate with complete AZFa or AZFb deletion.

Types

Type	Method	Use	Limits
Open biopsy	Small incision in the albuginea, pea-sized tissue	**Gold standard** for histology and retrieval	Open procedure
Percutaneous core biopsy	14 G biopsy gun under cord block	Fresh retrieval in OA; core of tissue	Blind; may injure epididymis or testicular artery; avoid in scarred scrotum
FNA (fine-needle aspiration), i.e. TESA	23 G needle, suction, several passes	OA retrieval; backup to epididymal methods	Few tubules, poor architecture; low yield in NOA
Microdissection biopsy (micro-TESE)	Operating microscope, plump tubules	NOA	Long operation

- **Also know:** multi-site FNA "mapping" in NOA.

Open biopsy technique

1. Local anaesthesia with **spermatic cord block** (or general/spinal); choose the larger testis.
2. 2 cm transverse scrotal incision; open tunica vaginalis.
3. Stay sutures; 3–4 mm albugineal incision avoiding surface vessels (loupes).
4. Excise extruding tubules with iris scissors, **no forceps**.
5. Fix in **Bouin's, Zenker's or glutaraldehyde, never formalin** (distorts histology).
6. **Touch prep** at once, then wet squash prep for sperm; if negative, contralateral biopsy.
7. Close albuginea and tunica vaginalis separately; no antibiotics.

Histology patterns

Pattern	Meaning
Normal spermatogenesis	Suggests obstruction
Hypospermatogenesis	All stages, fewer cells
Maturation arrest	Stops at a germ-cell stage
SCO (Sertoli cell–only)	No germ cells
Tubular sclerosis/hyalinisation	End-stage (e.g. Klinefelter)

Surgical sperm retrieval: indications

- **OA with normal spermatogenesis:** CBAVD, unreconstructable obstruction, failed or declined vasectomy reversal, back-up at reconstruction (vasoepididymostomy; frozen for IVF [in vitro fertilisation]), or failed epididymal aspiration.
- **NOA (testicular failure):** only testicular retrieval works; TESA gives poor yield; **micro-TESE (microdissection testicular sperm extraction)** preferred.
- **Anejaculation** (spinal cord injury): vibratory stimulation or electroejaculation first; retrieval after these fail.
- All retrieved sperm need **ICSI**, because sperm from obstructed systems have poor motility; cryopreserve spare sperm.

Techniques of sperm retrieval

Technique	Method / site	Best for	Merits, limits and notes
PESA (percutaneous epididymal sperm aspiration)	Fine needle into caput epididymis, local anaesthesia	OA	Simple; motile but few sperm; may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule opened under microscope at 16–25×; micropipette by **capillary action**, never suction	**OA (best)**	Many (most) motile sperm, multiple frozen vials; needs microsurgical skill
TESA	Testis, 23 G needle with suction, many passes	OA	Least invasive; immature, immotile sperm, small yield; poor in NOA
Percutaneous core biopsy	14 G biopsy gun	OA	Fresh retrieval; core of tissue; blind; vessel injury
Conventional TESE	Open biopsy, one or more sites	OA, NOA	When epididymis is absent or empty; gold standard for tissue diagnosis; blind sampling
Micro-TESE (Schlegel 1999)	Testis bivalved (opened in an avascular plane); **larger, opaque tubules** picked under the microscope	**NOA (best)**	Most sperm, least tissue; sperm in about half; long operation
Vasal aspiration	Vasotomy, testicular end flushed	At reconstruction	

MESA technique

1. Testis delivered; epididymis inspected at 16–25×.
2. Dilated tubule opened with a microknife; fluid checked at once.
3. No motile sperm: close and **move proximally** (start distally), up to the efferent ductules (proximal sperm are more motile; the most motile are often in the efferent ductules).
4. Collect by **capillary action** into micropipettes, never suction; 4–12 pipettes.
5. **Cryopreserve** in several aliquots; close tubule with 10-0 suture.

TESA technique

1. Spermatic cord block with local anaesthetic.
2. Testis fixed; **23 G needle** (or angiocatheter) on a high-suction syringe.
3. Multiple passes in different directions draw tubules into the needle (often 10–20 passes needed; P1 table: 3–4 passes).
4. Immediate wet examination for sperm; repeat until sperm are found. **Cryopreserve** spare sperm.

TESA: advantages and limitations

- **Advantages:** office procedure, no microsurgery, cheap, repeatable.
- **Limitations:** few tubules with poor architecture (not diagnostic histology); small numbers of immature, immotile sperm; often too few to freeze; blind.

Micro-TESE logic

- Spermatogenesis in NOA is **focal**. Plump, larger tubules hold sperm; thin white ones are Sertoli cell–only.
- Embryologist examines tissue in theatre; map positive sites; if negative, explore the other testis.
- It removes the least tissue for the most sperm, and is better than conventional TESE.

Choice: epididymal sperm first (MESA) in OA; testicular (TESA/TESE) if epididymal sperm are absent or retrieval fails; micro-TESE in NOA.

Results

- OA: retrieval almost always succeeds; good pregnancy rates.
- NOA: micro-TESE finds sperm in **about half**; fine-needle aspiration yields far fewer sperm than micro-TESE. Retrieval depends on the cause:

Condition	Micro-TESE retrieval
Post-orchidopexy	74%
AZFc deletion	70%
Klinefelter	68%
Post-chemotherapy	53%
Maturation arrest	40%
SCO	30%
AZFa/AZFb deletion	**0%: do not operate**

- **Before retrieval in NOA:** stop testosterone; repair clinical varicocele; off-label clomiphene, anastrozole or hCG (human chorionic gonadotropin).

Assisted reproduction and use of retrieved sperm

- **IUI (intrauterine insemination):** washed sperm into the uterus; mild male factor.
- **IVF:** oocytes and sperm combined in the laboratory.
- **ICSI:** one sperm injected into each oocyte; essential for surgically retrieved (testicular) sperm (poor motility).
- Frozen retrieved sperm equal fresh, except possibly in spinal cord injury.
- NOA pregnancies: more miscarriage (sperm aneuploidy, DNA damage); genetic counselling. No sperm: donor sperm or adoption.
- ICSI bypasses natural selection: risk of transmitting genetic defects. Offer karyotype, Y-microdeletion and CFTR testing as indicated before ICSI.
- Female burden: ovarian stimulation, multiple pregnancy.

Complications

- **Haematoma** (commonest), pain, infection (rare), testicular vascular injury (TESA, core biopsy).
- **Inadvertent epididymal biopsy** (most serious: causes obstruction); epididymal obstruction (PESA).
- Testicular atrophy or transient fall in testosterone (TESE).

Also know

- Testicular sperm may beat ejaculated sperm when DNA fragmentation is high.

- Post-mortem retrieval needs prior consent.
- Microfluidic sperm sorting.

Diagrams to draw

- Testis and epididymis showing PESA, MESA and TESA sites; testis with 23 G needle on a suction syringe and multiple passes; flow to ICSI.
- Open biopsy: small albugineal window with extruding tubules snipped off.
- Micro-TESE: bivalved testis with plump (dilated) spermatogenic tubules beside thin Sertoli cell–only tubules.

Key points to remember:

- Azoospermia: no sperm in centrifuged pellet on two samples.
- Biopsy indication: azoospermia, normal testes, vasa and FSH.
- Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
- OA: normal FSH and testes; MESA is best (capillary action, never suction; move proximally): many motile sperm, multiple vials.
- TESA: 23 G needle, suction, local anaesthesia; for OA; TESA/TESE when epididymal sperm absent.
- NOA: micro-TESE best (sperm in about half); TESA yield poor.
- AZFa/AZFb deletion: 0% retrieval, do not operate.
- All surgically retrieved sperm need ICSI; cryopreserve extras.
- Haematoma commonest; epididymal biopsy most serious.

Also, from the condensed book (not in the paper answer):

- Biopsy is usually bilateral and planned as testicular sperm extraction (TESE), with an andrology laboratory on site for wet examination and freezing.

ALGORITHM

1. Give a cord block; choose the larger testis (opposite a varicocele).
2. Make a 2 cm incision, open the tunica vaginalis, place stay sutures.
3. Incise the albuginea 3–4 mm, using bipolar on crossing vessels.
4. Take a pea-sized sample with iris scissors (no forceps).
5. Do a touch prep, then a wet prep. If negative, then biopsy the other side.
6. Close the tunica vaginalis with continuous 5-0; no antibiotics.

- A non-absorbable suture marks a positive site for future TESE.

Complication	Point
Haematoma — most common	Prevent with magnification, bipolar and continuous tunica vaginalis closure
Epididymal biopsy — most serious	If that tubule held sperm, obstruction is now certain

4 · Vasography

- **Absolute indications (all three)**: azoospermia, complete spermatogenesis on biopsy and at least one palpable vas. Relative: severe oligospermia with normal biopsy, high sperm-bound antibodies, and low volume with poor motility (partial ejaculatory duct obstruction (EDO)).
- Do it **only at planned reconstruction**, because of scar and stricture. It cannot diagnose epididymal obstruction, the commonest non-vasectomy site.

ALGORITHM

1. High scrotal incision; find the vas at the straight–convoluted junction; hemivasotomy at 25×.
2. **Examine vasal fluid first.** If motile sperm, then barbotage and cryopreserve **before any dye**.
3. Pass a 24-G sheath into the abdominal end and inject 1 mL saline. If flow is easy, then it is patent.
4. Rarely, give dilute indigo carmine; blue urine confirms.
5. Use formal contrast only for obstruction above the internal ring.
6. Close with 10-0 mucosa and 8-0 muscularis.

Vasal fluid / vasogram	Meaning and action
No sperm, abdominal end patent	Epididymal obstruction: vasoepididymostomy
Copious fluid, many sperm	Vasal obstruction or EDO: locate it (vasogram, or 2-0 Prolene threaded down)
Block at ejaculatory ducts; both vasa fill from one side	EDO or midline cyst: transurethral resection of the ejaculatory ducts (TURED)
Vas ends blindly	Partial vasal agenesis: cystic fibrosis gene (CFTR) testing; retrieve and freeze
Inguinal block	Inguinal or crossed vasovasostomy
Many sperm, normal vasogram	Retrograde ejaculation, emission failure, aperistalsis

- Use **indigo carmine, not methylene blue**, which kills sperm (use it only after banking).
- Complications are **stricture** (repeated punctures, poor closure, non-water-soluble contrast), devascularisation, haematoma and granuloma.
- **Transrectal seminal vesiculography** aspirates dilated vesicles under transrectal ultrasound (TRUS); sperm there show a patent vas and epididymis, allowing TURED without scrotal exploration.

5 · Vasovasostomy (vasectomy reversal)

Evaluation

- Prior fertility proves spermatogenesis. Check follicle-stimulating hormone (FSH) if testes are small or soft; raised FSH means poorer prognosis.
- An indurated epididymis predicts vasoepididymostomy. A **sperm granuloma at the testicular end gives a better prognosis**, because it vents pressure, whatever the interval.

- Operate only if able to do vasoepididymostomy.

Approach and preparation

- The approach is scrotal for most and inguinal for hernia or orchidopexy obstruction. **Never strip the perivasal sheath**.
- **Length**, in escalating steps: dissect to the external then internal ring; free the convoluted vas (4–6 cm; **do not unravel**); reroute beneath the inguinal floor; take the epididymis off the testis. This bridges gaps up to **10 cm**.
- A healthy cut end has a white mucosal ring, soft muscularis and a bleeding edge. **If the muscularis is gritty or bleeding poor, recut.**
- At the testicular end **aspirate, never flush**. Flush saline into the abdominal end to prove patency.

Intraoperative fluid grading — the decision

Vasal fluid	Microscopy	Operation
Copious, crystal clear, watery	No sperm	Vasovasostomy
Copious, cloudy, thin	Sperm with tails	Vasovasostomy
Copious, creamy yellow	Many sperm heads	Vasovasostomy
Scant, granuloma present	Barbotage: sperm	Vasovasostomy
Thick white **toothpaste**	No sperm/heads	**Vasoepididymostomy**
Scant white thin, or dry without granuloma	No sperm/heads	**Vasoepididymostomy**
Any	Grape-like clusters of heads	**Vasoepididymostomy**

> **KEY POINT**
>
> Clear watery fluid still gets vasovasostomy: the epididymis has not blown out, and sperm usually return. Toothpaste fluid means secondary epididymal blow-out; a discrete level (dilated above, collapsed below) confirms it. When in doubt or inexperienced, do vasovasostomy.

Anastomotic principles

Principle	Reason
Accurate mucosa-to-mucosa	Testicular lumen is 2–5× the abdominal one
Leakproof	Sperm are antigenic; fluid lacks clotting factors, so the seal depends on mucosal sutures
Tension-free	Tension gives early sperm, then falling counts, then azoospermia
Good blood supply; healthy mucosa/muscularis; atraumatic	Recut until bleeding and soft; redo if back-walled or torn

- **Multilayer microdot**: six mapped dots guide **six 10-0 mucosal sutures**, then 9-0 deep muscularis, 9-0 adventitia and 8-0 sheath. The two-layer alternative is 10-0 mucosa and 8-0 full thickness.
- Reinforcing vasal sheath sutures (ReVas) relieve tension and raise patency.
- **Crossed vasovasostomy** suits unilateral inguinal obstruction with a contralateral atrophic testis, or one unreconstructable side.

- **Varicocele with reversal**: vasal vessels are compromised, so stage varicocelectomy ≥6 months later.

Post-operative care and outcomes

- Check semen at **6 weeks**, then 6 months if no pregnancy. Poor motility gets a 4-week prednisone course. Azoospermia at 6 months means redo.
- An anastomotic granuloma heralds obstruction. **Late stricture** shows as falling motility, then count. **Bank sperm once motile sperm appear.**
- With sperm in vasal fluid on at least one side, sperm return in virtually all. Pregnancy is about half, depending on vasectomy interval and female age.

6 · Vasoepididymostomy

- The epididymis is a single tubule ~200 µm wide and 12–15 feet long, with its thinnest wall at the caput. So a **more distal anastomosis is easier and gives better pregnancy**.
- **Indications**: reversal by the fluid criteria; or non-vasectomy OA with proven spermatogenesis, no vasal sperm and a patent vas and ejaculatory ducts.

ALGORITHM

1. Hemitransect the vas at the straight–convoluted junction; sample fluid; confirm patency.
2. Open the tunica vaginalis; choose a site **above** the obstruction where tubules are dilated. If unclear, then puncture distal to proximal until sperm appear.
3. Make a 3–4 mm tunic window.
4. Bring the vas through the tunica vaginalis, **fixed with 5-0 Prolene to the albuginea (tension relief)**.
5. Place posterior tunic–vas 8-0, then intussusception sutures, then outer tunic–vas adventitia.

Technique	For	Against
Longitudinal intussusception (LIVE) — used for all	Two 10-0 longitudinal sutures placed in the **still-distended** tubule; incise between; tubule pulled into vas; bloodless, leakproof	Fluid not checked before sutures set
End-to-side	Bloodless; fluid checked first	Suturing a collapsed tubule
End-to-end	Rapid level finding; bridges gap	Haemostasis hard; inferior epididymal supply lost

- **No simultaneous varicocelectomy**; stage it after 6 months.
- Late shutdown occurs, so **bank sperm during surgery and once they appear**. If it fails, redo or retrieve sperm for ICSI.

7 · TURED

*See the paper answer **Ejacultory duct obstruction** (Section 9, Ch 69).*

Also, from the condensed book (not in the paper answer):

- For retrograde ejaculation give pseudoephedrine 60 mg 90 min before, or use alkalinised urinary sperm.

ALGORITHM

1. Use a 24-Fr resectoscope with a finger in the rectum.
2. Resect the **verumontanum** to open the dilated orifices or cyst, sparing bladder neck, sphincter and rectum.
3. **Blue efflux confirms** adequacy; coagulate minimally.
4. Catheter overnight; 7 days of antibiotics.

8 · Sperm retrieval

*See the paper answer **Azoospermia: testicular biopsy and surgical sperm retrieval techniques** (Section 9, Ch 70).*

Also, from the condensed book (not in the paper answer):

- Microsurgical epididymal sperm aspiration (MESA) gives most motile sperm and multiple vials. **Microdissection TESE (micro-TESE) is best in NOA**.

KEY POINT

Paradox: in obstruction the **most motile sperm are proximal** (efferent ductules); distal tubules hold senescent fragments and macrophages. Yet fertilisation is best when the longest epididymis is preserved.

ALGORITHM

1. Micro-TESE: general or regional anaesthesia.
2. At 10×, find the avascular plane on the anterior albuginea; make a wide equatorial incision and bivalve the testis.
3. Excise larger tubules; an andrologist in theatre examines them; map positives.
4. If negative, then explore the other testis.
5. Close the albuginea with 5-0 and the tunica vaginalis with 4-0.

9 · Cryopreservation

- Freeze at diagnostic biopsy, at every reconstruction (before any dye), and from the ejaculate once motile sperm return after reversal.

10 · Varicocelectomy

*See the paper answer **Varicocele: pathophysiology, controversies and surgical management** (Section 10, Ch 87).*

Also, from the condensed book (not in the paper answer):

- Conventional inguinal repair has hydrocele 3–15% and recurrence 5–15%, because it misses scrotal and periarterial veins.
- Ligate vasal veins only if >2.5 mm, keeping one set: **one intact deferential set guarantees venous return**.

ALGORITHM

1. Local anaesthesia with sedation; 2–3 cm incision over the external ring.
2. Deliver the cord on a Penrose, excluding the ilioinguinal nerve and genital branch.
3. **Map arteries with micro-Doppler before handling.**
4. Open the spermatic fasciae; work at 10–25× with 1% papaverine; loop the artery.
5. Ligate **all veins except vasal**; preserve lymphatics.
6. Deliver the testis; ligate external spermatic and gubernacular veins.
7. Do the impulse test. If there is no impulse, then a vein was missed.

- **End point**: only testicular and cremasteric arteries, lymphatics, and the vas with its vessels remain.

> **KEY POINT**
>
> Recurrence means a vein was missed; hydrocele, a lymphatic tied; atrophy, an artery tied. Microsurgery with testicular delivery addresses all three.

11 · Landmark trials

Trial	Design	Result	Take-home
Madgar (1995)	Randomised controlled trial (RCT): varicocelectomy vs no surgery in infertile men	1-year pregnancy 44% vs 10%	Repair of clinical varicocele with abnormal semen improves fertility

71 Physiology of Penile Erection and Pathophysiology of Erectile Dysfunction

1 · Definition and key concepts

- **Erectile dysfunction (ED)** is the consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction (American Urological Association (AUA) 2018).
- Erection needs **sinusoidal (trabecular) smooth muscle relaxation, arterial dilation and venous compression (veno-occlusion)**. Rigidity comes from trapping blood, not from inflow alone.
- Milestones MCQ: nitric oxide (NO) was identified in the 1990s, and oral phosphodiesterase type 5 inhibitors (PDE5Is) arrived in 1998.

2 · Functional anatomy

Corpora

- There are paired dorsal **corpora cavernosa** and a ventral **corpus spongiosum**, which surrounds the urethra and expands as the glans.
- The septum is **incomplete** in humans, so one injection fills both corpora.
- Sinusoidal oxygen tension (Po2) is venous when flaccid (**~35 mm Hg**) and arterial when erect (**~90 mm Hg**).

Tunica albuginea

- The **inner circular layer** gives off **intracavernous pillars** (absent at 5–7 o'clock). The **outer longitudinal layer** is thicker dorsally and absent in the ventral groove.
- The ventral tunica is ~0.8 mm against ~2.2 mm dorsally. So 5–7 o'clock is the weak zone for **penile fracture** and implant crossover or perforation.
- **Veno-occlusion**: emissary veins run obliquely between the layers and are compressed as the sinusoids stretch the tunica. Cavernous arteries escape because they lie in a periarterial sheath.
- The spongiosal tunica is thin, single-layered and absent in the glans, so spongiosum and glans are **low-pressure arteriovenous shunts** and never become rigid.

Arteries

- The internal iliac gives the **internal pudendal**, which becomes the common penile artery with three branches.
- The **cavernous** artery is the main corporal supply. Its **helicine arteries** are tortuous when flaccid and straight when erect, and open into sinusoids without capillaries.
- The **dorsal** artery fills the glans, and the **bulbourethral** artery supplies the bulb and spongiosum.
- **Accessory pudendal arteries** occur in ~one-third and may be dominant. Preserving them at radical prostatectomy speeds erectile recovery.

Veins

Sinusoids drain into the **subtunical venous plexus**, which exits through **emissary veins** in the tunica.

System	Drains	Destination
Deep dorsal vein (with or without circumflex and periurethral veins)	Glans and distal two-thirds of corpora	Periprostatic (Santorini) plexus
Cavernous and crural veins	Proximal (infrapubic) penis	Internal pudendal veins
Superficial dorsal vein	Skin, subcutaneous tissue	Saphenous veins

> **KEY POINT**
>
> Vascular anatomy varies widely between men, so the same "nerve-sparing" operation gives different erectile outcomes.

3 · Haemodynamics: phases of erection and detumescence

LAQ Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED

Paper answer · P1

Asked as:

- Write in detail about physiology of erection. How will you evaluate a diabetic with ED? — *Practice Set 5 Paper I Q2 (LAQ)*
- Neurotransmitters in erectile function — *Practice Set 12 Paper I Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(erection physiology also in Practice Set 5 Paper 1, Q2)

Introduction

- Erection is **cavernous smooth-muscle relaxation**: low cytosolic Ca^{2+} dilates helicine arteries and sinusoids; expanding sinusoids compress emissary veins (veno-occlusion).
- Pathways: parasympathetic **S2–S4** cavernous nerves (erection), sympathetic **T11–L2** (detumescence), pudendal (sensation, ischiocavernosus rigidity).

Functional anatomy

- **Corpora cavernosa:** sinusoids lined by endothelium, separated by **trabecular smooth muscle**. Flaccid tone depends on smooth-muscle contraction.
- **Tunica albuginea:** inner circular and outer longitudinal layers. **Emissary veins** pass obliquely between them and are compressed when the corpora expand: the **veno-occlusive mechanism**.
- **Arteries:** internal pudendal → common penile → **cavernous (helicine branches)**, dorsal and bulbourethral arteries. Accessory pudendal arteries in about one-third.
- Spongiosum and glans have a thin or absent tunica, so no veno-occlusion; they stay low-pressure.

Haemodynamic phases

Phase	Events	Intracavernous pressure
Flaccid	Smooth muscle contracted; low inflow	Low
Latent (filling)	Arterial dilation; **inflow peaks**	Barely rises
Tumescence	Sinusoids fill; venous outflow compressed	Rising

Phase	Events	Intracavernous pressure
Full erection	Maximal veno-occlusion	About 100 mmHg
Rigid	**Ischiocavernosus** contraction (pudendal nerve)	Several hundred mmHg
Detumescence	Initial, slow, fast phases	Falls to baseline

Molecular mechanism and peripheral neurotransmitters

- **Relaxation (erection):** NO (nitric oxide) from **nNOS (neuronal nitric oxide synthase)** in cavernous nerves initiates erection; endothelial NOS (eNOS) sustains it.
- NO → soluble guanylyl cyclase → **cGMP (cyclic guanosine monophosphate)** → protein kinase G (PKG) → K^+ channels open, Ca^{2+} channels close, Ca^{2+} sequestered → **low cytosolic Ca^{2+}** → relaxation.
- Second route: PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide) → **cAMP (cyclic adenosine monophosphate)** → protein kinase A.
- **PDE5 (phosphodiesterase type 5)** degrades (breaks down) cGMP and ends relaxation (ends / terminates erection); PDE5 inhibitors block it.

Peripheral pro-erectile transmitters

Transmitter	Source / mechanism
NO (nitric oxide)	Principal. **nNOS (neuronal NOS)** initiates; **eNOS (endothelial)** sustains. NO → soluble guanylyl cyclase → **cGMP** → PKG → K^+ channels open, Ca^{2+} channels close → relaxation
ACh (acetylcholine)	Indirect: endothelial NO release, inhibits adrenergic nerves; atropine does not block erection
PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide), CGRP (calcitonin gene-related peptide)	**cAMP** → PKA (protein kinase A); VIP role minimal in man
CO, H2S, EDHF (endothelium-derived hyperpolarising factor)	Gaseous/endothelial relaxants via cGMP or K^+ channels

- O2 is a NOS substrate: low flaccid Po2 (~35 mmHg) limits NO; erect Po2 ~90 mmHg.
- **Also know:** carbon monoxide and hydrogen sulfide also raise cGMP; acetylcholine acts indirectly through endothelial NO (atropine does not block erection).

Anti-erectile transmitters (flaccidity and detumescence)

- **Contraction (flaccidity): noradrenaline (α1)** (principal), endothelin-1, angiotensin II → raised Ca^{2+} → calmodulin → myosin light-chain kinase → contraction.
- **Noradrenaline (α1)**: principal; α2 prejunctional reduces noradrenaline and NO release.
- **Endothelin-1** (ET-A), angiotensin II (AT1), PGF2α, thromboxane A2: Gq → IP3 → Ca^{2+} rise → Ca^{2+}–calmodulin → MLC (myosin light-chain) kinase → contraction.
- **Rho-kinase** inhibits myosin (MLC) phosphatase (calcium sensitisation) and maintains (sustaining) tone.
- **Detumescence:** NO release stops, PDE5 degrades cGMP and sympathetic discharge contracts the smooth muscle, so emissary veins reopen.
- Gap junctions (connexin-43) synchronise the smooth muscle.

Neural control

Pathway	Origin	Role
Parasympathetic	**S2–S4** → pelvic plexus → cavernous nerves	Erection
Sympathetic	**T11–L2** → hypogastric nerves	Detumescence, emission
Somatic	Pudendal nerve (Onuf nucleus)	Penile sensation; ischiocavernosus rigidity

- **Three types: psychogenic** (brain, chiefly by reducing sympathetic tone), **reflexogenic** (genital touch, sacral arc; preserved in upper cord injury), **nocturnal** (REM [rapid eye movement] sleep).
- **Central:** MPOA (medial preoptic area) and **PVN (paraventricular nucleus)**. Dopamine and oxytocin promote; serotonin, prolactin, GABA (γ-aminobutyric acid) and opioids inhibit.

Central neurotransmitters (PVN [paraventricular nucleus], MPOA [medial preoptic area])

Promote	Inhibit
Dopamine (D2 in PVN; apomorphine)	**Serotonin** (5-HT1A, 5-HT2; but 5-HT2C pro-erectile)
Oxytocin	GABA (γ-aminobutyric acid)
Central NO	Opioids (high dose), cannabinoids
Central noradrenaline; melanocortins (MC4R)	**Prolactin**

- Dopamine and oxytocin converge on NO in the PVN. Noradrenaline is pro-sexual centrally but anti-erectile peripherally.

Clinical relevance

- PDE5 inhibitors, intracavernosal alprostadil (cAMP), papaverine, phenylephrine for priapism (α1), SSRIs (selective serotonin reuptake inhibitors) cause dysfunction, hyperprolactinaemia lowers libido.

Diagram to draw: NO–cGMP pathway in the cavernous smooth-muscle cell.

Evaluation of a diabetic with ED (erectile dysfunction)

Why diabetics: ED is commoner, earlier and more severe in DM (diabetes mellitus). Mechanisms: endothelial NO loss, autonomic neuropathy, arteriopathy, smooth-muscle loss and fibrosis (veno-occlusive failure), AGEs (advanced glycation end products) and low testosterone. ED is a **sentinel of cardiovascular disease**.

1. History

- Onset, duration (≥3 months), severity, morning erections, libido, ejaculation; partner.
- **IIEF-5 (International Index of Erectile Function, SHIM):** 5–7 severe, 8–11 moderate, 12–16 mild-to-moderate, 17–21 mild, 22–25 no ED.
- Diabetes: type, duration, **glycaemic control**, complications (retinopathy, nephropathy, neuropathy, foot disease), autonomic symptoms (postural giddiness, gastroparesis, bladder dysfunction).
- Cardiovascular: angina, exercise tolerance, smoking, hypertension, dyslipidaemia, nitrates.
- Drugs: antihypertensives, antidepressants. Depression screen.

2. Examination

- BMI, waist, blood pressure, peripheral pulses.

- Genitalia: Peyronie plaques, testicular size, gynaecomastia.
- Neurological: genital sensation, **bulbocavernosus reflex**, foot sensation. Penile neuropathy can exist without foot neuropathy.

3. Laboratory

- **HbA1c (glycated haemoglobin)** and fasting glucose, **lipid profile**, creatinine/eGFR (estimated glomerular filtration rate), urine albumin.
- **Morning total testosterone**; if low, repeat with free testosterone, LH (luteinising hormone), prolactin. TSH (thyroid-stimulating hormone) if indicated.

4. Cardiovascular risk stratification (Princeton Consensus)

- **High risk** (unstable angina, recent infarction, uncontrolled hypertension, some arrhythmias): defer sexual activity and treatment; cardiology referral. **Indeterminate:** cardiac assessment (e.g. stress test), then restratify. **Low:** start treatment.

5. Specialised tests (selected cases)

- **CIS (combined intracavernous injection and stimulation):** alprostadil 10–20 µg; a rigid erection excludes major vascular disease.
- **Penile duplex Doppler:** PSV (peak systolic velocity) **<25 cm/s = arterial insufficiency** (>35 normal); EDV (end-diastolic velocity) **>5 cm/s** and RI (resistive index) **<0.75 = veno-occlusive dysfunction**.
- **NPTR (nocturnal penile tumescence and rigidity), RigiScan:** separates psychogenic from organic ED.
- **Neurophysiological:** bulbocavernosus reflex latency, dorsal nerve conduction velocity, biothesiometry.
- DICC (dynamic infusion cavernosometry and cavernosography) and angiography: rarely, as diffuse diabetic vascular disease is not suited to revascularisation.

```
Diabetic with ED → history + IIEF-5 + examination → HbA1c, lipids,
testosterone, renal function → Princeton cardiac risk → treat risk factors →
PDE5 inhibitor trial → failure → duplex/CIS → ICI or prosthesis
```

> **Key points to remember:**
> - nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
> - Noradrenaline α1 is the main anti-erectile transmitter.
> - Dopamine/oxytocin promote; serotonin/prolactin inhibit.
> - Rigid phase: ischiocavernosus, suprasystolic pressure.
> - Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
> - Diabetic ED: check HbA1c, testosterone and cardiac risk.

Also, from the condensed book (not in the paper answer):

- Full-erection intracavernous pressure (ICP) plateaus at **~100 mm Hg**; the rigid phase is suprasystolic, driven by the pudendal nerve via ischiocavernosus.
- **Trap**: arterial inflow peaks in the latent phase, not at rigidity. This is why peak systolic velocity (PSV) on Doppler is measured early.
- Detumescence has three phases: initial (transient ICP rise), slow (veins gradually reopen) and fast (outflow restored).

4 · Neuroanatomy

Peripheral and spinal

	Sympathetic	Parasympathetic	Somatic
Origin	**T11–L2** (often T10–T12)	**S2–S4** intermediolateral columns	S2–S4; motor from **Onuf nucleus**
Route	Sympathetic chain, superior hypogastric plexus, hypogastric nerves, then pelvic plexus	Pelvic nerves (nervi erigentes) to the pelvic plexus	Pudendal nerve; afferent via dorsal nerve of penis
Role	**Detumescence**; emission	**Erection** (cavernous nerves)	Sensation; ischiocavernosus (rigidity), bulbocavernosus (ejaculation)

- **Cavernous nerves** arise from the pelvic plexus, carry both autonomic types and run posterolateral to the prostate. Pelvic ganglia lie along the bladder–prostate junction and seminal vesicles, so they are vulnerable in radical pelvic surgery.
- The **bulbocavernosus reflex** (glans squeeze contracts bulbocavernosus) has limited value in ED work-up.

Three types of erection

Type	Trigger	Mechanism	Spinal cord injury (SCI)
Psychogenic	Audiovisual, fantasy	Brain modulates spinal centres, chiefly by suppressing sympathetic tone	May persist with sacral lesions; absent with lesions above T9
Reflexogenic	Genital touch	Sacral reflex arc to the cavernous nerves	Preserved in upper cord lesions
Nocturnal	Rapid eye movement (REM) sleep	Cholinergic pontine tegmentum on; locus coeruleus (noradrenergic) and raphe (serotonergic) silent	—

Supraspinal (central)

- The **medial preoptic area (MPOA)** is dopaminergic and integrates hormonal and sensory cues. The **paraventricular nucleus (PVN)** sends pro-erectile **oxytocin** neurons to the lumbosacral nuclei, and the nucleus paragigantocellularis is the **serotonergic brake**.

Central neurotransmitters

*See the paper answer **Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED** (Section 9, Ch 71).*

Also, from the condensed book (not in the paper answer):

- Apomorphine is a D1/D2 agonist acting in the PVN, which releases oxytocin and then NO. Melanocortins act on the MC4 receptor.
- Serotonin 5-HT1A and 5-HT2 receptors inhibit erection but facilitate ejaculation.

> **KEY POINT**
>
> NO synthase (NOS) inhibitors block both apomorphine- and oxytocin-induced erections, so NO is the final common central pathway. **Trap**: noradrenaline is pro-sexual centrally but the principal anti-erectile transmitter peripherally.

5 · Molecular mechanisms of smooth muscle tone

Every erectogenic drug acts on one of these pathways.

Contraction (flaccid state)

- Corporal smooth muscle is **contracted at rest** through myogenic tone, adrenergic tone and endothelial constrictors.
- **Calcium pathway**: noradrenaline (α1), endothelin-1 (ETA), angiotensin II (AT1), PGF2α and thromboxane A2 act via Gq and phospholipase C to form inositol trisphosphate (IP3), raising Ca^{2+}. Ca^{2+}–calmodulin then activates **myosin light chain (MLC) kinase**, which phosphorylates MLC20.
- **Ca^{2+} sensitisation by RhoA/Rho-kinase**: it inhibits **MLC phosphatase (MLCP)**, giving sustained contraction without a further Ca^{2+} rise.
- Rho-kinase is raised in diabetes mellitus (DM), hypertension, hypercholesterolaemia, hypogonadism, ageing and nerve injury, and it also drives fibrosis.

Relaxation (erection)

- **NO–cGMP**: NOS converts L-arginine and O2 into NO and citrulline. **Neuronal NOS initiates and endothelial NOS (eNOS) sustains** erection.
- NO activates soluble guanylyl cyclase to make cyclic GMP (cGMP), which activates **protein kinase G (PKG-I)**. PKG lowers intracellular Ca^{2+}, inhibits Rho-kinase and activates MLCP, so the muscle relaxes. PKG-I knockout mice cannot erect, so PKG is the bottleneck.
- **cAMP pathway**: PGE1, vasoactive intestinal peptide (VIP) and calcitonin gene-related peptide (CGRP) act via Gs to raise cyclic AMP (cAMP), which activates protein kinase A (PKA). It is **NO-independent**, which is why intracavernosal alprostadil works when nerves or endothelium fail.
- **PDE5 (cGMP-specific) is the principal cavernous PDE**. PDE6 is retinal (visual side effects); PDE2, 3 and 4 degrade cAMP. Papaverine is nonspecific.
- **Gap junctions (connexin-43)** synchronise relaxation, so a local input gives a uniform erection.
- Oxygen acts as a switch: erect Po2 >50–60 mm Hg permits full NOS activity, while low flaccid Po2 suppresses it.
- Excess **adenosine** with increased A2B signalling causes prolonged erection in adenosine deaminase-deficient and sickle cell mice, the **priapism models**.
- Acetylcholine acts indirectly (endothelial NO, presynaptic adrenergic inhibition), so **atropine does not abolish** neural erection MCQ .

Second messenger	Activators	Kinase	Degraded by	Drug
cGMP	NO, CO, H2S, natriuretic peptides	PKG-I	PDE5	PDE5Is

Second messenger	Activators	Kinase	Degraded by	Drug
cAMP	PGE1, VIP, CGRP, adenosine (A2)	PKA	PDE2, 3, 4	Alprostadil; papaverine (both)

Detumescence

NO release stops and PDE5 hydrolyses cGMP. Sympathetic noradrenaline (α1; the principal transmitter of flaccidity), endothelin-1 and Rho-kinase then contract the trabeculae, so the veins reopen.

> **KEY POINT**
>
> Corporal tone is a tug-of-war over one phosphorylation site (MLC20): Ca^{2+}–calmodulin–MLC kinase and Rho-kinase keep it phosphorylated; NO–cGMP–PKG removes the phosphate. **Trap**: "MLCP" means the phosphatase.

6 · Classification of ED

Approach	Categories
Neurovascular mechanism	Failure to **initiate** (neurogenic, psychogenic, endocrine) · to **fill** (arterial) · to **store** (veno-occlusive)
Aetiological	Psychogenic · neurogenic · hormonal · arteriogenic · cavernosal (venogenic) · drug-induced · systemic disease · mixed (commonest)

Most organic ED has a psychogenic component.

7 · Risk factors and endothelial dysfunction

ED often signals vascular disease before the heart does.

- **Age** is the strongest demographic predictor, but DM, heart disease and hypertension carry more risk (Massachusetts Male Aging Study (MMAS)).
- **Endothelial dysfunction**: oxidative stress causes **NOS uncoupling**, while arginase and asymmetric dimethylarginine (ADMA) further lower NO.
- **Artery-size hypothesis**: smaller penile arteries become symptomatic first. ED preceded coronary artery disease (CAD) by a mean of **~2 years**, so ED is a **sentinel** and needs cardiac risk stratification (see Ch 72).
- The common end-path is raised Rho-kinase and transforming growth factor-β (TGF-β), which lower the smooth muscle-to-collagen ratio and cause veno-occlusive failure.

8 · Pathophysiology by type

Psychogenic

- The brain inhibits the spinal centres, and excess sympathetic tone contracts the muscle. Causes include performance anxiety, partner conflict, depression and post-traumatic stress disorder (PTSD).

Neurogenic

- **Brain** causes include Parkinson disease, stroke, tumours and dementia. **Cord** causes include SCI (commonest), spina bifida, disc disease and multiple sclerosis (MS).

Complete lesion	Reflexogenic erection	Psychogenic erection
Upper cord (above sacral centre)	Preserved in **~95%**; brief, need continuous stimulation	Lost
Lower cord (sacral)	Preserved in only **~25%**	May occur (thoracolumbar pathway); marginal tumescence

- **Iatrogenic**: radical prostatectomy, cystoprostatectomy, abdominoperineal resection (APR) and pelvic radiation. Most men have at least a transient decline despite nerve sparing, worse with baseline ED and age >60.
- Cavernous nerve injury causes apoptosis and fibrosis, leading to venous leak. This is the rationale for early penile rehabilitation.
- **Pelvic fracture** injures the cavernous nerves or pudendal artery. With posterior urethral injury, early realignment preserves potency better than delayed anastomosis.

Hormonal

- **Testosterone (T)** governs **libido** more than mechanics, but it maintains NOS, PDE5 and smooth muscle. Low libido is the commonest sexual symptom of low T.
- AUA: measure **morning, fasting** T in all men with ED. Low T is **<300 ng/dL**; treat only with ≥2 low levels and symptoms. Nocturnal erections need ~200 ng/dL.
- Nocturnal erections are androgen-dependent, but erections to visual stimuli are largely independent MCQ. Dihydrotestosterone (DHT) is the key penile androgen.
- **Hyperprolactinaemia** (prolactinoma, antipsychotics) suppresses gonadotrophin-releasing hormone (GnRH), causing low T and low libido.
- A low T:oestradiol ratio (obesity), hyperthyroidism and hypothyroidism also cause ED.

Vasculogenic – arteriogenic

- Atherosclerotic or traumatic occlusion of the hypogastric–pudendal–cavernous tree lowers perfusion pressure, giving a slower, softer erection.
- Risks are hypertension, hyperlipidaemia, smoking, DM, obesity, pelvic irradiation and blunt perineal trauma. Trauma causes focal stenosis in young men, who are revascularisation candidates.
- Chronic hypoxia raises TGF-β1 and collagen, so "inflow failure becomes storage failure".
- In hypertension, angiotensin receptor blockers (ARBs) and nebivolol protect penile tissue.

Vasculogenic – veno-occlusive (venous leak)

- Causes are **tunical** (ageing, DM, Peyronie disease, penile fracture), **fibroelastic trabecular change** preventing sinusoidal expansion, **insufficient relaxation** (anxiety, adrenergic tone, endothelial dysfunction), and **acquired shunts** after priapism shunt surgery. Congenital ectopic or crural veins cause primary ED.

> **KEY POINT**
>
> Veno-occlusion is an emergent property of healthy tissue (elastic tunica and fully relaxed trabeculae). So most "venous leak" is really cavernosal smooth muscle disease, which is why venous ligation surgery fails and is investigational.

Metabolic and systemic disease

- **DM** causes the earliest, most severe ED through arteriopathy, nitrergic neuropathy, endothelial dysfunction, advanced glycation end products (AGEs), raised Rho-kinase and hypogonadism. **Key**: the NO source is damaged but the response to exogenous NO is preserved, so PDE5Is work until fibrosis sets in.
- **Metabolic syndrome and obesity** improve with weight loss, Mediterranean diet, exercise and bariatric surgery. In **hyperlipidaemia**, oxidised low-density lipoprotein (LDL) inhibits NO, and statins are linked to better International Index of Erectile Function (IIEF) scores.
- **Chronic kidney disease**: up to **80%** of men with end-stage renal disease have ED, and it persists in ~half after transplant.
- Other associations include HIV, cirrhosis, chronic obstructive pulmonary disease, obstructive sleep apnoea and lower urinary tract symptoms.

Drug-induced

Class	Cause ED	Mechanism / note	Alternative
Antihypertensives	**Thiazides**, **nonselective β-blockers**, clonidine	Thiazide – unknown; β-blocker – prejunctional α2	ACE inhibitors, ARBs, nebivolol, α1-blockers (ejaculatory dysfunction), calcium channel blockers
Antidepressants	Selective serotonin reuptake inhibitors (SSRIs; paroxetine worst), tricyclics	Spinal 5-HT2/5-HT3 inhibition, less dopamine and NO; post-SSRI dysfunction may persist	Bupropion, mirtazapine, nefazodone, buspirone; add a PDE5I
Antipsychotics	Typical > atypical	Dopamine block causes hyperprolactinaemia	Newer agents
Antiandrogens	LHRH agonists/antagonists, androgen receptor antagonists, cyproterone, **5α-reductase inhibitors (5ARIs)**	5ARI: least T effect yet persistent ED after stopping	None
Recreational	Tobacco (dose–response, ≥20 pack-years), heavy alcohol, opioids	Endothelial damage, neuropathy, hypogonadism	Stop or moderate
Others	Cimetidine, spironolactone, carvedilol	—	—

Trazodone causes priapism, and among anticonvulsants carbamazepine and phenytoin are worse than lamotrigine.

Primary (lifelong) ED

- It is usually psychogenic, but exclude hypoplastic cavernous arteries, congenital venous leak, corporal atresia and micropenis.

KEY POINT

Why ED is a sentinel: nearly every systemic disease reaches the penis through the same three doors — endothelial NO failure, Rho-kinase-driven contraction and TGF-β-driven fibrosis.

72 Evaluation and Management of Erectile Dysfunction

1 · Definition and significance

- **Erectile dysfunction (ED)** (American Urological Association, AUA): a consistent or recurrent inability to attain and/or maintain an erection sufficient for sexual satisfaction. Minimum **3 months**, except after trauma or surgery.
- About **20%** of adult men have ED (up to 15% at 40–49 y; up to 40% at 60–69 y).
- **ED predicts cardiovascular disease (CVD)** through shared endothelial dysfunction, as strongly as smoking or a family history of myocardial infarction (MI). Younger onset means greater cardiovascular (CV) risk.

Independent association (multivariate)	Risk factors
Strongest	Antidepressant use, then antihypertensive use
Strong	Diabetes mellitus (DM), the strongest disease; obstructive urinary symptoms
Moderate	Hypertension, prostate enlargement, smoking, body mass index (BMI) ≥30, inactivity
Weak/null after adjustment	CVD, hypercholesterolaemia

- Drugs cause up to 25% of ED (see Ch 71); opioids double the risk.

2 · Management principles

- **Shared decision making** (AUA 2018): all non-contraindicated options are offered **in parallel, not ranked**, and the patient may choose any first. They are phosphodiesterase type 5 inhibitors (PDE5Is), the vacuum erection device (VED), intraurethral (IU) alprostadil, intracavernosal injection (ICI) and prosthesis.
- In testosterone (T) deficiency (symptoms and total T below 300 ng/dL), restoring T probably improves the other ED treatments.

Cardiac risk assessment – Princeton Consensus

*See the paper answer **Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED** (Section 9, Ch 71).*

Also, from the condensed book (not in the paper answer):

- Every man with ED is at potential CV risk; sex equals moderate exertion. **Refer** after failed therapy, for a young trauma patient, for deformity, in medicolegal cases, and for complex endocrine, psychiatric or vascular problems.

3 · Basic evaluation

History

- Ask about onset, pattern, morning erections, libido, ejaculation, curvature and prior treatment.
- Ask about DM, CVD, neurological disease, pelvic surgery or radiation, trauma and drugs. Screen for depression.

Psychogenic	Organic
Sudden onset	Gradual onset
Complete, immediate loss	Incremental progression
Situational	Global
Waking/nocturnal erections present	Poor or absent

> **KEY POINT**
>
> Preserved nocturnal/morning erections prove an intact neurovascular axis, the strongest historical pointer to psychogenic ED.

Questionnaires

- The **International Index of Erectile Function (IIEF)** has 15 items in 5 domains: erectile function, orgasm, desire, intercourse satisfaction and overall satisfaction.

*See the paper answer **Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED** (Section 9, Ch 71).*

Also, from the condensed book (not in the paper answer):

- The Sexual Health Inventory for Men (SHIM) has 5 items scored 1–5 over the past 6 months. Questionnaires grade severity and response but do **not** identify the cause.

Physical examination

- Look for hypogonadism (testes, gynaecomastia), vascular disease (blood pressure (BP), obesity, pulses), neuropathy (genital sensation, bulbocavernosus reflex) and structural problems (plaque, micropenis, chordee).

Laboratory tests

- **Recommended**: fasting glucose or glycated haemoglobin (HbA1c), lipids, serum chemistry, full blood count and **morning (7–11 am) total testosterone**.
- Repeat a low T, because about one-third of mildly low values normalise, and add luteinising hormone (LH) and prolactin. Check prostate-specific antigen (PSA) if prostate disease is suspected or before T therapy.

> **KEY POINT**
>
> The basic work-up finds the disease behind the ED (undiagnosed DM, dyslipidaemia, hypogonadism, CV risk); it does not prove ED.

4 · Hormonal evaluation

*See the paper answer **Male menopause (andropause, late-onset hypogonadism)** (Section 9, Ch 68).*

Also, from the condensed book (not in the paper answer):

- The AUA threshold is **300 ng/dL (10.4 nmol/L)**; above about 230 ng/dL, sexual function correlates only weakly with T. Order a pituitary magnetic resonance imaging (MRI) scan if T is **<150 ng/dL**.
- Hyperprolactinaemia is caused by antipsychotics, tricyclics, opiates, prolactinoma, hypothyroidism and renal failure.

5 · Specialised testing

These tests are for young trauma patients, primary ED, surgical candidates, medicolegal cases and failed therapy.

Test	Line	Answers	Indication
Combined intracavernosal injection and stimulation (CIS)	1st	Can the vessels produce a rigid erection?	Screening
Penile duplex Doppler	2nd	Arterial or veno-occlusive? Asymmetry?	Most reliable, least invasive
DICC (dynamic infusion cavernosometry/cavernosography)	3rd	Is there a venous leak, and where?	Trauma, lifelong ED, before venous surgery
Selective internal pudendal arteriography	3rd	Site of arterial lesion; is the inferior epigastric donor usable?	Young trauma patient before revascularisation

CIS test

- Inject alprostadil 10–20 µg or Bimix/Trimix 0.3 mL into the lateral base, then compress, stimulate and rate rigidity. A rigid, sustained erection means normal haemodynamics; anxiety or a low dose gives false negatives, so redose.
- If there is no detumescence by about 1 h, inject diluted phenylephrine (500 µg/mL) every 3–5 min.

Penile duplex Doppler

- Measure cavernous artery peak systolic velocity (PSV) and end-diastolic velocity (EDV) at the base every 5 min up to 20 min after ICI.

Parameter	Arterial insufficiency	Veno-occlusive dysfunction
PSV	**<25 cm/s** (25–35 borderline; >35 normal)	Stays >25 cm/s
EDV	—	**>5 cm/s**
Resistive index (RI) = (PSV − EDV)/PSV	—	**<0.75** (normal >0.9)
Other	Acceleration time >122 ms; side-to-side PSV difference >10 cm/s; artery dilation <75%	Rapid detumescence
When assessed	Within 5 min	15–20 min (needs full relaxation)

> **KEY POINT**
>
> Venous leak can be diagnosed only with adequate inflow and full smooth muscle relaxation; an anxious patient mimics veno-occlusive dysfunction, so redose first. Anatomical variants can falsely suggest arteriogenic ED.

DICC, NPTR and neurological tests

- DICC measures flow-to-maintain, pressure decay and cavernous artery systolic occlusion pressure after a vasodilator; cavernosography localises the leak.
- Nocturnal penile tumescence and rigidity (NPTR) testing in a sleep lab is the standard. The home **RigiScan** measures **radial** rigidity; normal is 4–5 erections a night, ≥60% rigid for

≥10 min. It suits obscure cause, medicolegal cases and suspected psychogenic ED.

6 · Treatment

ALGORITHM

1. Treat reversible causes (lifestyle, drugs, hormones, psychosocial) and stratify CV risk.
2. Shared decision: PDE5I, VED, IU alprostadil or ICI. Low-intensity extracorporeal shock wave therapy (Li-ESWT) is investigational.
3. If the response is inadequate, then optimise, combine or switch.
4. If still inadequate, then offer a penile prosthesis (see Ch 75).

Lifestyle and risk-factor modification

- Stopping smoking, exercise, weight loss and a Mediterranean diet improve function; control DM, lipids and BP.
- **Change medication**: replace thiazides or β-blockers with calcium channel blockers, angiotensin-converting enzyme inhibitors or angiotensin receptor blockers. For a selective serotonin reuptake inhibitor (SSRI), switch to bupropion or mirtazapine, or add a PDE5I.
- **Psychosexual therapy** works best combined with medical therapy.

Hormonal therapy

- Give T only for confirmed deficiency, as a ~3-month trial; it improves libido more than erection and is synergistic with PDE5Is.
- Monitor digital rectal examination, PSA, haematocrit, liver tests and lipids at baseline, 3–6 months and yearly. The main adverse effect (AE) is erythrocytosis; randomised trials show no causal CV link.
- Preparations include gels (transfer risk), intramuscular enanthate/cypionate (200–250 mg every 2–3 weeks) and undecanoate depot (750 mg at 0 and 4 weeks, then every 10 weeks). Avoid methyltestosterone because it is hepatotoxic.
- T does not correct hyperprolactinaemia: stop the drug, give bromocriptine for prolactinoma, and operate for failure or visual compromise.

PDE5 inhibitors (first-line oral)

They block cyclic guanosine monophosphate (cGMP) breakdown, so they **enhance, not initiate**, erection; sexual stimulation is obligatory.

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Doses (mg)	25, 50, 100	5, 10, 20	5, 10, 20 (2.5/5 daily)	50, 100, 200
Time to peak, T (h) max	0.8 (~1 h before)	0.7–0.9 (~1 h)	**2** (~2 h)	**0.3–0.5** (~30 min)
Onset (min)	15–60	15–60	15–120	15–60
Half-life (h)	3–5	4–5	**17.5**	3–5
Fatty food	Lowers absorption	Lowers absorption	**No effect**	Lowers absorption

	Sildenafil	Vardenafil	Tadalafil	Avanafil
Characteristic AE	Visual (blue tinge; PDE6), flushing	Visual, flushing; **QT prolongation, so avoid with class 1A/3 antiarrhythmics** and congenital long QT	**Back pain/myalgia** (PDE11)	Fewest visual effects
Nitrates	**Contraindicated (all)**			

- Headache is the commonest AE. Efficacy is equal across agents: about 70% achieve intercourse, and 40–50% in DM and after prostatectomy.
- **Nitrates** in any form, including amyl nitrite "poppers", cause profound hypotension. If angina occurs, give no nitrates for **24 h** (sildenafil, vardenafil) or **48 h** (tadalafil). There is no antidote; treat hypotension with head-down tilt, fluids and phenylephrine.
- Use caution with α-blockers (stabilise first, start low) and aortic stenosis; CYP3A4 inhibitors need a lower dose.
- They are not studied after MI or stroke within 6 months, with BP <90/50 or >170/100, or in retinitis pigmentosa. For sudden visual loss, stop the drug and seek care.
- **Before labelling failure**, check stimulation, timing, food, maximum dose and **up to 9–10 attempts**; correct hypogonadism, glycaemia and lipids; then switch agent or try daily tadalafil.
- **Penile rehabilitation** after prostatectomy is unproven, since REINVENT and other nightly-dosing trials were negative.

Other oral agents (yohimbine, apomorphine, trazodone) have little or no proven benefit.

Vacuum erection device

- Negative pressure fills the corpora and a base ring traps the blood; **remove it within 30 min**. The erection is hinged at the base, cool and cyanotic.
- Up to 90% get an erection but only 30–70% are satisfied; it suits older men in stable relationships. Take care with anticoagulants.

Intraurethral alprostadil (MUSE)

- A prostaglandin E1 (PGE1) pellet of 125–1000 µg is inserted into the urethra after voiding. About 50% respond, so it is a niche option: PDE5I failure after nerve injury, add-on to a PDE5I, or soft glans after prosthesis.
- Urethral pain affects about one-third; use a condom if the partner is pregnant.

Intracavernosal injection

ICI acts directly on smooth muscle, **independent of nerves and nitric oxide (NO)**, so it works when PDE5Is fail (neurogenic, post-prostatectomy, DM).

Agent	Mechanism	Notes
Alprostadil (PGE1) 5–40 µg	Adenylate cyclase raises cyclic AMP (cAMP)	Only FDA-approved ICI; ~70% efficacy; metabolised locally; **penile pain** commonest AE; least priapism and fibrosis
Papaverine	Nonspecific PDE inhibitor; Ca^{2+} channel block	Cheap; raises liver enzymes, causes priapism and fibrosis, so not used alone

Agent	Mechanism	Notes
Phentolamine	α-antagonist	Adjunct only; hypotension, reflex tachycardia
Trimix (alprostadil 10 µg, papaverine 30 mg and phentolamine 1 mg per mL)	Synergistic	≈90%; rescues monotherapy failures; less pain
Bimix	Papaverine with phentolamine, or alprostadil with phentolamine	≈90%

- Titrate in the office from a low dose, since neurogenic ED is very sensitive, and inject the lateral shaft.
- **Priapism** is highest with papaverine mixes and in neurogenic or young men; the patient must seek care beyond 4 h. Long-term use causes fibrosis and curvature.
- Contraindications: psychological instability, priapism risk (sickle cell), severe coagulopathy, unstable CVD, poor dexterity (the partner may inject), and monoamine oxidase (MAO) inhibitors, because phenylephrine reversal risks a hypertensive crisis.

Topical agents (alprostadil cream, the over-the-counter MD3000 gel) give modest benefit.

Restorative and investigational

Therapy	Proposed mechanism	Evidence/status
Li-ESWT	New vessels; nerve, endothelial, smooth muscle regeneration	RCT meta-analysis: IIEF better than sham in **mild–moderate vasculogenic** ED, the best evidence; still investigational (Sexual Medicine Society of North America: research only)
Platelet-rich plasma	Growth factors	Conflicting RCTs
Stem cells	Regeneration	Uncontrolled series only
Botulinum toxin ICI	Blocks sympathetic noradrenaline	Short-term benefit in refractory ED

Combination therapy

Monotherapy fails in nearly 40%. Useful pairs are daily tadalafil with on-demand sildenafil, a PDE5I with T, IU, ICI or VED, and a prosthesis with IU for a soft glans. PDE5I with ICI is off-label, so test it in the office.

Surgery

- A **penile prosthesis** suits men whose other therapy is contraindicated, ineffective or unwanted, and those with corporal fibrosis or Peyronie disease (see Ch 75).

LAQ Surgical management of ED and penile prostheses (types, biofilms)

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P2 LAQ "Surgical management of erectile dysfunction and penile prosthesis" (asked as "Discuss surgical management of erectile dysfunction", "Penile Prosthesis" and "Describe various penile prosthesis in ED") · P3 SAQ "Types of penile prosthesis. What are biofilms in penile prosthesis?" · P4 SAQ "Penile implants."

Introduction

- Surgery for ED (erectile dysfunction) means **penile prosthesis** (main operation) and, rarely, **penile vascular surgery**.
- Prosthesis: most effective, but last-line, treatment of ED; **satisfaction 80–90%**. First devices: Scott inflatable (1972), Small–Carrion rod (1978).
- Lue's goal-directed approach: surgery is for men who fail, cannot use or decline oral PDE5Is (phosphodiesterase type 5 inhibitors), intracavernosal injection, intraurethral alprostadil or a VED (vacuum erection device).
- Detailed vascular testing is needed only before vascular surgery.

Indications and contraindications for a prosthesis

- **Indications:** ED not responding to, contraindicated for, or unacceptable with (refusal of) oral PDE5Is, injections or vacuum devices; common after prostatectomy, pelvic radiotherapy, diabetes.
- Corporal (cavernosal) fibrosis: after ischaemic priapism (**>36 h**), infection or implant removal; early implant preserves length.
- **Peyronie disease with ED**; penile or pelvic trauma.
- **Contraindications:** active skin, urinary (UTI [urinary tract infection]) or systemic infection; uncontrolled diabetes; unrealistic expectations; reversible ED.
- In insensate men (spinal cord injury) use an inflatable device, because rods erode.

A. Penile vascular surgery

	Arterial revascularisation	**Venous surgery**
Defect	Arteriogenic (poor inflow)	Veno-occlusive (venous leak)
Work-up	Duplex; selective pudendal angiography (also checks the inferior epigastric donor)	DICC (dynamic infusion cavernosometry and cavernosography) locates the leak
Technique	Inferior epigastric artery to dorsal artery (revascularisation) or to deep dorsal vein (arterialisation)	Ligation or embolisation of deep dorsal, superficial dorsal or crural veins; crural plication
Selection	Age <55, non-smoker, non-diabetic, no venous leak, focal internal pudendal stenosis; best in young men after pelvic or perineal trauma	Not defined
Status	Strictly selected men	Investigational
Complications	Glans hyperaemia (commonest), shunt thrombosis, inguinal hernia	Glans numbness, shortening, curvature, skin necrosis

B. Penile prosthesis: types

Type	**Devices / examples**	**Features**	**Best for**
Malleable (semirigid) rods	Tactra (nitinol core), Genesis (braided silver), Rigi 10 (stainless steel)	Two bendable rods; simple, cheap; almost never fail (mechanical failure exceedingly rare; P3: no mechanical failure); always firm; erode more in insensate men	Poor manual dexterity; cost
Mechanical rods	Spectra (discontinued)	Articulating segments	–

Type	Devices / examples	Features	Best for
2-piece inflatable	Ambicor (discontinued)	Cylinders + scrotal pump; no reservoir; poor flaccidity	–
3-piece inflatable	AMS 700 CX/LGX/CXR; Coloplast Titan, Titan Touch, Titan Narrow; Rigicon Infla 10	Two cylinders, scrotal pump, abdominal reservoir; **best rigidity and true flaccidity**; most popular	Most men

Three-piece inflatable details

- **Expansion:** girth only (CX, Titan) or girth and length (LGX, up to 25%; Infla 10 AX). **Narrow** cylinders (CXR, Titan Narrow) for fibrotic corpora.
- **Material:** AMS and Rigicon use silicone–fabric–silicone layers; Titan uses polyurethane (Bioflex).
- **Coatings:** InhibiZone (rifampicin + minocycline) on AMS; hydrophilic coating on Titan, dipped in antibiotic.
- **Reservoir:** space of Retzius, or **ectopic submuscular** after pelvic surgery or cystectomy; lockout valve prevents autoinflation.

Consent

- The erection may be shorter (pseudocapsule scar); the aim is a firm penis for intercourse; sensation and ejaculation may reduce; infection or mechanical failure may need reoperation.

Preoperative preparation

- Treat phimosis and skin, fungal and urine infection; sterile urine; stop anticoagulants and aspirin.
- Blood glucose **<200 mg/dL on the day of surgery** (more important than HbA1c [glycated haemoglobin]).
- Shave in theatre; chlorhexidine–alcohol skin preparation (about 40% fewer wound infections than povidone-iodine); MRSA (methicillin-resistant *Staphylococcus aureus*) decolonisation if a carrier.
- IV vancomycin or a cephalosporin plus gentamicin, 1 hour before incision, for up to 24 hours.

Operative steps (3-piece, penoscrotal approach)

`Catheter → upper scrotal incision → expose corpora lateral to urethra → stay sutures → corporotomy → dilate proximally to ischial tuberosity, distally under glans (instruments dorsolateral; 7→13 mm) → goalpost test for crossover → measure → cylinders equal to or 1 cm shorter → test inflation → close corporotomies → pump in dependent subdartos (dartos) pouch → reservoir in space of Retzius → connect tubing → test inflation → irrigate → close in layers`

- Dilate with Hegar or Brooks dilators, instruments pointing dorsolaterally; check for crossover or perforation; cylinders sized to corporal length (equal to or 1 cm shorter than measured length).
- **Approaches:** penoscrotal (commonest; secures the pump; urethra at risk); infrapubic (reservoir placed under direct vision; dorsal nerve at risk); subcoronal or ventral penile for rods.

- Malleable rods are placed slightly shorter: 0.5–1 cm shorter than the measured length; girth chosen with the two-dilator test.
- **Surgical principles:** antibiotics, glucose control, no-touch technique.
- **Intraoperative problems:** crossover (redilate laterally); crural perforation (suture sling); distal urethral perforation or urethral injury (abandon that side for 5–6 months); fibrotic corpora (cavernotomes, narrow cylinders); curvature (modelling for Peyronie's disease); floppy glans (glans fixation).
- **Postoperative:** mummy wrap with glans visible; cylinders partly inflated; drain out by 24 hours; activation at 4–6 weeks.

Biofilm

- **Definition:** a structured microbial community in a self-made **polysaccharide matrix**, attached to the implant.
- **Stages:** 1. Conditioning film of host proteins coats the device within minutes. 2. Reversible, then irreversible bacterial adhesion. 3. Microcolonies. 4. Maturation into a 3D structure with water channels and quorum sensing. 5. Dispersal of free bacteria, causing clinical infection.
- **Organisms:** coagulase-negative staphylococci commonest; also *S. aureus*, Gram-negatives, *Candida*.
- **Why antibiotics fail:** the matrix blocks penetration, bacteria are slow-growing (persisters), and they survive up to **1000–1500 times** normal killing levels.
- **Relevance:** organisms from the first operation lie dormant; revising an uninfected device gives about 10% infection, cut to about 3% by **antiseptic washout**.

Complications

- **Mechanical failure** (commonest; inflatables): repair in about **15% at 5 years, 30% at 10 years**.
- **Infection (1–3%)**: biofilm on the device.
- **Prevention:** glucose <200 mg/dL; treat skin and urine; decolonise MRSA carriers; intravenous vancomycin or cephalosporin plus gentamicin (up to 24 h); chlorhexidine–alcohol preparation; **antibiotic-coated devices halve infection** (InhibiZone rifampin–minocycline; hydrophilic Titan dipped in antibiotic); **no-touch technique** (under 0.5%).
- **Infected implant:** pus, fixed pump or exposed parts. Remove **all** components.
- **Mulcahy salvage:** remove all parts; wash with dilute povidone-iodine (3 minutes) then saline or antibiotic; change gloves, drapes and instruments; insert a new device at once. Success **84%**; fails with early onset, aggressive organisms, cellulitis. Delayed reimplantation loses length.
- **Intraoperative:** crural perforation, crossover, urethral injury.
- **Others:** erosion or extrusion (catheterisation, rods in insensate men); reservoir erosion into bladder or bowel; reservoir herniation; pump migration; autoinflation; penile shortening; **glans necrosis** (diabetes, smoking, radiation, previous implant); glans hypermobility; cylinder aneurysm.

Special situations (also know)

- Ischaemic priapism >36 hours: early implantation preserves length.
- Transplant recipients: no extra infection.
- After phalloplasty: single cylinder, high erosion.

- Peyronie's disease: modelling.

Outcomes

- Patient and partner satisfaction **80–90%**, the highest of any ED treatment; the main complaint is shorter length.

Recent advances

- Minimal-dilation technique; flat ectopic reservoirs; penile transplantation. Regenerative therapies (Li-ESWT [low-intensity shock wave therapy], stem cells) remain experimental.

Diagrams to draw

- 3-piece inflatable prosthesis: two cylinders, scrotal pump, reservoir in the space of Retzius.
- Biofilm life cycle.

> **Key points to remember:**
> - Prosthesis is the last line but gives 80–90% satisfaction.
> - 3-piece inflatable gives the best rigidity and flaccidity.
> - Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
> - Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
> - Biofilm: microbes in a matrix; antibiotics fail.
> - Infection 1–3%; infected implant: remove all parts; Mulcahy salvage 84%.
> - Repairs 15% at 5 years.
> - Arterial revascularisation only for young men with post-traumatic focal stenosis.

7 · Landmark trials

Trial	Design	Result	Take-home
Prostate Cancer Prevention Trial (placebo arm)	Prospective cohort, men ≥55	ED gave 45% more cardiac events at 5 years	ED is a CV sentinel; stratify risk
REINVENT	RCT: nightly vs on-demand vardenafil vs placebo after nerve-sparing prostatectomy	No difference in spontaneous recovery	Scheduled "rehabilitation" PDE5I is unproven

73 Priapism

1 · Definition and classification

- **Priapism** is a full or partial erection lasting **≥4 h** beyond, or unrelated to, sexual stimulation. Untreated ischaemic priapism causes fibrosis and permanent erectile dysfunction (ED).
- It is classified by **oxygenation, not duration**: failed outflow causes time-critical ischaemia, while excess arterial inflow gives an oxygenated, painless erection that can wait.

Feature	Ischaemic (low-flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high-flow)
Share	**95%**	Subset of ischaemic	<5%
Lesion	Failed outflow, like a compartment syndrome	Repeated self-limiting episodes, often nocturnal	Fistula between cavernosal artery and sinusoids
Typical cause	Drugs, sickle cell disease (SCD), blood disorders	SCD	Blunt perineal/penile trauma, needle injury
Pain / rigidity	Painful (within 6–8 h); fully rigid, tender	Painful; rigid during episode	Painless; partially rigid
Blood gas / Doppler	Low oxygen, high CO_2, acidotic; absent or low flow	Ischaemic in each episode	Arterial gas; high flow, fistula or blush
Management	Emergency decompression	Treat episode as ischaemic, plus prophylaxis	Observe, then embolise
Fibrosis/ED	Common if untreated	Risk with each episode	Usually absent

> **KEY POINT**
>
> Trap: stuttering priapism is **ischaemic**, so any episode lasting ≥4 h needs the same first-line treatment.

2 · Aetiology

LAQ Priapism: diagnosis and management

Paper answer · P2 · P3 · merged from 2

Asked as: P3 LAQ "Priapism: Diagnosis and Management" (also asked as P3 SAQ "Priapism") · P2 SAQ "Management of priapism"

Definition

- Full or partial erection lasting **≥4 hours** beyond sexual stimulation, or unrelated to it (AUA/SMSNA [American Urological Association/Sexual Medicine Society of North America] 2021–2022).

Classification (types)

	Ischaemic (low flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high flow)
Share	**95%**	Subset of ischaemic	<5%

	Ischaemic (low flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high flow)
Lesion	Veno-occlusion	Repeated (recurrent) ischaemic episodes	Cavernosal artery–sinusoid (arterio-sinusoidal) fistula after trauma
Pain, rigidity	Painful, fully rigid	Painful, recurrent, often nocturnal	Painless, partial
Typical cause	ICI (intracavernosal injection), drugs, SCD (sickle cell disease)	SCD	Perineal or needle trauma
Urgency	**Emergency**	Emergency per episode + prophylaxis	Not an emergency

Aetiology

- **Ischaemic:** ICI (papaverine, phentolamine, PGE1 [prostaglandin E1]), including recreational use; α-blockers, trazodone, antipsychotics, cocaine; heparin or warfarin withdrawal; **SCD** (29–42% of affected men in their lifetime; 63% of paediatric ischaemic priapism); leukaemia (1–5%, high white cell count); penile metastases; scorpion and spider venom.
- **Non-ischaemic:** straddle or coital injury, pelvic fracture, needle injury. Onset is often delayed, when the clot over the arterial tear lyses.

Pathophysiology

- **Ischaemic:** failed outflow acts like a compartment syndrome. Hypoxia, hypercarbia and acidosis cause sinusoidal thrombosis, smooth-muscle necrosis and TGF-β (transforming growth factor beta)-driven fibrosis, so ED (erectile dysfunction) follows. Irreversible damage starts within hours.
- **Stuttering (molecular):** low basal NO (nitric oxide) from eNOS (endothelial NO synthase) dysfunction downregulates PDE5 (phosphodiesterase 5). When an erection starts, cGMP (cyclic guanosine monophosphate) is not broken down, so relaxation is unopposed. Low Rho-kinase and high adenosine add to this. This explains why daily PDE5 inhibitors prevent episodes (they reset PDE5 regulation).
- **Non-ischaemic:** unregulated, well-oxygenated arterial inflow, so fibrosis is unusual.

Diagnosis

- **History:** duration, pain, drugs, trauma, SCD or blood disease, prior episodes, baseline erectile function.
- **Examination:** rigid tender corpora with a soft glans and spongiosum (ischaemic); partial, non-tender (high flow); perineal bruising; vital signs before phenylephrine.
- **Corporal blood gas** (cornerstone; decides the type); high flow resembles arterial blood:

	Colour	pO2 (mmHg)	pCO2 (mmHg)	pH
Ischaemic	Dark red	**<30**	**>60**	**<7.25**
Flaccid (mixed venous)	Medium red	40	50	7.35
Non-ischaemic (arterial)	Bright red	>90	<40	7.40

- **PDUS (penile duplex ultrasound, colour Doppler)** when the gas is equivocal: ischaemic = absent cavernosal flow or PSV (peak systolic velocity) <50 cm/s with no diastolic flow;

high flow = PSV >50 cm/s, may show the fistula. Angiography only for embolisation.
- Blood count, reticulocytes, haemoglobin electrophoresis, coagulation, toxicology: in parallel, never delaying treatment.

Management of ischaemic priapism (AUA/SMSNA 2021)

``` Counsel on ED risk → dorsal penile block <36 h: aspiration (19 G, penoscrotal junction, 3 or 9 o'clock) ± saline irrigation + phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h (monitor blood pressure, pulse: reflex bradycardia) → fails → distal shunt (Winter, T-shunt, Al-Ghorab) ± tunnelling → proximal shunt (Quackels) or penoscrotal decompression

> 36 h or refractory → early (immediate) penile prosthesis (or analgesia and counselling) ```

- Aspiration alone succeeds in under 1 in 4. Cold (10 °C) saline irrigation was better than body temperature in an RCT (randomised controlled trial); avoid chilled saline in SCD.
- Ice, oral agents, hydration and IV (intravenous) fluids are adjuncts only; in SCD, intracavernosal treatment still comes first, with concurrent haematology care.

| Shunt | Method |
|---|---|
| **Winter** (distal) | Tru-Cut (biopsy) needle through glans; least effective |
| **Ebbehoj / T-shunt** (distal) | Scalpel through glans; T-shunt rotates blade 90° |
| **Al-Ghorab** (distal, open) | Dorsal glans incision; excise corporal (tunica) tips |
| **Snake tunnelling** | 7/8 Hegar dilator passed proximally through distal shunt |
| **Quackels / Sacher** (proximal) | Cavernosum to spongiosum at bulb; stagger ≥1 cm |
| **Grayhack / Barry** (proximal) | Cavernosum to saphenous / dorsal vein |
| **Penoscrotal decompression** | Proximal corporotomy, clot suction (evacuation); allows prosthesis |

- **ED by duration** (Bennett and Mulhall, SCD series): ≤12 h 0%; 12–24 h 22%; 24–36 h 56%; ≥36 h 100%.
- **Early prosthesis** (Zacharakis 2014): less shortening and fewer revisions than delayed insertion: shortening 3% vs 40%, revision 9% vs 27%, satisfaction 96% vs 60%.

### Stuttering priapism

- An episode lasting ≥4 hours is treated as ischaemic; home self-injection of phenylephrine.
- **Prophylaxis:** daily PDE5 inhibitor (Burnett 2014 RCT: primary endpoint missed, but promising), pseudoephedrine, etilefrine; hormonal agents (GnRH [gonadotropin-releasing hormone] agonists, antiandrogens, finasteride) at the cost of libido. **Also know:** digoxin; baclofen.

### Non-ischaemic priapism (AUA/SMSNA 2022)

- Observation, perineal compression and ice; about 60% resolve. Reassess at 4 weeks.
- Persistent: PDUS, then **selective cavernosal artery embolisation** (resolution ≥75%, recurrence 30–40%). For recurrence, repeat embolisation with non-resorbable PVA (polyvinyl alcohol), not surgery. Surgical ligation is a last resort (ED about 50%).

### Complications

- ED, corporal fibrosis, penile shortening; shunt-related oedema, haematoma, urethral injury or fistula, skin necrosis, infection.
```

Diagram to draw: distal (glans) and proximal shunt sites on a penile outline.

> **Key points to remember:**
> - Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
> - Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
> - ED risk 0% ≤12 h, 100% ≥36 h; early prosthesis beyond 36 h or if refractory.
> - High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

Also, from the condensed book (not in the paper answer):

- Metastases obstruct venous outflow.

Ischaemic – extra points

- Many other drugs, including oral phosphodiesterase type 5 inhibitors (PDE5Is), can cause it, so take a full drug history.
- In leukaemia, give cytotoxics and leukapheresis alongside penile treatment.
- **Recurrence after a shunt** may mean shunt thrombosis. Give **aspirin 325 mg + heparin 5000 U subcutaneously** before surgery, then aspirin 81 mg daily for 2 weeks.

Sickle cell disease

- Deoxygenated haemoglobin S polymerises, causing haemolysis and low nitric oxide (NO), which obstruct outflow.
- Lifetime risk is **29–42%**; SCD causes 23% of adult and **63% of paediatric** ischaemic priapism. Stuttering is commoner than major episodes.

Non-ischaemic

- Onset is often delayed ≥1 day after trauma, because the clot over the arterial tear disperses in a nocturnal erection.
- Transient post-ischaemic hyperaemia after treatment mimics it; colour Doppler distinguishes them.

3 · Molecular pathophysiology

- Low basal NO reduces cyclic guanosine monophosphate (cGMP) feedback, so **PDE5 is down-regulated**. A neural NO surge then produces cGMP that cannot be broken down, so relaxation is unopposed.
- Paradox: **a daily PDE5I restores PDE5 and prevents** recurrent episodes.

4 · Evaluation

*See the paper answer **Priapism: diagnosis and management** (Section 9, Ch 73).*

*See the paper answer **Doppler in urology** (Section 1, Ch 4).*

Also, from the condensed book (not in the paper answer):

- Record baseline erectile function, because it is needed for prosthesis counselling.
- The glans and spongiosum stay soft in both types; their involvement suggests another disease.

ALGORITHM

1. History, examination and corporal blood gas.
2. If ischaemic, then acute ischaemic algorithm.
3. If indeterminate, then penile/perineal PDUS.
4. If non-ischaemic, then expectant algorithm.

5 · Acute ischaemic priapism (AUA/SMSNA)

- Irreversible damage begins 4–12 h after presentation, so counsel about ED risk first. Conservative measures are **adjuncts only; never delay first-line treatment**. Oral terbutaline or pseudoephedrine is not recommended.

*See the paper answer **Priapism: diagnosis and management** (Section 9, Ch 73).*

ALGORITHM

1. If <36 h, then aspirate ± irrigate and inject intracavernosal phenylephrine (BP and cardiac monitoring).
2. If that fails, then distal shunt ± tunnelling; tunnel if not done.
3. If still erect, then proximal shunt or penoscrotal decompression; if refractory, prosthesis.
4. If >36 h, then counsel high ED risk with analgesia/follow-up, OR early penile prosthesis.

Aspiration / irrigation

- After a penile block, insert a **19-G needle** at the penoscrotal junction at **3 or 9 o'clock**; one side usually suffices. Aspirate, repeat and irrigate with saline.
- Aspiration alone works in fewer than 1 in 4. Cold (10 °C) irrigation gave 96% vs 60% in a randomised controlled trial (RCT), but **avoid chilled saline in SCD**.

Phenylephrine

- It is a selective α1-agonist with the least cardiac effect.
- Dilute to **100–500 µg/mL** and give **1-mL aliquots every 3–5 min** for up to **1 h** before declaring failure; the maximum is about 1 mg in 1 h (European Association of Urology).
- Toxicity includes hypertension, headache, arrhythmia and **reflex bradycardia**. Monitor, especially with cardiovascular disease or monoamine oxidase (MAO) inhibitors.

Shunts

*See the paper answer **Priapism: diagnosis and management** (Section 9, Ch 73).*

Also, from the condensed book (not in the paper answer):

- AUA: proximal shunts only after a failed distal shunt with tunnelling. Complications include urethral injury or fistula, necrosis, infection and pulmonary embolism.

Early penile prosthesis

- It is used for priapism **>36 h** or failed shunts, because oedematous corpora dilate easily, while delayed surgery meets dense fibrosis.
- Early vs delayed implantation: shortening 3% vs 40% and satisfaction 96% vs 60%.
- Prior distal shunts raise the risk of distal perforation (see Ch 75).

6 · Non-ischaemic priapism

*It is **not an emergency**.*

- About 60% resolve with observation, compression and ice.

ALGORITHM

1. Observe and reassess at 4 weeks.
2. If it persists, then PDUS (fistula site, size), counsel, and selectively embolise the internal pudendal/cavernosal artery.
3. If it recurs, then repeat embolisation with a non-resorbable agent.
4. Only after repeated failure, ligate surgically.

Agent	Type	Feature
Autologous clot	Temporary	Children; early recurrence
Gelatin sponge	Temporary (2–6 weeks)	Reabsorbed
Microcoils	Permanent	Theoretical ED risk
Polyvinyl alcohol particles	Permanent	**Preferred for repeat** (AUA)
N-butyl cyanoacrylate	Permanent	Second line for recurrence

- Embolisation succeeds in ≥75%, with 30–40% recurrence. Open ligation causes ED in about 50%.

7 · Stuttering priapism — prevention

- An episode ≥4 h is treated as acute ischaemic priapism. Short episodes are managed with fluids, walking and analgesia, and home self-injected phenylephrine for breakthrough.
- No prophylactic drug has robust evidence.

Agent	Dose	Note
PDE5I	Sildenafil 25–50 mg daily; tadalafil 5–10 mg	Most promising; start when flaccid; the only RCT missed its primary endpoint
Pseudoephedrine	60–120 mg at night	Off-label first-line α-agonist
Etilefrine	50–100 mg daily	α1-agonist; Cochrane: insufficient evidence
GnRH agonists, antiandrogens (bicalutamide, flutamide, cyproterone)	Depot / oral	Suppress libido and erections; avoid before puberty is complete
Finasteride	5 mg, tapered to 3 mg then 1 mg	One SCD series
Digoxin	0.25–0.5 mg	Inhibits Na^+/K^+-ATPase
Baclofen	Oral 40 mg at night; intrathecal	Neurogenic/reflex erections

SCD specifics

- First-line aspiration and phenylephrine are the same; give IV hydration and analgesia **concurrently, never instead**. Exchange transfusion is not a primary treatment (AUA); involve haematology and avoid chilled irrigation.

8 · Landmark trials

Trial	Design	Result	Take-home
Ateyah 2005	RCT, ICI-induced priapism, irrigation temperature	10 °C saline 96% vs 60% at 37 °C	Cold irrigation (not in SCD)
Burnett 2014	RCT, SCD, sildenafil vs placebo	Primary endpoint missed; open-label benefit	PDE5I prophylaxis plausible, unproven
Zacharakis 2014	Early vs delayed prosthesis	Shortening 3% vs 40%; satisfaction 96% vs 60%	Implant early when ED is inevitable

74 Disorders of Male Orgasm and Ejaculation

1 · Spectrum

- Ejaculatory dysfunction (EjD) is the **most common male sexual disorder**. It includes premature ejaculation (PE), delayed ejaculation (DE), anejaculation and retrograde ejaculation (RE). Painful ejaculation, anhedonia and postorgasmic illness syndrome (POIS) are rarer.
- Orgasm is a separate **cortical** event, so ejaculation and orgasm can dissociate.

2 · Physiology of ejaculation

- Lumbar interneurons at **L3–L5** form the **spinal ejaculation generator**.

Phase	Events	Mediation
Emission	Vas, seminal vesicles and prostate deliver semen to the posterior urethra	Autonomic, **T10–L2**
Ejection	Bulbospongiosus, ischiocavernosus, levator ani contract rhythmically; **bladder neck closes**; external sphincter relaxes	Somatic **S2–S4 (pudendal)** and a sympathetic reflex
Orgasm	Cerebral processing of pudendal input	Cortical

- **Dopamine (D2) promotes** and **serotonin (5-HT) inhibits** ejaculation.
- 5-HT2C stimulation delays and 5-HT1A shortens ejaculation, so lifelong PE is thought to be **5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity**.

> **KEY POINT**
>
> Serotonin brakes, dopamine accelerates: PE drugs raise synaptic 5-HT; DE drugs are dopaminergic, noradrenergic or anti-serotonergic.

3 · Premature ejaculation — definition and classification

Short latency alone is not PE; loss of control and distress are needed.

SAQ Premature ejaculation

Paper answer · P1 · P3 · merged from 2

Asked as: P1 SAQ "Premature Ejaculation" · P3 SAQ "Premature ejaculation."

Definition

- **ISSM (International Society for Sexual Medicine) unified definition (2014):** all three needed: 1. Ejaculation always or nearly always within **about 1 minute** of vaginal penetration (lifelong), or a clinically significant, bothersome fall in latency, often to **≤3 minutes** (acquired). 2. Inability to delay ejaculation on all or nearly all penetrations. 3. Negative personal consequences: distress, frustration, avoidance of intimacy.
- **AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2020:** lifelong PE (premature ejaculation) within about **2 minutes**.
- **Types:** lifelong, acquired, variable and subjective (normal latency).

Physiology and aetiology

- Ejaculation = **emission** (sympathetic T10–L2) then **expulsion** (somatic S2–S4, pudendal). **Serotonin brakes and dopamine accelerates**.
- **Lifelong PE:** neurobiological and genetic; 5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity.
- **Acquired:** performance anxiety, relationship problems, **ED (erectile dysfunction)**, chronic prostatitis, hyperthyroidism, drug withdrawal.

Evaluation

- Diagnosis is by **history**: latency, control, bother; lifelong or acquired.
- **IELT (intravaginal ejaculatory latency time):** self-estimate in clinic; stopwatch in trials. Population median 5.4 minutes.
- **PEDT (Premature Ejaculation Diagnostic Tool):** 5 items; score ≥11 suggests PE, 9–10 probable PE. Also PEP (Premature Ejaculation Profile) and IPE (Index of Premature Ejaculation).
- IIEF-5 (International Index of Erectile Function) for ED; examination; TSH (thyroid-stimulating hormone) only if suggested.

Office algorithm (diagram to draw):

1. Confirm PE.
2. Secondary to ED? → treat the cause.
3. Otherwise → classify lifelong or acquired.

Management

- **Psychosexual education** for all; behavioural therapy (stop–start, squeeze).
- Acquired PE: treat the cause first (ED, prostatitis, thyroid).

Drug	Dose and timing
Dapoxetine (short-acting SSRI [selective serotonin reuptake inhibitor]; only licensed oral drug)	30–60 mg, 1–3 h before intercourse
Paroxetine (strongest delay) 10–40 mg, sertraline, fluoxetine, citalopram	Daily, off-label; full effect 2–3 weeks; do not stop abruptly
Clomipramine	12.5–50 mg daily or 3–4 h before
Topical lidocaine–prilocaine spray (licensed in Europe 2014)	5–30 min before (P1: at least 5 min before); condom avoids partner numbness
PDE5Is (phosphodiesterase type 5 inhibitors)	With ED, alone or with SSRI
Tramadol	Low evidence; addiction risk

- SSRIs raise synaptic serotonin, which delays ejaculation. Avoid in bipolar disorder.
- **Also know:** high dropout from drug therapy.

> **Key points to remember:**
> - ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
> - Lifelong is neurobiological; acquired is usually secondary.
> - Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Also, from the condensed book (not in the paper answer):

- DSM-5 duration (≥6 months, 75–100% of occasions) and severity bands are in the table below.

	Lifelong	Acquired	Variable	Subjective
Onset	From first encounters, every partner	After previously normal latency	—	—
Latency	≤ ~1 min (mostly 30–60 s)	Markedly reduced, often ≤3 min	Occasionally short	Normal or long
Basis	Neurobiological, genetic	Medical/psychological	Normal variation	Psychological/cultural
Disorder?	Yes	Yes	No	No

- The International Society for Sexual Medicine (ISSM) requires ejaculation within **about 1 min** (lifelong) or a bothersome fall, often to **≤ about 3 min** (acquired). It is the first **evidence-based** definition, for vaginal intercourse only. **Latency alone never suffices**.

	ISSM	DSM-5 (2013)	AUA/SMSNA (2020)
Lifelong latency	≤ ~1 min	≈1 min	≈**2 min** since sexual debut
Acquired	Bothersome fall, often ≤3 min	Specifier only	Marked fall (2–3 min or ≥50%)
Duration/frequency	Always or nearly always	≥6 months, 75–100% of occasions	—
Severity	—	Mild 30–60 s; moderate 15–30 s; severe <15 s	—
Basis	Evidence	Same data	Expert opinion

- The AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2-min line trades specificity for sensitivity.
- Community stopwatch intravaginal ejaculatory latency time (IELT) is positively skewed, median **5.4 min**; ≤1 min is statistically abnormal.

4 · Premature ejaculation — aetiology and evaluation

- Lifelong PE affects younger men with few comorbidities; acquired PE affects older men with metabolic syndrome or diabetes.
- **PE with erectile dysfunction (ED)** affects up to half of men with ED, who rush before a partial erection is lost.
- PE is the main sexual symptom in **chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS)**, and it improves when **hyperthyroidism** is treated, though routine thyroid tests are not indicated.
- Examination is desirable in lifelong and **mandatory in acquired PE**, with digital rectal examination (DRE) for prostatitis.

5 · Premature ejaculation — treatment

ALGORITHM

1. Confirm PE.
2. If it is secondary to ED or another dysfunction, then treat that primary cause.
3. If lifelong, then SSRI ± behavioural therapy (patient preference).
4. If acquired, then treat the cause ± behavioural therapy/SSRI; try withdrawal after 6–8 weeks.
5. If variable or subjective, then reassurance, education, psychotherapy.

- Behavioural therapy is mainly an adjunct.

Drug	Dose / timing	Notes
Dapoxetine	30–60 mg, 1–3 h before	**Only licensed oral PE drug** (>50 countries, not FDA); short half-life; works from the first dose; no PDE5I interaction; many stop it
Paroxetine (daily)	10–40 mg	**Strongest delay**; off-label
Sertraline / fluoxetine / citalopram	50–200 / 20–40 / 20–40 mg daily	Off-label
Clomipramine (a tricyclic antidepressant, TCA)	12.5–50 mg daily or 3–4 h before	Off-label
Topical lidocaine/prilocaine	20–30 min before (spray ≥5 min)	Condom advised; level 1a evidence
Phosphodiesterase type 5 inhibitors (PDE5Is)	30–50 min before	Best with coexisting ED; add-on to SSRI
Tramadol	25–50 mg, 3–4 h before	Low evidence; addiction risk

- **Daily selective serotonin reuptake inhibitors (SSRIs)** act in 5–10 days, fully at **2–3 weeks**.
- Use caution in bipolar disorder, with aspirin or NSAIDs, and under 18 years. **Never stop them abruptly.**
- Meta-analyses show that a PDE5I beats placebo and that a PDE5I with an SSRI beats an SSRI alone.

On-demand dapoxetine	Daily off-label SSRI
Licensed; fewer side effects; little withdrawal; infrequent intercourse	Greater delay; full effect 2–3 weeks; must taper; established relationships

KEY POINT

The real limiter in PE is adherence, not efficacy: most stop SSRIs and topicals (cost, lost spontaneity, "antidepressant" stigma).

6 · Delayed ejaculation, anejaculation and anorgasmia

- DE (DSM-5) is a marked delay or absence of ejaculation despite adequate stimulation, on ~75–100% of occasions, with distress. There is **no validated cut-off**, but >25–30 min with distress suggests DE.
- **Anorgasmia** is absent perceived orgasm. DE is mostly acquired and rises with age; its hallmark is normal erections with **low subjective arousal**.

Category	Causes
Afferent	Ageing (penile afferents, Pacinian corpuscles)
Psychogenic	Inhibited ejaculation; performance anxiety; religious orthodoxy; **idiosyncratic vigorous masturbation**
Endocrine	Hypogonadism, **hypothyroidism** (DE; hyperthyroidism causes PE), hyperprolactinaemia
Neurogenic disease	Spinal cord injury (SCI), then diabetic autonomic neuropathy, then multiple sclerosis (MS)
Neurogenic surgery	Radical prostatectomy, rectal surgery, sympathectomy, aortic aneurysm repair, retroperitoneal lymph node dissection (RPLND)
Anatomical (causes RE)	Transurethral resection of prostate (TURP), bladder neck incision
Congenital / infective	Müllerian cyst, Wolffian anomaly, prune belly; urethritis, tuberculosis (TB), schistosomiasis
Drugs	**SSRIs** (~7-fold DE risk), TCAs, antipsychotics, α-methyldopa, thiazides, alcohol

- **Superior hypogastric plexus** injury at RPLND causes dry ejaculation, which **nerve-sparing RPLND** preserves in most men.
- Ask whether masturbation works and whether he has orgasm without semen (RE or obstruction). Examine the vasa and do a DRE; check morning testosterone (T), HbA1c and lipids.

ALGORITHM

1. Orgasm? If never or sometimes, then inhibited orgasm (psychosexual therapy) or ageing (reassure, change technique).
2. If always but no ejaculate, then look for **sperm in post-orgasm urine**; if present, RE.
3. If absent (aspermia), then failure of emission (neurogenic, drug, metabolic) or ejaculatory duct obstruction (EDO).

	Bilateral EDO	RE	Failure of emission
Antegrade volume	Very low	Low or absent	Absent
pH, fructose	Low (seminal vesicle fluid, ~80% of volume, is missing)	—	—
Post-orgasm urine	No sperm	**Sperm and fructose**	No sperm

Treatment

- Treat the cause, and counsel about anejaculation and sperm retrieval before pelvic surgery. Refer to a psychosexual therapist once organic and drug causes are excluded (AUA).
- Normalise T, change the offending drug and change masturbatory style. Drugs are all off-label, with about 40% success.

Drug	Action	Dose
Cabergoline	Central D2 agonist	0.5–2 mg every 3 days; **annual echocardiography** (valvular fibrosis)
Bupropion	Dopaminergic	150 mg once or twice daily
Pseudoephedrine	α1 agonist	120 mg 1–2 h before
Reboxetine	Noradrenaline reuptake inhibitor	4–8 mg daily
Cyproheptadine	5-HT antagonist	4–12 mg 3–4 h before (SSRI-induced DE); sedation
Amantadine, buspirone	—	Anecdotal
Intranasal oxytocin	Oxytocinergic	Both RCTs negative

7 · Retrograde ejaculation

*See the paper answer **Retrograde ejaculation** (Section 9, Ch 67).*

Also, from the condensed book (not in the paper answer):

- Diagnosis: **>5–10 sperm/high-power field** in urine passed after masturbation confirms RE.
- Aquablation preserves ejaculation better than TURP in randomised controlled trials.

8 · Sperm retrieval in SCI

- Reflex erection needs the sacral arc, but ejaculation needs thoracolumbar outflow, so it is rare (<5%) with complete upper motor neuron lesions; RE is common.

ALGORITHM

1. **Penile vibratory stimulation (PVS)** is first line, with up to ~70% success; it needs an intact reflex at the T12–S1 level.
2. If PVS fails, then **electroejaculation (EEJ)**: a rectal probe stimulates hypogastric sympathetic efferents; effective and safe.
3. If EEJ fails, then prostate massage, then surgical retrieval (TESE/TESA/MESA/PESA); testicular sperm extraction succeeds in most.

- Lesions above T6 risk **autonomic dysreflexia**: pretreat with **nifedipine** and monitor BP.

9 · Other disorders

Disorder	Key points
Painful ejaculation	Prostatitis/CPPS, urethritis, benign prostatic hyperplasia (BPH), seminal vesicle stones, EDO, α-blockers (least with alfuzosin); often none. Surgery (TURP, transurethral resection of ejaculatory ducts (TURED), pudendal neurolysis) weakly supported
Haematospermia	Usually benign (see pointer below). Seminal vesiculoscopy if refractory (see Ch 70); evaluate if >40 y or persistent
Climacturia	Leak at orgasm after radical prostatectomy. Steps: void first, condom, pelvic floor training, constriction ring, male sling
Ejaculatory anhedonia	Orgasm without pleasure; depression, hypogonadism, hyperprolactinaemia, SSRIs
Post-SSRI sexual dysfunction	Genital numbness, low desire, ED, anhedonia persisting months to years
POIS	Myalgia, fatigue, flu-like state within 30 min of orgasm
Orgasmic headache	Dull bilateral ache rising with excitement; sudden, intense at orgasm

SAQ Haematospermia

Paper answer · P1

Asked as:

- Hematospermia — *Winter 2025 Paper I Q3c (SAQ)*
- Hematospermia — *Practice Set 9 Paper I Q3d (SAQ)* (also answered in 2025-Winter Paper 1, Q3c)

Type: SAQ (10 marks)

Definition

- Blood in the ejaculate. It alarms patients but is usually **benign and self-limiting**, most often from non-specific prostatic or seminal vesicle inflammation.
- **Primary (idiopathic):** no cause found, often young men. **Secondary:** a cause is identified.
- **Pseudo-haematospermia:** blood from the partner (partner's blood; condom test) or haematuria.

Causes

Group	Examples
Iatrogenic (commonest identifiable)	Prostate biopsy (about 10–50%, lasts about a month), TURP (transurethral resection of the prostate), radiotherapy, BPH (benign prostatic hyperplasia) procedures
Inflammation and infection	Prostatitis, vesiculitis, urethritis, STI (sexually transmitted infection), **genitourinary tuberculosis**, schistosomiasis
Obstruction and cysts	Ejaculatory duct obstruction or calculi; seminal vesicle cyst (Zinner syndrome) or utricle cyst
Tumours	**Prostate cancer**; rarely urethral, bladder or seminal vesicle
Vascular	Prostatic varices, BPH vessels, AV (arteriovenous) malformation

Group	Examples
Systemic	Hypertension, bleeding disorders, anticoagulants, liver disease; also abstinence, trauma

Evaluation

- **History:** episodes, recent biopsy, LUTS (lower urinary tract symptoms), haematuria, TB (tuberculosis) contact, anticoagulants.
- **Examination:** blood pressure, genitalia, **DRE (digital rectal examination)**.
- **Basic tests:** urinalysis and culture, semen culture, STI screen, **PSA (prostate-specific antigen)** in men over 40, coagulation if indicated.

Who needs full work-up (full work-up if): age **≥40**, persistent beyond several weeks or recurrent, haematuria, abnormal DRE or PSA, risk factors for TB or cancer, or a distressed patient.

- **TRUS (transrectal ultrasound):** cysts, calculi, dilated seminal vesicles (>1.5 cm) or ejaculatory ducts (>2.3 mm) suggest obstruction.
- **MRI (magnetic resonance imaging) pelvis:** best for seminal vesicle haemorrhage, cysts and prostate cancer.
- **Cystoscopy and urine cytology**; early-morning urine for acid-fast bacilli.

Management

- **Young, single episode, normal tests: reassure**.
- **Treat the cause:** antibiotics or antitubercular drugs; control BP (blood pressure); correct coagulopathy.
- **Persistent prostatic bleeding:** 5-ARI (5-alpha-reductase inhibitor), e.g. finasteride.
- **Refractory or obstructive:** TURED (transurethral resection of ejaculatory ducts), cyst unroofing or aspiration, **seminal vesiculoscopy** (4–6 Fr small scope through the ejaculatory duct or utricle, irrigation, laser for stones).

Diagram to draw: algorithm: <40 and single episode → reassure; ≥40 or persistent → PSA, TRUS/MRI, cystoscopy.

> **Key points to remember:**
> - Usually benign; prostate biopsy is the commonest identifiable cause.
> - Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
> - Exclude prostate cancer and genitourinary tuberculosis.
> - TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

75 Surgery for Erectile Dysfunction

1 · Place of surgery and devices

A prosthesis is the last step for erectile dysfunction (ED).

*See the paper answer **Surgical management of ED and penile prostheses (types, biofilms)** (Section 9, Ch 72).*

Also, from the condensed book (not in the paper answer):

- Coloplast cylinders are polyurethane (Bioflex), so keep cautery **<35 W**; Boston and Rigicon silicone cylinders tolerate cautery.

2 · Selection and counselling

- **Consent is expectation management**: the device restores rigidity, not the former erection.
- The commonest disappointment is a **shorter erection**, because the pseudocapsule limits expansion (worse in Peyronie disease). Document it.
- With **phimosis**, circumcise first and implant about 6 weeks later.
- Raised glucose on the day needs intravenous (IV) insulin. Nasal methicillin-resistant Staphylococcus aureus (MRSA) carriers get chlorhexidine bathing and mupirocin 2% for 5 days.

3 · Surgical approaches

Approach	Advantages	Cautions
Penoscrotal	Majority choice; secure subdartos pump; good crural exposure	Urethra at risk, so Foley first; reservoir via inguinal canal
Infrapubic (Scott)	Reservoir under vision	Dorsal neurovascular bundle: stay sutures at 10 and 2 o'clock; patient pulls pump down daily
Subcoronal	Rods; allows straightening	Circumcise at the same time, or lymphoedema follows
Ventral penile	Rods without bending; foreskin kept	—

4 · Operative technique (three-piece)

ALGORITHM

1. Catheter; expose the tunica.
2. Two 2-0 polydioxanone (PDS) stay sutures per corpus; longitudinal corporotomy.
3. Dilate proximal, then distal (Metzenbaum, Hegar/Brooks 7→13 mm).
4. Goalpost test; measure (Furlow).
5. Cylinders in; test inflation 1. Close corporotomies; test inflation 2.
6. Reservoir, then pump; connect; test inflation 3.
7. Irrigate and close.

- Dilators point **dorsolaterally**, away from urethra and septum. The **goalpost test** (equal height and symmetry) detects crossover.

- The cylinder is equal to or 1 cm shorter than measured, because oversizing causes buckling and S-deformity; rear tip extenders adjust length.
- The pump sits in a dependent subdartos pouch, anterolateral to the testis.

Reservoir placement

- The **Retzius** space is entered with the bladder empty; the iliac vessels lie about 1 inch lateral.
- In a hostile pelvis (radiation, cystectomy, robotic prostatectomy, transplant) use an **ectopic** site, made safe by the **lockout valve**.
- **Submuscular suprafascial** is the commonest ectopic site. Fill it nearly full, because underfilling leads to migration.

5 · Intra-operative problems

- **Fibrotic corpora** follow priapism or explant (pelvic radiation does not add difficulty). Use cavernotomes or scar excision and **narrow cylinders**, always two.
- **Crossover** gives a lopsided erection and unequal measurements. Leave a large dilator in the common cavity and redilate the other side laterally; the Foley should lie midline between cylinders.
- **Crural perforation** (one dilator deeper, lengths differ by >1 cm) is repaired with a **suture sling** (2-0 Prolene through the rear tip, over the corporotomy).
- **Urethral perforation** at the corporotomy is closed and the operation proceeds. If distal (fossa navicularis, irrigant at the meatus), **abandon that cylinder**, place the contralateral one, leave no catheter and replace at 5–6 months.
- A **hypermobile glans** drooping ventrally is the supersonic transporter (**SST**) deformity. It is fixed with glans sutures (3-0 non-absorbable).
- **Residual curvature** is treated with high-rigidity cylinders (AMS 700 CX, Titan, Infla 10 X) and **modelling** (clamp tubing, bend 90° away for 90 s), then plication or incision with or without graft. Less than 20° is acceptable (see Ch 76).

> **KEY POINT**
>
> No catheter after distal urethral perforation: the urinary stream draws fluid into the urethra (Bernoulli), so urine does not enter the defect. Wait 5–6 months for scar to obliterate the false tract. Most intra-operative errors are **prevented by instrument direction** and **detected by symmetry** (goalpost test, midline Foley).

> **KEY POINT**
>
> After robotic prostatectomy the peritoneum is not replaced and bowel lies near the internal ring, so a blind inguinal Retzius reservoir can injure bowel. Choose an ectopic site or place it under vision.

6 · Complications

Infection usually costs the whole device.

Infection (1–3%)

- Risks include hyperglycaemia, revision, MRSA and scarred corpora.

- American Urological Association (AUA) prophylaxis is IV vancomycin, or a 3rd-generation cephalosporin plus gentamicin, 1 h before incision and for ≤24 h. Extended oral courses are unproven; in high-risk men add an antifungal, with or without piperacillin–tazobactam.
- Coatings (InhibiZone rifampin–minocycline; hydrophilic Titan/Infla) halve infection, and a **no-touch** technique gives <0.5%.
- Fever and erythema suggest infection; pus, an exposed device, progressive pain or a **pump fixed to the scrotal wall** are definitive. **Remove ALL components**, as retained parts harbour biofilm.

Option	Technique	Caveats
Explant, delayed replacement	Antibiotic irrigation drains for 3 days; reimplant at 2–6 months	Fibrosis, loss of several cm; vacuum erection device (VED) meanwhile
Mulcahy salvage	Remove all, antiseptic washes, new drapes, immediate new device	~84%; fails with onset <2 months, aggressive organisms, purulence with cellulitis
Delayed salvage	IV vancomycin and gentamicin, salvage at 3 days	Aggressive infection
Malleable salvage	Rods instead	Quick; convert later

- Washes use dilute povidone-iodine (0.35–3.5%) for 3 min, then saline or antibiotic; hydrogen peroxide is dropped as cytotoxic.
- An uninfected revision carries ~10% infection, falling to ~3% with antiseptic washout of all cavities.

Mechanical failure

- This is the **commonest complication**: leak, pump malfunction, autoinflation, kinking or cylinder aneurysm.
- If the device is >3 years old, replace **all** parts; check any retained part with the **ohmmeter test**. A rod that loses rigidity has a core fracture and is replaced.

Extrusion and erosion

Feature	Extrusion	Erosion
Position	Tip through tunica, still under skin/glans	Through skin or into urethra
Cause	Forceful dilation, oversizing, aggressive use	Infection, crossover, catheterisation; rods in insensate men (spinal cord injury)
Management	**Distal corporoplasty** (new dorsomedial plane behind pseudocapsule)	Remove all, washout, contralateral spacer, replace at 5–6 months

- **Reservoir erosion** into the bladder needs reservoir exchange and repair; into bowel, remove the whole device.
- **Penile necrosis** is linked to diabetes, vascular disease, smoking, prior implant and radiation. Avoid circumferential incisions and tight wraps, and remove cylinders promptly if it occurs.
- An early basal bulge is corporotomy disruption (repair); a late one is cylinder aneurysm (replace).

7 · Special situations

- After **priapism** >36 h, an **immediate implant** preserves length and relieves pain; avoid distally expanding cylinders (see Ch 73).
- In **Peyronie disease** length loss is greater, so use non-expanding high-rigidity cylinders with modelling.
- Transplant and HIV-positive men (adequate CD4) have no excess infection.

8 · Post-operative care

- It is day surgery. Leave cylinders **partially inflated** so the pseudocapsule forms large, remove the drain within 24 h and keep the penis upward for 4–6 weeks.
- Activation is at **4–6 weeks**, then the device is cycled daily. Dissatisfaction is mainly about length and rigidity.

9 · Penile revascularisation †

- †Only young men (<55) with a focal post-traumatic pudendal occlusion qualify (inferior epigastric to dorsal artery). It is not recommended for generalised atherosclerosis or venous leak.

76 Diagnosis and Management of Peyronie Disease

1 · Definition

- **PD** (American Urological Association, AUA 2015) is an acquired penile abnormality with **fibrosis of the tunica albuginea**, with or without pain, deformity, erectile dysfunction (ED) and distress.

> **KEY POINT**
>
> Myths corrected: PD is **not rare** (3–20%; higher in diabetes and ED). **Spontaneous resolution is rare** and up to half worsen untreated, so "wait 6–12 months" is a myth. **Calcification is not a marker of chronic disease**.

2 · Anatomy and pathophysiology

- The outer longitudinal tunica layer is **absent at 5–7 o'clock**, so the dorsum buckles; **60–70% of plaques are dorsal** with septal involvement.
- **Microtrauma** from bending during intercourse causes **delamination at the septal insertions**.
- **Transforming growth factor-β1 (TGF-β1)** is central. Collagen becomes disorganised and elastin decreases, so the penis **curves toward the plaque**.
- A **susceptible host** is needed, and PD is associated with **Dupuytren and Ledderhose** disease; only 16–40% recall trauma.

3 · Clinical features and phases

- Features are pain, deformity (curvature, indentation, **hourglass**, hinge, shortening), plaque and ED. Hourglass or hinge gives an **unstable penis**.
- Kelami grading: ≤30° mild, 31–60° moderate, >60° severe. Ventral and lateral curves are harder to manage than dorsal.

Feature	Active (acute)	Stable (chronic)
Pain	Present (flaccid, erection, intercourse)	Usually resolved
Deformity	Progressive	No progression
Aim	Prevent progression; non-steroidal anti-inflammatory drugs (NSAIDs) for pain; traction	Correct deformity (injection, surgery)
Surgery	Not appropriate	When **≥12 months from onset and stable ≥6 months**

- The differential includes congenital curvature, chordee, post-traumatic fibrosis and tumours (epithelioid sarcoma mimics PD).

4 · Evaluation

- The **key question** is: "If straight with current rigidity, would it be adequate for penetration?" If **no**, offer a prosthesis with straightening; the answer also predicts post-operative erectile function.
- Measure **stretched penile length (SPL)** from pubis to corona dorsally; this is essential, as patients overestimate loss.

- Extensive **calcification** on ultrasound predicts poor intralesional response and more surgery.

Levine calcification grade	Size	Surgery likelihood
1	<0.3 cm	Not increased
2	0.3–1.5 cm	Not increased
3	>1.5 cm, or multiple ≥1 cm	Increased

- **Intracavernosal injection (ICI) with or without penile duplex Doppler ultrasonography (PDDUS)** is required by the AUA **before any invasive treatment**. It measures curvature and gives peak systolic velocity, end-diastolic velocity and resistive index (RI).

5 · Non-surgical management

SAQ Management of Peyronie's disease

Paper answer · P1

Type: SAQ (10 marks)
Source: 2017-Winter Paper 1, Q3a

Definition

- **PD (Peyronie's disease)** (AUA (American Urological Association) 2015): acquired fibrosis of the tunica albuginea, with pain, deformity, ED (erectile dysfunction) and/or distress. Prevalence 3–20%; up to half worsen untreated.

Phases

- **Acute (active, first 12–18 months):** pain, worsening deformity; aim to stop progression. **Stable:** pain gone, deformity unchanged for ≥3–6 months; aim to straighten.

Evaluation

- History: pain, deformity, Dupuytren disease, ability to penetrate. **Key question:** "If your penis were straight, would your current rigidity allow intercourse?" (predicts postoperative function).
- Plaque, SPL (stretched penile length); **office intracavernosal injection ± duplex Doppler ultrasound** before any invasive treatment (curvature by goniometer, calcification, vascular status).
- **Kelami grading:** ≤30° mild, 31–60° moderate, >60° severe; note hourglass or hinge. PDQ (Peyronie's Disease Questionnaire).

Non-surgical treatment

Option	Evidence/position
NSAIDs (non-steroidal anti-inflammatory drugs)	For pain (only oral agent AUA supports)
Oral vitamin E, tamoxifen, potaba, pentoxifylline	No reliable curvature benefit; not recommended
CCH (collagenase *Clostridium histolyticum*)	Stable disease, curve 30–90°, intact erections. **IMPRESS I/II:** 34% improvement vs 18% placebo. Up to 4 cycles (2 injections of 0.58 mg) with modelling. Risk: corporal fracture

Option	Evidence/position
Verapamil, interferon α-2b (intralesional)	May offer; stabilise and modestly improve
Traction therapy	Gains length, reduces curvature; useful in acute phase
ESWT (extracorporeal shock wave therapy)	Pain only; no effect on curvature

Surgery

- **Indications:** stable disease (≥12 months, stable ≥6 months), deformity preventing intercourse, failed conservative therapy.

```
Erection adequate (± PDE5I (phosphodiesterase-5 inhibitor))? ├─ No → penile prosthesis (inflatable) + manual modelling; │ residual >30° → tunical incision; graft if defect >2 cm └─ Yes → curve <60–70°, no hourglass/hinge → plication (Nesbit, 16-dot, tunica albuginea plication) curve >60–70°, hourglass, hinge, short penis → plaque incision/partial excision + graft (pericardium, small intestinal submucosa)
```

- **Plication** shortens the convex side: simple, low ED risk, but loses length.
- **Grafting** restores length and girth but carries higher ED risk (5–54%).

Diagram to draw: Nesbit wedge excised opposite the plaque.

> **Key points to remember:**
> - Ask the "straight penis" question; duplex before surgery.
> - CCH: IMPRESS, stable 30–90° curve, intact erections.
> - Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

Also, from the condensed book (not in the paper answer):

- **Oral**: the AUA "should not offer" any except **NSAIDs for pain**; vitamin E, tamoxifen, colchicine and carnitine give no benefit.
- **Extracorporeal shock wave therapy (ESWT)** helps pain, not curvature; radiotherapy is harmful.

Collagenase Clostridium histolyticum (CCH, Xiaflex)

- It is the only PD drug approved by the Food and Drug Administration (FDA, 2013). It degrades collagen **types I and III** and spares elastin, vessels and nerves.
- **Protocol (IMPRESS)**: 2 injections of **0.58 mg** 24–72 h apart, then **model** the penis 24–72 h after the second. Give up to **4 cycles, 6 weeks apart** (max 8 injections). Modelling is essential.
- **AUA criteria** are stable disease, curvature **≥30° and <90°** and intact erections (with or without medication). IMPRESS excluded ventral plaques, extensive calcification and disease <12 months.
- Response is poor with calcification, curve ≥60°, severe indentation or hinge.
- Bruising and pain are common; the serious risk is **corporal rupture** (0–4.9%), so there is no intercourse until modelling ends.

Other intralesional agents

Agent	Mechanism	Evidence	AUA
Verapamil	Calcium channel blocker (CCB): reduces fibroblasts and extracellular matrix, increases collagenase	Randomised trials conflicting; stabilises; inferior to CCH; cheap	May offer
Interferon α-2b	Reduces fibroblasts and collagen, increases collagenase	Randomised controlled trial (RCT) vs saline: curvature reduced 27% vs 9%; flu-like symptoms (pre-treat with NSAIDs)	May administer
Nicardipine	CCB	Reduces pain and plaque; no curvature benefit	—

- Verapamil is given as 10 mg diluted (10–20 mL).

Mechanical therapy

- **Penile traction therapy (PTT)** in the acute phase reduces curvature, pain and the need for surgery, and gains length; it needs at least 3 h/day.
- The AUA makes no recommendation for traction, vacuum or combinations.

6 · Surgery

Indications

- Disease is **≥12 months from onset, with deformity stable ≥6 months**.
- Deformity or poor rigidity prevents intercourse, and conservative therapy has failed.

Consent

- The goal is a **functionally straight** penis (≤20°; European Association of Urology, EAU, ≤15°). Residual or recurrent curvature occurs in up to 16%.
- **Length loss** follows every procedure (plication more than grafting); 70–80% have already lost length, so document SPL.
- **New ED** is commoner after grafting than plication (~30% if pre-op rigidity is suboptimal); pre-op erection quality predicts it best.

Choice of procedure — algorithm

ALGORITHM

1. Ask: is the erection adequate (with or without a PDE5I)?
2. If **no**, place an inflatable prosthesis with modelling. If the curve persists, incise with or without a graft (graft if defect >2 cm).
3. If **yes**, do PDDUS. If the curve is simple and <60–70°, with no hourglass or hinge and acceptable length loss, **plicate** (if length loss is unacceptable, graft).
4. If **yes** but the curve is complex/multiplanar >60–70°, or there is severe hourglass, hinge or a short penis (<9 cm stretched), do **plaque incision/partial excision with graft** if the ED risk is accepted (otherwise prosthesis).

Plication (shortening)

- It shortens the **convex side opposite the plaque**. It is simple, with the least new ED, but does not fix hourglass, hinge or shortening.

Technique	Method	Point
Nesbit (1965)	Transverse elliptical full-thickness excision opposite the curve, closed transversely	Original; for a ventral curve, lift Buck fascia and take dorsal wedges
Yachia	Vertical 0.5–1.5 cm incision closed transversely (Heineke–Mikulicz)	No tunica removed; long incisions narrow the shaft
Essed–Schroeder	Non-absorbable figure-of-eight, buried knots	First incisionless plication
16-dot	Permanent extended Lembert, 4 dots per plication	Popular; no incision
Tunica albuginea plication (TAP, Levine)	Paired partial-thickness transverse incisions (outer layer only); buried permanent suture and 3-0 polydioxanone (PDS)	Cavernosal tissue undisturbed

Plaque incision/partial excision and graft (lengthening)

- It suits complex deformity with **strong erections** and no veno-occlusive dysfunction, and restores calibre and 0.5–3 cm of length.
- ED predictors are age >55, veno-occlusive dysfunction (RI <0.80), a large graft, a ventral curve and a curve >60°.
- **Key steps**: elevate Buck fascia with the neurovascular bundle, make an H/double-Y incision **crossing the septum**, and sew in the graft (pericardium or small intestinal submucosa) with running 4-0 PDS.

Prosthesis with straightening

- PD with PDE5I-refractory ED is treated in one operation; an **inflatable** device is preferred.
- **Manual modelling** (Wilson–Delk): inflate fully, clamp the tubing and bend opposite the curve for 60–90 s, with the hand on the shaft, not the glans. It straightens 86–100%.
- Residual curvature >30° needs a relaxing incision, with a graft if the defect is >2 cm. Length loss is the commonest complaint (see Ch 75).

7 · Landmark trials

Trial	Design	Result	Take-home
IMPRESS I and II	Phase III RCT, CCH vs placebo 2:1, curvature 30–90°	Curvature reduced 34% (17°) vs 18% (9.3°); bother improved	Basis of FDA approval; modelling essential
Interferon α-2b multicentre RCT	Intralesional interferon (IFN) vs saline	Curvature reduced 27% vs 9%	First RCT of intralesional therapy

★ Putting it together: andrology decisions

A · Infertile man: work-up

ALGORITHM

1. **History and examination**: testosterone use, mumps, cryptorchidism; testis volume, vasa, varicocele (see Ch 69).
2. **Two semen analyses** (abstinence 2–7 days). Low volume with acid pH suggests duct obstruction or congenital bilateral absent vas (CBAVD).
3. **Hormones**: follicle-stimulating hormone (FSH) and testosterone; if testosterone is low, add luteinising hormone (LH) and prolactin.
4. **Genetics** (azoospermia, severe oligozoospermia): karyotype, Y microdeletions; CFTR in CBAVD.

B · Azoospermia: obstructive or not

	Obstructive (OA)	**Non-obstructive (NOA)**
Testis	Normal volume	Small, soft
FSH	Normal (≤7.6 IU/L)	Raised (>7.6 IU/L)
Clues	Absent vasa, low volume, vasectomy or hernia repair	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy
Treatment	Vasovasostomy (VV) or vasoepididymostomy (VE), resection of ejaculatory ducts, or sperm retrieval for ICSI (see Ch 70)	Micro-TESE with ICSI; fix a clinical varicocele first; no retrieval if complete AZFa/b deletion

KEY POINT

† Reversal suits a younger partner wanting >1 child; ICSI suits an older partner or female factor. Copious vas fluid or sperm means VV; toothpaste-like fluid or none means VE (see Ch 70).

C · Erectile dysfunction ladder

ALGORITHM

1. **Assess:** IIEF-5 score, HbA1c, lipids, morning testosterone. ED is a cardiovascular sentinel, so stratify cardiac risk (see Ch 72).
2. **Modify** risk factors; treating hypogonadism rescues PDE5 inhibitor (PDE5I) failures.
3. **Shared decision:** PDE5I (never with nitrates), vacuum, intraurethral alprostadil or intracavernosal injection.
4. **Penile prosthesis** (inflatable three-piece) if others fail; highest satisfaction. (see Ch 75).
5. **Young, focal post-traumatic arterial occlusion** → revascularisation.

D · Priapism: ischaemic vs non-ischaemic

	Ischaemic (emergency)	Non-ischaemic (high-flow)
Clinical	Rigid, painful	Semi-rigid, painless; after perineal trauma
Cavernous blood gas	pO_2 <30, pCO_2 >60 mm Hg, pH <7.25	Like arterial blood: pO_2 >90, pCO_2 <40
Doppler	Little or no flow	High flow, fistula or pseudoaneurysm
Treatment	Aspirate with or without irrigation, then phenylephrine (100–500 µg every 3–5 min, up to 1 h), then distal shunt with or without tunnelling, then proximal shunt. >36 h or refractory: early prosthesis	Observe; selective embolisation if persistent

KEY POINT

Sickle cell: treat as above without waiting for transfusion. Stuttering priapism: daily PDE5I, nightly pseudoephedrine, or hormonal suppression after puberty (see Ch 73).

E · Peyronie disease

ALGORITHM

1. **Active**: analgesia, traction. Collagenase (CCH): stable dorsal/lateral 30–90°, intact erections; else verapamil, interferon α-2b.
2. **Surgery**: ≥12 months from onset, stable ≥6 months (see Ch 76).
3. If **erections are good**: simple <60–70°, no hourglass/hinge, adequate length → plication; complex, hourglass/hinge or short → incision/partial excision + graft (more ED).
4. If **erections are poor** despite PDE5I → inflatable prosthesis with modelling.

F · Premature ejaculation (PE)

KEY POINT

Diagnosis needs short latency (lifelong ≈1 min or less), poor control and distress. Treat ED first. Options: stop–start or squeeze, lidocaine–prilocaine spray, on-demand dapoxetine (only licensed oral drug outside the US), daily SSRI (paroxetine best), tramadol (off-label, addiction risk) (see Ch 74).

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ Oligospermia and oligoasthenozoospermia

Paper answer · P2 · P3 · merged from 2

Asked as: P3 SAQ "Oligospermia and oligoasthenozoospermia" (asked as "Oligoanaesthospermia" and "Oligospermia") · P2 SAQ "Oligospermia."

Assumed meaning (oligoanaesthospermia): oligoasthenozoospermia (low sperm count with poor motility).

Definition (WHO [World Health Organization] 2021, 6th edition, 5th centiles)

- **Oligozoospermia:** sperm concentration below the lower reference limit: **WHO 2021 (6th edition) 16 million/mL** (5th centile; total count 39 million per ejaculate). Earlier thresholds were 20 million/mL (1999, the legacy cut-off) and 15 million/mL (2010).
- **Severity:** mild to moderate 5–16 million/mL; **severe oligozoospermia <5 million/mL**; cryptozoospermia: sperm only in a centrifuged pellet.

Parameter (WHO 2021 5th centiles)	Lower reference value
Concentration	16 million/mL (total 39 million)
Volume	1.4 mL
Progressive motility	30%
Total motility	42%
Normal forms (strict)	4%
Vitality	54%

- These centiles come from fertile men; they are a reference, not pass/fail fertility cut-offs. Define oligospermia in context: read a single value with testis size, FSH (follicle-stimulating hormone) and repeated values.

Causes

- **Varicocele** (commonest correctable; heat and oxidative stress) and **idiopathic** (largest group).
- **Pre-testicular (hormonal):** hypogonadotropic hypogonadism, **exogenous testosterone or anabolic steroids**, obesity (raised oestradiol), hyperprolactinaemia, opioids.
- **Testicular: varicocele**, cryptorchidism, mumps orchitis, torsion, chemotherapy or radiotherapy, gonadotoxins; genetic (Klinefelter, Klinefelter mosaic, **AZFc [azoospermia factor c] deletion**).
- **Post-testicular (motility):** infection or **leucocytospermia** (over 1 million/mL), antisperm antibodies, partial ejaculatory duct obstruction (EDO; low volume), PCD (primary ciliary dyskinesia).
- **Lifestyle:** smoking, alcohol, cannabis, heat, drugs, gonadotoxins.

Evaluation

- **At least two semen analyses, 2–3 weeks apart**, same abstinence (WHO 2–7 days of abstinence; parameters peak at 1–2 days).
- History and examination: testis size/volume, varicocele, vasa, secondary sex characters.

- Morning **testosterone, LH (luteinising hormone), FSH**; oestradiol and prolactin if testosterone is low.
- **Under 5 million/mL** (especially with raised FSH or small testes): **karyotype and Y-microdeletion** testing (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine] 2020 guideline).
- Selected: vitality test (eosin or hypo-osmotic swelling), antisperm antibodies, semen culture, scrotal Doppler, TRUS (transrectal ultrasonography) if volume is low (<1.0 mL), DNA fragmentation.

Management

- **General:** stop smoking, weight loss, avoid heat; **stop exogenous testosterone** and gonadotoxins; treat infection.
- **Varicocelectomy** (microsurgical subinguinal) for a **palpable** varicocele with abnormal semen; it improves semen parameters and pregnancy rates. Never treat a subclinical one.
- **Medical:**
- Hypogonadotropic: hCG (human chorionic gonadotropin) ± FSH.
- Idiopathic with low testosterone: off-label clomiphene 25 mg daily, or anastrozole 1 mg daily if the testosterone:oestradiol ratio is low (<10).
- Antioxidants: weak evidence.
- **ART (assisted reproductive technology):** mild: IUI (intrauterine insemination); severe or failed IUI: **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Bank sperm in AZFc carriers because counts may fall.
- Spermatogenesis takes about 64 days, so judge response (reassess) after 3–6 months.

Diagram to draw: hypothalamic–pituitary–testis axis with sites of treatment.

> **Key points to remember:**
> - WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
> - Karyotype and Y-microdeletion if under 5 million/mL.
> - Repair palpable varicoceles only; never give testosterone.

SAQ Avanafil

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 21 SAQ Sheet, Q3e

Introduction

- Second-generation oral **PDE5I (phosphodiesterase type 5 inhibitor)**; **FDA (US Food and Drug Administration) approved 2012**; Europe 2013. Structure differs most from the other PDE5Is.

Mechanism

- Sexual stimulation releases NO (nitric oxide), which raises cGMP (cyclic guanosine monophosphate) and relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; avanafil blocks this.
- **Enhances, does not initiate, erection**: stimulation is needed.

- **More selective for PDE5** over PDE6 (retina) and PDE1 than sildenafil, so **no visual disturbance** in the comparison table.

Pharmacokinetics (Campbell Table 72.7)

Feature	Avanafil	Sildenafil	Tadalafil
Time to peak	**0.3–0.5 h (fastest)**	0.8 h	2 h
Half-life	3–5 h	3–5 h	17.5 h
Fatty food	Reduces absorption	Reduces	No effect
Doses	**50, 100, 200 mg**	25–100 mg	5–20 mg
Blue vision	No	Yes	Rare

- Taken about 15–30 minutes before sex; start 100 mg; maximum once daily. Metabolised by CYP3A4 (cytochrome P450 3A4).

Efficacy

- Effective in general ED (erectile dysfunction), **diabetes** and **after nerve-sparing radical prostatectomy** (REVIVE trial programme): improves IIEF (International Index of Erectile Function) erectile scores and successful-intercourse rates, some within 15 minutes. Efficacy equivalent across PDE5Is; no head-to-head superiority.

Adverse effects

- Headache, flushing, nasal congestion, dyspepsia; back pain rare.

Contraindications and interactions

- **Nitrates in any form**: absolute; no nitrate within 12 hours of a dose.
- Avoid strong CYP3A4 inhibitors (ketoconazole, ritonavir); maximum 50 mg with moderate inhibitors.
- **α-blockers:** start 50 mg once stable (additive hypotension).
- Caution: aortic stenosis, recent myocardial infarction or stroke, hypotension, retinitis pigmentosa. Not recommended in severe renal impairment, dialysis or severe hepatic impairment.
- **Also know:** proposed prophylaxis for recurrent ischaemic priapism; can cause priapism rarely.

Advantages

- **Fastest onset**, fewer visual and muscle effects; suits on-demand spontaneous use.

Diagram to draw: NO–cGMP pathway in cavernosal smooth muscle with the PDE5 block.

> **Key points to remember:**
> - Peak 0.3–0.5 h; doses 50, 100, 200 mg.
> - Highly PDE5-selective: no blue vision.
> - Nitrates absolutely contraindicated.

SAQ CRE & BEP

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3f

"CRE" is read as carbapenem-resistant Enterobacterales, and BEP as bleomycin, etoposide and cisplatin chemotherapy for germ cell tumours; the two are answered separately.

Part A: CRE (carbapenem-resistant Enterobacterales)

- **Definition:** Enterobacterales (Klebsiella pneumoniae, Escherichia coli, Enterobacter) resistant to at least one carbapenem, or producing a carbapenemase.

Mechanism	Example
Class A carbapenemase	KPC (Klebsiella pneumoniae carbapenemase)
Class B metallo-β-lactamase	**NDM (New Delhi metallo-β-lactamase)**: common in India
Class D	OXA-48
Non-enzymatic	ESBL (extended-spectrum β-lactamase) or AmpC plus porin loss

- **Urological risk:** catheters, stents, prolonged stay, prior carbapenem or fluoroquinolone use, transrectal biopsy, ICU (intensive care unit) care.
- **Treatment** (by culture and carbapenemase type; ID (infectious disease) input):
- KPC, OXA-48: ceftazidime–avibactam; meropenem–vaborbactam or imipenem–relebactam for KPC.
- **NDM:** ceftazidime–avibactam plus aztreonam, or cefiderocol.
- Cystitis: single-dose aminoglycoside, fosfomycin; polymyxins as last resort. Avoid tigecycline for UTI (urinary tract infection) (low urine levels).
- **Prevention:** antibiotic stewardship, contact precautions, rectal screening of high-risk patients, early catheter removal, transperineal rather than transrectal biopsy.

Part B: BEP (bleomycin, etoposide, cisplatin)

- Standard first-line chemotherapy for metastatic **GCT (germ cell tumour)**; replaced PVB (less neuromuscular toxicity).
- **Usual 21-day cycle:** cisplatin 20 mg/m² days 1–5, etoposide 100 mg/m² days 1–5, bleomycin 30 units days 1, 8, 15.

IGCCCG (International Germ Cell Cancer Collaborative Group) risk	Regimen
Good	BEP ×3 or EP ×4
Intermediate/poor	BEP ×4 (VIP ×4 if lungs poor)
Stage I NSGCT (non-seminomatous GCT), high risk	BEP ×1 adjuvant

- **Toxicity:** bleomycin lung fibrosis (keep FiO_2 low at later RPLND (retroperitoneal lymph node dissection)) and Raynaud phenomenon; cisplatin nephrotoxicity, ototoxicity, neuropathy; myelosuppression; infertility (bank sperm); late cardiovascular disease and second cancers.

Diagram to draw: BEP 21-day cycle timeline.

Key points to remember:

- CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
- BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
- Bleomycin lung: low FiO_2 and restricted fluids at surgery.

SAQ Tadalafil

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3b

Pharmacology

- Selective **PDE5 (phosphodiesterase type 5) inhibitor**; FDA (Food and Drug Administration) approved 2003.
- **Mechanism:** sexual stimulation releases NO (nitric oxide), raising cGMP (cyclic guanosine monophosphate), which relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; tadalafil blocks this. It **augments, not initiates,** erection.
- Structurally distinct from sildenafil; cross-reacts with **PDE11**, little with PDE6 (retina).

Property	Tadalafil (vs sildenafil)
Time to peak	2 h (0.8 h)
Half-life	**17.5 h** (3–5 h)
Fatty food	No effect (reduces absorption)
Window	Widest (up to 36 h)

Uses and doses

1. **ED (erectile dysfunction):** 10–20 mg on demand, or 2.5–5 mg once daily. Intercourse success about 70%; 40–50% in diabetes and after prostatectomy.
2. **BPH (benign prostatic hyperplasia) with LUTS (lower urinary tract symptoms), with or without ED: 5 mg daily**, the only PDE5 inhibitor licensed. IPSS (International Prostate Symptom Score) falls about 6 points (placebo 3.6) **without Qmax (peak flow rate) change**; mechanism is NO–cGMP relaxation, perfusion and reduced afferent activity, not de-obstruction. Equal to tamsulosin for IPSS. EAU (European Association of Urology) 2022 and AUA (American Urological Association) 2021: option for moderate–severe LUTS.
3. **Also know:** fixed finasteride–tadalafil (FDA 2021); penile rehabilitation (benefit unproven); prevention of stuttering priapism (PDE5I paradox); pulmonary arterial hypertension.

Adverse effects

- Headache, dyspepsia, **back pain and myalgia** (characteristic), flushing, nasal congestion. Visual effects rare.
- NAION (non-arteritic anterior ischaemic optic neuropathy): no proven link; stop if sudden visual loss.

Contraindications and interactions

- **Nitrates in any form: absolute.** After tadalafil, no nitrates for **48 h** (24 h for sildenafil).

- Caution: α-blockers (start lowest dose of the added drug), aortic stenosis, hypotension. CYP3A4 inhibitors (ketoconazole, ritonavir) raise levels.
- Not studied: MI (myocardial infarction) or stroke within 6 months, unstable angina, BP <90/50 or >170/100 mmHg, retinitis pigmentosa.

Diagram to draw: NO–cGMP–PDE5 pathway in cavernosal smooth muscle.

> **Key points to remember:**
> - Half-life 17.5 h; food-independent; back pain.
> - 5 mg daily for LUTS: IPSS improves, Qmax does not.
> - No nitrates for 48 h after a dose.

SAQ Sertoli cell only syndrome

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3a (non-obstructive azoospermia also answered in 2025-Winter Paper 2, Q3a)

Definition

- SCOS (Sertoli cell only syndrome), **germ cell aplasia** (Del Castillo syndrome): seminiferous tubules lined only by Sertoli cells, with **complete loss of spermatogonia**. Leydig cells usually normal.
- The worst histological pattern of NOA (non-obstructive azoospermia).

Types

Type	Feature
Pure (complete, congenital)	All tubules lack germ cells
Mixed (focal, acquired)	Some tubules keep foci of spermatogenesis, the basis of sperm retrieval

Aetiology

- **Developmental:** primordial germ cells fail to migrate into the testis.
- **Genetic: complete AZFa (azoospermia factor a) deletion** gives pure SCOS; **AZFc** may give SCOS with foci; Klinefelter (47,XXY).
- **Acquired:** cryptorchidism, chemotherapy or radiation, mumps orchitis, torsion; idiopathic.

Clinical features

- Normal virilisation; **azoospermia** with normal ejaculate volume; small soft testes.

Evaluation

- Semen: azoospermia on two centrifuged samples.
- **FSH (follicle-stimulating hormone) high**, **inhibin B low** (Sertoli marker); LH (luteinising hormone) normal or high; testosterone normal or low.
- **FSH >7.6 IU/L and testis long axis ≤4.6 cm** suggest NOA.
- **Karyotype and Y microdeletion** before surgery.
- **Histology** confirms, usually at sperm retrieval.

Pattern	Meaning	Micro-TESE retrieval
Hypospermatogenesis	All stages, fewer	Best
Maturation arrest	Stops at one stage	About 40%
SCOS	No germ cells	**About 30%**

Management

- **Counsel** on genetic findings.
- **Complete AZFa or AZFb deletion:** retrieval essentially nil; do not operate; donor sperm or adoption.
- **Micro-TESE (microdissection testicular sperm extraction):** at ×20–25, **larger, opaque, yellow tubules** (foci of spermatogenesis) are selected; SCOS tubules are thin, white and stringy. Live births are reported even in SCOS.
- Sperm used for **ICSI (intracytoplasmic sperm injection)**.
- Optimise first: stop exogenous testosterone; hormonal stimulation evidence weak.
- **Also know:** spermatogonial stem cell transplantation and in-vitro spermatogenesis are experimental.

Diagram to draw: tubule cross-section: normal layered epithelium versus Sertoli cells only.

> **Key points to remember:**
> - Complete loss of spermatogonia; high FSH, low inhibin B.
> - Complete AZFa deletion: pure SCOS, no retrieval.
> - Micro-TESE retrieval about 30%: worst histology, not hopeless.

SAQ Oxidative stress and male infertility

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3b

Definition

- **Oxidative stress (OS):** imbalance where ROS (reactive oxygen species) exceed the antioxidant capacity of semen. The term **MOSI (male oxidative stress infertility)** has been proposed.
- Low ROS levels are **physiological**: needed for capacitation, hyperactivation, acrosome reaction and sperm–oocyte fusion.

Sources of ROS

Source	Examples
Leucocytes (main)	Infection, prostatitis, epididymitis, leucocytospermia
Immature sperm	Excess residual cytoplasm
Varicocele	Heat and hypoxia
Lifestyle	**Smoking**, alcohol, obesity, heat
Others	Ageing, diabetes, pollutants, radiation

Why sperm are vulnerable

- Membrane rich in **polyunsaturated fatty acids** → lipid peroxidation.
- Little cytoplasm, so few antioxidant enzymes; limited DNA repair.
- **Seminal plasma antioxidants:** superoxide dismutase, catalase, glutathione peroxidase; taurine, vitamins C and E, zinc.

Effects

- Lipid peroxidation → loss of **motility** and membrane fluidity.
- **Sperm DNA fragmentation** → poor embryo quality, miscarriage, lower ART (assisted reproductive technology) success.
- Mitochondrial damage and apoptosis.

Evaluation

- Semen analysis: peroxidase-positive WBC (white blood cells) **>1 million/mL** = leucocytospermia.
- **Direct ROS:** chemiluminescence (luminol).
- **Indirect:** total antioxidant capacity, malondialdehyde (lipid peroxidation), ORP (oxidation-reduction potential).
- **DNA fragmentation tests:** TUNEL (terminal deoxynucleotidyl transferase dUTP nick-end labelling), SCSA (sperm chromatin structure assay), SCD (sperm chromatin dispersion), Comet.
- Use selectively (unexplained infertility, recurrent miscarriage, failed ART); not routine.

Management

- **Lifestyle:** stop smoking, reduce weight and alcohol, avoid heat.
- **Treat cause:** antibiotics for infection; **varicocelectomy** for clinical varicocele (reduces DNA fragmentation).
- **Antioxidants** (vitamins C and E, CoQ10 [coenzyme Q10], zinc, selenium, carnitine): Cochrane review, low-quality evidence; **MOXI trial (2020)** showed no gain in semen parameters, DNA fragmentation or live birth. Avoid excess: **reductive stress** also harms.
- High DNA fragmentation with failed ICSI (intracytoplasmic sperm injection): **testicular sperm** has less fragmentation.

Diagram to draw: balance of ROS sources against seminal antioxidants, with effects on sperm.

> **Key points to remember:**
> - Leucocytes and immature sperm are main ROS sources.
> - Lipid peroxidation and DNA fragmentation are the key injuries.
> - Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

SAQ Dhat syndrome

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3c

Definition

- **Dhat syndrome:** a **culture-bound syndrome** of South Asia. The man believes he is losing **semen ("dhat")** in urine or nocturnal emissions, and attributes weakness, fatigue and sexual problems to it.
- Term coined by **Wig** (India); listed in ICD-10 (International Classification of Diseases, 10th revision) and in the DSM-5 (Diagnostic and Statistical Manual, 5th edition) glossary of cultural concepts of distress.
- Related syndromes: **Shen-kui** (China), **Koro** (genital retraction).

Cultural basis

- Traditional belief that semen is a **precious vital fluid** made from many drops of blood; its loss is felt to drain body and mind.

Epidemiology

- Young men, often unmarried or newly married, rural, lower education; common in urology and sexual medicine clinics.

Clinical features

- **Somatic:** fatigue, weakness, aches, poor appetite.
- **Psychological:** anxiety, low mood, guilt (about masturbation), poor concentration, sleep loss.
- **Sexual:** fear of ED (erectile dysfunction) or PE (premature ejaculation), worry about penis size and fertility.
- Whitish urine is usually phosphaturia, oxaluria or normal secretions.
- **Comorbidity: depression** and **anxiety** in many; somatisation.

Evaluation

- Non-judgmental sexual history and belief system.
- Examination of genitalia; urine microscopy (phosphates, oxalates, sperm, pus cells).
- Exclude organic disease: **UTI (urinary tract infection), prostatitis, STI (sexually transmitted infection), diabetes**, true spermatorrhoea.
- Screen for depression and anxiety.

Management

1. Counselling (main)

- **Sex education:** semen is continuously produced; nocturnal emission and masturbation are normal and harmless.
- Reassure without dismissing.

2. Psychological therapy: CBT (cognitive behavioural therapy) for misbeliefs and anxiety.

3. Drugs

- **SSRIs (selective serotonin reuptake inhibitors)** for depression, anxiety or associated PE; short anxiolytic course if needed.
- Treat genuine ED or PE.

4. Avoid harm

- Avoid unnecessary "tonics", antibiotics and repeated tests; they reinforce the belief.
- Refer to psychiatry when depression is significant.

Prognosis

- Good with counselling; drop-out is common.

Diagram to draw: flowchart: complaint of semen loss → exclude organic cause → counsel → treat depression or anxiety.

> **Key points to remember:**
>
> - Culture-bound semen-loss anxiety of South Asia.
> - Depression and anxiety common; exclude UTI, prostatitis, STI.
> - Counselling and sex education first; SSRIs for comorbidity.

Rapid Revision Sheet · Section 9

Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias (LAQ)

- Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
- Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
- DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
- Optilume ROBUST III 74.6% vs 26.8%.
- Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
- About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

Spermatogenesis, azoospermia and non-obstructive azoospermia (LAQ)

- Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
- Confirm azoospermia on two centrifuged samples.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- AZFa/b complete deletion: do not operate (no retrieval).
- NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
- Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
- FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
- Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

Male reproductive immunology and immunobead test (SAQ)

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
- ASA: vasectomy commonest; ≥50% motile sperm bound is significant; head-bound IgA matters most.
- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
- Treatment of choice: IVF/ICSI.
- Leukocytospermia >1 million/mL.

Retrograde ejaculation (SAQ)

- TURP and diabetic neuropathy are the commonest causes.
- Post-ejaculatory urine: sperm and fructose confirm RE.
- Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

Male menopause (andropause, late-onset hypogonadism) (SAQ)

- Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
- SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

Discuss the role of urologist in reproductive endocrinology and male infertility practice. (LAQ)

- FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
- Exogenous testosterone is a contraceptive: never give it for infertility.
- Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
- Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

Semen analysis nomenclature. (SAQ)

- WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
- Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
- Two samples 2–3 weeks apart; abstinence 2–7 days.

Genetic causes of male infertility. Enumerate and classify. (SAQ)

- Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
- AZFa or AZFb complete deletion: do not attempt retrieval.
- CBAVD = genital form of cystic fibrosis; test both partners.

Ejacultory duct obstruction (SAQ)

- Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
- TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
- TURED: sperm return in about half; reflux and epididymitis are its price.

Klinefelter's syndrome. (SAQ)

- 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
- Small firm testes, high FSH and LH, azoospermia.
- Micro-TESE ~50% success; retrieve sperm before testosterone.
- Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

CBAVD (SAQ)

- CBAVD = mild CFTR mutation; test both partners.
- Low-volume, acidic, fructose-negative azoospermia; normal FSH.
- Renal ultrasound; MESA + ICSI.

Azoospermia: testicular biopsy and surgical sperm retrieval techniques (LAQ)

- Azoospermia: no sperm in centrifuged pellet on two samples.
- Biopsy indication: azoospermia, normal testes, vasa and FSH.
- Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
- OA: normal FSH and testes; ICSI essential.
- OA: MESA (capillary action); MESA is best: many motile sperm, multiple vials.
- Move proximally; capillary action, never suction.
- TESA: 23 G needle, suction, local anaesthesia; for OA with normal spermatogenesis.
- TESA/TESE when epididymal sperm absent.
- NOA: micro-TESE best; finds sperm in about half of NOA; TESA yield poor.
- AZFa/AZFb deletion: 0% retrieval (no sperm), do not operate.
- All surgically retrieved (testicular) sperm need ICSI; cryopreserve extras.
- Haematoma commonest; epididymal biopsy most serious.

Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED (LAQ)

- nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
- Noradrenaline α1 is the main anti-erectile transmitter.
- Dopamine/oxytocin promote; serotonin/prolactin inhibit.
- Rigid phase: ischiocavernosus, suprasystolic pressure.
- Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
- Diabetic ED: check HbA1c, testosterone and cardiac risk.

Surgical management of ED and penile prostheses (types, biofilms) (LAQ)

- Prosthesis is the last line but gives 80–90% satisfaction.
- 3-piece inflatable gives the best rigidity and flaccidity.
- Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
- Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
- Biofilm: microbes in a matrix; antibiotics fail.
- Infection 1–3%; coated devices and no-touch technique; infected implant: remove all parts; Mulcahy salvage 84%.
- Satisfaction 80–90%; repairs 15% at 5 years.
- Arterial revascularisation only for young men with post-traumatic focal stenosis.

Priapism: diagnosis and management (LAQ)

- Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
- Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
- ED risk 0% ≤12 h, 100% ≥36 h; consider early prosthesis beyond 36 h.
- Beyond 36 h or refractory: early penile prosthesis (or analgesia and counselling).
- High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

Premature ejaculation (SAQ)

- ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Haematospermia (SAQ)

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.
- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

Management of Peyronie's disease (SAQ)

- Ask the "straight penis" question; duplex before surgery.
- CCH: IMPRESS, stable 30–90° curve, intact erections.
- Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

Oligospermia and oligoasthenozoospermia (SAQ)

- WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
- Karyotype and Y-microdeletion if under 5 million/mL.

- Repair palpable varicoceles only; never give testosterone.

Avanafil (SAQ)

- Peak 0.3–0.5 h; doses 50, 100, 200 mg.
- Highly PDE5-selective: no blue vision.
- Nitrates absolutely contraindicated. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

CRE & BEP (SAQ)

- CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
- BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
- Bleomycin lung: low FiO_2 and restricted fluids at surgery. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Tadalafil (SAQ)

- Half-life 17.5 h; food-independent; back pain.
- 5 mg daily for LUTS: IPSS improves, Qmax does not.
- No nitrates for 48 h after a dose.

Sertoli cell only syndrome (SAQ)

- Complete loss of spermatogonia; high FSH, low inhibin B.
- Complete AZFa deletion: pure SCOS, no retrieval.
- Micro-TESE retrieval about 30%: worst histology, not hopeless.

Oxidative stress and male infertility (SAQ)

- Leucocytes and immature sperm are main ROS sources.
- Lipid peroxidation and DNA fragmentation are the key injuries.
- Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

Dhat syndrome (SAQ)

- Culture-bound semen-loss anxiety of South Asia.
- Depression and anxiety common; exclude UTI, prostatitis, STI.
- Counselling and sex education first; SSRIs for comorbidity. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 10

Testis, Penis & Urethra

78 Surgical, Radiographic and Endoscopic Anatomy of the Retroperitoneum

1 · Definition, boundaries and organs

LAQ **Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy.**

Paper answer · P1

Type: LAQ (25 marks)
Source: 2024-Winter Paper 1, Q1 (related: 2019-Summer Paper 1, Q2; 2020-Summer Paper 1, Q3b)

Anatomy: definition and boundaries

- The space behind the parietal peritoneum. **Anterior:** posterior peritoneal reflection; **posterior:** posterior abdominal wall; **cranial:** diaphragm; **caudal:** extraperitoneal pelvis.

Group	Contents
Primarily retroperitoneal (formed there)	Kidneys, ureters, adrenals, aorta, IVC (inferior vena cava)
Secondarily retroperitoneal (mesentery fused to the wall)	Ascending and descending colon, duodenum D2–D3, pancreas (not the tail)
Vessels, lymphatics, nerves	Lumbar nodes, cisterna chyli, sympathetic trunks, autonomic plexuses, lumbosacral plexus

Posterior abdominal wall

- **Flank layers:** external oblique, internal oblique, transversus abdominis, transversalis fascia. **Posterior:** psoas, quadratus lumborum, erector spinae.
- **Lumbodorsal (thoracolumbar) fascia:** three layers that fuse into one at the **tip of the 12th rib**, the standard entry point for retroperitoneoscopy.
- **Ribs:** access above the 12th rib risks the pleura; the intercostal bundle runs under the rib above.

Fasciae and spaces (three embryological strata)

Space	Boundaries	Contents
Anterior pararenal	Posterior peritoneum and anterior renal fascia	Colon, duodenum, pancreas
Perirenal	Anterior (Toldt) and posterior (Zuckerkandl) laminae of Gerota's fascia	Kidney, adrenal, perirenal fat, hilum, upper ureter, gonadal vessels
Posterior pararenal	Posterior lamina and transversalis fascia	Fat only

- The laminae fuse laterally as the **lateroconal fascia**. The perirenal cone is **open inferiorly**, so urine, blood or pus track into the pelvis.
- Perirenal fat is fine and pale; pararenal fat is coarse and yellow-orange.

Vessels, lymphatics and nerves

- **Aorta** enters at T12. Unpaired anterior branches supply the gut (coeliac, SMA (superior mesenteric artery), IMA (inferior mesenteric artery)); paired lateral branches supply adrenals, kidneys (L1–L2) and gonads; lumbar arteries go to the wall.
- At the hilum the **renal vein lies anterior to the artery**. The **left renal vein is long** (gonadal, adrenal and lumbar tributaries); the right is short.
- Lymph flows right to left; **cisterna chyli** at L1–L2.
- **Nerves at risk:** sympathetic trunk (medial to psoas), genitofemoral (on psoas), subcostal, iliohypogastric, ilioinguinal (on quadratus); hypogastric plexus (ejaculation).

Principles of laparoscopic retroperitoneal surgery

1. **No natural cavity:** the working space must be created (finger and balloon dissection) before insufflation.
2. **Psoas is the key landmark:** it forms the floor. Keep it horizontal on screen; the kidney in Gerota's fascia lies anterior, and the peritoneum is pushed medially.
3. **Direct posterior route to the hilum:** the renal artery, posterior to the vein, is met first, allowing early arterial control.
4. **Peritoneum stays intact:** no bowel mobilisation or adhesiolysis; no spill of tumour, urine or pus into the peritoneum.
5. **Small space:** closely spaced ports, limited triangulation, lens smudging.
6. **Gas behaviour:** CO_2 (carbon dioxide) absorbs through fat, so hypercarbia and subcutaneous emphysema are commoner; haemodynamic swings are smaller than transperitoneal.

Preferred for	Limitations
Adhesions, peritoneal dialysis, morbid obesity, pregnancy; posterior tumours, pyeloplasty, cyst surgery, renal biopsy	Small space, unfamiliar landmarks, harder suturing, learning curve

- **Evidence:** randomised trials and meta-analysis: blood loss, complications and oncology similar to transperitoneal; the consistent advantage is **shorter operative time**.

Salient steps: retroperitoneal laparoscopic nephrectomy

1. **Position:** full flank (90°), table flexed, axillary roll; surgeon behind the patient.
2. **Access (open Hasson):** 1.5–2.5-cm incision just below the 12th rib tip; split muscle, open the lumbodorsal fascia; finger confirms psoas behind and Gerota-covered lower pole in front.
3. **Space creation:** balloon dilator (commercial, or Gaur glove balloon with 600–800 mL saline) anterior to psoas, posterior to Gerota's fascia; blunt-tip balloon-cuff trocar seals the port.
4. **Ports:** two more trocars under vision, one off the 12th rib tip, one posterior and superior to the camera.
5. **Dissection:** along psoas, find the ureter and gonadal vein medially; lift ureter and lower pole to put the hilum on stretch; see arterial pulsation.
6. **Hilum:** clip or staple the **artery first**, then the vein; on the left a **lumbar vein** usually needs division.

7. **Mobilise:** upper pole (adrenal preserved or taken), then lateral and anterior attachments; radical = outside Gerota's fascia.
8. **Extract:** clip ureter, bag the specimen, remove via an extended port; close. - **Problems:** peritoneal tear (space collapses), emphysema, pneumothorax, vascular injury, flank hernia (intramuscular balloon), rhabdomyolysis (about 1%). **Bail-out:** open the peritoneum and convert.

Diagram to draw: transverse section showing the three spaces, Gerota's laminae and the balloon anterior to psoas.

> **Key points to remember:**
> - Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
> - Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
> - Psoas is the landmark; artery met first.
> - Main advantage: shorter operative time, peritoneum untouched.

Also, from the condensed book (not in the paper answer):

- Primary organs lie in the perirenal space; secondary organs lie in the anterior pararenal space.
- Secondary organs exclude the first part of the duodenum (**D1**) and the pancreatic tail; the **anal canal** is primarily retroperitoneal.

> **KEY POINT**
>
> Secondarily retroperitoneal organs lie **in front of** the kidney, so colon, duodenum and pancreas are always in the surgeon's path to the kidney.

2 · Posterior abdominal wall and surface landmarks

- The upper pole is at the 11th rib (left) and 12th (right, lower). The hilum is at **L1**; the lower pole at L3–L4.
- **Psoas major** (T12–L5 to lesser trochanter) flexes the hip with iliacus. **Quadratus lumborum** lies posterolateral to psoas and stabilises the 12th rib; psoas minor is inconstant.
- Dividing the **costovertebral (lumbocostal) ligament**, which tethers the 12th rib to L1–L2, frees the rib in posterior approaches.
- The **intercostal bundle** lies under the rib above (vein, artery, nerve from top down), so the nerve is least protected. Supracostal access may cross the pleura and has more complications than subcostal.
- **Dorsal lumbotomy** incises the lumbodorsal fascia vertically, lateral to erector spinae and quadratus, so **no muscle is cut**.

3 · Retroperitoneal fasciae and spaces

Collections spread along these spaces.

Three fetal strata form three fasciae and spaces: outer (transversalis; posterior pararenal), intermediate (Gerota; perirenal, genitourinary) and inner (fusion fascia; anterior pararenal,

gastrointestinal).

- The posterior Gerota lamina fuses medially with transversalis; this **must be incised to reach the hilum**.
- The Toldt (anterior) lamina is thinner than the Zuckerkandl (posterior) lamina.
- The perirenal cone may cross the midline below the hilum. Urinoma, haematoma and abscess are contained laterally but **track caudally**.
- The **anterior pararenal** space holds colon, pancreas and D2–D3, and is entered from the **white line of Toldt**.

4 · Gastrointestinal relations (operative hazards)

- **Right**: the pancreatic head lies in front of the inferior vena cava (IVC), and D2 lies on the right hilum, hence the **Kocher manoeuvre**. D3 crosses between the superior mesenteric artery (SMA) and the aorta.
- **Left**: pancreatic tail, spleen and stomach lie near the upper pole, and splenic vessels are at risk when the pancreas is lifted off renal fascia.

5 · Great vessels

Aorta and branches

The aorta enters at **T12**, left of the IVC, and bifurcates at **L4**.

Branch	Type	Level	Supplies
Inferior phrenic	Paired lateral	First, below hiatus	Diaphragm; superior adrenal
Celiac	Anterior	T12	Foregut
Middle adrenal	Paired lateral	SMA level	Adrenal
SMA	Anterior	L1–L2	Midgut
Renal	Paired lateral	L1–L2, below SMA	Kidney; inferior adrenal
Gonadal	Paired **anterolateral**	Below renals	Testis/ovary
Lumbar	Paired posterior	Four pairs, L1–L4	Wall, spine, cord
Inferior mesenteric (IMA)	Anterior	L3–L4	Hindgut

- Supernumerary renal arteries occur in ≥ ¼ (see Ch 84).
- **Ureteric supply**: proximal from renal; middle from aorta, common iliac and gonadal; distal from internal iliac. The abdominal ureter is supplied **medially**, the pelvic ureter **laterally**.
- IMA ligation is safe because the marginal artery of **Drummond** feeds the colon. **SMA ligation is not safe**: it causes severe bowel ischaemia.

IVC and veins

- The IVC passes the diaphragm at **T8**.
- The **left renal vein** crosses in front of the aorta below the SMA. It receives gonadal (right-angle entry), adrenal and ≥ 1 **lumbar vein** (avulsion risk). The **right** is short, without tributaries, and duplicated in one-sixth. Both enter the IVC at L1.
- The **right gonadal vein** enters the anterolateral IVC; the short **right adrenal vein** enters the posterolateral IVC.

- Left varicocele predominates (longer vein, perpendicular entry). A **new right varicocele** needs imaging for a retroperitoneal mass, e.g. renal cell carcinoma with venous thrombus.
- Dissect the right ureter **lateral** to the gonadal vein; medial dissection risks IVC injury and gonadal avulsion.
- The inferior mesenteric vein (**IMV**) lies in the left mesocolon and at reoperation is mistaken for the gonadal vein.

Variants (essential)

*See the paper answer **Retrocaval ureter and IVC anomalous development** (Section 4, Ch 52).*

Also, from the condensed book (not in the paper answer):

- † A retroaortic left renal vein (≈ 2–3%) is injured when lumbars are divided or the aorta rolled in retroperitoneal lymph node dissection (RPLND).
- † A circumaortic vein (≈ 5–7%) has a posterior limb that mimics a lumbar vein, so check the pre-operative CT.
- † Retrocaval ureter is a **venous** anomaly with a "reverse-J" on pyelogram, repaired by dismembered pyeloplasty in front of the IVC.

6 · Lymphatics

LAQ Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma

Paper answer · P1 · P2 · merged from 2

Asked as: P1 LAQ "Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications" (asked as "Lymphatic drainage of testis and principles for designing templates for RPLND" and "Anatomy of lymphatic drainage of testis and scrotum. Describe management implications of penile malignancies") · P2 LAQ "Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis."

The P1 second part (Summer 2021) is taken as the management implications of lymphatic drainage in penile cancer, i.e. management of the inguinal and pelvic nodes.

Embryological basis

- **Why retroperitoneal:** the testis develops at L1–L2 beside the mesonephros (near the kidney) and descends, carrying its vessels and lymphatics; its artery, vein and lymphatics stay retroperitoneal. So its lymph returns to where it began: the **retroperitoneal (lumbar) nodes near the renal hilum**, not the groin. The scrotum is skin and drains to the **groin**.

Lymphatic drainage of the testis

- **Route:** lymphatics of the testis and the caput and corpus of the epididymis run up the **spermatic cord** with the testicular (gonadal) vessels, through the internal ring and along the gonadal vessels to the lumbar nodes near the renal hilum (**para-aortic and interaortocaval nodes**). Lymph capillaries do not enter seminiferous tubules.
- **Onward:** lumbar trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta, behind the right crus; hence retrocrural nodes), then the **thoracic duct** to the left innominate vein. This explains **left supraclavicular** node metastases.
- Lymph flows **cephalad and right to left**.

- **Epididymis:** caput and corpus drain with the testis; **cauda epididymis** drains with the vas to the **external iliac** nodes.

Retroperitoneal nodal regions

Region	Extent
Right paracaval (with precaval, retrocaval)	IVC (inferior vena cava) midline to right ureter
Interaortocaval	IVC midline to aortic midline (between IVC and aorta)
Left para-aortic (with pre-aortic, retroaortic)	Aortic midline to left ureter

- **Also know:** ten mapped regions include right and left common iliac, interiliac and right and left suprahilar.

Primary landing zones

	Right testis	Left testis
Primary (landing zone)	**Interaortocaval** (below renal vessels)	**Left para-aortic** (and left hilar)
Secondary / next echelon	Right paracaval, precaval, then para-aortic	Interaortocaval; pre-aortic
Crossover / contralateral	Small but real to left para-aortic (another source: common; right to left occurs)	None significant; rare to the right, bulky disease only (another source: rare; implies bulk)

- **Crossover is only right to left**, matching the overall lymph flow.
- **Donohue mapping (1982):** contralateral and suprahilar spread is rare but rises with bulk; interiliac spread is very rare.
- **Pelvic, iliac and inguinal nodes** are involved only when drainage is altered (or with retrograde spread from bulky disease): previous scrotal or inguinal surgery (orchidopexy, hernia repair), scrotal violation, or tumour invading the tunica vaginalis or scrotum.
- The retroperitoneum is the first metastatic site in 70–80% (choriocarcinoma spreads by blood). CT (computed tomography) understages 25–35% of stage I NSGCT (non-seminomatous germ cell tumour); 5–9 mm landing-zone nodes are suspicious.

Lymphatic drainage of the scrotum

- Scrotal skin drains to the **ipsilateral superficial inguinal nodes**. Lymphatics, like vessels and nerves, **do not cross the median raphe**.

Clinical implications: testis and scrotum

- **Trans-scrotal orchidectomy or biopsy is contraindicated.** It leaves cord behind and opens scrotal skin lymphatic routes, so tumour can reach inguinal and pelvic nodes, causing local and inguinal or pelvic recurrence. Use **radical inguinal orchidectomy** with high cord clamping.
- Inguinal or pelvic spread from the testis means altered drainage: scrotal violation, previous orchidopexy or hernia repair, scrotal invasion.
- RPLND (retroperitoneal lymph node dissection) templates and para-aortic or "dog-leg" radiotherapy fields follow the landing zones.
- Scrotal skin cancer is staged at the ipsilateral groin.

Principles of designing RPLND templates

1. Map-based design: follow the side-specific landing zones to remove likely disease and spare the rest.

- Historical dissection: bilateral infrahilar + suprahilar + interiliac: most men anejaculatory.
- **Ray (1974)** first templated landing zones; **Donohue (1982)** mapping confirmed side-specific spread; **Weissbach and Boedefeld (1987)** reduced left template with frozen section.
- So a template must include the primary landing zone, the next echelon and the known crossover route.

2. Fixed boundaries

- **Superior:** renal hila (renal vessels). **Lateral:** ureters. **Inferior:** ureter crossing the common iliac artery. Always remove the **gonadal vein and cord stump** to the internal ring.

3. Omit suprahilar and interiliac regions in non-bulky disease, as spread there is rare. This reduces chylous ascites, renal, pancreatic and duodenal injury, and protects the hypogastric plexus.

4. Unilateral modified templates leave contralateral sympathetic fibres untouched. Because of right-to-left crossover, both templates take the para-aortic nodes **above the IMA (inferior mesenteric artery)**.

Template	Content
Right modified	Paracaval, precaval, retrocaval, interaortocaval, right common iliac; para-aortic **above IMA**; omit para-aortic below IMA
Left modified	Left para-aortic, preaortic, retroaortic, left common iliac; interaortocaval above IMA; omit paracaval/precaval/retrocaval
Full bilateral infrahilar	All three regions both sides (paracaval, precaval, retrocaval, interaortocaval, pre-aortic, para-aortic, retroaortic, both common iliacs), between renal vessels, ureters and ureteric crossing of iliacs
Historical extended	Plus suprahilar and interiliac (thoracoabdominal); most anejaculatory

- Omitting para-aortic above the IMA (right) or interaortocaval (left) is controversial.

5. Nerve sparing to preserve ejaculation: antegrade ejaculation (emission and bladder-neck closure) depends on **L1–L4 postganglionic sympathetic fibres** (about three each side) joining the **superior hypogastric plexus at or below the IMA** and running over the sacral promontory. Identify them prospectively before dividing tissue.

- **Unilateral templates** leave contralateral fibres untouched (antegrade ejaculation 75–97%). **Prospective nerve sparing** (Jewett 1988) identifies fibres: 90–100%.

6. Completeness ("duplicate open RPLND"): split-and-roll over the great vessels, **divide lumbar vessels** to clear retrocaval and retroaortic tissue, clip lymphatics at the cephalad limit (left renal vein) to prevent chylous ascites. Incomplete RPLND or mass excision gives worse outcomes (GETUG T93BP; 33% local relapse after post-chemotherapy mass-only resection).

7. Extend if needed: enlarged nodes at operation call for the full bilateral template (frozen section, Weissbach 1987). Convert to bilateral if nodes are suspicious at surgery or the primary has somatic-type malignancy.

Template by clinical setting

Setting	Template
Primary RPLND (high-risk stage I or low-volume stage II NSGCT, normal markers)	Default: modified unilateral nerve-sparing **or** full bilateral infrahilar nerve-sparing; no randomised trial. Modified allowed: low-volume, clinically negative nodes. Main risk: out-of-template nodes 3–23% (right > left)
PC-RPLND (post-chemotherapy), residual mass >1 cm, normal markers	**Full bilateral infrahilar + residual mass** (standard). Main risk: crossover with bulk; viable tissue unrecognisable
Modified PC template allowed only if	Mass ≤5 cm in landing zone, normal markers, IGCCCG (International Germ Cell Cancer Collaborative Group) good or intermediate risk

- Against unilateral templates: 3–23% of node-positive men have nodes outside the modified template, more with right tumours.
- *Sources differ:* Campbell testis-neoplasm chapter calls full bilateral nerve-sparing the standard; the surgery chapter accepts either.

RPLND outcomes and complications

- Primary RPLND: pN+ about 20–30%; cancer-specific survival (CSS) about 98.8–100%; retroperitoneal recurrence <2% after full template.
- **Antegrade ejaculation >90%** with templates plus nerve sparing (modified template alone: early 75–87%, refined 97%).
- Complications: ileus (commonest), wound infection, chylous ascites (0.2–2.1% primary; 2–12% post-chemotherapy), pulmonary (bleomycin lung toxicity; low FiO_2), ejaculatory loss.
- **Also know:** nephrectomy needed at PC-RPLND (left-sided mass >10 cm); GETUG T93BP showed incomplete RPLND predicts relapse.

Stage I seminoma: definition, staging and work-up

- Disease confined to the testis: any pT, N0, M0 on imaging. Stage IA = pT1; IB = pT2–4; IS = markers still raised after orchidectomy.
- **Work-up:** AFP (alpha-fetoprotein), β-hCG (human chorionic gonadotropin) and LDH (lactate dehydrogenase) before and after orchidectomy. CT abdomen and pelvis; chest CT or radiograph. Baseline testosterone; **offer sperm banking**. A raised AFP means non-seminoma.
- **Radical inguinal orchidectomy:** early cord control at the internal ring. It cures 80–85% of stage I seminoma.
- **Risk factors for relapse: tumour >4 cm** and **rete testis invasion**. Five-year relapse is 12%, 16% and 32% with 0, 1 and 2 factors. LVI (lymphovascular invasion) predicts relapse in non-seminoma, not seminoma.

Stage I seminoma: options after orchidectomy (cancer-specific survival about 100% with each)

	Surveillance	Carboplatin	Radiotherapy
Method	Clinical review, markers, CT	**One cycle, AUC (area under the curve) 7**	**20 Gy** in 10 fractions to the para-aortic strip
Relapse	14–20%	About 5%	3–5%

	Surveillance	Carboplatin	Radiotherapy
Relapse site	Retroperitoneum (84–100%)	Retroperitoneum	Pelvis, chest, left supraclavicular
Drawbacks	Late relapse, many scans, compliance	Underdosing; late effects unknown	Second cancers, GI (gastrointestinal) toxicity, oligospermia
Status	**Preferred**	If unwilling or unsuitable for surveillance; risk factors	If unwilling or unsuitable for surveillance

- **Carboplatin (Calvert):** 7 × (GFR [glomerular filtration rate] + 25) mg, using isotope GFR.
- **Trials: MRC TE18** (Medical Research Council): 20 Gy equals 30 Gy, with less toxicity. **MRC TE19/EORTC (European Organisation for Research and Treatment of Cancer) 30982:** one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE10: para-aortic field equals dog-leg, with 2% pelvic relapse.
- **Risk-adapted (EAU [European Association of Urology]):** surveillance for low risk; with risk factors discuss surveillance or carboplatin (treating all would overtreat two-thirds).

```
Inguinal orchidectomy + markers + CT │ Stage I seminoma │ Surveillance (preferred) ── unwilling / unreliable ──► Carboplatin AUC7 ×1(–2) │ (or radiotherapy 20 Gy) Relapse (mostly retroperitoneal) ├─ small nodes ► dog-leg radiotherapy with boost, or chemotherapy └─ bulky/distant ► BEP ×3 or EP ×4
```

- BEP: bleomycin, etoposide, cisplatin; EP: etoposide, cisplatin.

Stage I seminoma: relapse, IGCCCG and follow-up

- Relapses are almost always cured; median 14 months, but 10–20% occur after 4 years.
- **IGCCCG:** seminoma is good risk unless there are non-pulmonary visceral metastases (intermediate). **There is no poor-risk seminoma**, and markers do not change the risk group.
- **Surveillance follow-up:** examination and markers every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly; CT every 4–6 months (years 1–3), then 6-monthly to year 7, then yearly. CT finds >85% of relapses.

Lymphatic drainage of the penis

Superficial system	Deep system
Prepuce and shaft skin	Glans, frenulum, corpora
Dorsal network divides at the base and drains to both superficial inguinal groups	Trunks circle the corona and run under Buck's fascia to the presymphyseal plexus, then to femoral triangle nodes

- **Two rules:** drainage is **bilateral and unpredictable in side** (presymphyseal crossover), but **orderly in sequence**: superficial inguinal, then deep inguinal, then external iliac and obturator. **No skip metastases.**
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**; sentinel nodes are mostly **superomedial**, then central. **Deep nodes** (1–3) lie medial to the femoral vein; **Cloquet's node** is the highest.

Management implications in penile cancer

1. Prognosis and timing

- Nodal status is the **strongest prognostic factor**. Penile SCC (squamous cell carcinoma) has a long **locoregional phase**, so the groin is a **curable station**.
- **Kroon (2005):** early ILND (inguinal lymph node dissection) for impalpable nodes gave 3-year disease-specific survival about **84% vs 35%** when delayed until palpable.
- Manage both groins **within 3 months**; remove **at least 8 nodes per side**.

2. Bilateral management

- Crossover means **both groins** are staged or treated; with one palpable groin, the other holds micrometastasis in about 30%.

3. Impalpable groin (cN0): risk-based

Risk	EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023	Groin
Low	pTis, pTa, pT1a G1	Surveillance
Intermediate	pT1a G2	Surveillance if compliant, or surgical staging
High	**pT1b or higher**, G3	**DSNB (dynamic sentinel node biopsy)**; ILND if DSNB unavailable

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk and is staged surgically (ILND first).
- **DSNB:** ultrasound ± FNAC (fine-needle aspiration cytology) first, because a tumour-replaced node diverts flow (false negative). Then technetium-99m nanocolloid, blue dye and gamma probe.
- **No-skip spread and superomedial sentinels** justify **modified ILND (Catalona)**: smaller field, saphenous vein preserved; a positive frozen section leads to complete radical ILND.

4. Palpable nodes (cN+)

- About half are reactive, but do not delay with antibiotics. Ultrasound-guided **FNAC**; if positive, **radical ILND** on that side and stage the other.
- **Bulky (≥4 cm), fixed or bilateral nodes:** neoadjuvant **TIP** (paclitaxel, ifosfamide, cisplatin) × 4, then surgery in responders.

5. Pelvic nodes

- Stepwise spread means a negative groin needs no pelvic dissection.
- **PLND (pelvic lymph node dissection)**, ipsilateral, if **more than 2 inguinal nodes or ENE (extranodal extension)**. ENE in any node is pN3.

6. Complications of ILND

- Wound infection, **flap necrosis**, **lymphocele**, **lymphoedema**, DVT (deep vein thrombosis). Reduced by modified templates, saphenous sparing and **VEIL (video-endoscopic inguinal lymphadenectomy)**.
- **Also know:** adjuvant therapy for pN2–3; Cabanas sentinel biopsy abandoned.

Recent advances

- **Robotic RPLND** for low-volume disease (same templates, same rules).
- RPLND for small-volume stage II seminoma (**SEMS, PRIMETEST**).
- Serum **miR-371a-3p** microRNA for selection, and it detects relapse better than standard markers.

- MRI (magnetic resonance imaging) instead of CT for seminoma surveillance (TRISST trial) to reduce radiation.

Diagrams to draw

- Retroperitoneum showing great vessels with the three nodal regions, right (interaortocaval) and left (para-aortic) landing zones and the right-to-left crossover arrow; outline the right and left modified templates, IMA level and hypogastric plexus.
- Beside it, the femoral triangle with Daseler's zones and Cloquet's node.

Key points to remember:

- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Boundaries: renal hila, ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA, so unilateral templates spare ejaculation.
- PC-RPLND: full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum (ipsilateral superficial inguinal) and penis drain to inguinal nodes; never operate trans-scrotally.
- Seminoma relapse risk factors: >4 cm and rete testis invasion (not LVI).
- Stage I seminoma: surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.

Also, from the condensed book (not in the paper answer):

- Lumbar nodes are the primary drainage of kidney, adrenal, ureter and gonad.

KEY POINT

The testis is embryologically retroperitoneal, so it drains to where it began. Crossover is one-way (right to left), so right templates include interaortocaval and some para-aortic tissue. See Ch 80, 81.

7 · Nerves

Autonomic

- **Parasympathetic** outflow is craniosacral (III, VII, IX, X, S2–S4). **Sympathetic** outflow is thoracolumbar, **T1–L2**.
- **Sympathetic trunks** lie medial to psoas behind the great vessels, and are met when lumbar vessels are divided.
- **Ejaculation**: † postganglionic fibres from the **T12–L3** ganglia cross in front of the aorta and IVC and converge in the hypogastric plexus below the IMA. They drive **emission** and **bladder neck closure**, so extensive dissection causes anejaculation or retrograde ejaculation, while erection is spared. See Ch 81.

Lumbar (lumbosacral) plexus

Nerve	Roots	Course / motor	Sensory
Subcostal	T12	Below 12th rib	Skin over hip
Iliohypogastric	L1	Over quadratus; internal oblique, transversus	Gluteal, pubic
Ilioinguinal	L1	As above	Upper medial thigh, penile root, anterior scrotum
Genitofemoral	L1–L2	**On anterior psoas**; genital branch to cremaster	Anterior scrotum; upper anterior thigh
Lateral cutaneous thigh	L2–L3	Sensory only	Lateral thigh
Obturator	L2–L4	Medial to psoas, to obturator canal; adductors	Medial thigh
Femoral	L2–L4	Lateral to psoas; quadriceps	Anterior thigh, medial leg

Nerve	Setting	Result of injury
Genitofemoral	Psoas hitch, laparoscopic varicocelectomy, RPLND	Anterior scrotal/thigh numbness
Obturator	Lateral-wall TURBT (bladder tumour resection); pelvic node dissection	Adductor jerk causing bladder perforation
Femoral	Retractor blade on psoas; lithotomy	Loss of knee extension
Hypogastric plexus	Extensive retroperitoneal dissection	Anejaculation/retrograde ejaculation

8 · Imaging anatomy

- On axial CT the **duodenum lies against the right hilum** and the left renal vein in front of the aorta.
- † Before RPLND or nephrectomy, check accessory renal arteries, retro-/circumaortic left renal vein and nodal size by zone.

79 Neoplasms of the Testis

1 · Epidemiology and risk factors

- Over 95% are **germ cell tumours (GCTs)**: seminoma or non-seminomatous GCT (NSGCT); 5% are extragonadal. It is the commonest solid malignancy in men aged **20–40 y**.
- Cisplatin raised metastatic cure to **80–90%**, so the aim is **minimum toxicity without losing curability**.
- Bilateral in **2%**, mostly metachronous.
- **Risk factors**: White race, cryptorchidism, family history, personal history, germ cell neoplasia in situ (GCNIS) and infertility. Undescended testis (UDT) has a relative risk (RR) of **4–6**, falling to 2–3 with orchidopexy before puberty. A previous GCT gives a 12-fold contralateral risk, yet the 15-y cumulative incidence is only 2%.
- **GCNIS** is an arrested fetal gonocyte formed **before birth**. It lies beside invasive GCT in 80–90% and invades in **50% at 5 y** and 70% at 7 y. Contralateral GCNIS is found in 5–9%, and in ≈36% with atrophy, UDT or age <40.
- Microlithiasis alone needs no work-up; it matters only after a previous GCT.

> **KEY POINT**
>
> GCT is a disease of the arrested gonocyte: UDT, family history and atrophy mark an abnormal developmental field, so orchidopexy reduces but never abolishes risk.

2 · Pathology and classification

LAQ Classify testicular tumors and describe management of seminoma

Paper answer · P3

Type: LAQ (25 marks)
Source: Practice Set 1 Paper 3, Q1 (stage I seminoma also answered in 2018-Summer Paper 2, Q2)

Introduction

- Over 95% of testicular tumours are GCTs (germ cell tumours). Seminoma is 52–56% of GCNIS (germ cell neoplasia in situ)-derived GCTs, peaks in the fourth–fifth decade.
- **Risk factors:** cryptorchidism, previous contralateral GCT (twelvefold), family history, GCNIS, White race, infertility.
- All postpubertal GCTs except spermatocytic tumour carry **isochromosome 12p**.

WHO (World Health Organization) classification (2016, retained 2022)

Group	Tumours
1. GCNIS-derived GCTs	GCNIS; **seminoma**; **NSGCT (non-seminomatous GCT)**: embryonal carcinoma, postpubertal yolk sac tumour, choriocarcinoma and other trophoblastic tumours, postpubertal teratoma, teratoma with somatic-type malignancy; mixed GCT
2. Non-GCNIS-derived GCTs	**Spermatocytic tumour** (formerly spermatocytic seminoma); prepubertal teratoma, dermoid and epidermoid cyst; prepubertal yolk sac tumour

Group	Tumours
3. Sex cord–stromal tumours (<5%)	Leydig cell, Sertoli cell (including large cell calcifying), granulosa cell (adult, juvenile), fibroma–thecoma, mixed
4. Mixed germ cell–sex cord stromal	Gonadoblastoma (DSD [difference of sex development])
5. Others	Lymphoma (commonest testicular tumour in older men), rete testis and collecting duct tumours, adnexal and paratesticular tumours, metastases

- Any NSGCT element, however small, makes the tumour an NSGCT and is treated as such.
- **Seminoma pathology:** sheets of clear cells with lymphocyte-rich fibrovascular septa. Immunohistochemistry: CD117 and PLAP (placental alkaline phosphatase) positive, CD30 negative. Syncytiotrophoblasts (hCG [human chorionic gonadotropin]) in about 15%. "Anaplastic seminoma" is no longer recognised.
- **Spermatocytic tumour** (<1%): older men, no GCNIS, no i(12p), benign unless sarcomatous.

Evaluation and staging

- Scrotal ultrasound; AFP (alpha-fetoprotein), β-hCG and LDH (lactate dehydrogenase) before and after orchidectomy. **A raised AFP means NSGCT**, whatever the histology.
- **Radical inguinal orchidectomy** with early cord control at the internal ring; never trans-scrotal. Offer sperm banking and prosthesis.
- CT (computed tomography) of chest, abdomen and pelvis; TNMS (tumour, node, metastasis, serum markers) staging.
- About 80% of seminomas are clinical stage (CS) I; 15–20% CS II (nodes IIA ≤2 cm, IIB 2–5 cm, IIC >5 cm).

Management of seminoma by stage

CS I (cure by orchidectomy alone 80–85%; cancer-specific survival about 100% with any option)

	Surveillance (preferred)	Carboplatin	Radiotherapy
Method	Examination, markers, CT for at least 5 years	1 cycle **AUC (area under the curve) 7**, dose = 7 × (GFR [glomerular filtration rate] + 25) mg	**20 Gy** para-aortic strip
Relapse	14–20%	About 5% (1 cycle)	3–5%
Drawback	Late relapses (10–20% after 4 years)	Underdosing; late effects unknown	Second cancers, GI (gastrointestinal) toxicity

- **Relapse risk factors:** tumour **>4 cm** and **rete testis invasion** (relapse 12%, 16%, 32% with 0, 1, 2 factors). LVI (lymphovascular invasion) is not a factor in seminoma.
- **Trials:** MRC TE19/EORTC (Medical Research Council/European Organisation for Research and Treatment of Cancer) 30982: one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE18: 20 Gy equals 30 Gy.
- Surveillance relapses are 84–100% retroperitoneal and almost all salvaged.

CS IIA–B

- **Non-bulky (≤3 cm):** dog-leg radiotherapy with nodal boost (25–30 Gy in the chapter; EAU [European Association of Urology] uses 30 Gy for IIA and 36 Gy for IIB), or chemotherapy, or **primary RPLND (retroperitoneal lymph node dissection)** (SEMS trial).
- **Bulky (>3 cm) or multiple nodes:** BEP (bleomycin, etoposide, cisplatin) ×3 or EP (etoposide, cisplatin) ×4.

CS IIC–III: chemotherapy by IGCCCG (International Germ Cell Cancer Collaborative Group)

Risk	Seminoma definition	5-year survival	Regimen
Good (90%)	Any primary, **no NPVM (non-pulmonary visceral metastases)**, normal AFP, any hCG/LDH	86%	BEP ×3 or EP ×4
Intermediate (10%)	NPVM present	72%	BEP ×4
Poor	**None for seminoma**	–	–
- Carboplatin alone is inferior in advanced disease.			

Post-chemotherapy residual mass

```
Residual mass after chemotherapy ├─ <3 cm ──────────► observe (necrosis about 90%)
                                 └─ >3 cm ─► FDG-PET (fluorodeoxyglucose positron emission tomography) at ≥6 weeks
                                            ├─ negative ► observe
                                            └─ positive ► resection (full template PC-RPLND) or salvage
```

- Surgery is difficult because of desmoplasia; complete resection only 58–74%.

Relapse and follow-up

- After surveillance or radiotherapy: radiotherapy for small retroperitoneal relapse, otherwise first-line BEP. After chemotherapy: VeIP (vinblastine, ifosfamide, cisplatin) or high-dose chemotherapy.
- Surveillance schedule: review every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly. CT detects >85% of relapses.

Recent advances

- Serum **miR-371a-3p** microRNA: sensitive marker of residual disease and relapse.
- MRI (magnetic resonance imaging) for surveillance to cut radiation; primary RPLND for low-volume stage II.

Diagram to draw: WHO classification tree (GCNIS-derived vs non-GCNIS-derived vs sex cord–stromal) and the residual-mass flowchart.

> **Key points to remember:**
> - WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
> - CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
> - Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
> - No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

Also, from the condensed book (not in the paper answer):

- Isochromosome 12p, i(12p), proves GCT origin in a carcinoma of unknown primary and in somatic-type malignancy from teratoma.

Subtype	Markers	IHC / histology	Hallmark
Seminoma	Human chorionic gonadotrophin (hCG) ≈15%; **never alpha-fetoprotein (AFP)**	CD117+, CD30−, placental alkaline phosphatase (PLAP)+	4th–5th decade; can transform to NSGCT
Spermatocytic	None	OCT3/4−, PLAP−; three cell types	6th decade; never mixed, not with UDT; **benign** unless sarcomatous/anaplastic
Embryonal carcinoma (EC)	AFP/hCG or normal	AE1/AE3+, OCT3/4+, CD30+†, c-KIT−	Totipotent; spreads with normal markers; predicts occult spread
Yolk sac	AFP, not hCG	Schiller–Duval bodies (≈50%), hyaline globules	Pure form: children, mediastinum
Choriocarcinoma	Very high hCG	Syncytio- and cytotrophoblasts	Spreads by blood (lung, liver, brain); bleeds; poor risk
Teratoma	Normal/mild AFP	≥2 germ layers; maturity irrelevant after puberty	Benign histology, malignant genetics; **chemoresistant**

- Teratoma is present in ≈half of mixed GCTs. Unresected, it causes growing teratoma, somatic transformation or late relapse, so surgery is integral to cure.
- hCG shares its α-subunit with TSH and LH, so it can cause hyperthyroidism and gynaecomastia.

> **KEY POINT**
>
> Management follows the most chemoresistant element: EC drives occult metastasis, choriocarcinoma haemorrhage, teratoma surgery.

3 · Presentation

- A **painless mass** is commonest. Metastases are present at diagnosis in ≈2/3 of NSGCT and 15% of seminoma; 10–20% present with metastatic symptoms such as back pain or a left supraclavicular node.
- About 1/3 are first misdiagnosed, so re-examine epididymo-orchitis 2–4 weeks after antibiotics.

4 · Diagnosis and initial management

Scrotal ultrasound (US)

- GCT is **hypoechoic**: homogeneous suggests seminoma, heterogeneous suggests NSGCT.
- In metastatic GCT with normal palpation, US seeks a **burned-out primary**, which still needs orchidectomy; normal testes mean extragonadal GCT.

- An impalpable <10 mm lesion with normal markers is GCT in 20–50%: options are orchidectomy, inguinal exploration with frozen section, or serial US.

Serum tumour markers

*See the paper answer **Tumour markers in urology** (Section 1, Ch 3).*

Also, from the condensed book (not in the paper answer):

- Half-lives: AFP **5–7 days**, hCG **24–36 h**, LDH 24 h.
- After orchidectomy, a slow decline with negative imaging is monitored, but **rising markers mean chemotherapy**.
- An unstable patient with very high AFP/hCG gets chemotherapy first and orchidectomy later.

False positives (hepatic disease, hypogonadal LH cross-reactivity, marijuana) occur, and miR-371a-3p is a GCT-specific marker not raised by teratoma.

Radical inguinal orchidectomy

- Remove testis and cord to the internal ring within 1–2 weeks. **Trans-scrotal approach or biopsy is contraindicated**. Orchidectomy alone cures 80–85% of clinical stage (CS) I seminoma and 70–80% of CS I NSGCT.
- **Testis-sparing surgery** suits synchronous bilateral or solitary testis (<2–3 cm, adequate androgen) or a suspected benign lesion, with frozen section; GCNIS then gets ≥20 Gy.
- Biopsy the contralateral testis if atrophy, UDT, age <40 or suspicious US.

5 · Staging

Spread and imaging

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- CT abdomen–pelvis for all misses disease in **25–35% of CS I NSGCT**, so landing-zone nodes of 5–9 mm are suspicious. Equivocal nodes with normal markers are re-imaged in 6–8 weeks.
- Brain imaging if hCG >10,000 IU/L; no routine bone scan or FDG-PET at diagnosis.
- Chest CT if markers raised, abdominal disease, abnormal X-ray, or CS I NSGCT with LVI/EC.

Base decisions on CT and markers within 4 weeks of treatment, and offer **sperm cryopreservation** before RPLND, chemotherapy or radiotherapy.

TNM staging — the only TNM with a serum (S) category

Category	Definition
pTis	GCNIS
pT1	Testis/epididymis, **no LVI**; may invade tunica albuginea
pT2	LVI, or tunica vaginalis
pT3 / pT4	Spermatic cord / scrotum

Category	Definition
N1 / N2 / N3	≤2 cm / >2–5 cm / >5 cm (pN1 ≤5 nodes, pN2 >5 or extranodal; pN invalid after chemotherapy)
M1a / M1b	Non-regional nodes or lung / other viscera
S1	LDH <1.5×N and hCG <5000 mIU/mL and AFP <1000 ng/mL
S2	LDH 1.5–10×N or hCG 5000–50,000 or AFP 1000–10,000
S3	LDH >10×N or hCG >50,000 or AFP >10,000

Stage	Criteria
IA / IB	pT1 / pT2–4, N0 M0 S0
IS	Any pT, N0 M0, **S1–3** (raised markers, no visible disease)
IIA / IIB / IIC	N1 / N2 / N3, M0, S0–1
IIIA	M1a, S0–1
IIIB	N1–3 or M1a with S2
IIIC	N1–3 or M1a with S3, or **any M1b**

IGCCCG risk — metastatic disease

Risk	NSGCT	Seminoma
Good	Testis/retroperitoneal primary, no NPVM, AFP <1000, hCG <5000, LDH <1.5×N (56%; 5-y survival 92%)	Any primary, no NPVM, any hCG/LDH, normal AFP (90%; 86%)
Intermediate	Any of AFP 1000–10,000, hCG 5000–50,000, LDH 1.5–10×N (28%; 80%)	**NPVM** (10%; 72%)
Poor	**Mediastinal primary**, NPVM, AFP >10,000, hCG >50,000 or LDH >10×N (16%; 48%, now ≈71%)	**None**

NPVM = non-pulmonary visceral metastases.

> **KEY POINT**
>
> IGCCCG uses markers at the start of chemotherapy, not pre-orchidectomy values, and is invalid at relapse. Seminoma risk depends on NPVM alone; mediastinal NSGCT is always poor risk.

6 · Treatment principles

- Centralise care, give full-dose cisplatin and resect all residual disease.

	Seminoma	NSGCT
CS I / II / III	85 / 10 / 5%	≈1/3 each
Occult metastasis CS I	10–15%	25–35%
Radiosensitive	Exquisitely	No (except brain)
Metastatic teratoma	Rare	Frequent, so surgery needed
Markers	Rarely useful	Central

- Seminoma relapses contain NSGCT in 10–15%, so biopsy non-responders.

GCNIS management

- Options are orchidectomy, radiotherapy **≥20 Gy** (lower doses allow recurrence) or observation. Up to 40% later need testosterone after radiotherapy.
- With a normal contralateral testis and a wish for paternity, orchidectomy is preferred because scatter harms the other testis. Cisplatin only partly clears GCNIS.

7 · NSGCT management

Every CS I option cures ≈100%; the choice is about toxicity.

CS I NSGCT

- Adjuvant therapy overtreats 70–80%.
- **Risk factors** are **LVI** (strongest) and **EC predominance**. High-risk tumours relapse in ≈50%, low-risk in <20%.
- **Surveillance**: relapse ≈28%, >90% within 2 y and mostly retroperitoneal; relapses are chemo-naive and curable.
- **Primary RPLND** removes teratoma and finds pathological stage II in 19–28%, mostly cured by surgery alone. pN1 is observed; pN2–3 (relapse >50%) gets **2 cycles adjuvant** (relapse ≤1%) or observation.
- **Bleomycin–etoposide–cisplatin (BEP) ×1** gives the lowest relapse (≈2–3%) but leaves teratoma, exposes all to late toxicity, and relapses are chemoresistant.
- Full bilateral nerve-sparing template (ejaculation >90%) is standard; a modified template is acceptable for clinically negative nodes.

ALGORITHM

1. Assess CS I NSGCT for LVI and EC predominance.
2. If low risk (neither), then surveillance.
3. If high risk, then surveillance, BEP×1 or expert nerve-sparing RPLND, chosen by preference and toxicity.

KEY POINT

Surveillance maximises quality-adjusted survival when relapse risk is <≈1/3. All options cure ≈100%; they differ in overtreatment versus difficulty of salvage.

CS IS, IIA–B

- **CS IS** is treated with induction chemotherapy.
- **CS IIA–B** with raised markers or nodes **>3 cm** gets chemotherapy, with or without post-chemotherapy RPLND (PC-RPLND). With normal markers and nodes <3 cm, **primary RPLND** is used: 13–35% are pN0, ≈30% have teratoma, and half avoid chemotherapy.

CS IIC–III: induction by IGCCCG

P2 *CRE & BEP — BEP×3/EP×4 good, BEP×4 (VIP if lungs poor) intermediate/poor, toxicities*

- BEP×3 equals BEP×4; etoposide–cisplatin (EP) ×3 and carboplatin are inferior.
- Second line is TIP×4 (paclitaxel, ifosfamide, cisplatin), VeIP×4 or VIP; TIP is best for late relapse.
- High-dose chemotherapy (HDCT) failed in three randomised trials, so it is for specialised centres.

Post-chemotherapy residual masses (NSGCT)

- With normal markers and a residual mass **>1 cm**, do **PC-RPLND** with a full bilateral template. Histology is necrosis in ≈40%, teratoma in ≈45% and viable tumour in ≈15%.
- No model, imaging or **FDG-PET** predicts necrosis (teratoma is not avid), so always resect.
- After a complete response (≤1 cm), observe, or resect if pre-chemotherapy nodes were >3 cm (controversial).
- With multiple sites, resect all, retroperitoneum first, because its necrosis predicts necrosis elsewhere. Somatic transformation (6–8%) needs complete resection.
- After viable tumour, adverse factors are intermediate/poor risk, incomplete resection and >10% viable cells.

Relapsed NSGCT

SAQ Growing teratoma syndrome

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 20 SAQ Sheet, Q3a

Definition

- GTS (growing teratoma syndrome; Logothetis, 1982): metastatic masses of an NSGCT (non-seminomatous germ cell tumour) **enlarge during or after chemotherapy while AFP (alpha-fetoprotein) and hCG (human chorionic gonadotrophin) are normal or falling**. Resection shows **mature teratoma only**.
- Uncommon: about 2–8% of metastatic NSGCT.

Pathogenesis

- Teratoma is **histologically benign but genetically malignant** (i(12p), aneuploidy) and **chemoresistant**.
- Chemotherapy kills the sensitive elements (embryonal, yolk sac), so markers fall; the teratoma keeps growing, mostly by enlarging cysts. Chemotherapy-induced differentiation is a second theory.
- Main risk factor: teratoma in the orchidectomy specimen.

Clinical features

- Retroperitoneum commonest; also lung, mediastinum, supraclavicular nodes.
- Often silent; or compression: ureteric obstruction, IVC (inferior vena cava) compression or thrombosis, bowel obstruction, back pain.

Evaluation

- Markers before every cycle; **rising markers mean viable cancer, not GTS**.
- CT (computed tomography): enlarging, well-defined, **cystic** mass, sometimes with fat or calcification.
- **FDG-PET (fluorodeoxyglucose positron emission tomography) has no role**: teratoma is not FDG-avid.
- **Differential:** progressing viable GCT (germ cell tumour; markers rise); malignant transformation (somatic sarcoma or carcinoma); necrosis or fibrosis (mass shrinks).

Management

- **Stop chemotherapy**; do not escalate to salvage chemotherapy (ineffective).
- **Prompt complete surgical resection**: PC-RPLND (post-chemotherapy retroperitoneal lymph node dissection), with resection of every other site; adjacent vessel, kidney or bowel resection may be needed. Delay lets the mass become unresectable.
- Radiotherapy is ineffective.
- **Outcome:** favourable after complete resection (long-term survival 75–90% after resection of residual teratoma). Incomplete resection leads to regrowth, **malignant transformation** (rhabdomyosarcoma commonest) or **late relapse**.
- **Also know:** CDK4/6 (cyclin-dependent kinase 4/6) inhibitors studied for unresectable teratoma; lifelong CT and marker follow-up.

Diagram to draw: graph of mass size rising while marker curve falls during chemotherapy cycles.

> **Key points to remember:**
> - Enlarging mass plus normal or falling markers equals GTS.
> - Teratoma is chemoresistant and not FDG-avid.
> - Treatment is complete surgical resection, not more chemotherapy.

Also, from the condensed book (not in the paper answer):

- **Chemo-naive relapse** gets induction by IGCCCG, with cure >95%.
- **Early relapse (≤2 y) after chemotherapy** gets TIP/VeIP/VIP×4 or HDCT, then post-salvage surgery for residual masses.
- **Desperation surgery** is for rising markers with a single resectable site; survival is 33–57%.
- **Late relapse (>2 y)** occurs in ≈3%. Yolk sac tumour is commonest and chemoresistant, so **complete resection** is the treatment.

8 · Seminoma management

*See the paper answer **Classify testicular tumors and describe management of seminoma** (Section 10, Ch 79).*

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- CS I risk factors are tumour **>4 cm** and rete testis invasion (relapse 12/16/32% with 0/1/2), but they are not validated, so surveillance is preferred for all.
- IIA–B nonbulky (≤3 cm): dog-leg radiotherapy 25–30 Gy with a 5–10 Gy boost, BEP×3/EP×4, or primary RPLND (SEMS, COTRIMS). Bulky or multiple nodes get chemotherapy.
- Carboplatin ×2 gives relapse ≈1.5%. Seminoma relapses late (median 14 months), and radiotherapy carries second malignancy risk.

Residual seminoma masses are necrosis in ≈90%, and many regress over 12–18 months.

ALGORITHM

1. If the residual seminoma mass is <3 cm, then observe.
2. If >3 cm, then FDG-PET ≥6 weeks after chemotherapy (earlier scans give false positives).
3. If PET-negative, then observe.
4. If PET-positive, then template PC-RPLND (or repeat PET/observe). Mass-only excision gives 33% local relapse.

- Relapse after radiotherapy gets first-line chemotherapy; after chemotherapy, VeIP×4 or HDCT.

9 · Brain metastases

- Present in ≈1% at diagnosis, with **choriocarcinoma or very high hCG**; they can bleed fatally. Give BEP×4, then resect residuals.

10 · Treatment sequelae and follow-up

- RPLND can cause ejaculatory loss (>80% after full template without nerve sparing). Orchidectomy alone causes hypogonadism in 10–20%.
- Chemotherapy toxic death is 0–2.4% (good risk) and 3–4.4% (intermediate/poor). Azoospermia is near universal; ≈half recover by 2 y, 80% by 5 y.
- Late effects: Raynaud 20–45%, neuropathy 14–43%, ototoxicity, hypogonadism 15–40%, **cardiovascular disease** and **second malignancy** (60–100% excess after cisplatin or radiotherapy, 200% after both).
- Follow-up is most intensive in years 1–2 (NSGCT) or 1–3 (seminoma).

> **KEY POINT**
>
> Because cure is expected, late cardiovascular disease and second cancers, which rise with each added modality, decide between equally curative options.

11 · Non-germ-cell tumours†

- †**Leydig cell**: the commonest sex cord–stromal tumour, causing precocious puberty or gynaecomastia. ≈10% are malignant; these resist chemo- and radiotherapy, so nodes need RPLND. A small marker-negative lesion suits testis-sparing excision.
- †**Sertoli**: mostly benign; the large-cell calcifying variant occurs with Carney complex and Peutz–Jeghers.
- †**Lymphoma**: the commonest testicular tumour in men **>50–60 y** and the commonest bilateral one. It is treated by orchidectomy, then R-CHOP with CNS prophylaxis and contralateral testis radiotherapy.

12 · Landmark trials

Trial	Design	Result	Take-home
IGCCCG 1997	Pooled 5202 NSGCT, 660 seminoma	Markers, NPVM, mediastinal primary predict outcome	Risk groups select regimen
PVB vs BEP	RCT, advanced	Equal survival, less neurotoxicity	BEP standard

Trial	Design	Result	Take-home
BEP×3 vs ×4	RCT, good risk	Equal survival	BEP×3
VIP vs BEP×4	RCT, intermediate/poor	Equal, more toxicity	VIP only if lungs at risk
MRC TE19/EORTC 30982	Carboplatin ×1 vs radiotherapy, CS I seminoma	5-y RFS 94.7% vs 96%	Equivalent
MRC 20 vs 30 Gy	RCT, CS I seminoma	Equal relapse, less toxicity	20 Gy PA
Albers (German)	BEP×1 vs unilateral RPLND, CS I NSGCT	2-y PFS 99% vs 92%	Favours BEP×1
Williams	Adjuvant ×2 vs observation, pathological stage II	Relapse 6% vs 49%; same survival	Both acceptable
MRC surveillance	2 vs 5 CTs, CS I NSGCT	No difference	Fewer scans safe

RCT = randomised controlled trial; RFS = relapse-free survival; PFS = progression-free survival; PA = para-aortic.

80 Surgery of Testicular Tumours

1 · Evaluation of the testis mass

It is urgent, because delay escalates curative therapy.

- Examine supraclavicular nodes, breasts, abdomen and scrotum. Scrotal ultrasound (US) mainly checks the **contralateral testis** (bilateral synchronous ≈ **1%**). Markers (AFP, β-hCG, LDH) give diagnosis, stage and post-orchidectomy baseline.
- Discuss a prosthesis and † offer sperm banking.

2 · Radical inguinal orchidectomy

Treatment of choice: **early cord control** before handling the testis, removing testis, epididymis and cord to the internal ring. † A scrotal approach breaches a second (inguinal) lymphatic field and leads to local recurrence.

ALGORITHM

1. Make a 3–5 cm incision over the canal and open the external oblique.
2. **Preserve the ilioinguinal nerve**.
3. Mobilise the cord at the pubic tubercle and apply a **Penrose early clamp**.
4. Deliver the testis and divide the gubernaculum.
5. Dissect the cord to the internal ring.

- **High ligation**: ligate vas and gonadal vessels **separately** with **non-absorbable** suture, leaving a **1–2 cm** tail to mark the stump for excision at retroperitoneal lymph node dissection (RPLND).

KEY POINT

† **Scrotal violation**: inguinal exploration, high cord ligation and excision of the scrotal scar. Routine hemiscrotectomy is unnecessary; chemotherapy-era outcomes are not compromised.

Partial orchidectomy (testis-sparing surgery)

SAQ Testis sparing surgeries for CA testis

Paper answer · P4

Type: SAQ (10 marks)
Source: 2016-Summer Paper 4, Q3b

Definition and rationale

- **Partial orchidectomy**: inguinal excision of the tumour, preserving the rest of the testis. The standard remains **radical inguinal orchidectomy**; sparing aims to keep androgen production and avoid lifelong testosterone.

Indications

Suspected germ cell tumour (GCT), all of:

- Synchronous bilateral tumours or tumour in a **solitary testis**.
- Tumour **≤2 cm** (up to 2–3 cm, under 30% of testicular volume), preferably polar.

- Adequate preoperative testosterone.
- Not with a normal contralateral testis.

Suspected benign or indeterminate lesions:

- Small impalpable lesion (<10 mm) with normal markers: most benign (cyst, infarct, Leydig nodule), but 20–50% are small seminomas. Lesions <5 mm are rarely malignant.
- **Prepubertal**: teratoma and epidermoid cyst with normal AFP (alpha-fetoprotein): partial orchidectomy is the standard.

Preoperative work-up

- Ultrasound of both testes; AFP, beta-hCG (human chorionic gonadotropin), LDH (lactate dehydrogenase); testosterone, LH (luteinising hormone); **sperm banking**; consent for radical orchidectomy.

Technique

1. Inguinal incision; early cord control with a soft clamp.
2. Deliver the testis; isolate with towels.
3. **Intraoperative ultrasound** localises the lesion.
4. Cold ischaemia optional if resection <30 minutes.
5. Incise the tunica parallel to the segmental vessels; excise the mass with a rim of normal tubules.
6. **Frozen section** (benign versus malignant) and biopsies of adjacent parenchyma for GCNIS (germ cell neoplasia in situ).
7. Close the tunica; fix the testis in the scrotum.

Management of GCNIS

- **Radiotherapy about 18–20 Gy** to the remnant: prevents invasive GCT, keeps Leydig function, but causes sterility; up to 40% later need testosterone.
- German Testicular Cancer Study Group: no local recurrences with radiotherapy; most men who omitted it recurred.
- Radiotherapy may be deferred for paternity under close follow-up. Local recurrence → completion orchidectomy.

Complications and follow-up

- Haematoma, atrophy, hypogonadism, local recurrence. Follow with self-examination, ultrasound, markers and testosterone.

Diagram to draw: testis with polar tumour excised with a rim of normal tissue.

> **Key points to remember:**
> - Solitary or bilateral tumour, ≤2 cm, normal testosterone.
> - Frozen section plus adjacent biopsies for GCNIS.
> - GCNIS: 18–20 Gy keeps Leydig function but sterilises.

Also, from the condensed book (not in the paper answer):

- Needs **all three**: polar, **≤2 cm**, contralateral testis compromised/absent. **Never elective with a normal contralateral testis.**
- Masses **<5 mm** with negative markers have low malignancy, so selective surveillance is an option. Cold ischaemia can be omitted if resection takes **<30 min**.
- GCNIS options are completion orchidectomy, **18–20 Gy** or observation (local recurrence ≈ 7.5%). **Local recurrence always needs completion orchidectomy**.

Delayed orchidectomy

Life-threatening metastatic germ cell tumour (GCT) gets chemotherapy first, then **delayed orchidectomy in all**, because the testis is a sanctuary with **discordant response**. An apparent extragonadal retroperitoneal primary shows intratubular neoplasia in up to 42%, so lateralising nodes mean orchidectomy on that side.

3 · RPLND — terminology and indications

Term	Indication
Primary RPLND	High-risk clinical stage (CS) I or low-volume stage II non-seminomatous GCT (NSGCT), **normal markers**
Post-chemotherapy (PC-RPLND)	After induction; residual **>1 cm**, normal markers
Salvage PC-RPLND	After second-line chemotherapy
Desperation	Markers still elevated; resectable
Redo	Retroperitoneal recurrence after prior RPLND
Late relapse	**>24 months** after complete response

Each step down means more chemoresistant disease and harder surgery.

Preoperative planning

- CT within **6 weeks** and markers within **7–10 days**; note vascular variants; no bowel prep.
- After **bleomycin**, test lung function and ask anaesthesia to use **low FiO_2, restrictive fluids**.
- Total IVC occlusion by thrombus needs inferior vena cava (IVC) resection; partial needs reconstruction.

4 · Templates

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- The bilateral template is bounded by renal vessels, ureters and common iliac arteries. A modified post-chemotherapy template needs a lesion **≤5 cm**.

5 · Open RPLND technique

- The approach is **transperitoneal** or **extraperitoneal** (less ileus).

- Incise from the caecum to the inferior mesenteric vein (IMV) and along the line of Toldt to the foramen of Winslow, freeing duodenum and pancreas. Use **liberal clips** to prevent chyle leak.
- **Split and roll**: split at 12 o'clock on the aorta from the left renal vein down to the inferior mesenteric artery (IMA), roll packets off and ligate lumbars.
- Lumbar arteries are usually 3 pairs, with the **2nd pair at IMA level**. The **right sympathetic trunk**, posterolateral to the IVC, is mistaken for lymphatic tissue.

Nerve sparing

- Left fibres run along the lateral aortic border near the IMA. Right fibres emerge **beneath the IVC** and cross the aorta obliquely, so they are at risk when lumbar veins are ligated.

> **KEY POINT**
>
> Identify fibres **prospectively, before dividing anything**; loop them and pass the packet beneath. Avoid traction, which causes neuropraxia and transient anejaculation.

Postoperative care

- No routine drain; if one is used, remove it at **<100 mL/24 h** on diet. No nasogastric (NG) tube, low-fat diet early, and home on days 3–5.

6 · Auxiliary procedures (PC-RPLND 23–45%)

- **Nephrectomy** is commonest, usually **left**. The strongest predictor is a residual mass **>10 cm**.
- **Vascular** repair is planned in every PC-RPLND, using a patch or graft.
- **IVC**: narrowing **<25%** is repaired by venorrhaphy, >25% by patch or graft; ligation is acceptable. **The aorta is never ligated**.
- **Liver**: histology is discordant in ≈half; resect, or ablate if **<3 cm** and unresectable. **Thorax**: do RPLND first, then resect lesions >1 cm unless the retroperitoneum shows fibrosis only (thoracic necrosis then in ≈89%).

7 · Decision making

- **Complete remission (≈70%)**: generally observe, though ≤1 cm residuals hold teratoma in ≈ 1/5 and viable tumour in ≈ 1/20.
- **Necrosis is predicted by** no teratoma in the primary, normal pre-chemotherapy AFP/β-hCG, raised LDH, >50% shrinkage and residual <2 cm.
- **After primary RPLND**: pN0, teratoma only or pN1 are observed; pN2 gets 2 cycles or observation. Observation cures ≈70%.

8 · Histology of residual mass after chemotherapy

Finding	Frequency	Outcome/action
Fibrosis/necrosis	**≈40%**	Relapse-free (RFS) and cancer-specific survival (CSS) ≈95%; no adjuvant
Teratoma	**≈45%**	RFS 80–90%; must be resected
Viable GCT	**≈15%**	Survival 50–70%; consider 2 adjuvant cycles

> **KEY POINT**
>
> **Resect every NSGCT residual >1 cm**: no imaging (including PET †) or marker separates necrosis from teratoma, and teratoma is chemoresistant, grows, relapses late and transforms. Surgery is both staging and the only treatment.

- **Fizazi** model for viable GCT: incomplete resection, ≥10% viable cells and IGCCCG intermediate/poor risk. With 0, 1 or ≥2 factors, 5-y overall survival (OS) is 100/83/51%.
- Adjuvant chemotherapy: NCCN for incomplete resection or intermediate/poor risk; EAU for incomplete resection only. Not after salvage chemotherapy.

9 · High-risk PC-RPLND

- **Salvage**: viable GCT is at least doubled; OS 60–75%.
- **Desperation**: OS ≈54%. Operate if disease is confined and completely resectable, with markers declining or plateaued, or chemotherapy exhausted. Do **not** operate with rapidly rising markers or multisite progression. **Postoperative marker normalisation** is the only OS predictor.
- **Redo** is mostly for technical failure (retroaortic/retrocaval tissue), so do **complete great-vessel mobilisation the first time**.
- **Late relapse** (yolk sac, somatic-type malignancy) is treated by **surgery first**.

10 · Unique situations

- **Seminoma residual**: dense desmoplasia, viable in ≈10%. If **<3 cm**, observe; if **>3 cm**, do FDG-PET (≥6 weeks †). If avid, choose selected PC-RPLND, salvage chemotherapy or HDCT.
- **Sex cord stromal**: malignant features include >4 cm, necrosis, mitoses and lymphovascular invasion; RPLND is of unproven benefit.

11 · Outcomes, fertility and complications

- **Ejaculation**: template or nerve sparing preserves antegrade ejaculation in **>90%** (primary) at no oncological cost. † If lost, options include sympathomimetics and sperm retrieval.
- Complications: primary 10.6–24%, PC up to 56%. Mortality: primary ≈0, PC <1%.

Complication	Key facts / management
Pulmonary (bleomycin)	PC 3–5%, acute respiratory distress syndrome. Low FiO_2, restrictive fluids; accept SpO_2 **88–90%**
Ileus	Up to 26% (commonest); extraperitoneal approach, no NG tube, early feeding
Chylous ascites	Primary 0.2–2.1%, PC 2–12%; linked to suprahilar dissection (cisterna chyli L1–L2)
Lymphocele	Symptomatic ≤1.7%; drain if infected
Venous thromboembolism (VTE)	Primary <1%, PC 0.1–3.1%; drug prophylaxis if prior chemotherapy, VTE or obesity
Neurological	Femoral neuropraxia; paraplegia with bulky disease

*See the paper answer **Lymphocele and postoperative lymphatic leak (chylous ascites)** (Section 15, Ch 91).*

ALGORITHM

1. Paracentesis; drain if it reaccumulates.
2. Very-low-fat, high-protein diet with medium-chain triglycerides (MCT) and octreotide.
3. If no response, nil by mouth with total parenteral nutrition (TPN).
4. If refractory, lymphangiography with or without embolisation, peritoneovenous shunt or exploration.
5. Continue a low-fat diet ≥**4 weeks** after resolution.

12 · Landmark trials

Trial	Design	Result	Take-home
Williams (Testicular Intergroup)	RCT, pathological stage II: 2 adjuvant cycles vs observation	Adjuvant near-abolished relapse; equal survival	Observe pN1; individualise pN2
SEMPET	FDG-PET in post-chemotherapy seminoma residuals	>3 cm: specificity 100%, NPV 96%; ≤3 cm PPV 37%	PET for >3 cm only

81 Minimally Invasive Retroperitoneal Lymph Node Dissection

SAQ Minimally invasive and robotic RPLND

Paper answer · P3 · P4 · merged from 2

Asked as: P3 SAQ "Minimal invasive retroperitoneal lymph node dissection" · P4 SAQ "Robotic RPLND"

Introduction

- **MI-RPLND (minimally invasive retroperitoneal lymph node dissection)** for testicular GCT (germ cell tumour): laparoscopic (1992) or robotic (2006). **It must duplicate open RPLND exactly.**

Indications

- **Primary:** marker-negative stage I NSGCT (non-seminomatous GCT), especially high-risk IB; small-volume stage II NSGCT or seminoma (1–2 nodes ≤3 cm in the landing zone).
- **Postchemotherapy:** only a small solitary residual mass with little desmoplasia; **open** for large masses or vessel involvement.
- **Expanding indication:** primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy (SEMS, PRIMETEST).
- **Patient selection:** serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND.

Templates and nerve sparing

- Boundaries: renal hila, ureters, ureter crossing the common iliacs; plus spermatic cord.
- **Primary:** extended **unilateral modified template** with nerve sparing (right: paracaval, precaval, interaortocaval; left: para-aortic, preaortic; both take nodes above the IMA (inferior mesenteric artery)). If nodes are enlarged, the dissection becomes bilateral.
- **Postchemotherapy:** bilateral full template.
- **Nerve sparing:** identify the sympathetic chains and L1–L4 postganglionic fibres to the hypogastric plexus below the IMA.
- **Diagram to draw:** right and left modified templates.

Steps (robotic)

1. Positioning: **modified lateral decubitus** (unilateral) or **supine steep Trendelenburg with peritoneal hammock** (any template).
2. Mobilise colon; Kocherise duodenum (right); divide gonadal vein.
3. **Split-and-roll:** split over the great vessel from common iliac to renal vein; roll packets off.
4. **Divide lumbar vessels** to clear retrocaval and retroaortic tissue.
5. Clip lymphatics (left renal vein); bag each packet.

Feature	Laparoscopic	Robotic
Retraction	Assistant	Fourth arm
Instruments	Straight	Wristed
Suturing, vessel control	Difficult	Easier

Outcomes

- Stay 1–3 days, blood loss mostly ≤200 mL; conversion about 5%; **antegrade ejaculation ≥90%** (robotic nerve sparing after primary surgery); chylous ascites as open. Node yields approach open (about 30).
- **Major bleeding:** pressure, raise pneumoperitoneum, convert if uncontrolled (open set ready).

Concerns and guidelines

- Concerns: staging-only dissections, frequent adjuvant chemotherapy, unindicated surgery, catastrophic bleeding, follow-up <5 years; no randomised trials.
- **Guidelines:** AUA (American Urological Association): experienced surgeons may offer it; EAU (European Association of Urology): specialist centres only.

Recent advances

- Serum miR-371a-3p for selection of stage I patients with occult nodes.
- RPLND for small-volume stage II seminoma (SEMS, PRIMETEST) to avoid chemotherapy and radiotherapy.

> **Key points to remember:**
> - Same template, split-and-roll, lumbar division and nerve sparing as open.
> - Best for chemotherapy-naive low-volume disease.
> - Conversion about 5%; ejaculation preserved ≥90%.
> - Expanding indication: primary RPLND for small-volume stage II seminoma.
> - Serum miR-371a-3p may select stage I patients with occult nodes.

Also, from the condensed book (not in the paper answer):

- The benchmark recurrence-free rate is ≈**85%** for limited pN+ without adjuvant chemotherapy; low MI recurrence is **confounded** by high pN0 and adjuvant rates.

1 · Indications and rationale

- Goals in early-stage disease are to preserve cure, avoid **dual therapy** (surgery and chemotherapy) and minimise lifelong toxicity.
- **Primary** RPLND (retroperitoneal lymph node dissection) is for marker-negative clinical stage (CS) I non-seminomatous germ cell tumour (NSGCT), and small-volume stage II NSGCT and seminoma. It spares chemotherapy in the **20–30%** with enlarged nodes who prove pN0.
- **Post-chemotherapy (PC-RPLND)** is for NSGCT residual **>1 cm** and select seminomas.
- The robot adds fourth-arm retraction and **circumferential great-vessel dissection with lumbar control**.

> **KEY POINT**
>
> Much recovery gain came from Enhanced Recovery After Surgery (ERAS) pathways, which help every approach, so the benefit of access alone is smaller than it looks.

2 · Guidelines and controversy

- The AUA (early stage) says it **may be offered** by a surgeon expert in germ cell tumours and minimally invasive surgery (MIS), with the patient told long-term data are lacking. The EAU restricts it to **specialist centres** expert in open and MIS.
- With no noninferiority evidence, it is highly controversial. Concerns are staging-only dissection without lumbar division, excess adjuvant chemotherapy, operating on the wrong patients, conversion in **≈5%**, and short follow-up with reported peritoneal carcinomatosis.

3 · Patient selection

- There are no absolute contraindications; relative ones are extensive prior surgery and morbid obesity.
- **Suited to MIS**: chemo-naive high-risk IB; small-volume II (1–2 landing-zone nodes, none **>3 cm**); a post-chemotherapy solitary landing-zone mass with little desmoplasia.
- **Favour open**: most PC-RPLND, a large mass, large chemotherapy shrinkage, possible vessel reconstruction, somatic malignancy or retrocrural disease.

4 · Unchanging operative principles

- Imaging within **≈4 weeks** and markers within **1–2 weeks**.
- **MI-RPLND must completely duplicate open RPLND**: template, completeness and nerve-sparing are unchanged.

ALGORITHM

1. Safe entry and exposure.
2. **Split-and-roll** nodal removal.
3. Prospective nerve sparing.
4. **Lumbar vessel division**.
5. Spermatic cord removal, then closure.

- Chemo-naive patients may have an extended unilateral nerve-sparing template, converted to bilateral if enlarged nodes are found. Post-chemotherapy, the **bilateral full template** is standard.
- Incomplete RPLND or mass-only excision worsens survival.

5 · Preparation, positioning and ports

- Plan for haemorrhage with an **open set in theatre**; after bleomycin, manage fluids and oxygen carefully. A high-fat meal the day before highlights lymphatics.
- Positioning is lateral decubitus, laparoscopic supine, or robotic steep Trendelenburg with a **peritoneal hammock** that allows any template.

6 · Operative steps

- **Right**: incise the Toldt line to the foramen of Winslow and **Kocherise**; divide the gonadal vein at the inferior vena cava (IVC); split over the IVC; clip lymphatics; roll the paracaval packet, spare fibres and divide lumbars. **Do not ligate the right renal artery** in the interaortocaval space.

- **Left**: divide the gonadal vein at the left renal vein and control the **lumbar vein entering the left renal vein**.
- Lower the pneumoperitoneum to check haemostasis; **no routine drain**; discharge at **1–2 days**.

7 · Complications and management

ALGORITHM

1. Major vascular injury: apply pressure and raise the pneumoperitoneum.
2. Inform anaesthesia, get blood and call vascular help.
3. If uncontrolled, keep pressure, undock and convert.

- For MIS repair, **identify the injury before acting: no blind clipping**.
- Prevent chylous ascites with liberal clips rather than energy devices; rates match open, and it is treated by low-fat diet, paracentesis, octreotide and parenteral nutrition.
- Antegrade ejaculation is kept in **≥90%** after primary nerve-sparing.

8 · Outcomes vs open

- There is **no randomised trial**. MIS gives shorter stay and less blood loss but longer operative time; brisk bleeding is the unique risk.
- Robotic node yields approach the open benchmark of **≈30**, but adjuvant chemotherapy is still too frequent. Post-chemotherapy data are too small.

9 · Landmark trials

Trial	Design	Result	Take-home
GETUG T93BP	Good-risk NSGCT: BEP×3 vs EP×4	Most relapses/deaths without or with incomplete RPLND	Complete RPLND matters

82 Tumours of the Penis

1 · Overview

- Squamous cell carcinoma (SCC) is **~95%** of penile malignancy, and **0.4–0.6%** of male cancers in North America and Europe, but up to **10%** in parts of Asia, Africa and South America.
- Prognosis rests on **primary stage and histology** and on **nodal metastasis**.

2 · Premalignant lesions: penile intraepithelial neoplasia (PeIN)

*See the paper answer **Premalignant conditions in urology (including premalignant lesions of penis)** (Section 3, Ch 30).*

Also, from the condensed book (not in the paper answer):

- Erythroplasia of Queyrat (EQ) is a red velvety glans lesion in uncircumcised men that mimics balanitis; it progresses in **10–33%**.
- Bowen disease is a scaly shaft plaque mimicking psoriasis or eczema; it progresses in **~5%**. Bowenoid papulosis (HPV-linked shaft papules in young men) **does not progress**.

3 · Aetiology and risk factors of SCC

Factor	Key point
No neonatal circumcision	Neonatal circumcision virtually eliminates invasive cancer; it protects against invasive SCC, **not CIS** (carcinoma in situ). Adult circumcision gives little or no protection
Phimosis	Present in **25–75%**; **7–10×** risk. It mediates the non-circumcision risk, as smegma causes inflammation and fibrosis
HPV	Low risk 6, 11; high risk **16, 18**, 31, 33, 45, 52, 58; **HPV 16 commonest**. Present in **~50%**; wives have ~3× cervical cancer risk
Human immunodeficiency virus (HIV)	~4× risk of penile cancer death; PeIN progresses ~6 years earlier
Tobacco	All forms; dose-dependent and independent of phimosis
Lichen sclerosus (LS)	Cancer in **2.3–9%**; HPV-**independent** SCC
Trauma, obesity	Scarred shaft; obesity causes buried penis and functional phimosis
†**Psoralen and ultraviolet A (PUVA)**	† Genital exposure markedly increases SCC, so shield the genitalia

- **Prevention** is by neonatal circumcision (AAP 2012: benefits outweigh risks), hygiene, no tobacco and HPV vaccination (nonavalent adds 31/33/45/52/58).

> **KEY POINT**
>
> Two routes converge on the preputial sac: HPV-driven (E6/E7, p16-positive, basaloid/warty) and inflammation-driven HPV-independent (phimosis, smegma, LS). Neonatal circumcision, hygiene and vaccination each prevent part; adult circumcision prevents almost none.

4 · Natural history and spread

- Most arise on the glans or prepuce; **flat ulcerative tumours metastasise earlier**.
- **Buck fascia** is a temporary barrier; once it and the tunica are breached, tumour enters the corpora and spreads by blood.
- Distant metastases are uncommon (**1–10%**) and late. Untreated, death comes within **2 years** from femoral erosion haemorrhage or sepsis.

*See the paper answer **Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma** (Section 10, Ch 78).*

Also, from the condensed book (not in the paper answer):

- Pelvic stations are external iliac, internal iliac and obturator; distant sites (lung, liver, bone, brain) are rare without nodal disease.

5 · Presentation and diagnosis

- It presents as induration, a warty growth or an ulcer, often hidden by phimosis. Examine the lesion and **palpate both groins**.
- **Biopsy is mandatory before definitive therapy**, because depth and grade predict occult nodes and superficial biopsies mislead.
- **Hypercalcaemia without bone metastases** comes from PTH-related substances, tracks bulky inguinal disease and may resolve after node excision. Treat with saline, then bisphosphonate.

6 · Histology

Subtype (WHO 2022)		HPV	Features	Nodal risk
Usual (keratinising)	**45–65%**	Independent	Usually grade II; lymphovascular (LVI) or perineural invasion (PNI) in ~⅓	25–40%
Warty		Associated	Cauliflower, fibrovascular cores; slow	17–25%
Basaloid		Associated	Small cells, necrosis, deep	**>½ node positive**
Papillary		Independent	Low grade, no koilocytosis	≤12%
Verrucous		Independent	**Broad pushing border**	**No metastasis**; recurs if under-excised
Sarcomatoid		Independent	Spindle cells	**Most aggressive**
Rare: pseudoglandular, adenosquamous; clear cell, lymphoepithelioma-like (HPV)		—	Deep invasion	Pseudoglandular, clear cell worst

- **Any anaplastic cells make it grade III**. **LVI** predicts nodes in every study, and **PNI** does so strongly.
- **High-risk HPV or p16 positivity means better disease-specific survival (DSS)**, not better nodal status.

7 · Imaging

A negative scan never clears the groin.

- **Primary**: examination is reliable; **MRI** (with or without artificial erection) helps if equivocal and organ sparing is planned.
- **cN0 groin**: **no imaging detects micrometastases**, so stage surgically; use CT only if obese or after prior groin surgery.
- **cN+ or T3–T4**: CT chest, abdomen and pelvis. **FDG-PET/CT** is best for pelvic and distant disease.

8 · Staging: TNM 8th edition (2017)

T1a versus T1b decides who needs groin staging.

T	Definition
Tis	PeIN
Ta	Non-invasive localised SCC (incl. pure verrucous without destructive invasion)
T1	Glans: lamina propria · Foreskin: dermis/lamina propria/dartos · Shaft: tissue between epidermis and corpora
T1a	**No LVI, no PNI, not high grade**
T1b	**LVI and/or PNI, or high grade (grade III/sarcomatoid)**
T2	**Corpus spongiosum**, with or without urethra
T3	**Corpora cavernosa** (incl. tunica albuginea), with or without urethra
T4	Scrotum, prostate, pubic bone

N	Clinical (cN)	Pathological (pN)
1	Palpable mobile unilateral node	**≤2 unilateral**, no extranodal extension (ENE)
2	Mobile ≥2 unilateral or bilateral	**≥3** unilateral or bilateral
3	Fixed mass or pelvic nodes	**ENE or pelvic nodes**

Stage	TNM
I	T1a N0
IIA	T1b, T2 N0
IIB	T3 N0
IIIA / IIIB	T1–3 N1 / N2
IV	T4, any N3, any M1

- In the 8th edition, **urethral invasion no longer upstages**.

> **KEY POINT**
>
> Traps: T1b and T2 share stage IIA; any N3 or T4 is stage IV even without metastases; cN counts palpable nodes, pN counts metastases (pN1 allows two). The primary is staged by tissue, the groin by surgery.

9 · Management of the primary tumour

- **Penectomy** is the oncological gold standard (local recurrence **0–8%**), but over half of patients are ≤60, so preserve where safe.
- **Organ preservation** suits PeIN, Ta, T1 and grade I–II in **compliant** men. It is **contraindicated** by size **≥4 cm**, grade III, deep cavernosal invasion, or unfitness for salvage.
- Local recurrence is higher, but **early recurrence is salvaged without survival cost**, except in pT3 or high grade.
- **Margins**: microscopic spread is **~5 mm** for grade I–II and **~10 mm** for grade III. Only margins **<1 mm** raised recurrence, so a **negative frozen-section margin (>1 mm) is enough**; the 2-cm dogma is abandoned.

Stage / site	Treatment
PeIN, glans	Laser, resurfacing; topical alternative
Ta / PeIN — foreskin, shaft skin	Excision to negative margin; laser; topical (Tis only)
Ta, T1 glans	Excision, resurfacing, glansectomy, radiotherapy (RT; not Ta)
T2 glans, no gross cavernosal involvement	**Total glansectomy** with or without distal corporal transection; partial penectomy; RT
T2 with corporal invasion, T3	**Partial or total penectomy**
T4	Neoadjuvant chemotherapy, then surgery in responders
Recurrence after conservation	Re-excision or penectomy; superficial low-grade re-treated conservatively

- **Glans resurfacing** strips the epithelium and adds a split-skin graft, keeping erections.
- **Laser**: CO_2 reaches only ~0.1 mm (recurrence up to 50%), so **Nd:YAG** (up to 6 mm) is preferred, with base biopsies; recurrence is <10% for CIS and 10–25% for T1.
- † Partial penectomy leaves a stump for standing micturition; total penectomy needs a **perineal urethrostomy**.

> **KEY POINT**
>
> Margin width depends on grade, not a fixed 2 cm. Low-grade tumours spread millimetres and their recurrences are salvageable, so preservation is safe; in high-grade or cavernosal disease local relapse is lethal and amputation regains its place.

10 · Inguinal nodes (see Ch 84)

LAQ Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P2 LAQ "Importance and management of Inguinal lymph nodes in CA penis." · P3 SAQ "Inguinal Block Dissection?" · P4 SAQ "Videoendoscopic inguinal lymphadenectomy"

"Inguinal block dissection" is taken as ILND (inguinal lymph node dissection), mainly for penile cancer.

Importance

- Presence and extent of inguinal metastasis is the **strongest prognostic factor** in penile SCC (squamous cell carcinoma).
- A long **locoregional phase** precedes distant spread, so the groin is a **curable station**; lymphadenectomy alone can cure.
- **Kroon (2005)**: early ILND for impalpable nodes gave 3-year disease-specific survival about 84% vs about 35% when delayed until nodes were palpable (ENE [extranodal extension] about 20% vs 95%).
- Manage both groins **within 3 months** of diagnosis; aim for **at least 8 nodes per side**.
- About 80% 5-year survival with 2 or fewer unilateral nodes, no ENE, no pelvic nodes.

Lymphatic drainage and surgical anatomy

- **Superficial system** (prepuce, shaft skin) drains to superficial inguinal nodes. **Deep system** (glans, frenulum) drains to the corona, then within Buck's fascia to the presymphyseal plexus, then to femoral triangle nodes.
- Presymphyseal crossover makes drainage **bilateral** and unpredictable in side; with one palpable groin, the other has micrometastasis in about 30%.
- **Stepwise, no skip**: superficial inguinal, then deep inguinal, then external iliac; so superficial status gates deep dissection and inguinal status gates pelvic dissection.
- **Femoral triangle:** superior **inguinal ligament**, lateral **sartorius**, medial **adductor longus**; floor fascia lata; apex where the muscles cross.
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**:

Zone	Related vessel
Central	Saphenofemoral junction
Superolateral	Superficial circumflex iliac vein
Inferolateral	Lateral femoral cutaneous and superficial circumflex veins
Superomedial (sentinel nodes mostly here)	Superficial external pudendal and superficial epigastric veins
Inferomedial	Great saphenous vein

- **Deep nodes** (1–3) medial to the femoral vein; **Cloquet's node** is the highest.

Staging (TNM [tumour–node–metastasis] 8th edition)

N	Clinical	Pathological
1	One mobile unilateral node	2 or fewer unilateral, no ENE
2	Mobile, 2 or more unilateral or bilateral	3 or more (unilateral or bilateral)
3	Fixed mass or pelvic nodes	ENE or pelvic nodes

- **T1a**: no LVI (lymphovascular invasion), no PNI (perineural invasion), not high grade. **T1b**: LVI or PNI or high grade.

Evaluation

- **Careful bilateral palpation**: cN0 or cN+ (number, side, mobile or fixed). Imaging (US [ultrasound], CT [computed tomography], MRI) only if examination is unreliable (obesity, previous groin surgery).

- **cN+ or high risk**: FDG-PET/CT (fluorodeoxyglucose positron emission tomography–CT) or CT chest–abdomen–pelvis.
- About half of palpable nodes are reactive, yet an antibiotic trial is **no longer recommended** (delays treatment). US-guided **FNAC (fine-needle aspiration cytology)** is specific; a negative result never ends the pathway.

Management of the impalpable groin (cN0): risk-based

Risk (EAU-ASCO [European Association of Urology–American Society of Clinical Oncology] 2023)	Primary	Groin
Low	pTis, pTa, pT1a G1	Surveillance
Intermediate	pT1a G2	Surveillance (compliant) or surgical staging
High	**pT1b or more**, G3	**DSNB (dynamic sentinel node biopsy)**; ILND if unavailable or DSNB positive

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk, staged surgically (ILND first).
- **DSNB**: US ± FNAC first (a tumour-replaced node diverts flow and causes a false negative); then 99mTc (technetium-99m) nanocolloid, blue dye and gamma probe; serial sectioning. False-negative about 7% at expert centres.
- **Also know:** Cabanas anatomical sentinel biopsy abandoned.

Management of palpable nodes (cN+)

`Palpable node → US-guided FNAC positive → radical ILND same side + stage other side (superficial ILND, FS-ILND or DSNB) negative → excisional biopsy → negative: observe Bulky (≥4 cm), fixed, bilateral or pelvic nodes → neoadjuvant TIP × 4 → responders: ilioinguinal resection → cisplatin-unfit: chemoradiation`

- **PLND (pelvic lymph node dissection)**: if **more than 2 inguinal nodes or ENE** (pN2–3); ipsilateral; external iliac commonest, then obturator.
- **Neoadjuvant TIP** (paclitaxel, ifosfamide, cisplatin), **Pagliaro (2010)**: objective response 50%; survival mainly in responders. InPACT (International Penile Advanced Cancer Trial) is testing the sequence.
- **Adjuvant**: chemotherapy for pN2–3 (weak evidence); radiotherapy for multiple nodes or ENE (at least 50 Gy; 54 Gy for ENE).

Inguinal block dissection (ILND): indications

- Penile SCC: **proven cN+**; **cN0 high risk** (pT1b or more, G3) if DSNB is positive or unavailable. Also vulval cancer, melanoma.
- Within 3 months; at least 8 nodes per side. Bulky or fixed nodes: chemotherapy first.

ILND templates

Template	Extent
Radical (cN2 or more)	Superficial + deep; saphenous often ligated; sartorius transposed

Template	Extent
Fascia-sparing ILND (FS-ILND)	May be used for cN1
Modified (Catalona)	Smaller field: nothing lateral to femoral artery or below fossa ovalis; **saphenous spared**; no sartorius transposition; positive frozen section: convert to complete (radical) ILND
Superficial	Above fascia lata only

Technique: open radical ILND

1. Frog-leg position; incision **2 cm below and parallel** to the inguinal ligament.
2. Flaps **deep to Camper's fascia** (preserves the skin capillaries).
3. Boundaries: inguinal ligament, sartorius, adductor longus. Divide saphenous at apex; open fascia lata; clear along the femoral vessels; **never lateral to the femoral artery** (femoral nerve).
4. **Sartorius transposition** over the vessels; suction drains until <30 mL/day.
5. Pelvic dissection if more than 2 inguinal nodes or ENE.

Videoendoscopic inguinal lymphadenectomy (VEIL)

Definition and aim

- **VEIL:** ILND through thigh ports, beneath Scarpa's fascia; laparoscopic (Tobias-Machado) or robotic (**RVEIL**).
- **Aim:** open-surgery template without the groin incision that causes flap necrosis.

Indications

- Penile SCC, **impalpable nodes**, intermediate/high risk (pT1b+, G3, LVI).
- **Mobile, non-bulky palpable nodes.**
- **Not for** bulky, fixed (cN3) or skin-involving nodes: these need chemotherapy, then open surgery.
- Also vulval cancer, melanoma.
- DSNB still comes first for impalpable nodes (EAU-ASCO).

Technique

- Mark an inverted triangle on the skin **before insufflation**, because gas distorts landmarks. `Position: supine frog-leg / low lithotomy → 2 cm camera incision below the femoral triangle apex → finger dissection beneath Scarpa's fascia → two lateral and medial working ports (+ assistant port) → CO2 at 15 mmHg (higher briefly to create space) → develop space up to the inguinal ligament → remove nodal packet off fascia lata, apex to ligament → preserve saphenous vein if possible; clip tributaries → frozen section → deep dissection medial to femoral vein if positive → bag retrieval, suction drain in dependent part`
- **Template:** same as open: superficial ± deep; Daseler's five zones; Cloquet's node is the highest deep node. **Never dissect lateral to the femoral artery** (femoral nerve).

Outcomes vs open

	Open ILND	VEIL/RVEIL
Wound infection, necrosis, dehiscence	Higher	**Lower**

	Open ILND	VEIL/RVEIL
Hospital stay, drain time	Longer	Shorter
Operative time	Shorter	About 2 h (slightly longer per side)
Nodal yield, recurrence	Reference	**Similar**
Evidence	Retrospective	Retrospective, selection-biased

- Commonest complication of VEIL: **lymphocele**, then lymphoedema and cellulitis.

Robotic VEIL

- 3D vision, wristed instruments, ICG (indocyanine green) for lymphatics; adequate dissection on open re-exploration (phase I). No VEIL vs RVEIL comparison.

Complications of ILND

- Wound infection, **flap necrosis**, dehiscence, seroma, **lymphocele**, **lymphoedema**, DVT/PE (deep vein thrombosis/pulmonary embolism).
- Reduced by transverse incision, correct flap plane, modified template, saphenous sparing, drain removal by volume (<30 mL/day), thromboprophylaxis and minimally invasive access (VEIL); sartorius transposition increases them.

Follow-up

- Examination 3-monthly for 2 years, then 6-monthly to 5 years (recurrence mostly within 2 years); then self-examination.

Diagrams to draw

- Femoral triangle with Daseler's zones and Cloquet's node.
- Femoral triangle with VEIL port sites below the apex.

> **Key points to remember:**
> - Nodal status is the strongest prognostic factor; ILND can cure.
> - pT1b or higher: stage groins surgically (DSNB preferred) within 3 months; at least 8 nodes per side.
> - Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
> - PLND if more than 2 nodes or ENE; bulky nodes: neoadjuvant TIP (InPACT testing the sequence).
> - Femoral triangle: ligament, sartorius, adductor longus; flaps deep to Camper's; stay medial to femoral artery.
> - VEIL: same template, fewer wound complications, equal nodal yield; not for bulky, fixed or skin-involving nodes.

SAQ Prognostic index score in carcinoma penis

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3e

Assumed meaning: prognostic factors, risk-group systems and nomograms in penile SCC (squamous cell carcinoma). No single universally used "prognostic index score" exists.

Introduction

- Prognosis rests on primary histology and stage and, above all, **inguinal node status**.

1. Nodal status (most important factor)

Nodal finding	5-year survival
Node negative	About 73%
≤2 nodes, unilateral, no ENE (extranodal extension)	About 80%
>2 nodes	0–50%
ENE or **pelvic nodes**	Poor (pelvic about 14%)

- TNM (tumour-node-metastasis) 8th edition: pN1 (1–2 unilateral nodes) about 90% 3-year DSS (disease-specific survival); pN3 (ENE or pelvic) about one-third.

2. Primary tumour factors (predict occult nodes in cN0)

- **Depth:** corporal invasion → about 60% node positive.
- **Grade:** any grade III component = grade III.
- **LVI (lymphovascular invasion)** and **PNI (perineural invasion)**: strongest predictors; define pT1b.
- **Subtype:** verrucous and PeIN (penile intraepithelial neoplasia) do not metastasise; basaloid and sarcomatoid aggressive.

3. Risk-group systems

System	Low	Intermediate	High
Earlier EAU (European Association of Urology), after Solsona	Tis, Ta, T1 G1	T1 G2	T2 or more, or G3
Current EAU–ASCO (American Society of Clinical Oncology) / TNM	PeIN, Ta, **T1a** (nodes 0–18%)	—	**T1b or more** (one-third to over half)

- Low risk → groin surveillance; high risk → **DSNB (dynamic sentinel node biopsy)** or ILND (inguinal lymph node dissection).

4. Nomograms and indices

- **Ficarra nomogram:** predicts nodal involvement (thickness, growth pattern, grade, LVI, corporal invasion, clinical node status).
- **Kattan nomogram:** predicts cancer-specific survival.
- A pathological **prognostic index** (grade + anatomical level of invasion + PNI; Cubilla group) has been proposed.
- **Status:** none externally validated; not used routinely.

5. Molecular markers

- **HPV (human papillomavirus)/p16 positive:** better DSS (not node status).
- Adverse: p53, PD-L1 (programmed death-ligand 1), Ki-67, high **NLR** (neutrophil-to-lymphocyte ratio), raised CRP (C-reactive protein).

6. Other

- Inguinal recurrence after observation: only about one-third survive 5 years. Size >5 cm worse.

Diagram to draw: flowchart: primary pathology → low or high risk → surveillance or DSNB/ILND.

> **Key points to remember:**
> - Node status is the strongest prognostic factor.
> - LVI, PNI, grade III define high-risk T1b.
> - Nomograms (Ficarra, Kattan) unvalidated; p16 favourable.

Also, from the condensed book (not in the paper answer):

- Cure is ~**80%** with ≤2 unilateral nodes and no ENE or pelvic nodes; with positive pelvic nodes, 5-year survival after surgery alone is ~**14%**.
- Occult node rates are **0–18%** for PeIN/Ta/T1a, one-third to over half for T1b and above, and ~60% with corporal invasion.

ALGORITHM

1. **cN0, low risk**: surveillance. If a node appears, do fine-needle aspiration cytology (FNAC); if positive, inguinal lymph node dissection (ILND) and contralateral staging.
2. **cN0, high risk (T1b+)**: bilateral dynamic sentinel node biopsy (**DSNB**) or superficial/fascia-sparing/minimally invasive ILND. If positive, completion ILND.
3. **cN1–2, mobile, <4 cm**: biopsy, then ILND and contralateral staging. If >2 nodes or ENE, add pelvic lymph node dissection (PLND).
4. **Fixed, ≥4 cm or pelvic**: neoadjuvant TIP (cisplatin-eligible) or chemoradiation, then resect responders.
5. **Adjuvant therapy** if >2 nodes, bilateral nodes, ENE or pelvic nodes.

- DSNB refinements (pre-operative ultrasound with or without FNAC, serial sectioning) cut false negatives from ~18% to ~7%. Morbidity is greatest with radical ILND, then superficial or fascia-sparing, then minimally invasive, and least with DSNB.
- No pelvic skip metastases occur, so PLND is mainly staging.

11 · Radiotherapy

- **Circumcise first** to expose the lesion and prevent oedema and phimosis. Surgical salvage remains curative.
- **External-beam RT (EBRT)**: **≥66 Gy in 2-Gy fractions over ~6.5 weeks** with **bolus**; ~55% are recurrence-free at 5 years.
- **Brachytherapy** (Ir-192) suits **T1–T2 glans tumours <4 cm**: ~60 Gy over ~5 days gives ~79% relapse-free survival and ~85% preservation at 5 years, with better erections than EBRT.
- **Late effects** are **meatal stenosis** (10–45%, at 18–24 months) and ulceration; **1 in 5 local recurrences occur after 5 years**.
- For the groins, give **no prophylactic RT** for cN0. Adjuvant RT for **≥2 nodes or ENE** is ≥50 Gy, with **54 Gy for ENE and 57–60 Gy for gross residual**. Chemoradiation (cisplatin if GFR >50 mL/min) treats inoperable nodes.

12 · Systemic therapy

- No randomised trials are completed. **TIP** (paclitaxel, ifosfamide, cisplatin) is most active; **Bleomycin regimens are discouraged**.
- **Neoadjuvant TIP ×4** for bulky N2–N3 or pelvic nodes gives ~50% response, and ~⅔ of responders stay disease-free, so **response selects for surgery**.
- Adjuvant chemotherapy has no proven survival gain but may be considered for pN3 or pelvic nodes.
- Palliative platinum is first line; checkpoint inhibitors help ~1 in 6 in trials.

13 · Follow-up

Groin relapse after observation is usually fatal.

- After organ preservation, use close follow-up and self-examination.
- Observed groins are checked 2–3-monthly for 2 years, then less often. Only ~⅓ of cN0 men with groin relapse survive 5 years.

14 · Non-squamous penile tumours (~5%)

Tumour	Key features / treatment
BCC	Proximal shaft; almost never metastasises. Treat by excision or Mohs
Melanoma	<1%; glans; spreads by blood and lymph; AJCC cutaneous staging. Bilateral modified ILND if Breslow ≥1 mm, ulceration or Clark IV–V (DSNB reasonable); adjuvant nivolumab for stage IIIB–IV
Sarcoma	Malignant are **proximal**; vascular commonest; **local recurrence characteristic**. Superficial: wide excision or partial; deep corporal: total. No ILND unless palpable
Extramammary Paget	Penoscrotal eczematoid plaque; CK7+, CEA+, S-100−; ~½ have another cancer, so screen. Wide excision (up to 3 cm) guided by **mapping biopsies**
Lymphoma	Usually systemic: R-CHOP even if isolated
Metastases	From **bladder, prostate, rectum**; **priapism** is the commonest sign; most die within a year; palliate

15 · Landmark trials

Trial	Design	Result	Take-home
Neoadjuvant TIP (MD Anderson)	Phase II; TIP ×4, cN2–N3 M0	Response 50%; median OS 17.1 months; durable mainly in responders	Standard before surgery for bulky/pelvic nodes
Netherlands Cancer Institute early vs delayed	Early DSNB vs dissection after surveillance relapse	3-y survival 84% vs 35%; ENE near-universal if delayed	Stage high-risk cN0 groins early

84 Inguinal Node Dissection

1 · Why the groin matters in penile cancer

Groin surgery alone can cure node-positive disease.

- Penile squamous cell carcinoma (SCC) has a **locoregional phase**, so the groin is a **curable station**.
- **Nodal status is the strongest prognostic factor**. Extranodal extension (ENE), even in one node, makes it pN3.
- Early versus delayed inguinal lymph node dissection (ILND) for impalpable nodes gave ENE ≈20% vs ≈95% and 3-year disease-specific survival ≈84% vs ≈35%. So manage the groins within **3 months** of diagnosis.
- An adequate dissection removes ≥**8 nodes per side**.

2 · Indications and risk groups

	Low risk	Intermediate	High risk
NCCN	pTis, pTa, pT1a	pT1b G1–2	≥pT2, or G3/G4
EAU-ASCO	pTa, pTis, G1 pT1a	G2 pT1a	≥pT1b
cN0 — NCCN	Surveillance	Modified/radical bilateral ILND or DSNB	Same
cN0 — EAU-ASCO	Surveillance	Surveillance (compliant) vs DSNB	DSNB; ILND if unavailable

NCCN = National Comprehensive Cancer Network; EAU-ASCO = European Association of Urology–American Society of Clinical Oncology; DSNB = dynamic sentinel node biopsy; G = grade. **Trap**: pT1b G2 is intermediate (NCCN) but high (EAU-ASCO).

- For the impalpable high-risk groin, **NCCN leads with ILND** and **EAU-ASCO leads with DSNB**. Select on **primary pathology only**, not HPV or nomograms.
- **cN+**: about half of palpable nodes are metastatic. Do fine-needle aspiration cytology (FNAC), then excisional biopsy if negative. If any is positive, do **bilateral radical ILND**. Give no antibiotic trial.
- With unilateral palpable nodes, the other side has micrometastasis in ≈**30%**, so stage it.
- **Bulky or fixed (cN3)**: give cisplatin–taxane chemotherapy first, then surgery. Resect upfront only if chemotherapy-ineligible or vessel invasion is imminent; involved skin needs a myocutaneous flap.

3 · Staging work-up

- Palpable nodes or high risk need 18FDG-PET/CT or CT chest–abdomen–pelvis. Imaging misses deposits <1 cm, so it has no role in the impalpable groin.
- **FNAC** is specific but insensitive: if positive, do ILND; if negative, do DSNB or ILND and never discharge.
- **DSNB**: do ultrasound, with or without FNAC, first, because a **tumour-replaced node** reroutes tracer and gives a false negative. Then use ^{99m}Tc nanocolloid and blue dye with serial sectioning. Positive: ILND †; negative: surveillance.

KEY POINT

Every groin test fails toward **false negatives**; a negative result never ends the pathway.

4 · Lymphatic anatomy

*See the paper answer **Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy** (Section 10, Ch 82).*

Also, from the condensed book (not in the paper answer):

- The **femoral triangle** is bounded by the inguinal ligament, **adductor longus** medially and **sartorius** laterally.
- Raise flaps deep to **Camper's fascia**, which holds the capillary network; too thin causes necrosis and too thick leaves nodes.
- In the fossa ovalis the order, lateral to medial, is **nerve, artery, vein**, so **never dissect lateral to the femoral artery**.

5 · Templates

Template	Extent	Features
Radical	Whole femoral triangle, superficial and deep	Fascia lata opened; saphenous ligated; sartorius transposed
Modified (Catalona)	Excludes lateral to femoral artery and caudal to fossa ovalis	Shorter incision, saphenous preserved, thicker flaps, no transposition
Superficial	Above fascia lata only	Fascia lata preserved

Modified ILND nodes go to frozen section; if **positive, convert to radical** †.

6 · Open radical ILND

ALGORITHM

1. Frog-leg position; incision ≈2 cm below and parallel to the inguinal ligament.
2. Raise flaps 3 cm above the ligament to the apex; divide the saphenous at the apex (spare if disease minimal).
3. Open fascia lata over sartorius and adductor longus; dissect cephalad on the femoral vessels.
4. Divide the saphenous at the junction; clear deep nodes up to the femoral canal.
5. **Sartorius transposition**; close the femoral canal.
6. Suction drains; tack the flaps.

7 · Pelvic lymph node dissection (PLND)

- If the inguinal frozen section is positive, do PLND at the same sitting; otherwise stage it.
- † EAU: with ≥2 positive inguinal nodes or ENE (pN2–3), do ipsilateral PLND, open or minimally invasive.

8 · Video-endoscopic ILND (VEIL)

*See the paper answer **Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy** (Section 10, Ch 82).*

Also, from the condensed book (not in the paper answer):

- The commonest robotic complication is lymphocele.

9 · Complications and post-op care

Fear of complications must not cause omission.

Complication	Cause	Prevention / management
Wound infection, breakdown (commonest)	Skin flora, dead space	Asepsis, tack flaps, incisional negative pressure
Flap necrosis	Breach of Camper's capillary network	Correct plane, modified template, VEIL; debride, then flap or negative pressure †
Lymphocele / seroma	Unsealed lymphatics	Sealing devices; drain until **<30 mL/day**; drain symptomatic or infected collections †
Lymphoedema	Lymphatic interruption	Saphenous sparing, modified template, compression hose ≥1 month
Others	Phlebitis, neurapraxia, pulmonary embolism (PE), femoral bleeding	Thromboprophylaxis; stay medial to artery

Mobilise early. NCCN advises antibiotic prophylaxis, though the authors do not use it routinely.

10 · Follow-up

Most groin relapses occur within 2–3 years (median ≈1 year). EAU-ASCO advises examination 3-monthly for 2 years, then 6-monthly to 5 years, with imaging only if node-positive; NCCN images everyone.

83 Tumours of the Urethra

1 · Benign urethral tumours

Lesion	Who / where	Clue	Treatment
Leiomyoma	Reproductive-age women; posterior wall	Firm suburothelial mass; grows in pregnancy	Excision; may recur
Haemangioma	Men, any age	Bluish sessile lesion, painless bleeding	Fulguration/laser; open excision if extensive
Fibroepithelial polyp	Boys; bulbar/posterior	Polyp on a stalk; filling defect	Transurethral resection

2 · Anatomy, histology and lymphatics

Tumour site predicts its type and its nodal basin.

Feature	Male (~20 cm)	Female (~4 cm)
Segments	Anterior: fossa navicularis, pendulous, bulbar. Posterior: membranous, prostatic	Anterior = distal ⅓; posterior = proximal ⅔
Epithelium	Urothelium in the prostatic urethra, columnar in the anterior urethra, squamous in the fossa navicularis	Proximal ⅓ urothelium; distal ⅔ stratified squamous
Histology	Urothelial carcinoma (UC) 50–80%, squamous cell carcinoma (SCC) 10–30%, adenocarcinoma 5–10%. **Anterior: up to 80% SCC; posterior: mostly UC**	UC 28–45%, SCC 19–29%, **adenocarcinoma 28–38%** (near-equal)
Lymphatics (both sexes)	Anterior drains to **inguinal** nodes (bulbar also external iliac); posterior drains to **pelvic** nodes	

KEY POINT

Site predicts histology (distal SCC, proximal UC), histology predicts behaviour, and the same anterior–posterior line fixes the nodal basin. In women it is also the **continence boundary**: the distal third can be excised without incontinence.

3 · Staging: AJCC 8th edition

T	Penile urethra and female urethra	Prostatic urethra
Ta / Tis	Non-invasive papillary / carcinoma in situ (CIS)	Tis also includes prostatic duct involvement **without stromal invasion**
T1	Subepithelial connective tissue	Subepithelial connective tissue
T2	Corpus spongiosum or periurethral muscle	Prostatic stroma
T3	Corpus cavernosum or anterior vagina	Periprostatic fat
T4	Other adjacent organs (bladder)	Bladder, rectum
N	N1 single, N2 multiple regional nodes (inguinal, pelvic, presacral): **number, not size**	

- **Evaluation** is by examination under anaesthesia with cystourethroscopy, bimanual palpation and biopsy, plus **MRI** for local extent and CT chest if invasive.

- **Palpable inguinal nodes (20–30%) are almost always metastatic**, so, unlike penile cancer, no antibiotic trial is given.

4 · Male primary urethral cancer (PUC)

Distal tumours are curable by surgery; bulbar tumours need multimodal therapy.

- Patients are >60 years. **Urethral stricture** is the commonest risk factor (up to 50%); others are lichen sclerosus, pelvic radiotherapy (RT) and HPV 16.
- † Biopsy a stricture that bleeds or recurs rapidly.
- **Prognosis** depends on stage and location: distal tumours do 2–3-fold better, and bulbar survival is as low as 20–30%.

Pendulous / fossa navicularis (distal)

- Surgery alone controls low-stage disease, ranging from transurethral resection to **distal urethrectomy with or without partial penectomy**, to total urethrectomy with perineal urethrostomy, with or without total penectomy.
- **Penile preservation** is endorsed by EAU and NCCN, using options such as topical 5-fluorouracil (5-FU) for distal CIS, distal urethrectomy, or glansectomy. A positive margin needs re-excision or adjuvant RT.
- A **margin of 5 mm** is now standard (1 cm traditionally).
- **Primary RT** is only for early anterior lesions in men refusing surgery, and it is ineffective for bulbar PUC.

Bulbar (proximal anterior)

- The few low-stage tumours are treated by transurethral resection, or segmental excision and anastomosis.
- **Advanced** disease needs radical cystoprostatectomy, pelvic lymphadenectomy, total urethrectomy and penectomy. Surgery alone gives ~¼ 5-year survival, so use **multimodal therapy**.

Inguinal nodes (both sexes)

- There is **no proven benefit of prophylactic inguinal dissection in cN0**. For cN1, give neoadjuvant chemotherapy or chemoradiotherapy, then lymphadenectomy.

5 · Male posterior (prostatic) urethral cancer

Only stromal invasion needs cystectomy.

- It is usually **secondary to bladder UC**, found in 20–40% at radical cystectomy (RC), so always evaluate the bladder and upper tracts.
- Diagnose by TURP biopsy; it is often CIS. BCG for high-grade superficial disease gives ≥70% response.

Involvement	Treatment	5-year survival
Mucosal	TURP + BCG	67–74%
Ductal / acinar (still Tis)	TURP + BCG, or RC with or without urethrectomy	67–74%
Stromal invasion	**RC + urethrectomy + pelvic lymphadenectomy**, with or without neoadjuvant chemotherapy	~36%; follows bladder stage

> **KEY POINT**
>
> The only question is "stroma breached?" TURP before BCG debulks, improves contact and stages. Ductal disease is not T2, so do not jump to cystectomy.

6 · Female urethral cancer

- It is ~⅓ of PUC, in women >60 years. Risks include chronic irritation, HPV and **urethral diverticulum (6–10% harbour malignancy**, mostly adenocarcinoma).
- Distal tumours are SCC; proximal ones are UC or adenocarcinoma, which is **mucinous** or **clear cell** (in diverticula).
- **Location decides continence-sparing excision**; stage with MRI pelvis.
- **Stage dominates prognosis**: distal tumours do better than proximal, which do better than pan-urethral.

Treatment

Situation	Management
Small non-invasive	Endoscopic resection or laser, with or without BCG
Low-stage distal	**Distal urethrectomy** (transvaginal) **or RT** (brachytherapy with or without external beam, ~60 Gy): equal, ≥70% cured. Stress incontinence (>40%), meatal stenosis; ~1 in 5 recur locally. RT complications 20–50% (stricture, necrosis, fistula)
Proximal / pan-urethral	**Anterior exenteration**: RC, total urethrectomy, extended pelvic lymphadenectomy and anterior vaginal wall, with or without hysterectomy and pubic rami. Alone, most die, so use **multimodal** therapy
Palpable groin nodes	Chemoradiotherapy, chemotherapy or lymphadenectomy; often curable. Pelvic nodes are poor

- RT local control depends on the **length of urethra involved**, not histology or size.

7 · Multimodal therapy (both sexes)

- For advanced or proximal disease, give **cisplatin-based neoadjuvant chemotherapy** (EAU, NCCN) or **chemoradiotherapy**, then consolidative surgery. Response predicts survival.
- **Chemoradiotherapy** is 45–55 Gy to the genitalia, perineum and groins with 5-FU and mitomycin C. It is preferred for advanced SCC, though every responder develops a stricture.

Histology	Regimen
UC	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin); gemcitabine–cisplatin
UC or SCC	Paclitaxel, ifosfamide, cisplatin (TIP); cisplatin, gemcitabine, ifosfamide
Adenocarcinoma	Gemcitabine, 5-FU, leucovorin, cisplatin

Stage (NCCN, anterior male / female)	First line
Tis, Ta, T1	Repeat transurethral resection, with or without intraurethral BCG or chemotherapy
Male T2 pendulous	Partial urethrectomy with or without partial penectomy; if margin positive, chemoradiotherapy, surgery or RT

Stage (NCCN, anterior male / female)	First line
Male T2 bulbar	Urethrectomy with or without RC; if ≥pT3/N1, adjuvant chemotherapy or chemoradiotherapy
Female T2	RC + urethrectomy, or distal urethrectomy by location, or chemoradiotherapy
≥T3 N0	Chemoradiotherapy with or without surgery; or neoadjuvant chemotherapy, then surgery or RT
N1	Chemoradiotherapy (preferred for SCC) or systemic therapy, with or without consolidative surgery

- Recurrence after resection of invasive disease is ~50%, so survey the urethra, groins and body.

8 · Urethral recurrence (UR) after radical cystectomy

Urethral risk decides whether a neobladder is safe.

Male

- Risk is **1–15%** (~5%), mostly within 2 years.
- **Risk factors** are **prostatic urethral involvement** (15% vs 5%), **positive urethral margin**, cutaneous diversion, CIS and multifocality. An orthotopic neobladder (ONB) is protective (~5% vs 9%).
- Routine prophylactic urethrectomy is abandoned, because 64–100% of specimens are negative.
- **Before ONB**, confirm a negative intraoperative frozen section of the urethral margin.
- **Urethrectomy** is absolute for a positive margin or gross urethral tumour, and relative for diffuse CIS, prostatic urethral involvement or multifocal disease. Delayed urethrectomy is as good as simultaneous.

Female

- The main risk is **bladder neck involvement**, but only 30–50% of these have urethral tumour, so it is not an absolute contraindication.
- ONB is allowed if the bladder neck, urethra and anterior vaginal wall are uninvolved and frozen section is negative; UR after ONB is <5%.

Surveillance and treatment

- Urethral wash cytology has no proven survival benefit; if positive, do urethroscopy and biopsy.
- **Total urethrectomy is treatment of choice**, including the meatus; after ONB, convert to cutaneous diversion. **Intraurethral BCG treats CIS** (not papillary or invasive). Invasive UR has median survival ~17 months.

85 Benign Disorders of the Penis and Urethra

1 · Reconstructive surgical technique

- **Cautery**: use **bipolar on the urethra** to limit spongiosal damage, and monopolar on subcutaneous tissue and muscle.
- A **bougie à boule** calibrates. **Goodwin sounds** dilate over a guidewire, so they are safer than van Buren sounds.
- **Sutures** are always **absorbable**, because permanent ones form a stone nidus; monofilament polydioxanone (PDS) is preferred. Support is needed for **30–60 days**, so avoid chromic, which lasts 14 days. Use **5-0** for most urethral work and **6-0** for a single-layer mucosal anastomosis (non-transecting excision and primary anastomosis, EPA).

2 · Tissue transfer: grafts and flaps

LAQ Grafts in urology and buccal mucosa

Paper answer · P1 · P3 · merged from 2

Asked as: P1 LAQ "Enlist various grafts used in urology and role of BMG in urology" · P3 SAQ "Uses of buccal mucosa in urology"

Graft versus flap

	Graft	Flap
Blood supply	Separated; gets a new supply from the bed	Moved with its own pedicle
Depends on	A well-vascularised host bed	Its pedicle
Example	BMG (buccal mucosal graft)	Penile fasciocutaneous island flap

Graft take (about 96 hours)

Phase	Duration	Mechanism
Imbibition	about 48 h	Graft "drinks" nutrients from the bed; graft is cooler than core
Inosculation	about 48 h	Bed and graft vessels connect; true microcirculation returns

- A graft lives on its bed, so it does best on thick vascular beds. Take fails with poor beds (scar, radiation), haematoma, seroma, infection or shear. Hence **quilting or pie-crusting** and **immobilisation for 5–10 days** (bolster or vacuum dressing).

Grafts used in urology

Group	Graft	Main uses
Skin	**STSG** (split-thickness skin graft), meshed	Staged urethroplasty only; penile and scrotal cover (Fournier gangrene, buried penis)
	FTSG (full-thickness skin graft)	Fossa navicularis, penile urethra (avoid in lichen sclerosus)
	Posterior auricular (Wolfe) FTSG	Staged hypospadias repair
Mucosa	**Buccal, lingual, labial (oral mucosa)**	Urethra, hypospadias, ureter
	Bladder epithelium (mucosa)	Urethra, hypospadias; meatal prolapse

Group	Graft	Main uses
	Rectal (colonic) mucosa	Panurethral stricture when oral mucosa is unavailable
Other autologous	Dermis	Tunica albuginea in Peyronie disease
	Tunica vaginalis	Peyronie disease, urethral cover; easy harvest
	Saphenous vein	Peyronie defects; vascular repair
	Rectus fascia, fascia lata	Pubovaginal sling
Allograft / xenograft	Pericardium, porcine SIS, porcine dermis, cadaveric fascia	Peyronie grafting ("off the shelf"); slings
Synthetic	Polypropylene (macroporous type I mesh)	Mid-urethral sling, prolapse
	Dacron, PTFE (polytetrafluoroethylene)	Vascular grafts; not advised for Peyronie (infection, fibrosis)
Newer	Collagen fleece (TachoSil); acellular matrix, tissue-engineered oral mucosa	Peyronie; experimental urethra

- Vein is not a true graft: it survives by direct endothelial perfusion and its vasa vasorum.
- **Peyronie grafts:** pericardium (thin, strong, does not contract) and SIS (cut about 25% oversize; contracts) are commonest; AUA gives no preference.
- **STSG versus FTSG:** STSG carries little dermis, takes easily, can be meshed (up to sixfold) but contracts. FTSG carries full dermis, contracts less, looks better but is more "fastidious" in take.

Buccal mucosa: history and why it works

- **BMG** is the workhorse graft for urethral reconstruction. **Humby 1941** (hypospadias); **Bürger 1992** and **El-Kasaby 1993** (stricture).
- Thick, **non-keratinised**, elastin-rich, hairless epithelium used to a **wet environment**; resists infection.
- Thin, **highly vascular lamina propria** with a panlaminar plexus, so imbibition and inosculation are rapid and take is excellent.
- Little contraction; handles and holds sutures well; low donor morbidity.

Harvest

1. Nasotracheal intubation; mouth gag; mark the graft (about 4–5 × 2 cm per cheek; P3: 4–6 × 2–2.5 cm).
2. Stay **1 cm from the commissure** and avoid **Stensen's (parotid) duct** opposite the upper second molar.
3. Infiltrate, incise, raise the mucosa off buccinator; haemostasis.
4. Donor site: leave open or close; both acceptable.
5. Back table: **defat** the graft, cut about 10% oversize, fenestrate with a needle; quilt to the bed. - **Lingual** (under the tongue) and **labial (lip)** grafts are alternatives. Buccal and lingual give equal urethroplasty success; lip mucosa is thinner, suits the glans.

Uses in urethroplasty

Technique	Principle	Best for
Dorsal onlay (Barbagli, 1995)	Urethra rotated, dorsal urethrotomy; graft spread-fixed on corpora	Bulbar; reliable bed
Ventral onlay	Ventral urethrotomy; graft covered by spongioplasty	Bulbar with healthy spongiosum
Dorsal inlay (Asopa)	Ventral approach (urethrotomy); dorsal plate incised; graft inlaid	Penile and bulbar, no mobilisation
Augmented anastomosis	Excise narrowest part, floor anastomosis, graft roof	Long, dense bulbar stricture
Kulkarni (2012)	One-sided dissection keeps ventral blood supply; penile invagination; dorsal graft (onlay)	Panurethral, one stage
Staged (Johanson, Bracka)	Stage 1 lay open and graft; stage 2 tubularise after 6 months or more	**Lichen sclerosus**, failed hypospadias
BMG perineal urethrostomy	Graft completes dorsal wall to skin	Panurethral or failed repairs

- Dorsal and ventral results are similar (dorsal equals ventral); graft equals flap; onlay beats tubes.
- **Skin is avoided in lichen sclerosus** because the disease recurs in skin; use staged BMG.

Other uses of BMG

- **Ureter:** BMG onlay ureteroplasty (first human report 1999), now robotic with an **omental wrap**; for long proximal or mid strictures in dense scar. CORRUS (Collaborative of Reconstructive Robotic Ureteral Surgery) failure about 15–25% by site.
- Redo and staged hypospadias; urethral closure in rectourethral fistula; bladder neck contracture; female urethral stricture; Peyronie grafting (small defects only).

Complications

- **Donor:** pain, swelling (first month), numbness (up to 6 months), mouth tightness, salivary duct injury.
- **Recipient:** graft failure, restricture, sacculation, fistula, ED.

Outcomes

- Bulbar BMG urethroplasty about 85–90% success; most failures within 12 months.
- Results fall with time and in penile, redo and lichen sclerosus cases.

Recent advances

- Tissue-engineered oral mucosa; acellular matrix; robotic BMG ureteroplasty.

Diagram to draw: bulbar cross-section showing dorsal onlay, ventral onlay and dorsal inlay (Asopa) positions.

Key points to remember:

- Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
- BMG: non-keratinised, thin vascular lamina propria (excellent take), wet-adapted.
- Avoid Stensen's duct; buccal equals lingual in success.
- Staged BMG, not skin, in lichen sclerosus.
- Bulbar success about 85–90%.

Also, from the condensed book (not in the paper answer):

- **Extragenital full-thickness skin graft (FTSG)** fails in the urethra because its greater mass makes take ischaemic. The exception is the **posterior auricular (Wolfe) graft**, which behaves like a split-thickness skin graft (STSG).
- Oral mucosa graft (OMG) has a **panlaminar plexus** and sacculates less than skin. Buccal onlay failures occur in the **first 12 months**.

KEY POINT

Graft trade-off: a dense fine plexus and low mass favour take, while carried dermis or lamina propria gives durability and resists contraction. This explains STSG vs FTSG, the failure of extragenital FTSG, the Wolfe exception, and why OMG is ideal.

Flaps

- Flaps are classified by **composition** (fasciocutaneous, fascial, musculocutaneous, muscle or visceral), **blood supply** (axial on a named vessel, or random as in a Z-plasty), **movement** (advancement, rotation or transposition) and **location** (local, distant pedicled or free).

3 · Surgical anatomy

Fascial layers decide flap planes and how blood spreads.

- Layers, deep to superficial, are **tunica albuginea, Buck's fascia, dartos, skin**. Dartos is fat-free, carries the skin vessels and fuses with Colles fascia.
- **Skin supply**: the superficial external pudendal artery feeds a subdermal plexus, which is the basis of the McAninch circular flap.
- **Deep supply**: the internal pudendal artery continues as the common penile artery, giving the **bulbourethral, dorsal and cavernosal** arteries.

Situation	Spread
Buck's intact (penile fracture)	Confined to the shaft, giving an **eggplant deformity**
Buck's ruptured (straddle)	Within Colles–Scarpa: scrotum, perineum, abdominal wall; limited on the thigh by fascia lata
Fournier gangrene	From perineum and scrotum up the abdominal wall via Scarpa

KEY POINT

Two envelopes, two vascular worlds: dartos carries the skin supply (so skin flaps are raised on dartos); Buck's encloses the erectile bodies fed by the common penile artery.

4 · Lichen sclerosus (LS)

*See the paper answer **Lichen sclerosus (BXO, balanitis xerotica obliterans)** (Section 3, Ch 30).*

Also, from the condensed book (not in the paper answer):

- Malignant transformation (mostly SCC) reaches up to **8%**.
- **Steroid**: clobetasol propionate or betamethasone 0.05% once or twice daily for 1–2 months, then on alternate days for 3–4 months.
- Meatal stenosis is treated by extended meatotomy or BMG onlay, and early repair may avert a panurethral stricture.

> **KEY POINT**
>
> LS is a progressive skin disease sparing the proximal urethra: hence meatus as sentinel, retrograde urethrography (RUG) before meatotomy, oral mucosa not skin, high recurrence, and easy perineal urethrostomy.

5 · Other urethral lesions

Lesion	Key features	Management
Urethral haemangioma	Bloody discharge; often extends beyond view, so image with **MRI**; benign	Observe; small: laser or triamcinolone; large: excision, staged repair
Reactive arthritis (Reiter)	Arthritis, conjunctivitis, urethritis after Chlamydia/Shigella/Salmonella; HLA-B27 in ~2/3; **circinate balanitis**	Urethritis is self-limiting
Amyloidosis	Urethral mass, obstruction, haematuria; needs biopsy	Observe; dilate or urethroplasty if obstructive
Urethrocutaneous fistula	Mostly iatrogenic (hypospadias, phalloplasty)	Treat the **underlying process** as well as the tract
Male urethral diverticulum	Congenital (all layers) or acquired (no smooth muscle; behind stricture, catheter, trauma); dribbling, UTI, stone	Cystourethroscopy and RUG; repair and treat the cause
Enlarged utricle	Müllerian remnant, with proximal hypospadias	A catheter may enter the utricle; UTI, post-void dribble

6 · Prepuce and meatus

- **Paraphimosis (emergency)**: a trapped foreskin causes oedema and can lead to glans necrosis. Reduce by steady compression, with the thumbs pushing the glans and the fingers pulling the foreskin. If irreducible, do an emergency **dorsal slit**; offer elective circumcision for recurrence.
- **Phimosis** is physiological in infants (resolving at 1–8 years); in adults it leads to balanitis and cancer. Treat with steroid, dorsal slit or circumcision.
- **Balanitis**: **Candida** is commonest. Treat with hygiene and antifungal, and **check blood glucose**.

SAQ Meatal stenosis and submeatal stricture in males

Paper answer · P1 · P2 · merged from 2

Asked as: P2 SAQ "Management of Urethral meatal stenosis." · P1 SAQ "Treatment of submeatal stricture urethra in males"

Definition

- **Meatal stenosis:** narrowing of the external urethral meatus below the normal calibre for age.
- **"Submeatal stricture"** is read as a stricture of the meatus and fossa navicularis (glanular urethra). In adults, meatal stenosis is usually the tip of a **fossa navicularis stricture**.

Age	Normal calibre
Under 4 months	6–10 Fr
1–12 months	10–12 Fr
1–10 years	12–14 Fr
Over 10 years	14–18 Fr

Causes

- **Children:** almost always after **neonatal circumcision**: wet nappies irritate the exposed meatus (ammoniacal meatitis) and the ventral lips fuse into a thin membrane. Rarely congenital with hypospadias.
- **LS (lichen sclerosus, formerly BXO [balanitis xerotica obliterans]):** main inflammatory cause in adults today; begins at the glans and meatus and progresses proximally.
- **Iatrogenic:** catheter or instrument trauma, TURP (transurethral resection of prostate).
- **Failed hypospadias repair.**

Clinical features

- Thin, forceful stream deflected **upwards**; spraying; straining, frequency, incomplete emptying; pain at the start of voiding.

Evaluation

- Child: watch the child void; calibrate with a bougie à boule. Uroflowmetry. Urethrography only for UTI (urinary tract infection) or suspected proximal lesion.
- Adult: uroflowmetry; **RUG (retrograde urethrogram) before any meatal surgery**, as a proximal stricture often coexists, especially in LS.
- Biopsy suspicious or atypical lesions: LS turns malignant (squamous cell carcinoma) in up to 8%.

Treatment

Option	Method	Indication / comment
Topical steroid	Clobetasol 0.05% once or twice daily for 1–2 months, then alternate days; gentle dilation and nappy hygiene in children	Meatitis and early narrowing in a child; early LS
Dilation or self-calibration	Meatal dilator lubricated with clobetasol	Elderly, unfit; seldom durable

Option	Method	Indication / comment
Ventral meatotomy (child)	EMLA (eutectic mixture of local anaesthetics) cream in clinic, or general anaesthesia; parents spread the edges with a dilator tip 3 times daily for 7–10 days	Symptomatic stenosis in a child
Ventral meatotomy (adult)	Crush and incise ventrally; **three sutures** (apex and each side), mucosa to glans skin; dilate afterwards	Simple stenosis; in LS extend it with an inverted-V relaxing incision (extended meatotomy)
Dorsal meatotomy (Y-V plasty)	Excise the scar ridge, advance a flap	Scarred or recurrent; effective but cosmesis poorer
Fossa navicularis reconstruction (flaps)	**Jordan flap:** ventral urethrotomy through the stenosis; **transverse ventral penile skin island on dartos** transposed and inverted into the defect. **De Sy:** ventral longitudinal island advanced	Fossa navicularis involved, non-LS; excellent function and cosmesis; not in LS
Graft	**BMG (buccal mucosal graft)** inlay or onlay, one or two stages; glans wings developed	**LS: oral mucosa, never genital skin**
Perineal urethrostomy	Bypass	Pan-anterior LS, failed repairs, elderly

- LS affects genital skin, so skin flaps recur; it spares the proximal urethra, so perineal urethrostomy works.

Complications

- Bleeding, restenosis (commoner in LS), spraying, poor cosmesis.

Diagrams to draw

- Ventral meatotomy: crush, incise ventrally, suture mucosa to glans at apex and both sides.
- Jordan flap: transverse ventral skin island on dartos inverted into the ventral meatotomy.

> **Key points to remember:**
> - Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
> - Adult: RUG first; adult meatal stenosis is usually the tip of a fossa navicularis stricture.
> - Non-LS: meatotomy or Jordan/De Sy skin island flap.
> - LS: clobetasol, buccal mucosa (never genital skin), perineal urethrostomy if extensive.

Adult circumcision (sleeve)

ALGORITHM

1. Penile block. Foreskin retracted: mark the cuff incision across the frenular base; incise to Buck's.
2. Foreskin advanced: mark the coronal incision.
3. Excise the sleeve using **bipolar**, because adult vessels do not seal by pressure.
4. Quadrant dartos sutures, then skin closure.

- **Human immunodeficiency virus (HIV)**: the foreskin has many CD4+ cells and microtears. Circumcision lowers female-to-male HIV by ~**60%**, and also lowers herpes simplex and HPV.

7 · Penile curvature

*See the paper answer **Congenital Penile curvature..** (Section 2, Ch 55).*

Also, from the condensed book (not in the paper answer):

- Congenital curvature is a smooth whole-shaft curve, whereas chordee without hypospadias shows a hooded prepuce and ventral inelasticity.

Devine–Horton type	Abnormal tissue
I	Spongiosum deficient (chordee without hypospadias, I–III)
II	Dysgenetic Buck's and dartos; urethra/spongiosum normal
III	Dartos only
IV	All normal; asymmetric tunica = **congenital curvature**
V	Congenital short urethra; its existence is doubted

ALGORITHM

1. Excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum.
2. Artificial erection; ventral midline septotomy.
3. If still curved, do dorsal tunical ellipse excision or plication under the elevated neurovascular bundle.

- **Acquired non-Peyronie curvature** (see Ch 76) follows deep urethrotomy, neglected or **subclinical fracture**, or a septal tear. Erections are normal, with no shortening. Treat by **scar excision and graft**, not contralateral plication.

8 · Penile transplantation

A composite allotransplant restores appearance, erection and voiding after penile loss, but needs lifelong immunosuppression; the alternative is phalloplasty.

9 · Adult acquired buried penis (AABP)

It is self-perpetuating and usually needs surgery.

- Risks are **obesity**, aggressive circumcision, scrotal lymphoedema and LS.
- It is a **vicious cycle**: the escutcheon tethers the penis, and trapped urine causes inflammation, LS and scarring, which trap it further. So **weight loss alone rarely cures**.
- The **Pariser** classification grades the repair needed: I skin flap, II graft, III scrotoplasty, IV escutcheonectomy, V panniculectomy.
- **Evaluate the urethra**, because a severe stricture needs urethroplasty **before** grafting.

ALGORITHM

1. Slit the phimotic ring and place glans traction.
2. Circumcising incision; excise non-viable skin.
3. Do escutcheonectomy (harvest the graft from it; keep 2–3 cm of skin at the base).
4. Tack Scarpa to the rectus fascia or pubis; place drains.
5. Cover with STSG (0.012–0.020 inch), fibrin glue and a bolster for 5–7 days.

- **Scrotal lymphoedema** may be congenital, infective (**Wuchereria bancrofti**, lymphogranuloma venereum), malignant or after node dissection or radiation. Resect **all** involved tissue and cover with spared perineal or lateral scrotal skin.
- **Foreign body injection** (silicone, paraffin) causes fibrosis, necrosis and abscess. Excise it all and graft, and screen for **body dysmorphic disorder**.

10 · Landmark trials

Trial	Design	Result	Take-home
African circumcision trials (South Africa, Kenya, Uganda)	Three randomised controlled trials (RCTs), adult male circumcision vs none	~60% reduction in female-to-male HIV acquisition	Circumcision is an HIV-prevention strategy

86 Urethral Stricture Disease

1 · Definitions, anatomy, aetiology

LAQ Urethral stricture: etiology, classification, DVIU, management and failed VIU

Paper answer · P2 · P3 · merged from 2

Asked as: P2 LAQ "Urethral stricture disease: etiology, presentation and management (including recent advances)" (asked as "Describe etiology, clinical presentation and management of urethral strictures" and "Describe etiology, clinical presentation and recent advances in management of Urethral Structure Disease") · P3 LAQ "Urethral stricture: classification, DVIU and surgery after failed VIU" (asked as "Classify urethral strictures. Discuss in brief, surgical management of stricture after failed VIU" and SAQ "DVIU (Direct Visual Internal Urethrotomy)")

"Structure" is taken to mean urethral stricture disease.

Definition

- A **urethral stricture** is a fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing the mucosa. It is an epithelial scar, often with variable **spongiofibrosis**.
- Posterior narrowing is named differently (WHO [World Health Organization] 2004, ICUD [International Consultation on Urological Diseases] 2010): after pelvic fracture, **PFUI (pelvic fracture urethral injury)** or distraction defect; after prostate surgery or radiation, **stenosis or contracture**.
- Area falls with the square of diameter: halving calibre (30 to 15 Fr) loses about three-quarters of the lumen area, so symptoms appear late.

Anatomy relevant to surgery

- **Bulbar urethra lies eccentric (dorsal)** in the spongiosum; the penile urethra is central.
- **Dual blood supply** (bulbar arteries plus retrograde flow from the glans) lets the bulb be divided and mobilised.

Aetiology

Category	Examples and mechanism
Idiopathic (commonest, about equal to iatrogenic)	No cause found
Iatrogenic	Catheters (very common in LMICs [low- and middle-income countries]), TURP (transurethral resection of prostate), TURBT (transurethral resection of bladder tumour), cystoscopy, hypospadias repair, radiotherapy. Mechanism: mucosal injury, pressure ischaemia, infection
Traumatic	**Straddle injury**: bulb crushed against the pubis; pelvic fracture (PFUI) gives a posterior obliterative defect
Inflammatory	Gonorrhoea (historically commonest, now rare); **lichen sclerosus (LS)**. Chlamydia: no proven link
Congenital	Rarest; infants only (short anterior stricture)

- **LS**: chronic skin disease starting at the glans and meatus. High-pressure voiding forces urine into Littre glands, causing inflammation and deep spongiofibrosis. It **recurs in any genital skin** used to repair it.

Classification

1. By site (WHO 2004 / ICUD 2010 terminology)

Segment	Term
Anterior: meatus, fossa navicularis, penile (pendulous), bulbar	**Stricture**
Membranous urethra after pelvic fracture	**PFUI** / distraction defect
Posterior urethra after prostatectomy or radiotherapy; bladder neck	**Stenosis or contracture**

2. By aetiology: idiopathic, iatrogenic, traumatic, inflammatory, congenital (table above).

3. By length: short (up to about 2 cm), long, multifocal, **panurethral** (whole anterior urethra).

4. By depth (Jordan's spectrum): (A) mucosal fold, (B) iris constriction, (C) full-thickness with minimal spongiofibrosis, (D) full-thickness (deep) spongiofibrosis, (E) fibrosis beyond the spongiosum (periurethral fibrosis), (F) complex stricture with fistula or abscess.

5. By complexity: simple (short, bulbar, first treatment) versus complex (long, multiple, LS, radiation, failed previous treatment, fistula).

- **PFUI** (Ch 125): major distraction, minor distraction, partial injury.
- **Also know: LSE system** (Lumen calibre, Segment involved, Etiology) for standard reporting.

Clinical presentation

- Poor stream, straining, spraying, incomplete emptying, post-void dribble; UTI (urinary tract infection), prostatitis, epididymitis.
- **Retention** after long-tolerated symptoms; periurethral abscess, fistula ("watering-can perineum"), stones, upper-tract damage.

Evaluation (define the 4 Ls: location, length, depth, density)

Test	Role
Uroflowmetry and PVR (post-void residual)	Low, flat plateau-shaped flow curve; screen and follow-up
Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)	Site and length; steep oblique views; done by the surgeon
Flexible urethroscopy	Calibre, appearance; paediatric scope sees beyond
Sonourethrography	Length and depth of spongiofibrosis
MRI (magnetic resonance imaging)	PFUI defect alignment only; not routine

- **Urethral rest** (suprapubic catheter 6–8 weeks) shows the true length. Avoid blind dilation and instrumentation before surgery. In retention, place a **suprapubic catheter** rather than force a dilation.
- Before urethroplasty: treat infection; look for LS at the meatus; examine the mouth; record erectile function.

Management principles

- First agree the aim: **palliation or cure**.
- The old "reconstructive ladder" (repeat endoscopy before surgery) is abandoned (archaic), because repeated endoscopy makes later urethroplasty harder.

1. Dilation

- Soft or balloon dilation over a guidewire. It stretches, not tears, the scar. Cures only epithelial strictures without spongiofibrosis.

2. DVIU (direct vision internal urethrotomy, VIU)

Principle

- Incision of the stricture scar into healthy tissue under vision (Sachse cold-knife urethrotome), so the lumen heals enlarged by **secondary intention**: a **race between epithelialisation and wound contraction**. The stricture recurs when contraction wins, which is usual when spongiofibrosis is deep. Deep spongiofibrosis cannot be cured endoscopically.

Indications

Favourable	Unfavourable
Bulbar	Penile, fossa navicularis, any non-bulbar
Short (<1.5–2 cm)	Longer
Superficial, no dense spongiofibrosis	Dense scar (straddle, LS)
First procedure	Repeat; luminal narrowing >74%

Technique

1. RUG and VCUG define length and site.
2. Saline irrigation; **guidewire** through the stricture.
3. **Cold-knife incision traditionally at 12 o'clock** through scar to healthy spongiosum. Campbell notes the spongiosum is thinnest from 10 to 2 o'clock; deep cuts enter the corpora (ED [erectile dysfunction], fistula).
4. Catheter **3–7 days**; longer catheterisation adds no benefit. - **Laser** (holmium, Nd:YAG [neodymium:yttrium–aluminium–garnet], others): **no advantage over cold knife** (laser = cold knife).

Adjuncts

- **CISC (clean intermittent self-catheterisation)/self-dilation**, tapering over 3–6 months: delays recurrence, which returns when stopped.
- Mitomycin C injection (reduces recurrence), steroid injection (limited data); removable stents; permanent UroLume stents withdrawn.

Complications

- **Recurrence** (commonest); bleeding (often with erections); **irrigant extravasation**; UTI, sepsis; ED from deep cuts; spongio-cavernous fistula; incontinence if sphincter incised.

Why repeat VIU is futile after failure

- VIU succeeds only in **bulbar strictures <1.5 cm without dense spongiofibrosis, at the first attempt**. Recurrence shows deep spongiofibrosis, which an incision cannot remove.

- **Steenkamp 1997, Heyns 1998 (randomised trials):** dilation = urethrotomy; a second urethrotomy helps few, a third none. **A second urethrotomy adds virtually nothing.**
- **Santucci & Eisenberg (2010):** about **8% stricture-free** long term after a first DVIU; repeat procedures almost always fail.
- Each repeat dilation or urethrotomy lengthens the scar and lowers later urethroplasty success; urethroplasty is more **cost-effective** after one failure.
- **AUA (American Urological Association) guideline (2016)** and EAU (European Association of Urology): DVIU or dilation may be offered for **short (<2 cm) bulbar** strictures; offer **early urethroplasty, not repeated endoscopy**, after one failed dilation or DVIU; no permanent stents (no stent if urethroplasty is possible).

3. Urethroplasty (including surgery after failed VIU)

Stricture	Operation	Principle and key points
Bulbar, short (1–2 cm, up to 3–5 cm proximally)	**EPA (excision and primary anastomosis)**: gold standard, about 90–95% success	Excise all scar, spatulate both ends widely, tension-free anastomosis. **Non-transecting** or **vessel-sparing** variants keep bulbar arteries
Bulbar, longer	**BMG (buccal mucosal graft) onlay**: dorsal (**Barbagli**), ventral, dorsal inlay (**Asopa**)	Graft spread-fixed on a well-vascular bed (corpora or spongiosum)
Bulbar with an obliterated segment; long dense	**Augmented anastomosis**	Excise the narrowest part, anastomose the floor, graft the roof
Penile	One-stage BMG inlay or onlay; **penile skin flap** (Orandi, Quartey, circular)	Skin island carried on dartos; not in LS; avoid EPA here (chordee)
LS, failed hypospadias, infection	**Staged repair** (Johanson): stage 1 open and graft; tubularise after about 6 months or more	Oral mucosa, not genital skin, in LS
Panurethral	Kulkarni one-sided, one-stage dorsal graft, or staged	—
Elderly, unfit, complex, multiple failures	**Perineal urethrostomy** (BMG-augmented)	Simple, durable bypass
Meatal stenosis; fossa navicularis	Meatotomy; flap or graft	Topical clobetasol for early LS meatal disease
PFUI (including after failed VIU)	Suprapubic catheter, then delayed **perineal anastomotic urethroplasty** after 3 months or more	Progressive steps: bulbar mobilisation, intracrural separation, inferior pubectomy, rerouting of the urethra

- **Onlay is better than tube** repairs. Graft and flap results are equal; grafts are simpler.
- **Avoid genital skin in LS** (use oral mucosa).
- **Not fit or declines surgery: Optilume** paclitaxel-coated balloon; intermittent self-dilation; long-term suprapubic catheter.
- **Urethral stents** (UroLume) are withdrawn.

Recent advances in management

1. **Ladder abandoned**: repeated endoscopy lowers later urethroplasty success, so **early urethroplasty** is offered after one failed DVIU (AUA and EAU guidelines).
2. **Optilume paclitaxel-coated balloon** (FDA [Food and Drug Administration] 2023): paclitaxel inhibits fibroblasts. **ROBUST III**: stricture-free 74.6% vs 26.8% at 1 year after (repeated) failed endoscopy; IPSS (International Prostate Symptom Score) 9 vs 19.9.
3. **Anti-scarring adjuncts after DVIU**: mitomycin C injection reduces recurrence; intermittent self-dilation; removable stents. Permanent UroLume stents withdrawn.
4. **Vessel-sparing and non-transecting EPA**: preserve bulbar arteries, theoretically less ED.
5. **Augmented anastomosis**: excise the narrowest segment, graft the rest.
6. **Kulkarni one-sided dorsal BMG** through a perineal incision for panurethral strictures.
7. **Staged oral mucosa repair for LS**; avoid skin. **BMG perineal urethrostomy** reduces neomeatal stenosis.
8. **Robotic reconstruction** of bladder neck contracture and posterior stenosis (Y-V plasty, BMG).
9. **Tissue-engineered oral mucosa** (investigational). **Prevention**: silicone and hydrophilic catheters, smaller endoscopes.
10. **PROMs (patient-reported outcome measures)**: USS-PROM (Urethral Stricture Surgery PROM) is increasingly used to judge success. - **Also know:** dartos-island fossa repair; lingual graft; clobetasol for early LS.

Complications of urethroplasty

- Recurrence, ED (overall similar to circumcision; higher with long repairs), chordee, post-void dribble, graft-site pain, numbness.

Follow-up

- Voiding cystourethrogram at catheter removal (about 3 weeks); then uroflow, PVR and symptom score; RUG, cystoscopy or urethroscopy if flow falls.
- Most EPA recurrences occur in the first 6 months; graft results decline with time.

Diagrams to draw

- RUG of a bulbar stricture.
- Cross-section of dorsal onlay versus ventral onlay.
- EPA with spatulated ends.
- Urethrotome cutting at 12 o'clock over a guidewire.

> **Key points to remember:**
>
> - Idiopathic and iatrogenic are commonest; LS recurs in skin.
> - Define location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
> - DVIU only once, for bulbar strictures <1.5–2 cm at first attempt; do not repeat it.
> - Santucci: about 8% long-term success; repeat DVIU futile.
> - AUA: urethroplasty after failure; laser = cold knife; ladder abandoned.
> - EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
> - Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
> - ROBUST III: Optilume 74.6% vs 26.8% stricture-free at 1 year.

Also, from the condensed book (not in the paper answer):

- Straddle strictures often present late; prostate cancer treatment causes posterior stenosis.

Site	Correct term
Anterior urethra	Stricture (epithelial scar with or without spongiofibrosis)
Membranous urethra after pelvic fracture	Distraction defect; pelvic fracture urethral injury (PFUI)
Posterior urethra after trauma/radical prostatectomy	Posterior urethral stenosis
Prostatic urethra, bladder neck	Contracture or stenosis

- The common penile artery supplies the corpus spongiosum. The bulbar urethra lies dorsally, so a deep 12 o'clock urethrotomy in the bulb is dangerous.
- In lichen sclerosus (LS), microabscesses form in the glands of Littre; antibiotics relieve symptoms, not the stricture.

> **KEY POINT**
>
> Retrograde glans flow allows the bulb to be transected and mobilised. Where it is lost (hypospadias, common penile artery injury), mobilisation causes ischaemia and PFUI repair failure.

> **KEY POINT**
>
> Aetiology predicts biology: iatrogenic and straddle strictures are focal scars suited to excision; LS is a progressive skin disease that recurs in any skin used to repair it.

2 · Presentation and evaluation

- **Acute retention**: if a catheter fails, do a retrograde urethrogram (RUG) and place a **suprapubic cystostomy**, not blind dilation. Pass any guidewire under flexible endoscopic vision.
- Four determinants guide treatment: **location, length, depth, density**.
- The treating surgeon does a dynamic RUG and voiding cystourethrogram (VCUG). Do not dilate at first urethroscopy. MRI is for tumours and PFUI only.
- **Hydraulic trap**: voiding pressure holds the urethra above the stricture open, so a segment that cones down is scarred.
- **Urethral rest** with a suprapubic tube for **6–8 weeks** reveals the true length.

3 · Decision making

- First agree on **palliation or cure**. The **reconstructive ladder is archaic**: go to the procedure most likely to cure.

ALGORITHM

1. If bulbar, <1.5 cm, without dense spongiofibrosis and untreated, then dilate or do one internal urethrotomy; if it recurs, then urethroplasty.
2. If bulbar 1–2 cm (up to 3–5 cm proximally), then excision and primary anastomosis (EPA).
3. If longer bulbar, then oral mucosa graft (OMG) onlay, with or without augmented anastomosis.
4. If penile, then onlay (graft or flap); if hostile bed, LS or failed hypospadias, then staged OMG.
5. If panurethral, options exhausted or elderly, then perineal urethrostomy.

4 · Dilation and internal urethrotomy (DVIU)

*See the paper answer **Urethral stricture: etiology, classification, DVIU, management and failed VIU** (Section 10, Ch 86).*

Also, from the condensed book (not in the paper answer):

- DVIU does worse when luminal narrowing exceeds **74%**; the safest dilation is a **balloon** over an endoscopically placed guidewire.

Favourable	Unfavourable
Bulbar	Penile, fossa navicularis
<1.5 cm	Longer
No dense deep spongiofibrosis	Dense (e.g. straddle)
First procedure	Repeat

- Self-catheterisation tapers over **3–6 months**; the stricture recurs when it stops. A 6-week catheter is no better than 3–7 days.

KEY POINT

No endoscopic adjunct removes spongiofibrosis, so endoscopy cures only the short superficial bulbar stricture; each repetition spends the patient's reconstructive capital.

5 · Excision and primary anastomosis (EPA)

- EPA is the **gold standard** for short bulbar strictures; recurrence is rare. Rules: **excise all fibrosis, spatulate widely, keep it tension-free**.
- Length comes from spongiosal mobilisation, freeing Buck's fascia and the intracrural space, and detaching bulbospongiosus from the perineal body. **The more proximal the stricture, the longer the excisable segment.**

	Transecting	Vessel-sparing transecting	Non-transecting
Bulbar arteries	Divided	Preserved	Preserved
Steps	Excise, spatulate, anastomose	Divide at membranous–bulbar junction, dorsal spatulation	Dorsal stricturotomy, excise scar from within, close transversely
Use	Most	Proximal bulbar	Proximal

Vessel-sparing protects men with poor retrograde supply (hypospadias, trauma, prior surgery); its other benefits are unproven.

6 · Substitution urethroplasty

- **Buccal mucosa graft (BMG)** is the workhorse, one-stage or staged. Skin grafts are not used in LS. Grafts do best in the **bulb**, which has a thick vascular bed.
- Graft position may be ventral (with spongioplasty), lateral, or **dorsal (Barbagli)**, spread-fixed on the corpora.
- **Augmented anastomosis** often beats pure onlay. **Kulkarni** is a one-stage panurethral dorsal onlay via the perineum with **penile invagination**.
- **Staged**: stage 1 lays the urethra open and quilts the graft to dartos. Stage 2 tubularises it to about 28 Fr after at least 6 months. It is used for LS and failed hypospadias.

> **KEY POINT**
>
> A graft lives on its bed: it succeeds where the bed is thick and vascular (bulb, dorsal corpora), and is staged when the bed, or skin disease, cannot be trusted in one step.

- **Flaps** use hairless genital skin on dartos, such as the **Orandi** ventral longitudinal flap or a **circular island** flap up to **15 cm**; fossa navicularis strictures use a Jordan or De Sy island.
- Grafts equal flaps; flaps suit irradiated or multiply operated urethras. **Onlay beats tube**.
- **LS**: skin repairs fail, so use **oral mucosa only**, often staged.
- **Perineal urethrostomy** opens the urethra dorsally to healthy mucosa, sutures it to skin, and can add a BMG to complete the dorsal wall.

7 · Meatal stenosis

- **Child**: post-circumcision meatitis needs ventral meatotomy and parental dilation; early meatitis responds to 0.05% clobetasol.
- **Adult**: the fossa is usually involved, so evaluate it as a stricture before ventral meatotomy or dorsal Y-V plasty.

8 · Pelvic fracture urethral injury (PFUI)

*See the paper answer **Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management** (Section 11, Ch 125).*

Also, from the condensed book (not in the paper answer):

- PFUI occurs in about **10%** of pelvic fractures. Repair at **3–6 months**; endoscopic core-through is not curative.
- Failure is mostly **ischaemic stenosis** of the mobilised bulb; internal pudendal status decides the order of surgery (table below).

ALGORITHM

1. On-table endoscopy; exaggerated lithotomy; perineal incision.
2. Mobilise the bulb; pass a sound via the suprapubic tract; excise fibrosis.
3. Spatulate to **≥30 Fr**; place 8–10 tension-free 4-0 PDS sutures over a silicone stent.
4. If tension remains, then add steps in order: mobilisation, intracrural separation, infrapubectomy, corporal rerouting.

- Success is in the high **90s %**.

Internal pudendal status	Potency	Action
Intact ≥1 side	Often potent	Reconstruct
Reconstituted only	Never potent	Reconstruct; revascularise for ED later
Bilateral occlusion	—	**Revascularise first** (ischaemic stenosis risk)

Screen at-risk men with **duplex ultrasound, then angiography if abnormal**.

9 · Vesicourethral stenosis and fistula

- Stenosis after prostate cancer treatment is incised deep at 3 and 9 o'clock, adding mitomycin C if recalcitrant. Obliteration needs robotic or perineal repair, often followed by an artificial urinary sphincter.
- **Rectourethral fistula**: a simple one (<2 cm, non-irradiated) is repaired perineally or by York-Mason. A complex (irradiated) one needs faecal diversion and **gracilis interposition, with or without BMG**.

10 · Complications and follow-up

- Recurrence is early after EPA and within 12 months after grafts. ED is uncommon, higher with long repairs and straddle injuries, and often improves.
- Follow up with symptoms, uroflow and post-void residual, and urethroscopy if suspicious.

11 · Landmark trials

Trial	Design	Result	Take-home
ROBUST III	Phase III RCT, Optilume (paclitaxel balloon) vs endoscopic control after multiple failed endoscopic treatments	1-year stricture-free 74.6% vs 26.8%	Drug-coated balloon beats repeat endoscopy; role as first-line undefined

87 Surgery of the Scrotum and Seminal Vesicles

1 · Surgical anatomy of the scrotum

- Lax, rugated skin gives **poor tamponade**, so haematomas keep enlarging and haemostasis must be meticulous.
- Dartos keeps necrotising fasciitis off the testes. **Transdermal absorption** is about **40×** that of other skin, and the skin is rich in 5α-reductase.

Territory	Artery	Parent
Scrotum, anterior	Superficial and deep external pudendal	Femoral
Scrotum, posterior	Posterior scrotal	Perineal (internal pudendal)
Testis, primary	Testicular	Aorta
Testis, secondary	Deferential	Superior vesical, from internal iliac
Testis, secondary	Cremasteric	Inferior epigastric, from external iliac

KEY POINT

Three trunks (aorta, internal iliac, external iliac) anastomose in the cord and at the cauda. So the testicular artery can be ligated in Fowler–Stephens orchidopexy, and the superior epididymal artery sacrificed in vasectomy reversal.

2 · Peri-operative principles and access

- Shave with a **razor on the day of surgery**. The American Urological Association (AUA) advises a single IV cephalosporin or clindamycin **only with risk factors**.
- Benign disease uses a scrotal incision. **Possible malignancy** needs an inguinal incision, with the cord clamped before the testis is handled.
- **Always close the dartos**, because most haematomas come from inadequate dartos closure.

3 · Scrotal wall: excision, scrotectomy, reconstruction

- **Fournier gangrene** is the commonest indication for partial scrotectomy. Give fluids and broad-spectrum antibiotics, and **debride urgently**, because it spreads **2–3 cm/h**; mortality is about **20%**. Re-debride every **24–48 h**.
- Scrotal violation in non-seminomatous germ cell tumour (NSGCT) calls for partial scrotectomy of the scar.
- If more than **50–60%** of skin is lost, do not reconstruct at once; protect the testes in thigh pockets or under a vacuum dressing (VAC).

ALGORITHM

1. If <50% and tension-free, then close primarily.
2. If <50% with tension, then local scrotal advancement or secondary intention.
3. If >50%, then VAC, then stretched scrotal advancement or delayed primary closure.
4. If no local tissue, then fasciocutaneous thigh flap; meshed split-thickness skin graft (STSG) last.

> **KEY POINT**
>
> The scrotum works because it is lax. Graft contracture defeats that, whereas residual scrotal skin stretches enormously and is the best donor.

4 · Hydrocele

- **Congenital** hydrocele comes from a patent processus; about **80%** resolve by 18 months. **Acquired** hydrocele reflects a production–absorption imbalance.
- **Colour Doppler ultrasound first** excludes tumour and decides the scrotal or inguinal route.

Technique	Steps	Best for
Lord plication	Small opening; radial plicating sutures; no dissection	Thin-walled, smaller; least haematoma, no drain
Jaboulay (Winkelman)	Excise part of parietal layer, evert remainder behind testis	Thin, floppy sacs
Bottleneck	Trim to 2-cm cuff, tack edges	Excisional
Excision	Excise sac, oversew edges	Thick, large, multiloculated; lowest recurrence

> **KEY POINT**
>
> The more tunica excised, the lower the recurrence but the more haematoma. Recurrence after any successful repair is **0–6%**.

- **Aspiration and sclerotherapy** suit the unfit; aspiration alone recurs, so add a sclerosant such as doxycycline. With **retreatment**, resolution is **85–96%**, but results are poor if multiloculated or more than **750 mL**.

5 · Spermatocele and epididymal surgery

- Operate on a spermatocele only for symptoms, and **counsel about obstructive azoospermia**.
- A suspected malignant epididymal tumour is approached inguinally; a proven benign one by scrotal microsurgical excision.
- **Epididymectomy** gives **poor results for chronic epididymitis** (>6 weeks), so it is a last resort; it works better with a **palpable lesion**.

6 · Varicocele repair

SAQ Varicocele: pathophysiology, controversies and surgical management

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 SAQ "Varicocele: controversies and present status of treatment" (asked as "Controversies in management of varicocele" and "Present status of varicocele treatment") · P1 SAQ "Varicocele – Pathophysiology" · P2 SAQ "Surgical management of bilateral varicocele"

Definition and epidemiology

- Abnormal dilatation and tortuosity of the **pampiniform plexus** of the internal spermatic veins.
- About **15% of men, 35% with primary and 75–81% with secondary infertility** (P3: up to 80%). Appears after age 10, peaks at Tanner 3. About 90% left-sided.
- With sensitive tests most are bilateral; the clinical significance of subclinical ones is uncertain.
- **Venous anatomy:** anterior (internal spermatic), middle (deferential) and posterior (cremasteric) groups; the last two give return after repair and cause recurrence.

Aetiology: why it develops

1. **Left-sided anatomy:** the left vein enters the left renal vein at a right angle, 8–10 cm higher than the right enters the IVC (inferior vena cava). The taller column overloads its valves. The right vein enters obliquely, which acts as a valve.
2. **Valve incompetence** or absence; continuous reflux predicts progression.
3. **Nutcracker:** left renal vein compressed between aorta and SMA (superior mesenteric artery).
4. **Collateral pathways**; **genetic** (4–8 times commoner in first-degree relatives); tall, thin habitus. - **Secondary:** an isolated **right-sided or non-decompressing** varicocele (one that does not decompress lying down) suggests a retroperitoneal or renal mass (tumour), so image the abdomen.

Pathophysiology: how it damages the testis

Mechanism	Explanation
Heat (primary)	Lost countercurrent heat exchange; the testis normally runs 2–4 °C below core
Oxidative stress	Raised reactive oxygen species cause sperm DNA fragmentation
Hypoperfusion and stasis	Hypoxia, toxin accumulation
Apoptosis	Germ cell loss
Leydig dysfunction	Lower testosterone than controls

- The injury is **progressive and duration-dependent** (atrophy), and may affect **both** testes.

Consequences

- **Testicular hypotrophy** (asymmetry above about 20% guides adolescent repair); poor semen; hypogonadism (lower testosterone); ache on standing (2–11%).
- Repair improves semen, DNA fragmentation, testosterone and pregnancy.

Grading

Grade	Finding
Subclinical	Doppler only (not treated)
I	Palpable with Valsalva only
II	Palpable, not visible
III	Visible

Controversies and present status of treatment (indications)

Question / situation	Evidence	Current position
Does repair improve fertility? Infertile couple, **palpable** varicocele, abnormal semen	Madgar RCT (randomised controlled trial, 1995): pregnancy 44% versus 10% at 1 year; meta-analyses mixed	**Repair clinical** varicocele with **abnormal semen** in an infertile couple with normal female partner (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine])
Subclinical varicocele?	No benefit for semen or pregnancy	**Do not treat**
Normal semen?	No proven benefit	Not for fertility
Adolescent varicocele	Testis volume is a surrogate; link to adult fertility unproven	Observe; repair for **testis asymmetry (hypotrophy) above 20%**, pain, or abnormal semen (Tanner 5 or 18 years)
NOA (non-obstructive azoospermia)	Small series: sperm returns to ejaculate in some	No definitive evidence; may be considered before micro-TESE (microsurgical testicular sperm extraction)
Before IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)	Repair may improve live-birth rate or allow less invasive ART (assisted reproductive technology)	Discuss, especially with oligozoospermia
High DNA fragmentation	Repair lowers fragmentation	Reasonable indication
Hypogonadism (low testosterone)	Repair raises testosterone in hypogonadal men	Emerging indication
Pain	Dull ache on standing; relief in most after microsurgery	Repair after other causes excluded

- **Bilateral disease: do not repair subclinical** sides; repair each palpable side.

Choice of technique

Technique	Recurrence	Hydrocele	Comment / bilateral use
Microsurgical subinguinal/inguinal	Lowest, **0.5–1%**	**About 0**	**Best, preferred**: artery and lymphatics preserved; two small incisions, local or regional anaesthesia
Retroperitoneal (Palomo)	High with artery spared (15–25%)	High with mass ligation (about 7%)	Children (artery ligation acceptable)
Laparoscopic	3–15%	Up to 12%	Both sides through same ports; general anaesthesia; serious morbidity possible
Embolisation / sclerotherapy	4–25% (P2: 15–25%)	Nil	Outpatient; femoral (left) or jugular (right) access; technical failure

Microsurgical steps (each side)

1. Subinguinal incision over the external ring; deliver the cord.
2. Microscope and **micro-Doppler** to map the testicular artery; papaverine for spasm.
3. Ligate all internal and external spermatic veins (all veins except vasal); preserve **artery, lymphatics, vas and one set of vasal veins**.
4. Deliver the testis to ligate gubernacular and external spermatic veins (lower recurrence).
5. **Impulse test** confirms no missed vein.

Bilateral varicocele: specific points

- **Never ligate both testicular arteries**: in adults this has caused azoospermia and atrophy. So artery-sparing microsurgery is safest bilaterally.
- Image for a renal or retroperitoneal mass if a varicocele is isolated right-sided or non-decompressing.
- Repair any coexisting hydrocele.

Complications and outcome

- **Hydrocele** (lymphatic ligation; commonest after non-microsurgical repair); testicular artery injury (atrophy); recurrence (diagnose by upright colour Doppler showing reflux).
- **Predictors of poor result:** recurrence, artery ligation, hydrocele. Earlier repair recovers more.

Diagrams to draw

- Left and right testicular veins draining into renal vein and IVC.
- Spermatic cord cross-section at microsurgery: artery, lymphatics and vas preserved, veins ligated.

> **Key points to remember:**
> - Left vein: right-angled entry, 8–10 cm higher column; 90% left.
> - Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage; injury is progressive.
> - Right or non-decompressing varicocele: image for a renal mass.
> - Repair only palpable varicocele (each palpable side) with abnormal semen; never subclinical.
> - Madgar 1995: 44% versus 10% pregnancy.
> - Adolescents: observe; repair for asymmetry above 20%.
> - Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).
> - Never ligate both testicular arteries.

Also, from the condensed book (not in the paper answer):

- †Embolisation suits recurrence after surgery. For orchialgia, **microsurgical repair** gives more complete relief.

7 · Vasectomy

- **Counsel**: the man is not sterile at once, so contraception continues until post-vasectomy semen analysis (PVSA). Repeat vasectomy is needed in <**1%**; pregnancy occurs in about **1**

in 2000 even after an azoospermic PVSA; chronic pain affects about **1–2%**.

- Give **no routine antibiotics**. **Isolation** determines morbidity; **occlusion** determines efficacy.

Feature	Conventional (CV)	Minimally invasive (MIV)	No-scalpel (NSV)
Skin opening	1.5–3 cm	≤**10 mm**	Puncture <10 mm
Status	Difficult cases	**Recommended**: less pain, bleeding, infection	Original MIV (Li Shunqiang, China, 1974)

- Open on the **straight vas**, because the convoluted vas is harder now and at reversal.

AUA-recommended occlusion (failure ≤1%)	Note
Mucosal cautery (MC) and fascial interposition (FI)	FI = internal spermatic fascia over one end
MC alone (no clips/ligatures)	Cautery of lumen 0.5–1.5 cm
Open-ended: abdominal end MC and FI, testicular end open	Less congestive epididymal pain; granuloma linked to better reversal

KEY POINT

Every recommended method rests on **mucosal cautery**; ligatures and clips alone are not core methods. Division is required; segment excision is optional (**1 cm** is adequate).

ALGORITHM

1. Do PVSA at **8–16 weeks**.
2. If azoospermia or rare non-motile sperm (RNMS, ≤**100,000/mL**), then stop contraception.
3. If motile sperm, then repeat at 6 months; if still motile, then it is a failure, so repeat the vasectomy.

- **Early failure** is technical, such as a missed vas; **late failure** is recanalisation after a successful PVSA.
- **Sperm granuloma** is expected rather than a complication. Post-vasectomy pain affects about **2%**; if a cord block helps, then denervation or reversal follows.

8 · Orchidectomy and prosthesis

- **Bilateral simple orchidectomy** treats hormone-sensitive metastatic prostate cancer. It is cheaper than androgen deprivation therapy but has no proven survival benefit.
- **Unilateral** orchidectomy is for trauma or non-salvageable torsion, and for orchialgia only after all conservative measures. **Epididymis-sparing** orchidectomy gives a better contour.
- **For orchialgia, use the inguinal route**, because it removes more of the proximal ilioinguinal, iliohypogastric and genitofemoral nerves.
- **Prosthesis**: offer it to every man lacking a testis, before cancer orchidectomy. Pierce the fill port ≤**5 times**, never after insertion. A **high scrotal subdartos** pouch has the least infection and erosion.

9 · Testicular torsion and fixation

- †**Acute torsion**: if suspicion is high, explore without waiting for imaging. Detorse and assess, then fix or remove; **always fix the contralateral testis**.
- Intermittent torsion (pain <2 h) and postpubertal retractile testis need elective orchidopexy.

Fixation	Method	Note
Suture fixation	Albuginea to dartos (3-0 absorbable)	Re-torsion still possible; transparenchymal sutures damage spermatogenesis
Dartos pouch	Subdartos pouch with a purse-string at the cremaster exit	No transparenchymal suture, so **preferred when fertility matters**

10 · Chronic scrotal pain (>3 months)

ALGORITHM

1. History, examination, duplex ultrasound; if a cause is found, then treat it.
2. If none, then exclude extrascrotal causes (abdomen, pelvis, spine, hip, prostatitis, pelvic floor).
3. If non-specific, then give support, NSAIDs, tricyclics or gabapentin for ≥3 months.
4. If epididymal, then epididymectomy.
5. If testicular or post-vasectomy, then do a cord block: if relief >**50%**, then microsurgical denervation (or reversal); if <50%, then pain clinic, with inguinal orchidectomy last.
6. If post-hernia, then ilioinguinal block; if relief >**90%**, then neurolysis.

- **Microsurgical denervation of the spermatic cord (MDSC)** is done through a subinguinal incision. A micro-Doppler helps **preserve testicular and cremasteric arteries, lymphatics, and the vas with its vessels**; everything else, including the trifecta nerve complex, is divided.

11 · Complications of scrotal surgery

- **Haematoma** is commonest; drain it if large, infected or enlarging. A falling haematocrit after orchidectomy suggests a slipped cord ligature with a **retroperitoneal bleed**, which needs re-ligation through a retroperitoneal extension.

12 · Seminal vesicles

- The seminal vesicles (SV) are Wolffian derivatives that bud at about **12 weeks** and make about **60%** of seminal fluid. The **neurovascular bundles lie lateral to their tips**.
- **MRI is most accurate**. On transrectal ultrasound (TRUS), suspect obstruction if the SV is >**1.5 cm** or the ejaculatory duct >**2.3 mm**; confirm by **aspiration showing sperm**.
- **Zinner syndrome** is an SV cyst with **ipsilateral renal agenesis** and ejaculatory duct obstruction (EDO), often with an ectopic ureter into the cyst.
- Observe or aspirate small cysts. **Transurethral unroofing** is preferred for cysts, EDO and stones; large symptomatic cysts need laparoscopic or robotic excision.

Benign	Malignant
Cystadenoma, papillary adenoma, hydatid cyst, amyloid	Adenocarcinoma, mixed epithelial–stromal, phyllodes, sarcoma, carcinoid, seminoma, yolk sac

- Image tumours with MRI and biopsy by TRUS if inconclusive. Secondary invasion is far commoner. Adenocarcinoma needs radical excision and lymphadenectomy.

13 · Surgical approaches to the seminal vesicles

Approach	Use	Main risk
Transvesical transtrigonal	Large benign cyst; ectopic ureter into cyst	Urinary leak, ureter
Perivesical	Children, large cyst with or without nephroureterectomy	Ureter, vas
Transperitoneal (laparoscopic/robotic)	Bilateral lesions; dominant, because it opens neither organ	Access injuries
Perineal	At perineal prostatectomy	Neurovascular bundle
Transcoccygeal	Hostile abdomen/perineum	Rectum

- A rectal injury needs two-layer repair; a ureteric injury needs reimplantation over a stent.

★ Putting it together

A · Testicular mass: the first week

ALGORITHM

1. Ultrasound.
2. Tumour markers AFP, hCG and LDH **before** orchidectomy; sperm bank.
3. **Radical inguinal orchidectomy** (never trans-scrotal).
4. Repeat markers (AFP half-life 5–7 days, hCG 1–1.5 days); CT chest, abdomen and pelvis.

B · Germ cell tumour by stage

Stage	Seminoma	Non-seminoma (NSGCT)
CS I	Surveillance (occult spread 10–15%); single-dose carboplatin	Low risk: surveillance. High risk (lymphovascular invasion, embryonal-predominant): surveillance, BEP ×1 or nerve-sparing RPLND
II	≤3 cm: radiotherapy, chemo or RPLND; >3 cm: chemo	Normal markers and small nodes: RPLND; otherwise chemo
Advanced (IGCCCG)	Full-dose cisplatin. Good: BEP ×3 or EP ×4. Intermediate/poor: BEP ×4 (VIP ×4 if lung concern).	
Residual mass	>3 cm: FDG-PET (≥6 weeks post-chemotherapy)	Any >1 cm: post-chemotherapy RPLND (teratoma)

C · Penile cancer: primary and groin

ALGORITHM

1. **Primary:** if PeIN, then topical 5-FU/imiquimod, laser or glans resurfacing. If invasive, then organ-sparing surgery, glansectomy, or partial/total penectomy; margins <1 cm are acceptable.
2. **cN0:** if low risk (pTis/pTa/pT1a), then surveillance; if intermediate/high, then DSNB or modified ILND; if positive, then radical.
3. **cN+:** if FNA positive, then bilateral radical ILND; if cN3, then neoadjuvant TIP first. Add pelvic dissection if ≥2 inguinal nodes or extranodal extension.

D · Anterior urethral stricture

Stricture	Treatment
Bulbar <1.5 cm, first time	Dilation or DVIU once; if it recurs, urethroplasty
Bulbar 1–2 cm (3–5 cm proximally)	EPA
Longer bulbar	BMG onlay or augmented anastomosis
Penile	Onlay graft/flap; staged if hostile bed, LS, failed hypospadias
Panurethral, frail	Perineal urethrostomy
Pelvic fracture urethral injury	Suprapubic catheter, then delayed perineal anastomosis

In retention, place a suprapubic catheter; never blind-dilate.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ Surgical management of primary scrotal cancer

Paper answer · P1 · P4 · merged from 2

Asked as: P1 SAQ "Surgical management of primary scrotal cancer" · P4 SAQ "Surgical management of primary scrotal cancer."

Introduction and aetiology

- Rare. **SCC (squamous cell carcinoma)** is commonest. **Percivall Pott (1775)** linked it to chimney soot in chimney sweeps: the first occupational cancer.
- **Risks:** soot, tar, mineral oils, arsenic, PUVA (psoralen + ultraviolet A), HPV (human papillomavirus), chronic inflammation.
- **Other types:** EMPD (extramammary Paget disease), BCC (basal cell carcinoma), melanoma, sarcoma.
- Lymph drains to **superficial inguinal nodes, crossing the midline**: both groins at risk.

Evaluation

- Painless nodule or ulcer on the anterolateral scrotum, often presenting late; **biopsy** confirms.
- Examine both groins (lymphatics cross the midline); FNAC (fine-needle aspiration cytology) of palpable nodes; CT (computed tomography) of abdomen and pelvis for staging.

Staging (Lowe modification of Ray and Whitmore)

Stage	Extent
A1	Confined to scrotum
A2	Invades testis, cord, penis, pubis or perineum
B	Inguinal nodes only
C	Pelvic nodes, no distant spread
D	Distant metastases

Management of the primary

1. **Wide local excision with a 2 cm margin**, full thickness including dartos; frozen sections for margins.
2. Testes and cord preserved unless invaded; hemi- or total scrotectomy for large tumours.
3. Stage A2: en bloc removal of involved testis, penis, perineum or pubis.
4. Positive margin: re-excise.

Reconstruction (Chapter 87)

- Scrotal laxity usually allows primary closure.

Defect	Management
<50%, no tension	Primary closure
<50%, tension	Scrotal advancement flap or secondary intention

Defect	Management
>50%	Testes in thigh pockets (pouches); local or thigh flaps; later STSG (split-thickness skin graft) or flap

Management of nodes (extrapolated from penile cancer)

- **Palpable, cytology positive:** ILND (inguinal lymph node dissection) ± pelvic dissection. Nodal status is the main prognostic factor.
- **Impalpable:** surveillance; dynamic sentinel node biopsy or modified ILND for high-grade/invasive tumours.
- **Bulky or fixed:** neoadjuvant cisplatin-based chemotherapy, then surgery; RT (radiotherapy) for palliation.
- Routine prophylactic dissection is not advised.

Other histologies

- **EMPD:** wide excision (up to 3 cm) guided by **mapping biopsies**; screen for associated GU (genitourinary) and other cancers.
- **Also know:** BCC, excision or Mohs surgery; melanoma, excision by Breslow depth ± sentinel node; sarcoma, wide excision.

Diagram to draw: elliptical wide excision with 2 cm margin and bilateral inguinal node drainage.

> **Key points to remember:**
> - Pott 1775: chimney sweeps, first occupational cancer.
> - Wide local excision, 2 cm margin; testes spared unless invaded.
> - Nodes: managed as in penile cancer; nodes decide prognosis.

SAQ Perineal urethrostomy and eponymous urethroplasty techniques

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 SAQ "Eponymous urethroplasty techniques and perineal urethrostomy" · P1 SAQ "Perineal urethrostomy." · P2 SAQ "Perineal urethrostomy."

Introduction: urethroplasty options

- Options: **EPA (excision and primary anastomosis)**, the gold standard for short bulbar strictures; **graft or flap onlay**; **staged repair**.
- **Principles:** grafts need a vascular bed; onlay beats tube; graft equals flap.

Eponymous urethroplasty techniques

Eponym	Technique	Use
Johanson	Two-stage: stricture laid open, edges sutured to skin (urethrostomy); tubularised later	LS (lichen sclerosus), failed hypospadias, infection
Blandy	Inverted-U scrotal or perineal skin flap into a ventral urethrotomy	Bulbar strictures; classic perineal urethrostomy (PU)
Turner-Warwick	Two-stage scrotal inlay; transpubic approach, omental wrap	Complex, posterior
Webster	Progressive perineal approach: mobilisation, crural separation, inferior pubectomy, rerouting	PFUI (pelvic fracture urethral injury)

Eponym	Technique	Use
Barbagli	**Dorsal onlay** BMG (buccal mucosal graft) spread-fixed to corpora	Bulbar
Asopa	**Dorsal inlay** graft through a ventral urethrotomy, no mobilisation	Penile and bulbar
Kulkarni	One-sided dorsolateral onlay; panurethral repair via perineum with penile invagination	Panurethral (LS)
Orandi	Ventral longitudinal penile skin island flap on dartos	Pendulous
Quartey	Showed skin island is a passenger on a fascial (dartos) flap	Basis of all genital skin flaps
McAninch	Circular fasciocutaneous penile skin flap	Long anterior strictures (up to 15 cm)
Jordan	Vessel-sparing EPA; ventral transverse island for fossa	Proximal bulbar; fossa navicularis

- **Also know:** Monseur dorsal urethrotomy; Andrich–Mundy non-transecting anastomosis; Bracka two-stage BMG.
- **Diagram to draw:** bulbar cross-section showing dorsal onlay (Barbagli), dorsal inlay (Asopa) and ventral onlay.

Perineal urethrostomy: definition

- PU (perineal urethrostomy): the proximal bulbar urethra is opened onto perineal skin as a permanent or staged new meatus; the patient voids sitting. It **bypasses**, rather than reconstructs, a diseased anterior urethra.

Indications

- **Complex anterior strictures:** LS panurethral disease, multiple failed urethroplasties, failed hypospadias repair.
- **Elderly or high-risk men** unwilling or unfit for staged reconstruction.
- **First stage** of a staged (Johanson-type) urethroplasty.
- After **penectomy** or distal urethrectomy for penile or urethral cancer.
- Temporary: stricture after phalloplasty; Fournier gangrene; resectoscope access in morbid obesity.
- AUA (American Urological Association) guideline (2016): PU is an acceptable long-term alternative to urethroplasty.
- LS spares the proximal urethra, so PU suits LS and is easy in LS.

Contraindications and limits

- Disease reaching the proximal bulbar or membranous urethra; prior pelvic radiation and LS raise restricture.

Techniques

1. Classical (Blandy) PU: inverted-U perineal skin flap (a random-pattern advancement flap) advanced into a ventral bulbar urethrotomy (the opened bulbar urethra).

2. BMG PU, steps:

1. High lithotomy; inverted-U incision; split (divide) bulbospongiosus in the midline.
2. Free the spongiosum and **transect** it; **dorsal urethrotomy** into healthy mucosa.

3. Bring the healthy ventral urethra to perineal skin; calibrate proximally to **30 Fr**; close the distal urethra.
4. Quilt a BMG on the corpora to complete the dorsal wall up to the skin. - The graft prevents the circumferential scar that narrows a deep PU.

Diagram to draw: sagittal perineum with the inverted-U flap sutured into the opened bulbar urethra.

Complications and outcomes

- **Stenosis of the urethrostomy** (main failure): revision or periodic dilation.
- Spraying, dribbling, perineal wetness, wound infection, body-image concerns.
- Compared with staged reconstruction it is a shorter, single operation with less morbidity.
- High mid-term success and satisfaction in selected patients.

> **Key points to remember:**
> - Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
> - Orandi and McAninch flaps; Quartey flap anatomy.
> - Johanson staged; Blandy inverted-U PU.
> - PU bypasses, rather than reconstructs, the diseased anterior urethra; patient voids sitting.
> - Ideal for LS, failed repairs and elderly patients.
> - BMG PU prevents circumferential scarring; calibrate to 30 Fr.

SAQ Testicular microlithiasis

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3a

Definition

- **TM (testicular microlithiasis):** multiple small (<3 mm) bright echogenic foci **without acoustic shadowing** scattered in the testicular parenchyma on ultrasound.
- **Classic TM:** ≥5 microliths per testis (or per image); **limited TM:** <5, of no clinical significance.

Pathology

- Laminated calcified bodies **inside seminiferous tubules**, formed on degenerated tubular cells that Sertoli cells fail to clear. Affected tubules are often obliterated.

Prevalence

Population	Prevalence
General population	5–10% (population-based 4.2%, rising with age)
Multicentre review, >37,000 males	2.9%
Boys with cryptorchidism	3.5%
Previously cryptorchid testis vs contralateral	14% vs 19% (no difference)

Associations

- Cryptorchidism, subfertility, **GCNIS (germ cell neoplasia in situ)** and TGCT (testicular germ cell tumour), Klinefelter syndrome, Down syndrome.

Cancer significance

- Tumours were found in 4.64% with TM vs 0.33% without, but cryptorchidism was not accounted for.
- **TM is not clearly an independent risk factor** for TGCT. An EAU (European Association of Urology) paediatric panel review of several hundred children found only one cancer through puberty.
- **Exception:** contralateral TM in a man with previous GCT (germ cell tumour) signals a higher GCNIS risk.

Management

- **Isolated TM, no risk factors:** reassure; **no further evaluation, no routine ultrasound follow-up**.
- **TM with risk factors** (previous GCT, cryptorchidism, infertility, atrophic testis): teach **monthly self-examination from puberty**; ultrasound only for a palpable change.
- **Biopsy:** not for TM alone. Contralateral biopsy at orchidectomy is considered when GCNIS risk is high: atrophic testis (<12 mL), cryptorchidism, age <40.
- Counsel carefully: incidental TM causes anxiety and unnecessary tests.

Differential diagnosis on ultrasound

- Coarse calcification, "burned-out" tumour scar, stellate scar (these need orchidectomy if intratesticular), scrotal pearls, rete testis ectasia.

Also know: routine screening ultrasound for infertility is discouraged because of incidental TM; follow-up schedules remain undefined.

Diagram to draw: ultrasound sketch of a testis with scattered non-shadowing echogenic dots.

> **Key points to remember:**
> - Classic TM ≥5 microliths; limited <5.
> - Not an independent cancer risk factor in isolation.
> - Self-examination only with risk factors; no routine imaging.

SAQ Muscle assisted urethroplasty

Paper answer · P4

Asked as:

- Muscle assisted urethroplasty — *Winter 2023 Paper IV Q3b (SAQ)*
- Muscle assisted urethroplasty — *Practice Set 5 Paper IV Q3a (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3b)

Read as muscle-flap-assisted graft urethroplasty: a vascularised muscle flap used as bed or cover for a graft. "Muscle-sparing" bulbar urethroplasty is noted at the end.

Rationale

- A free graft (e.g. **BMG [buccal mucosal graft]**) has no blood supply. It survives by **imbibition** (about 48 h, nutrients from the bed), then **inosculation** (new microcirculation). So **"a graft lives on its bed"**.
- **Poor beds:** radiation, multiple failed repairs, fistula, pelvic-fracture scar, infection. Here a **muscle flap brings blood supply**, fills dead space and separates suture lines.

Muscle flaps used

Flap	Supply	Use
Gracilis (muscle-only, axial, multiple pedicles)	Medial circumflex femoral branch, proximal thigh	Irradiated posterior urethra, rectourethral fistula, complex perineal repair
Bulbospongiosus	Perineal branches	Covers a ventral graft; support in bulbar repair
Rectus abdominis (VRAM, vertical rectus abdominis myocutaneous)	Inferior epigastric	Large pelvic or perineal defects

Technique (gracilis-assisted BMG)

1. Lithotomy; perineal exposure; excise scar or fistula.
2. Close rectal side; urethra opened.
3. Gracilis harvested through a medial thigh incision; pedicle preserved; tunnelled to perineum.
4. Muscle interposed between rectum and urethra, or used as the graft bed.
5. **BMG quilted onto the muscle** as onlay; urethra closed over a catheter.
6. Catheter drainage; urethrogram before removal.

Results (Chapter 86)

- Gracilis with or without BMG for rectourethral fistula: **non-irradiated, all closed**, no later strictures. **Irradiated:** most closed in one stage, but more strictures and permanent bowel diversion.
- Muscle interposition is integral; BMG is invaluable for urethral closure.

Complications

- Thigh wound problems; graft contraction, restricture, fistula recurrence.

"Muscle-sparing" bulbar urethroplasty (Barbagli)

- The **bulbospongiosus muscle, central perineal tendon and perineal nerves** are left intact; dorsal BMG onlay through a lateral approach.
- Aim: less post-void dribbling and ejaculatory dysfunction.

Diagram to draw: gracilis flap tunnelled to the perineum, lying between rectum and urethra with BMG on it.

> **Key points to remember:**
>
> - Graft survival: imbibition then inosculation.
> - Muscle flap rescues a poor (irradiated) bed.
> - Gracilis is axial, robust, easy to tunnel.
> - Muscle-sparing: preserves bulbospongiosus and nerves.

SAQ Gracilis myocutaneous flap in Uro.

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 4, Q3d (gracilis-assisted urethroplasty also in 2023-Winter Paper 4, Q3b)

Definition

- A pedicled **axial** flap of gracilis muscle, with or without skin, rotated from the medial thigh to bring **vascularised bulk** to the perineum, groin or vagina.

Anatomy

Feature	Detail
Origin–insertion	Inferior pubic ramus and body of pubis → medial proximal tibia (pes anserinus)
Blood supply	**Mathes–Nahai type II**: dominant pedicle from the **medial circumflex femoral artery** (profunda femoris), entering about 8–10 cm below the pubic tubercle; minor distal pedicles from the superficial femoral artery
Nerve	Anterior branch of the **obturator nerve**
Skin paddle	Reliable over the proximal two-thirds only
Function	Expendable (other adductors compensate)

Why it is used

- Brings blood supply into **irradiated or scarred** fields, fills dead space and **separates suture lines**. Near the perineum, easy to tunnel.

Uses in urology

1. **Rectourethral fistula** (perineal): non-irradiated all closed; irradiated mostly closed, but more strictures and permanent colostomy.
2. **Complex or irradiated VVF (vesicovaginal fistula)** with tissue loss.
3. **Urethroplasty** in a poor bed: muscle as bed for BMG (buccal mucosal graft).
4. Extensive perineal or scrotal loss after **Fournier gangrene**.
5. **Groin cover** of femoral vessels after inguinal node dissection with skin excision.
6. Vaginal reconstruction after exenteration.

Technique

1. Lithotomy; line from pubic tubercle to medial femoral condyle; gracilis lies just behind it.
2. Medial thigh incision; skin paddle over the proximal muscle if needed.
3. Divide the distal tendon; ligate minor pedicles; **preserve the dominant pedicle**.

4. Tunnel subcutaneously to the perineum; rotate without twisting.
5. Interpose and fix; drain the donor site.

Complications

- Donor seroma, dehiscence, **distal skin-paddle necrosis**, thigh numbness; flap loss rare.
- **Also know:** alternatives: Martius, omentum, VRAM (vertical rectus abdominis myocutaneous), Singapore flap.

Diagram to draw: gracilis tunnelled on its proximal pedicle between urethra and rectum.

> **Key points to remember:**
> - Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
> - Skin paddle unreliable over the distal third.
> - Main uses: rectourethral fistula, complex VVF, Fournier, groin cover.

Rapid Revision Sheet · Section 10

Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. (LAQ)

- Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
- Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
- Psoas is the landmark; artery met first.
- Main advantage: shorter operative time, peritoneum untouched.

Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma (LAQ)

- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Boundaries: renal hila (renal vessels), ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA, so unilateral templates spare ejaculation.
- PC-RPLND (post-chemotherapy): full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum to ipsilateral superficial inguinal nodes; scrotum and penis drain to inguinal nodes; never operate trans-scrotally.
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.
- Seminoma risk factors: >4 cm and rete testis invasion (not LVI).
- Stage I seminoma: surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).

Classify testicular tumors and describe management of seminoma (LAQ)

- WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
- CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
- Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
- No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

Growing teratoma syndrome (SAQ)

- Enlarging mass plus normal or falling markers equals GTS.
- Teratoma is chemoresistant and not FDG-avid.
- Treatment is complete surgical resection, not more chemotherapy. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Testis sparing surgeries for CA testis (SAQ)

- Solitary or bilateral tumour, ≤2 cm, normal testosterone.
- Frozen section plus adjacent biopsies for GCNIS.
- GCNIS: 18–20 Gy keeps Leydig function but sterilises.

Minimally invasive and robotic RPLND (SAQ)

- Same template, split-and-roll, lumbar division and nerve sparing as open.

- Best for chemotherapy-naive low-volume disease.
- Conversion about 5%; ejaculation preserved ≥90%.
- Expanding indication: primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy.
- Robotic nerve sparing preserves antegrade ejaculation in ≥90% after primary surgery.
- Serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND (retroperitoneal lymph node dissection).

Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy (LAQ)

- Nodal status is the strongest prognostic factor; ILND can cure.
- pT1b or higher: stage groins surgically (DSNB preferred) within 3 months.
- Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
- PLND if more than 2 nodes or ENE; bulky nodes: neoadjuvant TIP (InPACT [International Penile Advanced Cancer Trial] testing the sequence).
- Femoral triangle: ligament, sartorius, adductor longus.
- Flaps deep to Camper's; stay medial to femoral artery.
- Within 3 months; at least 8 nodes per side.
- VEIL: same template as open; fewer wound complications; equal nodal yield.
- VEIL not for bulky, fixed or skin-involving nodes.

Prognostic index score in carcinoma penis (SAQ)

- Node status is the strongest prognostic factor.
- LVI, PNI, grade III define high-risk T1b.
- Nomograms (Ficarra, Kattan) unvalidated; p16 favourable. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Grafts in urology and buccal mucosa (LAQ)

- Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
- BMG: non-keratinised, wet-adapted; thin vascular lamina propria gives excellent take.
- Avoid Stensen's duct; buccal equals lingual in success.
- Staged BMG, not skin, in lichen sclerosus.
- Bulbar success about 85–90%.

Meatal stenosis and submeatal stricture in males (SAQ)

- Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
- Adult: RUG first; adult meatal stenosis is usually the tip of a fossa navicularis stricture.
- Non-LS: meatotomy or Jordan/De Sy skin island flap.
- LS: steroid (clobetasol), buccal mucosa (never genital skin), perineal urethrostomy if extensive.

Urethral stricture: etiology, classification, DVIU, management and failed VIU (LAQ)

- Idiopathic and iatrogenic are commonest; LS recurs in skin.
- Define location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
- DVIU (VIU) only once, for short bulbar strictures (<1.5–2 cm) at first attempt; do not repeat it.
- Santucci: about 8% long-term success; repeat DVIU futile.

- AUA: urethroplasty after failure; laser = cold knife; early urethroplasty, ladder abandoned.
- EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
- Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
- Optilume (ROBUST III): 74.6% vs 26.8% stricture-free at 1 year.

Varicocele: pathophysiology, controversies and surgical management (SAQ)

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage.
- Right or non-decompressing varicocele: image for a renal mass.
- Injury is progressive; subclinical varicocele is not treated.
- Repair only palpable varicocele (each palpable side) with abnormal semen; never subclinical.
- Madgar 1995: 44% versus 10% pregnancy.
- Adolescents: observe; repair for asymmetry above 20%.
- Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).
- Never ligate both testicular arteries.

Surgical management of primary scrotal cancer (SAQ)

- Pott 1775: chimney sweeps, first occupational cancer.
- Wide local excision, 2 cm margin; testes spared unless invaded.
- Nodes: managed as in penile cancer; nodes decide prognosis.

Perineal urethrostomy and eponymous urethroplasty techniques (SAQ)

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU.
- PU bypasses, rather than reconstructs, the diseased anterior urethra; patient voids sitting.
- Ideal for LS, failed repairs and elderly patients.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

Testicular microlithiasis (SAQ)

- Classic TM ≥5 microliths; limited <5.
- Not an independent cancer risk factor in isolation.
- Self-examination only with risk factors; no routine imaging.

Muscle assisted urethroplasty (SAQ)

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

Gracilis myocutaneous flap in Uro. (SAQ)

- Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
- Skin paddle unreliable over the distal third.
- Main uses: rectourethral fistula, complex VVF, Fournier, groin cover. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 11

Stones, Obstruction & Trauma

93 Management of Upper Urinary Tract Obstruction

1 · Ureteropelvic junction obstruction (UPJO): presentation

- Trauma: see Ch 94, Ch 125; stones: see Ch 95–100.
- UPJO is usually congenital. **Neonates** show prenatal hydronephrosis on ultrasonography, with azotaemia if bilateral or solitary.
- **Older children and adults** have intermittent flank pain with vomiting, haematuria after minor trauma, urinary tract infection (UTI) and, rarely, hypertension.

2 · Diagnosis

Prove impaired drainage before operating.

> **KEY POINT**
>
> **Dilation is not obstruction.** Anatomical imaging (ultrasonography, computed tomography [CT], retrograde) shows where; functional tests (renography, Whitaker) show whether it matters. Combine one of each.

Modality	Key point
Ultrasonography (first in neonates)	Large medial pelvis, rounded calyces joined by dilated infundibula (unlike multicystic kidney); ureter not seen
Non-contrast CT (acute pain)	Hydronephrosis to UPJ without stone; excludes radiolucent stone, tumour
Contrast CT	Shows **crossing vessels**; can substitute for renography
Diuretic renography	Differential function, drainage; best follow-up study
Retrograde pyelography	**At the time of repair** (avoids introducing infection)
Nephrostomy and antegrade study	When retrograde fails; tests recoverability; permits Whitaker

SAQ Radionuclide studies and diuretic renography in PUJ obstruction

Paper answer · P1

Asked as:

- Radionuclide studies for PUJ obstruction. — *Summer 2021 Paper I Q3e (SAQ)*
- Radionuclear studies in PUJ obstruction — *Practice Set 4 Paper I Q3e (SAQ)*
- Diureteric renography — *Practice Set 2 Paper I Q3f (SAQ)*

Type: SAQ (10 marks)

(Practice Set 4: also answered in 2021-Summer Paper 1, Q3e. Practice Set 2: PUJ obstruction renography also in 2021-Summer Paper 1, Q3e; tracers: see Practice Set 2 Paper 1, Q3e)

"Diureteric" is read as diuretic renography.

Introduction / principle

- **Dilatation is not obstruction.** PUJ (pelviureteric junction) obstruction is impaired drainage that damages the kidney. **Diuretic renography** is the functional standard (functional gold standard for PUJ obstruction) and best follow-up test (best follow-up study).
- A capacious pelvis retains tracer passively and furosemide washes it out; an obstructed system cannot empty even under diuresis.

Tracers

Tracer	Handling	Use
^{99m}Tc-**MAG3** (technetium-99m mercaptoacetyltriglycine)	Tubular secretion	**Agent of choice** (tracer of choice): split function and drainage; good images (good curves) even with poor function
^{99m}Tc-**DTPA** (diethylenetriaminepentaacetic acid)	Glomerular filtration	GFR (glomerular filtration rate); poor in renal impairment; an alternative to MAG3
^{99m}Tc-**DMSA** (dimercaptosuccinic acid)	Cortical retention	Static split function and scars; no drainage data

"Well-tempered" diuretic renogram (standard protocol)

Element	How	Why
Hydration	Oral, or IV (intravenous) fluid in children (about 15 mL/kg)	Dehydration mimics obstruction
Bladder catheter	Urethral catheter	Removes full-bladder (back-pressure) and reflux (VUR, vesicoureteral reflux) effects
Furosemide	0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children	Maximal diuresis
Timing	**F+20** (classic; 20 min after tracer, when the pelvis is full; Campbell authors use F+15), **F0** (with tracer) or **F−15** (15 min before, so the system is already in diuresis)	Diuretic before filling (early diuretic into an unfilled system) misleads; always state timing
Regions of interest	Regions of interest over cortex (for split) and pelvis (dilated pelvis for drainage); then upright images	Wrong placement gives spurious values

- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- First study after about 4–6 weeks of age (immature function).

Outputs

- **DRF (differential renal function)** from counts at 1–3 min (1–3 minute counts); normal 45–55%.
- **Drainage half-time (T½, half-clearance time)** after furosemide, from maximal filling, not from injection:

T½ after furosemide	Interpretation
<10 min (minutes)	Not obstructed
10–20 min (minutes)	Equivocal (indeterminate)
>20 min (minutes)	Obstructed

- **O'Reilly curves:** I normal; II obstructed (keeps rising); IIIa dilated, non-obstructed (falls after furosemide); IIIb equivocal.
- **Also know:** P40 (activity remaining / left at 40 min) raises sensitivity; output efficiency; NORA (normalised residual activity); a **biphasic curve** and poor upright (post-void) drainage predict progression.

Indications

- PUJ (pelviureteric junction) obstruction, antenatal hydronephrosis, megaureter, after pyeloplasty, transplant ureter.

Indications for pyeloplasty from renography

- Consider surgery for:
- **DRF <40%**, **fall in DRF** on serial comparable scans (**>10%** on serial scans; ≥5% in children; 10% in older series), **T½ >20 min** (minutes) or upright retention; also (poor drainage with) symptoms, infection, increasing dilatation.
- Adult DRF <15%: likely non-salvageable; nephrostomy and reassess, or nephrectomy.

Pitfalls

- Dehydration, early diuretic (mistimed diuretic), full bladder, wrong ROI (region of interest), very poor function, very capacious pelvis.
- One Campbell chapter calls T½ >40 min obstructed; children have no agreed (paediatric T½) cut-off (false positives up to 44%; neonatal false-positive rates reach 44%); one Campbell chapter uses **T½ >40 min** in adults, so **serial trends and curve shape** matter / decide.

Follow-up and alternatives

- After pyeloplasty: renogram 1 month after stent removal; spontaneous drainage means success.
- Equivocal (cases) → **Whitaker test** (>15–20 cm H_2O obstructed) or MRU (magnetic resonance urography).

Diagram to draw: normal, obstructed and dilated non-obstructed renogram curves (O'Reilly curves I, II, IIIa and IIIb) with furosemide arrow.

> **Key points to remember:**
>
> - MAG3 is the tracer of choice; hydrate and catheterise.
> - Well-tempered: hydrate, catheter, standard diuretic dose and timing.
> - Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
> - DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
> - Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

Also, from the condensed book (not in the paper answer):

- **Renography**: hydrate, then give furosemide **0.3–0.5 mg/kg** at **20 min**; mercaptoacetyltriglycine (MAG3) is preferred.
- **Obstruction is a half-time (T½) >40 min**. After successful pyeloplasty, drainage occurs even before furosemide.

ALGORITHM

1. **Whitaker test**
2. Place a nephrostomy and a bladder catheter; perfuse at **10 mL/min**.
3. If pressure stays ≤12–15 cm H2O, there is no obstruction; if it exceeds **15–20 cm H2O**, there is.
4. It is invasive, so use it only when other tests are non-diagnostic.

3 · Indications and choice of intervention

- **Indications**: symptoms; impaired or falling function; stones or infection; rarely hypertension.
- **Neonates** are selectively observed; operate for progressive hydronephrosis or a **10%** fall in differential function. **Asymptomatic adults** with indeterminate results have serial renography.
- The **procedure of choice** is dismembered pyeloplasty, now robotic at most major centres; endopyelotomy fails more.
- Drain before repair only for infection, azotaemia (solitary or bilateral) or unrelenting pain.

Endopyelotomy subgroup (landmark series)	Success
Crossing vessel present vs absent	**42% vs 86%**
High- vs low-grade hydronephrosis	60% vs 81%
Both excluded	95% (≈ pyeloplasty)

- A crossing vessel is **prognostic, not a contraindication**, but it pushes primary UPJO towards pyeloplasty.
- **Secondary UPJO: cross over.** Failed endopyelotomy is treated by pyeloplasty; failed pyeloplasty by endourology first.
- **Nephrectomy** suits a thin shell with a normal contralateral kidney, stones with chronic infection, or multiple failed repairs. Adults with <**15%** differential function are non-salvageable; if uncertain, stent or drain and re-scan.

4 · Endopyelotomy

- **Principle**: a full-thickness **lateral** incision into peripelvic fat heals over a stent; lateral is **free of crossing vessels**.
- Pyeloplasty (≈**95%**) wins because it makes a wide dependent funnel; endourology wins only on recovery.
- **Contraindications**: segment >**2 cm**, active infection, uncorrected coagulopathy. Urine must be sterile first.

Percutaneous antegrade

- Best for **UPJO with pelvicalyceal stones**. Puncture the **mid-posterior or superolateral calyx**, not the lower pole, whose acute infundibular angle blocks passage.
- Stent with a 14/7-Fr endopyelotomy stent, traditionally for **6 weeks**; 2 and 4 weeks are equal.
- **Clear stones before incising**, because fragments migrate periureterally and cause fibrosis.

Retrograde ureteroscopic (standard)

SAQ Retrograde endopyelotomy

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 9 Paper 1, Q3f

Definition and principle

- Endoscopic **full-thickness incision of the UPJ (ureteropelvic junction)** from inside the ureter, through to peripelvic fat, followed by stenting so the ureter heals with a wider lumen (Davis "intubated ureterotomy" concept, 1943).
- **Retrograde ureteroscopic holmium laser endopyelotomy** is now the standard endourological route: direct vision without a percutaneous tract.

Indications and contraindications

Favourable	Unfavourable / contraindicated
Functionally significant UPJO (UPJ obstruction); short segment	**Long stenotic segment**
Secondary UPJO (failed pyeloplasty)	**Upper-tract stones** (treat percutaneously instead)
Moderate hydronephrosis, fair function	**Crossing vessel**, **gross hydronephrosis**, poor function
- Pre-operative CT (computed tomography) angiography or Doppler to look for crossing vessels.	

Technique

1. General anaesthesia; retrograde pyelogram; hydrophilic safety wire into the pelvis.
2. Access sheath; flexible ureteroscope to the UPJ (semirigid in some women). Dilate a narrow intramural ureter or pre-stent.
3. Holmium laser, 200-µm fibre, 0.8–1.2 J, 10–15 Hz.
4. **Lateral incision** (no crossing vessels laterally), deepened into fat and extended into normal ureter and pelvis.
5. Optional balloon dilation to 24 Fr; DJ (double-J) stent and Foley catheter.
6. Stent out after some weeks; **diuretic renogram 4 weeks later**; follow up for 24–36 months.

Results

- Holmium series 70–80% at 5 years (range 44–89%). Excluding crossing vessels >4 mm raises success to about 90%.
- Crossing vessel present versus absent: 42% versus 86% (antegrade series).

- Still below pyeloplasty (about 95%), but most cost-effective endourological option.

Complications

- Urine leak, stent migration, infection, bleeding (lateral cut spares vessels), stricture (rare with small scopes); ureteroarterial fistula reported.
- **Also know:** antegrade (percutaneous) endopyelotomy for UPJO with stones; cautery wire balloon (more haemorrhage); balloon dilation alone (about 42% success); failure then needs pyeloplasty.

Diagram to draw: UPJ with lateral full-thickness incision and crossing vessel lying anterior/posterior.

> **Key points to remember:**
> - Lateral full-thickness incision over a stent.
> - Crossing vessel and gross hydronephrosis predict failure.
> - Success 70–80%, below pyeloplasty (about 95%).

Also, from the condensed book (not in the paper answer):
- Control bleeding with haemostatic laser settings or 10-min balloon tamponade.

ALGORITHM
1. General anaesthesia; working and safety wires; flexible ureteroscope via a sheath.
2. With a 200-µm holmium fibre, incise **laterally** into fat and into normal ureter.
3. Leave a stent and Foley; do a renogram 4 weeks after stent removal.

- Percutaneous endopyeloplasty (endoscopic Heineke–Mikulicz) fails in secondary UPJO.

Complications and follow-up

ALGORITHM
1. **Percutaneous bleeding**
2. Bed rest and transfusion.
3. **Do not irrigate** the nephrostomy, so clot tamponades.
4. If bleeding persists, do **selective embolisation**.

- Renogram at 1 and 6 months, then yearly for **2–3 years**. Pyeloplasty 85/75% vs endopyelotomy 63/41% at 3/10 years.
- After failure, get CT angiography; a significant crossing vessel needs **pyeloplasty, not repeat incision**.

5 · Pyeloplasty

> **KEY POINT**
>
> **Principles (open = minimally invasive)**: widely patent, watertight, tension-free anastomosis; funnel-shaped, **dependent** transition.

LAQ Laparoscopic pyeloplasty.

Paper answer · P3

Type: LAQ (25 marks)
Source: 2016-Summer Paper 3, Q2

Introduction

- **PUJO (pelviureteric junction obstruction)** is impaired flow from pelvis to ureter that, if untreated, causes progressive renal damage (Koff).
- Causes: **intrinsic** adynamic segment (abnormal circular muscle), **extrinsic** crossing lower-pole vessel, high insertion; secondary (failed repair, stone).
- **Anderson–Hynes (1949) dismembered pyeloplasty** is the gold standard by any route. First laparoscopic pyeloplasty **1993 (Schuessler)**; robotic **2002**. Success **over 90–95%**, equal to open, with less morbidity.

Indications

- Symptoms (pain, Dietl's crisis), infection, stones, hypertension (rare).
- Impaired split function or **fall of 10% or more** on serial renography; rising APD (anteroposterior diameter); obstructed solitary kidney or bilateral disease.
- Children: most antenatal hydronephrosis resolves; about one-third need surgery.
- **Contraindications**: uncorrected coagulopathy, untreated UTI (urinary tract infection), cardiopulmonary disease barring pneumoperitoneum.
- **Nephrectomy instead** if split function under 15% in adults with a normal other kidney.

Preoperative evaluation

- **USG (ultrasonography)**: large pelvis, small calyces, ureter not seen.
- **MAG3 (mercaptoacetyltriglycine) diuretic renogram**: split function and an obstructive curve that does not wash out after furosemide. Usual teaching: washout half-time under 10 minutes normal, 10–20 minutes equivocal, over 20 minutes obstructed (Ch 93 quotes over 40 minutes). Split function under 40% supports repair.
- **CT (computed tomography) angiography**: anatomy, **crossing vessels**, stones.
- **Retrograde pyelogram** at surgery: confirms the level, excludes ureteric stricture.
- **Whitaker test** if equivocal: perfusion 10 mL/min; pelvic pressure over 15–20 cm H2O means obstruction.
- Sterile urine. Drain beforehand (stent or nephrostomy) only for infection, azotaemia or severe pain.

Approaches

	Transperitoneal	**Retroperitoneal**
Space	Large, familiar landmarks	Small, created by balloon
Bowel	Colon reflected (left: **transmesocolic** window)	Not entered; leak stays extraperitoneal
Use	Commonest; easier suturing	Avoids bowel; harder orientation
- One paediatric randomised trial favoured retroperitoneal (shorter stay); overall outcomes similar.		

Steps (transperitoneal Anderson–Hynes)

1. Cystoscopy, retrograde pyelogram, catheter.

2. 45–60° flank; pneumoperitoneum 12–15 mmHg in adults (under 12 mmHg in children); 3–4 ports, camera at umbilicus.
3. Reflect colon medially, or open the mesocolon over the pelvis.
4. Find the ureter; trace it to the PUJ with its periureteric tissue; minimal diathermy. Look for a **crossing vessel**.
5. Hitch stitch on the pelvis; transect the pelvis above the PUJ; **reduce** a redundant pelvis if needed.
6. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a handle (no-touch).
7. **Transpose** the PUJ in front of any crossing vessel.
8. Anastomose the spatulation apex to the most dependent pelvis with 4-0 absorbable (5-0/6-0 in infants): back wall, stent, front wall, then close the pelvis. Watertight, tension-free, funnel-shaped.
9. **Antegrade double-J** over a wire; clamp the catheter: urine from the stent side holes confirms the bladder coil.
10. Excise the narrow segment; drain near (not on) the suture line.

Crossing vessel

- It lowers endopyelotomy success (42% vs 86%), so pyeloplasty is preferred. **Never divide** a lower-pole vessel (segmental infarction).
- **Vascular hitch (Hellström)**: vessel moved up onto the pelvis, only if the PUJ is intrinsically normal and peristalsing.

Stenting and postoperative care

- Double-J for **4–6 weeks** (children 1–4 weeks: no difference). Externalised stent in infants avoids a second anaesthetic.
- Foley for 24–48 hours (bladder pressure refluxes up the stent); drain out when dry. If drain output rises after Foley removal, replace the Foley.
- **MAG3 renogram** after stent removal, then ultrasound; follow for years.

Robotic pyeloplasty

- 3D vision, wristed suturing, tremor filtration, **shorter learning curve**. Now first-line at major centres.
- Success 94.7–100%, the same as laparoscopy; advantage is ergonomic, at higher cost.

Outcomes

- Laparoscopic: operating time 2–4 hours; complications 2–16%; conversion 0–5.5%; stay about 2.7 vs 4.2 days (open); success 87–99%.
- **Up to 30% of failures appear after 2 years**, so follow-up must be long.

Complications

- Urine leak and urinoma, stent symptoms or migration, UTI, haematuria, ileus, bowel or vessel injury, port-site hernia, recurrent obstruction.

Management of failure

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Early (clot, oedema) -> nephrostomy, wait Persistent obstruction -> CT
angiography (missed vessel?) short, no vessel -> endopyelotomy (about 70%)
otherwise -> redo laparoscopic/robotic pyeloplasty (about 89% after failed
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open; buccal graft onlay) small intrarenal pelvis -> ureterocalicostomy poor function -> nephrectomy

Also know: Y-V plasty (high insertion); Culp spiral flap (long narrowing); stones removed through the pyelotomy with a flexible scope; safe in horseshoe and pelvic kidneys.

Recent advances

- Single-port robot (2018); indocyanine green fluorescence for redo cases; buccal mucosa ureteroplasty.

Diagram to draw: Anderson–Hynes steps: transection, lateral spatulation, transposition over the crossing vessel, dependent anastomosis.

> **Key points to remember:**
> - Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
> - Transpose over a crossing vessel; never divide it.
> - Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
> - Robotic equals laparoscopic in outcome, with easier suturing.

LAQ Congenital PUJ obstruction, failed pyeloplasty and salvage procedures

Paper answer · P1 · P2 · merged from 3

Asked as: P1 LAQ "Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management." · P1 SAQ "Management of failed pyeloplasty" · P2 SAQ "Salvage procedures for Complicated PUJO."

Definition

- PUJO (pelviureteric junction obstruction) is impaired flow of urine from the pelvis into the ureter. **Koff:** obstruction is "any restriction to urinary outflow that, if left untreated, will lead to progressive kidney damage". **Dilatation is not obstruction.**
- About 1 in 1500 live births; commonest cause of antenatal hydronephrosis. Male and left predominance; **bilateral in 10–40%**.

Pathogenesis

Type	Lesion	Typical patient
Intrinsic (commonest)	Short adynamic segment: interrupted circular muscle, excess collagen, fewer pacemaker (interstitial Cajal-like) cells. The peristaltic wave cannot cross, so the pelvis empties poorly	Neonate or infant, antenatal finding
Extrinsic	**Lower-pole crossing vessel** anterior to the PUJ; the ureter kinks over it	Older child or adult with intermittent pain
Secondary	Kinking by a tortuous ureter in high-grade VUR (vesicoureteral reflux); fibroepithelial polyp; high insertion	Any age

- **Crossing vessels** are persistent embryonic renal arteries. They occur at a fifth to a quarter of **normal** PUJs, so a vessel alone does not prove cause.
- **Also know:** proposed incomplete ureteric recanalisation; horseshoe kidney and duplex lower moiety associations.

Effects of obstruction (pathophysiology)

- Pelvic pressure rises; filtration and perfusion fall.
- **Immature kidney:** apoptosis and impaired nephrogenesis. **Mature kidney:** hypertrophy, then fibrosis (TGF-β [transforming growth factor-beta] signalling). Fibrosis, not dilatation, causes loss of function.
- **Also know:** concentrating defect, salt wasting; rarely renin-mediated hypertension.

Clinical features

- **Antenatal ultrasound** (commonest); infants: UTI (urinary tract infection), flank mass, failure to thrive.
- Older children and adults: **intermittent loin pain with vomiting (Dietl crisis)**, haematuria after minor trauma, stones, infection.
- **Associated anomalies (up to 50%):** contralateral PUJO, dysplasia or MCDK (multicystic dysplastic kidney), VUR (up to 40%).

Diagnosis

1. Antenatal grading (APD, anteroposterior renal pelvic diameter)

	Second trimester	Third trimester
Mild	4 to <7 mm	4 to <9 mm
Moderate	7–10 mm	9–15 mm
Severe	>10 mm	>15 mm

- **UTD (urinary tract dilation) 2014 consensus:** six parameters (APD, central/peripheral calyces, parenchymal thickness and appearance, ureter, bladder). Prenatal A1 (low) or A2–3; postnatal **P1** (APD 10–<15 mm, central calyces), **P2** (≥15 mm or peripheral calyces or ureter), **P3** (abnormal parenchyma).
- **SFU (Society for Fetal Urology) grades 0–4:** 1 pelvis just seen; 2 pelvis plus major calyces; 3 all calyces, normal parenchyma; **4 parenchymal thinning**.

2. Postnatal ultrasound after **48 hours** (neonatal dehydration under-reads dilatation): large pelvis, rounded calyces, ureter not seen.

3. MAG3 (technetium-99m mercaptoacetyltriglycine) diuretic renogram ("well-tempered": hydrated, catheterised, furosemide)

- **DRF (differential renal function)** (normal 45–55%); drainage half-time <10 min normal, 10–20 equivocal, **>20 min obstructed** (adult chapter: >40 min).

4. Others: VCUG (voiding cystourethrogram) by UTI risk; MRU (magnetic resonance urography) or CT (computed tomography) angiography for vessels; **Whitaker test** (10 mL/min; >15–20 cm H_2O obstructed) if equivocal.

Primary management

Conservative: most resolve, about **one-third need surgery**; serial ultrasound and renogram.

Indications for pyeloplasty

- **DRF <40%** or a **fall of >10%** on serial scans.
- **Increasing APD** or SFU 4 / P3 with poor drainage.
- **Breakthrough UTI**, symptoms, stones, hypertension.

- Bilateral or solitary kidney.

Anderson–Hynes dismembered pyeloplasty: gold standard (~95% success)

1. Expose the PUJ; free the lower pole and look for a crossing vessel.
2. Stay sutures; transect the pelvis; reduce a redundant pelvis.
3. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a "no-touch" handle.
4. **Transpose** the PUJ anterior to any vessel (never divide it).
5. Watertight, tension-free, dependent anastomosis (5-0/6-0 absorbable); excise the segment.
6. Double-J or externalised stent (1–4 weeks equivalent); Foley 24–48 h with a double-J.

- **Route:** open preferred in infants; laparoscopic equals open with less pain; **robotic** (easier suturing, shorter learning curve) is now primary at most major centres.
- **Alternatives:** Foley Y-V plasty (high insertion); Culp spiral or vertical flap (long segment); **vascular hitch** (crossing vessel with normal PUJ); **endopyelotomy** (73% overall; **42% with vs 86% without a crossing vessel**), mainly for secondary PUJO.
- **Nephrectomy:** DRF **<15% in adults** (children: no agreed cut-off).

Diagram to draw: Anderson–Hynes steps: transection, lateral spatulation, transposition anterior to the crossing vessel, dependent anastomosis.

Complications after pyeloplasty

- Urine leak, stent problems, UTI.

Failed pyeloplasty: definition

- Persistent pain, UTI or stones; **obstructive diuretic renogram** (T½ >20 minutes); unchanged or increasing hydronephrosis; falling split function.
- Dilatation settles slowly; judge by renography.
- About 1 in 9 children need a secondary procedure. Most failures occur within 1–2 years (P2: within 2 years), but up to 30% occur later.

Causes of failure

- **Ischaemia** (over-stripping) and **tension** at the anastomosis.
- **Missed crossing vessel**; inadequate excision of the narrow segment; non-dependent anastomosis; kinking.
- **Urine leak** or infection causing periureteric fibrosis.
- Missed distal stricture or high-grade VUR.

Early failure (first days or weeks)

- Pain, fever, increasing dilatation: usually **oedema or clot**; temporary nephrostomy or stent, and most settle.

Evaluation of late failure

- **MAG3 diuretic renogram:** obstruction (drainage) and split function.
- **CT angiography:** crossing vessel.
- Antegrade or retrograde pyelography (stricture length); **Whitaker test** if equivocal.
- Drain first (stent or nephrostomy) if infected or function is uncertain, then reassess.

Salvage options

Option	When	Results/notes
Endopyelotomy (antegrade or retrograde ureteroscopic, laser)	Short stricture, no significant crossing vessel, moderate hydronephrosis; first choice after failed open repair	About 70% after failed pyeloplasty; lateral incision; stent 4–6 weeks
Redo pyeloplasty (open, laparoscopic, robotic)	Most failures; long or dense stricture, crossing vessel; failed endopyelotomy	About 86–89% (P2: about 83–89%); best choice in most children; wide mobilisation of kidney and ureter; transperitoneal (scarred retroperitoneum)
Ureterocalicostomy	Scarred, small intrarenal pelvis; dilated lower-pole calyx	About 70% in redo series; **resect** lower-pole parenchyma (not nephrotomy)
Buccal mucosa onlay graft	Long scarred PUJ/UPJ (ureteropelvic junction) or proximal ureter, dense scar	Robotic with omental wrap; failure about 15–25% (CORRUS)
Appendiceal onlay flap	Right-sided proximal stricture	Pedicled on appendiceal artery
Flaps (spiral, vertical), downward nephropexy	Gain length	Combine with redo repair
Ileal ureter	Long defect / long ureteric loss	Contraindicated if creatinine >2 mg/dL, bladder dysfunction, IBD (inflammatory bowel disease), radiation
Autotransplantation	Solitary or poor contralateral kidney; other repairs impossible; long ureteric loss	About 90% long term
Nephrectomy	Split function <15% with normal contralateral kidney; multiple failures	Definitive; laparoscopic

- Rule of thumb: failed pyeloplasty: try endourology first if anatomy suits; failed endopyelotomy: redo pyeloplasty. Redo pyeloplasty is the most reliable.
- **Also know:** robotic near-infrared fluorescence (indocyanine green, ICG) to identify the ureter and perfusion; Davis intubated ureterotomy (historical).

Principles of redo surgery

- Transperitoneal route in children (retroperitoneum is scarred).
- Ureteric catheter; wide mobilisation of kidney and ureter; **excise all scar**; spatulate; **tension-free, watertight** anastomosis over a stent; wrap with omentum or perirenal fat.
- **Also know:** near-infrared fluorescence shows perfusion; downward nephropexy.

Diagram to draw: ureterocalicostomy: lower-pole parenchyma excised, spatulated ureter sewn to lower calyx.

Follow-up

- After primary pyeloplasty: renogram and ultrasound after stent removal.
- After redo: stent out at 4–6 weeks; renograms for 2–3 years (late failures).

Recent advances

- Robotic redo with buccal graft and ICG fluorescence; pyramidal thickness ≤3 mm predicts surgery; urinary biomarkers.

Key points to remember:

- Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.
- Causes of failure: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable (about 83–89%); endopyelotomy about 70%.
- Ureterocalycostomy for small scarred pelvis: resect lower-pole parenchyma; about 70% in redo series. Buccal onlay emerging.
- Split function <15% with normal other kidney: nephrectomy.

Also, from the condensed book (not in the paper answer):

- Dismembered (Anderson–Hynes 1949) repair suits any insertion, allows pelvic reduction and **transposition over crossing vessels**, and alone excises the abnormal UPJ.
- Open approach is flank extraperitoneal by default; anterior for horseshoe or pelvic kidney.

ALGORITHM

1. Dissect the ureter with periureteral tissue; place a lateral marking stitch.
2. Excise the UPJ; spatulate **laterally**; bring the apex to the inferior pelvis, anterior to any crossing vessel.
3. Do a watertight absorbable anastomosis, with or without reduction; leave a stent and drain.

- Remove the **stent** at **4–6 weeks**. Reserve an added nephrostomy for secondary UPJO.
- Laparoscopic and robotic success is >**90–95%**, equal to open; the robot's advantage is ergonomic.

ALGORITHM

1. Remove the Foley at 24–36 h, then the drain if output is negligible.
2. If output rises after Foley removal (reflux along the stent), **replace the Foley for 7 days**.
3. Remove the stent at 4–6 weeks; then renography.

6 · Other UPJ reconstructions

> **KEY POINT**
>
> Every UPJ repair asks **what tissue bridges the gap while keeping blood supply**. Dismemberment excises disease but devascularises; flaps keep blood supply but carry disease forward; the calyx is used when the pelvis is too small.

Procedure	Indication	Note
Foley Y-V-plasty	**High insertion**	Not with crossing vessels or need for reduction
Culp–DeWeerd spiral flap	Large pelvis, long proximal narrowing	Longest flap; length:width ≤**3:1**
Scardino–Prince vertical flap	Dependent UPJ, box-shaped pelvis	Shorter flap
Davis intubated ureterotomy	Long strictures, with spiral flap	Heals over stent (secondary intention)
Ureterocalicostomy	Small intrarenal pelvis, horseshoe, failed pyeloplasty	Ureter to lower-pole calyx

- Outline flaps longer than needed; trim the apex, never the base.

SAQ Ureterocalycostomy

Paper answer · P2 · P3 · merged from 2

Asked as: P3 SAQ "Indication, procedure and complications of Uretero-Calycostomy?" · P2 SAQ "Uretero-calycostomy."

Definition

- End-to-side anastomosis of the spatulated proximal ureter to a dilated **lower-pole calyx**, after **resecting lower-pole parenchyma**, bypassing the pelvis and PUJ (pelviureteric junction).

Indications

- **Failed pyeloplasty** with a scarred pelvis and short ureter (commonest).
- **Small intrarenal pelvis** with PUJ obstruction.
- **Rotation or fusion anomalies** (horseshoe kidney).
- Proximal ureteric or pelvic loss: trauma, iatrogenic, **TB (tuberculosis) PUJ stricture**.
- **Prerequisites:** dilated lower calyx, ureter reaching it without tension, worthwhile function. Cortex <5 mm with poor function predicts failure.

Procedure (open, laparoscopic or robotic)

`Flank position (45–60°); pre-placed double-J stent → mobilise ureter with generous periureteral tissue → mobilise kidney; partial lower-pole nephrectomy to expose the calyx widely (not simple nephrotomy) → close (ligate) the pelvic outlet; spatulate ureter laterally → apex stitch to lateral calyx wall; second at 180° → interrupted mucosa-to-mucosa sutures, all placed then tied → capsule closed over cut surface, away from anastomosis → cover with perinephric fat, peritoneum or omentum; drain`

- **Why resect:** parenchyma or capsule left round the anastomosis contracts and causes stenosis.

Postoperative care and follow-up

- Stent out at 4–6 weeks; urogram about 1 month later, then renogram.
- Later endoscopy needs **antegrade upper-pole access**.

Complications and outcome

- **Anastomotic stricture** (main failure; mostly within the first year).
- Urine leak, urinoma, bleeding from the lower pole, infection, loss of the renal unit.
- Success about **70%** in the largest (mostly redo) series; better in small paediatric (robotic) series. The access route does not affect outcome.

Alternatives

- Endopyelotomy, redo pyeloplasty, flap repair, Davis intubated ureterotomy, ileal ureter, buccal graft, autotransplantation, nephrectomy (poor kidney); calicovesicostomy for a low-lying kidney.

Diagram to draw: kidney with resected lower pole and ureter sewn to the calyx.

> **Key points to remember:**
> - Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
> - Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
> - Stricture is the main failure; success about 70% in redo series.

7 · Retrocaval ureter

*See the paper answer **Retrocaval ureter and IVC anomalous development** (Section 4, Ch 52).*

8 · Ureteral stricture — evaluation and endourology

Exclude cancer; endoscopy suits only short, well-vascularised strictures.

- **Causes**: malignancy, stone, radiation, surgical ischaemia, periureteral fibrosis, endoscopy, infection (tuberculosis, schistosomiasis).
- **Exclude malignancy first** with CT, with or without ureteroscopy and biopsy.
- Stricture follows ureteroscopy (URS) in ≈**3%** vs 1.5% after shock-wave lithotripsy; an impacted stone >2 months gives ≈1 in 4.
- Endourology needs ipsilateral function ≥**25%** (the algorithm uses 20%).
- **Stents** suit intrinsic strictures; extrinsic compression needs tandem or metallic stents. Chronic stenting suits unfit or cancer patients.
- **Balloon dilation** is inflated until the **waist disappears**, then stented 2–4 weeks. Contraindications: infection, stricture >**2 cm**. It works best (≈85%) in short, recent, non-anastomotic, well-vascularised strictures; anastomotic ≈50%.
- **Endoureterotomy** (retrograde holmium) is first line under 2 cm. Cut full thickness into **2–3 mm of normal ureter** each side, over a safety wire.

Level	Incision	Avoids
Distal/mid ureter	**Anteromedial**	Iliac vessels

Level	Incision	Avoids
Proximal ureter	**Lateral/posterolateral**	Great vessels

- Complete obliteration in high-risk patients can be crossed by **cutting toward the light**, then stenting 8–10 weeks.

ALGORITHM

1. If <2 cm and function >20%, then endoscopic incision.
2. If >2 cm and function >20%, then reconstruction.
3. If function <20%, then nephrectomy (children may still merit repair).
4. Transplant ureter: consider balloon. Follow with renography for 2 years.

KEY POINT

Each failure predictor (long, old, anastomotic, poorly vascularised) is a proxy for **ischaemic scar** that an incision cannot heal across.

9 · Ureteral stricture — reconstruction ladder

Defect length and level choose the repair.

Technique	Defect bridged	Level
Ureteroureterostomy	**2–3 cm**	Upper/mid
Ureteroneocystostomy	4–5 cm	Distal
Psoas hitch	**6–10 cm**	Below pelvic brim
Boari flap	**10–15 cm**	Lower/mid; spiral may reach pelvis
Renal descensus	5–8 cm	Upper; adds to any repair

- **Ureteroureterostomy** suits short defects only, because tension causes stricture. Preserve adventitia, spatulate 180° apart, stent and cover with omentum; success is >90%.

ALGORITHM

1. **Persistent leak**
2. X-ray the stent position.
3. Pull the drain off the anastomosis and stop suction.
4. Place a Foley and give an anticholinergic; if it persists, place a nephrostomy.

SAQ Boari flap

Paper answer · P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3d

Definition

- A **tubularised flap of anterior bladder wall**, based posterolaterally on the side of the lesion, swung up to bridge a distal or mid-ureteric gap. Described in dogs 1894, humans 1947; laparoscopic 2001, robotic 2008.

Reconstructive ladder (distal ureter)

Option	Gap bridged
Ureteroneocystostomy	Distal 4–5 cm
+ Psoas hitch	Adds up to 5 cm (total 6–10 cm)
Boari flap (+ hitch)	**10–15 cm**; spiral flap may reach pelvis (right)
Downward nephropexy	Adds up to 5–8 cm
- Beyond: TUU (transureteroureterostomy), ileal ureter, autotransplantation.	

Indications

- Long distal or mid-ureteric loss: gynaecological injury (commonest), stricture, failed reimplantation, distal TCC (transitional cell carcinoma) ureter, tuberculosis.

Contraindications

- **Small, contracted or poorly compliant bladder** (radiation, tuberculosis); uncorrected outlet obstruction or neurogenic bladder. Not usually for acute unstable trauma.
- Preoperative: urodynamics or cystogram for capacity; antegrade and retrograde studies for length.

Technique

1. Mobilise bladder; divide contralateral superior vesical pedicle if needed.
2. Mark flap on the **non-distended** bladder: **base ≥4 cm, apex ≥3 cm, length:base ≤3:1**; length = gap + 3 cm if tunnelled.
3. Raise flap; **psoas hitch** its base (sutures superficial, parallel to fibres: spare genitofemoral and femoral nerves).
4. Anastomose spatulated ureter: **non-refluxing** 3 cm submucosal tunnel, or **refluxing** direct (apex about 3 times ureteric diameter).
5. Stent (double-J); tubularise in two layers 2-0 absorbable; drain, catheter 7–14 days. - Laparoscopic and robotic: transperitoneal, refluxing anastomosis, results excellent.

Complications

- **Recurrent stricture** (commonest; ischaemia or tension), reduced capacity with frequency and urgency, leak, nerve palsy, iliac vessel injury, pseudodiverticulum.
- eGFR (estimated glomerular filtration rate) fell in 44% within 2 years after reimplantation: counsel.
- **Alternative:** bladder advancement flap, similar reach and simpler.

Diagram to draw: bladder outline with oblique flap (4 cm base, 3 cm apex), hitched and tubularised to ureter.

> **Key points to remember:**
>
> - Bridges 10–15 cm; combine with psoas hitch.
> - Base ≥4 cm, length:base ≤3:1, measure bladder empty.
> - Contraindicated in small contracted bladder.

SAQ Trans Uretero Ureteric anastomoses

Paper answer · P2

Type: SAQ (10 marks)
Source: 2016-Summer Paper 2, Q3a

Definition

- **TUU (transureteroureterostomy)**: the injured (donor) ureter is brought across the midline and joined **end-to-side** to the opposite (recipient) ureter. First clinical use in 1934.

Indications

- Lower ureteric loss that reimplantation, psoas hitch or Boari flap cannot bridge: small, scarred or irradiated bladder; long ureteric loss; pelvic phlegmon; associated rectal, vascular or extensive bladder injury.
- One ureter short while the other is reimplanted with a psoas hitch (hitch only one side).
- Undiversion and reconstruction in children (Hendren).
- Usually a **delayed** repair, not acute.

Contraindications

Absolute	**Relative (disease that may involve both ureters)**
Donor ureter too short to reach without tension	Stone disease, **urothelial carcinoma**
Diseased recipient ureter	Retroperitoneal fibrosis, pelvic radiation
	Chronic pyelonephritis, genitourinary tuberculosis
- Recipient reflux must be corrected at the same time, so do a preoperative VCUG (voiding cystourethrogram). Less favoured in children (long life, both kidneys at risk).	

Technique (steps)

1. Midline transperitoneal incision; reflect the colon.
2. Mobilise the **donor ureter with its adventitia** (blood supply); divide it just above the diseased segment.
3. Expose about 5 cm of recipient ureter with minimal mobilisation.
4. Tunnel the donor ureter **behind the sigmoid mesentery, above the IMA (inferior mesenteric artery)** to avoid kinking.
5. **Anteromedial ureterotomy** of the recipient (at least 2 cm); **spatulated, tension-free end-to-side** anastomosis with fine absorbable interrupted sutures.
6. **Double-J stent** from donor kidney to bladder; drain. (Laparoscopic and robotic routes are feasible.)

Complications

- Urine leak (commonest), urinoma, stricture or obstruction, kinking under the IMA.
- **Main risk: one unilateral problem becomes bilateral obstruction.**
- Later endoscopic access is difficult; use the antegrade route.

Results

- Success 90–97% in adults; complications (mostly leak) in about one-quarter of complex cases.

Diagram to draw: donor ureter crossing above the IMA to an end-to-side anastomosis, with stent.

> **Key points to remember:**
> - End-to-side, spatulated, tension-free; cross above the IMA.
> - Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
> - Risk: unilateral injury becomes bilateral.

Also, from the condensed book (not in the paper answer):

- In adults, refluxing and non-refluxing reimplants are equal. Treat outlet obstruction first.
- **Renal descensus** rotates the kidney inferomedially and fixes its lower pole; the renal vein limits descent.
- **Ileal ureter** is isoperistaltic. Contraindications: creatinine >**2 mg/dL** (acidosis in ≈half), bladder dysfunction, inflammatory bowel disease, radiation enteritis. Start endoscopic surveillance from year 3.
- **Buccal mucosa onlay** suits dense scar. **Autotransplantation** suits an absent or poor contralateral kidney, or when other repairs are impossible.

> **KEY POINT**
>
> Ladder: bring the bladder up, the kidney down, borrow the other ureter, or substitute (ileum, buccal). Tension and ischaemia cause failure at every rung.

10 · Ureteroenteric anastomotic stricture

- Strictures occur in 4–9% of conduits, so **upper-tract surveillance is lifelong**.
- **The cause is ischaemia**. The **left is commoner**, because it needs extra length, passes under the sigmoid mesentery and angulates at the inferior mesenteric artery.
- Risks: **non-refluxing anastomosis** (≈13% vs 2–3%), carcinoma in situ at the margin, obesity and major complications.
- Conduit dilation is often non-obstructive, so confirm with renography; rapid progression with a mass suggests recurrent cancer.
- **Open revision is the gold standard** (≈91% vs 26% endoscopic); robotic is equal. Endourology is **antegrade**. Endoscopic success is ≈50% under 1 cm and better on the right.

ALGORITHM

1. If <2 cm and function >20%: right side, endoscopic incision; **left side, formal repair** (sigmoid mesentery bleeding).
2. If >2 cm, then formal repair.
3. If function <20%, then nephrectomy or observation.

11 · Retroperitoneal fibrosis

LAQ Retroperitoneal fibrosis: diagnosis and (medical) management

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P3 LAQ "Discuss the pathology, etiology, Clinical features diagnostic evaluation and treatment of retroperitoneal fibrosis." · P1 SAQ "Retroperitoneal fibrosis" · P2 SAQ "Retro peritoneal fibrosis medical management"

Introduction

- RPF (retroperitoneal fibrosis) is a rare inflammatory fibrosis (fibrous plaque) encasing the aorta, IVC (inferior vena cava) and ureters, centred on the **distal aorta at L4–L5**.
- Incidence about 1 per 200,000–500,000 a year; age 40–60; male:female 2–3:1.

Etiology

Idiopathic (about 70%)	Secondary (about 30%)
Part of **chronic periaortitis**, a large-vessel vasculitis	**Drugs** (commonest identifiable cause): methysergide and other ergots, β-blockers, phenacetin
Proposed antigen: **ceroid** (oxidised lipid in plaque)	**Malignancy:** lymphoma (commonest), myeloma, carcinoid, pancreas, prostate, sarcoma
Probably autoimmune: membranous glomerulonephritis, ankylosing spondylitis, granulomatosis with polyangiitis; ANA (antinuclear antibody) positivity predicts relapse	Radiotherapy; asbestos
Part of multifocal fibrosclerosis: sclerosing mediastinitis and cholangitis, orbital pseudotumour, Riedel thyroiditis	Infection: tuberculosis, actinomyces, gonorrhoea, schistosomiasis

- Many idiopathic cases are now classed as **IgG4 (immunoglobulin G4)-related disease** (raised serum IgG4, IgG4-positive plasma cells, storiform fibrosis).

Pathology

- **Gross:** flat, tan, dense plaque centred on the **distal aorta at L4–L5**, wrapping the ureters; may invade ureter or psoas.
- **Histology changes with stage:**
- Early (active): collagen with capillary proliferation and lymphocytes, plasma cells and fibroblasts (steroid-responsive).
- Late (burnt-out): acellular, avascular sheets of collagen.
- **Trap:** malignant RPF looks identical except for small islands of tumour cells, so take **deep, multiple biopsies**.
- **Mechanism of obstruction:** extrinsic compression and loss of ureteric peristalsis; the lumen stays patent.

Clinical features

- **Dull lower back or flank pain** (commonest); weight loss, anorexia, nausea, malaise, fever, hypertension.
- About half present with **uraemia** or oliguria/anuria; symptoms last 4–6 months before diagnosis.

- **Extension:** IVC (leg oedema, DVT [deep venous thrombosis]); renal vein (gross haematuria); spermatic cord and scrotum. Examination is usually unrevealing.

Diagnostic evaluation

Laboratory

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein) in half to two-thirds; leucocytosis; anaemia; raised creatinine (renal failure); ANA; serum IgG4.

Imaging

Test	Findings and role
CT (computed tomography) / CT urography	Homogeneous soft-tissue mass enveloping aorta, IVC and ureters; **medial deviation** of proximal and mid-ureters with smooth tapering
Retrograde pyelography	When renal function is poor
MRI (magnetic resonance imaging)	Low T1; **high T2 signal means active disease**; enhancement falls with treatment
PET-CT (positron emission tomography)	**Most sensitive test of activity**; guides biopsy and follow-up
Renography	Split function (repair or nephrectomy)

- **Biopsy** (CT-guided or at ureterolysis) if atypical or malignancy is suspected; may be omitted with no malignancy history, no lymphadenopathy and typical imaging.

Treatment algorithm

```
RPF with hydronephrosis -> Relieve obstruction (stents; nephrostomy if anuric or septic) -> Treat post-obstructive diuresis -> Stop culprit drug; exclude malignancy -> Active disease: steroids (+/- second-line drugs) -> Failure or burnt-out fibrosis: ureterolysis
```

Initial management

- Emergent **stents or PCN (percutaneous nephrostomy)**; nephrostomy if critically ill, uraemic and anuric with little urine. Replace fluid during **post-obstructive diuresis**.
- **Stop the culprit drug;** treat any secondary cause.

Medical management

Agent	Role and regimen
Prednisone (first line)	About 60 mg daily (about 1 mg/kg), tapered to 5 mg, total course up to 2 years. About 80% respond; relapse on tapering 25–50%
Mycophenolate mofetil (with steroids), azathioprine, cyclophosphamide, ciclosporin, colchicine	Second line or steroid-sparing
Tamoxifen; medroxyprogesterone	Thought to inhibit fibroblasts
Rituximab	Refractory or IgG4-related disease

- Steroids work on the cellular, active phase: pain and symptoms settle in days, ESR falls and the mass shrinks. Active inflammation predicts response; burnt-out fibrosis responds poorly.
- **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% with prednisone vs 50% with tamoxifen, so steroids remain first line.

- Drainage plus medical therapy clears obstruction in about two-thirds; only about **15% need ureterolysis**.

Surgical ureterolysis

- For failed medical therapy, burnt-out (fibrotic) disease or tissue diagnosis. **Bilateral**, even if hydronephrosis is unilateral; open, laparoscopic or robotic.
- Steps: 1. Pre-place stents to identify the ureters. 2. Mobilise the colon; take deep biopsies for frozen section. 3. Start at the distal, non-dilated ureter. 4. "Split and roll": right-angle clamp between ureter and plaque; incise the fibrosis over it. 5. Close any ureterotomy with absorbable suture. 6. **Separate the ureter from the fibrotic bed so it cannot be re-encased.**

Protection method	How
Intraperitonealisation	Ureter lateralised; peritoneum closed behind it
Lateral retroperitoneal placement (lateralisation)	Ureter displaced laterally (equivalent results)
Omental wrap	Omentum split on the gastroepiploic arteries, wrapped round each ureter; adds blood supply; best for extensive disease

- Postoperative steroids may reduce recurrence.
- **Outcomes:** open about 94% stent-free at 1 year; laparoscopic about 90%; robotic with omental wrap about 86%.

Other options

- Long-term metallic stents (unfit); autotransplantation (rarely needed) if ureterolysis is impossible; nephrectomy for an unrecoverable (dead) kidney.

Follow-up and monitoring

- Symptoms, ESR/CRP, creatinine, ultrasound; periodic CT/MRI or PET-CT for activity; remove stents once the ureters drain.
- Watch for relapse and steroid side effects (diabetes, osteoporosis).

Diagram to draw: periaortic plaque at L4–L5 drawing both ureters medially, with an omental wrap.

> **Key points to remember:**
> - Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
> - Medially deviated ureters; PET-CT best shows activity; biopsy deep.
> - Drain first, then prednisone (about 80% respond).
> - Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
> - Rituximab for IgG4-related or refractory disease.
> - About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

ALGORITHM

1. If uraemic, decompress with a **nephrostomy** or stents and manage the **post-obstructive diuresis**.
2. Stop any culprit drug; biopsy.
3. Give steroids; if they fail, do ureterolysis.

- Steroids work while the lesion is cellular; second line is immunosuppressants or **tamoxifen**. Ureterolysis with an omental wrap is most definitive.

94 Upper Urinary Tract Trauma

LAQ Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management

Paper answer · P2

Asked as:

- Traumatic renal vascular parenchymal injuries : diagnosis and management. — *Summer 2017 Paper II Q2 (LAQ)*
- Diagnosis and Treatment of renal trauma — *Practice Set 9 Paper II Q1 (LAQ)*
- Grade 4 Renal trauma. — *Practice Set 3 Paper II Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 9 Paper II Q1: also answered in 2017-Summer Paper 2, Q2; grade IV in Practice Set 3 Paper 2, Q3c) (Practice Set 3 Paper II Q3c: renal trauma also answered in 2017-Summer Paper 2, Q2)

Assumed meaning: diagnosis and management of traumatic renal parenchymal and renal vascular injuries.

Introduction

- Kidney is the **commonest urological organ injured** (up to 5% of all trauma); **blunt injury causes 80–90%**; firearms cause most penetrating injuries.
- **Rapid deceleration** stretches the kidney at its fixed points (hilum and UPJ, ureteropelvic junction): arterial intimal tear and thrombosis, vein disruption, pedicle or UPJ avulsion.
- Main renal artery injury is rare (about 0.05% of blunt trauma).

Classification: AAST (American Association for the Surgery of Trauma) 2018

Grade	Injury
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration ≤1 cm, no extravasation
III	Laceration >1 cm without collecting-system injury; **any vascular injury or active bleeding contained within Gerota's fascia**
IV	Laceration into collecting system with extravasation; renal pelvis laceration and/or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's fascia; segmental or complete infarction (thrombosis)
V	**Main renal artery or vein laceration or hilar avulsion**; devascularised kidney with active bleeding; shattered kidney
- Advance one grade for multiple injuries, up to grade III. Change from 1989: main vessel injury moved from IV to V; pseudoaneurysm and AVF (arteriovenous fistula) now grade IV.	

Definition: AAST 2018, grade IV in detail

- Laceration into the **collecting system with urinary extravasation**.
- **Renal pelvis laceration and/or complete UPJ (ureteropelvic junction) disruption.**
- **Segmental artery or vein injury** (pseudoaneurysm, AVF [arteriovenous fistula]).
- **Active bleeding beyond Gerota's fascia.**
- **Segmental or complete infarction** from thrombosis without active bleeding.
- Change from 1989: main-vessel injury moved to grade V; segmental vascular and UPJ injuries added to grade IV.

Clinical diagnosis

History and examination

- Mechanism is the most important (key) history; mechanism matters. Signs: flank bruise or tenderness, **fractured ribs 9–12**, flank or low chest wound.

Haematuria

- Test the **first urine sample**. Significant: gross haematuria, or microscopic (microhaematuria) (>5 RBC [red blood cells] per high-power field, hpf) with **shock (SBP [systolic blood pressure] <90 mmHg)**.
- Degree does **not** match severity; it may be **absent in pedicle injury** and UPJ avulsion (in about half of UPJ injuries), and in grade IV.

Indications for imaging (AUA (American Urological Association) and EAU (European Association of Urology))

1. Blunt trauma with gross haematuria.
2. Blunt trauma with microhaematuria and **SBP (systolic blood pressure) <90 mmHg** at any time.
3. Rapid deceleration (high-speed collision, fall), even without haematuria.
4. Penetrating trauma with likely renal injury, any haematuria.
5. Children: image liberally (hypotension appears late). - Microhaematuria without shock or deceleration (microhaematuria alone without shock): observe without imaging.

Imaging

- **Contrast-enhanced CT (computed tomography)** is the gold standard. Arterial/nephrographic phase (~80 s) shows bleeding; **delayed excretory phase** (5–10 min / minutes) shows urinary extravasation (urine leak).
- **CT red flags:**

Finding	Suggests
Medial haematoma	Pedicle (vascular) injury
Medial extravasation, no contrast in distal ureter (without distal ureteric contrast)	Renal pelvis or UPJ avulsion
Absent enhancement, cortical rim sign	Main renal artery occlusion
Active contrast blush (vascular contrast blush)	Arterial bleeding needing AE (angioembolisation)

Finding	Suggests
Haematoma rim >4 cm (another source: >3.5–4 cm), medial laceration	High risk of intervention; bleeding likely to need intervention (≥2 of rim, blush, medial laceration predict intervention)
- **One-shot IVP (intravenous pyelography)** on table (2 mL/kg, one film at 10 min) in the unstable patient: confirms a **functioning contralateral kidney**. FAST (focused assessment with sonography for trauma) is non-specific. Renal vein injury is poorly seen on CT.	

Management

`Unstable despite resuscitation → laparotomy (± damage control) (or angioembolisation / AE if team ready, else laparotomy) Stable → stage with CT Grade I–III → non-operative Grade IV–V → non-operative in selected; AE for active bleeding Active blush / pseudoaneurysm / AVF → superselective AE Extravasation → observe (90% heal); stent ± nephrostomy if urinoma enlarges, fever, obstruction, or UPJ/pelvis injury`

- **Non-operative**: bed rest until gross haematuria clears, serial haemoglobin (**every 4–6 h**); success >90% in blunt injury and selected penetrating ones. **Non-operative success** about 85% for grade IV–V even in penetrating series; most failures need only AE, stent or nephrostomy.
- Failure: instability, or falling haemoglobin despite 2–3 units of blood.
- Urinary extravasation heals spontaneously in **76–90%** (another source: up to 90%).
- **Angioembolisation**: superselective coils for active bleeding, pseudoaneurysm, AVF or delayed bleeding; repeat AE often succeeds.
- **Reimaging** (reimage / repeat CT at ~48 hours) for grade IV–V with deep laceration (deep lacerations), or new fever, pain or falling haematocrit (routine reimaging debated).

Indications for exploration

Absolute	Relative
Haemodynamic instability, no or transient response	Extravasation with devitalised parenchyma (>25%; lowers the threshold)
Expanding or pulsatile retroperitoneal haematoma	Arterial thrombosis
Suspected pedicle avulsion	Associated colon or pancreatic injury
UPJ avulsion	Incomplete staging

Operative technique (operative steps)

- Midline transabdominal (laparotomy); life-threatening injuries first.
- **Vascular control before opening Gerota's fascia**: lift small bowel; incise retroperitoneum over the aorta above the inferior mesenteric artery → find the left renal vein crossing the aorta → loop vein and both arteries; then reflect colon, open Gerota's fascia.
- **Renorrhaphy**: debride → suture-ligate vessels → watertight collecting-system closure → close defect over bolster → omental cover → drain. Polar injury: partial nephrectomy.

- **Nephrectomy** if unreconstructable or unstable. **Damage control**: pack, resuscitate in ICU (intensive care unit), relook (return) at 24–48 h.

Renovascular injury

- **Main artery thrombosis** (intimal tear): unilateral with normal other kidney → observe (revascularisation rarely succeeds). **Solitary or bilateral → revascularise** (warm ischaemia ideally <120 min); endovascular stent for (or stent an) intimal flap.
- **Vein**: left renal vein can be ligated near the IVC (inferior vena cava) (gonadal, adrenal collaterals); right cannot. Repair with 5-0 vascular suture; segmental veins may be ligated.

Complications

- **Delayed bleeding** (within 3 weeks; usually within 21 days): AE (embolise).
- Urinoma or abscess: stent, nephrostomy, drain.
- **Hypertension**: Goldblatt (arterial stenosis), **Page kidney** (haematoma compression), segmental ischaemia, AVF.
- Hydronephrosis, loss of function; **also know:** AKI (acute kidney injury).

Follow-up

- Blood pressure, urinalysis, creatinine; EAU: DMSA (dimercaptosuccinic acid) scan for salvaged grade III–V kidneys; imaging if new symptoms.

Diagram to draw: left renal vein crossing the aorta, with vessel loops on the vein and artery. For grade IV: kidney with laceration into a calyx, urinoma and segmental pseudoaneurysm.

> **Key points to remember:**
> - AAST 2018: segmental vessel injury grade IV; main vessel grade V.
> - Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
> - Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
> - Delayed CT phase is essential; blush means AE.
> - Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
> - Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
> - Revascularise only a solitary kidney or bilateral injuries.

Also, from the condensed book (not in the paper answer):

- Polar injury is treated by partial nephrectomy with omental cover.

1 · Renal injury — mechanism and presentation

- The kidney is the commonest urological organ injured. Blunt trauma causes 80–90%; most are grade I–III and heal conservatively.
- **Deceleration** injures the fixation points (hilum and ureteropelvic junction, UPJ), causing artery thrombosis, vein disruption, and pedicle or UPJ avulsion.
- **Penetrating** injury causes more bleeding, nephrectomy and complications; high-velocity missiles cause delayed necrosis.
- Signs: flank ecchymosis and **ipsilateral rib 9–12 fractures** (strongest association).

2 · Haematuria

- Significant haematuria is **gross**, or **microhaematuria (>5 red blood cells per high-power field) with systolic blood pressure (SBP) <u>≤90 mm Hg</u>**.
- Degree does not match severity; haematuria is absent in ~⅓ of blunt renovascular injuries and **~½ of UPJ injuries**. Test the first aliquot.

> **KEY POINT**
>
> Haematuria triggers imaging; it does not measure injury. The worst injuries (pedicle, UPJ) may not bleed into the urine, so mechanism carries equal weight.

3 · AAST kidney injury scale (2018)

The AAST grade, set on CT, predicts intervention and nephrectomy.

Grade	AAST 2018
I	Subcapsular haematoma and/or contusion, no laceration
II	Perirenal haematoma within Gerota's fascia; laceration **<u>≤1 cm</u>**, no extravasation
III	Laceration >1 cm, no collecting-system rupture; **any vascular injury or active bleeding contained within Gerota's**
IV	Collecting system laceration with extravasation; pelvis laceration or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's; segmental/complete infarction without active bleeding
V	**Main renal artery or vein laceration** or hilar avulsion; devascularised kidney with active bleeding; shattered kidney

- Advance one grade for bilateral or multiple injuries, up to III; grade by the highest finding. MCQ Main-vessel injury was grade IV in 1989 and is now V; segmental pseudoaneurysm or arteriovenous fistula (AVF) is now IV.

4 · Indications for imaging (AUA = EAU)

- **Penetrating** trauma with **any haematuria** needs CT, if the patient is stable.
- **Blunt** trauma needs CT with rapid deceleration, gross haematuria, or microhaematuria with SBP <90 at any time; otherwise blunt microhaematuria is **observed without imaging**.
- **Children** hold blood pressure until ~50% blood loss, so image liberally.

5 · Imaging studies

Without a delayed phase, urinary extravasation is missed.

- Gold standard: **contrast CT with arterial/nephrographic and delayed (~<u>10 min</u>) phases**. A medial haematoma with normal arteries suggests vein injury.
- Focused assessment with sonography for trauma (FAST) cannot separate haematoma from urine.

CT finding	Suggests
Medial haematoma	Vascular pedicle injury
Medial extravasation; no contrast in distal ureter	Renal pelvis/UPJ avulsion

CT finding	Suggests
Absent global enhancement (cortical rim sign)	Main renal artery occlusion
Contrast blush	Brisk bleeding; strongly predicts need for angioembolisation (AE)
≥2 of: rim distance >4 cm, medial laceration, blush	High-risk grade IV

- The **MiGUTS nomogram** predicts bleeding intervention; its largest contributor is **haematoma rim distance**.

ALGORITHM

1. **One-shot intravenous pyelogram (IVP)** (unstable, no CT): **2 mL/kg contrast → single film at 10 min**. It confirms a working contralateral kidney.

Perinephric collection	Hounsfield units (HU)	Feature
Urinoma	0–20	Contrast pooling on delayed phase
Haematoma	>30	—
Abscess	>20	Rim enhancement

6 · Nonoperative management

- Stable, staged **grade I–IV and selected V** injuries are observed, whatever the mechanism. Success is >90% in blunt injury, even with extravasation; early mobilisation is safe.
- **Failure triggers**: instability; transfusion of **2–3 units**; new pain or guarding; enlarging urinoma with fever. Most failures need only stent, nephrostomy or AE.
- **Penetrating** injury breaches Gerota's fascia, losing tamponade, so more fail; stabs do better than gunshots.
- **Reimaging** (AUA): CT at ~48 h for **grade IV–V deep lacerations** or deterioration, not for uncomplicated I–III.

7 · Urinary extravasation

- Parenchymal leaks **heal in 76–90%**. **Drain early** when urine cannot pass antegrade: pelvis or UPJ avulsion, ureteric clot, UPJ obstruction.
- A symptomatic or enlarging urinoma gets a retrograde stent and catheter; if that is impossible, a nephrostomy.

> **KEY POINT**
>
> Parenchymal leaks heal because antegrade flow is preserved; anything stopping urine reaching the bladder converts an observable leak into one that must be drained.

8 · Intervention

Needless exploration releases tamponade and costs kidneys.

Absolute indications	Relative (increasingly observed)
Instability, no or transient response	Extravasation with significant devascularisation

Absolute indications	Relative (increasingly observed)
Expanding or pulsatile haematoma	Colon or pancreatic injury
Suspected pedicle avulsion; UPJ avulsion	Arterial thrombosis; parenchymal extravasation

- **AE** is superselective, raises salvage and can be repeated. **Failure predictors**: instability, grade V, urinary extravasation, gross haematuria.
- At laparotomy, exploration is **not mandatory** for staged grade I–III. The only absolute indication is a **pulsatile or expanding haematoma**.

ALGORITHM

1. **Vascular control first**
2. Eviscerate the small bowel; incise peritoneum over the aorta above the inferior mesenteric artery (IMA), up to the ligament of Treitz.
3. Follow the aorta to the left renal vein; loop and retract it; loop both renal arteries.
4. Reflect the colon and open Gerota's fascia; occlude the vessels if bleeding.

- **Damage control** (hypothermia, acidosis, coagulopathy): pack, close temporarily, resuscitate and return at 24–48 h.

ALGORITHM

1. Expose and gain vascular control; debride only devitalised tissue.
2. Suture-ligate vessels; close the collecting system **watertight**.
3. Approximate parenchyma over a bolster; cover with Gerota's fascia or **pedicled omentum**; drain.

- Colon or pancreatic injury no longer mandates nephrectomy.
- **Revascularise** only a solitary kidney or bilateral injury, with warm ischaemia **<120 min**; unilateral repair salvages only ~⅓.
- The **left renal vein may be ligated; the right cannot**.
- Hypertension follows thrombosis repair in ≥43%, so monitor blood pressure for a year.

9 · Complications and follow-up

- Delayed bleeding comes within 21 days and is treated by AE.
- **Page kidney** (subcapsular compression) is treated with an ACE inhibitor or receptor blocker, then drainage if refractory; post-traumatic AVF needs AE.
- EAU: no imaging for grade I–II; salvaged grade III–V get a dimercaptosuccinic acid (DMSA) scan.
- Bleeding risk peaks at **2–6 weeks**; a solitary kidney does not bar contact sport.

10 · Ureteral injury

Often silent; late diagnosis means fistula or kidney loss.

LAQ Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury

Paper answer · P3

Asked as:

- Iatrogenic ureteric injury — *Practice Set 2 Paper III Q1 (LAQ)* (upper ureteric injury also answered in 2023-Winter Paper 3, Q2)
- Discuss management of ureteric injuries occurring during gynec surgeries — *Practice Set 4 Paper III Q1 (LAQ)* (also answered in Practice Set 2 Paper 3, Q1)
- Discuss Etiology, Diagnosis and Management of Upper ureteric injury. — *Winter 2023 Paper III Q2 (LAQ)*

Type: LAQ (25 marks)

Assumed scope: surgical (iatrogenic) injury at any level of the ureter, in gynaecological, colorectal, vascular and urological surgery. Assumed meaning (upper ureteric injury): injury to the upper (proximal) ureter, from the PUJ (pelviureteric junction) to the pelvic brim, from external trauma or surgery.

Introduction

- Ureteric trauma is uncommon: 1–2.5% of genitourinary trauma. **Iatrogenic injury is commoner overall** than external trauma.
- **The proximal ureter is the commonest site of external injury. The distal ureter is the commonest (usual) iatrogenic site** in non-urological surgery.
- Gynaecological surgery has historically caused about half of all iatrogenic ureteric injuries; hysterectomy is the main culprit.
- It is mainly a problem of **recognition** (the core problem): thermal, ligation and devascularisation injuries are invisible at operation and declare themselves days to weeks later. Injury is often **missed** and declares itself later as urinoma, sepsis, stricture or fistula.
- Late diagnosis brings urinoma, sepsis, AKI (acute kidney injury), fistula and renal loss, and needs nephrostomy about ten times more often.

Aetiology and incidence (EAU (European Association of Urology) trauma guideline data)

1. External trauma

- **Penetrating** (most): gunshot wounds up to about 90%. Blast damages the intramural blood supply up to 2 cm from the wound.
- **Blunt** (up to about 40%): rapid deceleration avulses the ureter at the **PUJ**; linked to lumbar transverse process fractures.
- **Over 90% have associated injuries** (small bowel commonest, colon, vessels, kidney 10–28%). Mortality 6–9% reflects these.

2. Iatrogenic

Source	Key facts (upper-ureter relevance)
Gynaecological	Historically about half. Hysterectomy: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery highest (1.7–4.3%). Endometriosis and placenta accreta raise risk
Colorectal	0.15–10%; laparoscopic equals open
Colorectal, spinal, retroperitoneal surgery	Ligation, transection, cautery

Source	Key facts (upper-ureter relevance)
Urological / Ureteroscopy	Ureteroscopy is the commonest (iatrogenic) cause in some series: abrasion, perforation 0.2–2%, avulsion under 0.5% (another source: <0.3%) (proximal third most at risk); stricture about 3% (another source: 0.3–3%). Risk: proximal stone, impacted stone, long operation, large sheath, inexperience. Also radical prostatectomy, partial nephrectomy or ablation (late proximal stricture)
Partial nephrectomy, thermal ablation	Proximal stricture months later (lower-pole and hilar tumours)
Vascular (aortic) surgery	Aorto-iliac bypass: transient hydronephrosis 12–20%; up to 85% unrecognised (missed initially). Devascularisation. Rare ureteroarterial fistula (chronic stent plus pelvic surgery or radiation)

Gynaecological incidence (EAU trauma guideline data)

Procedure	Ureteric injury (%)
Caesarean section (emergency)	0.01–0.06
Vaginal hysterectomy	0.02–0.5
Abdominal hysterectomy	0.03–2.0
Minimally invasive hysterectomy	0.13–6.0 (laparoscopic = robotic, about 0.9)
Prolapse surgery	**1.7–4.3 (highest)**

- **Danger sites in the pelvis (applied anatomy):** pelvic brim near (at) the infundibulopelvic ligament; **under the uterine artery** ("water under the bridge"); beside the uterosacral ligament; vaginal fornix / vaginal cuff near the UVJ (ureterovesical junction).
- Up to a third are recognised during open surgery, fewer during laparoscopy; the uterosacral region is hardest to see.
- **Mechanisms:** ligation, crush, laceration, transection, avulsion, thermal injury, **devascularisation**. About 10% of women get a large share of distal ureteric blood from the uterine artery (the uterine artery supplies much of the distal ureter), so some (ischaemic) injuries are unavoidable.
- **Risk factors:** previous surgery, adhesions, malignancy, radiotherapy, heavy bleeding, long operation, learning curve; endometriosis (encasement, adhesions, medial deviation), placenta accreta.
- **URS (ureteroscopy) risks:** long operation, 12/14-Fr sheath, impacted or proximal stone, inexperience.

Classification: AAST (American Association for the Surgery of Trauma) ureter scale

Grade	Injury (description)
I	Haematoma or contusion, no devascularisation (without devascularisation)
II	Laceration <50% of circumference
III	Laceration ≥50%
IV	Complete transection, <2 cm devascularised (with <2 cm devascularisation)
V	Avulsion, >2 cm devascularised (with >2 cm devascularisation)

- Advance one grade for bilateral injury, up to grade III.

Prevention

- See (identify) the ureter at the **pelvic brim** (crossing the common iliac artery) and trace it before ligating pedicles. Good haemostasis and exposure.
- **Prophylactic stents** (or lighted stents) help recognition in high-risk (complex) cases, but do not clearly reduce (prevent) injury. **ICG (indocyanine green)** instilled into the ureter (intraureteric) makes it fluoresce (fluoresces) for hours.
- **Cystoscopy** after minimally invasive hysterectomy and prolapse surgery, with oral phenazopyridine or IV (intravenous) dye to show efflux.
- URS: safety wire; ≤14-Fr sheath in an unstented ureter; fragment before basketing; "stop, stent, stage".

Diagnosis and recognition

Suspicion (trauma)

- Penetrating flank or abdominal wound; high-energy deceleration.
- **Haematuria is absent in 25–45%**, so its absence does not exclude injury.

Intraoperative (best outcome)

- Inspect the ureter (at laparotomy); IV (intravenous) indigo carmine or methylene blue; cystoscopy for efflux (jets), with retrograde (on-table) pyelogram or stent. Check the other ureter and bladder. About 11% are missed at trauma laparotomy.
- Methylene blue is an MAO (monoamine oxidase) inhibitor: avoid with SSRIs/SNRIs (selective serotonin / serotonin–noradrenaline reuptake inhibitors; serotonin toxicity), G6PD (glucose-6-phosphate dehydrogenase) deficiency and pregnancy.

Delayed presentation (most cases) / postoperative

- Mean diagnosis about day 10: fever, ileus, flank pain, leucocytosis, rising creatinine, rising drain output, drain or wound urine leak, urinoma, anuria (bilateral), or **ureterovaginal fistula** (UVF): normal voiding plus constant leak, because the other kidney fills the bladder, usually within 4 weeks.

Imaging and tests

Test	Role
CT (computed tomography) urography with 10-minute delayed images (both ureters)	Best test in stable patients. Signs: extravasation, non-opacified distal ureter, delayed nephrogram, urinoma
Retrograde pyelography	Most sensitive; shows site; allows stenting at once
Antegrade nephrostogram	When retrograde fails or gap is wide
One-shot IVU (intravenous urography)	Non-diagnostic in 33–100%; never rely on it

- **Drain-fluid (or vaginal fluid) creatinine** above (far above) serum creatinine confirms urine.
- **Double-dye test** separates VVF (vesicovaginal fistula) from ureterovaginal fistula: oral phenazopyridine plus intravesical methylene blue. Orange pad = UVF; blue = VVF; both = both.

Management

Principles

- **Stability decides timing; level and length decide the operation.**

- Repair: mobilise sparing (preserve) adventitia; debride to bleeding tissue (edges); **spatulated, tension-free, watertight, stented** repair (anastomosis) with fine absorbable suture; drain; retroperitonealise; omentum (omental wrap) for thermal, radiation, blast or vascular injury; non-tunnelled reimplant.
- **No bowel interposition or autotransplantation acutely.**

By timing ``` Recognised at surgery -> repair at once (stable patient) Unstable -> ligate + later nephrostomy, or exteriorised stent Recognised late -> try retrograde stent (or nephrostomy + antegrade stent)

|-- succeeds -> stent 4-8 weeks, then image |-- fails -> nephrostomy; early repair if <=7 days, otherwise delayed repair after about 6 weeks Uncontrolled sepsis / collections -> drain; repair regardless of timing Ureteric injury suspected

|-- Unstable -> damage control: ligate + later PCN, | or exteriorised stent / cutaneous ureterostomy |-- Stable, found at surgery -> immediate repair over stent |-- Found late / postoperatively -> Attempt retrograde stent (+/- PCN, drain) -> Fails -> PCN; delayed definitive repair Suspected late injury -> retrograde stent (or nephrostomy + antegrade stent)

```
|-- succeeds -> stent 4-8 weeks, then image
|-- fails -> nephrostomy
     |-- found <=7 days -> early repair
     |-- later -> repair after about 6 weeks
```

Sepsis / uncontrolled collections -> drain; repair regardless of timing ```

At surgery, by injury

Injury	Action
Suture through wall, minor thermal	Remove suture; stent
Partial tear	Close transversely (transverse closure, 4-0/5-0) over stent, 4–8 weeks
Ligation	Remove ligature (deligate); stent if viable, excise and repair if not
Transection, lower ureter (usual)	**Ureteroneocystostomy ± psoas hitch** (95–100% success)
Long distal gap	**Boari flap** + hitch (gaps up to 12 cm)
Transection above pelvic brim	Ureteroureterostomy
Unstable patient	Ligate + later nephrostomy, or exteriorised stent

Repair by grade

- **Grade I–II (contusion, partial):** double-J stent for about 6 weeks; or close a laceration transversely over a stent. Excise non-viable contused ureter.
- **Grade III–V:** formal reconstruction.

By level (complete injury)

Level	Repair
Upper	Ureteroureterostomy; ureteropyelostomy (PUJ (pelviureteric junction) avulsion); ureterocalicostomy
Mid	Ureteroureterostomy ± downward nephropexy; TUU (transureteroureterostomy) when delayed

Level	Repair
Lower (commonest iatrogenic)	**Ureteroneocystostomy, non-tunnelled ± psoas hitch** (95–100% success); **Boari flap** for gaps up to 12 cm
Long loss (delayed only)	Ileal ureter or Yang–Monti; buccal graft ureteroplasty; autotransplantation (last resort)

Options for the upper ureter

Procedure	Key points
Ureteroureterostomy	Commonest acute repair; short defects (2–3 cm); spatulate 5–6 mm, 180° apart; 5-0 absorbable over double-J; up to 90% success; leak 10–24%, stricture up to 12%
Ureteropyelostomy (reimplant into pelvis)	PUJ avulsion; spatulated
Ureterocalicostomy	Pelvis or PUJ destroyed; lower-pole parenchyma resected
Renal descensus	Mobilise kidney downwards; gains up to 8 cm
TUU (transureteroureterostomy)	Delayed; long loss when the bladder cannot reach; 90–97% success. Avoid with stones, RPF (retroperitoneal fibrosis), TB (tuberculosis), urothelial cancer or radiation
Ileal ureter / Yang–Monti	Long or panureteric loss; 81–100% success. Avoid if creatinine >2 mg/dL, bladder dysfunction, inflammatory bowel disease, radiation enteritis. Acidosis about 20%
Appendiceal interposition	Right side; about 92% short-term success
Buccal mucosa graft ureteroplasty	Delayed long strictures; onlay with omental wrap, often robotic; about 87% success
Autotransplantation	Last resort before nephrectomy; 5–10% renal units lost
Nephrectomy	Rarely; unstable with unreconstructable kidney, or poor function

Iatrogenic endoscopic injury

- **URS perforation:** stop the procedure and place a **stent** about 4 weeks (± nephrostomy), image later; later reconstruction if stricture develops. **Avulsion:** nephrostomy, delayed reconstruction; avoid nephrectomy.

Fistulae

- **Ureterovaginal fistula (UVF):** stent first (best / ideally within 2 weeks; most heal); if stenting fails, reimplant with psoas hitch or Boari flap (success ≥90%). Early and delayed repair do equally well.
- **Ureteroarterial fistula:** endovascular covered stent.
- **Also know:** TUU (transureteroureterostomy), ileal ureter, buccal graft ureteroplasty and autotransplantation for long delayed defects; robotic reconstruction (repair) for delayed repair matches open; avoid nephrectomy.

Complications and follow-up

- Urinoma, abscess, leak, fistula, stricture, loss of renal function.
- Persistent leak: check stent position, take drain off suction, catheterise the bladder, then nephrostomy.

- **Follow-up:** ultrasound or diuretic renogram at 4–12 weeks and (at) 1 year; strictures (late stenosis) can be silent and may appear up to 30 months.

Medicolegal

- Disclose and document; malpractice claims most often follow gynaecological surgery.

Diagram to draw: psoas hitch with non-tunnelled reimplant; ureter's relation to the uterine artery (ureter passing under the uterine artery); spatulated ureteroureterostomy over a double-J stent.

> **Key points to remember:**
> - Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
> - Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
> - AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
> - CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
> - Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
> - Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
> - Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

Also, from the condensed book (not in the paper answer):

- Delayed upper-ureter repair may use appendix (right) or buccal graft.

AAST ureter grade	Description
I	Contusion/haematoma without devascularisation
II	Laceration **<50%** transection
III	Laceration ≥50% transection
IV	Complete transection, **<2 cm** devascularisation
V	Avulsion, >2 cm devascularisation

- Iatrogenic injury is commoner (distal); external is mostly gunshot (proximal). **Haematuria is absent in 25–45%**.
- The **distal ureter** is the commonest iatrogenic site; prolapse surgery is highest-risk, and minimally invasive hysterectomy is riskier than open.
- **Ureteroscopy**: stricture ~3%, avulsion <1%. At a tear, stop basketing and stent.
- Preoperative stents aid **recognition**, not proven prevention. Check efflux at cystoscopy after minimally invasive hysterectomy and prolapse surgery. **Methylene blue** is a monoamine oxidase inhibitor; avoid it in pregnancy and G6PD deficiency.

Modality	Role
CT urography, **10-min delays**	Best in a stable patient; trace both ureters
Retrograde ureterography	**Most sensitive**; allows stenting
Antegrade study	When retrograde fails
One-shot IVP	Often non-diagnostic; never rely on it

- CT signs: **no contrast distal to the injury**, medial extravasation (UPJ avulsion), delayed nephrogram, urinoma.

> **KEY POINT**
>
> Ureteral injury is diagnosed by absent contrast beyond it, so time the study for excretion, follow both ureters, and never treat a non-diagnostic study as negative.

ALGORITHM

1. If stable at laparotomy, repair now; if unstable, do damage control and repair later.
2. If contusion, stent, or excise and repair.
3. If found late or partial, stent; if stenting fails, place a nephrostomy and repair later.
4. If ureterovaginal fistula, try a stent first.

- **Damage control**: ligate the proximal stump with long silk and place a nephrostomy once stable, or exteriorise a single-J stent.
- Principles: spare adventitia; debride to bleeding edges; make it **spatulated, tension-free, stented and watertight**; drain. **No bowel or autotransplantation acutely.**
- In trauma, reimplant **non-tunnelled**, because tunnelled reimplants stricture more.
- Psoas hitch beats distal ureteroureterostomy, because the distal stump's supply is tenuous; spare the genitofemoral and femoral nerves.

95 Urinary Lithiasis: Etiology, Epidemiology and Pathogenesis

1 · Overview and epidemiology

- **Removal is surgical, but recurrence is metabolic**. Prophylaxis: see Ch 99.
- Lifetime prevalence is **1–15%**; in the United States ≈ **10%** and rising.
- **Sex**: males outnumber females 2–3 : 1, but the gap is narrowing. **Oestrogen is protective**, hence a second female peak at menopause.
- **Race**: highest in White, lowest in Asian people. **Age**: peak in the 4th–6th decades; paediatric stones are rising.
- **Climate and occupation**: heat lowers urine volume and pH, sunlight causes vitamin D-mediated hypercalciuria, and sedentary work causes hypercalciuria.
- **Obesity**, **metabolic syndrome** and **diabetes** raise risk. **Insulin resistance** unifies these: it lowers urine pH, causing uric acid stones, and hyperinsulinaemia causes hypercalciuria.
- Hypertension, cardiovascular disease and chronic kidney disease (CKD) are associated. Fluid intake is inversely related (proven by randomised controlled trial, RCT).

2 · Physicochemistry and stone formation

Prevention lowers supersaturation or aids inhibition.

LAQ Urolithiasis: etiopathogenesis, medical management and prevention of recurrence

Paper answer · P1 · P2 · P4 · merged from 3

Asked as: P4 LAQ "Describe etiopathogenesis of stone formation and medical management of urolithiasis" / "Describe medical management of recurrent stone disease and discuss recent advances in instrumentations of RIRS" · P2 LAQ "Describe etiopathogenesis of stone formation and medical management of urolithiasis" (also P2 SAQ "Describe medical management of renal calculi") · P1 LAQ "Discuss pathophysiology of renal stones and also describe how to prevent its recurrence"

Introduction

- Urolithiasis is a **chronic metabolic disease** punctuated by stone episodes; recurrence is up to **50% within 5 years** (another source: about half recur within 5–10 years). Removal is surgical; recurrence is metabolic. Recurrent formers need a **metabolic evaluation** and targeted therapy.
- **Composition**: calcium oxalate about 60%, mixed with apatite about 20% (calcium oxalate with or without apatite about 80% in all), uric acid and struvite about 7% each, cystine 1–3%.
- RIRS (retrograde intrarenal surgery) is flexible ureteroscopy inside the kidney.

Aetiology: risk factors

- **Epidemiological**: male sex (gap narrowing), hot climate and occupation (low urine volume), diet (animal protein, salt), family history.
- **Systemic**: **metabolic syndrome**, obesity, diabetes (insulin resistance lowers urine pH), gout, primary hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass,

sarcoidosis.

- **Anatomical (stasis)**: PUJ (pelviureteric junction) obstruction, horseshoe kidney, calyceal diverticulum, medullary sponge kidney (stasis plus metabolic defects).
- **Drugs**: direct (indinavir, triamterene); indirect (acetazolamide, topiramate, vitamin C and D excess).

Physicochemistry: state of saturation

State	Concentration product (CP)	What happens
Undersaturated	Below Ksp (solubility product)	No crystals; stones may dissolve
Metastable	Between Ksp and Kf (formation product)	No spontaneous nucleation, but growth on existing crystals; inhibitors act here. **Zone of therapy**
Unstable	Above Kf	Spontaneous nucleation, growth, aggregation

- Crystals form in metastable urine by local high CP, stasis, or **heterogeneous nucleation** on cells, debris or other crystals (needs less energy).

Steps of stone formation

1. **Supersaturation:** low volume; high calcium, oxalate or uric acid; low citrate.
2. **Nucleation:** homogeneous, or **heterogeneous** on debris or other crystals (needs less energy).
3. **Growth and aggregation:** aggregates, not single crystals, can occlude a tubule within the 5–7 minute nephron transit (**free-particle theory**).
4. **Retention (fixed-particle theory): Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) point to a secondary cause: enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria.

Inhibitors, promoters and matrix

- **Inhibitors**: **citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm–Horsfall protein**, osteopontin, bikunin. There is **no inhibitor of uric acid**.
- **Matrix** (about 2.5% of stone weight; matrix substance A): role uncertain.

Pathophysiology by urinary abnormality

Abnormality (definition)	Mechanism
Hypercalciuria (commonest; >200 mg/day, or >4 mg/kg/day)	Absorptive (gut), renal leak (secondary high PTH [parathyroid hormone]), resorptive (primary hyperparathyroidism): Pak classification
Hyperoxaluria (>40 mg/day)	**Primary:** hepatic enzyme defect (PH1, AGXT gene); **enteric:** malabsorbed fat binds calcium, freeing oxalate; dietary
Hyperuricosuria (>600 mg/day)	Urate promotes calcium oxalate nucleation
Hypocitraturia (<320 mg/day)	Distal RTA, diarrhoea, hypokalaemia, animal protein (acid load, intracellular acidosis)
Uric acid stones	**Low urine pH (<5.5)** more than excess uric acid; pKa 5.5

Abnormality (definition)	Mechanism
Cystine	Autosomal recessive defect of tubular cystine reabsorption
Struvite	**Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate and carbonate apatite; biofilm

Medical management: aims

- Relieve pain, dissolve suitable stones and **prevent recurrence** (up to 50% at 5 years).

Acute colic and MET (medical expulsive therapy)

- **NSAIDs (non-steroidal anti-inflammatory drugs)** first line; opioids only if refractory. Urgent drainage if infection with obstruction.
- **α-blocker** (tamsulosin; silodosin faster) for **distal ureteric stones 5–10 mm** after migration from the kidney: relaxes ureteric muscle. **SUSPEND (2015)** was negative because it included all sizes and sites. Trial of passage 4–6 weeks.

Dissolution (chemolysis)

- **Uric acid** stones (radiolucent) dissolve with oral potassium citrate to alkalinise (EAU [European Association of Urology]: pH 7.0–7.2 for dissolution); allopurinol if hyperuricosuric.
- **Cystine:** alkalinise to pH 7.0–7.5 with thiol drugs. **Struvite:** acetohydroxamic acid occasionally dissolves.

Prevention of recurrence

1. Screen every first-time former (AUA [American Urological Association], EAU)

- History, serum electrolytes, creatinine, calcium, uric acid; urine pH and culture; **stone analysis**; imaging. This sorts patients into high and low risk.

2. General measures (every former)

- **Fluid for urine ≥2.5 L/day** (drink about 3 L). **Borghi 1996**: urine >2 L cut 5-year recurrence (12.1% vs 27%; recurrence halved).
- **Normal calcium**, low sodium (≤2.3 g/day), less animal protein, more fruit and vegetables (DASH [Dietary Approaches to Stop Hypertension]-type diet). **Borghi 2002**: this diet halved recurrence compared with a low-calcium diet. Restricting calcium raises oxalate absorption.
- Limit high-oxalate foods (spinach, nuts, chocolate); weight loss.
- Citrus juices (lemonade, orange) are adjuncts only.

3. Metabolic evaluation that guides therapy (high-risk or recurrent formers, children) (see 2015-Summer Paper 2, Q2)

- **Stone analysis** (infrared spectroscopy or X-ray diffraction); serum chemistry: serum calcium, uric acid, creatinine, electrolytes, PTH if calcium is high-normal.
- One or **two 24-hour urines** on usual diet, at least 4 weeks after an episode: volume, pH, calcium, oxalate, citrate, uric acid, sodium, potassium, creatinine, cystine. Therapy targeted to these results reduces stone events.
- Targets: volume >2.5 L, calcium <250 mg (women) or <300 mg (men), oxalate <40 mg, citrate >320 mg, uric acid <600 mg.

4. Selective drug therapy

Abnormality	Drug and dose	Rationale and notes
Hypercalciuria	**Thiazide**: hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg	Distal calcium reabsorption plus volume contraction. Add potassium citrate: thiazide lowers potassium, and hypokalaemia causes hypocitraturia
Primary hyperparathyroidism	Parathyroidectomy	Cures resorptive hypercalciuria
Hypocitraturia, distal RTA, chronic diarrhoea	**Potassium citrate 30–60 mEq/day**	Alkali load reduces tubular (proximal) citrate reabsorption; avoid urine pH ≥7
Hyperuricosuria / hyperuricosuric calcium oxalate stones	Allopurinol 300 mg/day	Watch hypersensitivity; reduce dose in renal impairment
Uric acid stones	Potassium citrate to **pH 6.0–6.5** (avoid >7.0: calcium phosphate); allopurinol only for hyperuricosuria or failed alkalinisation	A pH disease: pKa 5.5; dissolution possible
Enteric hyperoxaluria	Low fat and oxalate; **calcium with meals**; liquid potassium citrate; magnesium; cholestyramine	Calcium binds oxalate in the gut
Primary hyperoxaluria type 1	Pyridoxine; **lumasiran** (RNA [ribonucleic acid] interference against glycolate oxidase)	ILLUMINATE-A: 84% vs 0% reached near-normal urine oxalate
Cystinuria	Urine ≥3 L/day (another source: >3 L); low sodium and protein; potassium citrate to pH **7.0–7.5**; then **tiopronin** (or D-penicillamine)	Aim cystine <250 mg/L; thiols form soluble mixed disulfides
Struvite	**Complete clearance**; culture-specific antibiotics; AHA (acetohydroxamic acid) 250 mg three times daily (another source: two to three times daily) if surgery impossible	Urease inhibitor; risk of thromboembolism

- **Also know:** recurrent calcium stones with a normal work-up: thiazide or potassium citrate (AUA: even without abnormality); citrus juices are adjuncts.
- **Trials**: Ettinger 1988 (chlorthalidone) and Barcelo 1993 (potassium citrate) reduced new stones. **NOSTONE (2023)**: hydrochlorothiazide 12.5–50 mg did not reduce recurrence, which questions the older trials.

5. Monitoring and follow-up

- Repeat 24-hour urine within **6 months** of starting therapy, then yearly; reanalyse any new stone.
- Serum potassium 1–2 weeks after starting a thiazide or citrate; imaging for new stones.

Diagram to draw: saturation zones (Ksp, Kf) and Randall plaque on a renal papilla.

Diagram to draw: flowchart from stone analysis and 24-hour urine to abnormality to selective drug therapy.

Recent advances in prevention

- **ROKS (Recurrence of Kidney Stone) nomogram** predicts individual risk.
- **Also know:** NOSTONE (2023) negative; citrus juices are adjuncts; *Oxalobacter formigenes* loss; lumasiran for PH1.

Recent advances in RIRS instrumentation

Advance	Principle	Benefit or limitation
Digital flexible ureteroscopes	CMOS (complementary metal-oxide semiconductor) chip at the tip; LED (light-emitting diode) light	Sharper image that does not degrade; tip larger (~8.5 Fr); no more durable
Single-use scopes (LithoVue, 2016)	7.7 Fr tip, ≥270° deflection, new each case	No repair or reprocessing; cheaper below ~99 cases a year; for lower pole and difficult cases
Thulium fibre laser (about 1940 nm)	Long pulse, very high frequency	Fine dust, minimal retropulsion; faster than Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) at low settings
Pulse-modulated (Moses) Ho:YAG	Pulse shaped through a vapour bubble	Less retropulsion, faster dusting
Access sheaths (12/14 Fr)	Outflow channel	Lower intrarenal pressure and temperature; severe ureteric injury in ~1 in 8
Suction sheaths and FANS (flexible and navigable suction access sheath)	Vacuum at the sheath; flexible tip follows the scope into calyces	Aspirates dust; lower pressure; better stone-free rates in early studies
Intrarenal pressure monitoring	Sensor in the wire or scope (LithoVue Elite)	High pressure causes backflow, SIRS (systemic inflammatory response syndrome) and sepsis; safe limits not yet defined
Robotic flexible ureteroscopy (Avicenna Roboflex)	Seated console controls the scope	Ergonomics, less radiation to surgeon

- **Laser safety**: tissue damage above 43 °C, so keep power ≤10 W in the ureter and ≤25 W in the kidney, with continuous cool irrigation.
- **Also know:** dusting versus fragment-and-basket; nitinol tipless baskets; anti-retropulsion devices; Monarch robotic platform for PCNL (percutaneous nephrolithotomy) access.

Key points to remember:

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor for idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi); Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide (plus potassium citrate) for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

Also, from the condensed book (not in the paper answer):

- Free-particle theory (crystals trapped as plugs) vs fixed-particle theory (anchored on plaque).
- Stones form by **supersaturation, then nucleation, growth and aggregation, then retention**. The concentration product (CP) multiplies ion concentrations; the solubility product (Ksp) is CP at saturation; the formation product (Kf) is CP at spontaneous crystallisation.

State	CP	What happens
Undersaturated	< Ksp	No crystals; stones may dissolve
Metastable	Ksp–Kf (most urinary salts)	No homogeneous (spontaneous) nucleation; growth and heterogeneous nucleation possible; **inhibitors effective, so this is the zone of therapy**
Unstable	> Kf	Spontaneous nucleation; inhibitors ineffective

- The **relative saturation ratio (RSR)** is CP ÷ Ksp. Below 1, stones dissolve; above 1, they grow.
- Changing pH lowers the saturation of **phosphate, uric acid and cystine, but not oxalate**. **Calcium and oxalate contribute equally** to calcium oxalate saturation.
- Urine holds calcium oxalate at ≈ 4× its solubility in water; with inhibitors it precipitates only at **7–11×**.
- Nephron transit is too short for one crystal to block a tubule, so aggregation occludes.
- **Randall plaque** (MCQ 1937) is **calcium apatite**. It starts in the **thin-limb basement membrane**, spreads through the interstitium and erodes the papilla; calcium oxalate then grows on it.

Stone former	Randall plaque	Ductal plugs
Idiopathic calcium oxalate	Present, as the anchor	Uncommon, mild
Intestinal bypass	Absent	Apatite plugs in inner medullary collecting ducts; injury, fibrosis
Brushite	Present	Apatite plugs; interstitial fibrosis

Stone former	Randall plaque	Ductal plugs
Distal renal tubular acidosis (RTA), hyperparathyroidism, cystinuria	—	Present

> **KEY POINT**
>
> **Plaque vs plug**: a plaque is an idiopathic interstitial apatite anchor; a plug is intratubular obstruction with injury, caused by systemic disease. **Ductal plugs should prompt a search for a secondary cause.** Infection or shock-wave lithotripsy may convert a calcium oxalate former into a brushite former.

3 · Inhibitors and promoters

- Inhibitors raise the supersaturation needed for nucleation. **Uric acid and cystine have no inhibitor**; only pH and concentration matter. Promoters: low volume and citrate; high calcium, oxalate, phosphate, uric acid.

Inhibitor	Action	Key feature
Citrate	Complexes calcium; inhibits precipitation, agglomeration and urate-induced nucleation	Most important small-molecule inhibitor
Magnesium	Complexes oxalate	Synergy with citrate, phytate
Glycosaminoglycans	Bind crystal surface	Lower in stone formers
Nephrocalcin	Inhibits calcium oxalate monohydrate nucleation, growth, aggregation	Formers excrete isoforms lacking γ-carboxyglutamic acid
Tamm-Horsfall protein	Inhibits aggregation, not growth	Most abundant urinary protein, made in the thick ascending limb (TAL); **a promoter in acid urine**
Osteopontin	Inhibits nucleation, growth, aggregation, epithelial binding	Inducible

- **Matrix** is ≈ **2.5%** of stone weight.

4 · Mineral metabolism

- **Calcium**: 30–40% is absorbed in the small bowel, transcellularly (calcitriol-dependent) at low intake and paracellularly at high intake.
- **Oxalate-rich food lowers calcium absorption, and luminal calcium lowers oxalate absorption**.
- **Calcitriol**, made in the proximal tubule under **parathyroid hormone (PTH) and hypophosphataemia**, is the most potent stimulator of gut calcium absorption.
- **PTH** causes bone resorption, raises renal calcium reabsorption and calcitriol, and lowers phosphate reabsorption; it has **no direct gut effect**.

Segment	Filtered calcium reabsorbed	Route
Proximal tubule	**60–65%**	Paracellular; claudin-2
TAL	25–30%	Paracellular, lumen-positive; claudin-16/19; the calcium-sensing receptor (CaSR) acts through claudin-14 to block it

Segment	Filtered calcium reabsorbed	Route
Distal convoluted tubule (DCT)	8–10% (fine control)	Transcellular: entry by TRPV5 channel, then carried by calbindin-D28k

- **Oxalate**: 6–14% is absorbed, unregulated, and excreted almost entirely in urine; half is dietary, half hepatic. **Ascorbic acid** is the largest endogenous source; Oxalobacter formigenes degrades oxalate.

5 · Classification

Stone	Frequency
Calcium oxalate (pure or with apatite)	**60%**
Hydroxyapatite	20%
Brushite	2%
Uric acid	≈ 7% (8–10% US, rising)
Struvite	≈ 7% (falling)
Cystine	1–3% (up to 10% children)
Triamterene, silica, 2,8-dihydroxyadenine (2,8-DHA)	< 1% each

Condition	Defect	% of formers
Absorptive hypercalciuria (incl. phosphate leak)	Increased gut calcium absorption	20–40
Renal hypercalciuria	Renal calcium leak	5–8
Resorptive hypercalciuria	Primary hyperparathyroidism (PHPT)	3–5
Hyperuricosuric calcium	Purine excess	10–40
Hypocitraturic calcium	Idiopathic, diarrhoea, distal RTA, thiazide	10–50
Hyperoxaluric calcium	Primary, enteric, dietary	2–15
Hypomagnesiuric calcium	Reduced gut magnesium absorption	5–10
Idiopathic low urine pH	Low pH	15–30
Low urine volume	Inadequate intake	10–50
Infection; cystinuria	Urease; cystine transport	1–5; < 1
No abnormality	—	< 3

> **KEY POINT**
>
> Classify by the **urinary abnormality, not the stone**: composition starts the work-up. Fewer than 3% have no abnormality. Work-up: see Ch 99.

6 · Hypercalciuria

Fasting urine and PTH reveal the source.

- It is the commonest abnormality (35–65%).

- **Definition**: > **200 mg/day** after 1 week on a 400 mg calcium, 100 mEq sodium diet; or > 4 mg/kg/day.
- The **Pak classification** (gut, kidney, bone) is an exam framework, not a therapy guide.

Feature	Absorptive	Renal	Resorptive
Defect	Increased gut absorption	Renal leak	PHPT
Serum calcium	Normal	Normal	**High**
PTH	**Suppressed**/normal	**High** (secondary)	High
Fasting urinary calcium	**Normal** (< 0.11 mg/dL glomerular filtrate)	**High**	High
Serum phosphate	Normal (low in phosphate leak)	Normal	Low
Sequence	Hyperabsorption suppresses PTH and raises the filtered load	Renal loss raises PTH, which raises calcitriol	PTH drives bone resorption and calcitriol, causing hypercalcaemia

- Absorptive **type I** persists on calcium restriction; **type II** normalises.
- **CYP24A1** loss stops calcitriol inactivation, causing hypercalciuria and nephrocalcinosis. In **renal phosphate leak**, hypophosphataemia raises calcitriol and gut calcium absorption.

Disorder	Gene	Features
Dent disease (X-linked)	ClC-5	Hypercalciuria, **low-molecular-weight proteinuria**, nephrocalcinosis, renal failure
Familial hypomagnesaemia with hypercalciuria and nephrocalcinosis (FHHNC)	Claudin-16, -19	TAL magnesium/calcium loss
Bartter	NKCC2, ROMK, ClC-Kb, barttin; type V activating CaSR	Salt wasting, hypokalaemic alkalosis, hypercalciuria

> **KEY POINT**
>
> **Fasting urine and PTH separate gut from kidney**: in renal leak, fasting calcium and PTH are high; in absorptive hypercalciuria, fasting calcium is normal and PTH suppressed.

- **PHPT** causes ≈ 5% of stones, and stones occur in < 5% of PHPT. Suspect it if serum calcium is > **10.1 mg/dL**; repeat with intact PTH, and use ionised calcium for normocalcaemic PHPT. A **thiazide challenge** unmasks it.

Cause	Mechanism	PTH	Clue
PHPT	Autonomous PTH	High/inappropriately normal	Commonest in outpatients
Sarcoidosis	Macrophage 1α-hydroxylase	Suppressed	Corrects with **corticosteroids**
Malignancy	PTH-related protein (PTHrP), osteolysis (RANKL)	Suppressed	Commonest in inpatients

Cause	Mechanism	PTH	Clue
Glucocorticoid excess	Increased bone resorption	Raised	Cushing syndrome

7 · Hyperoxaluria

- **Definition**: > **40 mg/day**. Types: primary, enteric, dietary and idiopathic.

SAQ Primary Hyperoxaluria

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 10 Paper 2, Q3f

Definition

- Rare **autosomal recessive** errors of **hepatic glyoxylate metabolism**: glyoxylate is not converted to glycine and is oxidised to oxalate, an end product cleared only by the kidney.
- Hyperoxaluria = urinary oxalate >40 mg/day; in PH (primary hyperoxaluria) often **>100 mg/day**.

Types

	PH1 (~70%)	PH2 (~10%)	PH3 (~10%)
Enzyme (gene)	**AGT** (alanine-glyoxylate aminotransferase; **AGXT**), liver peroxisome, **pyridoxine (B6) cofactor**	GRHPR (glyoxylate reductase/hydroxypyruvate reductase)	HOGA1 (4-hydroxy-2-oxoglutarate aldolase), mitochondria
Urine marker	**Glycolate**	**L-glycerate**	4-hydroxy-2-oxoglutarate
Course	**Most severe**; ESKD (end-stage kidney disease) in 2nd–3rd decade	Milder, can reach ESKD	Childhood stones; rarely ESKD

Pathophysiology

- High urinary oxalate → calcium oxalate crystals in tubules → **nephrocalcinosis** and recurrent stones.
- **Vicious circle:** falling GFR (glomerular filtration rate) reduces oxalate clearance → plasma oxalate rises → **systemic oxalosis** (bone, retina, heart, vessels, skin).

Clinical features

- Spectrum from **infantile oxalosis with renal failure** to occasional adult stones; some diagnosed only after a transplant fails.
- Clues: childhood stones, nephrocalcinosis, family history, **calcium oxalate monohydrate** stones.

Diagnosis

`24-h urinary oxalate ↑ (child: spot oxalate:creatinine ratio) → urinary glycolate / L-glycerate / 4-hydroxy-2-oxoglutarate → genetic testing (AGXT, GRHPR, HOGA1) all normal → secondary (enteric, dietary) hyperoxaluria`

- Plasma oxalate when GFR is low; screen siblings and related living donors.

Management

- **Hyperhydration** (about 3 L/m²/day), **potassium citrate**, normal calcium intake.
- **Pyridoxine** trial in all PH1: responsive mainly with the Gly170Arg mutation; it restores AGT activity and peroxisomal targeting.
- **Lumasiran** (RNA interference against glycolate oxidase, which cuts glyoxylate supply): **ILLUMINATE-A**: 84% vs 0% reached urinary oxalate ≤1.5 × upper limit of normal. **Also know:** nedosiran.
- Stones: ureteroscopy or PCNL (percutaneous nephrolithotomy); limit SWL (shock wave lithotripsy).
- **ESKD:** intensive dialysis, then **combined liver–kidney transplant for PH1** (only the liver replaces the enzyme; a lone kidney would be re-damaged). Kidney alone usually suffices for PH2.

Diagram to draw: hepatic glyoxylate pathway showing the AGT block (PH1) and the lumasiran target (glycolate oxidase).

> **Key points to remember:**
> - PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.
> - Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
> - Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

ALGORITHM

1. **Enteric hyperoxaluria**
2. Fat malabsorption leaves fatty acids in the gut lumen.
3. Fatty acids saponify calcium and magnesium, so less calcium binds oxalate and more oxalate is free.
4. Free oxalate undergoes **colonic hyperabsorption**, helped by bile salts.

- **Enteric** is the commonest acquired form: small-bowel resection or Crohn disease, jejunoileal bypass and malabsorptive bariatric surgery (not band or sleeve).
- It **needs an intact colon**; with an ileostomy, uric acid or ammonium urate stones form instead. Diarrhoea also causes hypocitraturia, hypomagnesiuria and low urine pH.
- **Dietary**: nuts, chocolate, tea, spinach, rhubarb and vitamin C. **Severe calcium restriction is counter-productive**, as less calcium binds gut oxalate.

8 · Hyperuricosuria, hypocitraturia, hypomagnesiuria

- **Hyperuricosuria** is > **600 mg/day**. Urate promotes calcium oxalate stones by **adsorbing inhibitors, "salting out" and epitaxy**; allopurinol cut recurrence (trial below).
- **Hypocitraturia** is < **320 mg/day**; **acid–base status is the main determinant**. Causes: distal RTA, chronic diarrhoea, animal protein, thiazides (through hypokalaemia), ACE inhibitors.

> **KEY POINT**
>
> Citrate excretion reads the proximal-tubular **intracellular pH**: acidosis, hypokalaemia or an acid diet make the tubule reclaim citrate. Hence potassium citrate is the treatment (see Ch 99).

- Hypomagnesiuria: **two RCTs of magnesium showed no benefit**. **Low urine pH** (< 5.5) causes uric acid and calcium oxalate stones.

9 · Renal tubular acidosis

SAQ Renal tubular acidosis (distal and type 2)

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P2 SAQ "Renal tubular acidosis" · P1 SAQ "Distal Renal tubular acidosis." · P3 SAQ "Type 2 Renal tubular acidosis?"

Definition

- **RTA (renal tubular acidosis)**: hyperchloraemic, normal AG (anion gap) metabolic acidosis from defective tubular H^+ secretion or HCO_3^- reabsorption.
- **Distal (type 1) RTA:** the collecting duct cannot secrete H^+ despite systemic acidosis. Result: **hypokalaemic, hyperchloraemic, normal anion gap metabolic acidosis** with **urine pH >6**: the only acidosis with alkaline urine.
- **Proximal (type 2) RTA:** failure of the proximal tubule to reclaim bicarbonate, causing hyperchloraemic, normal anion gap metabolic acidosis.

Comparison of types

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4 (hyperkalaemic)
Defect	H^+ secretion (α-intercalated cell)	Proximal HCO_3^- reabsorption	Aldosterone deficiency or resistance
Urine pH in acidosis	**Always >6**	<5.5 at steady state	Low
Potassium	Low	Low	**High**
Serum HCO_3^-	Low, may be very low	15–18 mEq/L (P3: 14–18 mEq/L)	Mildly low
Urine calcium	High	Normal or mildly high	Low
Urine citrate	**Profoundly low**	Near normal	–
Stones	**Common: up to 70%; calcium phosphate, nephrocalcinosis**	Uncommon	Uncommon
Causes	Sjögren, lupus, inherited	Fanconi	Diabetes, obstruction
Treatment	**Potassium citrate** 1–2 mEq/kg/day	Alkali 10–15 mEq/kg/day plus potassium	Fludrocortisone, low-potassium diet, stop offending drugs

Distal (type 1) RTA

Normal acidification and the defect

- **α-intercalated cells** secrete H^+ by apical H^+-ATPase; bicarbonate exits by the basolateral Cl^-/HCO_3^- exchanger (AE1). Normally urine pH can fall to 4.5.
- In distal RTA, potassium is secreted in place of H^+, causing hypokalaemia.

Causes

Inherited	Acquired (secondary)
SLC4A1 (AE1): mostly dominant	Sjögren syndrome, SLE (systemic lupus erythematosus)
ATP6V1B1, ATP6V0A4 (H^+-ATPase): recessive, **sensorineural deafness**	Obstructive uropathy, pyelonephritis, medullary sponge kidney
CA2: mixed proximal–distal (carbonic anhydrase II deficiency)	Hypercalciuria, hyperparathyroidism, amphotericin B

Why stones form (urological importance)

- **Stones in up to 70%**: **calcium phosphate** stones and **nephrocalcinosis**. Stones lead to diagnosis in up to half.
- **Profound hypocitraturia** (most important): acidosis increases proximal citrate reabsorption.
- **Hypercalciuria:** bone buffering releases calcium. **Fixed alkaline (high) urine pH** precipitates calcium phosphate.

Clinical features

- Stones, nephrocalcinosis, weakness (hypokalaemia), osteomalacia or rickets; children fail to thrive (growth failure).
- **Incomplete distal RTA:** normal bicarbonate, but urine pH cannot fall below 5.5 after an acid load.

Proximal (type 2) RTA

Pathophysiology

- The proximal tubule reclaims **85–90% of filtered bicarbonate** (luminal carbonic anhydrase; exit by the basolateral Na–HCO_3 cotransporter).
- **Self-limiting:** bicarbonate wasting lowers plasma HCO_3^- until the filtered load drops below the lowered threshold; the intact distal tubule then copes, giving a **steady state at about 14–18 mEq/L (P2: 15–18 mEq/L) with urine pH <5.5**.
- **Hypokalaemia** from distal sodium bicarbonate delivery; worsens on alkali.

Causes

Isolated	Generalised (Fanconi syndrome)
Carbonic anhydrase inhibitors (acetazolamide, **topiramate**)	**Cystinosis** (commonest in children), Lowe, Wilson, galactosaemia
Carbonic anhydrase II deficiency (mixed)	**Ifosfamide, cisplatin**, aminoglycosides, valproate
	Myeloma, amyloidosis, lead

- **Fanconi:** glycosuria, aminoaciduria, phosphaturia, uricosuria.

Clinical features

- Children: growth failure, **rickets**; adults: osteomalacia, weakness.
- **Stones uncommon**: urinary citrate stays near normal and urine is acid at steady state. Topiramate and acetazolamide are exceptions (alkaline urine, calcium phosphate stones).

Type 4 (hyperkalaemic) RTA

- Hyperkalaemia, low urine pH; diabetes, obstructive uropathy. Amiloride and trimethoprim cause hyperkalaemic distal RTA.

Diagnosis

- Bloods: low bicarbonate, high chloride, low potassium (types 1 and 2); urine pH >6 on a fasting morning sample in distal RTA; urine can acidify below 5.5 in proximal RTA.
- **UAG (urine anion gap) = (Na^+ + K^+) − Cl^-** reflects urinary ammonium: **positive** (low ammonium excretion) in renal causes; negative in GI (gastrointestinal) loss such as diarrhoea.
- **Ammonium chloride load** (0.1 g/kg orally, hourly urine pH for about 6 h): **failure to fall below 5.5** confirms distal RTA. The **furosemide–fludrocortisone test** is a better-tolerated alternative.
- **Bicarbonate loading:** urine pH rises; **fractional bicarbonate excretion >15%** indicates proximal RTA. Look for Fanconi features; drug history.
- 24-hour urine: low citrate, high calcium (distal); ultrasound for nephrocalcinosis.

Treatment

- **Distal RTA: potassium citrate** (about 1–2 mEq/kg/day in divided doses): corrects acidosis, hypokalaemia and hypocitraturia, reduces calcium excretion and stone formation. Sodium alkali is less suitable because sodium raises urinary calcium. Treat the cause; manage stones endourologically.
- **Proximal RTA:** treat the cause; **large alkali doses (about 10–15 mEq/kg/day)** as potassium citrate or mixed citrate, replacing potassium; thiazide; phosphate and vitamin D for rickets.
- **Type 4:** fludrocortisone, low-potassium diet, stop offending drugs.

Urological relevance

- Obstruction causes hyperkalaemic distal RTA, reversible by relief.
- Topiramate and acetazolamide mimic RTA (alkaline urine, low citrate, calcium phosphate stones).
- Potassium citrate beats sodium alkali, which raises calcium excretion.
- **Also know:** CKD (chronic kidney disease) acidosis; amiloride and trimethoprim (hyperkalaemic distal RTA); mixed type (carbonic anhydrase II deficiency); medullary sponge kidney; laxative abuse (ammonium urate stones); inherited forms (recessive with deafness); growth failure in children.

Diagram to draw: nephron with defect sites; α-intercalated cell with H^+-ATPase, AE1 and the stone mechanism; proximal tubular cell with carbonic anhydrase.

> **Key points to remember:**
> - Type 1 (distal): urine pH >6 with acidosis, hypokalaemia, hypocitraturia (key), positive UAG; calcium phosphate stones in up to 70%.
> - NH_4Cl load: pH fails to fall below 5.5 in distal RTA.
> - Potassium citrate is the treatment for distal RTA.
> - Type 2 (proximal): bicarbonate wasting; HCO_3^- 14–18 mEq/L (P2: 15–18); urine pH <5.5 at steady state; usually Fanconi (ifosfamide, cisplatin).
> - Fractional HCO_3^- excretion >15%; stones uncommon in type 2.
> - Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
> - Positive UAG = renal cause.

Also, from the condensed book (not in the paper answer):

- Genetics: SLC4A1 (dominant); ATP6V1B1/ATP6V0A4 (recessive, with **sensorineural hearing loss**).

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4
Defect	Distal H^+ secretion	Proximal HCO_3^- reabsorption	Aldosterone deficiency/resistance
Serum K^+	Low	Low	**High**
Urine pH	**Always > 6.0**	< 5.5 at steady state	Low
Urinary citrate	**Profoundly low**	Near normal	—
Stones	**Up to 70%**: calcium phosphate, nephrocalcinosis	Uncommon	Uncommon
Associations	Sjögren, lupus, obstruction; genetic	Fanconi	Diabetic nephropathy

- **Incomplete distal RTA**: electrolytes are normal, but urine pH cannot fall below 5.5 after an ammonium chloride load.
- In distal RTA, **profound hypocitraturia** is probably the most important stone factor.

> **KEY POINT**
>
> Stone risk in RTA follows citrate and pH: only distal RTA combines fixed alkaline urine with profound hypocitraturia and hypercalciuria, the recipe for **calcium phosphate** stones.

10 · Uric acid stones

LAQ Uric acid calculi: etiopathogenesis, medical management and allopurinol

Paper answer · P2

Asked as:

- Discuss etiopathogenesis of uric acid calculi. Describe rationale of medical management, — *Practice Set 9 Paper II Q2 (LAQ)*
- Allopurinol — *Practice Set 10 Paper II Q3a (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction

- Uric acid stones form **8–10%** of stones in the USA and are increasing with obesity and diabetes. They are **radiolucent** and can be **dissolved medically**.

Etiopathogenesis

1. Why humans form uric acid stones

- Humans and apes lack hepatic **uricase**, which converts uric acid to allantoin (10–100 times more soluble). Blood and urine uric acid are therefore high.

2. Physical chemistry: pH is the key

- Uric acid is a weak acid, **pKa about 5.35–5.5**. At pH 5.5 half is free uric acid and half is urate.

- Free uric acid is poorly soluble (about 96–100 mg/L); sodium urate is about 20 times more soluble. At **pH 6.5, over 90% is ionised**.
- **No natural inhibitor** of uric acid crystallisation is known, so stone formation depends almost entirely on supersaturation.

3. The three determinants

Determinant	Causes	Importance
Low urine pH (<5.5)	Insulin resistance, obesity, type 2 diabetes, gout, high animal-protein diet, chronic diarrhoea, ileostomy	**Most important**; most formers have normal uric acid excretion
Low urine volume	Hot climate or occupation, diarrhoea, dehydration	Concentrates uric acid
Hyperuricosuria (>600 mg/day)	Purine-rich diet, gout, myeloproliferative disorders, tumour lysis, uricosuric drugs, inherited defects	Least often the driver in idiopathic cases

4. Why urine is too acid: insulin resistance

- Insulin normally stimulates **ammonia production** from glutamine in the proximal tubule and the Na^+/H^+ exchanger NHE3.
- Insulin resistance reduces ammonium excretion. Acid is then buffered as titratable acid, so urine pH falls.
- **Renal lipotoxicity:** fat accumulates in the proximal tubule, which then burns fatty acids instead of glutamine, so ammoniagenesis falls further.
- Endogenous acid production is also higher. Uric acid stones are thus a **renal manifestation of the metabolic syndrome**; about a third of stones in diabetics are uric acid.

5. Aetiological categories

- **Congenital:** HGPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency (Lesch–Nyhan), PRPP (phosphoribosylpyrophosphate) synthetase overactivity, hereditary renal hypouricaemia (URAT1, the urate transporter).
- **Idiopathic:** low urine pH without gout ("gouty diathesis").

6. Uric acid and calcium stones

- In **hyperuricosuric calcium oxalate stone disease**, urate promotes calcium oxalate by removing inhibitors, by "salting out" and by **epitaxy** (urate crystals act as a nidus).
- **Also know:** ammonium acid urate stones (laxative abuse, diarrhoea, ileostomy); theobromine (cocoa) inhibits uric acid crystallisation in vitro.

Diagnosis in brief

- Radiolucent on X-ray; CT **200–400 HU (Hounsfield units)**; **dual-energy CT** identifies uric acid.
- Urine pH persistently <5.5; serum uric acid; 24-hour urine; screen for diabetes.

Rationale of medical management

Principle: uric acid stones are a **"pH disease" more than a uric acid disease**. Raising pH turns insoluble uric acid into soluble urate. Diluting urine lowers supersaturation. Lowering uric acid production helps only when excretion is high.

1. Fluids

- Urine output **at least 2.5 L/day** lowers uric acid concentration.

2. Diet and lifestyle

- Limit non-dairy **animal protein** (it gives purines and an acid load; dietary purines provide about 30% of urinary uric acid).
- More fruit and vegetables (alkali load). Weight loss, exercise and glucose control correct the underlying insulin resistance.

3. Urinary alkalinisation (mainstay)

- **Potassium citrate 30–60 mEq/day** in two or three doses. Citrate is metabolised to bicarbonate, raising urine pH.
- **Target pH 6.0–6.5** for prevention (EAU [European Association of Urology]: 6.2–6.8); **avoid pH >7.0**, which risks calcium phosphate stones.
- **Why potassium, not sodium:** sodium alkali raises urinary calcium and can cause calcium stones. Sodium bicarbonate or sodium citrate is used if potassium is unsafe (renal impairment, hyperkalaemia).
- Patients **monitor urine pH with dipsticks**.

4. Dissolution (oral chemolysis)

- Pure uric acid stones dissolve over weeks to months. **EAU: pH 7.0–7.2**; stent or nephrostomy if obstructed or infected; monitor by imaging.

5. Xanthine oxidase inhibitors

- **Allopurinol 300 mg/day** lowers uric acid production. It is **not first-line**, because low pH, not hyperuricosuria, is the usual cause.
- Indications: hyperuricosuria with hyperuricaemia (e.g. gout), recurrence despite alkalinisation (AUA, American Urological Association), hyperuricosuric calcium oxalate stones.
- Reduce the dose in renal impairment. Risks: rash; **allopurinol hypersensitivity syndrome** (2–8% have reactions; may be fatal); Stevens–Johnson syndrome.
- **Febuxostat** is an alternative. **Rasburicase** (recombinant uricase) prevents uric acid nephropathy in tumour lysis.

Landmark trial: Ettinger (1986), randomised placebo-controlled trial: allopurinol more than halved recurrence in hyperuricosuric, normocalciuric calcium oxalate stone formers.

Allopurinol (in detail)

Introduction and mechanism

- A **purine (hypoxanthine) analogue** that inhibits **xanthine oxidase (xanthine dehydrogenase)**. This blocks hypoxanthine → xanthine → uric acid, lowering serum and urinary uric acid.
- Metabolised in the liver to **oxypurinol**, the active metabolite. It has a long half-life and is cleared by the kidney, so the dose is cut in renal impairment.

Uses in urology

Indication	Rationale
Recurrent calcium oxalate stones with hyperuricosuria (>600 mg/day) and normal urinary calcium	Urate "seeds" calcium oxalate (heterogeneous nucleation, salting out, inhibitor adsorption). **Ettinger (1986) RCT (randomised controlled trial): recurrence more than halved**; AUA (American Urological Association) recommends it
Uric acid stones	**Not first line:** low urine pH is the main cause, so alkalinise with potassium citrate to pH 6.0–6.5. Allopurinol is added for hyperuricosuria, gout or failure of alkalinisation
2,8-DHA (dihydroxyadenine) stones (APRT (adenine phosphoribosyltransferase) deficiency)	Blocks adenine → DHA; febuxostat more effective
Tumour lysis prophylaxis; gout; HPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency, PRPS (phosphoribosylpyrophosphate synthetase) overactivity	Prevents urate nephropathy

Dose

- Adult **300 mg/day** (start 100 mg, titrate); child 4–10 mg/kg/day (maximum 300 mg). Reduce for low creatinine clearance.

Adverse effects

- Rash (commonest), myalgia, gastrointestinal upset; gout flare on starting; interstitial nephritis.
- **Hypersensitivity in 2–8%**, from skin reactions to **allopurinol hypersensitivity syndrome** (fatal in up to 27%) and Stevens-Johnson syndrome. Risk is higher with high dose, renal failure, diuretics and statins. **HLA-B*58:01** carriers (some Asian populations) have higher risk.
- **Xanthine stones** (radiolucent): rare, mainly with high doses in Lesch-Nyhan syndrome.

Drug interactions

- **Azathioprine and 6-mercaptopurine:** xanthine oxidase metabolises them, so toxicity (marrow suppression) rises. Reduce their dose sharply (important after transplant).

Alternatives

- **Febuxostat:** non-purine xanthine oxidase inhibitor, no dose change in mild to moderate renal impairment. Carries a cardiovascular-death warning.
- **Also know:** rasburicase (urate oxidase) for tumour lysis.

Diagram to draw:

- Sigmoid curve of uric acid solubility against urine pH, marking pKa 5.5 and target 6.0–6.5.
- Purine pathway (hypoxanthine → xanthine → uric acid) showing the xanthine oxidase block.

Key points to remember:

- Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
- Insulin resistance reduces ammoniagenesis, so urine is acid.
- Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
- Allopurinol only for hyperuricosuria or failed alkalinisation.
- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

Also, from the condensed book (not in the paper answer):

- Uric acid forms about one-third of stones in type 2 diabetes.

Determinant	Causes	Importance
Low urine pH	Insulin resistance/obesity, animal protein, diarrhoea, gout	**Single most important**
Low volume	Diarrhoea, heat	Concentrates uric acid
Hyperuricosuria	Purine diet, gout, myeloproliferative disease, uricosurics	Least often the driver

- Humans lack hepatic **uricase**, so uric acid is not converted to soluble allantoin.
- The pKa of uric acid is **5.35**. Undissociated uric acid dissolves only to ≈ 100 mg/L, but at pH 6.5 > 1200 mg/L stays soluble.
- Stones are radiolucent on plain film but visible on non-contrast CT.

ALGORITHM

1. **Why the urine is acid in insulin resistance**
2. Insulin resistance and lipotoxicity lower proximal ammoniagenesis and sodium–hydrogen exchanger 3 (NHE3) activity.
3. H^+ is excreted as titratable acid instead of ammonium.
4. Urine pH stays **persistently low**, uric acid stays undissociated, and stones form.

KEY POINT

Uric acid stones are a disease of **urine pH, not uric acid**. No inhibitor exists, so raising pH above the pKa removes the driver; alkalinisation can dissolve stones (see Ch 99).

11 · Cystine and struvite stones

Both recur aggressively without lifelong prevention or complete clearance.

SAQ Cystinuria

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 21 SAQ Sheet, Q3a

Definition and genetics

- **Autosomal recessive** defect of proximal tubular (and jejunal) reabsorption of the dibasic amino acids **COLA: cystine, ornithine, lysine, arginine**. Only cystine is insoluble, so only

cystine forms stones.

- Transporter subunits and ICC (International Cystinuria Consortium) types:

Gene	Subunit	Chromosome	Type
SLC3A1	rBAT (heavy)	2p	A
SLC7A9	$b^{0,+}$AT (light)	19q	B
Both	—	2 and 19	AB

- Fractional excretion of cystine rises from about 0.4% to up to 100%. Incidence 1 in 1000 to 1 in 17,000; **1–3% of adult stones, up to 10% of paediatric stones**; males more severe.

Pathophysiology

- **Pure supersaturation disease**: no specific inhibitor. Solubility is pH dependent: about 300 mg/L at pH 5, 400 mg/L at pH 7, 1000 mg/L at pH 9.
- Coexisting hypocitraturia (commonest), hyperuricosuria or hypercalciuria cause mixed calcium–cystine stones.

Clinical features

- First stone at mean about 12 years; recurrent, bilateral, staghorn stones; CKD (chronic kidney disease) from repeated surgery (about a quarter of adults have eGFR [estimated glomerular filtration rate] <60).

Diagnosis

- **Hexagonal crystals** in urine (always abnormal); cyanide–nitroprusside screen; 24-hour urinary cystine (normal <60 mg/day/1.73 m^2; homozygotes often >400).
- Stone: yellow, waxy, faintly radiopaque (sulphur); **resists SWL (shock wave lithotripsy)**.

Management

- **Fluids:** urine output **≥3 L/day**, including at night.
- **Diet:** sodium ≤2300 mg/day; limit animal protein (not in children).
- **Alkalinise to pH 7.0–7.5** with potassium citrate 30–60 mEq/day; avoid >7.5 (calcium phosphate stones) and sodium alkali (raises cystine excretion).
- **Thiols** if cystine stays >250 mg/L: thiol–cystine mixed disulfide is soluble.
- **Tiopronin** (choice): 300 mg twice daily, up to 1200 mg/day.
- **D-penicillamine:** 250 mg/day, titrated; rash, proteinuria, nephrotic syndrome, pancytopenia.
- Captopril: weak evidence.
- Monitor blood count, liver tests, urine protein.
- **Surgery:** ureteroscopy or PCNL (percutaneous nephrolithotomy) with laser; nephron-sparing approach.

Diagram to draw: proximal tubule cell with rBAT/$b^{0,+}$AT transporter and hexagonal crystal.

> **Key points to remember:**
>
> - Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
> - Hexagonal crystals; target cystine <250 mg/L.
> - Urine ≥3 L, pH 7.0–7.5, then tiopronin.

SAQ Struvite stones.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2018-Summer Paper 4, Q3a

Definition and epidemiology

- **Infection stones:** magnesium ammonium phosphate hexahydrate ($MgNH_4PO_4 \cdot 6H_2O$), usually mixed with **carbonate apatite**.
- 5–15% of stones (falling); female:male ~2:1; highest in India and Pakistan. Risk: neurogenic bladder, spinal injury, diversion, catheters, obstruction, diabetes, elderly.

Pathogenesis

```
Urease-producing bacteria Urea + H2O -> 2NH3 + CO2 NH3 -> NH4+ (urine pH > 7.2)
CO2 -> carbonate; phosphate -> PO4(3-) + Mg -> struvite; + Ca -> carbonate
apatite
```

- Sterile urine has no urease, so infection is essential. Ammonia damages the GAG (glycosaminoglycan) layer; bacteria live inside the stone in a **biofilm**, so antibiotics cannot sterilise a retained stone.
- **Organisms: Proteus mirabilis** (commonest), Klebsiella, Pseudomonas, Providencia, Staphylococcus, **Ureaplasma** (easily missed). E. coli is rarely urease-positive.

Clinical features

- Recurrent UTI (urinary tract infection), fever, flank pain, haematuria; often few symptoms. Grow fast into **staghorn** calculi. Untreated: pyonephrosis, XGP (xanthogranulomatous pyelonephritis), sepsis, renal loss.

Evaluation

- Urine pH >7; culture (urine and stone cultures often differ). NCCT (non-contrast computed tomography): burden and anatomy; DMSA (dimercaptosuccinic acid) or renogram for split function. Stone analysis; 24-h urine (many have hypercalciuria or hypocitraturia).

Management

- **Complete stone clearance** is the goal: each fragment is an infected nidus.
- Culture-specific antibiotics before surgery; drain an obstructed infected kidney first.
- **PCNL (percutaneous nephrolithotomy): first line** for staghorns (AUA, American Urological Association). Add flexible nephroscopy, ECIRS (endoscopic combined intrarenal surgery) or second look. SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery) for small stones only.
- Anatrophic nephrolithotomy rarely; nephrectomy for a poorly functioning kidney.
- **Prevention:** surveillance cultures, suppressive antibiotics, fluids, correct stasis or metabolic defects.
- **AHA (acetohydroxamic acid)** 250 mg three times daily (urease inhibitor), only when surgery is exhausted. Side effects in up to 30% (haemolytic anaemia, DVT (deep vein thrombosis)); avoid if creatinine >2.5 mg/dL or pregnancy.
- **Also know:** hemiacidrin irrigation (historical); matrix stones (radiolucent, Proteus).

Diagram to draw: urease cascade; staghorn with PCNL tract.

Key points to remember:

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

Also, from the condensed book (not in the paper answer):

- **Cystinuria genes**: SLC3A1 (type A), SLC7A9 (type B), or both (type AB). It is autosomal recessive, and with **no inhibitor** cystine is soluble only at alkaline pH (≈ 1000 mg/L at pH 9).
- **Struvite**: **Proteus mirabilis** is commonest; **Ureaplasma** is easily missed. **Less than half of staghorns are infection stones**, and bacteria within the stone demand complete clearance.

ALGORITHM

1. **Urease cascade**
2. Urease splits urea into ammonia (NH_3) and CO_2.
3. Ammonia forms NH_4^+ and OH^-, raising urine pH to 7.2–8.0.
4. With magnesium, phosphate forms **struvite**; with calcium, carbonate apatite.

12 · Miscellaneous and drug-related stones

	Xanthine	2,8-DHA
Enzyme deficient	Xanthine dehydrogenase (XDH)	Adenine phosphoribosyltransferase (APRT)
Radiology	Both radiolucent, so mistaken for uric acid	
Treatment	—	Allopurinol or febuxostat (more effective)

- **Ammonium acid urate** stones are radiolucent and follow diarrhoea, ileostomy, **laxative abuse** and infection.

Route	Drug	Mechanism / stone
Direct	**Indinavir**	Insoluble at physiological pH; stones 4–13%; radiolucent even on CT
Direct	Triamterene; guaifenesin/ephedrine; silicates	Drug stones
Indirect	Acetazolamide, **topiramate**, zonisamide	Mimic distal RTA, causing calcium phosphate stones
Indirect	Loop diuretics	Hypercalciuria (up to 64% of low-birth-weight infants on furosemide)
Indirect	Steroids, vitamin D, thiazides, laxatives, cytotoxics	Hypercalciuria, hypocitraturia, urate

13 · Anatomic predisposition and pregnancy

Anomaly	Stones	Metabolic findings
Ureteropelvic junction obstruction (UPJO)	≈ 20%; children ~70× risk	Common (hypercalciuria first); recurrence after repair, often contralateral

Anomaly	Stones	Metabolic findings
Horseshoe kidney	≈ 36% adults	All tested abnormal; **hypocitraturia**; calcium stones ≈ 89%
Calyceal diverticulum	Up to 40%	Abnormal in half; UTI up to 40%
Medullary sponge kidney	Frequent, with nephrocalcinosis	Hypercalciuria, hypocitraturia; distal RTA on acid loading

> **KEY POINT**
>
> Anatomy sets the stage but metabolism writes the stone: anatomical stone formers still need **metabolic evaluation**, and pyeloplasty alone does not prevent recurrence (see Ch 99).

- **Pregnancy**: stones occur in the 2nd–3rd trimester. Physiological hydronephrosis affects up to 90%, **right more than left**.
- Raised GFR and placental calcitriol promote stones, while raised citrate and magnesium inhibit them; the **net risk is unchanged**, but stones skew to **calcium phosphate**.

14 · Landmark trials

Trial	Design	Result	Take-home
Borghi water trial	RCT, high fluid vs no advice, first-time idiopathic calcium formers	Recurrence 12% vs 27%	Urine volume is the one proven universal modifier (see Ch 99)
Ettinger allopurinol trial	RCT, allopurinol vs placebo, hyperuricosuric normocalciuric calcium oxalate formers	> 2-fold fewer recurrences	Allopurinol for hyperuricosuric calcium oxalate stones (see Ch 99)

96 Acute Care of Urinary Lithiasis

1 · Presentation and diagnosis

Fever with obstruction is an emergency.

- **Renal colic** is flank-to-groin pain; a UVJ stone causes penile-tip pain and frequency.
- Haematuria occurs in ~**90%**, so it is absent in >10%.
- **Fever means an infected obstructed system**, needing urgent intervention.
- Send bloods, urinalysis and a **pregnancy test** in women of reproductive age. Urine >50 WBC/hpf or nitrites suggest infection.

Modality	Role	Limitation
Non-contrast CT (NCCT)	Gold standard (sens 97%, spec 95%); shows alternative diagnoses	8–16 mSv; misses matrix and indinavir stones
Low-dose NCCT (≤**3 mSv**)	Equal for stones ≥3 mm	Weak <3 mm and in obesity
Ultrasound (US)	First line in pregnancy, children, known formers	Up to 40% still need CT
Plain X-ray (KUB)	**Avoid** for acute diagnosis	Lowest accuracy
MRI	Second line in pregnancy and children	Poor stone detection

> **KEY POINT**
>
> Unknown diagnosis: NCCT answers everything at once. Known stone: US showing hydronephrosis suffices.

2 · Pain management

- **NSAIDs** block prostaglandin release and match opioids with less vomiting.

Agent	Place	Note
Ketorolac 30 mg	**First line** (EAU, NICE)	Avoid in renal failure, GI bleed, pregnancy
IV paracetamol	Second line	NSAID barred or insufficient
Opioids	Intractable pain	**Continue the NSAID**: morphine plus ketorolac beats either
Antispasmodics; forced fluids	**Ineffective**	Fluids only for vomiting

3 · Medical expulsive therapy (MET)

SAQ Medical expulsive therapy (MET) in urinary calculi

Paper answer · P2

Asked as:

- Medical expulsive therapy in urinary calculi. — *Summer 2021 Paper II Q3d (SAQ)*
- Medical expulsive Rx for urinary calculi. — *Practice Set 11 Paper II Q3e (SAQ)*

Type: SAQ (10 marks)
(also answered in 2021-Summer Paper 2, Q3d)

Definition

- MET (medical expulsive therapy): drugs that help a ureteric stone pass into the bladder. **α-blockers** are the main agents.

Rationale: natural history

- Spontaneous passage depends on **width**, size and site:

Width	Spontaneous passage	Location	Spontaneous passage
0–3 mm	98%	Vesicoureteric junction (small stones)	92%
5 mm	65%	Distal ureter	75%
>6.5 mm	9%	Proximal ureter	48%

Mechanism

- **α1 receptors** (mainly **α1A and α1D**) are abundant in (dominate in) the distal ureter and VUJ (vesicoureteric junction). Blockade reduces spasm and the force and frequency of peristalsis (peristaltic force and frequency), so a larger urine bolus pushes the stone / builds higher pressure above the stone.
- Others: calcium-channel blockers (nifedipine); PDE5 (phosphodiesterase-5) inhibitors under study. **NSAIDs (non-steroidal anti-inflammatory drugs) relieve pain but do not aid passage.**

Drugs

- **Tamsulosin 0.4 mg daily** (most studied); **silodosin 8 mg daily** (better expulsion in meta-analyses but twice the retrograde ejaculation); alfuzosin, doxazosin.

Evidence

Study	Result
SUSPEND (Pickard, 2015): 1136 patients; tamsulosin, nifedipine or placebo	No difference in need for intervention at 4 weeks (19% vs 20%)
STONE (Meltzer, 2018): stones <9 mm (up to 9 mm)	Tamsulosin no better than placebo
Meta-analyses and a large distal-stone RCT (randomised controlled trial)	Benefit for **distal stones >5 mm**

- Stratified, the trials agree: benefit lies in **distal stones of 5–10 mm** (smaller ones pass anyway; larger rarely pass).

Indications and guideline position (AUA [American Urological Association] and EAU [European Association of Urology])

- AUA and EAU: offer MET for **ureteric stones <10 mm**, especially distal, when immediate intervention is not needed (EAU: distal >5 mm).
- **After SWL (shock wave lithotripsy)** for faster fragment clearance (to clear fragments). Children: off-label but effective.

Contraindications (need urgent drainage instead; drain urgently)

- Infection with obstruction or sepsis, solitary or bilateral obstruction, renal deterioration (impairment), refractory pain or vomiting.

Protocol

- NSAID analgesia; counsel on off-label use, dizziness, hypotension, retrograde ejaculation.
- Periodic KUB (kidney–ureter–bladder radiograph) or ultrasound follow-up; **intervene after 4–6 weeks** if not passed.

Diagram to draw: ureter with α1 receptors concentrated distally (at the distal ureter and VUJ [vesicoureteric junction]) and a stone above it (above the vesicoureteric junction).

> **Key points to remember:**
> - Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
> - SUSPEND 2015: no benefit in an unselected population.
> - Trial of passage 4–6 weeks, then intervene.

Also, from the condensed book (not in the paper answer):

- Measure **transverse width**, not length; a 6 mm wide stone passes in ≈ 33%.
- Trial of passage, with or without MET, lasts **4–6 weeks**.

4 · Urgent intervention

- **AUA indications**: obstruction with infection, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Stent and nephrostomy (PCN)** are equal for sepsis; stents cause dysuria and haematuria, PCN a longer stay.
- **Stent failure** is predicted by P-CUT (ureteric diameter at the stone minus stone width) ≥**7.5 mm**, stone ≥9.5 mm and multiple stones; prefer PCN then.
- Acute SWL (proximal) or URS (distal <10 mm) is done **only without infection**.

ALGORITHM

1. Give analgesia; get NCCT.
2. If infection or an AUA criterion, then drain only (stent/PCN); treat the stone later.
3. If not and stone <10 mm, then trial with or without MET for 4–6 weeks.
4. If it fails, then URS or SWL.

5 · Special populations

SAQ Stones in pregnancy

Paper answer · P2 · P3 · merged from 2

Asked as: P2 SAQ "Management of Stones in pregnancy." / "Urolithiasis in pregnancy" · P3 SAQ "Management of lower ureteric stone of size less than 1 cm in pregnancy?"

Background

- 1 in 200 to 1 in 1500 pregnancies; renal colic is the **commonest non-obstetric cause of admission**; mostly second or third trimester.
- Linked with preterm delivery, pre-eclampsia and premature rupture of membranes.
- **Why calcium phosphate stones:** raised GFR (glomerular filtration rate) and placental calcitriol cause hypercalciuria; urine is alkaline. Raised citrate, magnesium and urine flow protect, so stone rates barely change.

- **Physiological hydronephrosis** (up to 90% by third trimester, right-sided) mimics obstruction.

Evaluation

- Urinalysis, **urine culture always**, blood count, creatinine; obstetric review.
- **US (ultrasound) first line**; **transvaginal US** for the distal ureter; a rise in Doppler **RI (resistive index)** suggests acute obstruction.
- Second line: **MRI (magnetic resonance imaging)** (stone as a filling defect) or **low-dose CT (computed tomography)**. Keep total fetal dose **≤50 mGy** (under 50 mGy).

Management algorithm

```
Symptomatic stone in pregnancy (including lower ureteric stone <1 cm) ├─ sepsis, solitary kidney, renal failure, uncontrolled pain │ → urgent stent or nephrostomy (+ antibiotics) ├─ pain controlled, no infection → observe (50–80% pass) └─ failed observation → ureteroscopy + holmium laser or temporary stent/nephrostomy until delivery
```

Conservative management

- Hydration, paracetamol, small doses of morphine (opioids).
- **NSAIDs (non-steroidal anti-inflammatory drugs) are contraindicated** (ductus arteriosus closure in the third trimester).
- **MET (medical expulsive therapy)** with tamsulosin is off-label.

Ureteroscopy

- **URS (ureteroscopy) with holmium laser:** first-line intervention and definitive treatment for a non-complex stone in any trimester; ideal for a distal (lower ureteric) stone; second trimester is the safest window.
- Neuraxial anaesthesia, lateral tilt, minimal fluoroscopy (lead shield beneath the patient, US guidance), fetal monitoring.
- Third-trimester URS may raise preterm labour risk.

Temporising drainage

- **Stent or PCN (percutaneous nephrostomy):** temporising. Encrustation is rapid: change stents about **every 4 weeks** (P3: every 4–6 weeks; 4–6 weeks in series), nephrostomy about every 3 weeks; definitive treatment after delivery.

Other modalities

- **Asymptomatic renal stones:** observe until after delivery.
- **SWL (shock wave lithotripsy) contraindicated. PCNL (percutaneous nephrolithotomy) deferred** until after delivery except in expert centres.

After delivery

- Clear residual stones; **metabolic evaluation** postpartum.

Diagram to draw: the management flowchart above.

Key points to remember:

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks (P3: every 4–6 weeks).

Also, from the condensed book (not in the paper answer):

- Resistive index difference ≥**0.06** means obstruction.
- About 30% need intervention; change stent/PCN every **4–6 weeks**. URS is safe in any trimester (ACOG 2019).

Population	Pitfall	Key point
Elderly	Atypical pain; first stone after 60 unusual	Always CT to exclude aortic/cardiac mimics; drain early
Transplant kidney	Denervated, so **painless**	Fever or renal failure; endoscopic, else percutaneous

6 · Landmark trials

Trial	Design	Result	Take-home
SUSPEND	RCT, tamsulosin vs placebo, any ureteral stone, n = 1136	Intervention at 4 weeks 19% vs 20%	Unstratified; does not abolish MET for distal 5–10 mm

97 Surgical Management of Ureteral Calculi

LAQ Management of upper ureteric stones

Paper answer · P2

Type: LAQ (25 marks)
Source: Practice Set 2 Paper 2, Q2 (SWL also answered in 2019-Summer Paper 2, Q2 and 2021-Summer Paper 1, Q1; flexible URS in 2019-Summer Paper 2, Q3b)

Introduction

- **Upper (proximal) ureter:** UPJ (ureteropelvic junction) to the upper border of the sacroiliac joint.
- Proximal stones pass least often, so many need intervention. Dangers: renal damage and sepsis.

Evaluation

- **NCCT (non-contrast CT)**: gold standard; gives size, density (HU, Hounsfield units), SSD (skin-to-stone distance) and impaction. KUB (kidney–ureter–bladder radiograph) for SWL follow-up; creatinine, urine culture, coagulation.

1. Emergency: drain first

- **AUA (American Urological Association) indications for urgent intervention:** infection with obstruction, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Obstructed infected kidney: drain only** (double-J stent or PCN, percutaneous nephrostomy; equal for sepsis); definitive treatment later. PCN suits large stones (stent fails if transverse diameter ≥9.5 mm).

2. Observation and MET (medical expulsive therapy): limits

Spontaneous passage	Rate
By size	1–4 mm 78%; 5–7 mm 60%; 8–10 mm 39%
By site	UVJ (ureterovesical junction) 92%; distal 75%; mid 60%; **proximal 48%**

- **Observation** suits small (<5–6 mm) uncomplicated stones: NSAIDs (non-steroidal anti-inflammatory drugs) first.
- **MET (alpha-blockers, tamsulosin or silodosin):** reduce spasm so a larger urine bolus pushes the stone. Benefit is clear only for **distal stones 5–10 mm**; meta-analysis shows **no benefit for proximal stones**. SUSPEND (2015) showed no overall benefit.
- Trial of passage is limited to **4–6 weeks**, with imaging.

3. Choice of active treatment

- Factors: stone (size, density, impaction), patient (habitus, anticoagulation, preference), surgeon and equipment.

EAU (European Association of Urology) recommendation for proximal ureteric stones

Size	First choice	Alternative
<10 mm	SWL or URS (ureteroscopy)	–
>10 mm	URS (retrograde or antegrade)	SWL

Size	First choice	Alternative
Large (>15–20 mm), impacted	PCNL with antegrade URS	Laparoscopic ureterolithotomy

SWL versus URS

	SWL (shock wave lithotripsy)	URS
Stone-free rate (AUA/EAU meta-analysis, approximate)	<10 mm about 90%; >10 mm about 70%	<10 mm about 80%; >10 mm about 80%
Sessions	More retreatment	Single session; higher SFR at 2 weeks, equal at 3 months
Anaesthesia	Sedation or none	Usually general
Complications	Fewer	Perforation, stent pain
Anticoagulation	Must stop	Laser URS can continue

- **SWL fails** with size >10–15 mm, impaction (CT "ring" sign), HU >1000, SSD >10 cm, cystine or COM (calcium oxalate monohydrate) stones.
- **SWL technique:** supine; 60–90 shocks/min; ramp energy; no routine pre-stent. Contraindicated in pregnancy, bleeding disorder, untreated UTI (urinary tract infection), distal obstruction, uncontrolled hypertension.

4. Ureteroscopy for proximal stones

- **Flexible URS** preferred: it follows a stone that migrates into the kidney.
- Safety wire; access sheath (12/14 Fr); tight ureter: pre-stent and return after 2–4 weeks.
- **Laser:** holmium:YAG (2120 nm) or thulium fibre laser; dusting or fragmenting; keep **≤10 W in the ureter** to avoid thermal injury.
- **Retropulsion** into the kidney: reduced by low-energy settings and antiretropulsion devices (Stone Cone, NTrap); pneumatic lithotripsy works poorly above the brim.
- **Push-back:** stone deliberately pushed into the renal pelvis with a catheter, then treated by SWL, RIRS (retrograde intrarenal surgery) or PCNL; now largely replaced by in-situ flexible URS.
- Stent only for perforation, impaction, infection or a planned second stage.

5. Antegrade approach (PCNL with antegrade URS)

- Indications: large (>15–20 mm), hard or **impacted proximal stones**, failed retrograde access, urinary diversion, transplant, concomitant renal stones.
- **Upper or mid-pole posterior calyx** gives a straight line to the UPJ; flexible nephroscope or antegrade URS down the dilated ureter; high SFR.

6. Laparoscopic (or robotic) ureterolithotomy

- Indications: large impacted stones, failed SWL/URS, where endourology is unavailable, or with concomitant UPJ obstruction (pyeloplasty).
- Steps: retroperitoneal or transperitoneal access; identify the bulge; **clamp the ureter above the stone** to prevent migration; longitudinal ureterotomy; remove; double-J stent; close with 4-0 absorbable sutures; drain.
- High single-stage SFR; risk of urine leak. **Open ureterolithotomy** is now rare.

Complications of treatment

Complication	Points
Perforation	0–4%; stop, stent about 4 weeks
Avulsion	0.06–0.5%; **proximal third most at risk**; nephrostomy and delayed reconstruction
Stricture	3–6%; image all after URS
Sepsis	About 5% after URS
SWL	Steinstrasse, haematoma, colic

Also know: pregnancy: conservative, then stent or URS (no SWL).

Recent advances

- Single-use digital scopes; pressure sensors; thulium fibre laser; robotic URS; **BWL (burst wave lithotripsy)**.

Diagram to draw: algorithm for proximal stone: drain if infected; <10 mm SWL or URS; >10 mm URS; large impacted antegrade or laparoscopic.

> **Key points to remember:**
> - Proximal stones pass in only 48%; MET does not help proximal stones.
> - <10 mm: SWL equals URS; >10 mm: URS first (EAU).
> - Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
> - Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

Also, from the condensed book (not in the paper answer):

- SWL **within 48 h** of an acute obstructing stone beats delayed SWL.
- Little is gained beyond 2 SWL sessions; **forced diuresis is useless**.

1 · Background and indications

- Non-contrast CT (NCCT) is the gold standard; ultrasound misses mid-ureteric stones. US with a plain film (KUB) suits follow-up.
- Medical expulsive therapy is controversial after SUSPEND; intervene after **4–6 weeks**.
- The ladder is SWL, then ureteroscopy (URS), then percutaneous nephrolithotomy (PCNL) with antegrade URS.

Tier	Trigger
Elective	Failed trial of passage
Urgent	Unrelenting pain; obstruction-related renal dysfunction
Extremely urgent	Obstructive pyelonephritis or sepsis: **drain only**, never manipulate the stone

- Work-up: creatinine and **urine culture**; exclude coagulopathy and untreated UTI.

Modality	Antiplatelet/anticoagulant	Bridging
PCNL	Stop if feasible	Short-acting heparin if high risk
SWL	Stop if feasible	Not recommended; choose another modality
Laser URS	**Need not stop**	None; more haematuria and stent symptoms

KEY POINT

Only URS tolerates anticoagulation: SWL and PCNL injure parenchyma where bleeding is unseen; laser URS works inside the lumen under vision.

2 · Choosing treatment

- Upper ureter ≤2 cm and distal ureter: stone-free rate (SFR) is similar, but SWL needs more sessions.
- **Impaction** (thick wall around the stone on NCCT) defeats SWL, so treat endoscopically.
- A large hard proximal stone needs PCNL with antegrade URS; if unfit, staged URS.

Modality	Size guidance	
SWL	Historical limit 2 cm; SFR falls above	**10–15 mm**
URS	No definite limit	
PCNL with or without antegrade URS	>15–20 mm, large hard proximal	

KEY POINT

Constant trade-off: SWL means fewer complications but more sessions; URS means one session with slightly more morbidity.

3 · Ureteroscopes

	Semirigid	Fibreoptic flexible	Digital flexible	Single-use digital
Image	Stable unless bent	Degrades as fibres break	No degradation	New each case
Calibre	Tapered	Smallest tip	~8.5 Fr	7.7 Fr
Cost	Cheap	Repair ~every **15** cases		Low volume

- Semirigid scopes suit the distal and mid ureter; flexible scopes suit proximal stones.
- Digital scopes are **not more durable**; keep a fibreoptic scope for a narrow ureter.

4 · Access

Most injuries happen at access, so be safe before fast.

SAQ Flexible ureteroscopy / RIRS and accessories

Paper answer · P1 · P2 · P3 · merged from 3

Asked as: P2 SAQ "RIRS (Retrograde intrarenal surgery)" / "RIRS" / "Flexible Ureteroscopy." / "Flexible URS" · P1 SAQ "Flexible Ureterorenoscopy" · P3 SAQ "Accessories for RIRS"

Introduction

- RIRS (retrograde intrarenal surgery) is flexible ureterorenoscopy (fURS; flexible ureteroscopy) inside the kidney; it inspects and treats the whole upper tract retrogradely by the natural route.
- Marshall first reported it in 1964; a working channel and active deflection made it therapeutic. Active deflection, miniaturised digital scopes and the holmium laser made it practical.

Instrument

Feature	Details
Tip	4.5–7.5 Fr, tapered (shaft larger) (digital tips up to about 8.5 Fr)
Working channel	**3.6 Fr** (laser fibre, basket), shared by accessories and irrigation
Deflection	**≥270° both ways** (reaches lower pole); lower pole angle averages about 140° (up to 175°)
Fibreoptic	Smallest tip; image degrades as fibres break (black spots); repair about every 15 cases
Digital ("chip-on-tip", CCD [charge-coupled device] or CMOS [complementary metal-oxide semiconductor] chip)	Better image; larger tip (about 8.5 Fr); not more durable
Single-use (LithoVue, 2016)	7.7 Fr; no repair, reprocessing or sterilisation cost; cheaper below about 99 cases a year

Indications

- **Stones:** renal stones **<2 cm**; lower pole <1 cm (or 1–2 cm / larger as an alternative to PCNL [percutaneous nephrolithotomy]); patients who must stay on anticoagulants (first choice when anticoagulants cannot be stopped); failed SWL (shock wave lithotripsy); obesity; bleeding disorders; staged treatment of larger stones.
- **Others:** diagnosis and laser ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects, positive cytology; short strictures and endopyelotomy; calyceal diverticulum; infundibular stenosis; essential haematuria; foreign bodies.

Technique

1. Sterile urine; single-dose antibiotic; α-blocker or **pre-stenting** if the ureter is tight.
2. Cystoscopy, **safety guidewire**, retrograde pyelogram.
3. **Ureteral access sheath (UAS)** (usually 12/14 Fr) over a stiff wire under fluoroscopy: lowers intrarenal pressure and temperature and allows repeated passes, but causes ureteric injury in nearly half (severe in about 1 in 8).
4. Low-pressure irrigation. **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) or thulium fibre laser**: dusting (low energy, high frequency) or fragment-and-basket. Keep power ≤10 W in the ureter and ≤25 W in the kidney (heat above 43 °C damages tissue).
5. Lower pole stones: relocate the stone to an upper calyx with a basket.
6. Inspect the ureter on withdrawal; stent unless uncomplicated. **Omit stent** if the case was uncomplicated with normal anatomy, normal renal function, no ureteric injury and no second stage (AUA [American Urological Association]). - **Difficult (unfavourable) lower pole:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum (EAU [European Association of Urology] factors against SWL; they also hinder fURS).

Accessories

Guidewires (0.018–0.038 inch, 145–260 cm)

Type	Construction	Use	Weakness
Hydrophilic	Nitinol core, slippery when wet	Bypassing impacted stones, tortuous ureter	Displaces easily

Type	Construction	Use	Weakness
PTFE (polytetrafluoroethylene)-coated	Steel or nitinol	Safety and working wire	Damages scope channel on backloading
Hybrid	Floppy hydrophilic tip, stiff PTFE shaft	Both roles	—

- **Safety wire** keeps access and allows stenting after injury (AUA, EAU).
- **Also know:** open-ended or dual-lumen catheter (pyelogram, second wire); stiff wire for the sheath; balloon dilators.

UAS (ureteral access sheath)

- Hydrophilic sheath with tapered obturator; usually **12/14 Fr** (inner/outer); tip below the PUJ (pelviureteric junction).
- **Pros:** lower intrarenal pressure and temperature; repeated passes; less scope wear.
- **Cons:** ureteric injury in nearly half, severe in about **1 in 8** (less if pre-stented); no proven fall in infection.
- **Suction sheaths and FANS (flexible and navigable suction access sheath):** aspirate dust and keep pressure low.

Retrieval devices

- **Nitinol tipless baskets** (about 1.5–2.4 Fr): kink-resistant, open flush in calyces; relocate lower-pole stones. Small baskets allow more flow but are fragile. Graspers release easily.

Laser and fibres

- Ho:YAG or thulium fibre laser; small-core fibres (about 200 µm) keep deflection and flow; **≤10 W in the ureter**.

Irrigation and antiretropulsion

- Gravity, pressure bag, hand or foot pumps (hand pumps: more fever); pressure sensors emerging.
- Stone Cone, NTrap, BackStop gel: mainly ureteric stones.

Stents

- **Double-J**; AUA: may omit after uncomplicated URS (ureteroscopy).

Outcomes

- Stone-free rate is higher than SWL for renal stones <2 cm but falls with size and lower pole location. Larger stones may need staged procedures.
- Reported rates depend on the imaging used (CT strictest).

Complications

- Stent symptoms (commonest); **sepsis about 5%**; stricture about 3%; perforation (a few per cent); bleeding or subcapsular haematoma <1%; **avulsion <1%** (nephrostomy, delayed reconstruction); scope damage.

Recent advances

- Single-use digital scopes; **thulium fibre laser** and pulse-modulated (Moses) holmium (less retropulsion, finer dust); **suction access sheaths and FANS**; intrarenal pressure sensors

(pressure-sensing scopes); robotic flexible ureteroscopy; AI (artificial intelligence) navigation.

Diagram to draw: flexible scope through an access sheath over a wire, deflected into the lower pole calyx, with a tipless basket in the lower calyx.

> **Key points to remember:**
> - 3.6 Fr channel; ≥270° deflection.
> - 12/14 Fr access sheath lowers pressure but severely injures the ureter in about 1 in 8.
> - Laser power ≤10 W ureter, ≤25 W kidney.
> - RIRS: stones <2 cm; first choice when anticoagulants cannot be stopped.
> - Sepsis about 5%, stricture about 3%.
> - Hydrophilic wire passes; PTFE wire stays.
> - Nitinol tipless basket for calyces.

LAQ Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

Paper answer · P3

Type: LAQ (25 marks)
Source: 2015-Summer Paper 3, Q1

Introduction

- A stent keeps the upper tract draining past obstruction, injury or an anastomosis. **Zimskind 1967** placed the first indwelling splint; **Finney 1978** introduced the self-retaining **double-J (double-pigtail)** stent.

Types and materials

Feature	Details
Size	3–8.5 Fr (adult usually 6 Fr); adults 20–30 cm by height, children about age + 10 cm
Polymers	Polyurethane, silicone (least encrusting, soft); hydrocoated silicone causes less pain
Metallic (Resonance)	Resists extrinsic compression: malignant obstruction, retroperitoneal fibrosis
Others	Hybrid metal-reinforced, thermo-expandable nitinol (Memokath), biodegradable (experimental)
Designs	Single-pigtail diversion stent; open-ended catheter to identify the ureter; modified bladder ends (Polaris loop: no benefit)

Also know: anatomy: three narrowings (PUJ (pelviureteric junction), iliac vessels, intramural ureter, the narrowest at 3–4 mm). Placement: retrograde pyelogram first, stent over a wire under fluoroscopy; antegrade route if retrograde fails.

Diagram to draw: double-J stent in situ with both coils and side holes.

Indications

- Obstruction: infected obstructed kidney (stent equals PCN (percutaneous nephrostomy)), malignancy, stricture, pregnancy.
- After URS (ureteroscopy), pyeloplasty, reimplantation, transplant; injury, leak, fistula; pre-stenting a tight ureter.

Stent symptoms and complications

- About **4 in 5** have urgency, frequency, haematuria or flank pain, measured by the **USSQ (Ureteral Stent Symptom Questionnaire)**. Cause: **intravesical material irritating the bladder and reflux up the stent**. Diameter and length matter little.
- Relief: **alpha-blockers and antimuscarinics**; intraureteral placement; shortest dwell; opioid-free ERAS (enhanced recovery after surgery) analgesia.
- **Misplacement** (submucosal) and **migration** (too short).
- **Stent failure** from compression, debris or encrustation: worst with malignant or irradiated ureter, bilateral obstruction; then PCN.
- **Encrustation** rises with dwell time, infection and stone-forming urine. **Forgotten stent** causes sepsis and renal loss; keep a register.
- **FECal (Forgotten, Encrusted, Calcified) grading**: I minimal linear encrustation; II circular encrustation; III partial encrustation of one or both coils; IV complete encrustation of one coil; V both coils.
- **Encrusted stent**: never pull by force; NCCT (non-contrast computed tomography); clear the **bladder coil**, then **ureter**, then **renal coil** (flexible URS or PCNL (percutaneous nephrolithotomy)).

Ureteroscopy

	Semirigid	Flexible
Tip	7 Fr or less	4.5–7.5 Fr; 3.6 Fr channel
Use	Distal and mid ureter	Upper ureter, kidney
Deflection	None	Over 270°

- Segura's dictum: always a safety wire. Tight ureter: stent and return after **2–4 weeks**; routine pre-stenting is unnecessary.
- **UAS (ureteral access sheath)**, usually 12/14 Fr: lowers intrarenal pressure and eases repeated passes, but **severe injury in about 1 in 8**; less after pre-stenting.
- **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm)** or **TFL (thulium fibre laser, about 1940 nm)**. **Dusting** (low energy, high frequency) is faster but leaves fines; **fragment and basket** confirms stone clearance and gives stone for analysis. Keep pressure and temperature low with outflow.

Steps of URS: (1) sterile urine, antibiotic dose; (2) cystoscopy and **safety guidewire**; (3) semirigid inspection, dilate only if needed; (4) access sheath over a working wire under fluoroscopy (for flexible URS); (5) low-pressure irrigation, laser, basket; (6) inspect the ureter on withdrawal; (7) decide on stent.

- **Also know:** anti-retropulsion devices; digital and single-use scopes; URS suits patients who cannot stop anticoagulants.
- Complications: sepsis about 5% (treat positive urine culture first), stricture about 3%, bleeding under 1%, perforation (stop and stent), avulsion under 1% (PCN, delayed repair).

Stent after URS?

- **Denstedt 2001** randomised trial and a Cochrane review: URS is safe without a stent.
- **AUA (American Urological Association)**: stent may be omitted when anatomy and renal function are normal, there is no ureteric injury and no second stage is planned. **EAU**

(European Association of Urology) agrees.

- **Stent needed**: perforation, stricture or tight ureter, infection, planned second stage, solitary kidney, impacted stone.
- Removal at 3 days gave more unplanned visits than at 7 days; a string allows home removal. Stenting at SWL (shock wave lithotripsy) does not improve stone passage.

```
Uncomplicated URS -> no stent (or short string stent) Complicated -> double-J:
injury 2-6 wk; second stage until re-look
```

Recent advances

- Suction sheaths, intrarenal pressure sensors, biodegradable stents.

> **Key points to remember:**
> - Finney 1978 double-J; symptoms from intravesical stent and reflux.
> - Encrusted stent: bladder coil, ureter, then kidney.
> - Denstedt 2001: no routine stent after uncomplicated URS.

Also, from the condensed book (not in the paper answer):

- **α-Blockers** ~1 week pre-URS give easier access, higher SFR and less stent pain.
- Reserve the sheath for stones **above the iliac vessels**.
- For a difficult ureter, place a stent and return after **2–4 weeks**.

ALGORITHM

1. If the wire will not pass, place an open-ended catheter 1–2 mm short of the stone.
2. Do a retrograde pyelogram.
3. Use contrast hydrodissection with a hydrophilic wire.
4. Advance the catheter past the stone, then exchange for a stiff wire.

- If the catheter abuts the stone, the wire exits at 90° and causes **perforation**. If the ureter is inaccessible, **stent and stage**.
- If the scope will not enter the orifice, use **railroading** between two wires, a dual-lumen catheter or scope rotation.

Dilation method	Best for
Indwelling stent	Whole ureter narrow; gentlest, but commits to a second procedure
Serial/coaxial dilators	Long narrow segments
Balloon	Focal stenosis; inflate until the waist resolves, hold 1–2 min
Semirigid scope	Distal ureter

- **Pre-stenting is not needed** for ureteral stones; it helps only renal stones.
- The access **sheath** helps with large burden, dusting and high power, lowering intrarenal pressure (IRP) and heat.

5 · Lithotripsy

- **Basketing**: keep the stone **5–10 mm from the tip**, **never basket blind**, and when in doubt **fragment first**.

Source	Key points
Electrohydraulic	High ureteral trauma; abandoned
Pneumatic	~12 Hz; **rigid/semirigid only**; best below brim; retropulsion
Holmium laser (Ho:YAG)	Short pulse fragments; long or modulated pulse lowers retropulsion; dusting is low energy, high frequency
Thulium fibre laser (TFL)	Low peak power and long pulse give fine dust; minimal retropulsion; >40 Hz misses

*See the paper answer **LASERs in urology: holmium, thulium, KTP and newer laser technologies** (Section 1, Ch 17).*

Also, from the condensed book (not in the paper answer):

- Impacted stone: target the centre first, so the sides fall away from the wall.

Setting	Ho:YAG	TFL
Fragmentation	0.6–1.0 J × 6–10 Hz	0.6 J × 6 Hz short pulse (3.6 W)
Dusting	0.2–0.3 J × 40–80 Hz	0.2 J × 40 Hz (8 W)

- **Settings, not only the laser, decide outcome**.
- **Thermal** damage begins above **43 °C**, and **heat tracks power, not laser type**.
- Cap at **≤10 W in ureter** and ≤25 W in kidney, with continuous irrigation, because stopping it causes thermal stricture.

6 · After URS

Most need neither long antibiotics nor a stent.

- Give a **single perioperative intravenous antibiotic dose**; extend only after urosepsis this episode.
- Stents cause symptoms in ~4/5; stentless uncomplicated URS is safe.
- Stent for perforation, stricture, tight ureter, infection or a second stage.

7 · SWL for ureteral calculi

- SWL is an option at all levels.

Absolute contraindications	Relative
Pregnancy; uncontrolled bleeding disorder; anticoagulation that cannot stop (for treatment and 48 h); distal obstruction; untreated UTI; uncontrolled hypertension	Skeletal deformity or obesity preventing focusing; adjacent aneurysm

- **Predictors**: size matters most, and URS is better ≥10 mm. Impaction (CT "ring" sign) is possibly the strongest.
- Cystine, brushite and calcium oxalate monohydrate resist SWL. Density >1000 HU and skin-to-stone distance >10 cm are proven for renal stones only.
- **No routine pre-stenting**. Hold aspirin, NSAIDs and anticoagulants for treatment and 48 h. Give antibiotics only with stents, struvite or recent UTI.
- Position the upper ureter supine, the mid-ureter **prone** and the distal ureter supine; **60–90 shocks/min** beats 120. α-Blockers raise SFR.

8 · Special situations

- **Pregnancy**: treat conservatively first; **SWL is contraindicated**; URS is safe with shielded fluoroscopy. Stents encrust, so change them **every 4 weeks**.
- **Diversion or reimplant**: use SWL (prone) or **antegrade flexible URS**. Transplant: both routes are safe.

9 · Complications

- **Perforation** (0–4%): stop, stent for ~4 weeks and drain any urinoma.
- **Avulsion** (0.06–0.5%) follows forceful extraction, mostly in the **proximal third**; place a **nephrostomy and repair later**.
- **Stricture** occurs in 3–6% and is silent, so image every patient.
- **Sepsis** (~5%) is linked to positive culture, pre-stenting, diabetes and long surgery.

> **KEY POINT**
>
> Nearly every serious URS complication is force without vision: blind basketing, forced sheaths, persistence in a tight ureter. Remedy: see, fragment, and when in doubt stop, stent and stage.

10 · Landmark trials

Trial	Design	Result	Take-home
Stentless URS	RCT, stent vs none after uncomplicated URS	Safe without stent	Stent only for named indications
TFL vs Ho:YAG	RCT, low TFL settings	TFL faster; equal ureteral SFR	Low settings optimal for TFL

98 Surgical Management of Upper Urinary Tract Calculi

1 · Choosing the approach

- The framework is the 2016 AUA/Endourological Society guideline, built on index patients and shared decision making.

Domain	Factors
Stone	Size, number, location, composition
Renal anatomy	Obstruction or stasis, hydronephrosis, ureteropelvic junction obstruction (UPJO), calyceal diverticulum, horseshoe or ectopic kidney, lower pole
Patient	Infection, obesity, habitus deformity, coagulopathy, age, hypertension, renal failure or transplant, solitary kidney, diversion, pregnancy

> **KEY POINT**
>
> Only the stone domain is changed by choosing a modality; anatomy and patient factors decide which modality is available. Obesity, coagulopathy and pregnancy each exclude a modality, not surgery.

Asymptomatic renal stones

- The natural history is not benign: **~50%** have a stone event within **5 years**. **Size predicts intervention**; risk rises above 5 mm and above 10 mm.
- With **recurrent UTI and a non-obstructing stone**, remove it whatever its composition, and **culture the stone**.
- Treat residual fragments **>4 mm** after ureteroscopy (URS).
- **Treat small asymptomatic contralateral stones** during surgery on the symptomatic side.

Preoperative evaluation

- AUA: non-contrast CT (NCCT) **should** precede percutaneous nephrolithotomy (PCNL), and **may** guide the choice of shock wave lithotripsy (SWL) versus URS.
- SWL works best when density is **<900 HU** and skin-to-stone distance (SSD) is **<10 cm**.
- Everyone needs urine culture within 1 month.

Modality selection

Scenario (AUA)	Offer	Caveat
<2 cm, non-lower pole, symptomatic	SWL or URS	SWL has fewer complications; URS has higher stone-free rate (SFR)
Total burden >2 cm	**PCNL**; staged URS an option	SWL not first-line
Lower pole >1 cm	PCNL (highest SFR) or URS	SWL not first-line, because fragments clear poorly
Lower pole <1 cm	SWL or URS	—
Staghorn	PCNL	—
Cannot stop antiplatelet/anticoagulant	**URS**	Avoid SWL and PCNL

- EAU: for >20 mm use PCNL, then URS or SWL; for **10–20 mm, SWL and endourology are in equipoise**; for <10 mm use SWL or URS, then PCNL.

Stone morphometry scores

SAQ Stone complexity scores

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 20 SAQ Sheet, Q3b

Purpose

- Grade stone and kidney complexity, chiefly to **predict stone-free rate (SFR) after PCNL (percutaneous nephrolithotomy)**. Used for counselling, choosing the surgeon, auditing and comparing series.

Main scores

Score	Variables and scoring	Predicts
Guy's Stone Score (Thomas, 2011)	**Grade I:** single stone in mid or lower pole, or pelvis, simple anatomy. **II:** single upper-pole stone, or multiple stones with simple anatomy, or single stone with abnormal anatomy. **III:** multiple stones with abnormal anatomy, calyceal diverticulum stone, or partial staghorn. **IV:** complete staghorn, or any stone in spina bifida or spinal injury	SFR on KUB (kidney, ureter, bladder radiograph); not complications
S.T.O.N.E. nephrolithometry	**S**ize (area), **T**ract length (≤100 or >100 mm), **O**bstruction (hydronephrosis), **N**umber of involved calyces, **E**ssence (density, ≤950 or >950 HU [Hounsfield units]); total 5–13	SFR, blood loss, operative time, stay; needs CT (computed tomography)
CROES nomogram (Clinical Research Office of the Endourological Society)	Stone burden, location, staghorn, number, prior treatment, **annual centre volume**	SFR; only multicentre derivation, but SFR on KUB
S-ReSC (Seoul National University Renal Stone Complexity)	Number of **9 sites** involved on CT: pelvis, upper and lower major calyces, anterior and posterior minor calyces of the three groups; score 1–9	SFR; not complications
RUSS (Resorlu–Unsal Stone Score)	1 point each: stone >20 mm; lower-pole stone with infundibulopelvic angle <45°; stones in more than one calyx; abnormal anatomy; score 0–4	SFR after **RIRS** (retrograde intrarenal surgery)

Limitations

- Each score added what the previous lacked: anatomy, then CT depth and density, then centre factors, then calyceal distribution.
- Mostly predict clearance, **not complications**. Meta-analysis: heterogeneous; no single generalisable score.
- **Also know:** Triple D score predicts SWL (shock wave lithotripsy) success, not PCNL.

Diagram to draw: kidney outline showing the 9 S-ReSC sites.

Key points to remember:

- Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
- S.T.O.N.E.: five variables, score 5–13.
- CROES includes centre volume; RUSS is for RIRS.

2 · Shock wave lithotripsy

LAQ ESWL: mechanism, indications, factors affecting outcome, complications and current status

Paper answer · P2

Asked as:

- Describe indication, mechanism of stone fragmentation and current status of ESWL? — *Summer 2019 Paper II Q2 (LAQ)*
- Describe factors affecting outcome of ESWL. Discuss complications of ESWL and strategies to avoid them. — *Summer 2021 Paper II Q1 (LAQ)* (Mechanism, indications and current status are answered in 2019-Summer Paper 2, Q2.)
- Desc. factors influencing outcome of ESWL. Discuss complic. & strategies to avoid them. — *Practice Set 13 Paper II Q2 (LAQ)* (also answered in 2021-Summer Paper 2, Q1; mechanism and indications in 2019-Summer Paper 2, Q2)

Type: LAQ (25 marks)

Introduction

- ESWL (extracorporeal shock wave lithotripsy), now SWL, breaks stones with focused acoustic shock waves generated outside the body (generated at F1 and focused on the stone at F2). First human use 1980 (Dornier HM1).
- A shock wave is a short pulse: peak positive pressure 30–100 MPa, then a negative tail of about −10 MPa.
- Stones break by spallation, shear, squeezing and **cavitation**. Cavitation also causes most tissue injury, so steps that improve outcome usually reduce complications too.
- Outcome = SFR (stone-free rate) and need for retreatment.

Components of a lithotripter

- Energy source at F1 (first focus); focusing system; coupling (water or gel); fluoroscopy and/or ultrasound to place the stone at F2. A focal zone wider than the stone improves breakage.

Energy source	Principle	Advantages	Disadvantages
Electrohydraulic (HM3)	Underwater spark; ellipsoid reflector	Most effective	Electrode wear; variable shocks
Piezoelectric	Ceramic elements on a spherical dish	Long life; little pain	Low energy; tiny focal zone
Electromagnetic (modern machines)	Coil moves a metal membrane; lens focuses	Consistent; high energy at F2	—

Mechanisms of stone fragmentation

- Each shock creates stress inside the stone. Above a critical value, cracks form; repeated shocks make them grow and join (**fatigue**).

Mechanism	How it works
Spallation	Wave reflects and inverts at the back surface of the stone; tension breaks off the distal face
Shear and tear (superfocusing)	Waves inside the stone reflect and focus; cracks form where shear peaks
Quasistatic squeezing	Wave travels faster in stone than fluid; a ring of pressure compresses the stone
Dynamic squeezing	Shear waves at the stone edges combine; strongest force in initial breakage
Cavitation	Negative phase forms bubbles; collapse erodes the surface. Excess bubbles shield the stone and injure tissue

Indications

- **Renal (AUA/EAU):** <2 cm outside the lower pole and lower pole <1 cm (SWL or URS [ureteroscopy]); EAU: 10–20 mm equal to endourology; >20 mm only after PCNL (percutaneous nephrolithotomy).
- **Ureteric:** any level, best <10 mm; emergency SWL within 48 hours works well. Patients avoiding anaesthesia and stents; children.
- **Poor choices:** >2 cm or staghorn; lower pole >1 cm (**Lower Pole I**: stone-free 21% SWL vs 91% PCNL); cystine, brushite, calcium oxalate monohydrate; infection stones.

Contraindications

- **Absolute:** pregnancy; uncorrected coagulopathy or (ongoing) antithrombotics; untreated UTI (urinary tract infection); obstruction distal to the stone; nearby arterial aneurysm (aneurysm near the stone; aortic >4 cm); severe uncontrolled (untreated) hypertension.
- **Relative:** obesity or skeletal deformity preventing targeting; pacemaker.

Factors affecting (influencing) outcome / predictors of success

1. Stone factors

Factor	Favourable	Unfavourable
Size (most important)	<10 mm	Renal >20 mm (PCNL [percutaneous nephrolithotomy] first); ureteric ≥10 mm favours URS (ureteroscopy)
Density on NCCT (non-contrast computed tomography / non-contrast CT [computed tomography])	<900 HU (Hounsfield units) clears best	>1000 HU predicts failure
Composition	Calcium oxalate dihydrate, uric acid	Cystine, brushite, calcium oxalate monohydrate (dense, homogeneous)
Impaction (ureter)	Recent, mobile	Chronic impaction (soft-tissue ring of thick ureteral wall around the stone on CT, unchanged position): may be the strongest predictor

Factor	Favourable	Unfavourable
Burden and site	Single, renal pelvis or upper calyx	Multiple sites; lower pole >1 cm

- Infection stones and stones in several sites: prefer a more efficient modality.

2. Renal anatomical factors

- **Lower pole (lower pole anatomy):** fragments drain against gravity. **Lower Pole I (Albala, 2001):** stones >1 cm, SFR **21% SWL vs 91% PCNL**. A steep infundibulopelvic angle and a long (>10 mm), narrow (<5 mm) infundibulum reduce clearance.
- Obstruction distal to the stone (UPJO [ureteropelvic junction obstruction], infundibular stenosis, calyceal diverticulum) traps fragments. Horseshoe and ectopic kidneys drain poorly.

3. Patient factors

- **SSD (skin-to-stone distance) >10 cm** (obesity) predicts failure. Skeletal deformity can prevent targeting.
- Respiratory movement of 10–15 mm moves the stone out of F2.
- **Triple D score:** stone density, stone volume and skin-to-stone distance together predict success.

4. Machine and technique factors (technique to optimise results)

1. **Energy source:** electrohydraulic (HM3) breaks best; piezoelectric has low energy and a tiny focal zone; modern machines are electromagnetic. A focal zone wider than the stone breaks better.
2. Treat infection; stop antithrombotics; control blood pressure. No routine stent: it does not improve clearance.
3. **Analgesia:** good analgesia, GA (general anaesthesia) or sedation keeps the stone in focus, because stone movement reduces breakage.
4. **Coupling:** gel as a mound, free of air; air pockets in the gel cut fragmentation by up to two-fifths.
5. **Rate 60–90 shocks/min** beats 120/min, because the bubble cloud clears between shocks (bubbles clear between shocks), so less shielding and renal injury.
6. **Ramping:** start at low energy and step up (ramping energy up also helps); this causes protective vasoconstriction.
7. Positioning (prone for mid-ureteric stones over the sacrum), continuous imaging (monitor with imaging), experienced team. Little gain beyond two sessions (more than two sessions add little).
8. **Adjuncts:** α-blockers (faster fragment clearance); potassium citrate; percussion, diuresis and inversion for lower pole fragments.

Complications

Group	Complication	Comment
Fragment-related	Renal colic; **steinstrasse** ("stone street" of fragments in the ureter); residual fragments	Steinstrasse rises several-fold with burden >2 cm^2
Infective	UTI (urinary tract infection), pyelonephritis, urosepsis (sepsis)	Infection stones, stents

Group	Complication	Comment
Renal tissue injury	Haematuria (transient, common); **subcapsular or perirenal haematoma** (minority; rarely embolisation)	Risk: age, hypertension, obesity, diabetes, coronary disease, thrombocytopenia, anticoagulation, high energy, many shocks, narrow-focus machines
Other organs	Liver, spleen, pancreas, colon / bowel, lung (children), skeletal muscle	Very rare
Cardiac	Arrhythmia during treatment	Pause or ECG (electrocardiogram) gating
Skin	Petechiae, bruising	Minor

- **Refuted long-term effects** (**not** proven): new hypertension, diabetes, decline in (loss of) renal function and infertility.

Strategies to avoid complications

1. **Select.** Absolute contraindications as above. Send large, dense, lower pole or impacted stones to URS or PCNL.
2. **Prepare.** Urine culture; treat bacteriuria. Prophylaxis only with risk factors (stent, nephrostomy, struvite, recent UTI, bowel reconstruction). Stop antithrombotics before and for the day and (about) 48 hours after. Control blood pressure; reprogramme / check pacemakers.
3. **Protect the kidney.** Slow rate; **slow ramping** (ramp over at least 4 minutes) (induces protective renal vasoconstriction and lowers haematoma risk); fewest shocks and lowest effective energy.
4. **Prevent steinstrasse.** PCNL or URS for large burdens. Routine stenting does not improve SFR, though it may reduce steinstrasse with large stones. α-blockers after SWL.
5. **Treat early.** Steinstrasse: MET (medical expulsive therapy), SWL to the lead fragment or URS; **nephrostomy if fever or obstruction**. Haematoma: bed rest, transfusion, embolisation if bleeding continues.
6. **Follow up** with KUB (kidney-ureter-bladder radiograph) and ultrasound.

Current status

- **Declining use worldwide.** URS has grown with small digital scopes and lasers, giving higher single-session stone-free rates. Newer lithotripters are also less effective than the HM3.
- **SWL vs URS:** SWL has fewer complications, no stent or general anaesthesia, lower cost, but more retreatment. Guidelines still list it first-line for small renal and ureteric stones.

Recent advances

- **Burst wave lithotripsy:** bursts of 5–100 ultrasound cycles at 4–8 MPa; less cavitation; done awake (no anaesthesia); uniform fragments.
- **Ultrasonic propulsion** moves fragments to help them pass.

Diagram to draw: ellipsoid reflector with F1 (spark) and F2 (stone), stone and gel coupling, with spallation and squeezing arrows on the stone; box of failure predictors.

Key points to remember:

- Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 cm^2; no routine stent.
- Use is declining as URS grows; burst wave lithotripsy is coming.

Also, from the condensed book (not in the paper answer):

- **Absolute contraindications** are pregnancy, uncorrected coagulopathy or antithrombotics, untreated UTI, severe untreated hypertension, aortic or renal artery aneurysm (aorta >**4 cm**), obstruction distal to the stone, and inability to target the stone.
- No prophylactic antibiotic is needed if urinalysis is negative and there are no risk factors (AUA, EAU).
- The electrohydraulic HM3 (Dornier, 1980) remains the efficacy benchmark MCQ.

3 · Ureteroscopy (retrograde intrarenal surgery, RIRS)

URS is the workhorse for small stones and antithrombotic patients.

- Pre-treat bacteriuria and give a single antibiotic dose.
- **Antithrombotics**: URS is the modality for those who must continue them; antiplatelets increase admissions.

Instruments and principles

- **Safety wire**: AUA and EAU recommend it, although a multicentre study found no outcome difference. Never force a wire past an impacted stone.
- The **ureteral access sheath (UAS)** lowers intrarenal pressure but gives **no clear fall in infection**. It injures the ureter in nearly half, less if **pre-stented**.
- **Anti-retropulsion** devices reduce stone migration.
- AUA allows **stent omission** only when all four apply: normal anatomy, normal renal function, no ureteral injury and no planned second stage.

Success

"Clinically insignificant fragments" cause reintervention in up to a fifth, so report zero-fragment rates.

Complications

Complication	Rate	Risk factors / management
Sepsis	~5%	Pre-stent, positive culture, ischaemic heart disease, age, long surgery, diabetes; treat the urine first
Subcapsular/perirenal haematoma	<1%	Hydronephrosis, thin cortex, long surgery, hypertension, UTI

Complication	Rate	Risk factors / management
Stricture	~3%	UAS injury, impacted stone; ultrasound surveillance (rate underestimated)
Perforation	Few %	Stop and stent
Avulsion	<1%	Proximal ureter; immediate nephrostomy, delayed reconstruction

4 · Percutaneous nephrolithotomy

PCNL clears the most stone; safe access is the key skill.

- Landmarks MCQ: Fernström and Johansson performed the first percutaneous stone removal in 1976.
- For 1–2 cm stones PCNL gives higher SFR than SWL or URS. It is the treatment of choice for **staghorn** and most stones >2 cm.

PCNL first-line	Applications
Burden	Stones >2 cm, staghorn, matrix stones
Lower pole	>1 cm; lower pole diverticulum or infundibular stenosis
Poor retrograde route	Urinary diversion, inaccessible lower tract, prior ureteral stricture, injury or reconstruction
Coexisting pathology	UPJO; infected stones in non-ambulatory patients

Preoperative

- Stop anticoagulants with the prescriber (bridge if high risk); **low-dose aspirin is safe**.
- **NCCT is the gold standard**, showing calyceal anatomy, SSD, HU and the surrounding organs.
- For a poorly functioning kidney, get a renogram; below 10–20% function, nephrectomy may suit.
- Give single-dose, culture-directed prophylaxis. **Tranexamic acid** reduces transfusion without more thromboembolism.

Positioning

SAQ PCNL patient positioning: prone vs supine

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "Prone vs Supine PCNL: Indications, positioning, technique, specific complications, advantages and disadvantages of each." · P2 SAQ "Describe various pt positions in PCNL"

Introduction

- PCNL (percutaneous nephrolithotomy) position decides calyceal access, working space, airway safety and simultaneous retrograde access.

Indications for PCNL

- Renal stones **>2 cm**, staghorn, lower pole >1 cm, calyceal diverticulum, failed retrograde route or urinary diversion, UPJ (ureteropelvic junction) obstruction.

Positions

Position	Set-up	Advantages	Disadvantages
Prone (Goodwin, 1955)	Lithotomy for ureteric catheter, then turned prone; chest and pelvic bolsters; horizontal bolsters at nipple line and ASIS (anterior superior iliac spine) bring the kidney down more than vertical ones; arms ≤90°, face and knees padded	**Posterior calyx and upper pole** access; large working area; multiple tracts; familiar	Turning (repositioning); airway hard to reach; pressure and positioning injuries; retrorenal colon commoner prone; needs split-leg table for ECIRS
Prone split-leg / prone-flexed	Legs apart for flexible cystoscopy; hip flexion opens the costovertebral space	ECIRS (endoscopic combined intrarenal surgery) prone	Needs split-leg table; prone cystoscopy learning curve
Supine (Valdivia-Uría, 1987)	Ipsilateral flank at table edge on a lumbar bolster; arm across chest	**Airway retained**; no repositioning; single position, shorter time; **gravity drains fragments**, low intrarenal pressure; safer anaesthesia	Smaller working area; collapsed system; anterior calyces; longer tract; mobile kidney; harder upper pole
Galdakao-modified supine Valdivia (Scoffone)	Supine flank tilt with ipsilateral leg extended and other leg in lithotomy	**Simultaneous retrograde access (ECIRS)** in one position	Longer, more horizontal tract
Bart's flank-free modified supine (Bach)	Bolster under the thorax, flank free	More flank space	Fewer data
Lateral decubitus (flank)	Kidney-up flank position	Obese, kyphoscoliosis, poor respiratory reserve	Unfamiliar fluoroscopic orientation

Technique: prone vs supine

Feature	Prone	Supine
Puncture and tract	Posterior calyx puncture under fluoroscopy or ultrasound; tract more vertical; large working area	Puncture in posterior axillary line; tract more horizontal and longer; kidney more mobile; air outlines anterior calyces
Drainage	Against gravity; higher pelvic pressure	Gravity drains fragments, low intrarenal pressure
Anaesthesia	Airway access difficult; turning needed	Airway retained; no repositioning

Positioning safety (prone)

- Bolster shins so toes float; pad knees; arms ≤90° to the torso (brachial plexus); axillary and pectoral rolls; padded face and eyes.

Specific complications

- **Prone:** ocular injury, neuropraxia (brachial plexus), pressure injury, endotracheal tube dislodgement while turning.
- **Supine: colon injury** (if puncture too anterior), access failure to upper pole, kidney displacement during dilatation.
- **Both:** bleeding (transfusion now about 1%), pleural injury above the 12th rib, sepsis, bowel, spleen or liver injury.

Choosing a position

- **Prone:** standard anatomy, upper pole or multiple posterior tracts.
- **Supine variants:** obesity, cardiopulmonary disease, kyphosis, ECIRS, ileal conduit or continent channel, combined ureteric stone.
- **Spinal deformity:** individualise; pad every pressure point.
- **Evidence:** stone-free rates and complications similar; supine often shortens operating time; choice is mainly surgeon preference.

Diagram to draw: axial CT cross-section showing prone (vertical) and supine (horizontal) tracts.

> **Key points to remember:**
> - Prone: posterior and upper pole access, large working area.
> - Supine: airway safe, ECIRS, gravity drainage, no turning; Galdakao-modified supine gives one position for ECIRS.
> - Horizontal bolsters drop the kidney more.
> - Similar outcomes; choose by anatomy and experience.

Access: the most important step

- The **ideal** target is a posterior calyx with a short, wide infundibulum.
- **Supracostal (above the 12th rib)** upper pole access reaches most of the system but raises chest complications. Lower pole access risks colonic injury.

Step	Eye-of-the-needle (bulls-eye)	Triangulation
Align	C-arm rotated 15–30° toward surgeon; needle seen end-on	Mediolateral in the oblique view, cephalocaudal in the anteroposterior (AP) view
Depth	Rotate back past AP	Short advances, alternating views

- Confirm entry by urine return, not contrast.
- **Ultrasound access** avoids radiation and shows adjacent organs.
- In **endoscopic combined intrarenal surgery (ECIRS)**, a ureteroscope watches the puncture, giving fewer tracts and higher clearance.

Dilatation and tract size

- Dilate with Alken metal, Amplatz serial or **balloon** dilators (one pass, 20–30 atm), keeping a safety wire.
- Multiple tracts raise bleeding and reduce function, especially in chronic kidney disease.

SAQ Mini-PCNL and newer PCNL techniques

Paper answer · P3 · P4 · merged from 2

Asked as: P4 SAQ "Miniperc" / "Newer PCNL techniques and nomenclature of miniaturised PCNL" · P3 SAQ "Mini Perc."

Definition and why miniaturise

- PCNL (percutaneous nephrolithotomy) through a **smaller tract** than standard, aiming to reduce bleeding and renal trauma. First "mini-perc" by **Jackman (1998)**, 13 Fr, in children.
- In PCNL, bleeding rises with tract size. Smaller tracts mean less bleeding and pain, at the cost of longer operating time and poorer vision.

Nomenclature (by tract or outer sheath size)

Name	Tract size	Features
Standard	24–34 Fr	Ultrasonic or pneumatic lithotripsy; graspers
Mini	15–22 Fr	First "mini-perc": Jackman 1998, 13 Fr; irrigation washes fragments out ("vacuum-cleaner" effect)
Super-mini	About 10–14 Fr	Sheath with **active suction** of fragments
Ultra-mini	11–14 Fr	3 Fr inline telescope in 11–13 Fr sheath
Micro-perc	<10 Fr	Single-step "all-seeing needle" with optics; laser only, fragments pass spontaneously

- **Problem:** names overlap and cut-offs are contested (paediatric chapter: mini <20–24 Fr), so many authors report the **actual sheath size in Fr**.
- The **working channel**, not sheath size, decides which instruments pass: the smallest systems take only a laser fibre.

Instruments and technique

- Mini nephroscopes (e.g. 12 Fr telescope, 15–18 Fr sheath, 6 Fr channel); laser (holmium/thulium) or small pneumatic probes.
- **Fragment removal**: irrigation outflow around the scope washes fragments out (**"vacuum-cleaner" effect**), or **active suction sheaths** (super-mini PCNL).
- Ultrasound or fluoroscopic access; supine or prone; ECIRS (endoscopic combined intrarenal surgery) possible.
- Exit: often **tubeless** or small nephrostomy (≤10 Fr: less pain).

Newer PCNL techniques

1. **Supine PCNL and ECIRS:** Galdakao-modified supine; simultaneous retrograde flexible ureteroscopy watches puncture and dilatation; fewer tracts.
2. **Ultrasound-guided access:** less radiation; shows pleura and viscera.
3. **Single-step or balloon dilatation.**
4. **Suction sheaths:** lower pressure, faster clearance.
5. **Lasers:** high-power and pulse-modulated holmium, TFL (thulium fibre laser) for dusting through small tracts.
6. **Flexible nephroscopy** after bulk removal: higher stone-free rate.
7. **Tubeless or small (≤10 Fr) nephrostomy:** less pain.

- **Also know:** robotic or artificial intelligence-assisted puncture under study.

Indications and contraindications

- Renal stones **1–2 cm**, **lower pole stones** >1 cm (alternative to RIRS [retrograde intrarenal surgery]), failed SWL (shock wave lithotripsy) or RIRS, **paediatric stones**, calyceal diverticular stones; selected larger stones or extra staghorn tracts.
- **Contraindications**: as for PCNL (untreated infection, uncorrected coagulopathy, pregnancy).

Advantages over standard PCNL

- Less bleeding and haemoglobin drop, **fewer transfusions**, shorter stay; less parenchymal damage; stone-free rates non-inferior to standard PCNL.
- Children: a 24–30 Fr tract in a child equals about a 72 Fr tract in an adult, so miniaturisation matters most here.

Disadvantages

- **Longer operative time**, poorer vision, slower fragment clearance.
- **Intrarenal pressure**: high-pressure irrigation through a tight sheath, especially when the scope nearly fills it, raises pelvic pressure and causes pyelovenous backflow, **fever and sepsis**. Keep outflow free, use low-pressure irrigation, or suction sheaths.

Evidence

- Mini vs standard (meta-analysis): **non-inferior stone-free rate**, less bleeding (fewer transfusions), shorter stay, longer operation; same complications.
- Stones >2 cm: equal stone-free rate and fever; mini bleeds less than **30 Fr** tracts, but **every advantage disappears against a 24 Fr tract**.
- Mini-PCNL versus RIRS, large matched study: similar stone-free and retreatment rates; RIRS shorter stay.

Diagram to draw: scale bar of tract sizes compared (micro, ultra-mini, mini, standard).

> **Key points to remember:**
> - Definition by sheath size: standard 24–34 Fr; mini-PCNL 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr; Jackman 1998: first mini-perc (13 Fr).
> - Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.
> - Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.
> - Working channel decides instruments; suction lowers pressure.

Completion and exit

- Finish with **flexible nephroscopy or URS**.

ALGORITHM

1. Confirm the collecting system and ureter are clear.
2. Withdraw the sheath under vision, with the safety wire out last.
3. Compress the tract.
4. If bleeding persists, re-inflate the balloon or place a **Kaye tamponade catheter**, and alert the anaesthetist.
5. If it still persists, arrange **selective angioembolisation**.

- Drain with a nephrostomy (≤10 Fr means less pain) or a double-J stent, or go tubeless.

Complications

- Cochrane (42 trials): **PCNL equals RIRS for major complications**, with fewer secondary procedures.
- Complications include haemorrhage (delayed bleeding from arteriovenous fistula or pseudoaneurysm needs angioembolisation), visceral injury, hydrothorax or pneumothorax, and sepsis.

5 · Special anatomy

Anatomy	Key adaptation
Horseshoe (stones in ~1 in 5)	SWL: long SSD. PCNL: access is medial and caudal; the upper pole posterior calyx is usually subcostal; more colonic injury; UPJO is frequent
Calyceal diverticulum/infundibular stenosis	Treat only if pain, UTI or haematuria. URS for upper or interpolar (laser-incise or dilate the ostium); lower pole: PCNL. Fulguration controversial
Pelvic kidney	SWL or URS <2 cm; PCNL with ultrasound, with or without CT or laparoscopic guidance
Urinary diversion	Antegrade wire access most reliable
Transplant	Always remove stones. The denervated graft presents with fever, oliguria and haematuria, not colic. Ultrasound-guided PCNL
Spinal deformity	SWL poor; URS good; PCNL equal clearance but more infection

- **Laparoscopic or robotic** surgery is only for when SWL, URS and PCNL fail or are expected to.

6 · Landmark trials

Trial	Design	Result	Take-home
Lower Pole I	Randomised controlled trial (RCT), PCNL vs SWL, lower pole >1 cm	SFR 91% vs 21%	SWL not first-line for lower pole >1 cm
Contralateral stone RCT (2022)	Remove vs observe small asymptomatic stones during surgery	Relapse 16% vs 63%	Treat contralateral stones

Trial	Design	Result	Take-home
Cochrane PCNL vs RIRS (2023)	42 trials	No difference in major complications; fewer secondary procedures	PCNL no longer "more morbid"

99 Prevention of Urinary Stone Disease

1 · Stone disease as a chronic condition

Recurrence is the disease, so prevention is lifelong.

- Up to **50%** recur within 5–10 years. The **ROKS** nomogram predicts a second stone.
- **Low urine volume** is the single most important risk; others are young age, male sex, obesity, family history and low citrate.

2 · Screening and metabolic evaluation

LAQ Metabolic evaluation in stone disease

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Discuss metabolic evaluation in urinary calculus disease." / "Describe metabolic evaluation in stone disease" / "Discuss evaluation for urolithiasis and management of bilateral renal staghorn calculi" · P1 SAQ "Metabolic Evaluation in Stone Disease"

Introduction and rationale

- Stone disease is a chronic metabolic disorder: first-time formers have about a **50% risk of recurrence within 10 years** (another source: up to 50% recur within 5–10 years).
- Evaluation finds a treatable cause and directs targeted therapy; therapy matched to 24-hour urine results reduces stone events. The **ROKS (Recurrence of Kidney Stone) nomogram** estimates individual risk.

Physicochemical basis (why we measure these)

- Stones form when urine is **supersaturated**: nucleation, growth, aggregation and retention, often on **Randall plaque**.
- In the **metastable zone** (between solubility and formation products) crystals grow on existing nuclei but do not form spontaneously; above it nucleation is spontaneous.
- **Promoters** (calcium, oxalate, uric acid, low volume) raise supersaturation; **inhibitors** (citrate, magnesium, Tamm–Horsfall protein) raise the supersaturation needed to crystallise. The 24-hour urine measures both.

Who needs evaluation

- **Screening evaluation** for every first-time former (AUA [American Urological Association], EAU [European Association of Urology]).
- **24-hour urine for high-risk and recurrent formers:** recurrent formers, **children**, solitary kidney, bilateral or large burden, GI (gastrointestinal) disease, bariatric surgery or bypass, obesity, metabolic syndrome, diabetes, primary hyperparathyroidism, RTA (renal tubular acidosis), sarcoidosis, gout, family history; uric acid, cystine or infection stones. Bilateral staghorns are high risk by definition.

Step 1: Screening evaluation (every first-time former)

Component	What is assessed
History	Colic, UTI (urinary tract infection) and recurrent UTI, previous stones, fluid and diet; GI disease or bariatric surgery, gout, sarcoidosis, family history; lithogenic drugs: calcium, vitamins C and D, acetazolamide, topiramate, steroids, triamterene, indinavir

Component	What is assessed
Blood	Creatinine, sodium, potassium (electrolytes), calcium, uric acid, bicarbonate (if RTA suspected); **PTH (parathyroid hormone)** if calcium is high or high-normal
Urine	Urinalysis: pH, crystals (hexagonal = cystine), pyuria; culture; cyanide–nitroprusside cystine screen, confirmed by quantitative 24-hour cystine
Stone analysis	Infrared spectroscopy or X-ray diffraction: uric acid points to metabolic syndrome, cystine to a genetic cause, struvite to infection
Imaging	NCCT: burden predicts future events; uric acid usually <500 HU (Hounsfield units)

- Screening is a **triage**: low risk gets general advice; high risk goes on to 24-hour urine.

Imaging in evaluation for urolithiasis

Imaging	Role
NCCT (non-contrast CT)	Gold standard: sensitivity 97%, specificity 95%. Size, site, burden, density (uric acid 200–400 HU; calcium oxalate 600–1200 HU), skin-to-stone distance
Low-dose NCCT (≤3 mSv)	Equal accuracy for stones ≥3 mm
Ultrasound	First line in pregnancy and children
KUB (plain X-ray)	Follow-up of radio-opaque stones only
CT urography	Anatomy before PCNL (percutaneous nephrolithotomy)
MAG3 (mercaptoacetyltriglycine) renogram or DMSA (dimercaptosuccinic acid) scan	Split function

Step 2: Extended metabolic evaluation (24-hour urine)

- **One or two 24-hour urines** on the usual diet, at least 4 weeks after the episode, stone-free and infection-free. Creatinine checks completeness (men about 20–25, women about 15–20 mg/kg/day).
- Measure volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine (cystine if suspected).

Parameter	Desirable per day	Abnormality and meaning
Volume	**>2.5 L**	Low volume: commonest, easiest to fix
Calcium	<250 mg women, <300 mg men (or <4 mg/kg)	**Hypercalciuria**: absorptive (PTH suppressed), renal leak (PTH high, serum calcium normal), resorptive (primary hyperparathyroidism)
Oxalate	**<40 mg**	Dietary; **enteric** (fat malabsorption, gastric bypass: fat binds calcium, freeing oxalate; often 60–70 mg); **primary** (hepatic, often >100 mg, childhood; type 1 (AGXT gene) ~70%, type 2 (GRHPR), type 3 (HOGA1))
Citrate	**>320 mg**	Hypocitraturia: distal RTA, diarrhoea, hypokalaemia, high animal protein
Uric acid	<600 mg (labs: <750 women, <800 men)	Hyperuricosuria promotes calcium oxalate stones
pH	5.8–6.2	<5.5 uric acid (metabolic syndrome); >6.5–7 calcium phosphate, RTA, infection

Parameter	Desirable per day	Abnormality and meaning
Sodium	<100 mmol	High sodium raises calcium excretion
Cystine	Negative screen	Cystinuria; keep <250 mg/L

- **Uric acid stones** depend mainly on pH: below the pKa (5.5) uric acid is undissociated and insoluble.
- **Infection stones**: urease-splitting organisms, pH >7, struvite.
- **Also know:** magnesium, potassium, phosphate; hypomagnesiuria; RSS (relative supersaturation) is calculated from these and guides therapy; medullary sponge kidney and calyceal diverticulum (stasis plus metabolic defects); 2,8-dihydroxyadenine stones (APRT [adenine phosphoribosyltransferase] deficiency).

Interpreting hypercalciuria (Pak classification)

Type	Defect	Serum calcium	PTH	Fasting urine calcium
Absorptive (20–40%)	Increased gut absorption	Normal	Normal or suppressed	Normal
Renal leak (5–8%)	Tubular calcium loss	Normal	High (secondary)	High
Resorptive (3–5%)	Primary hyperparathyroidism	High	High	High

- Fast and calcium-load subtyping is no longer advised, because it does not change treatment.

Special tests

- **Distal RTA:** urine pH >6, low citrate, calcium phosphate stones. Incomplete forms: oral **ammonium chloride load** (0.1 g/kg) or furosemide–fludrocortisone test fails to lower urine pH below 5.5.
- **Cystinuria:** autosomal recessive transport defect (cystine, ornithine, lysine, arginine); quantitative 24-hour cystine; solubility about 300 mg/L at pH 5 and 400 mg/L at pH 7, so aim for <250 mg/L.
- **Primary hyperoxaluria:** genetic testing (type 1, AGXT gene, about 70%).

Evaluation algorithm

`Stone episode → screening evaluation Low risk → general advice High risk → 24-h urine ×1–2 → targeted therapy → repeat within 6 months → yearly`

Diagram to draw: supersaturation zones (undersaturated, metastable, unstable) and the evaluation flowchart (first stone, screening, low risk with general advice or high risk with 24-hour urine and targeted therapy).

Treatment linked to abnormality

- **General**: urine >2.5 L, sodium ≤2.3 g/day, normal calcium, less animal protein, more fruit and vegetables.

Abnormality	Treatment
Low volume	Fluid for urine >2.5 L/day
Hypercalciuria	**Thiazide** (hydrochlorothiazide 25 mg twice daily, chlorthalidone, indapamide) plus potassium citrate; plus sodium restriction

Abnormality	Treatment
Primary hyperparathyroidism	**Parathyroidectomy**
Hypocitraturia, distal RTA	**Potassium citrate** 30–60 mEq/day
Enteric hyperoxaluria	Low fat and oxalate; **calcium with meals**; magnesium
Primary hyperoxaluria type 1	Pyridoxine, **lumasiran** (RNAi [ribonucleic acid interference]); liver ± kidney transplant
Hyperuricosuria	**Allopurinol** 300 mg/day
Uric acid stones	**Potassium citrate** to urine pH 6.0–6.5
Cystinuria	Urine ≥3 L/day; pH **7.0–7.5**; **tiopronin**
Struvite	Complete clearance; AHA (acetohydroxamic acid) if surgery impossible

- **Borghi 1996**: urine >2 L/day cut 5-year recurrence (12% vs 27%). **Borghi 2002**: low-salt, low-protein, normal-calcium diet halved recurrence versus low calcium. **NOSTONE (2023)**: hydrochlorothiazide did not reduce recurrence.

Special groups and follow-up

- **Children:** all evaluated (spot ratios if not toilet-trained; consider Dent disease, monogenic causes). **Pregnancy:** defer 24-hour urine (physiological hypercalciuria).
- **Follow-up (AUA):** repeat 24-hour urine **within 6 months** of starting therapy, then yearly; blood tests for drug side effects (potassium on thiazide or citrate); re-analyse every new stone. Adherence is poor; citrus juices are only adjuncts.

Recent advances

- **Lumasiran** (RNA interference) for primary hyperoxaluria type 1; **NOSTONE (2023):** hydrochlorothiazide did not reduce recurrence; ROKS nomogram.

Staghorn calculus (bilateral staghorn question)

- **Definition:** a branched stone filling the renal pelvis and extending into the calyces. **Partial:** pelvis and some calyces; **complete:** pelvis and all major calyces.
- **Composition:** classically struvite (magnesium ammonium phosphate) with carbonate apatite. Bacterial **urease** (Proteus, Klebsiella, Ureaplasma) splits urea into ammonia, so urine pH rises above 7.2 and struvite crystallises inside a bacterial biofilm. Fewer than half of modern staghorns are infection stones; calcium phosphate, cystine and uric acid also occur.
- **Why treat:** untreated staghorns destroy the kidney and cause sepsis, XGP (xanthogranulomatous pyelonephritis) and death.

Staghorn: work-up before surgery

- NCCT with coronal and 3D views: burden, calyceal anatomy, colon and spleen.
- **Renal function:** eGFR (estimated glomerular filtration rate) and split function, because both kidneys are at risk.
- **Infection:** culture-specific antibiotics before surgery; drain an obstructed or infected kidney (nephrostomy through a calyx usable later for PCNL). Culture the stone.
- **Guy's stone score (2011)** predicts stone-free rate:

Grade	Stone
I	Single stone in mid or lower pole or pelvis, simple anatomy
II	Single upper-pole stone; multiple stones in simple anatomy; or single stone in abnormal anatomy
III	Multiple stones in abnormal anatomy; calyceal diverticulum; partial staghorn
IV	Staghorn; or any stone in spina bifida or spinal injury

- **Also know:** STONE score, CROES nomogram, S-ReSC score.

Staghorn: treatment options (AUA, EAU)

- **Goal: complete clearance**, because each fragment is an infected nidus; also clear infection and preserve function.

Option	Role
PCNL	**First line.** Multiple tracts if needed; upper-pole access reaches pelvis, lower pole and ureteropelvic junction. Flexible nephroscopy raises clearance; second-look for residue
ECIRS (endoscopic combined intrarenal surgery)	PCNL (usually supine) plus retrograde flexible ureteroscopy: fewer tracts, less bleeding, higher clearance
Staged RIRS (retrograde intrarenal surgery)	Selected partial staghorns; patients who cannot stop anticoagulants
SWL (shock wave lithotripsy)	Not as monotherapy; adjunct for residual fragments
Anatrophic nephrolithotomy (Boyce–Smith)	Incision along Brödel's avascular line, artery clamped, cold ischaemia, infundibula reconstructed. Now rare; when endoscopy cannot clear the stone in a reasonable number of sessions
Nephrectomy	Poorly functioning kidney (split below about 10–20%) that is infected or painful; XGP

Bilateral staghorns: sequence

``` Sepsis or obstruction on either side? Yes -> drain + antibiotics -> treat that side once sterile No -> treat the better-functioning (or symptomatic) kidney first

|

NCCT: stone-free? -> second look if residue

|

Other kidney after recovery (selected: same-session bilateral PCNL, or PCNL + contralateral RIRS) ```

- **Why the better kidney first:** it secures renal function early; some start with the simpler side. No trial defines the order.
- **Same-session bilateral surgery:** feasible at experienced centres if the first side was short and uncomplicated; saves an anaesthetic but risks bilateral bleeding or obstruction.
- Check creatinine after each side. Remove a non-functioning kidney only after the other side is cleared.
```

Staghorn: complications and follow-up

- PCNL risks: bleeding (transfusion about 1%; angioembolisation), sepsis, pleural, colon, spleen or liver injury.
- NCCT for stone-free status; treat residual fragments >4 mm. Surveillance cultures.
- 24-hour urine even in struvite formers, then targeted therapy; urine ≥2.5 L/day.
- **AHA** 250 mg three times daily, a urease inhibitor, only when surgery is exhausted. Avoid if creatinine >2.5 mg/dL or in pregnancy; causes DVT (deep vein thrombosis), haemolytic anaemia.

Diagram to draw: staghorn with an upper-pole PCNL tract and a retrograde flexible scope (ECIRS).

> **Key points to remember:**
> - Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
> - Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
> - Distal RTA: ammonium chloride (NH4Cl) fails to lower pH below 5.5; cystine <250 mg/L at pH 7.0–7.5.
> - Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months of therapy.
> - NCCT gold standard (97%/95%).
> - Staghorn: PCNL first line; complete clearance is the goal.
> - Bilateral: drain an infected side first, then the better kidney.
> - AHA only when surgery is not possible.

Also, from the condensed book (not in the paper answer):

- A 24-hour urine is **not** part of screening.
- Lithogenic drugs include calcium, vitamins C and D, acetazolamide and steroids.

3 · General measures (all formers)

- **Fluid** is the cornerstone: aim for urine ≥**2.5 L/day**.
- **Limit nondairy animal protein and sodium**, which raise urinary calcium and lower citrate.
- Keep **calcium intake normal**, because restriction increases stones.

> **KEY POINT**
>
> Why normal calcium? Dietary calcium binds oxalate in the gut; restricting it frees oxalate for absorption.

4 · Hypercalciuria: thiazides

- The AUA makes thiazides the **preferred drug** for hypercalciuria with recurrent calcium stones, **and** for recurrent calcium stones with a normal evaluation.
- **Hypokalaemia causes intracellular acidosis, which leads to hypocitraturia**. So add **potassium citrate**, and check potassium 1–2 weeks after starting or changing the dose.
- With amiloride, **never add potassium**. Avoid triamterene, as it forms stones.
- Side effects include gout, diabetes and sexual dysfunction.

Target	Drug and dose	Key side effects
Calcium	Hydrochlorothiazide 25 mg twice daily; chlorthalidone 25–50 mg; indapamide 2.5 mg	Hypokalaemia, hyperuricosuria, hypocitraturia
Citrate	Potassium citrate 30–60 mEq/day	Gastrointestinal (GI) upset, hyperkalaemia
Uric acid	Allopurinol 300 mg	Rash, myalgia, hypersensitivity
Oxalate	Magnesium; pyridoxine 100 mg	Diarrhoea (magnesium); ataxia, numbness (pyridoxine)
Cystine	Tiopronin; D-penicillamine 250 mg titrated	See thiol table
Urease	Acetohydroxamic acid 250 mg three times daily	Heaviest burden (below)

5 · Hyperoxaluria

- Stone risk rises continuously with urinary oxalate. Avoid large portions of spinach, nuts, chocolate and tea.
- **Enteric** hyperoxaluria follows malabsorption or Roux-en-Y gastric bypass; urine oxalate is >40 mg/day. Unabsorbed fat binds calcium, which frees oxalate for colonic absorption.
- Treat with **calcium up to 1 g/day with meals**, very high fluid, **liquid** potassium citrate, magnesium and a low-oxalate diet.

*See the paper answer **Primary Hyperoxaluria** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- Lumasiran is approved for PH1.

6 · Hypocitraturia

- AUA offers **potassium citrate** 30–60 mEq/day for recurrent calcium stones with low citrate, **and** for those with no abnormality.
- Sodium citrate does not lower urinary calcium. Use caution in renal impairment.
- Over-alkalinising to **pH ≥7** creates **calcium phosphate** risk.
- **Alkali citrate, not total citrate, matters**.

7 · Uric acid stones: a pH disease

*See the paper answer **Uric acid calculi: etiopathogenesis, medical management and allopurinol** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- Give allopurinol only for hyperuricaemia with hyperuricosuria, or failed alkalinisation.

8 · Cystine stones

ALGORITHM

1. **Dilute**: urine ≥**3 L/day**, including at night; sodium ≤2300 mg/day; moderate protein (not in children).
2. Aim for cystine <**250 mg/L**.
3. If not enough, **alkalinise** with potassium citrate to pH **7.0–7.5**.
4. If still forming stones, add a **thiol** (tiopronin).

- pH >7.5 risks calcium phosphate, and sodium alkali raises cystine excretion.
- Thiols form a soluble cystine–drug disulfide.

	D-Penicillamine	Tiopronin	Captopril
Status	First thiol; largely replaced	**Thiol of choice**; fewer stop it	Weak evidence
Dose	250 mg/day, titrate	300 mg twice daily, up to 1200 mg/day	—
Toxicity	Rash, fever, arthralgia, pancytopenia, nephrotic syndrome	Asthenia, GI upset, rash, joint aches	—

9 · Infection (struvite) stones

- **Clear the stone completely**, because every fragment harbours bacteria.
- **Acetohydroxamic acid (AHA)** is a urease inhibitor. AUA: use it only for residual or recurrent struvite after surgical options are exhausted.
- AHA toxicity is ~30%, with **deep venous thrombosis** the most severe. It is contraindicated if creatinine is >**2.5 mg/dL** or in pregnancy.
- Most struvite formers are also hypercalciuric or hypocitraturic, so do a 24-hour urine.

10 · Monitoring

ALGORITHM

1. Start therapy.
2. Repeat the 24-hour urine within **6 months**.
3. Then check urine and blood (for side effects) at least yearly.
4. If a new stone forms, **re-analyse** it.
5. Image periodically.

11 · Landmark trials

Trial	Design	Result	Take-home
Borghi 1996	Randomised controlled trial (RCT), high fluid, calcium oxalate formers	5-year recurrence 12% vs 27%	Urine >2 L/day prevents stones
Borghi 2002	RCT, hypercalciuric formers	Low protein and salt with normal calcium halved events vs low calcium	Never restrict calcium

Trial	Design	Result	Take-home
Ettinger 1988	Double-blind RCT, chlorthalidone	~90% fewer stones	Thiazides prevent calcium stones
Barcelo 1993	RCT, potassium citrate	Fewer new stones at 3 years	Citrate prevents recurrence
ILLUMINATE-A	Phase 3 RCT, lumasiran, PH1	84% vs 0% reached near-normal urine oxalate	RNA interference for PH1

100 Lower Urinary Tract Calculi

1 · Bladder calculi

- Bladder stones are ~5% of urinary stones. Types are **migrant**, **primary** (endemic) and **secondary**.

	Primary (endemic)	Secondary
Patient	Children; peak **2–4 years**; boys far more than girls	Men >60 with outlet obstruction
Cause	Cereal diet low in animal protein and phosphate	Bladder outlet obstruction (BOO), foreign body, catheter, infection
Composition	Ammonium acid urate, calcium oxalate, uric acid	**Struvite**
Recurrence	Solitary; rarely recurs	Recurs unless the cause is removed

- Foreign bodies act as a nidus: a migrated intrauterine device, sutures, balloon fragments or hair during clean intermittent catheterisation (CIC).

> **KEY POINT**
>
> Primary stone is a disease of the urine (normal bladder); secondary is a disease of the bladder (stasis, nidus, infection). Do not hunt for outlet obstruction in the endemic child.

- Adults present with **terminal haematuria** and lower urinary tract symptoms.

2 · Management of bladder stones

The route, not the stone, dictates modality.

Option	Indication / key point
Chemodissolution	Adjunct only; for catheter encrustation
Extracorporeal shock wave lithotripsy (ESWL)	Artificial sphincter, penile prosthesis, neobladder, high-risk patient; works poorly in continent reservoirs
Transurethral cystolithotripsy	Most popular; holmium laser of choice; risks urethral stricture
Percutaneous cystolithotomy	Difficult urethra (bladder neck reconstruction or closure); spares the urethra
Open cystolithotomy	Small-capacity bladder, stricture, no safe endoscopic route

3 · Special situations

Treat the cause along with the stone.

- **BOO** is present in 45–79%. **A bladder stone secondary to BOO is an absolute indication for BPH surgery**, but **prove BOO** first.
- **Diversion or augmentation** stones arise from acidosis-driven hypercalciuria, enteric hyperoxaluria, urease producers and **mucus**.
- A **continent pouch: go percutaneous**, which protects continence. A conduit with stomal stenosis needs stone clearance and stomal revision.
- **Spinal cord injury**: risk peaks **3 months** after injury; **CIC beats indwelling catheters**. Stones ≤4 cm get transurethral laser; larger ones percutaneous or open.

- **Transplant**: **remove all exposed suture**. **Children**: use percutaneous cystolithotripsy for large stones.

4 · Urethral calculi

- Urethral stones are 0.3–1% of stones, in males. Most are **migratory**, and the commonest site is the posterior urethra.
- They cause **acute painful retention**; anterior stones give a palpable mass. ~98% are radio-opaque.

Site	Treatment
Posterior	**Push back** into the bladder, then laser; else treat in situ
Anterior	**No pushback**; in situ holmium laser; urethrotomy for large stones
Fossa navicularis	Forceps; ventral meatotomy
Diverticular / stricture	Extraction with diverticulectomy or stricture repair, same sitting or staged

5 · Unusual sites

- **Preputial** stones follow phimosis; treat by circumcision and send the **excised skin to histopathology**.
- **Prostatic** stones are linked to BPH, **not cancer**.
- **Seminal vesicle** stones cause haematospermia; **TRUS is most accurate**.

54 Management of Paediatric Kidney Stone Disease

1 · Epidemiology

Children carry stone disease for life.

- Early onset means stones before **24 years**. About 50% recur within 3–5 years.

Feature	Children	Adults
Sex ratio	Female predominance	Male (historically 3 : 1)
Calcium oxalate (CaOx)	≈ **80%**, mostly idiopathic, on Randall's plaques	75–80%
Calcium phosphate (CaP) / uric acid	CaP commoner and uric acid rarer, because urine is more alkaline	≈ 5% each
Lifetime stakes	Chronic kidney disease (CKD), hypertension, low bone mineral density (BMD); cumulative radiation	Shorter

- **High-risk** groups include bladder augmentation (≈ 1%/year) and malabsorption.

> **KEY POINT**
> Childhood is the bone-accrual window, so seek low BMD early.

2 · Diagnosis and imaging

ALGORITHM

1. If a stone is suspected, do grey-scale **ultrasound (US)** with colour Doppler.
2. If US is negative but suspicion is high, or there is hydronephrosis, then do **low-dose non-contrast computed tomography (NCCT)**.
3. If US is negative and suspicion is low, stop.

- US shows the **twinkle artefact** (sensitivity > 70%, specificity > 95%) but overestimates size by 1–2 mm. NCCT is ≈ 100% accurate.

3 · Metabolic evaluation: every child

An abnormality raises recurrence up to **fivefold**, so do a **full evaluation after the first stone**.

- **History**: ketogenic diet, vitamin D, furosemide, topiramate, prematurity.
- **Examination clues**: dysmorphism suggests Williams syndrome, rickets suggests Dent disease or HHRH, and tetany suggests FHHNC.
- Do **stone analysis** first: cystine, struvite, 2,8-dihydroxyadenine and xanthine are each diagnostic.
- With hypercalcaemia and hypercalciuria, check for **CYP24A1** loss.

Metabolite	Normal (child)
Calcium	< **4 mg/kg/day**; spot Ca : creatinine < 0.8 (0–6 mo), < 0.6 (7–12 mo), < **0.21** (> 2 y)
Oxalate	< 40–50 mg/1.73 m²/day

Metabolite	Normal (child)
Citrate (lower limit)	Hypocitraturia is < 128 mg/g creatinine (male), < 300 (female)
Cystine	< 50–60 mg/1.73 m²/day
Uric acid	< 815 mg/1.73 m²/day; spot < 0.56 mg/dL glomerular filtration rate

> **KEY POINT**
>
> Never use adult limits: infants excrete more calcium, so the school-age 0.21 ratio overcalls them. Citrate's abnormality is too little.

4 · Urinary abnormalities

Hypercalciuria: commonest (30–50%)

Serum calcium	Causes
High	Hyperparathyroidism, CYP24A1, hypervitaminosis D, immobilisation, Williams
Low	Hypoparathyroidism; autosomal dominant hypocalcaemic hypercalciuria (calcium-sensing receptor gain)
Normal	Prematurity, loop diuretics, topiramate, ketogenic diet, Dent, Bartter, FHHNC, distal renal tubular acidosis (dRTA), HHRH; if none, **idiopathic**

Hyperoxaluria

*See the paper answer **Primary Hyperoxaluria** (Section 11, Ch 95).*

Also, from the condensed book (not in the paper answer):

- Only 10–30% of primary hyperoxaluria type 1 (PH1) respond to pyridoxine.
- Thiols and lumasiran are the only drugs approved for paediatric stone prevention.

Others

- **Hypocitraturia**: incomplete dRTA needs an acid-load test. **Cystinuria** may present in infancy with a staghorn.

Monogenic disease	Inheritance	Signature
Dent (CLCN5)	X-linked	Low-molecular-weight proteinuria, nephrocalcinosis, CKD
Bartter	Recessive	Salt wasting, hypokalaemic **alkalosis**, hypercalciuria
FHHNC (CLDN16/19)	Recessive	Hypomagnesaemia, tetany, nephrocalcinosis
dRTA	Dominant/recessive	Hypokalaemic **acidosis**, hypocitraturia, nephrocalcinosis

5 · Management principles

- Tamsulosin is off-label (≤ 6 weeks).
- The residual fragment threshold is **4 mm on US**: 2 mm on CT plus the ≈ 2 mm US overestimate.
- Give antibiotic prophylaxis for all URS and PCNL, but for SWL only with infection risk; limit it to ≤ 24 h.

Scenario (AUA/Endourological Society 2016)	Recommendation
Uncomplicated ureteric ≤ **10 mm**	Observation with or without an α-blocker
Ureteric, failed or unlikely to pass	URS or SWL (strong)
Before URS	No routine pre-stenting
Renal ≤ **20 mm**	SWL or URS
Renal > 20 mm	PCNL or SWL (with stent or nephrostomy); authors prefer PCNL
Before PCNL	Low-dose CT (strong)
Asymptomatic non-obstructing renal	Surveillance with US
Open/laparoscopic/robotic	Only with anatomic abnormality

6 · Ureteroscopy

The small ureter is the main limit, so never force access.

- Clearance is ≈ 90%, but worse under 6 years and in the lower pole.

ALGORITHM

1. Confirm a sterile culture; place in lithotomy; do cystoscopy.
2. Do a retrograde pyelogram and place a safety wire.
3. Calibrate the orifice with an 8/10 Fr coaxial dilator.
4. **If it will not pass, then place a 4.6–4.8 Fr stent for ≥ 14 days and return later.**
5. If it passes, do URS.
6. Stent if the case was long, needed many passes, or caused trauma or oedema.

- **No balloon or active orifice dilatation**; rely on passive stent dilatation.

7 · Shockwave lithotripsy

- SWL equals URS for small stones.
- Favourable: < 10 mm, non-lower-pole, < 1000 HU. Unfavourable: cystine, brushite, infundibulopelvic angle > 45°.
- Use < 20 kV and ≤ 2000–3000 shocks (< 2000 in infants); **80/min is better than 120/min**.
- Absolute contraindications are bleeding diathesis, untreated UTI and pregnancy.

8 · Percutaneous nephrolithotomy

- PCNL is first-line for stones > 20 mm. A 24–30 Fr tract in a child equals ≈ 72 Fr in an adult, so **miniaturise**.

ALGORITHM

1. Prevent hypothermia; place a ureteric catheter.
2. Position prone (latex precautions in myelomeningocele).
3. Puncture a posterior calyx under US or fluoroscopy.
4. Dilate serially to an 18 Fr sheath (24 Fr for staghorn).
5. Fragment and check with a flexible scope.
6. Place a nephrostomy (7 Fr infant to 16 Fr adolescent).

9 · Secondary prevention

- **Fluid**: at least maintenance; > 2–2.5 L/day in adolescents; 1.5–2 L/m² in cystinuria or PH.
- **Sodium** < 2–3 mEq/kg/day. **Calcium is not restricted**, because it binds gut oxalate.
- Target urine pH > 6.5 for uric acid and > 7 for cystine; avoid > 6.5 in calcium formers.

Drug	Indication	Child dose / caution
Hydrochlorothiazide	Hypercalciuria despite salt restriction	1–2 mg/kg/day; for hypokalaemia add potassium citrate
Potassium citrate	Hypocitraturia, uric acid, cystine	2–4 mEq/kg/day; CaP risk if pH > 6.5
Tiopronin	Cystinuria failing fluid and alkali	Fewer adverse effects than penicillamine
Allopurinol	HPRT, PRPS, adenine phosphoribosyltransferase deficiency; hyperuricosuric CaOx	4–10 mg/kg/day; can cause xanthine stones
Pyridoxine	PH1 (trial in all)	10–30% respond
Lumasiran	PH1	RNA interference against glycolate oxidase

PH1 with end-stage disease needs liver–kidney transplantation.

10 · Landmark trials

Trial	Design	Result	Take-home
ILLUMINATE-A	Randomised, lumasiran vs placebo, PH1	Urinary oxalate lowered by 65%	RNA interference therapy for PH1

125 Genital and Lower Urinary Tract Trauma

1 · Principles

- Key decision: **immediate reconstruction vs deliberate delay**. Testicular rupture is repaired at once; pelvic fracture urethral injury (PFUI) gets a suprapubic tube now and urethroplasty later.
- Imaging: ultrasonography for scrotum and equivocal penile fracture; computed tomography (CT) cystography for bladder; retrograde urethrography (RUG) for urethra.

2 · Penile fracture

SAQ Fracture penis

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 15 Paper 2, Q3c

Definition and mechanism

- **Penile fracture** is a tear of the **tunica albuginea** with rupture of the corpus cavernosum, in the **erect** penis.
- Abnormal bending of the erect penis causes an abrupt rise in intracavernosal pressure beyond tunical strength → **transverse tear**.
- The tunica (inner circular, outer longitudinal layers) is **thinnest ventrolaterally**, so most tears are ventral or lateral, proximal to mid shaft.
- **Causes:** intercourse (commonest; rear-entry), masturbation, rolling over; self-inflicted "taqaandan" (Middle East); collagenase injection for Peyronie disease.
- **Bilateral tears in about 10%**; urethral injury about 20% in the West vs about 3% in Asia and the Middle East.

Clinical features

- **Cracking or popping sound → pain → rapid detumescence → swelling and discoloration.**
- **"Eggplant deformity"** when Buck's fascia is intact; if torn, haematoma spreads to scrotum, perineum and suprapubic area.
- Penis deviates **away from the tear**; tunical defect may be palpable.
- **Urethral injury:** blood at meatus, haematuria, retention.

Investigations

- **Diagnosis is clinical** (AUA [American Urological Association] Urotrauma Guideline).
- **Ultrasound** for equivocal cases (localises the tear); **MRI** (magnetic resonance imaging) if still equivocal (100% negative predictive value). Cavernosography is discouraged.
- Urethra: **intraoperative flexible cystoscopy** (AUA and EAU [European Association of Urology] preference) or RUG (retrograde urethrogram).
- **"False fracture":** dorsal vein or artery rupture.

Management

- **Immediate surgical repair** (less curvature, ED [erectile dysfunction] and nodules than conservative care). 1. Incision: ventral penoscrotal or lateral over the defect; subcoronal degloving if site unclear (preputial ischaemia risk). 2. Evacuate haematoma; find the tear (artificial erection helps). 3. Close the tunica with **interrupted 2-0/3-0 absorbable sutures**; avoid debriding erectile tissue. 4. Urethra: partial → close over catheter; complete → tension-free repair; catheter 2–3 weeks.
- Abstinence about 1 month. Delay up to 7 days does not worsen outcome. After collagenase injection without urethral injury, conservative care is acceptable.

Complications

- About 20%: nodule, curvature, ED, painful erection, infection, stricture or fistula.

Diagram to draw: cross-section of penis showing tunical tear, Buck's fascia and haematoma spread.

> **Key points to remember:**
> - Snap, detumescence, eggplant deformity.
> - Urethral injury about 20% (West); suspect if bilateral.
> - Immediate exploration and interrupted absorbable tunical closure.

Also, from the condensed book (not in the paper answer):

- Bilateral tears (~10%) carry more urethral injury, and absence of blood at the meatus does not exclude it.
- Collagenase ruptures mostly occur after the 14-day observation window; manage them conservatively unless the urethra is injured.

ALGORITHM

1. Explore promptly. Through a ventral penoscrotal (or small lateral) incision, evacuate haematoma and find the tear (artificial erection).
2. Irrigate; close the tunica with interrupted **2-0/3-0 absorbable** sutures.
3. If the urethra is injured, repair it.

- A breached Buck's fascia gives a butterfly haematoma, and a negative MRI reliably excludes a fracture. Avoid deep corporal ligation and excessive debridement.

3 · Penetrating, bite, amputation, zipper and strangulation

- **Gunshot**: explore immediately without waiting for imaging. Debride, give antibiotics and close the tunica.
- Urethral injury occurs in **15–50%** of penile gunshots, so check with RUG or intraoperative dye. Viable tissue is repaired primarily; high-velocity tissue loss needs suprapubic diversion and staged repair.
- **Dog bites**: irrigate, debride and close **primarily**, with amoxicillin–clavulanate and tetanus and rabies cover. **Human bites** present late and infected, so **leave them open** and give amoxicillin–clavulanate or moxifloxacin.

Penile amputation

- It is usually self-mutilation, so **always get psychiatric consultation**. **Microsurgical replantation** is preferred; otherwise repair urethra and corpora macroscopically.
- Transport the part **"double bagged"**: saline gauze in a sterile bag, inside an outer bag of ice slush.
- Erections return in >50% with either technique, but microvascular repair gives less stricture and sensory loss. If the part is lost, formalise the stump; total loss needs phalloplasty.

ALGORITHM

1. Suprapubic cystostomy; debride.
2. Tunica with 2-0 absorbable; urethra over a catheter with 4-0.
3. Anastomose dorsal artery with **11-0**, dorsal vein with **9-0** and dorsal nerve with **10-0** nylon.
4. Provide skin cover.

- **Zipper** injury (children): under penile block, try mineral oil once; if that fails, cut the slider's **median bar** with bone cutters.
- **Strangulation**: in children think of hair tourniquet and **child abuse**. Escalate from lubrication to the **string technique**, then to cutting tools with saline cooling in theatre.

4 · Testicular injury

- About 80% of penetrating scrotal wounds also involve neighbouring structures, including the **femoral vessels**.
- Consider rupture in every blunt injury; haematoma size does not predict it. **Ultrasonography** sensitivity is only ~64–75%, so an **equivocal scan must not delay exploration**.
- A non-palpable testis after trauma suggests **dislocation**, commonest to the superficial inguinal pouch (50%). Trauma does not exclude torsion.

AAST testis grade	Injury
I	Contusion/haematoma
II	Subclinical laceration of tunica albuginea
III	Tunical laceration with <50% parenchymal loss
IV	Major laceration with ≥50% parenchymal loss
V	Total destruction or avulsion

ALGORITHM

1. Explore; remove necrotic and extruded tubules.
2. Close the tunica with small absorbable sutures.
3. If the defect is large, use a tunica vaginalis flap or graft (never synthetic); drain.

- Explore a **significant haematocele** (up to 80% hide rupture) and an **intratesticular haematoma** (delayed orchidectomy 15%).
- Always explore penetrating injury; ligate an injured vas and reconstruct later.
- Salvage is **>90%** if repaired within **72 h**; non-operative care leads to infection, atrophy and late orchidectomy.

5 · Genital skin loss

- Fournier gangrene debridement is the commonest cause. Treat it with antibiotics, resuscitation and **repeated debridement**, then negative pressure wound therapy; hyperbaric oxygen is unproven.

	Penis	**Scrotum**
Graft	**Thick, non-meshed** split-thickness skin graft (STSG) 0.012–0.015 inch, because meshed graft contracts	**Meshed** STSG after local flaps (exudate escapes, so better take)
Key step	Excise residual subcoronal skin, which otherwise causes lymphoedema; never reuse avulsed skin	Suture the testes together to avoid a bifid scrotum; ≤50% loss closes primarily
Note	Graft is insensate, but sexual function is preserved	Thigh pouches not recommended

- Graft 5–7 days after the last debridement, once the wound is granulating, under a bolster or vacuum dressing.

6 · Bladder injury

Intraperitoneal tears need surgery; most extraperitoneal tears heal on a catheter.

LAQ Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management

Paper answer · P2 · P3 · merged from 2

Asked as: P2 LAQ "Classification and treatment of lower track trauma in # pelvis" (also P2 SAQ "Pelvic # assessment and management" and P2 SAQ "Early surgical intervention in urethral injuries") · P3 LAQ "Describe early and late management of PFUD injuries"

PFUD is taken as pelvic fracture urethral distraction defect, i.e. PFUI (pelvic fracture urethral injury). Lower urinary tract trauma means bladder and urethral injury in pelvic fracture.

Introduction

- Pelvic fracture accompanies **83–95% of bladder ruptures**, yet only **5–10% of pelvic fractures** injure the bladder. Urethral injury (PFUI) occurs in about **10% of male** and up to 6% of female pelvic fractures.
- 80–94% have other major (associated) injuries, which cause the deaths; long-term disability comes from wrong management.

Mechanism

- **EPR (extraperitoneal rupture):** pelvic disruption (shearing) tears the bladder at its fascial attachments, or bone spicules lacerate it.
- **IPR (intraperitoneal rupture):** a blow on a **full bladder** bursts the dome, the weakest part, covered by peritoneum.
- **PFUI:** the posterior urethra is fixed to the pubis (puboprostatic ligaments, perineal membrane), so displacement shears it, usually at the **bulbomembranous junction**. In prepubertal boys the tear may extend into the prostate and bladder neck.
- Highest risk: straddle fracture of all four pubic rami, vertically unstable fractures, symphyseal diastasis.
- The sphincter is usually avulsed but stays functional, so most men stay continent after repair.
- The gap fills with scar: an obliterative **distraction defect**, not a true stricture.

Classification

Bladder: AAST (American Association for the Surgery of Trauma) grades I–V

Grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal wall laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Practical: contusion, EPR (simple or complex), IPR, combined.

Urethra: Goldman I–V

Type	Injury
I	Posterior urethra stretched, intact
II	Disruption above the urogenital diaphragm (perineal membrane)
III	Disruption of membranous urethra through the urogenital diaphragm, extending below the membrane into the bulb (commonest)
IV	Bladder neck injury extending into urethra (IVa: bladder-base injury mimicking IV)
V	Pure anterior urethral injury

- **Campbell-Walsh practical grading: partial injury; minor distraction** (a cystoscope passes retrogradely into the bladder); **major distraction** (a cystoscope cannot pass into the bladder). Practically partial vs complete disruption. Stricture risk and need for urethroplasty rise from partial to major.
- Anterior urethra: contusion, incomplete or complete.

Assessment and clinical features

- **ATLS (Advanced Trauma Life Support)** resuscitation first; associated injuries take priority; pelvic binder and stabilisation.
- Bladder: **gross haematuria (nearly all)**, suprapubic pain, inability to void, free fluid.
- Urethra: **triad: blood at the meatus, inability to void, palpable bladder**. A "high-riding" prostate and butterfly perineal haematoma are unreliable.
- **Vaginal and rectal examination**; vaginal examination in every female.

Investigations

- **Blood at the meatus mandates RUG (retrograde urethrogram) before any catheter** (AUA [American Urological Association]). Method: 16 Fr catheter tip in fossa navicularis, 3 mL balloon, penile traction, oblique position, 20 mL contrast, film during injection. Extravasation with bladder filling means partial injury; without filling, complete. Females: urethroscopy.
- **Cystography:** absolute indication: **gross haematuria + pelvic fracture** (rupture in ~29%). Relative: gross haematuria alone; microscopic haematuria with fracture.
- **CT cystography:** retrograde fill with **350 mL** of 2–4% contrast; no drainage film needed.
- Plain cystogram: scout, full-bladder and **drainage films** (posterior leaks otherwise missed).
- Clamping the catheter after IV contrast is inadequate.

- **EPR:** flame-shaped pelvic contrast. **IPR:** contrast outlines bowel loops.

Bladder injury: management

- **Uncomplicated EPR:** 22 Fr urethral catheter 2–3 weeks; cystogram before removal; repair if leak persists beyond 4 weeks.
- **Immediate repair:** IPR; penetrating injury; bladder neck injury; bone fragments in the bladder; rectal or vaginal injury; poor drainage or clots; open pelvic fixation or laparotomy for other reasons.
- IPR: open or laparoscopic two-layer closure; inspect ureteric orifices; cystogram at 7–10 days. At laparotomy, repair EPR from inside the bladder, leaving the pelvic haematoma undisturbed.
- Suprapubic tube adds nothing over urethral catheter (AUA). Antibiotics only around IPR repair.
- Rectal or vaginal injury: separate the walls, no overlapping suture lines, interpose tissue.

Posterior urethra (PFUI): early management

- Key decision: **immediate repair versus deliberate delay**. Clean, localised injuries are repaired at once; PFUI is drained and reconstructed later.

```
Blood at meatus → RUG Partial → one gentle catheter attempt (confirm position) Complete → SPC (suprapubic catheter): percutaneous 16 Fr or open (if bladder repair needed) ± brief primary realignment if stable (flexible cystoscopy, antegrade + retrograde) Bladder neck/rectal injury → immediate repair Delayed (2–3 months) → up-and-down-o-gram → perineal excision and primary anastomosis (corporal split → pubectomy → rerouting)
```

1. ATLS resuscitation; associated injuries take priority.
2. **SPC (suprapubic cystostomy): the standard of care.** Percutaneous (peel-away sheath, 16 Fr Foley; avoid small "punch" tubes) or open through a lower midline incision, which allows bladder inspection (needed if bladder repair is required). - SPC is safe with pelvic plating (AUA Urotrauma guideline); place and tunnel it high, away from hardware.
3. **Primary realignment (PR)** (AUA Urotrauma option): a brief attempt in a stable patient, acutely or within days, by combined antegrade and retrograde flexible cystoscopy over a guidewire (avoid prolonged attempts). Catheter out at 4–6 weeks; keep the SPC and remove it 7–14 days later if voiding well.
4. Partial tear: one gentle catheter attempt allowed.
5. **Immediate open exploration/repair** only for **bladder neck or rectal injury**, or when laparotomy is needed anyway. - **Also know:** female urethral injuries are usually repaired early; in boys tears may extend into the bladder neck.

SPC versus primary realignment

	SPC + delayed urethroplasty	Primary realignment
Advantage	Simple; haematoma untouched	Partial tears may heal; aligned ends ease repair
Drawback	Complete injuries need urethroplasty	Most still form a 1–2 cm stenosis
Evidence	Prospective data: **no significant difference** in need for urethroplasty	Modern PR does not add ED (erectile dysfunction) or incontinence

Anterior urethra: early management

Injury	Early management
Contusion, incomplete straddle tear	Urethral catheter alone
Major straddle injury	SPC (standard); primary realignment for lesser injuries; delayed anastomotic urethroplasty
Low-velocity gunshot or stab	**Immediate primary repair:** debride minimally, spatulate, tension-free anastomosis over a catheter
High-velocity or close-range shotgun	SPC, staged repair
Penile fracture with urethral tear	**Immediate repair:** partial tear oversewn; complete tear debrided and anastomosed; catheter 2–3 weeks

- Penile fracture: urethral injury in about 20% of Western and 3% of Asian series; **intraoperative flexible cystoscopy** before catheterisation. Limit spongiosal debridement.

Late management of PFUI

Timing: delayed repair at 2–3 months (P3: after **3 months**), once the scar is stable and the patient is ambulant.

Evaluation

- **"Up-and-down-o-gram":** cystogram plus RUG while the patient tries to void, to measure the gap length; antegrade flexible cystoscopy via the SPC tract if the bladder neck stays closed / does not open. Bladder neck appearance does **not** predict its later function.
- **MRI (magnetic resonance imaging)** for complex cases: gap length, prostatic displacement (cephalad displacement predicts pubectomy).

Endoscopic: DVIU (direct vision internal urethrotomy) only for short, incomplete stenosis. "Cut-to-the-light" or repeated DVIU for gaps ≥1 cm fails (false passages).

Perineal excision and primary anastomosis (EPA; delayed anastomotic urethroplasty): treatment of choice

1. Exaggerated lithotomy (under 5 hours); sounds via meatus and SPC tract.
2. Midline or lambda incision; mobilise the bulbar urethra up to the penoscrotal junction (not beyond: chordee).
3. Divide the urethra at the scar; excise **all** fibrosis until healthy prostatic mucosa appears over the sound (identify the prostatic apex).
4. Spatulate both ends to 30 Fr; 8–10 interrupted 4-0 absorbable sutures; tension-free mucosa-to-mucosa anastomosis over a silicone catheter.

Progressive perineal approach (each step shortens the path to bridge a longer gap):

Step	Manoeuvre
1	Bulbar urethral mobilisation (enough for most)
2	Separation of the corporal bodies (corporal splitting, intracrural space)
3	Inferior wedge pubectomy (Kerrison rongeur)
4	Supracrural rerouting of the urethra around one corpus

- **Abdominoperineal or transpubic approach:** urethrorectal fistula, failed repair, bladder neck injury, children.
- **Postoperative:** voiding study at 21–28 days; cystoscopy at about 6 months.

Outcomes and complications

- Urethroplasty success above 95%. Restenosis 5–15%; often DVIU-treatable because the scar is already excised.
- **ED** up to 82%, mostly from the injury (cavernous nerves, pudendal arteries; nerve, arterial, venous). **Incontinence <4%** (uncommon; bladder neck).
- Missed IPR: ascites, acidosis, azotaemia (peritoneal reabsorption).
- Fistula, abscess, neurogenic voiding dysfunction; follow up ≥1 year.
- **Also know:** chordee, penile shortening, ejaculatory dysfunction, lithotomy neuropathy, penile Doppler before repair in impotent men.

Diagram to draw: pelvis with EPR (flame-shaped) vs IPR on cystogram; RUG showing partial vs complete posterior urethral disruption; sagittal pelvis showing the bulbomembranous gap and the four progressive perineal steps.

> **Key points to remember:**
> - Blood at meatus: RUG before catheter.
> - Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
> - EPR: catheter 2–3 weeks; IPR: immediate repair.
> - PFUI: SPC is the standard initial step, then delayed perineal anastomotic urethroplasty at 2–3 months (P3: after 3 months).
> - Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
> - Progressive perineal steps: bulbar mobilisation, crural separation, pubectomy, rerouting.
> - Immediate repair: bladder neck or rectal injury, penile fracture urethra, low-velocity penetrating injury.
> - ED mostly from the injury; incontinence <4%.

Also, from the condensed book (not in the paper answer):

- Underfilling the bladder gives false-negative cystograms.
- Bone spicules in the wall prevent healing, so repair them; a ureteric orifice injury needs a stented reimplant.

AAST bladder grade	Injury
I	Contusion, intramural haematoma, partial-thickness laceration
II	Extraperitoneal laceration <2 cm
III	Extraperitoneal ≥2 cm, or intraperitoneal <2 cm
IV	Intraperitoneal laceration ≥2 cm
V	Laceration extending into bladder neck or ureteric orifice (trigone)

- Intraperitoneal leakage causes chemical peritonitis and, once reabsorbed, **acidosis and azotaemia**.

KEY POINT

Immediate repair: intraperitoneal, penetrating, bladder neck, rectal/vaginal injury, bone fragments in bladder, clots/poor drainage, iatrogenic urological injury, laparotomy for other reasons.

7 · Posterior urethral injury (PFUI)

Drain now and reconstruct later.

*See the paper answer **Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management** (Section 11, Ch 125).*

Also, from the condensed book (not in the paper answer):

- PFUI complicates ~10% of male and up to 6% of female pelvic fractures, mostly **straddle (four-rami)** and unstable fractures.
- ED is predicted by symphyseal diastasis, lateral displacement and a long gap. Endoscopic "cut-to-the-light" fails for gaps ≥1 cm.

AAST urethra grade	Injury
I	Contusion — blood at meatus, normal urethrogram
II	Stretch — elongation without extravasation
III	Partial disruption — extravasation, contrast reaches bladder
IV	Complete disruption, <2 cm separation
V	Complete, >2 cm separation or extension into prostate/vagina

ALGORITHM

1. If there is blood at the meatus, do RUG before any catheter.
2. Place a 16-Fr catheter in the fossa navicularis and inflate the balloon with 3 mL.
3. Apply penile traction, position oblique, inject 20 mL contrast and image during injection.

- In females, examine the vagina in every pelvic fracture. A suprapubic tube is safe with pelvic hardware if tunnelled high in the midline.

8 · Anterior urethral injury

- Most are **straddle injuries** crushing the bulbar urethra against the pubis. Contusion or incomplete tear needs a catheter; major injury a suprapubic tube.
- Later, the 1.5–2 cm obliterated bulbar segment is treated by **excision and primary anastomosis** (>95% success). Urethrotomy through an obliteration is futile.

9 · Landmark trials

Trial	Design	Result	Take-home
Multi-institutional PFUI study	Prospective, suprapubic tube vs early endoscopic realignment	No difference in stenosis or urethroplasty need	Suprapubic tube remains standard; realignment optional, brief

62 Paediatric Genitourinary Trauma

1 · Renal injury: how the child differs

- The kidney is the commonest site of paediatric genitourinary (GU) trauma. Most injuries are **blunt** and grade 1–3; half have other injuries.
- Congenital anomalies such as ureteropelvic junction (UPJ) obstruction are commoner than in adults.

Feature	Child	Adult
Protection	Less perirenal fat, weaker wall, kidney below the ribs	Better cushioned
Blood pressure	Maintained until ≈ **25–30% volume loss** ; hypotension is late	Falls earlier
Haematuria	May be absent (up to 2/3 of grade ≥ 2 injuries)	Classic trigger
Imaging decision	**Mechanism-led**; keep radiation as low as reasonably achievable (ALARA)	Led by haematuria and hypotension

2 · Indications for imaging

Setting	Indication (American Urological Association [AUA])
Stable blunt	Gross haematuria
Stable blunt	Microscopic haematuria with systolic blood pressure (SBP) < 90 mm Hg
Mechanism / examination	Rapid deceleration; significant flank blow; rib fracture; flank ecchymosis
Penetrating	Any injury to abdomen, flank or lower chest, even without haematuria

- Triggers: fall > **10 feet (3 m)**, high-speed collision, flank blow; rib, spine, femur or pelvic fracture. All four markers together (haematuria, hypotension, associated injury, mechanism) give > **98% sensitivity**.
- Paediatric hypotension (1–10 years) is SBP < 70 + (2 × age) mm Hg.

> **KEY POINT**
>
> Trap: "no haematuria, so no imaging". In children both adult markers fail, so mechanism drives the decision.

3 · Imaging studies

- Renal ultrasound (US) detects grade 3–5 without radiation and suits screening, but **contrast computed tomography (CT)** is preferred. Add a **delayed (excretory) phase** if a leak is suspected.

4 · Grading

American Association for the Surgery of Trauma (AAST): grades 1–3 are **low** and 4–5 are **high**.

Grade	1989 AAST	2011 revision
1	Contusion or non-expanding subcapsular haematoma	Subcapsular haematoma/contusion

Grade	1989 AAST	2011 revision
2	Perirenal haematoma; cortical laceration < 1 cm	Same, within Gerota fascia
3	Laceration > 1 cm, no collecting-system injury	Same, into medulla
4	Laceration into collecting system; main artery/vein injury with contained haemorrhage	Collecting system with extravasation; segmental vessel injury; pelvis laceration or **UPJ disruption**; shattered kidney
5	**Shattered kidney**; hilar avulsion	Main artery/vein laceration, avulsion or thrombosis

KEY POINT

Grades 1–3 differ only in depth (1 cm) and never involve the collecting system; entry makes grade 4. Shattered kidney is grade 5 (1989) but grade 4 (2011).

High-risk CT criteria (grade 4–5)

Criterion	Predicts
Perirenal haematoma > 3.5 cm adult, > **2.2 cm child**	Bleeding needing angioembolisation
Vascular contrast extravasation; medial laceration	Bleeding
> 25% parenchyma devitalised	Bleeding and open surgery
Medial urinary extravasation	UPJ disruption
Extravasation with no contrast in distal ureter	Urine leak needing intervention

With 0–1 factor, < 10% need intervention; with 2–3, > 50% do (grade 4a vs 4b).

5 · Non-operative management (NOM)

- NOM is the **default at every grade**, including grade 4–5 and stable penetrating injury, provided the child is stable.

ALGORITHM

1. Confirm stability; grade the injury.
2. Bed rest, with serial haemoglobin and abdominal examination.
3. When gross haematuria resolves and counts are stable, mobilise.

- Grade 1–2 needs only **23-hour observation**. Repeat imaging: AUA CT at 48 h (grade 4–5), EAU US at 2–3 days; but **symptoms, not imaging, predict complications**.
- Grade 3 (devitalised) to 5: ≈ **25%** need embolisation, ≈ **15%** urinoma intervention, ≈ 5% exploration.

6 · Complications during NOM

- **Urine leaks** occur only in grade 4–5, and up to **85%** resolve. A urinoma shows **flank pain, ileus and low-grade fever**.
- If no contrast reaches the distal ureter, do retrograde pyelography (RPG). Escalate from **stent and Foley catheter** (AUA first line, ≈ 90%; out at 4–6 weeks) to percutaneous drain, then nephrostomy, then open repair.

Bleeding	Persistent	Delayed
Mechanism	Vessels fail to tamponade	Fistula or pseudoaneurysm ruptures as the haematoma liquefies
Timing	From injury	Typically **1–2 weeks** (up to 1 month)
Stable	Allow 2–3 transfusions, then embolise	Early embolisation unless mild
Unstable	Urgent angioembolisation	

- Superselective embolisation succeeds in > 80%.

7 · Operative management

- Operate for **instability from renal bleeding**, pulsatile or expanding haematoma, or failed embolisation. A stable non-pulsatile haematoma found at laparotomy is **left alone**.
- A **one-shot pyelogram** (2 mL/kg, film at 10–15 min) confirms a **contralateral kidney**.

*See the paper answer **Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management** (Section 11, Ch 94).*

Also, from the condensed book (not in the paper answer):

- Early vascular control did not lower nephrectomy in the one randomised trial.

ALGORITHM

1. Confirm a contralateral kidney.
2. Open Gerota fascia; control the vessels; debride; close the collecting system, then parenchyma over a bolster.
3. Drain. If the injury is penetrating, inspect the ureter.

- **UPJ disruption** follows deceleration, often without haematuria; CT shows **medial extravasation, empty distal ureter and often no laceration**. Repair early if stable; otherwise nephrostomy and repair at 2–3 months.

8 · Follow-up

- Scarring: grade 1–2 none, grade 3 ≈ 50–60%, grade 4–5 ≈ 100%, so do **US at 2–4 months**.
- **Hypertension** is rare and renin-mediated, so use renin–angiotensin blockade. Check blood pressure yearly after grade 3–5.

9 · Ureteral trauma

- Most are penetrating. **Haematuria is absent in 25–50%** and ≈ 10% are missed at laparotomy, so diagnose by **delayed-phase CT** or RPG.

AAST grade	Injury
I	Contusion/haematoma, no devascularisation
II / III	< 50% / ≥ **50%** transection
IV	Complete transection, < 2 cm devascularisation
V	Avulsion, > **2 cm** devascularisation

- Found within 1 week: repair if stable; ligate and place a nephrostomy if unstable. Found after 5–7 days: stent first; if that fails, nephrostomy and reconstruction at 6–12 weeks.

*See the paper answer **Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury** (Section 11, Ch 94).*

Also, from the condensed book (not in the paper answer):

- **Avoid transureteroureterostomy in children**, because a long life puts the other kidney at risk.

10 · Bladder injury

Feature	Child	Adult
Bladder position	**More intraperitoneal**, so isolated injury is commoner (≈ half have no fracture)	Pelvic organ
Pelvic fracture	Half as frequent, so it implies higher energy and more urethral injury	Commoner
Posterior urethral injury	Prostatic; **bladder neck injury commoner**	Bulbomembranous

- **Gross haematuria or retention** calls for **retrograde cystography** with retrograde urethrography (RUG); delayed-phase CT may miss a rupture.
- AAST: I contusion; II extraperitoneal < 2 cm; III extraperitoneal ≥ 2 cm or intraperitoneal < 2 cm; IV intraperitoneal ≥ 2 cm; V bladder neck or trigone.
- Extraperitoneal tears get a catheter, but are repaired for bone fragments, hardware, rectal or vaginal tear, or bladder neck injury. Intraperitoneal tears are **always repaired**. Keep the catheter 1–2 weeks, then do a cystogram.

> **KEY POINT**
>
> Bladder neck avulsion is commoner in children; girls depend more on the bladder neck for continence, so outcome is poor and many need secondary procedures.

11 · Urethral injury

Grade/type	AAST	Goldman
I	Contusion: meatal blood, normal RUG	Posterior urethra stretched, intact
II	Stretch, no extravasation	Rupture above intact urogenital diaphragm
III	Partial: extravasation, contrast reaches bladder	Rupture through diaphragm into bulbar urethra
IV	Complete, no bladder filling, < 2 cm gap	Bladder rupture extending into proximal urethra
V	≥ 2 cm gap or into prostate/vagina	Anterior urethral injury

- Do **RUG before any catheter**; if a catheter will not pass, place a **suprapubic tube**.
- Realignment is a brief attempt only; EAU advises against immediate urethroplasty. **Bladder neck injury is repaired at once.**

ALGORITHM

1. Divert with a suprapubic tube and wait several months.
2. Assess with cystogram and RUG, with or without endoscopy.
3. Do transperineal excision and primary anastomosis; if there is tension, add pubectomy.
4. Catheter for 2–3 weeks; cystogram; remove the suprapubic tube.

- Restricture ≈ 10%, incontinence ≤ 20%, and erectile dysfunction up to half (gap > 2.5 cm).

12 · Genital injury

> **KEY POINT**
>
> Most are accidental, but **abuse must always be assessed** (≈ 1 in 7 boys, 1 in 3 girls abused). Involve social services before discharge.

- Penis AAST: I skin; II Buck fascia; III avulsion, glans/meatus laceration, defect < 2 cm; IV partial penectomy or ≥ 2 cm; V total penectomy.
- **Circumcision** is the commonest cause; the **Mogen clamp** causes the worst (glans amputation).
- Testis AAST: I contusion; II subclinical tunica tear; III < 50% loss; IV ≥ 50% loss; V destruction. If US shows tunica discontinuity, haematocele or focal flow loss, **explore and close the tunica**.
- Hymenal, vaginal or perianal injury means **penetration: evaluate for abuse**; a **normal examination does not exclude abuse**.

★ Putting it together

A · Renal colic

ALGORITHM

1. Give a non-steroidal anti-inflammatory drug (NSAID; ketorolac 30 mg) unless there is renal failure, bleeding or pregnancy. Confirm with non-contrast CT, or ultrasound in pregnancy and children.
2. With infection, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration or refractory pain, **decompress now** (stent or nephrostomy; equal). Never lithotripsy while infected.
3. **Otherwise**, if the stone is <10 mm, try passage with or without an α-blocker (distal 5–10 mm) for 4–6 weeks; if that fails, do ureteroscopy (URS) or shock wave lithotripsy (SWL).

B · Stone operation

Stone	First choice	Notes
Ureteric, any level	URS or SWL	URS clears more, SWL has fewer complications; hard proximal >15–20 mm: consider percutaneous nephrolithotomy (PCNL)
Renal <2 cm, not lower pole	SWL or URS	SWL works best <900 Hounsfield units and skin-to-stone <10 cm
Lower pole >1 cm	PCNL or URS	SWL not first line (Lower Pole I)
Burden >2 cm, staghorn	PCNL	Staged URS an option

KEY POINT

SWL absolute contraindications: pregnancy, coagulopathy or antithrombotics, untreated infection, uncontrolled hypertension, arterial aneurysm, and obstruction distal to the stone. No routine stent after SWL or uncomplicated URS.

C · Prevention

ALGORITHM

1. **All patients:** urine ≥2.5 L/day, normal calcium, low sodium and less animal protein.
2. **If high risk**, give therapy guided by 24-hour urines, repeated within 6 months.

Abnormality	Treatment
Hypercalciuria	Thiazide and potassium citrate (hypokalaemia lowers citrate)
Hypocitraturia	Potassium citrate
Uric acid	Alkalinise to pH 6.5–7; allopurinol only for hyperuricosuria/gout or failed alkali
Cystine	≥3 L/day fluid, pH 7.0–7.5, then tiopronin
Struvite	Complete clearance; check metabolism

D · Renal trauma

ALGORITHM

1. **Image** with delayed-phase CT for gross haematuria, microhaematuria with shock, deceleration or penetrating injury.
2. If stable with grade I–IV (or selected V), manage **non-operatively** (>90% succeed). Embolise a contrast blush; stent a symptomatic leak.
3. If unstable or expanding, **explore**; a one-shot pyelogram (2 mL/kg, 10 min) confirms the other kidney.

E · Bladder and urethra

Injury	Management
Extraperitoneal bladder rupture	22 Fr catheter, cystogram at 2–3 weeks; repair if at laparotomy, bone spicule, or unhealed at 4 weeks
Intraperitoneal bladder rupture	Surgical repair
Pelvic fracture urethral injury	Suprapubic tube (or brief realignment) now, delayed urethroplasty later
Penile fracture	Prompt tunical repair; check the urethra
Testicular rupture	Immediate repair

Retrograde cystography must fill to ≥350 mL.

F · Ureteric defects

Defect	Repair
2–3 cm, upper/mid	Spatulated ureteroureterostomy
Lower ureter, 6–10 cm	Psoas hitch reimplant
10–15 cm	Boari flap
Longer	Transureteroureterostomy, ileal ureter (creatinine <2 mg/dL), autotransplant, nephrectomy

G · Children

KEY POINT

Stones: ultrasound first; full metabolic evaluation after one stone. Renal ≤20 mm: SWL or URS; >20 mm: PCNL. **Trauma:** image by mechanism (fall >3 m, deceleration); a perirenal haematoma >2.2 cm predicts embolisation.

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

SAQ Renal autotransplantation

Paper answer · P1 · P3 · merged from 2

Asked as: P3 SAQ "Auto-transplantation of Kidney?" / "Role of auto transplant of kidney in Urology." / "Renal Autotransplantation" · P1 SAQ "Autotransplantation"

Definition and principle

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) surgery** if needed. First done in **1963** for proximal ureteric injury.
- **Principle:** bringing the kidney to the bladder replaces the ureter without bowel; the bench gives a cold, bloodless field.

Role (indications)

Group	Examples
Ureteric loss (commonest)	Long-segment loss or stricture; long iatrogenic or traumatic loss; repeated failed repairs (**delayed setting only, never acute**)
RPF (retroperitoneal fibrosis)	When ureterolysis fails or is impossible and function is satisfactory
Tumour	Total ureterectomy for UTUC (upper tract urothelial carcinoma) in a solitary kidney; bench partial nephrectomy in a solitary kidney; left RCC (renal cell carcinoma) needing suprarenal caval resection (new drainage for the right kidney)
Vascular	Complex branch renal artery aneurysm, stenosis or AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker syndrome
Pain	Selected loin pain–haematuria syndrome

- **Preferred over ileal ureter** when the contralateral kidney is absent or poor, or bowel is unsuitable (creatinine >2 mg/dL, inflammatory bowel disease, radiation enteritis).
- **Contraindications:** poor renal function, severe iliac atherosclerosis, unfit patient.

Procedure

`CT (computed tomography) angiography, renogram, ureteric study → harvest as for live donor nephrectomy (open/laparoscopic/robotic): maximal vessel length, preserve ureteric blood supply → cold perfusion on bench ± ex vivo repair → Gibson incision, extraperitoneal iliac fossa → renal vein to external iliac vein (end-to-side); renal artery to external iliac (end-to-side) or internal iliac (end-to-end) → drainage: pyelovesicostomy (± Boari flap), ureteroneocystostomy, or native ureter left in continuity → stent, catheter, drain`

- Laparoscopic harvest plus Gibson implantation: less analgesia, faster recovery; **robotic autotransplantation** emerging.

Complications and outcome

- Arterial or venous **thrombosis**, bleeding, urine leak, anastomotic stricture, lymphocele, infection.
- **Renal unit loss 5–10%** (lower in transplant centres); major (Clavien–Dindo ≥III) complications about 15–20%.
- **Long-term success about 90%.**

Follow-up

- Doppler ultrasound early; renogram and blood pressure later.

Alternatives

- Boari flap, TUU (transureteroureterostomy), ileal ureter, Yang–Monti, buccal ureteroplasty, renal descensus.

Diagram to draw: kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

> **Key points to remember:**
> - For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
> - Delayed setting only; bench surgery for complex vascular or tumour cases.
> - Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
> - Unit loss 5–10%; success about 90%.

SAQ Newer methods of stone analysis

Paper answer · P3 · P4 · merged from 2

Asked as: P3 SAQ "Newer methods of stone analysis" / "Newer techniques of stone analysis" · P4 SAQ "Newer methods of stone analysis and their significance in management of calculus disease."

Why analyse stones

- AUA (American Urological Association) and EAU (European Association of Urology): analyse a stone at least once in every former (every first stone); **repeat if stones recur on treatment** (evolving metabolic condition), or on early recurrence after clearance.
- Composition points to the cause: **uric acid** suggests metabolic disease, **cystine** a genetic one, **struvite** infection. Cystine, struvite, 2,8-DHA (2,8-dihydroxyadenine) and xanthine are diagnostic in themselves.
- Composition can change over time (e.g. calcium oxalate converting to brushite), so re-analyse recurrent stones.

Why wet chemical analysis is obsolete

- Detects ions, not **crystal phases**, so it cannot separate COM (calcium oxalate monohydrate, whewellite) from COD (dihydrate, weddellite), or apatite from brushite.
- High error rates; misses rare stones (2,8-DHA, drugs); needs a large sample.

Newer methods

Method	Principle	Comment
FTIR (Fourier-transform infrared) spectroscopy	Each compound absorbs a characteristic infrared spectrum	Commonest reference method; tiny sample; detects proteins and drugs
XRD (X-ray diffraction)	Crystal lattice diffracts X-rays in a unique pattern	Gold standard for crystals; misses amorphous material
Polarising microscopy	Crystal optics (birefringence)	Cheap; needs an experienced observer (observer-dependent)
Morpho-constitutional analysis (Daudon)	Stereomicroscopy of surface and section, plus FTIR	Subtypes point to cause (e.g. type Ia COM: low urine volume; Ic: primary hyperoxaluria)
SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray)	Surface structure and elements	Research
Micro-CT (micro-computed tomography)	3-D non-destructive structure	Research
Raman spectroscopy	Scattered light spectrum	Research

In vivo prediction

- **DECT (dual-energy computed tomography):** two tube energies (e.g. 80 and 140 kVp) separate **uric acid from non–uric acid stones** with near-perfect accuracy; it estimates composition in vivo, mainly uric acid versus calcium stones.
- **Hounsfield units:** SWL (shock wave lithotripsy) works best below about 900–1000 HU; >1000 HU predicts SWL failure; cystine and brushite are dense.
- **Endoscopic recognition** of morphology (lost when stones are dusted). Dusting loses the specimen: collect fragments.

Significance in management

- **Uric acid:** oral dissolution by urinary alkalinisation (potassium citrate).
- **Struvite:** complete clearance, culture-directed antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin; avoid SWL.
- **COM, brushite, cystine:** hard, SWL-resistant; prefer endoscopy.
- **Brushite:** can convert from calcium oxalate after SWL or infection; avoid over-alkalinisation.
- **2,8-DHA:** APRT (adenine phosphoribosyltransferase) deficiency; treat with allopurinol or febuxostat. **Drug stones** (indinavir, triamterene): stop the drug.

Diagram to draw: FTIR spectrum outline and COM (dumbbell) versus COD (envelope) crystal shapes.

Key points to remember:

- FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemistry is obsolete.
- DECT (dual-energy computed tomography) separates uric acid stones in vivo, mainly uric acid versus calcium stones.
- Composition guides dissolution, SWL choice and prevention.
- Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

SAQ What are nanobacteria? Describe their role in urolithiasis.

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 1, Q3e

Definition

- **Nanobacteria**, now called **CNPs (calcifying nanoparticles)**, are spherical particles far smaller than bacteria with a **calcium apatite shell**, described in bovine and human serum by **Kajander and Çiftçioglu (1998)** as *Nanobacterium sanguineum*.
- **Their living nature is disputed.** Proposed as the smallest self-replicating bacteria, but later work showed "replication" is mineral nucleation, and that they are **mineralo-protein complexes** (apatite with serum proteins such as fetuin-A). Their bacterial DNA proved to be contamination. Most now regard them as **non-living**.

Claimed properties

Property	Claim
Size	Pass through bacteria-retaining filters
"Growth"	Very slow, in serum-containing medium
Shell	Apatite; resistant to heat and gamma irradiation
Detection	Electron microscopy, von Kossa and Hoechst staining, specific antibodies, culture
Susceptibility	Tetracycline and EDTA (ethylenediaminetetraacetic acid) inhibit them

Proposed role in urolithiasis

- **Nidus theory:** CNPs in blood or urine are filtered or excreted, deposit apatite in renal tubular cells and interstitium, and act as a **nucleating nidus** for crystal growth.
- **Link to Randall plaque:** idiopathic calcium oxalate stones grow on **Randall plaque**, an interstitial **apatite** deposit that starts in the basement membrane of the thin loop of Henle and erodes through the papilla. CNPs are **associated with**, but **not proven to cause**, Randall plaque (Campbell, Chapter 95).
- Early studies found apatite nanoparticles in a large majority of stones; rat studies reported renal calcification after injection.
- Also implicated in atherosclerotic calcification, polycystic kidney cysts and chronic prostatitis.

Therapeutic implications (unproven)

- Tetracycline-based regimens with EDTA were tried in stones and chronic prostatitis/chronic pelvic pain syndrome; no RCT (randomised controlled trial) evidence supports routine use.

Current position

- Not in stone guidelines; pathogenesis is explained by supersaturation, inhibitors, Randall plaque and ductal plugs.

Diagram to draw: renal papilla with interstitial apatite (Randall plaque) eroding through urothelium, CNPs as nidus, and calcium oxalate stone attached.

> **Key points to remember:**
> - Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
> - Proposed nidus for apatite and Randall plaque; association, not causation.
> - Tetracycline/EDTA therapy unproven.

SAQ Tear drop bladder

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 1, Q3a

Definition

- A **tear-drop (pear-shaped or inverted-pear) bladder** is a bladder that is symmetrically **compressed from both sides** and elongated upwards on cystogram, IVU (intravenous urogram) or CT (computed tomography).
- It is a radiological sign of a **bilateral perivesical mass or collection**, not a disease itself.

Causes

Group	Causes
Collections	**Pelvic haematoma** (pelvic fracture: commonest in trauma); urinoma (extraperitoneal leak); bilateral lymphocele (after lymphadenectomy or transplant); pelvic abscess
Fat	**Pelvic lipomatosis**
Muscle	**Iliopsoas hypertrophy** (muscular young men)
Vascular	Dilated pelvic venous collaterals (IVC [inferior vena cava] occlusion); bilateral iliac artery aneurysms
Nodes and tumour	Pelvic lymphadenopathy (lymphoma, metastases); pelvic retroperitoneal fibrosis; pelvic tumours

Pelvic lipomatosis (the classic cause)

- Overgrowth of benign, non-encapsulated fat in the perivesical and perirectal spaces; mainly in men, more in Black men.
- **Presentation:** LUTS (lower urinary tract symptoms), suprapubic pain, constipation; often incidental.
- **Imaging:**
- Plain film: pelvic lucency. IVU: elevated, elongated pear-shaped bladder; **medial deviation of both ureters**; hydronephrosis.
- Barium enema: straightened, elongated rectosigmoid.

- **CT or MRI (magnetic resonance imaging):** fat density is diagnostic.
- **Cystoscopy:** high bladder neck, difficult passage; often **cystitis glandularis**, which can progress to adenocarcinoma. Biopsy and surveillance are needed.
- **Management:** observe if no obstruction. Upper-tract obstruction → stent or nephrostomy, ureteric reimplantation or urinary diversion. Weight loss rarely helps; surgical excision of fat is difficult.

Evaluation of a tear-drop bladder

- History: trauma, pelvic surgery or transplant, malignancy.
- **CT** shows the cause (blood, fat, fluid, muscle, nodes, vessels). Ultrasound for collections; aspiration (creatinine for urinoma, lymph for lymphocele).

Management

- Treat the cause: pelvic haematoma usually observed; lymphocele drained or marsupialised (laparoscopic fenestration); urinoma drained and leak repaired.

Diagram to draw: cystogram outline of a bladder squeezed into a tear drop by bilateral pelvic haematoma.

> **Key points to remember:**
> - Tear-drop bladder = bilateral extrinsic compression.
> - Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
> - Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

SAQ Calyceal diverticulum – classification and management

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 20 SAQ Sheet, Q3c

Definition

- Cystic cavity in the renal cortex, **lined by transitional epithelium**, connected to a calyx (or pelvis) by a **narrow neck**. Fills passively with urine.
- About 4.5 per 1000; equal in children and adults; no side or sex preference.

Classification

	Type I	Type II
Frequency	Commoner	Less common
Connects to	Minor calyx (usually upper pole)	Renal pelvis or major calyx
Size, symptoms	Small, usually silent	Larger, often symptomatic

Aetiology

- **Congenital** (favoured): persistent third- and fourth-generation ureteric bud branches.
- **Acquired:** cortical abscess draining into a calyx, infundibular stenosis, trauma, reflux.

Clinical features

- Stasis: **stones in up to 40%**, milk of calcium, infection, pain, haematuria.

Evaluation

- Ultrasound mimics a simple cyst. **CT (computed tomography) urography with delayed images shows contrast pooling**; milk of calcium layers and shifts with position.
- Retrograde pyelography maps the neck before intervention.
- **Differential:** hydrocalycosis (dilated major calyx from infundibular obstruction), megacalycosis, simple cyst.

Management

- **Asymptomatic:** observe with ultrasound.
- **Treat** pain, recurrent infection, haematuria, abscess, symptomatic stones.

Approach	Technique	Best for
Ureteroscopy	Find the ostium, laser-incise or balloon-dilate the neck, remove stones	Upper or mid-pole, small stones
PCNL (percutaneous nephrolithotomy)	Direct puncture onto stone, clear stones, fulgurate lining, dilate or create neck	Posterior, lower-pole, larger stones; long-term results good
Laparoscopic or robotic	Unroof, remove stones, close neck, obliterate cavity, pack with fat	Anterior, thin-parenchyma, failed endoscopy
Partial nephrectomy	Excision	Rare; destroyed segment

- Endoscopy keeps the cavity draining by widening the neck; laparoscopy abolishes it by closing the neck.
- **Also know:** SWL (shock wave lithotripsy) gives symptom relief but low stone clearance; fulguration of lining is debated.

Diagram to draw: kidney with upper-pole diverticulum, narrow neck to calyx, stone inside.

> **Key points to remember:**
> - Transitional lined, narrow neck; type I calyceal, type II pelvic.
> - Stones in up to 40%; delayed CT shows pooling.
> - Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

SAQ Pelvic lipomatosis

Paper answer · P1

Type: SAQ (10 marks)
Source: Practice Set 20 SAQ Sheet, Q3e

Definition

- Rare benign **overgrowth of non-encapsulated mature fat** in the perivesical and perirectal spaces, compressing the bladder, ureters, rectum and pelvic veins.
- Mostly men aged 20–60; associated with obesity and hypertension. Cause unknown; obesity alone does not explain it (weight loss rarely shrinks the fat). Chronic inflammation and venous obstruction are proposed.

Clinical features

- LUTS (lower urinary tract symptoms), haematuria, recurrent UTI (urinary tract infection), suprapubic or perineal pain, constipation.
- Leg oedema or venous thrombosis from venous compression.

- **Hydronephrosis and renal failure** from ureteric compression in a substantial minority.
- **Proliferative cystitis (cystitis glandularis)** is common; it can transform to **adenocarcinoma**.

Evaluation

- **IVU (intravenous urography) or cystogram:** elevated, elongated **"pear-" or "teardrop-shaped" bladder**; medially displaced pelvic ureters; high prostate.
- **Barium enema:** elongated, narrowed rectosigmoid.
- **Ultrasound:** echogenic pelvic fat; hydronephrosis.
- Medial deviation of **both** pelvic ureters also occurs in retroperitoneal fibrosis and after abdominoperineal surgery.
- **CT (computed tomography) or MRI (magnetic resonance imaging): diagnostic**; homogeneous fat density surrounding pelvic organs, separates it from tumour.
- **Cystoscopy:** elevated trigone, long prostatic urethra (rigid scope may be difficult); biopsy for cystitis glandularis.
- **Differential of pear-shaped bladder:** pelvic haematoma, lymphadenopathy, liposarcoma, iliopsoas hypertrophy, venous collaterals, pelvic fibrosis.

Management

- **Surveillance** (often progressive): ultrasound for hydronephrosis, renal function, periodic cystoscopy for cystitis glandularis.
- Weight loss, antibiotics and steroids give limited or no benefit; COX-2 (cyclo-oxygenase-2) inhibitors for cystitis glandularis are case reports only.
- **Ureteric obstruction:** stents or nephrostomy; ureteric reimplantation (difficult, dense fat).
- **Advanced or failed cases:** urinary diversion; cystectomy if adenocarcinoma or unmanageable bladder.
- Surgical excision of fat is difficult: no tissue planes, heavy bleeding.
- Resect cystitis glandularis transurethrally and treat obstruction or inflammation.

Diagram to draw: cystogram outline of pear-shaped bladder with medially deviated ureters.

> **Key points to remember:**
> - Pear-shaped bladder with medial ureters; CT shows fat.
> - Cystitis glandularis common; adenocarcinoma risk.
> - Watch for hydronephrosis; diversion if obstruction progresses.

SAQ Physiology of UPJ in relation to urine transport

Paper answer · P2

Type: SAQ (10 marks)
Source: 2023-Winter Paper 2, Q3f

UPJ = ureteropelvic (pelviureteric) junction.

Anatomy relevant to function

- The UPJ is the apex of a funnel-shaped pelvis; its spiral smooth muscle is continuous with calyces and ureter.

Pacemaker and electrical activity

- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves carry pain and only modulate.
- **Primary pacemaker: atypical smooth muscle cells** at the pelvicalyceal border; they fire spontaneously (HCN, hyperpolarisation-activated cyclic nucleotide-gated, channels).
- **ICC-like (interstitial cells of Cajal-like) cells** (c-Kit positive) in the pelvis and UPJ relay the signal to typical smooth muscle and act as back-up pacemakers.
- Signal spreads through **gap junctions**. Calcium entry gives a plateau action potential; calcium–calmodulin activates myosin light-chain kinase, causing contraction.

Bolus formation and transport

1. Urine fills the pelvis at low pressure; the UPJ is closed.
2. Pelvic contraction pushes urine through the opening UPJ into the upper ureter.
3. The UPJ closes behind it, forming a **bolus**.
4. Ureteric walls **coapt** behind the bolus; the wave (2–6 per minute, 20–80 cm H_2O) carries it to the bladder. - At **low flow**, the UPJ acts as a filter: several pelvic contractions occur for each ureteric contraction. - As **flow rises**, ureteric frequency reaches 1:1, then bolus volume rises. - At **very high flow**, boluses merge into a continuous column and the ureter dilates **without obstruction**.

Effect of obstruction

- **Acute:** stronger, more frequent contractions.
- **Chronic:** dilatation, walls fail to coapt, peristalsis becomes ineffective; raised pelvic pressure lowers GFR (glomerular filtration rate).
- **Intrinsic UPJO (UPJ obstruction):** abnormal muscle, excess collagen, fewer ICC-like cells, so an **aperistaltic segment** cannot transmit the wave.
- **Also know:** endotoxin and progesterone reduce peristalsis. Diuretic renogram and Whitaker test (chapter: pressure difference >12 cm H_2O suggests obstruction; classic teaching: <15 unobstructed, >22 obstructed) assess transport.

Diagram to draw: pelvis, UPJ and ureter showing bolus formation and coaptation.

> **Key points to remember:**
> - Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
> - Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
> - Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

SAQ Complications of PCNL, including bleeding and persistent haematuria after PCNL

Paper answer · P2

Asked as:

- Complications of PCNL — *Practice Set 2 Paper II Q3e (SAQ)*
- What are the causes of bleeding in PCNL? How will you manage persistent hematuria 6 days after PCNL!? — *Practice Set 6 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)
(mini-PCNL complications in 2022-Summer Paper 1, Q1) (PCNL complications also in Practice Set 2 Paper 2, Q3e)

Introduction

- PCNL (percutaneous nephrolithotomy) is the standard for renal stones >2 cm. Most complications arise from **access and tract dilatation**.
- Graded by the **Clavien–Dindo** system (I: no intervention; II: drugs or transfusion; III: intervention; IV: ICU [intensive care unit]; V: death).
- Bleeding is the commonest serious complication of PCNL. Transfusion was historically ~7%, now nearer 1% with smaller tracts. Angioembolisation is needed in <1%.

Incidence (systematic review, Seitz 2012, quoted by EAU [European Association of Urology])

Complication	Rate
Fever	10.8%
Transfusion	7% (nearer 1% with modern and mini tracts)
Thoracic	1.5%
Sepsis	0.5%
Organ injury	0.4%
Angioembolisation	0.4%
Death	0.05%

Main complications

Complication	Cause	Prevention and management
Acute bleeding	Infundibular or segmental vessel injury	Fornix puncture; place sheath, tamponade with balloon or **Kaye catheter**, clamp nephrostomy; transfuse
Delayed bleeding	Pseudoaneurysm, AV (arteriovenous) fistula	**Selective angioembolisation**
Infection, SIRS (systemic inflammatory response syndrome), sepsis	Infected stone, high intrarenal pressure	Sterile urine, culture-specific antibiotics, low-pressure irrigation; stop and drain if pus
Pleural (pneumothorax, hydrothorax)	Supracostal access, highest above 11th rib	Subcostal where possible, puncture in expiration; chest drain
Colon	Retrorenal colon, left lower pole	CT review; extraperitoneal: double-J stent and withdraw nephrostomy into colon (colostomy tube)
Spleen, liver	Upper-pole access, organomegaly	Ultrasound guidance; conservative or surgery
Collecting system	Perforation, extravasation	Stop, drain; later infundibular stenosis
Fluid absorption	Long operation, perforation	**Normal saline**; hyponatraemia, overload

Also know: venous air embolism, DVT (deep vein thrombosis) and pulmonary embolism, neuropraxia and eye injury (prone), residual fragments, loss of renal function (multiple tracts), urinoma, nephrocutaneous fistula, retained hardware.

Risk factors

- Staghorn, multiple tracts, supracostal access, large tract, long operation, CKD (chronic kidney disease), diabetes, positive urine culture, inexperience.
- **Bleeding risk factors:** infundibular or upper-pole puncture, multiple or large tracts, staghorn stone, long operation, antithrombotic drugs, coagulopathy, diabetes.

Reducing risk

- **Tranexamic acid** reduces transfusion (Cochrane).
- Ultrasound-guided access, mini-PCNL, ECIRS (endoscopic combined intrarenal surgery).
- **Bleeding prevention:** puncture through the **calyceal fornix** along a posterior calyx (Sampaio: no arterial injury); correct clotting; mini-tracts.

Bleeding in PCNL

Causes

Timing	Cause
Intraoperative	Parenchymal or venous tract bleeding; infundibular artery injury; through-and-through puncture; posterior segmental artery injury (upper-pole access); torn infundibulum or pelvis
Early postoperative	Tract bleeding after sheath removal; clot retention
Delayed (about 1 week)	**Pseudoaneurysm**, **AV (arteriovenous) fistula**

Mechanism of delayed bleeding

- **Pseudoaneurysm:** clot seals an arterial tear, then ruptures intermittently.
- **AV fistula:** artery and vein heal together; the thin vein cannot withstand arterial pressure.

Management at day 6 (persistent haematuria)

- At day 6, assume an **arterial lesion** until proved otherwise. `Resuscitate: IV access, crossmatch, transfuse; Hb, clotting → Stop anticoagulants; catheter, irrigate clots → Stable, mild: bed rest, clamp nephrostomy, serial Hb → Persistent or recurrent bleeding or Hb fall → CT angiography (arterial blush) or direct angiography → SUPERSELECTIVE ANGIOEMBOLISATION (coils) → Fails or unstable after embolisation → open exploration, partial or total nephrectomy`
- **Angiography** diagnoses >90% and treats in the same sitting. It may miss intermittent bleeds; non-selective embolisation loses renal tissue.
- Clot colic or obstruction: double-J stent or nephrostomy.
- Follow-up: Hb and creatinine; check for infection and abscess.

Intraoperative ladder (for completeness)

- Pressure on tract → nephrostomy clamped (tamponade) → balloon reinflation or **Kaye tamponade catheter** → angioembolisation.

Diagram to draw: kidney cross-section with posterior calyx, Brödel line, pleura and retrorenal colon; angiogram sketch: pseudoaneurysm before and after coiling.

> **Key points to remember:**
> - Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
> - Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
> - Early bleeding venous: tamponade.
> - Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
> - Treatment: superselective angioembolisation.
> - Supracostal access increases pleural injury; colon injury at left lower pole.

SAQ Steinstresse.

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 7 Paper 2, Q3e

Assumed meaning: steinstrasse.

Definition

- **Steinstrasse** (German, "stone street"): a column of stone fragments in the ureter that fails to pass and may obstruct. Typically after **SWL (shock wave lithotripsy)**; occasionally after PCNL (percutaneous nephrolithotomy) or URS (ureteroscopy).
- Incidence is low overall (about 4–7% after SWL, EAU [European Association of Urology]) but rises several-fold with stone burden **>2 cm²**.

Risk factors

- Large burden (>2 cm), renal pelvic stones, hard stones that break into large fragments, high energy, impaired ureteric drainage.

Classification (Coptcoat)

Type	Composition
I	Small fragments only
II	Larger lead fragment (4–5 mm) followed by small fragments
III	Large fragments throughout

Clinical features

- Asymptomatic (silent obstruction), colic, nausea; fever and sepsis if infected; anuria and AKI (acute kidney injury) in a solitary kidney. Most lie in the distal ureter.

Evaluation

- KUB (kidney, ureter, bladder) X-ray, ultrasound, or non-contrast CT (computed tomography); creatinine, urine culture, blood counts.

Management

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Steinstrasse ├─ infection / sepsis → urgent PCN (percutaneous nephrostomy) + antibiotics ├─ asymptomatic, no obstruction → MET + analgesia, imaging every 1–2 weeks └─ failure, pain, obstruction: ├─ lead fragment (type II/III) → SWL to lead fragment └─ URS with laser lithotripsy (most definitive)
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- **MET (medical expulsive therapy):** α-blockers (tamsulosin) improve fragment clearance and reduce pain.
- **PCN** is preferred in infected obstruction: drains without manipulating the ureter, and fragments often pass once the kidney is decompressed. A stent is an alternative.
- Retained fragments with poor progress: URS clears the column.

Prevention

- Choose PCNL or RIRS (retrograde intrarenal surgery) for large burdens.
- Energy ramping, slower shock rate; MET after SWL.
- **Stent before SWL:** may reduce steinstrasse with large stones but does not improve stone-free rate and adds stent symptoms. The chapters disagree: Chapter 97 says it may reduce steinstrasse; Chapter 98 advises against stenting at SWL.

Diagram to draw: ureter with lead fragment and trailing column (Coptcoat types I–III).

> **Key points to remember:**
> - Risk rises several-fold with burden >2 cm^2.
> - Coptcoat I–III by fragment size.
> - Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

SAQ Treatment of inferior calyceal calculus

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 2, Q3b

Introduction

- Lower pole is the **commonest site** of renal stones. Gravity-dependent position makes fragments hard to clear, so **SWL (shock wave lithotripsy) works poorly** here.

Factors in decision

- **Stone:** size, density (<900 HU [Hounsfield units] favours SWL), composition.
- **Anatomy unfavourable for SWL clearance:** acute **infundibulopelvic angle**, long and narrow infundibulum, skin-to-stone distance >10 cm.
- **Patient:** obesity, anticoagulants (favour URS [ureteroscopy]), pregnancy, preference.
- Evaluation: NCCT (non-contrast CT), urine culture, renal function.

Treatment by size

Size	Options
Asymptomatic, small	Observation acceptable; about half have an event in 5 years. Treat if growing, symptomatic, infected or occupational
<10 mm	SWL or **RIRS** (retrograde intrarenal surgery). AUA (American Urological Association): either
10–20 mm	EAU (European Association of Urology): SWL only if favourable anatomy; otherwise **RIRS or PCNL** (percutaneous nephrolithotomy)
>10 mm (AUA)	**PCNL** (highest SFR [stone-free rate]) or URS; SWL not first-line
>20 mm	PCNL first; staged RIRS an alternative

Evidence

- **Lower Pole I (Albala 2001):** stones >1 cm: SFR 91% PCNL vs 21% SWL.
- Lower Pole II: stones ≤1 cm, URS vs SWL: no significant SFR difference.

Technique points

- **RIRS:** flexible ureteroscope (deflection up to 270°); relocate the stone to an upper calyx with a basket before laser (holmium or thulium fibre) lithotripsy; access sheath.
- **PCNL:** direct lower pole posterior calyx puncture; mini-PCNL reduces bleeding (transfusion near 1%).
- **SWL adjuncts:** percussion, diuresis, inversion therapy and alpha-blockers improve fragment clearance.
- **Also know:** lower pole calyceal diverticulum or infundibular stenosis favours PCNL; residual fragments >4 mm after URS need treatment.

Follow-up

- Imaging for residual fragments; metabolic evaluation if recurrent.

Diagram to draw: infundibulopelvic angle and infundibular length/width on an IVU (intravenous urogram) tracing.

> **Key points to remember:**
> - Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
> - <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
> - Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
> - Reposition stone during RIRS.

SAQ Ovarian vein syndromes

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 15 Paper 2, Q3e

Definition

- **Ovarian vein syndrome** (Clark, 1964) is extrinsic compression of the **right ureter** by a dilated or thrombosed **right ovarian vein**, causing intermittent hydroureteronephrosis. It is a **controversial** entity; many cases are now attributed to other causes.
- The plural also covers ovarian vein thrombosis and pelvic congestion.

Anatomy (basis)

- Ovarian veins run in the infundibulopelvic ligaments. The **right** enters the IVC (inferior vena cava) obliquely; the **left** enters the left renal vein at a right angle.
- Gonadal vessels **cross anterior to the ureter**; ovarian veins dilate greatly in pregnancy.

Types

Syndrome	Mechanism	Features
Ovarian vein (ureteric) syndrome	Dilated right ovarian vein compresses the ureter at the crossing	Multiparous women; right flank pain (premenstrual or in pregnancy); recurrent UTI (urinary tract infection)

Syndrome	Mechanism	Features
Ovarian vein thrombosis	Postpartum hypercoagulability and venous stasis; mostly right-sided	Fever and right lower-quadrant or flank pain after delivery; may obstruct ureter; risk of pulmonary embolism
Pelvic congestion syndrome	Incompetent, refluxing ovarian veins with pelvic varices; sometimes secondary to nutcracker (left renal vein compression)	Chronic pelvic pain worse on standing, dyspareunia, post-coital ache

Controversy

- Right-sided dilatation in pregnancy is mostly **physiological hydronephrosis** (progesterone plus uterine compression; stops at the pelvic brim). Venous compression is rarely proven, so diagnosis is by **exclusion**.

Investigations

- Ultrasound; **CT urography** or IVU (intravenous urography): right hydroureter to a smooth oblique extrinsic indentation, normal ureter below.
- **Diuretic renogram** to prove obstruction; retrograde pyelography.
- CT or MR (magnetic resonance) venography, Doppler; **gonadal venography** (gold standard for reflux).
- **Exclude:** stone, retrocaval ureter, iliac vessel crossing, retroperitoneal fibrosis, endometriosis.

Management

- **Asymptomatic:** observe. **Obstruction:** stent or nephrostomy if infected or pregnant; definitive **ureterolysis with ligation and division of the ovarian vein** (open or laparoscopic).
- **Thrombosis:** anticoagulation and antibiotics.
- **Pelvic congestion:** progestogens or GnRH (gonadotropin-releasing hormone) analogues; **coil or sclerotherapy embolisation**; ligation.

Diagram to draw: right ovarian vein crossing anterior to the right ureter, with proximal hydroureteronephrosis.

> **Key points to remember:**
> - Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
> - Postpartum ovarian vein thrombosis: right side, anticoagulate.
> - Treatment: ureterolysis with vein division, or embolisation.

SAQ Post traumatic bladder neck injury: acute and delayed management?

Paper answer · P3

Type: SAQ (10 marks)
Source: 2021-Summer Paper 3, Q3c

Introduction

- The bladder neck is the **internal sphincter**. It is torn mainly in **pelvic fracture**, often with prostatic urethral, rectal or vaginal injury; **commoner in children**.
- Missed injury causes **incontinence, fistula and stricture**.

Diagnosis

- Pelvic fracture with gross haematuria, retention or blood at the meatus; **vaginal examination** in females.
- **RUG (retrograde urethrogram)**, then **retrograde CT or plain cystography** (350 mL).
- Grading: AAST (American Association for the Surgery of Trauma) bladder **grade V** = laceration into the bladder neck or ureteric orifice; Goldman type IV = bladder rupture extending into the proximal urethra.

Acute management

- Resuscitate; if unstable, damage control (drainage), repair later.
- **Bladder neck injury needs immediate repair** (AUA (American Urological Association) and EAU (European Association of Urology)), unlike isolated posterior urethral distraction (suprapubic tube, delayed repair). `Lower midline incision → anterior cystotomy → identify ureteric orifices (stent if near) → minimal debridement; re-approximate neck to prostatic urethra, layered absorbable sutures over a 22 Fr catheter → rectal or vaginal tear: separate layers, no overlapping suture lines, interpose omentum (colostomy for rectum) → remove bone spicules; drain ± suprapubic tube`
- Catheter 2–3 weeks; cystogram before removal.

Delayed management (3–6 months)

Problem	Management
Associated urethral distraction	Cystogram + RUG + endoscopy; **anastomotic urethroplasty first**, not combined with bladder neck surgery
Bladder neck contracture	Endoscopic incision or resection; open Y-V plasty if refractory
Incontinence (both sphincters damaged)	Urodynamics; **AUS (artificial urinary sphincter)**; bladder neck reconstruction (Young–Dees–Leadbetter) or fascial sling; last resort **bladder neck closure + Mitrofanoff** ± augmentation
Fistula	Layered repair with tissue interposition
- Bladder neck appearance on imaging **does not predict** continence; restore continuity first, then treat incontinence.	
- Continence is poor whenever the bladder neck is involved (paediatric series: incontinence up to 20% overall).	

Diagram to draw: pelvic fracture with bladder neck tear and repair over a catheter.

> **Key points to remember:**
> - Bladder neck injury = immediate repair.
> - Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
> - Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

SAQ Role of open and laparoscopic stone surgeries in modern era of endourology.

Paper answer · P4

Type: SAQ (10 marks)
Source: 2021-Summer Paper 4, Q3f

Introduction

- SWL (shock wave lithotripsy), URS (ureteroscopy) and PCNL (percutaneous nephrolithotomy) treat almost all stones. Open surgery is now a small minority in centres with full endourology, but remains in resource-limited settings.
- **AUA (American Urological Association)/Endourological Society 2016:** consider laparoscopic, robotic or open surgery **only when SWL, URS and PCNL fail or are expected to fail**; not first line.

Indications

Category	Examples
Endourology failed or unlikely	Very complex staghorn needing many PCNL sessions; failed multiple procedures
Anatomical problem treated together	**PUJ (pelviureteric junction) obstruction + stone:** pyeloplasty with pyelolithotomy; anterior calyceal diverticulum; infundibular stenosis
Difficult access	Pelvic or ectopic kidney (laparoscopy-assisted PCNL); morbid obesity; skeletal deformity
Poor renal function	**Nephrectomy** for a non-functioning kidney; partial nephrectomy for a non-functioning pole
Large impacted ureteric stone	Laparoscopic ureterolithotomy after failed URS/SWL
Concomitant surgery	Other open or laparoscopic procedure
Bladder	Very large stone; with open prostatectomy, diverticulectomy or augmentation; small bladder; urethral stricture

Procedures

- **Pyelolithotomy:** simple, extended (Gil-Vernet, sinus dissection) or coagulum.
- **Anatrophic nephrolithotomy (Smith and Boyce):** for complex staghorn. Clamp renal artery; cool with ice slush; give mannitol; incise along **Brödel's avascular line**; remove stone; repair infundibula; close.
- Radial nephrotomy, ureterolithotomy, cystolithotomy.

Laparoscopic and robotic surgery

- Transperitoneal or retroperitoneal pyelolithotomy and ureterolithotomy; robotic anatrophic nephrolithotomy reported.
- Best for large pelvic stone in an extrarenal pelvis: high single-stage clearance, less bleeding than PCNL. Drawbacks: longer operation, calyceal stones need a flexible scope.

Feature	Open	Laparoscopic/robotic
Morbidity	Large incision, pain, long stay	Less pain, shorter stay
Kidney	Ischaemic loss of function (anatrophic)	Similar repair, less wound morbidity

Feature	Open	Laparoscopic/robotic
Later surgery	Adhesions make redo difficult	Fewer

Complications

- Bleeding, urinary leak, infection, loss of renal function, residual fragments.

Diagram to draw: Brödel's line incision for anatrophic nephrolithotomy.

> **Key points to remember:**
> - Open/laparoscopic only when endourology fails or is unsuitable.
> - Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
> - Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

SAQ Diet in stone disease

Paper answer · P4

Asked as:

- Diet in stone disease. — *Winter 2023 Paper IV Q3f (SAQ)*
- Diet in stone disease. — *Practice Set 5 Paper IV Q3f (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3f)

Principle

- Diet lowers urinary **supersaturation** by **dilution**, by reducing promoters (calcium, oxalate, uric acid, sodium) and by raising inhibitors (citrate).
- General advice for all; recurrent formers tailored to **24-hour urine**.

General measures

Measure	Target	Mechanism and evidence
Fluid	Urine **≥2.5 L/day** (about 3 L intake), spread through the day	Most important single factor. **Borghi 1996 RCT (randomised controlled trial):** recurrence 12.1% vs 27% at 5 years
Normal calcium	1000–1200 mg/day, with meals	Binds oxalate in the gut. **Restriction increases stones**. **Borghi 2002:** normal-calcium, low-salt, low-protein diet halved stone events vs low-calcium diet
Low sodium	Salt about 4–5 g/day	Sodium increases urinary calcium, lowers citrate
Limit animal protein	0.8–1.0 g/kg/day	Acid and purine load: hypercalciuria, hypocitraturia, hyperuricosuria
Fruit and vegetables (DASH [Dietary Approaches to Stop Hypertension] diet)	Daily	Alkali and citrate; DASH linked to fewer stones
Weight	BMI (body mass index) 18–25	Obesity: low urine pH, more oxalate and uric acid

- **Citrus:** lemonade (most evidence) and orange juice raise citrate; orange juice adds alkali. **Volume matters more than drink type.**

- Avoid sugar-sweetened drinks and **high-dose vitamin C** (converted to oxalate).

Stone-specific diet

Stone or abnormality	Diet
Hyperoxaluria	Avoid large portions of **spinach, beets, chocolate, nuts, tea**; calcium with meals
Enteric hyperoxaluria	Strict low oxalate and **low fat**; **calcium up to 1 g/day with meals**; fluid; magnesium
Uric acid	Less animal protein (purines give about 30% of urinary uric acid); fruit and vegetables; weight loss
Cystine	Fluid **3–5 L/day** including night; sodium **≤2300 mg/day**; less animal protein (not in children)
Struvite	Infection control; diet has little role

Adherence

- Poor: only half reach 2.5 L at 6 months; apps and smart bottles help.

Diagram to draw: "stone-prevention plate": water, fruit and vegetables, normal dairy, little meat and salt.

> **Key points to remember:**
> - Urine ≥2.5 L/day (Borghi 1996).
> - Never restrict calcium (Borghi 2002).
> - Low salt, less animal protein, more fruit and vegetables.
> - Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

SAQ ECIRS (endoscopic combined intrarenal surgery)

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3b

Definition

- **ECIRS:** PCNL (percutaneous nephrolithotomy) and retrograde flexible ureteroscopy performed **at the same time, in one position**, by two surgeons.
- Term and technique popularised by **Scoffone (2008)** in the **GMSV (Galdakao-modified supine Valdivia) position**. Combined access dates back to the 1980s (Lawson's retrograde needle placement, 1983).

Rationale

- Each route covers the other's blind spots: the retrograde scope guides and watches the puncture, and reaches calyces the rigid nephroscope cannot.
- Result: one tract, better clearance.

Positioning

Position	Features
GMSV (preferred)	Supine; flank on a bolster at the table edge; ipsilateral leg extended, contralateral leg in lithotomy
Bart's modified supine	Variant of supine
Prone split-leg	Easier posterior calyces; needs a split-leg table

- **Supine advantages:** airway access, gravity drainage of fragments, no turning, simultaneous retrograde access.

Technique

1. Cystoscopy; UAS (ureteral access sheath) placed; flexible ureteroscope maps the stones and chooses the target calyx.
2. Ultrasound or fluoroscopic puncture **watched endoscopically** (double bull's-eye technique).
3. Wire grasped and brought out: through-and-through access.
4. Tract dilated under vision (balloon) and sheath placed.
5. Antegrade lithotripsy and suction; the retrograde scope **relocates stones** from inaccessible calyces into the nephroscope's view, or lasers them.
6. Final check with both scopes; stent or tubeless exit. - Mini-ECIRS uses a mini-PCNL tract.

Indications

- Staghorn and complex multi-calyceal stones, combined renal and ureteric stones, anomalous kidneys, failed previous treatment, urinary diversion, obese or cardiopulmonary patients (supine).

Advantages (meta-analysis)

- Single position, **fewer tracts, less bleeding**, fewer ancillary procedures, **higher stone-free rate**; less fluoroscopy; the UAS keeps intrarenal pressure low.

Disadvantages

- Two surgeons and two sets of equipment; cost; longer set-up; ureteric injury from the sheath; less working space in supine.

Diagram to draw: GMSV position with nephroscope from the flank and flexible ureteroscope from below in the same calyx.

> **Key points to remember:**
> - Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
> - Retrograde scope guides the puncture and relocates stones.
> - Fewer tracts, less bleeding, higher stone-free rate.

Rapid Revision Sheet · Section 11

Radionuclide studies and diuretic renography in PUJ obstruction (SAQ)

- MAG3 is the tracer of choice; hydrate and catheterise.
- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
- DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
- Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

Retrograde endopyelotomy (SAQ)

- Lateral full-thickness incision over a stent.
- Crossing vessel and gross hydronephrosis predict failure.
- Success 70–80%, below pyeloplasty (about 95%).

Laparoscopic pyeloplasty. (LAQ)

- Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
- Transpose over a crossing vessel; never divide it.
- Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
- Robotic equals laparoscopic in outcome, with easier suturing.

Congenital PUJ obstruction, failed pyeloplasty and salvage procedures (LAQ)

- Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.
- Causes of failure: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable (about 83–89%); endopyelotomy about 70%.
- Ureterocalycostomy for small scarred pelvis: resect lower-pole parenchyma; about 70% in redo series. Buccal onlay emerging.
- Split function <15% with normal other kidney: nephrectomy.

Ureterocalycostomy (SAQ)

- Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
- Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
- Stricture is the main failure; success about 70% in redo series.

Boari flap (SAQ)

- Bridges 10–15 cm; combine with psoas hitch.
- Base ≥4 cm, length:base ≤3:1, measure bladder empty.
- Contraindicated in small contracted bladder.

Trans Uretero Ureteric anastomoses (SAQ)

- End-to-side, spatulated, tension-free; cross above the IMA.
- Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.

- Risk: unilateral injury becomes bilateral.

Retroperitoneal fibrosis: diagnosis and (medical) management (LAQ)

- Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
- Medially deviated ureters; PET-CT best shows activity; biopsy deep.
- Drain first, then prednisone (about 80% respond).
- Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
- Rituximab for IgG4-related or refractory disease.
- About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management (LAQ)

- AAST 2018: segmental vessel injury grade IV; main vessel grade V.
- Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
- Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
- Delayed CT phase is essential; blush means AE.
- Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
- Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
- Revascularise only a solitary kidney or bilateral injuries.

Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury (LAQ)

- Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
- Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
- AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
- CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
- Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
- Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
- Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

Urolithiasis: etiopathogenesis, medical management and prevention of recurrence (LAQ)

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor for idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi); Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide (plus potassium citrate) for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

Primary Hyperoxaluria (SAQ)

- PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.

- Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
- Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

Renal tubular acidosis (distal and type 2) (SAQ)

- Type 1 (distal): urine pH >6 with acidosis, hypokalaemia, hypocitraturia (key), positive UAG; calcium phosphate stones in up to 70%.
- NH_4Cl load: pH fails to fall below 5.5 in distal RTA.
- Potassium citrate is the treatment for distal RTA.
- Type 2 (proximal): bicarbonate wasting; HCO_3^- 14–18 mEq/L (P2: 15–18); urine pH <5.5 at steady state; usually Fanconi (ifosfamide, cisplatin).
- Fractional HCO_3^- excretion >15%; stones uncommon in type 2.
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause.

Uric acid calculi: etiopathogenesis, medical management and allopurinol (LAQ)

- Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
- Insulin resistance reduces ammoniagenesis, so urine is acid.
- Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
- Allopurinol only for hyperuricosuria or failed alkalinisation.
- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

Cystinuria (SAQ)

- Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
- Hexagonal crystals; target cystine <250 mg/L.
- Urine ≥3 L, pH 7.0–7.5, then tiopronin. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Struvite stones. (SAQ)

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

Medical expulsive therapy (MET) in urinary calculi (SAQ)

- Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
- SUSPEND 2015: no benefit in an unselected population.
- Trial of passage 4–6 weeks, then intervene.

Stones in pregnancy (SAQ)

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks (P3: every 4–6 weeks).

Management of upper ureteric stones (LAQ)

- Proximal stones pass in only 48%; MET does not help proximal stones.

- <10 mm: SWL equals URS; >10 mm: URS first (EAU).
- Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
- Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

Flexible ureteroscopy / RIRS and accessories (SAQ)

- 3.6 Fr channel; ≥270° deflection.
- 12/14 Fr access sheath lowers pressure but severely injures the ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.
- RIRS: stones <2 cm; first choice when anticoagulants cannot be stopped.
- Sepsis about 5%, stricture about 3%.
- Hydrophilic wire passes; PTFE wire stays.
- Nitinol tipless basket for calyces.

Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
- Encrusted stent: bladder coil, ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS.

Stone complexity scores (SAQ)

- Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
- S.T.O.N.E.: five variables, score 5–13.
- CROES includes centre volume; RUSS is for RIRS.

ESWL: mechanism, indications, factors affecting outcome, complications and current status (LAQ)

- Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 cm^2; no routine stent.
- Use is declining as URS grows; burst wave lithotripsy is coming.

PCNL patient positioning: prone vs supine (SAQ)

- Prone: posterior and upper pole access, large working area.
- Supine: airway safe, ECIRS, gravity drainage, no turning; Galdakao-modified supine gives one position for ECIRS.
- Horizontal bolsters drop the kidney more.
- Similar outcomes; choose by anatomy and experience.

Mini-PCNL and newer PCNL techniques (SAQ)

- Definition by sheath size: standard 24–34 Fr; mini-PCNL 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr; Jackman 1998: first mini-perc (13 Fr).
- Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.

- Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.
- Working channel decides instruments; suction lowers pressure.

Metabolic evaluation in stone disease (LAQ)

- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride (NH4Cl) fails to lower pH below 5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months of therapy.
- NCCT gold standard (97%/95%).
- Staghorn: PCNL first line; complete clearance is the goal.
- Bilateral: drain an infected side first, then the better kidney.
- AHA only when surgery is not possible.

Fracture penis (SAQ)

- Snap, detumescence, eggplant deformity.
- Urethral injury about 20% (West); suspect if bilateral.
- Immediate exploration and interrupted absorbable tunical closure.

Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management (LAQ)

- Blood at meatus: RUG before catheter.
- Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
- EPR: catheter 2–3 weeks; IPR: immediate repair.
- PFUI: SPC is the standard initial step, then delayed perineal anastomotic urethroplasty at 2–3 months (P3: after 3 months).
- Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
- Progressive perineal steps: bulbar mobilisation, crural separation, pubectomy, rerouting.
- Immediate repair: bladder neck or rectal injury, penile fracture urethra, low-velocity penetrating injury.
- ED mostly from the injury; incontinence <4%.

Renal autotransplantation (SAQ)

- For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for complex vascular or tumour cases.
- Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
- Unit loss 5–10%; success about 90%.

Newer methods of stone analysis (SAQ)

- FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemistry is obsolete.
- DECT (dual-energy computed tomography) separates uric acid stones in vivo, mainly uric acid versus calcium stones.
- Composition guides dissolution, SWL choice and prevention.

- Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

What are nanobacteria? Describe their role in urolithiasis. (SAQ)

- Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
- Proposed nidus for apatite and Randall plaque; association, not causation.
- Tetracycline/EDTA therapy unproven.

Tear drop bladder (SAQ)

- Tear-drop bladder = bilateral extrinsic compression.
- Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
- Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

Calyceal diverticulum – classification and management (SAQ)

- Transitional lined, narrow neck; type I calyceal, type II pelvic.
- Stones in up to 40%; delayed CT shows pooling.
- Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

Pelvic lipomatosis (SAQ)

- Pear-shaped bladder with medial ureters; CT shows fat.
- Cystitis glandularis common; adenocarcinoma risk.
- Watch for hydronephrosis; diversion if obstruction progresses.

Physiology of UPJ in relation to urine transport (SAQ)

- Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
- Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
- Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

Complications of PCNL, including bleeding and persistent haematuria after PCNL (SAQ)

- Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
- Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
- Early bleeding venous: tamponade.
- Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
- Treatment: superselective angioembolisation.
- Supracostal access increases pleural injury; colon injury at left lower pole.

Steinstresse. (SAQ)

- Risk rises several-fold with burden >2 cm^2.
- Coptcoat I–III by fragment size.
- Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

Treatment of inferior calyceal calculus (SAQ)

- Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
- <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
- Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
- Reposition stone during RIRS.

Ovarian vein syndromes (SAQ)

- Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
- Postpartum ovarian vein thrombosis: right side, anticoagulate.
- Treatment: ureterolysis with vein division, or embolisation.

Post traumatic bladder neck injury: acute and delayed management? (SAQ)

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

Role of open and laparoscopic stone surgeries in modern era of endourology. (SAQ)

- Open/laparoscopic only when endourology fails or is unsuitable.
- Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
- Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

Diet in stone disease (SAQ)

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

ECIRS (endoscopic combined intrarenal surgery) (SAQ)

- Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
- Retrograde scope guides the puncture and relocates stones.
- Fewer tracts, less bleeding, higher stone-free rate. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

SECTION 12

Bladder Cancer & Diversion

129 Tumours of the Urinary Bladder

1 · Epidemiology (headline)

- It is the **fourth most prevalent** cancer and the **costliest per patient over a lifetime**.
- The median age at diagnosis is ***73***, and the **male:female ratio is about 3:1**.
- **Women present later and do worse stage for stage**, as haematuria is mistaken for infection.

2 · Risk factors

The exposure predicts the histology.

LAQ CA bladder, etiology, newer investigations and treatment.

Paper answer · P2

Type: LAQ (25 marks)
Source: 2016-Summer Paper 2, Q1

Introduction

- Commonest urinary tract cancer; over 90% **UC (urothelial carcinoma)**. Men 3 times commoner; women present later. About 70% are **NMIBC (non-muscle-invasive bladder cancer)**; it recurs often.

Aetiology (risk factors)

Factor	Key facts
Smoking	30–40% of UC; risk 2–3 times. Aromatic amines form DNA adducts; **NAT2 (N-acetyltransferase 2) slow acetylators** at higher risk. Risk nears never-smokers >15 years after quitting
Occupation (5–10%)	Aromatic amines (dye, rubber, leather, painters, hairdressers); polycyclic hydrocarbons (aluminium, chimney sweeps)
Environment	**Arsenic** in water (West Bengal, Bangladesh, Taiwan)
Radiation	Pelvic radiotherapy 1.5–4 times; latency 15–30 years
Drugs	**Cyclophosphamide** (dose-dependent, above 20 g); pioglitazone (conflicting)
Chronic inflammation	*Schistosoma haematobium*, long-term catheter, stones → **SCC (squamous cell carcinoma)**; exstrophy → adenocarcinoma
Genetic	First-degree relatives 2 times; **Lynch syndrome** (MSH2); GSTM1 deletion
- **Also know:** obesity raises risk; diet has no proven role.	

Pathology and staging

- **WHO (World Health Organization) 2004 grading**: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade. Grade predicts progression.
- **Papillary pathway** (FGFR3; recurs, rarely progresses) versus **flat CIS (carcinoma in situ)** (TP53, RB; progresses).

TNM (tumour, node, metastasis), 2017	Extent
Ta / Tis	Papillary non-invasive / flat CIS
T1	Lamina propria
T2a / T2b	Inner / outer half of muscle
T3a / T3b	Perivesical fat, microscopic / macroscopic
T4a / T4b	Prostatic stroma, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	One pelvic node / several / common iliac

Clinical features and standard work-up

- **Painless gross haematuria (85%)**; irritative symptoms suggest CIS.
- **Cystoscopy and biopsy (gold standard)**; **CT (computed tomography) urography**; **urine cytology** (sensitivity 31–62%, specificity 94–100%, detects high grade; **Paris System** reporting).

Newer investigations

Test	Principle	Value
Blue-light cystoscopy (PDD, photodynamic diagnosis) with HAL (hexaminolevulinate)	Porphyrins in tumour fluoresce red under blue light	About one-third more CIS; recurrence benefit inconsistent
NBI (narrow-band imaging)	415 and 540 nm light absorbed by haemoglobin highlights vessels	More tumours; many false positives
Urine biomarkers	NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), **UroVysion FISH (fluorescence in situ hybridisation)** (chromosomes 3, 7, 17, 9p21), ImmunoCyt, Cxbladder	More sensitive, less specific than cytology; **none replaces cystoscopy**
mpMRI (multiparametric MRI) with VI-RADS (Vesical Imaging-Reporting and Data System)	Muscle-line integrity scored 1–5	**≥3 suspicious for muscle invasion**; about 80–90% accurate; before TURBT
En bloc TURBT (transurethral resection of bladder tumour)	Tumour removed in one piece with detrusor	Better staging and muscle sampling, fewer perforations; recurrence similar
- **Also know:** optical coherence tomography; urinary tumour DNA; PET-CT (positron emission tomography) for doubtful metastases.		

Treatment

1. TURBT: complete, with muscle in the specimen. **Re-TURBT within 6 weeks** for T1, incomplete resection or no muscle. **Single chemotherapy instillation within 6 hours** (mitomycin C, epirubicin or gemcitabine) for low-risk tumours; never BCG (bacillus Calmette–Guérin) at this time.

2. NMIBC by risk (AUA (American Urological Association))

Risk	Definition	Treatment
Low	Solitary low-grade Ta ≤3 cm; PUNLMP	Single instillation; surveillance
Intermediate	Low-grade Ta recurrent, >3 cm or multifocal; low-grade T1; high-grade Ta ≤3 cm	Chemotherapy or BCG; maintenance 1 year
High	High-grade T1; large, multifocal or recurrent high-grade Ta; any CIS; variant histology; lymphovascular invasion	**BCG induction + 3-year maintenance**; early cystectomy if very high risk
- **BCG** induces a Th1 immune response; six weekly doses from 2–4 weeks after TURBT, then **SWOG (Southwest Oncology Group) maintenance** (recurrence-free survival 76.8 vs 35.7 months). Only agent that reduces progression.		
- **BCG-unresponsive**: radical cystectomy; if unfit, **pembrolizumab** (KEYNOTE-057), **nadofaragene firadenovec**, gemcitabine–docetaxel, TAR-200.		

3. MIBC (muscle-invasive bladder cancer), cT2–T4a N0M0

- **Cisplatin-based NAC (neoadjuvant chemotherapy)** (dose-dense MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) or gemcitabine–cisplatin) → **radical cystectomy + pelvic lymphadenectomy**. NAC adds 5–8% absolute survival (SWOG 8710: pT0 38% vs 15%). Cisplatin-ineligible: GFR (glomerular filtration rate) <60 mL/min, poor performance status, hearing loss, neuropathy, heart failure.
- **Diversion**: ileal conduit or orthotopic neobladder. Robotic equals open (**RAZOR**).
- **Trimodality therapy**: complete TURBT + chemoradiation (5-FU (fluorouracil) + mitomycin C, **BC2001**). For solitary T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- **Adjuvant nivolumab** for ≥pT3 or N+ (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months). Partial cystectomy for selected solitary dome tumours.

4. Metastatic: **enfortumab vedotin + pembrolizumab** (**EV-302**: nearly doubled survival versus platinum). Otherwise platinum then **avelumab maintenance**; **erdafitinib** for FGFR3 alterations.

Follow-up and recent advances

- Cystoscopy at 3 months, then by risk; upper-tract imaging in high risk.
- Molecular subtypes (basal tumours respond to cisplatin); **NIAGARA** (2024): perioperative durvalumab with NAC improved survival.

Diagram to draw: bladder wall layers with Ta, Tis, T1–T4 marked.

> **Key points to remember:**
> - Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
> - Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
> - High-risk NMIBC: BCG with 3-year maintenance.
> - MIBC: cisplatin NAC + cystectomy; trimodality in selected.

Also, from the condensed book (not in the paper answer):

- **Cyclophosphamide** is the only proven chemotherapeutic carcinogen, via **phosphoramide mustard** (acrolein causes haemorrhagic cystitis, not cancer). Risk is ~4.5× above 20 g, with latency peaking at 10–14 years.
- **Radiation**: external beam radiotherapy for cervical cancer raises risk 2–4×, and for prostate cancer 1.5–2×.
- **Exstrophy** carries a malignancy risk of 3.3–7.5%, and **>90% is adenocarcinoma**.
- **Smoking** causes **30–40%** of urothelial carcinoma (UC) and raises risk **2–3×**.
- **N-acetyltransferase 2 (NAT2) slow acetylators** carry the highest risk, because acetylation detoxifies aromatic amines. Risk falls to near that of never-smokers **>15 years** after quitting.
- **Schistosoma haematobium** causes chronic inflammation and fibrosis, which leads to **squamous cell carcinoma (SCC)**. Long-term catheters in spinal cord injury and stones also cause SCC.
- Germline variants (MSH2, BRCA1/2) occur in 13–24% of UC, so test selected high-grade patients.

3 · Pathology

Benign lesions (reactive mimics)

Lesion	Hallmark	Potential / action
Squamous/glandular metaplasia	Trigone; squamous in up to 40% of females	None; no biopsy
Inverted papilloma	Trigone; endophytic nests, no atypia; <1% of tumours	Benign, ~1% recur; resect
Urothelial papilloma	Delicate stalks; FGFR3 mutant, TP53/RB wild type	None; resect
Nephrogenic adenoma	Tubule-like glands after injury	None; re-resect if it recurs
Leukoplakia	Keratinising squamous metaplasia; floating white flakes	**Not premalignant**, so no follow-up
Cystitis cystica/glandularis	From proliferating von Brunn nests	Glandularis: low adenocarcinoma risk, so resect, treat cause and survey

Flat (precursor) lesions

Loss of polarity means dysplasia; loss of umbrella cells means carcinoma in situ (CIS).

Lesion	Key features	Significance
Hyperplasia	≥10 layers, no atypia	May precede low-grade papillary tumour (FGFR3)
Reactive atypia	Atypia proportionate to inflammation; polarity kept	Coexists with dysplasia/CIS, so investigate
Atypia of unknown significance	Atypia disproportionate to inflammation	Limited utility
Dysplasia	Loss of polarity, umbrella cells kept; CK20, p53	14–19% progress; **60%** progress to CIS if there was prior cancer
CIS	Flat, high grade, noncohesive pleomorphic cells; umbrella cells lost	Precursor of invasive high-grade disease

UC grade (2004 WHO) – noninvasive spectrum

Non-muscle-invasive bladder cancer (NMIBC) makes up about **70%** of UC and includes Ta, CIS and T1.

	PUNLMP*	Low-grade Ta	High-grade papillary	CIS
Morphology	Papillary >7 cells, minimal atypia, solitary, trigone	Branching stalks, some atypia	Fused papillae, many mitoses	Flat, severe atypia
Genetics	TERT promoter, FGFR3	Chromosome 9 loss; FGFR3, PI3K, RAS	TP53/RB	RB, TP53, PTEN; **no FGFR3**
Behaviour	Recur ~20% at 5 y; progress <1%	Recur 48–71%; progress 2–12%	>80% invade if untreated	With invasive tumour: 45–65% 5-y mortality

*Papillary urothelial neoplasm of low malignant potential. **Grade, not recurrence, predicts danger.**

Histological variants of UC

About 80% have mixed (mostly squamous) differentiation. **Altered growth patterns behave worse than their stage.**

Variant	Hallmark	Chemosensitive?	Management
Micropapillary (0.7–2.2%)	Tight clusters with retraction artefact; **any amount counts**; ERBB2 amplification	No	**Early radical cystectomy (RC) even for NMIBC**
Sarcomatoid (0.3–0.6%)	Large infiltrative mass	Poor	Immediate RC
Plasmacytoid (<1%; includes signet ring)	Eccentric nuclei; **intraperitoneal spread**; CDH1 truncation	Yes, but relapse is common	Neoadjuvant chemotherapy (NAC) whenever possible
Nested (M:F 6:1)	Mimics von Brunn nests and inverted papilloma	No	Often upstaged; stage-matched outcome as UC
Squamous (16–22%, up to 60%)	Basal subtype	NAC active in mixed tumours	Linked to recurrence and progression
Glandular (~10%)	Gland-like spaces	–	Outcome as conventional UC

Non-urothelial malignancy

	Small cell	SCC	Adenocarcinoma	Urachal adenocarcinoma
Setting	Rare; dedifferentiated UC	2–5% in West; majority where schistosomiasis is endemic	**Secondary is commoner than primary** (colon, prostate, uterus, cervix); exstrophy	~1/3 of bladder adenocarcinomas; from the allantoic remnant
Hallmark	Chromogranin, synaptophysin; paraneoplastic syndromes	Locally advanced, fewer metastases	Cribriform/glandular; **colonoscopy** to exclude a primary	Dome/anterior wall; Sheldon staging
Treatment	**Cisplatin–etoposide first**, then RC or radiation	RC; preoperative radiation worsened survival	RC; 5-fluorouracil–cisplatin by analogy	**En bloc dome with urachal ligament and umbilicus**

> **KEY POINT**
>
> Small cell is systemic disease: primary cystectomy gives poor disease-free survival that adjuvant chemotherapy does not rescue, whereas neoadjuvant chemotherapy markedly improves survival. SCC in spinal cord injury fell after the switch to intermittent self-catheterisation, showing that inflammation drives it.

Urachal carcinoma (2016 WHO criteria): the epicentre is in the dome or anterior wall, there is no widespread cystitis glandularis, and a secondary source is excluded. TNM staging is unsuitable.

Sheldon stage	Extent
I	Urachal mucosa
II	Confined to urachus
IIIA / IIIB / IIIC	Bladder / abdominal wall / peritoneum
IV	Metastatic

4 · Molecular biology

	Papillary pathway	Flat pathway
Origin and product	Hyperplasia gives low-grade papillary tumour	Dysplasia gives CIS, then high-grade invasive tumour
Genetics	**FGFR3** (up to 80% of low-grade Ta vs 10–20% of MIBC), chromosome 9 deletion, PIK3CA	**TP53** (up to 76% of MIBC), RB1, CDKN2A, PTEN loss
Behaviour	Recurs, rarely progresses	Progresses

- HRAS/KRAS mutations are **mutually exclusive with FGFR3**, because they act in the same pathway.

The Cancer Genome Atlas (TCGA) subtype	Luminal	Basal/squamous
Markers	FGFR3, GATA3, KRT20, PPARG, FOXA1	CK5, CK6, CK14; low GATA3 and FOXA1

The Cancer Genome Atlas (TCGA) subtype	Luminal	Basal/squamous
Behaviour	Less aggressive; papillary	Aggressive; squamous; women; T3/T4
Targets	FGFR3 inhibitors, ERBB2	Cisplatin NAC; checkpoint inhibitors; EGFR

- The **consensus classification (2020) has six classes**. Luminal papillary has the best survival, **basal/squamous (35%) is the commonest**, and neuroendocrine-like (3%) is the worst. Luminal-infiltrated tumours respond poorly to cisplatin NAC.

5 · Clinical presentation and evaluation

About one in five visible haematurias hides a cancer.

- **Painless gross haematuria occurs in 85%**. Irritative symptoms suggest CIS.
- Evaluate every gross haematuria, even on anticoagulants. Check prostate-specific antigen, because ~10% of recurrent gross haematuria is prostate cancer.
- After a negative work-up, repeat it at 3–5 years only if haematuria persists.

	Gross haematuria	Asymptomatic microhaematuria
Malignancy	~20%	2.6–5.2%
Cystoscopy	All adults	Age ≥35 years
Upper tract	Computed tomography urogram (CTU)	CTU (AUA 2012); ultrasound may replace it
Cytology/markers	Reasonable	Avoid routinely

The dipstick has sensitivity 91–100% but specificity only ~65%, so **confirm with microscopy**.

Cystoscopy and enhanced imaging

Cystoscopy with biopsy is the gold standard; flexible equals rigid. White light misses small tumours and CIS.

	Blue light (HAL–BLC)	Narrow-band imaging (NBI)
Principle	Hexaminolevulinate (HAL) makes porphyrins build up in tumour, which fluoresce red under blue light	No dye; 415 and 540 nm light is absorbed by haemoglobin, so vessels stand out
Benefit	Finds more Ta and CIS; CIS is seen only on BLC in ~¼	Better CIS detection; lower 12-month recurrence in low risk
Status	Adjunct; only HAL is FDA approved	Adjunct

Urine cytology and biomarkers

*See the paper answer **Urine cytology** (Section 1, Ch 3).*

*See the paper answer **Microscopic / asymptomatic haematuria and urinary biomarkers** (Section 1, Ch 2).*

ALGORITHM

1. Cytology shows high-grade cells but cystoscopy is negative.
2. Take random bladder biopsies and prostatic urethral biopsies.
3. If these are negative, take selective upper-tract washings for cytology, with or without ureteroscopy.

6 · Staging and risk groups

Stage	TNM, American Joint Committee on Cancer (AJCC) 2017
0a / 0is	Ta (noninvasive papillary) / Tis (CIS)
I	T1 – lamina propria (subepithelial connective tissue)
II	T2a inner / T2b outer half of muscularis propria, N0
IIIA	T3a (microscopic perivesical) / T3b (macroscopic) / T4a (prostatic stroma, uterus, vagina) N0; or T1–T4a N1 (single true pelvic node)
IIIB	T1–T4a N2 (≥2 true pelvic nodes) or **N3 (common iliac)**
IVA / IVB	T4b (pelvic/abdominal wall) or M1a (distant nodes) / M1b (organs)

- **The 2017 change**: regional nodes without distant metastasis are now stage III (previously IV). MCQ Common iliac nodes are N3, not M1a.
- **T1a/T1b substaging** is recommended, because ~50% of T1 are upstaged to ≥T2 at cystectomy.
- **Prostatic urethra**: mucosal CIS is not adverse in itself but escapes intravesical therapy. Only **prostatic stromal invasion is T4a** (5-year survival historically <25%).

NMIBC risk group	Definition
Low	Primary, solitary, low-grade Ta, <3 cm
Intermediate	Low-grade Ta that is multiple and/or recurrent and/or >3 cm
High	Any high grade, CIS or T1

130 Management Strategies for Non–Muscle-Invasive Bladder Cancer

1 · Definition and natural history

- **Non-muscle-invasive bladder cancer (NMIBC)** is urothelial carcinoma that has not invaded the detrusor: Ta, T1 and carcinoma in situ (CIS). It makes up **70–80%** of new bladder cancers.
- Recurrence occurs in **40–60%**; progression ranges from ~6% in low-grade (LG) Ta to 17% in high-grade (HG) T1.
- **Grade, not stage, is the strongest predictor of progression**, and CIS comes second. Primary CIS progresses in **>50%**.
- **Understaging**: cystectomy for presumed NMIBC shows worse disease in **one-third to one-half**.

> **KEY POINT**
>
> NMIBC is two diseases. LG disease is a **recurrence** problem; HG disease is a **progression** problem. Chemotherapy treats recurrence, bacille Calmette-Guérin (BCG) treats progression, and cystectomy is for progression that cannot be controlled.

2 · Risk stratification

*Risk is **dynamic**, so reassess it at every recurrence.*

American Urological Association/Society of Urologic Oncology (AUA/SUO) 2016 (updated)

Risk	Criteria	Intervention
Low	LG solitary Ta **≤3 cm**; papillary urothelial neoplasm of low malignant potential (PUNLMP)	Single postoperative chemotherapy instillation
Intermediate	LG Ta recurring within 1 yr; LG Ta solitary >3 cm or multifocal; **LG T1**; **HG Ta ≤3 cm**	6-week induction of chemotherapy or BCG; maintenance to 1 yr
High	HG T1; recurrent HG Ta; HG Ta >3 cm or multifocal; **any CIS**; any BCG failure in HG; **any variant histology**; **any lymphovascular invasion (LVI)**; any HG prostatic urethral involvement	Induction BCG and maintenance to **3 yr**. If BCG-unresponsive: radical cystectomy (RC), pembrolizumab, nadofaragene or gemcitabine–docetaxel

EORTC and CUETO scoring models

Factor	EORTC	CUETO
Number, prior recurrence rate, T stage, concurrent CIS, grade (WHO 1973)	✓	✓
Tumour diameter	✓	—
Age, sex	—	✓

- EORTC **overestimates** risk, because its patients had little maintenance BCG. CUETO, derived in BCG-treated patients, fits better. MCQ Both use **WHO 1973** grading.

3 · Endoscopic management

TURBT: principles

- Transurethral resection of bladder tumour (TURBT) quality depends on **complete resection** and **detrusor in the specimen**; missing muscle means higher mortality.
- Image the upper tracts at diagnosis. Upper tract tumours occur in **<5%**, more often with trigonal tumour, CIS and high risk.
- **Diverticular tumour**: there is no detrusor, so invasion beyond the lamina propria is **T3a**. For HG, strongly consider partial cystectomy or RC.
- **En bloc resection** gives fewer perforations, more detrusor and better staging, but the **same recurrence**.

Repeat (second-look) TURBT

SAQ Re-look (second-look) TUR bladder tumour

Paper answer · P4

Asked as:

- Re-look TUR bladder tumour. — *Summer 2017 Paper IV Q3d (SAQ)*
- Re-look TUR bladder tumour. — *Practice Set 5 Paper IV Q3d (earlier version) (SAQ)*

Type: SAQ (10 marks)

Read as second-look (repeat or restaging) TURBT.

Definition

- A **planned second (restaging) TURBT (transurethral resection of bladder tumour)** of the primary site, including detrusor muscle, a few weeks after the first. It completes resection and corrects staging. It is a quality indicator of NMIBC (non-muscle-invasive bladder cancer) care.

Rationale

- **TURBT is a staging operation first.** Completeness and muscle in the specimen are the quality markers.
- **Incomplete resection** — residual tumour is common:
- At the resection site in **≥40%** when re-resected within weeks; tumour at first check cystoscopy in up to 45%.
- **HG (high-grade) Ta:** residual up to 50%, upstaging up to 15%.
- **Understaging:** in **HG T1**, re-resection shows worse disease or upstaging to muscle invasion in **25–30%**. This rises to about half if the first specimen had **no muscle**.
- Understaging is about twice as likely without muscle.

Indications (AUA [American Urological Association] / EAU [European Association of Urology])

Recommended	Consider	Not needed
Incomplete first (initial) resection	HG Ta, large or multifocal (AUA) (residual up to 50%, upstaging up to 15%)	Low-grade Ta

Recommended	Consider	Not needed
All T1 tumours	Outside resection with no slides for review	Primary CIS (carcinoma in situ)
No detrusor muscle in specimen (except low-grade Ta and primary CIS)		Clear muscle invasion
		Cystectomy already planned

Timing

- **2–6 weeks** after the first TURBT (EAU); **within 6 weeks** (AUA), before BCG (bacille Calmette–Guérin). This allows healing yet avoids delaying intravesical BCG.

Technique

- Resect the **primary-site scar and base deep into detrusor**, plus any residual tumour; send the base / specimens separately. Use blue light or NBI (narrow-band imaging) if available; it helps.

Benefits / evidence

- Corrects stage and **changes treatment in about one-third**.
- **Divrik RCT (randomised controlled trial, 2010)** in T1: re-TUR improved recurrence-free and progression-free survival (lower recurrence and progression).
- Better BCG response with minimal residual tumour / when BCG follows re-TUR.

Result-based plan (decision after re-TUR)

- **Residual T1 at re-resection** means (predicts) progression in up to **80%**: consider early cystectomy. `No tumour (T0) / Ta → BCG induction + maintenance Residual HG T1 → early cystectomy (or BCG with close review) Muscle invasion (≥T2) → treat as muscle-invasive: neoadjuvant chemotherapy + cystectomy`

Recent advances

- **En bloc resection** gives muscle (more detrusor) more often, less residual tumour, and may reduce re-TUR. It is not yet a proven substitute.
- Expert pathology review changes management in about one-third.
- **Also know:** maximal TURBT before trimodality therapy.

Diagram to draw: flowchart: T1 or no muscle → re-TUR 2–6 weeks → T2: cystectomy pathway / residual T1: consider cystectomy / T0–Ta: BCG; and bladder wall showing first resection scar and deep second resection into detrusor.

> **Key points to remember:**
>
> - Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
> - Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
> - Residual T1 → progression (progresses) in up to 80%.

Recommended (within 6 weeks, include muscularis propria)	Consider
Incomplete resection; all T1 (especially without muscle); no muscle in specimen	High-risk HG Ta (multifocal and/or large)

Biopsies and office management

- Random biopsies are **not** needed in low risk; consider them with positive cytology or high risk. Biopsy the prostatic urethra if there is CIS or a bladder neck/trigone tumour.
- Small (**≤1 cm**) LG recurrences can be fulgurated in the office, but this needs a **prior LG diagnosis and negative cytology**. **Active surveillance** also suits selected patients.

Enhanced cystoscopy

	Blue-light (hexaminolevulinate, HAL)	Narrow-band imaging (NBI)
Principle	Porphyrins fluoresce red under blue light; needs intravesical instillation	415 and 540 nm light is absorbed by haemoglobin; no agent
Gain	~**one-third more CIS**; smaller papillary tumours	More papillary tumours and CIS
Weakness	Cost; no 3-yr recurrence benefit in a pragmatic trial	False positives ~60%

4 · Single post-operative instillation

LAQ Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation

Paper answer · P2 · P3 · P4 · merged from 3

Asked as: P4 LAQ "Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy" · P2 SAQ "Intravesical therapy for TCC of bladder" · P3 SAQ "Immediate post TURBT intravesical Therapy for TCC"

Introduction

- NMIBC (non-muscle-invasive bladder cancer: Ta, T1, CIS [carcinoma in situ]) is 70–80% of bladder cancers. "Urothelial carcinoma" now replaces "transitional cell carcinoma" (TCC).
- Two problems: **recurrence** (40–60%) and **progression** (about 6% in low-grade Ta to 17% in high-grade T1). Intravesical therapy follows complete TURBT (transurethral resection of bladder tumour).
- **Rule:** chemotherapy reduces recurrence only. BCG (bacille Calmette-Guérin) is the only agent that also reduces progression.

Risk-adapted use (AUA [American Urological Association]/SUO)

Risk	Criteria	Therapy
Low	LG (low-grade) solitary (primary) Ta ≤3 cm; PUNLMP	Single post-op instillation only
Intermediate	LG Ta recurrent, >3 cm or multifocal; LG T1; HG (high-grade) Ta ≤3 cm	6-week chemotherapy or BCG induction; maintenance 1 year
High	HG T1; HG Ta large, multifocal or recurrent; any CIS; LVI (lymphovascular invasion); variant histology	BCG induction + maintenance 3 years

- EAU (European Association of Urology) adds a "very high" group; EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables.

1. Single immediate instillation after TURBT

Rationale

- After TURBT, NMIBC recurs in 40–60%. Many early recurrences come from **floating cells implanting** on the raw resection bed. A single chemotherapy instillation (SI) kills them

before implantation.

Timing

- Implantation occurs within hours, so instil **as early as possible: within 6 hours** (P3: ideally within 2–6 h, in theatre or recovery). Guidelines allow up to 24 h, but MMC given at 24 h lost its benefit (no benefit at 24 hours).

Agents

Agent	Dose	Note
MMC (mitomycin C)	40 mg in 20–40 mL	Alkylating; skin rash
Gemcitabine	2 g in 100 mL, 1 h	Standard in USA; mild toxicity
Epirubicin	50 mg	Chemical cystitis
- **Also know:** pirarubicin works; thiotepa does not.		

Technique: exclude perforation; instil via catheter; retain 1–2 h; drain.

Evidence

- **Sylvester 2016 individual-patient meta-analysis:** about 35% relative reduction in recurrence; **no benefit** with >1 recurrence/year or EORTC recurrence score ≥5.
- **SWOG (Southwest Oncology Group) S0337:** gemcitabine 2 g versus saline in suspected low-grade NMIBC: 4-year recurrence 35% vs 47%; no excess toxicity.
- No effect on progression.

Who benefits

- **Low risk** (primary solitary low-grade Ta ≤3 cm): SI alone; greatest benefit.
- Intermediate risk: SI then adjuvant course. High risk: BCG follows.

Contraindications

- **Suspected perforation**, extensive resection, bleeding needing irrigation: withhold.
- **Never BCG immediately**: intravasation of live bacteria causes sepsis and death.

Complications

- Irritative symptoms (commonest), chemical cystitis, skin desquamation (MMC), contracted bladder; serious harm mainly after perforation.
- **Also know:** continuous saline irrigation may also reduce recurrence.

2. Adjuvant chemotherapy

- **MMC** (alkylating), 40 mg in 20 mL weekly ×6. Optimise it (Au, 2001): empty bladder, fluid restriction, sodium bicarbonate (alkalinised urine), concentrated dose. Skin rash.
- **Gemcitabine** (nucleoside analogue): minimal absorption, well tolerated. **Anthracyclines** (epirubicin, doxorubicin) cause chemical cystitis.
- Intermediate risk: 6-week induction, then monthly maintenance for 1 year.
- **Gem/Doce:** gemcitabine 1 g, drain, then docetaxel 37.5 mg; weekly ×6, then monthly for 2 years. Used in BCG-unresponsive disease and during BCG shortage (details under combination therapy).

3. BCG immunotherapy

- **Mechanism:** BCG binds **fibronectin**. Cytokines follow (IFN-γ [interferon-gamma], IL [interleukin]-2, IL-12, TNF-α [tumour necrosis factor-alpha]). A **Th1 (T-helper 1)** response

gives cell-mediated tumour kill; TRAIL release is important.

- **Technique:** 1 vial in 50 mL saline. Start **2–4 weeks** after TURBT (re-epithelialisation). Urinalysis first; retain 1–2 hours.
- **SWOG schedule (Lamm, 2000):** 6 weekly induction doses, then 3 weekly doses at 3 and 6 months, then every 6 months to 3 years. Median recurrence-free survival 76.8 vs 35.7 months.
- **EORTC 30962:** full dose; 1 year for intermediate, 3 years for high risk. **NIMBUS:** reduced frequency was inferior.
- CIS complete response up to 84%; BCG beats MMC only with maintenance. Only maintenance BCG reduces progression.
- **Contraindications:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk intravasation); previous BCG sepsis; total incontinence; immunosuppression (now relative). Relative: active UTI (urinary tract infection), liver disease, previous TB (tuberculosis).

BCG toxicity

Grade	Features	Management
1	Cystitis, fever <38.5 °C, <48 h	Anticholinergics, analgesics, NSAIDs (non-steroidal anti-inflammatory drugs)
2	Severe or >48 h; persistent fever	Culture; isoniazid + rifampicin; resume at ½–⅓ dose
3	Sepsis, organ involvement	Stop BCG; isoniazid + rifampicin + ethambutol (triple therapy); steroids only with cover

- BCG is resistant to **pyrazinamide**. BCG sepsis 0.5%.

BCG failure

- **Adequate BCG:** ≥5 of 6 induction + ≥2 of 3 maintenance doses.
- **BCG-unresponsive:** persistent or recurrent CIS or HG Ta/T1 despite adequate BCG, or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Also know:** refractory, intolerant, exposed, delayed relapse.

BCG-unresponsive options

- **Timely radical cystectomy** is the standard (EAU, AUA).
- If unfit or refusing:
- **Pembrolizumab** (KEYNOTE-057): IV every 3 weeks; CIS complete response 41%.
- **Nadofaragene firadenovec:** adenovirus with interferon-α2b gene, every 3 months; CIS complete response 53%.
- **N-803:** IL-15 superagonist with BCG; CIS complete response 71%.
- Gem/Doce; valrubicin; **TAR-200** (gemcitabine-releasing device); cretostimogene (trials).

Device-assisted therapy

- **EMDA (electromotive drug administration):** current increases MMC tissue delivery 4–7-fold. Sequential BCG + EMDA-MMC improved outcomes in T1.
- **Chemohyperthermia:** RITE (radiofrequency, 42 °C); HIVEC (heated recirculating MMC).
- **Also know:** UGN-102 MMC gel; interferon with low-dose BCG.

Combination intravesical therapy

Rationale

- Single agents fail: chemotherapy reduces **recurrence** only; BCG reduces **progression** but fails in about a third.
- Combinations add **different mechanisms**, overcome resistance, and are useful during **BCG shortage** and in **BCG-unresponsive** NMIBC.

1. Sequential chemotherapy (doublets)

Regimen	Method	Results
Gem/Doce (gemcitabine–docetaxel)	Gemcitabine 1 g in 50 mL (60–90 min), drain, then docetaxel 37.5 mg (60–120 min); weekly ×6, then monthly for 2 years	High-grade recurrence-free survival about 65% at 1 year, 52% at 2 years; CIS complete response 60% at 1 year; **84% avoid cystectomy**
Gem/MMC	Gemcitabine 1 g, then MMC 40 mg	Durable recurrence-free about 38%

- **Why it works:** gemcitabine (antimetabolite) and docetaxel (microtubule stabiliser) act on different cell-cycle phases.
- Oral sodium bicarbonate and antiemetics aid tolerance. No level 1 evidence; phase III BRIDGE compares it with BCG in BCG-naive patients.

2. Chemotherapy with BCG

- Sequential MMC–BCG: **no benefit** over BCG alone.
- **Exception: BCG alternating with EMDA-MMC** in T1: lower recurrence and progression than BCG alone. A later trial was non-significant.
- Chemohyperthermia (RITE, radiofrequency-induced thermochemotherapeutic effect) potentiates BCG.

3. BCG with immunomodulators

- **BCG + interferon-α (α2b):** no gain over BCG; allows BCG dose reduction to one-third.
- **N-803 (nogapendekin alfa)** = an IL-15 (interleukin-15) superagonist activating CD8 T and NK (natural killer) cells. With BCG in BCG-unresponsive CIS: complete response 71%, 45% at 12 months. FDA (Food and Drug Administration)-approved in 2024 (chapter: "under review").

4. BCG with checkpoint inhibitors (BCG-naive high risk)

- **CREST** (subcutaneous sasanlimab + BCG) and **POTOMAC** (durvalumab + BCG): improved event-free or disease-free survival over BCG alone.
- **ALBAN** (atezolizumab + BCG): negative. KEYNOTE-676 (pembrolizumab) ongoing.

5. Device-assisted and novel combinations

- **Chemohyperthermia** (MMC at 42 °C: RITE; HIVEC, hyperthermic intravesical chemotherapy) increases uptake.
- **TAR-200:** gemcitabine-releasing intravesical device (± cetrelimab); very high early complete response in BCG-unresponsive CIS (SunRISe-1).
- **Cretostimogene** (oncolytic adenovirus) + pembrolizumab: about two-thirds complete response at 12 months.

Toxicity of combinations

- Additive cystitis; immune-related events with checkpoint inhibitors.

Diagrams to draw: BCG mechanism (fibronectin, cytokines, Th1, kill) with SWOG schedule timeline; Gem/Doce sequence timeline (instil, dwell, drain, instil); floating cells implanting on the resection bed and killed by instillation.

Key points to remember:

- Single instillation within 6 h (at most 24 h); never BCG immediately; withhold if perforation; best for low-risk solitary tumours.
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Only maintenance BCG reduces progression; SWOG 3-year schedule; NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab (41% CR), nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

Also, from the condensed book (not in the paper answer):

- Give mitomycin C (MMC) within **6 h** (level 1 evidence). After perforation it risks chemical cystitis, contracture and rare deaths.

5 · Intravesical chemotherapy

*See the paper answer **Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation** (Section 12, Ch 130).*

Also, from the condensed book (not in the paper answer):

- Chemotherapy reduces **recurrence**, with **no proven effect on progression**. In intermediate risk, MMC or gemcitabine equals BCG. Give a 6-week induction, then monthly for 1 yr.
- A quinolone for 24–48 h after each BCG dose reduces cystitis without loss of efficacy.

Agent	Class	Dose	Signature toxicity
Mitomycin C	Alkylating	40 mg/20 mL weekly ×6	Rash, severe skin reaction, contracted bladder 5%
Gemcitabine	Nucleoside analogue	2 g/100 mL	Mild; minimal absorption
Doxorubicin / epirubicin	Anthracycline	50 mg/50 mL	Chemical cystitis, contracted bladder
Valrubicin	Anthracycline	800 mg/75 mL	FDA approved for BCG-refractory CIS; complete response (CR) ~1/5
Thiotepa	Alkylating	60 mg/30 mL	Myelosuppression
Docetaxel	Taxane	75 mg/100 mL	Myelosuppression

- **Gemcitabine–docetaxel (Gem/Doce)** is increasingly used in BCG-naive HG disease.

6 · Bacille Calmette-Guérin (BCG)

BCG prevents progression, but only with maintenance.

Mechanism

BCG binds fibronectin, releasing cytokines that recruit immune cells and produce a **T-helper 1 (Th1)** cytotoxic response. No strain is superior.

Administration

ALGORITHM

1. Wait **2–4 weeks** after TURBT (re-epithelialisation).
2. Check urinalysis before each dose to exclude infection and gross bleeding.
3. Instil 1 vial in 50 mL saline by catheter and retain it for 1–2 h, restricting fluids beforehand.
4. If catheterisation is traumatic, delay several days.

Contraindications

Absolute	Relative
Immediately after TUR; gross haematuria; traumatic catheterisation; previous BCG sepsis; immunosuppression*; total incontinence	Urinary tract infection; liver disease (isoniazid cannot then be given for sepsis); previous tuberculosis; anti-TNF drugs; poor performance status; advanced age

- Most absolute contraindications mean a **breached urothelium, allowing intravasation**. *Transplant and steroid patients may receive it, with weaker response.

Efficacy

- BCG is the **only intravesical agent that reduces progression**, and only with **maintenance**.
- In CIS, the initial CR is up to **84%**, and about half are durable.
- A second induction for persistent CIS gives a further 30–50% CR, but beyond 2 courses progression risk rises to 20–50%. **The response at 6 months predicts prognosis.**

Schedule

ALGORITHM

1. **SWOG (Southwest Oncology Group; Lamm 2000)**: give induction weekly ×6.
2. Then give 3 weekly instillations at **3 and 6 months, then every 6 months to 3 yr**.
3. If high risk, give full dose for 3 yr. If intermediate risk, give 1 yr (EORTC 30962).

Toxicity management

Grade	Features	Management
1	Cystitis, mild haematuria, fever <38.5°C, lasting <48 h	Symptomatic; consider culture
2	Severe symptoms or >48 h	Culture, chest X-ray, liver tests; isoniazid, rifampicin and pyridoxine; resume at ½–⅓ dose

Grade	Features	Management
3	Haemodynamic change, persistent fever; allergic (arthralgia, rash); organ involvement (liver, lung, kidney)	**Stop BCG**; isoniazid, rifampicin and ethambutol; prednisone only **with** antimycobacterial cover; infectious disease input

- MCQ BCG is resistant to **pyrazinamide**. Asymptomatic granulomatous prostatitis needs no treatment.

BCG shortage

ALGORITHM

1. Prioritise high-risk patients.
2. Reduce the dose, especially in maintenance, and split vials.
3. Use chemotherapy: single agent for intermediate, Gem/Doce for high risk. Early RC for the highest risk.

7 · Recurrence after BCG

SAQ BCG therapy and BCG failure

Paper answer · P1 · P2 · merged from 2

Asked as: P1 SAQ "BCG therapy in Bladder Cancer" · P2 SAQ "BCG failure"

Introduction

- BCG (bacille Calmette-Guérin), a live attenuated mycobacterium (Morales, 1976), is standard for high-risk NMIBC (non-muscle-invasive bladder cancer) and the only intravesical agent that reduces both **recurrence and progression**.

Mechanism

- BCG binds **fibronectin** on the urothelium; cytokines (IFN-γ [interferon-gamma], IL-2 [interleukin-2] and TNF-α [tumour necrosis factor-alpha]) and immune cells follow.
- A **Th1 (T-helper 1)** response gives cell-mediated tumour kill. It needs repeated stimulation, which is why **maintenance** is essential.

Indications

- **High-risk NMIBC:** HG (high-grade) T1, large/multifocal/recurrent HG Ta, **CIS (carcinoma in situ)** (initial complete response up to 84%).
- **Intermediate risk:** option. Not for low risk.

Technique and schedule

- Start **2–4 weeks after TURBT (transurethral resection of bladder tumour)**; urinalysis first; atraumatic catheter; retain **1–2 hours**.
- **Induction:** weekly × 6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** weekly × 3 at 3 and 6 months, then every 6 months to **3 years** (recurrence-free survival 76.8 vs 35.7 months).
- **EORTC (European Organisation for Research and Treatment of Cancer) 30962:** intermediate risk, full dose for 1 year; high risk, full dose for 3 years. **NIMBUS:** reduced frequency was inferior.

Contraindications

- **Absolute:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk **intravasation**); previous BCG sepsis; total incontinence; immunosuppression (now debated).
- **Relative:** UTI (urinary tract infection), liver disease, prior TB (tuberculosis).

Toxicity

Grade	Features	Management
1	Cystitis, fever <38.5 °C, <48 h	Symptomatic
2	Severe or >48 h	Isoniazid + rifampicin; resume at ½–⅓ dose
3	Sepsis, organ involvement	Stop BCG; add ethambutol; steroids only with cover
- BCG is resistant to **pyrazinamide**.		

BCG failure: definitions

- "Failure" was poorly defined; consensus categories now guide treatment.

Category	Definition	Implication
Adequate BCG	≥5 of 6 induction + ≥2 of 3 maintenance, or ≥2 doses of a second induction; for T1, ≥5 of 6 induction alone	Threshold for "unresponsive"
BCG-unresponsive	Persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within **6 months (Ta/T1) or 12 months (CIS)** of last dose; **HG T1 at first 3-month check**	More BCG futile; cystectomy
BCG-resistant	CIS or HG Ta at 3 months after induction	Reinduction: 30–50% respond
Delayed relapse	HG relapse beyond these windows	Rechallenge may work
BCG-intolerant	Toxicity prevented adequate BCG	Salvage chemotherapy

- **Why BCG fails:** inadequate dosing; PD-L1 (programmed death-ligand 1) upregulation on tumour cells.

Evaluation of BCG failure

- Cystoscopy, cytology, **re-TURBT** and biopsies to exclude invasion.
- **Extravesical disease:** prostatic urethral biopsy (24–30%) and CT urography (upper tract 0.8–25%).

Management of BCG failure

1. **Timely radical cystectomy** is standard (EAU [European Association of Urology], AUA [American Urological Association]): HG T1 at 3 months progresses in 82%; CIS failure in about 50%.
2. **Bladder sparing** (unfit or unwilling):

Option	Key result (CIS complete response, CR)
Pembrolizumab IV (KEYNOTE-057)	41% at 3 months, 19% at 12 months; grade 3–4 toxicity 13%
Nadofaragene firadenovec (adenovirus carrying IFN-α2b [interferon alpha-2b] gene) every 3 months	53% at 3 months, 24% at 12 months
Gemcitabine–docetaxel sequential	60% at 1 year; 84% avoid cystectomy (retrospective)
N-803 + BCG (IL-15 [interleukin-15] superagonist)	45% at 12 months

- Salvage CR rates halve by 12 months.
- **Also know:** BCG-refractory (older term); gemcitabine, gemcitabine–mitomycin, valrubicin, chemohyperthermia, EMDA (electromotive drug administration), chemoradiation; TAR-200; cretostimogene.

Recent advances

- PD-1/PD-L1 (programmed death-1/ligand-1) inhibitor + BCG in BCG-naive high risk improved event-free survival (CREST, POTOMAC).
- TAR-200 gemcitabine-releasing device.

Diagrams to draw: SWOG schedule timeline; flowchart: HG recurrence after BCG, was BCG adequate?, unresponsive, then cystectomy or bladder sparing.

> **Key points to remember:**
> - Only BCG reduces progression; maintenance is essential.
> - Never immediately after TURBT; start at 2–4 weeks.
> - Adequate BCG = 5/6 induction + 2/3 maintenance.
> - Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
> - Timely cystectomy is standard; salvage CR rates halve by 12 months.

Also, from the condensed book (not in the paper answer):

- After failure, check the prostatic urethra and upper tracts, because recurrence there develops in about half on long follow-up.

> **KEY POINT**
>
> **Trap:** HG T1 at 3 months after induction alone counts as unresponsive. CIS or HG Ta at 3 months is "resistant", so give more BCG.

Bladder-sparing options (unfit for or declining RC)

Agent	Mechanism / schedule	CIS CR at 3 mo, then 12 mo	Cystectomy-free
Pembrolizumab (KEYNOTE-057)*	Anti-PD-1; 200 mg intravenously every 3 weeks up to 2 yr	41%, then 19%	63%; grade 3–4 toxicity 13% (immune)
Nadofaragene firadenovec*	Adenoviral interferon-α2b gene; intravesical 3-monthly	53%, then 24%	71%; grade 3–4 4%

Agent	Mechanism / schedule	CIS CR at 3 mo, then 12 mo	Cystectomy-free
N-803 + BCG	Interleukin-15 superagonist	71% (any time), then 45%	~85%
Gem/Doce	Weekly ×6, monthly ×2 yr	—, then 60% (43% at 2 yr)	84%; no level 1 evidence

*FDA-approved. CR roughly halves by 12 months, and none matches cystectomy.

8 · Early radical cystectomy in high-risk NMIBC

- T1 failing BCG at 3 months progresses in **82%**, and CIS in ~50%.
- Immediate RC for clinical T1 gives 10-yr disease-free survival of **92%** (64% if understaged). Even a 12-week delay worsens survival.
- **Patients who progress to muscle invasion under surveillance do worse than those with de novo muscle-invasive disease.**

ALGORITHM

1. **Strongly consider RC** if: BCG-unresponsive HG; persistent HG T1 on re-resection; T1 with CIS, LVI or variant histology; deep lamina propria invasion; diffuse CIS.
2. Also if: diverticular HG tumour; distal ureteric or prostatic ductal involvement; incompletely resectable tumour; patient preference.

9 · Surveillance

Cystoscopy with cytology is the gold standard. **Tumour at the 3-month cystoscopy** and number at first resection best predict recurrence.

Risk	Cystoscopy	Cytology	Upper tract imaging
Low	At 3 months, then annually from 12 months; consider stopping at **≥5 yr**	Consider	Only if haematuria
Intermediate	3–6-monthly for 1–2 yr, then 6–12-monthly for 2 yr, then annually	Consider	Consider every 1–2 yr
High	**3–4-monthly for 2 yr**, then 6-monthly for 2 yr, then annually **for life**	Same schedule	Annually for 2 yr, then lengthen

- Restart the clock at each recurrence, using the risk of the **latest resected** tumour.
- Positive cytology with negative cystoscopy needs biopsies and upper-tract work-up.
- Urinary markers (UroVysion FISH is the most specific) do not replace cytology.
- **Extravesical recurrence**: upper tract UC occurs in <1% at low risk and ~10% at high risk. The prostatic urethra is involved in 20–40% of high-risk patients at 10 yr; HG ductal disease needs cystoprostatectomy.

10 · Prevention

- **Advise stopping smoking** at every visit, although its effect on recurrence is uncertain.

11 · Landmark trials

Trial	Design	Result	Take-home
SWOG S0337	Immediate gemcitabine 2 g vs saline, suspected LG	4-yr recurrence 35% vs 47% (LG 34% vs 54%); no extra toxicity	Gemcitabine is the standard single instillation
SWOG maintenance (Lamm 2000)	Induction with or without maintenance BCG, recurrent Ta/T1/CIS	Median RFS 76.8 vs 35.7 months; only 16% completed	Maintenance is essential
EORTC 30962	Full vs ⅓ dose; 1 vs 3 yr	Toxicity stops ~7–8%	Intermediate: full dose 1 yr; high: full dose 3 yr
NIMBUS	Reduced-frequency (9 doses) vs standard, HG	Recurrence doubled; stopped early	Do not reduce frequency
KEYNOTE-057	Single-arm pembrolizumab, BCG-unresponsive CIS	CR 41% (3 mo), 19% (12 mo)	First systemic therapy approved for NMIBC

131 Management of Muscle-Invasive and Metastatic Bladder Cancer

1 · Natural history and histology

MIBC kills through micrometastases, so cure needs systemic treatment too.

- **20–30%** of new bladder cancers are muscle-invasive (MIBC), and **~12%** are node-positive or metastatic.
- **Secondary** MIBC (**~20%**), which has progressed from non-muscle-invasive disease, does worse than **primary** MIBC, with more understaging and less response to neoadjuvant chemotherapy (NAC).
- Most deaths after local therapy occur within **2 years**, from micrometastases present at diagnosis.
- Urothelial carcinoma (UC) is 90%. **Radical cystectomy (RC) is standard whatever the histology**; subtype changes only the use of NAC and radiation.

Variant	Key points	Treatment
Squamous (5%)	Schistosomiasis, chronic cystitis	Poor chemo- and radiosensitivity, so up-front RC with lymphadenectomy
Adenocarcinoma (2%)	Commonest in exstrophy; urachal; exclude colorectal or breast metastasis	RC (urachal: partial cystectomy)
Small cell	High stage; ectopic ACTH, hypercalcaemia	**Chemosensitive**, so NAC with etoposide and cisplatin, then RC
Micropapillary	Frequent lymphovascular invasion (LVI)	If predominant, up-front RC
Mixed UC + variant	Quantify the variant	UC-predominant: NAC; variant-predominant: RC

2 · Clinical staging

Stage by resection depth, examination under anaesthesia (EUA), computed tomography (CT) with excretory phase, then chest imaging. In cT2, pathology upstages **40–54%**; LVI, variant histology, female sex and secondary MIBC predict upstaging.

TURBT

- Resect all visible tumour if safe, with a separate deep specimen; this is needed for bladder preservation, and pT0 at RC improves survival.
- If high-grade T1 or no muscle, repeat resection within **6 weeks**.
- **Prostatic urethral biopsy** (trigone or bladder-neck tumour, carcinoma in situ (CIS), multifocal disease) is a full-thickness loop from mid-prostate to verumontanum at **5 and 7 o'clock**.
- **Frozen urethral margin at cystoprostatectomy** is mandatory if the prostatic biopsy was positive, and advisable whenever a neobladder is planned; a neobladder is made only if it is negative.

Bimanual EUA

Examine before and after resection with the bladder empty; EUA more often understages than overstages.

cT	Bimanual finding
T2a	Nonpalpable
T2b	Induration, no three-dimensional mass
T3a	Mobile three-dimensional mass
T4a	Invades prostate, vagina, rectum
T4b	Fixed to pelvic sidewall

Imaging and bloods

- **Image before resection**, as post-resection inflammation mimics T3. CT nodal sensitivity is only **30–50%**; hydronephrosis suggests extravesical disease.
- Magnetic resonance imaging stages T best. The **Vesical Imaging-Reporting and Data System (VI-RADS)** grades the muscularis line: intact in 1–3, interrupted in 4, into fat in 5. A score **≥3** is suspicious.
- FDG PET/CT is reserved for equivocal imaging.
- Chest CT is needed for all; bone or brain imaging only with symptoms or raised alkaline phosphatase. Renal function decides NAC and orthotopic diversion.

3 · Pathological staging (AJCC 8th) and prognostic markers

Category (American Joint Committee on Cancer)	Definition
T2a / T2b	Inner / outer half of muscularis propria
T3a / T3b	Perivesical fat microscopic / macroscopic
T4a / T4b	Prostatic stroma, seminal vesicles, uterus, vagina / pelvic or abdominal wall
N1 / N2 / N3	Single true-pelvis node (perivesical included) / multiple / **common iliac**
M1a / M1b	Nodes beyond common iliacs / non-nodal
Stage II / IIIA / IIIB	T2N0 / T3–T4a N0 or T1–T4a N1 / N2–N3
Stage IVA / IVB	T4b any N, or M1a / M1b

- Traps: common iliac is **N3, not M1**; T4b N0 is stage IVA. pT4a needs **stromal** invasion, since urethral or ductal CIS does not upstage.
- **Subtypes**: luminal survives best, **basal gains most from NAC**, and p53-like is chemoresistant.
- **Circulating tumour DNA**: over a third stay positive after RC and do badly; in IMvigor010, adjuvant atezolizumab helped only them.

4 · Radical cystectomy: principles and extent

- **RC with bilateral pelvic lymph node dissection (PLND) is standard for resectable MIBC**, ideally after cisplatin NAC. Delay >**12 weeks** worsens survival.
- **Men**: bladder, perivesical tissue, prostate and seminal vesicles are removed. **Women** (anterior exenteration): bladder, urethra, uterus, cervix, ovaries, tubes and anterior vaginal wall.

- **Male organ-sparing** needs no bladder-neck disease and negative prostatic biopsies; incidental prostate cancer is found in ¼–½.
- **Female organ-sparing** needs no base or trigone tumour. Keeping the uterus and anterior vagina supports a neobladder; keep ovaries premenopause unless there is breast or ovarian cancer risk.

> **KEY POINT**
>
> Spare an organ only where it is oncologically silent. The gatekeepers are, in men, the prostatic urethra and bladder neck; in women, the bladder neck, base and anterior vaginal wall.

5 · Pelvic lymph node dissection

Nodal status is the strongest predictor of survival.

LAQ Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy

Paper answer · P2 · P4 · merged from 2

Asked as: P2 LAQ "Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types" · P4 LAQ "Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer?"

Introduction

- MIBC (muscle-invasive bladder cancer; TCC, transitional cell carcinoma, now called urothelial carcinoma) is cT2–T4a disease.
- Nearly half of patients treated by RC (radical cystectomy) alone develop metastases from occult micrometastases, so cure needs local plus systemic therapy, planned by an MDT (multidisciplinary team).
- Cure improves with **systemic therapy before surgery** and a **meticulous lymph node dissection** (accurate staging, possible therapeutic value).
- UTUC (upper tract urothelial carcinoma) behaves alike, but RNU (radical nephroureterectomy) removes a kidney, changing chemotherapy timing.

Staging

- Complete TURBT / Re-TURBT (transurethral resection of bladder tumour) with detrusor muscle; **MRI (magnetic resonance imaging) with VI-RADS** (Vesical Imaging-Reporting and Data System; ≥3 suggests invasion); CT (computed tomography) urogram and chest CT; bone scan if symptomatic.
- Assess fitness and cisplatin eligibility.

Management algorithm

```
cT2–T4a N0 M0 ├─ Cisplatin-eligible → cisplatin NAC (neoadjuvant chemotherapy) │ → RC + bilateral PLND (pelvic lymph node dissection) ├─ Cisplatin-ineligible → RC + PLND directly │ (pT3–4 or N+ after either → adjuvant nivolumab or chemotherapy) ├─ Selected, motivated → trimodality therapy (TMT) └─ Unfit → chemoradiation / palliative radiotherapy cN+ → induction chemotherapy → consolidation RC in responders M1 → systemic therapy
```

1. Cystectomy pathway: radical cystectomy (standard of care for cT2–T4a N0 M0)

- **Cisplatin-based NAC:** ddMVAC (dose-dense methotrexate, vinblastine, doxorubicin, cisplatin) or GC (gemcitabine–cisplatin) ×4 (details below).
- **Cisplatin-ineligible:** direct cystectomy.
- **RC + standard bilateral PLND:** obturator, external and internal iliac nodes. **SWOG S1011:** extended PLND gave no survival benefit (details below).
- Men: cystoprostatectomy. Women: anterior exenteration; organ-sparing in selected cases.
- Do not delay beyond **12 weeks** from diagnosis (worse survival). Diversion: ileal conduit, orthotopic neobladder or continent pouch.
- **Adjuvant:** cisplatin chemotherapy if no NAC; **nivolumab** for pT3–4 or N+ (details below).

2. Bladder preservation: TMT (trimodality therapy)

- Maximal (visibly complete) TURBT + RT (radiotherapy) (64 Gy, or 55 Gy in 20 fractions) + radiosensitiser. **BC2001:** mitomycin C + 5-FU (fluorouracil) improved 2-year locoregional disease-free survival (67% vs 48%).
- Best candidates: cT2, solitary tumour, **no CIS (carcinoma in situ), no hydronephrosis**, good bladder function.
- 5-year cancer-specific survival 50–84%; salvage cystectomy 10–30%.
- **Partial cystectomy:** solitary dome or urachal tumour, no CIS.
- **Also know:** chemotherapy alone or radical TURBT alone is not recommended.

3. Metastatic disease

- First line: **EV (enfortumab vedotin) + pembrolizumab** (EV-302: about doubled overall survival (OS) vs platinum). Alternatives: platinum chemotherapy then **avelumab maintenance** (JAVELIN Bladder 100); erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

Neoadjuvant chemotherapy (NAC) in MIBC

Rationale

- Treats micrometastases early, while tumour burden is low; better tolerated before surgery, so more patients complete it.
- Downstages; **pT0 at cystectomy** gives about 90% 5-year recurrence-free survival.
- Drawback: delays surgery in non-responders.

Eligibility: cT2–T4a N0 M0 and cisplatin-fit. **Cisplatin-ineligible** if any one: GFR (glomerular filtration rate) <60 mL/min, ECOG (Eastern Cooperative Oncology Group) performance status ≥2, grade ≥2 hearing loss, grade ≥2 neuropathy, NYHA (New York Heart Association) class III–IV failure.

Trial	Regimen	Result
SWOG 8710 (Southwest Oncology Group)	MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×3 then RC vs RC	pT0 38% vs 15%; median survival 77 vs 46 months (P4: 5-year OS 57% vs 43%); **P = 0.06**
BA06 30894 (EORTC/MRC, European Organisation for Research and Treatment of Cancer/Medical Research Council)	CMV (cisplatin, methotrexate, vinblastine) ×3	16% relative reduction in mortality; 10-year OS 36% vs 30%
ABC meta-analysis (Advanced Bladder Cancer)	Cisplatin combinations	OS HR (hazard ratio) 0.86: **5% absolute 5-year survival gain**

Trial	Regimen	Result
VESPER	ddMVAC ×6 vs GC ×4	ddMVAC better than GC: organ-confined response 77% vs 63%; 3-year PFS (progression-free survival) 66% vs 56%; more toxicity
NIAGARA	Perioperative durvalumab + GC, then adjuvant durvalumab, vs GC	Better event-free survival (HR 0.68) and OS (HR 0.75)

- **Guidelines (EAU, European Association of Urology; AUA, American Urological Association; NCCN, National Comprehensive Cancer Network):** cisplatin-based NAC for all eligible patients; ddMVAC 3–6 cycles or GC 4 cycles.
- **No survival benefit:** single-agent cisplatin, carboplatin regimens, neoadjuvant radiotherapy.
- **Cisplatin-ineligible:** proceed straight to RC; neoadjuvant pembrolizumab (PURE-01) and atezolizumab (ABACUS) are investigational.
- The chapter calls neoadjuvant immunotherapy unproven; NIAGARA has since made perioperative durvalumab + GC an option.

Adjuvant therapy in MIBC

	Neoadjuvant	Adjuvant
Selection	Clinical stage (understaging risk)	True pathological stage, but still overtreats
Delivery	Better tolerated	Post-RC morbidity; a quarter never receive it
Evidence	Level 1	Weaker: small, early-closed trials

Adjuvant chemotherapy

- Trials were small or closed early. **ABC 2022 update:** cisplatin-based adjuvant chemotherapy improves 5-year OS by about 6%. Benefit greatest in node-positive and extravesical disease.
- Use: cisplatin-naive patients (no NAC) with **pT3–4 or N+**: ddMVAC or GC.

Adjuvant immunotherapy

- **CheckMate 274:** nivolumab for 1 year after RC (pT3–4a or N+, with or without prior NAC): median disease-free survival **20.8 vs 10.8 months**. It also included UTUC.
- **IMvigor010** (atezolizumab): negative overall (included pT2); benefit in ctDNA (circulating tumour DNA)-positive patients.
- Use nivolumab if cisplatin-ineligible or declining chemotherapy, or with ypT3 or worse after NAC.
- **Also know:** adjuvant pelvic radiotherapy (Cairo trials) improves pelvic control in pT3–4; ctDNA-guided therapy is emerging.

Lymphadenectomy in MIBC

- **Primary drainage:** obturator, internal and external iliac, presacral. **Secondary:** common iliac, para-aortic, paracaval.
- Node positivity: pT2 8–26%, pT3 33–50%, pT4 36–80%.

Template	Extent
Standard	Bilateral external and internal iliac and obturator nodes; genitofemoral nerve laterally, node of Cloquet caudally, up to the common iliac bifurcation
Extended	Adds common iliac and presacral nodes, up to the aortic bifurcation
Super-extended	Up to the inferior mesenteric artery

- **Evidence:** German phase III (limited vs extended): no survival benefit. **SWOG S1011** (most had NAC): median nodes 24 vs 39, **no difference** in disease-free survival (HR 1.10) or OS; more grade ≥3 events. So **standard bilateral PLND** is the standard of care.
- **Node count:** fewer than 10 nodes doubled the risk of death (SWOG 8710 reanalysis). **Also know:** lymph node density, extranodal extension (prognostic).
- Unexpected nodes at RC: frozen section, proceed if resectable within the pelvis. Clinically node-positive: chemotherapy first, then RC in responders.

Neoadjuvant and adjuvant therapy in UTUC

- **Why neoadjuvant:** CKD (chronic kidney disease) is common and GFR falls after RNU, losing cisplatin eligibility. Give platinum **while both kidneys work**.
- Biopsy under-stages, so select by high-risk features (high grade, invasion on imaging, hydronephrosis).
- **EA8141** (phase II): accelerated MVAC, or gemcitabine–carboplatin if CrCl (creatinine clearance) 30–49; complete response 14%, below ypT2N0 60%.
- **AUA/SUO (Society of Urologic Oncology):** neoadjuvant platinum is a standard option for high-risk UTUC, especially if postoperative CrCl will fall below 50.
- **Adjuvant: POUT** (phase III, pT2–T4 or N+ after RNU): gemcitabine–platinum vs surveillance; disease-free survival HR 0.45, 3-year 71% vs 46%. Weaker with carboplatin, favouring neoadjuvant cisplatin. Adjuvant nivolumab (CheckMate 274) also included UTUC.

Lymphadenectomy in UTUC

- Node positivity 12–25%. Lymph drains **right to left and top to bottom**, so the template depends on side and level.

Primary site	Right	Left
Renal pelvis, upper two-thirds ureter	Hilar, paracaval, retrocaval, **interaortocaval**	Hilar, para-aortic
Lower third ureter	Ipsilateral common, external, internal iliac, obturator, presacral	Same

- Value: pN0 better than pNx; ≥8 nodes improved disease-specific survival above pT1. Indicated for high-grade or invasive disease; robotic yield equals open. Risk: chylous leak.

Urinary diversion options

- Ileal conduit (simplest, commonest), continent cutaneous pouch, or **orthotopic neobladder**. No diversion is proven superior for QoL (quality of life).

Neobladder: principles

1. **Reliable outlet:** intact rhabdosphincter, tumour-free urethra (frozen section).

2. **Low pressure:** complete **detubularisation** and cross-folding abolish peristalsis; a **sphere** gives most volume (Laplace).
3. **Capacity:** 44 cm of folded ileum holds <200 mL at first, **500–600 mL by 1 year**. - **Contraindications:** positive urethral margin, stricture, eGFR (estimated GFR) <35–40 mL/min, liver failure, inability to self-catheterise. Age, obesity and nodes are not bars.

Critical evaluation of neobladder types

Type	Bowel; ureteric technique	Strengths	Weaknesses
Studer	40–44 cm ileum + isoperistaltic afferent limb; refluxing	Simple, no staples, takes short ureters; stricture <3%; good long-term function	Night incontinence; reflux on cystogram (harmless)
Hautmann W	About 70 cm ileum; refluxing since 1997	Largest capacity, early continence; stricture fell 9.5% → 1%	More retention, acidosis, B12 loss
T pouch (Stein–Skinner)	44 cm + tunnelled T-limb flap valve	Antireflux; accepts dilated ureters	Complex; more secondary procedures (USC [University of Southern California] RCT)
Abol-Enein serous tunnel	40 cm W, serosal troughs	Antireflux; stricture about 3.8%	Long ureters needed; no dilated ureters
Hemi-Kock	Intussuscepted nipple	Good early results	Late nipple stenosis, stones; abandoned
Camey II; Padua (VIP)	65 cm or 42 cm ileum; Le Duc tunnel	Early designs	Le Duc stricture 7–29%
Mainz III; right colon; sigmoid (Reddy)	Colon ± ileum	When ileum short or diseased	High-pressure contractions, poor compliance, more leakage

Critical points

- **Reflux vs obstruction:** Valsalva voiding pressurises the whole system equally, so reflux is benign; every antireflux valve can obstruct. **USC RCT** (484 patients): Studer = T pouch for renal function and infection; T pouch needed more secondary procedures. Hence **Studer and Hautmann** are the most popular.
- **Segment:** ileum is most compliant; colon and stomach give high pressures.
- **Capacity trade-off:** a larger pouch gives earlier continence but more retention and metabolic load.
- **Robotic intracorporeal neobladder:** comparable outcomes; long-term data awaited.

Neobladder outcomes, complications and follow-up

- **Day continence 80–90%; night 20–50%.** Retention: men 4–10%, **women 20–30%**.
- **Late:** hyperchloraemic metabolic acidosis, B12 deficiency, mucus, stones, ureteroileal stricture, pouch rupture, urethral recurrence (men about 7% at 5 years).
- **Follow-up:** blood gas, B12, upper-tract imaging, urethral cytology if CIS.

Diagrams to draw: bladder wall T-staging (T2a, T2b, T3, T4) beside the standard PLND template; standard vs extended pelvic templates, and right vs left UTUC templates; Studer pouch with afferent limb and urethral anastomosis.

Key points to remember:

- NAC: cisplatin combinations only; ABC 5% absolute 5-year survival gain; SWOG 8710 pT0 38% vs 15%, P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

Also, from the condensed book (not in the paper answer):

- About **25%** are node-positive at RC. Recurrence is **70–80%** if node-positive vs ~30% if node-negative.
- Standard PLND reaches the common iliac bifurcation; extended reaches the aortic bifurcation. About 10 nodes is the usual cut-off for pelvic recurrence risk.

6 · Intraoperative decision making

- If **clinically node-positive**, give cisplatin chemotherapy first, then **consolidative RC** in responders.
- **Unexpected nodes**: proceed only if completely resectable within the true pelvis. **Abort RC** if nodes are unresectable, the bladder is fixed to the sidewall or the rectosigmoid is invaded.
- **Ureteric margin** is involved in 6–8% and raises upper-tract recurrence (4–10%, mostly at 2–4 years). Frozen section is not mandated, but send one for gross tumour.
- **Male urethral recurrence** is 1–8% and highest with **prostatic stromal invasion** (~30%), so consider urethrectomy then. A positive pre-op prostatic biopsy does **not** bar a neobladder; the **frozen urethral margin decides**.
- **Female urethra** is involved in 2–12%. Urethrectomy is mandatory only for T4 disease involving urethra or vagina; send a proximal urethral frozen section in every neobladder candidate.

7 · Outcomes after RC

Stage	5-year disease-specific survival
≤pT1 / pT2 / pT3 / pT4	~90% / 70–81% / 40–52% / 16–44%
N0 / N+	62–80% / 21–35%

- **~50%** ultimately die of bladder cancer, mostly from metastases within 2 years.
- **Pelvic recurrence**: low risk (≤pT2) 5–8%; intermediate (pT3–4, negative margins, ≥10 nodes) 15–21%; high (pT3–4 with positive margins or <10 nodes) **39–46%**.

8 · Neoadjuvant therapy

Cisplatin NAC gives a level 1 survival gain.

*See the paper answer **Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy** (Section 12, Ch 131).*

Also, from the condensed book (not in the paper answer):

- About half of patients are cisplatin-ineligible, and they go straight to local therapy.
- Neoadjuvant immunotherapy (IO) phase II trials: **ABACUS** (atezolizumab) complete response 31%; **PURE-01** (pembrolizumab) pT0 42%.

Criterion	Cisplatin-ineligible (any one; ~50%)	Platinum-ineligible
Renal	Glomerular filtration rate **<60 mL/min**	Creatinine clearance <30 mL/min
ECOG performance status	≥2	≥2 (text) / ≥3 (table)
Hearing loss / neuropathy	Grade ≥2 / ≥2	— / grade ≥3
Heart failure	NYHA III–IV	NYHA III–IV; multiple high-grade comorbidities

- The NCCN prefers dose-dense methotrexate, vinblastine, doxorubicin and cisplatin (ddMVAC).
- **Neoadjuvant IO** blocks the PD-1/PD-L1 checkpoint and was not yet recommended in the source. †NIAGARA has since shown that perioperative durvalumab with gemcitabine–cisplatin (GC) improves event-free and overall survival (OS).
- Neoadjuvant radiation downstages but gives no survival gain.

9 · Adjuvant therapy

- The target is **pT3–4 or N+**. Post-RC morbidity prevents delivery in ~¼, so neoadjuvant treatment is preferred.
- **Chemotherapy** is not standard, but discuss ddMVAC or GC in **cisplatin-naive** pT3–4/N+ patients.
- **Nivolumab for 12 months** is given for ≥pT3/N+ after NAC, or if cisplatin-ineligible or declining. IMvigor010 (atezolizumab) was negative.
- **Radiation** may be considered for pT3/N+, where local failure is 20–40%.

10 · Bladder preservation

Selected patients keep their bladder with similar survival.

Trimodality therapy (TMT)

LAQ Bladder-preserving treatment of TCC

Paper answer · P2 · P4 · merged from 2

Asked as: P2 LAQ "Classification of bladder cancer and bladder-preserving treatment of TCC" · P4 LAQ "Discuss bladder preservation surgery for TCC of urinary bladder."

Read as: classification of bladder cancer, and bladder-preserving treatment of TCC (transitional cell carcinoma, urothelial carcinoma), including TURBT (transurethral resection of bladder tumour), radical TURBT, partial cystectomy and trimodality therapy.

Classification of bladder cancer

1. Histological

Type	Share	Note
Urothelial carcinoma	~90%	Variants: micropapillary, plasmacytoid, sarcomatoid, nested; squamous or glandular differentiation
Squamous cell carcinoma	~5%	Schistosomiasis, catheter, stones
Adenocarcinoma	~1–2%	Urachal (dome), exstrophy
Small cell / others	<1%	Neuroendocrine; sarcoma

2. Grade (WHO 2004/2016, World Health Organization)

- PUNLMP (papillary urothelial neoplasm of low malignant potential), low-grade and high-grade carcinoma. WHO 1973 (G1–G3) is still reported by some.
- Low-grade (FGFR3 pathway) recurs but rarely progresses. High-grade and CIS (carcinoma in situ; TP53/RB pathway) progress.

3. Stage: TNM 2017

T	Definition
Ta / Tis	Non-invasive papillary / flat CIS
T1	Lamina propria
T2a / T2b	Inner / outer half of muscle
T3a / T3b	Perivesical fat, microscopic / macroscopic
T4a / T4b	Prostatic stroma, uterus, vagina / pelvic or abdominal wall

- N1 one true-pelvic node; N2 ≥2 pelvic nodes; N3 common iliac; M1a distant nodes; M1b organs.

4. Clinical groups

- **NMIBC (non-muscle-invasive, ~70%)**: Ta, Tis, T1. Risk groups: low (solitary primary low-grade Ta <3 cm), intermediate, high (high-grade, T1 or CIS).
- **MIBC (muscle-invasive)**: T2–T4a. Metastatic.
- **Molecular:** luminal papillary versus basal/squamous subtypes.

Why bladder preservation (introduction)

- Standard for MIBC (muscle-invasive bladder cancer) is cisplatin NAC (neoadjuvant chemotherapy) plus RC (radical cystectomy) with PLND (pelvic lymph node dissection).
- RC is morbid and patients are often elderly. Bladder preservation suits **(1) unfit patients** and **(2) fit, motivated patients with limited disease**.
- **Why single modalities fail:** they leave either the remaining urothelium or occult nodes untreated (cT2 understaged to pT3 in up to about 40%; occult nodes about 25%).

Options

Option	Use / status
Radical TURBT alone	Not a primary treatment
Partial cystectomy + PLND	Highly selected solitary tumour **<3 cm** in mobile bladder (dome), **no CIS**, 1–2 cm margin, frozen sections; **urachal adenocarcinoma**
RT (radiotherapy) alone	Inferior; quasi-palliative
Chemotherapy alone	Inferior; not recommended
TMT (trimodality therapy)	**Best-studied, guideline-accepted**

TURBT for NMIBC: the standard

- Complete resection with **detrusor muscle in the specimen**; separate base biopsies; cold-cup biopsy of suspicious areas.
- **En bloc resection** (loop or laser) keeps architecture, gives more detrusor and fewer perforations.
- **Repeat TURBT within 6 weeks** for all T1, incomplete resection or no muscle. Residual tumour in ≥40%.
- Then intravesical chemotherapy or BCG (bacillus Calmette–Guérin) by risk group.

Radical TURBT for MIBC

- Rationale: about 10% are pT0 at cystectomy. Only for tumour limited to superficial muscle, with a negative re-resection.
- About a third later progress to MIBC on long follow-up; selected series report 5-year CSS (cancer-specific survival) about 82%. Not a primary treatment.

Partial cystectomy

Ideal candidate	Contraindication
Small (**<3 cm**), **solitary** tumour in a mobile part of bladder (dome)	**CIS**, multifocal tumour
No CIS, no hydronephrosis	Trigone or bladder neck tumour
1–2 cm margin achievable away from the ureteric orifice and trigone; adequate residual capacity	Small bladder; prostatic urethral involvement
Urachal adenocarcinoma (treatment of choice: en bloc dome, urachus and umbilicus)	

Technique (steps):

1. NAC (neoadjuvant cisplatin chemotherapy) if cisplatin-eligible.
2. PLND (always).
3. Cystotomy away from tumour.
4. Full-thickness excision with perivesical fat, **frozen-section margins**.
5. Ureteric reimplantation if needed.
6. Watertight closure; irrigation (prevents seeding); drain.
7. Cystogram before catheter removal.

Outcomes: 5-year OS (overall survival) 67–70%; intact bladder 65–74%; salvage RC up to a quarter. Late recurrence, so lifelong follow-up.

Trimodality therapy (TMT)

Components: (1) **maximal, visibly complete TURBT**; (2) **RT** to bladder ± pelvic nodes; (3) concurrent **radiosensitising chemotherapy**.

Selection criteria:

- cT2 (selected cT3–4a), small solitary tumour.
- **No hydronephrosis, no extensive CIS (no CIS)**, visibly complete TURBT.
- Good bladder function; motivated, compliant patient.
- Only 6–19% of MIBC meet strict criteria.

Radiosensitising regimens:

Regimen	Evidence
Mitomycin C + 5-FU (fluorouracil)	**BC2001**: 2-year locoregional DFS (disease-free survival) **67% vs 48%** vs RT alone, no excess late toxicity
Cisplatin (± 5-FU or paclitaxel)	MGH (Massachusetts General Hospital)/RTOG (Radiation Therapy Oncology Group) protocols
Gemcitabine	Cisplatin-ineligible
Carbogen + nicotinamide (BCON)	Hypoxia modification

Radiation dose and schedules: 64–66 Gy in 32 fractions or **55 Gy in 20** (non-inferior).
`Split course (MGH/RTOG): TURBT → chemoRT ~40 Gy → cystoscopy + biopsy CR → consolidation to 60–66 Gy Residual tumour → immediate cystectomy Continuous (current): TURBT → chemoRT 64 Gy/32 fr or 55 Gy/20 fr → cystoscopy + biopsy and imaging at 3–4 months ├─ Complete response → surveillance (NMIBC recurrence: TURBT + BCG) └─ Persistent / recurrent MIBC (invasive) → salvage cystectomy (10–30%)`

Response assessment: cystoscopy with biopsy of tumour site, cytology, CT (computed tomography)/MRI (magnetic resonance imaging). **CR (complete response)** leads to surveillance. Non-response leads to salvage RC.

Salvage cystectomy:

- Needed in **10–30%** for persistent or recurrent muscle-invasive disease; it is the designed safety net.
- NMIBC (non-muscle-invasive) recurrence: TURBT and intravesical BCG.

Outcomes:

- 5-year CSS 50–84%, OS 36–74%.
- MGH/RTOG: CR 69%; 5-year DSS (disease-specific survival) 71%; 80% of 5-year survivors keep the bladder.
- Propensity-matched RC vs TMT: equal 5-year metastasis-free survival (about 75%) and similar CSS (81% vs 84%).
- Low late toxicity; good quality of life.

Comparison with radical cystectomy

	Radical cystectomy	Bladder preservation (TMT)
Staging	Pathological	Clinical only

	Radical cystectomy	Bladder preservation (TMT)
Nodes	Removed	Irradiated or not treated
Function	Diversion; sexual dysfunction	Native bladder; RT (bowel/bladder) toxicity
Follow-up	Upper tract, urethra	**Intensive cystoscopy: 3-monthly for 2 years, to 10 years**, plus CT
Evidence	Standard	No completed RCT (SPARE closed for poor accrual)

Follow-up after preservation

- Cystoscopy every 3–4 months for 2 years, 6-monthly in years 3–4, then yearly to 10 years.
- CT urogram and chest CT every 3–6 months for 2 years, then yearly to 5 years; cytology.

Recent advances

- **Immunotherapy + chemoradiation:** PD-1/PD-L1 (programmed death-1 / ligand-1) inhibitors with TMT under trial (e.g. SWOG/NRG S1806 with atezolizumab; KEYNOTE-992 with pembrolizumab); results awaited.
- Hypofractionation 55 Gy/20 non-inferior (BC2001–BCON meta-analysis: same toxicity).
- Bladder sparing after complete response to NAC (± checkpoint inhibitor) with biomarker selection (DNA repair gene mutations, circulating tumour DNA); RETAIN: about half recurred.
- Robotic partial cystectomy; molecular subtypes.
- **Also know:** RT alone is quasi-palliative; chemotherapy alone not recommended.

Diagram to draw: TNM T-stage diagram of the bladder wall, with the TMT flowchart from TURBT to chemoradiation, response check, surveillance or salvage cystectomy.

Key points to remember:

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
- TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
- BC2001: MMC/5-FU (mitomycin C/5-FU) + RT, 2-year locoregional DFS 67% vs 48%.
- Salvage cystectomy 10–30%; follow up the bladder for 10 years.

Also, from the condensed book (not in the paper answer):

- **Dose**: a continuous course to **60–66 Gy** with restaging at 3–4 months is commonest; **55 Gy in 20 fractions** is non-inferior.
- Concurrent chemotherapy beats radiation alone in two randomised trials.

KEY POINT

TMT contains cystectomy rather than replacing it. Salvage RC is the built-in safety net, so selection and lifelong cystoscopy decide success. Complete resection cuts the need for cystectomy (~11% vs 42%).

Single modalities (none is primary therapy)

Modality	Use	Why it fails
Chemotherapy alone	Complete responders declining RC	Early intravesical recurrence (~half by 5 months)
Radiation alone (64–66 Gy/32 or 55 Gy/20 fractions)	Unfit for surgery and chemoradiation; palliation	Inferior to chemoradiation; local control 27% in T3
Radical TUR	Superficial muscle invasion only	Understaging; ~25% have occult nodes
Partial cystectomy with PLND	<3 cm solitary, no CIS, margin away from orifices; **urachal adenocarcinoma (treatment of choice)**	Late recurrence, so lifelong follow-up; convert to RC if the frozen margin is positive

11 · Metastatic urothelial carcinoma

First-line choice hinges on platinum eligibility.

- Bajorin adverse factors are Karnofsky status <80% and visceral metastasis.
- **Cisplatin-eligible**: GC or ddMVAC. **GC equals MVAC** and is less toxic (von der Maase), and dose-dense beat standard MVAC (EORTC 30924).
- **Cisplatin-ineligible, platinum-eligible**: gemcitabine–carboplatin (EORTC 30986) or enfortumab vedotin (EV) with pembrolizumab. †EV-302 later showed EV with pembrolizumab roughly doubled OS vs platinum, making it the preferred first line.
- **Platinum-ineligible**: pembrolizumab (KEYNOTE-052) or atezolizumab if PD-L1-positive. Trap: **first-line IO alone is only for platinum-ineligible patients.**
- **Maintenance avelumab** is given if there is no progression after platinum (JAVELIN Bladder 100). IO combined with chemotherapy is not recommended.
- **Second line** is **pembrolizumab** (KEYNOTE-045, regardless of PD-L1); atezolizumab failed (IMvigor211). Platinum can be rechallenged if the last dose was ≥12 months earlier.
- **Antibody–drug conjugates**: **EV** (Nectin-4; neuropathy, rash) beat chemotherapy after platinum and IO (EV-301). **Sacituzumab govitecan** (Trop-2; neutropenia) is third line.
- **Erdafitinib**, an oral FGFR inhibitor, treats FGFR3/FGFR2 alterations after platinum, with 40% response; it causes **hyperphosphataemia**.
- **IO toxicity** includes rash, colitis, hepatitis, pneumonitis, myocarditis and endocrinopathies, which are often **irreversible**.

ALGORITHM

1. If platinum-eligible, give GC or ddMVAC (or gemcitabine–carboplatin, or EV with pembrolizumab).
2. If there is response or stable disease, give avelumab maintenance.
3. On progression, give pembrolizumab, or erdafitinib if FGFR-altered.
4. Then give EV, then sacituzumab govitecan.
5. If platinum-ineligible, give pembrolizumab, then EV.

12 · Follow-up

- **After RC**: CT with excretory phase and chest CT 6-monthly for 2–3 years, then yearly to 5 (EAU, AUA). Check renal function at each visit and vitamin B12 yearly from year 2.
- Upper-tract recurrence (4–10%) is the commonest **late** one; pelvic and distant recurrences come mostly within 2 years.
- **After bladder preservation**: cystoscopy 3–4-monthly in years 1–2, 6-monthly in years 3–4, then yearly to **10 years**, with CT urogram and chest CT 3–6-monthly for 2 years, then yearly to 5.

13 · Landmark trials

Trial	Design	Result	Take-home
SWOG 8710	MVAC ×3, then RC, vs RC	Median OS 77 vs 46 months; pT0 38% vs 15%	pT0 is a surrogate; OS P = .06
BA06 30894 (EORTC/MRC)	Cisplatin, methotrexate, vinblastine (CMV) ×3, then RC or radiotherapy	10-year OS 36% vs 30%	Largest positive NAC trial
ABC meta-analysis	11 randomised NAC trials	OS hazard ratio (HR) 0.86; +5% at 5 years	Combination platinum only
VESPER	ddMVAC ×6 vs GC ×4	Organ-confined 77% vs 63%; 3-year progression-free survival 66% vs 56%	ddMVAC better but more toxic
SWOG S1011	Standard vs extended PLND	No difference in disease-free survival (DFS) or OS	Standard PLND suffices
CheckMate 274	Adjuvant nivolumab vs placebo, pT3–4/N+	DFS 20.8 vs 10.8 months	Adjuvant standard
BC2001	Radiation with or without mitomycin C/5-fluorouracil	Locoregional DFS 67% vs 48%	Sensitise radiation
von der Maase	GC vs MVAC, metastatic	OS 14.0 vs 15.2 months (not significant)	Equal efficacy, GC less toxic
EORTC 30986	Gemcitabine–carboplatin vs MCAVI, cisplatin-ineligible	Equal OS, less toxicity	Gem–carbo standard
JAVELIN Bladder 100	Avelumab maintenance vs none after platinum	OS 21.4 vs 14.3 months (HR 0.69)	Maintenance standard
KEYNOTE-045	Pembrolizumab vs chemotherapy, second line	OS 10.3 vs 7.4 months	Second-line standard
EV-301	EV vs chemotherapy after platinum and IO	OS 12.9 vs 9.0 months	ADC third line

132 Surgical Management of Bladder Cancer: Transurethral, Open and Robotic

1 · Framing

- Transurethral resection (TUR) stages every patient and cures low-grade non-muscle-invasive bladder cancer (NMIBC); muscle-invasive disease (MIBC) needs radical cystectomy (RC) and pelvic lymph node dissection (PLND), after cisplatin neoadjuvant chemotherapy (NAC) if eligible.

2 · Transurethral resection of bladder tumour (TURBT)

Complete resection with muscle in the specimen makes staging correct.

Pre-operative evaluation

- **Continue aspirin**; hold other anticoagulants and P2Y12 inhibitors by half-life. **Treat active infection before resection**.
- Image the upper tracts by CT urogram, MRI, or ultrasound with retrograde study. Iodinated contrast needs a glomerular filtration rate (GFR) **≥60** mL/min; gadolinium needs GFR **>30**.

Set-up and inspection

ALGORITHM

1. Antibiotic within 1 h (avoid fluoroquinolones).
2. Inspect with 30° and 70° lenses (120° if incomplete).
3. Map tumours; add blue-light or narrow-band imaging if available (more tumours and carcinoma in situ (CIS) found, less recurrence).

	Monopolar	Bipolar
Irrigant	Sterile water or 1.5% glycine	0.9% saline
Electrolyte disturbance	Higher risk	Lower risk
Obturator reflex	Present	Reduced

Resection principles

- Remove all visible tumour with **detrusor in the specimen** and a **~2 cm** margin of normal tissue where possible.
- Then examine under anaesthesia (EUA) with the bladder empty: a **mass palpable after resection is cT3b**.
- **En bloc resection** keeps orientation and detrusor, and improves T1 staging.
- Maximal resection is therapeutic in NMIBC, but before RC for MIBC it gives no survival gain; residual tumour is only prognostic.

Difficult locations

Site	Hazard	Countermeasure
Lateral wall	**Obturator jerk**: nerve stimulation makes the leg adduct and the loop perforates	Low distension; bipolar; muscle relaxant (general anaesthesia)

Site	Hazard	Countermeasure
Ureteric orifice	Coagulation causes stricture and hydronephrosis	**Pure cutting current**; routine stent not needed; short-term stent after large resection or heavy cautery is reasonable
Dome	Intraperitoneal perforation; heat injury to bowel	Low distension; suprapubic pressure; straight loop; minimal cautery

Bladder perforation

	Extraperitoneal	Intraperitoneal
Site / frequency	Away from dome; commoner	Dome, posterior wall; rare, large ones morbid
Clues	—	Bladder will not distend; defect seen; abdomen distends
Confirm	Cystogram at same sitting	
Management	**Foley drainage and observation**	**Laparotomy**, inspect bowel, repair; catheter and drain

> **KEY POINT**
>
> Perforation can follow adequate resection of advanced tumour. It **rules out immediate intravesical chemotherapy**, because the drug would leak outside the bladder.

Post-operative intravesical agents

*See the paper answer **Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation** (Section 12, Ch 130).*

Also, from the condensed book (not in the paper answer):

- Single mitomycin C or epirubicin **within 24 h** gives a ~**13%** absolute fall in recurrence (meta-analysis); gemcitabine (SWOG S0337) cut recurrence by 12% at 4 years. They reduce recurrence, not progression.
- **BCG (bacille Calmette–Guérin) is never given immediately**: wait **2–4 weeks** with no gross haematuria, because of the risk of systemic absorption.

3 · Radical cystectomy: preparation

- **Mark a stoma for every patient** (sitting and standing) and consent for a conduit.
- **Avoid routine mechanical bowel prep** and oral antibiotics; omitting them speeds recovery and cuts wound complications.
- **Alvimopan** (peripheral μ-opioid antagonist), given 30–90 min pre-op and then twice daily, lessens ileus and shortens stay.
- Give an IV antibiotic within 1 h covering skin and bowel flora, e.g. **cefoxitin**.
- Start **venous thromboembolism (VTE) prophylaxis** (compression and drug) **before induction** and extend it to **4 weeks**: 28 rather than 8 days gives fewer major VTE without extra bleeding. Enoxaparin is standard; apixaban is equal, with better compliance.

4 · Open radical cystectomy: technique

ALGORITHM

1. Lower midline; Retzius space; divide urachus and vas or round ligament.
2. Mobilise caecum and sigmoid; make the **retrosigmoid window** (left ureter).
3. Free and divide ureters with adventitia.
4. Retractor (check vena cava); PLND; staple, clip or seal vesical pedicles.

- **Distal ureteric frozen section** is controversial: a positive margin predicts upper-tract recurrence, but clearing it has no proven benefit.

Male

ALGORITHM

1. Incise peritoneum over seminal vesicles; separate rectum; incise Denonvilliers fascia.
2. Control posterior pedicles; open endopelvic fascia.
3. Ligate dorsal venous complex (DVC); divide urethra.

- **Rectal injury is repaired primarily**, with or without faecal diversion.
- **Urethral frozen section** before a neobladder has a very high negative predictive value. **A positive result rules out orthotopic neobladder**, so warn the patient beforehand.
- **Nerve sparing** leaves few men potent at 1 year, and local recurrence occurs even in pT2. Avoid it in advanced disease or pre-existing erectile dysfunction.
- **Prostate sparing** improves potency and continence, but occult prostate cancer is common. It needs normal DRE and prostate-specific antigen (PSA), a TURP and a negative frozen section.

Female

- The standard is **anterior exenteration**, essential if EUA is positive or cT4a. Discuss **ovarian sparing** if disease is not locally advanced.

ALGORITHM

1. Ligate ovarian vessels; incise posterior fornix; control pedicles; excise urethra.
2. Watertight **clamshell vaginal closure** (keeps girth, loses length).

- **Vagina and urethra sparing for neobladder** needs ≤cT2, an uninvolved bladder neck and a negative urethral frozen section; preserve the striated sphincter.

5 · Pelvic lymphadenectomy

Template	Boundaries	Evidence
Standard	Lateral genitofemoral nerve; medial internal iliac artery; distal Cooper ligament; proximal ureter crossing common iliac	Reference
Extended	Standard plus common iliac and presacral	Better staging; possible survival gain
Superextended	Up to inferior mesenteric artery (preaortic)	No benefit over extended

- **≥10 nodes** removed goes with better survival (NMIBC at RC: >20).
- A randomised trial found **extended not superior to limited** for recurrence-free or cancer-specific survival, but its staging value is certain.

6 · Partial cystectomy

- Indications are a **solitary small MIBC without CIS** and **urachal adenocarcinoma** (en bloc umbilicus, urachus and dome with frozen margins).
- Oncology is as for RC: PLND, NAC if eligible, and consent for RC. Muscle-invasive recurrence needs salvage RC.

ALGORITHM

1. PLND; open the bladder away from tumour.
2. Full-thickness excision with fat and a 1–2 cm frozen-section margin; reimplant ureter if needed.
3. Watertight closure; warm water irrigation (against seeding); drain.
4. Cystogram day 7, then remove catheter.

7 · Robotic radical cystectomy (RARC)

KEY POINT

No effective salvage exists: urine spillage, poor PLND or a positive soft-tissue margin leads to near-uniformly fatal recurrence, so robotics is judged on oncology first.

- Indications match open surgery; prior surgery and obesity are relative contraindications. A **15-mm left assistant port** allows stapling and bagging.

ALGORITHM

1. Release sigmoid; develop presacral plane; open cul-de-sac to prostate apex.
2. Open Retzius; control anterior pedicle; free ureters (**never grasp**).
3. **Extended PLND with ureters intact**.
4. Clip ureters, send margin, pass left ureter under sigmoid.
5. Posterior pedicles, DVC, urethra (clip proximally only if neobladder).

- PLND before ureteric division keeps urine output as a resuscitation guide. Bag each packet whole; the obturator nerve is **not** the posterior limit, so go down to the hypogastric vein.

8 · Complications and post-operative care

- Gastrointestinal complications lead, then infection. For **ileus**, use alvimopan, early feeding and opioid-sparing analgesia, with **no routine nasogastric tube**.
- Ureteric devascularisation causes ischaemic stricture (keep adventitia).

9 · Landmark trials

Trial	Design	Result	Take-home
Grossman 2003	NAC and RC vs RC	Median survival 46 vs 77 months	Offer cisplatin NAC

Trial	Design	Result	Take-home
SWOG S0337	Single gemcitabine vs saline after TUR	12% fewer recurrences at 4 years	Immediate gemcitabine for low grade
RAZOR (2018)	Non-inferiority RARC vs open; open diversion	2-year progression-free survival 72.3% vs 71.6%; less blood loss, shorter stay	Robotic oncologically equivalent
iROC	RARC with intracorporeal diversion vs open	More days alive and out of hospital at 90 days; fewer VTE, wound complications	Recovery advantage

133 Use of Intestinal Segments in Urinary Diversion

1 · Surgical anatomy

A diversion depends on its pedicle's blood supply.

- **Stomach** is supplied by the right gastroepiploic (gastroduodenal, from hepatic) and left gastroepiploic (from splenic) along the greater curve. The graft is antrum or a fundal wedge, which must **never reach the pylorus or lesser curve**.
- **Small bowel** is ~22 ft: the proximal 2/5 is jejunum (wider, single arcades) and the distal 3/5 ileum (multiple arcades). Bowel >**8 cm** from a straight vessel may not survive, so clear ≤8 cm of mesentery at a segment end.
- **Avoid pelvic small bowel after radiotherapy**, including the fixed terminal ileum. A right colic arising from the ileocolic lets the ascending colon reach the pelvis.
- **Colon** is supplied by the superior mesenteric artery (SMA: ileocolic, right colic, middle colic) and inferior mesenteric artery (IMA: left colic, sigmoid and superior haemorrhoidal). The internal iliac gives the middle haemorrhoidal; all link through the **marginal artery of Drummond**.

Watershed	Site	Between
Griffiths point	Splenic flexure	Middle colic and ascending left colic
Sudeck's point	Rectosigmoid	Last sigmoid and superior haemorrhoidal

> **KEY POINT**
>
> Watershed zones get the smallest end branches of two arteries, so they go ischaemic first in hypotension or embolism; site anastomoses to one side.

2 · Segment selection

LAQ Intestinal segments in urology and orthotopic neobladder

Paper answer · P1 · P2 · merged from 2

Asked as: P2 LAQ "Use of intestinal segments in urology" / "Various uses of intestines in Uro" / "Principles of uses of bowel segments in urology. Describe and compare techniques of orthotopic neobladder." (also P2 SAQ "Neobladder") · P1 SAQ "Instestinal segments in urology"

Introduction

- Bowel is used to divert urine (conduit, continent pouch, neobladder), store urine, carry it to the skin, enlarge the bladder (augmentation), replace the ureter, make catheterisable channels and line a neovagina.
- Choice depends on renal function, previous surgery or irradiation, bowel disease, mesenteric length and the reconstruction needed.

Uses at a glance

Use	Segment and operation
Incontinent diversion	**Ileal conduit** (10–15 cm ileum, 10–15 cm from the ileocaecal valve); colon conduit (transverse) after pelvic radiation; jejunal conduit only if nothing else (avoid)

Use	Segment and operation
Continent cutaneous pouch	Indiana (right colon, plicated ileum as channel), Mainz I, Kock (ileal nipple valves); with appendix (Mitrofanoff) or tapered ileum channel
Orthotopic neobladder	Studer (40–44 cm ileum plus afferent limb), Hautmann (~70 cm ileum, W shape); about 50–70 cm detubularised ileum joined to the urethra
Rectal reservoir	Ureterosigmoidostomy, Mainz II: now rare (acidosis, cancer)
Augmentation cystoplasty	"Clam" ileocystoplasty (20–40 cm ileum; ~25 cm ileum), sigmoid, caecum; gastrocystoplasty in renal failure. For refractory neurogenic or small contracted bladder; needs CIC (clean intermittent catheterisation)
Ureteric replacement	**Ileal ureter**, isoperistaltic ileum, for long defects; Yang–Monti tubes
Catheterisable channels	**Mitrofanoff** appendicovesicostomy; **Yang–Monti** ileal tube (1–2 cm ileum retubularised transversely) when the appendix is absent; MACE (Malone antegrade continence enema) for bowel
Neovagina	**Sigmoid** preferred (wide lumen, mobile mesentery, less stenosis); ileum if pelvis irradiated; MRKH (Mayer–Rokitansky–Küster–Hauser), cloaca, gender-affirming vaginoplasty

- Ileal ureter contraindications: creatinine >2 mg/dL, bladder outlet obstruction, inflammatory bowel disease or radiation enteritis.

Choice of segment

Segment	Electrolyte disturbance	Advantages / use	Drawbacks
Stomach	Hypochloraemic, hypokalaemic metabolic alkalosis	Secretes H^+ and Cl^- (acid); less mucus; usable for renal failure, short gut, dense adhesions	**Haematuria–dysuria syndrome**, ulcers
Jejunum	Hyponatraemic, hypochloraemic, **hyperkalaemic acidosis**	Outside radiation field	Severe salt loss; avoid
Ileum	**Hyperchloraemic metabolic acidosis**	Default: mobile, compliant, constant (reliable) blood supply; default conduit, ureteric replacement	B12 and bile-salt loss; obstruction ~10%
Colon	Hyperchloraemic metabolic acidosis, hypokalaemia	Easy antireflux tunnel; transverse colon outside (spared by) the pelvic radiation field	Diarrhoea if ileocaecal valve used; obstruction ~5%
Appendix	—	Mitrofanoff catheterisable channel	—

Technical principles of using bowel

- **Preparation:** ERAS (enhanced recovery after surgery); routine mechanical bowel preparation not needed. Assess renal function. Mark the stoma site before surgery when a stoma is possible, through the rectus.
- **Mesentery:** transilluminate, preserve a major mesenteric arcade, wide base; clear ≤8 cm of mesentery from a segment end; avoid watershed areas (Griffiths point at splenic flexure, Sudeck point at rectosigmoid).
- **Bowel anastomosis:** well-vascularised, no tension, no spillage, close the mesenteric window.

- **Detubularisation** (Kock): opening the bowel on its antimesenteric border and refolding it **abolishes coordinated (peristaltic) contractions** and gives more volume for the same surface area; a **sphere** gives the largest volume for its surface area; by Laplace's law (pressure ∝ wall tension/radius) a larger radius means lower pressure, so pressure stays low.

Ureterointestinal anastomoses

Technique	Type	Stricture
Bricker	Refluxing, end to side	4–22% (mean 6%)
Wallace	Refluxing, ureters joined end to end	~3% (lowest)
Leadbetter–Clarke, Goodwin	Nonrefluxing (antireflux) tunnel in colon	8–14%

- Principles: spatulate, preserve periureteric tissue, tension-free, mucosa to mucosa, soft stents (leak 3–5% without). Bring the bowel to the ureter, not the ureter to the bowel.
- Refluxing (Bricker, Wallace; stricture about 3–6%) versus antireflux tunnels (more strictures).
- Fix and retroperitonealise the anastomosis. The left ureter passes behind the sigmoid mesentery; avoid angulation under the IMA (inferior mesenteric artery), a common stricture site.
- Reflux with normal unobstructed ureters has not been shown to harm the adult kidney; antireflux tunnels stricture more. Obstruction, not low-pressure reflux, damages kidneys.

Stomas

- **Protruding (nipple, "rosebud") stoma** for appliance wearers: less stenosis, better appliance fit. **Flush stoma** for continent pouches that are catheterised.
- **Loop-end (Turnbull) stoma** for obese patients with a short mesentery. Site is marked before surgery, through the rectus muscle.

Selecting conduit versus continent diversion

- **Renal function needed for continent diversion:** creatinine <2 mg/dL or GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; urine osmolality ≥600 mOsm/kg after water deprivation; minimal proteinuria; normal liver function; ability to self-catheterise.
- **Contraindications to continent diversion:** poor renal or hepatic function, inability or unwillingness to self-catheterise, inflammatory bowel disease or short gut. A neobladder also needs a tumour-free urethra and a competent sphincter.
- A conduit is simplest and fastest; continent options improve body image but carry more reoperations and metabolic problems.

Complications of bowel in the urinary tract

Metabolic (worse with longer segment, longer contact time, poorer renal function)

- **Hyperchloraemic acidosis (ileum, colon):** NH_4^+ (ammonium) replaces Na^+ on the Na^+/H^+ exchanger, coupled with Cl^-/HCO_3^- exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate (potassium citrate if hypokalaemic); chlorpromazine or nicotinic acid limit chloride transport.
- **Hypokalaemia:** worse with colon; replace potassium before correcting acidosis (paralysis risk).
- **Jejunal syndrome:** treat with sodium chloride and thiazide. **Gastric alkalosis:** H2 blockers or PPI (proton-pump inhibitor).

- **Altered sensorium:** hyperammonaemia (drain the pouch/diversion, lactulose), hypomagnesaemia.
- **Drug reabsorption:** phenytoin, methotrexate; drain pouches during chemotherapy.
- **Osteomalacia:** acidosis, vitamin D resistance, calcium loss; growth retardation in children.
- **Nutrition:** B12 deficiency (give parenteral B12 after 5–7 years of ileal diversion), bile-salt diarrhoea, short-gut effects.

Other

- Infection (bacteriuria in about three-quarters of conduits; treat only symptoms or urea-splitting organisms, or pure Proteus or Pseudomonas).
- Mucus.
- **Stones:** colon conduit 3–4%, ileal conduit 10–12%, continent (caecal) reservoirs ~20% (mucus, staples).
- **Augment perforation (pouch rupture):** surgical emergency; regular CIC prevents it.
- Ureterointestinal stricture (lifelong risk), renal deterioration, bowel obstruction, stomal stenosis and parastomal hernia.
- **Secondary malignancy:** ureterosigmoidostomy 6–29% after 10–20 years (endoscopy from 5 years); augments (endoscopy from 10 years). Neovagina: stenosis, mucus discharge, adenocarcinoma reported.

Orthotopic neobladder: definition and principles

- An **orthotopic** detubularised bowel (usually ileal) reservoir joined to the native urethra after radical cystectomy, when the urethra is tumour-free, to allow **volitional voiding** per urethra without a stoma. 80–90% of men and about 75% of women are medically eligible.
- **Principles:** 1. **Reliable outlet:** intact rhabdosphincter (pudendal nerve), healthy, tumour-free urethra. 2. **Low pressure:** complete **detubularisation** abolishes coordinated contractions; folding into a **sphere** gives most volume for surface area. 3. **Adequate capacity:** ≥500 mL when mature (44 cm double-folded ileum: <200 mL early, **500–600 mL** by 1 year). 4. Unobstructed upper tracts.

Neobladder: selection and contraindications

- **Absolute contraindications:** tumour at the **urethral margin** (positive urethral margin on frozen section), urethral stricture, poor renal function (eGFR [estimated glomerular filtration rate] <35–40 mL/min), severe liver disease, inability to self-catheterise.
- **Relative:** prostatic stromal invasion (men); bladder-neck or anterior vaginal tumour (women); poor sphincter; prior radiation (irradiated pelvis); frailty. Age, obesity and CIS (carcinoma in situ) alone are not bars.

Neobladder techniques compared

Design	Segment and shape	Ureters	Advantage	Drawback
Studer	40–44 cm ileum, double-folded U + afferent limb	Refluxing, isoperistaltic limb	Simple, no staples, short ureters; stricture <3%	Reflux on cystogram (benign)
Hautmann W	About 70 cm ileum, W	Refluxing (since 1997)	Largest early capacity, early continence	More late retention, more absorption
T pouch	44 cm + 15 cm T limb	Antireflux flap valve	Accepts dilated or short ureters	More reoperations than Studer

Design	Segment and shape	Ureters	Advantage	Drawback
Abol-Enein	40 cm ileum, W	Serous-lined tunnel	Durable antireflux	Needs long, non-dilated ureters
Camey II	65 cm ileum, U	Le Duc antireflux tunnel	Early detubularised design	Le Duc strictures 7–29%
Hemi-Kock	61 cm ileum	Intussuscepted nipple	Good early results	Nipple stenosis, staple stones
Padua (VIP)	About 42 cm ileum, funnel	Le Duc	Anatomical funnel; robotic	—
Mainz III	10–15 cm caecum + 20–30 cm ileum (ileocaecal)	Submucosal tunnel	Good compliance	Loses ileocaecal valve
Sigmoid (Reddy)	35 cm sigmoid colon, U	Tunnelled	Spares ileum (short bowel)	Poor compliance, urgency

- **Hautmann vs Studer:** larger initial capacity gives earlier continence but more late retention and absorption.
- **Studer steps (brief):** isolate ileum 15–20 cm proximal to the ileocaecal valve; open distal 40–44 cm antimesenterically; sew into a U and fold into a sphere; stented ureters end-to-side to the afferent limb; dependent point to urethra.
- **USC (University of Southern California) RCT (randomised controlled trial):** Studer vs T pouch, no difference in renal function at 3 years; T pouch needed more secondary procedures.

Neobladder outcomes

Outcome	Result
Day continence	80–90% (3–6 months)
Night incontinence	20–50% (no sphincter reflex, mucosal water secretion)
Retention needing CIC	Men 4–10%; women 20–30%: men leak, women fail to empty
Urethral recurrence (men)	About 7% at 5 years

Neobladder complications

- **Early:** urine leak, ileus, infection, mucus plugging.
- **Late: hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, ureteroileal stricture (<3%), stones, pouch rupture, incisional hernia, urethral recurrence.

Care and follow-up

- Neobladder: mucus irrigation, bicarbonate; voiding training (sit, relax, strain), timed voiding, night alarm. Annual metabolic panel, B12, upper tracts; urethral cytology if CIS.
- Lifelong for all bowel reconstructions: electrolytes, bicarbonate, renal function, B12, upper-tract imaging (ultrasound or CT) for stricture and stones; endoscopy of colonic or augmented segments as above.

Recent advances

- Robotic intracorporeal diversion (complications similar to open in experienced hands); robotic intracorporeal neobladder comparable to open; tissue-engineered bladder

(experimental) and alternatives to bowel (ureterocystoplasty, autoaugmentation) remain limited.

- **Also know:** neobladder uptake is low despite 80–90% of men being eligible.

Diagram to draw: ileal conduit (loop) with mesenteric arcade and Wallace anastomosis.

Diagram to draw: Studer U-pouch with afferent limb, ureters and urethral anastomosis.

Key points to remember:

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
- Wallace has the lowest stricture rate; detubularisation gives low pressure.
- Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
- Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
- Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

Also, from the condensed book (not in the paper answer):

- Losing **ileum** causes B12, bile-salt and fat malabsorption; losing the **ileocaecal valve** causes bacterial overgrowth, diarrhoea and bicarbonate loss. **B12** liver stores last **3–5 yr**, so give empirical parenteral B12 after 5–7 yr of ileal diversion.
- **Stomach** suits renal failure, short gut and dense adhesions, but causes the **haematuria–dysuria syndrome** (treat with H2 blockers). Use **jejunum** only if nothing else works.

3 · Bowel preparation and enhanced recovery

SAQ Methods of preoperative bowel preparation.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2022-Summer Paper 3, Q3d

Rationale

- Bowel used in urinary diversion is opened into a sterile urinary tract. Faecal spillage causes wound infection, abscess, leak and sepsis.
- Flora: aerobes (*E. coli*, *Enterococcus faecalis*); anaerobes (*Bacteroides*, *Clostridium*).

Segment	Organisms per gram
Jejunum	$10–10^5$
Distal ileum	$10^5–10^7$
Ascending colon	$10^6–10^8$
Descending colon	$10^{10}–10^{12}$

Methods

1. Dietary: clear liquids the day before; clear drinks until 2 hours before surgery.

2. Mechanical bowel preparation (MBP) reduces faecal **amount**, so total bacteria, not concentration.

- **PEG (polyethylene glycol)–electrolyte** lavage: non-absorbable; preferred in elderly, renal or heart failure, cirrhosis.
- Sodium phosphate, magnesium citrate; enemas.
- Historical: 3-day purgation, whole-gut irrigation by nasogastric tube (9–12 L saline), mannitol (feeds bacteria).
- Harms: dehydration, electrolyte loss; liquid stool may spill more.

3. Oral antibiotic preparation reduces bacterial **concentration**.

- **Neomycin + erythromycin base** (Nichols–Condon, 1977: septic complications 43% with MBP alone vs 9% with added antibiotics); neomycin + metronidazole; kanamycin.
- Harms: diarrhoea, *C. difficile* colitis, candidiasis; useless in obstruction.

4. Intravenous prophylaxis (most important): anaerobic plus aerobic cover (e.g. cephalosporin + metronidazole) within 1–2 hours before incision; it must be in tissue before contamination.

5. ERAS (enhanced recovery after surgery): no routine MBP; prehabilitation, multimodal opioid-sparing analgesia, chewing gum, **alvimopan** (peripheral μ-opioid antagonist, 12 mg before surgery), early feeding and mobilisation.

Evidence

- Colorectal meta-analyses and Cochrane: MBP does not reduce leak or abscess; may raise SSI (surgical site infection).
- Oral antibiotics added to MBP reduce SSI and leaks in large colectomy series.
- Small cystectomy trials: no difference without preparation.

Who still needs full preparation

- Salvage cystectomy after radiation, or cystectomy for benign disease (radiation cystitis, fistula, neurogenic bladder); immunosuppression.

Also know: wound protectors reduce SSI; enema before transrectal biopsy.

Diagram to draw: bacterial load along the gut.

> **Key points to remember:**
>
> - Mechanical lowers amount; antibiotics lower concentration.
> - Nichols–Condon: neomycin + erythromycin.
> - Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

Also, from the condensed book (not in the paper answer):

- The descending colon holds **10$^{10–12}$/g** bacteria; oral antibiotics cut this to ~10^2/g. The **IV antibiotic**, given 1–2 h before incision and covering anaerobes, matters most.
- **Alvimopan** (peripheral μ-opioid antagonist) is given as 12 mg 30 min–5 h pre-op, then twice daily, limited to 15 doses for inpatients.
- **Enhanced recovery after surgery (ERAS)** uses **no routine bowel prep**; postoperative ileus is the commonest cystectomy complication.
- Salvage or benign cystectomy (radiation, fistula, neurogenic) still needs full mechanical and antibiotic prep.

4 · Intestinal anastomoses

1. **Get exposure** and wall off the field; check the **blood supply** (transilluminate, bleeding edges, no tension).
2. **Prevent spillage** with non-crushing clamps, and irrigate the segment.
3. Join **serosa to serosa, inverted** and watertight, with **no strangulating sutures** and the **mesentery aligned**.

- Breakdown follows ischaemia, faecal sepsis, a **drain on the anastomosis** or irradiated bowel (most lethal). Sutured and stapled are equal. Leak presents late (2nd week); a neobladder–vaginal fistula causes continuous incontinence.
- **Obstruction** occurs in ~10% with ileum or stomach and ~5% with colon, mostly from adhesions. Prevent it with unirradiated bowel, closed mesenteric apertures, omentum and **reperitonealisation**.
- **Ogilvie syndrome** is a dilated caecum; exclude obstruction with a water-soluble enema. If the caecum is **>12 cm** or white count rises, give neostigmine or decompress by colonoscopy; operate for ischaemia or peritonitis.

5 · Ureterointestinal anastomoses

- Mobilise minimally, keep the periadventitia, spatulate, and sew mucosa to mucosa. **Bring the bowel to the ureter**, retroperitonealise, and use soft stents (they cut leak from 3–5% to near zero).

Technique	Type	Key feature	Stricture	Leak
Leadbetter–Clarke	Nonrefluxing	Colon; tunnel under taenia (2.5–3 cm)	8–14%	2.5%
Goodwin	Nonrefluxing	Transcolonic tunnel made from inside (3–4 cm)	No reliable data	
Bricker	Refluxing end-to-side	Any segment; simple	4–22% (mean 6%)	~3%
Wallace	Refluxing end-to-end	Ureters joined together (side-by-side, Y, head-to-tail)	**~3% (lowest)**	~2%

KEY POINT

Conduit pressure is damped by ureteric peristalsis. Reflux harms the kidney only with impaired peristalsis plus infection or obstruction. Antireflux tunnels stricture more and do not stop colonisation, so a stented refluxing anastomosis is the default; reflux also allows a loopogram.

- **Stricture** follows ischaemia, leak, radiation or infection, so surveillance is lifelong. The classic site is the **left ureter under the IMA**.
- Open reanastomosis is most durable; endoincision works in ~50–60% at 2 yr. Endourology fails more if the stricture appears <1 yr post-op, is **≥1.5 cm**, or is left-sided.

6 · Conduit diversion

- The **ileal conduit** is simplest, with fewest early complications; avoid it in short bowel, inflammatory small-bowel disease and irradiated ileum.

ALGORITHM

1. Take **10–15 cm** of ileum, 10–15 cm from the ileocaecal valve, keeping a major arcade.
2. Restore continuity by ileoileostomy; close the mesenteric window; flush the segment.
3. Pass the left ureter behind the sigmoid mesentery; make stented anastomoses.
4. Fix the base retroperitoneally in the right lower quadrant (RLQ); make the stoma.

- Intractable stomal bleeding suggests portal-hypertensive varices, treated by TIPS or embolisation.
- A **jejunal conduit** (15–25 cm from Treitz, left upper quadrant stoma) is used only if ileum and colon are unusable.

	Transverse	Sigmoid	Ileocaecal
Best for	**Extensive pelvic irradiation**; intestinal pyelostomy	Exenteration with colostomy	Long ureteric replacement
Supply	Middle colic	Sigmoid vessels	Ileocolic
Avoid	—	Sigmoid disease; ligated hypogastrics with rectum in situ; pelvic irradiation	—

- Avoid colon conduits in inflammatory bowel disease and chronic diarrhoea. An **ileal vesicostomy** suits neurogenic bladder and is reversible.
- Remove stents at days 4–6; drain retroperitoneally, **never intraperitoneally**.

Complication (%)	Ileal conduit	Colon conduit	Continent
Bowel obstruction	10	4	0.8
Ureterointestinal obstruction	8	Antireflux 20	4
Acidosis needing treatment	16	5	Ileum 8; colon 27
Renal deterioration	18	15	—

A gastric conduit causes no acidosis (0%).

7 · Stomas

- Mark sitting and supine; go **through the rectus** (lateral stomas herniate); stay ≥5 cm from the incision and avoid creases, scars and irradiated skin.
- **Keep the subcutaneous fat**, cut the fascia in a cross, and make the aperture admit two fingers.
- A **nipple (rosebud)** stoma suits an appliance (2–3 cm spout, less stenosis); a **flush** stoma suits a continent catheterisable channel.

	End stoma	**Loop (Turnbull)**
Stenosis	Higher	Lower
Parastomal hernia	1–4%	4–20%
Best for	Standard	**Obese** (thick, short mesentery)

- Parastomal hernia occurs in 2–6.6% of ileal conduits. Stomal stenosis occurs in 20–24% of ileal and 10–20% of colon conduits, and in >50% of paediatric catheterisable stomas.

8 · Renal function

- Renal deterioration after conduits is 10–74% (ileal conduit 18%), and ~6% die of renal failure. After ureterosigmoidostomy it is the leading cause of death beyond 15 yr.
- Continent diversions demand more of the kidney than conduits. **Tubular acidification matters as much as glomerular filtration rate (GFR).**

Test	**Threshold for continent diversion**
Creatinine and urine protein	**<2.0 mg/dL**, no proteinuria
GFR	**>35 mL/min**
Ammonium chloride load	Urine pH **≤5.8**
Water deprivation	Osmolality **≥600 mOsm/kg**

- **Bowel invalidates** urine osmolality and pH and serum urea and creatinine, because they are reabsorbed.

9 · Metabolic complications

Absorption depends on the **segment, surface area, dwell time, solute concentration, renal function and pH**.

Segment	**Na**	**K**	**Cl**	**pH**	**Associated**	**Treatment**
Stomach	—	Low	Low	High	Raised aldosterone; lethargy, seizures, arrhythmia	H2 blocker, PPI; arginine HCl; remove segment
Jejunum	Low	High	Low	Low	Raised renin, angiotensin; vomiting, dehydration	IV saline, bicarbonate, thiazide
Ileum/colon	—	Low	High	Low	Total-body K depletion; hypocalcaemia	Alkali; chlorpromazine, nicotinic acid

> **KEY POINT**
>
> Stomach secretes HCl (alkalosis, low Cl); jejunum secretes NaCl (low Na and Cl, high K); ileum and colon absorb NH4Cl (high Cl). **Only jejunum is hyperkalaemic**.

- **Gastric alkalosis** is usually trivial, but severe with renal failure or dehydration. Above a **gastrin** of **120 ng/L**, small gastrin rises cause large bicarbonate rises. Treat stepwise: rehydration and H2 blocker, arginine HCl if life-threatening, omeprazole, then segment removal.
- In the **jejunal syndrome**, NaCl loss causes hypovolaemia and activates renin–aldosterone, so the kidney keeps losing K-rich urine. It is worse with proximal segments, can be fatal, and needs long-term oral salt and thiazide.
- **Hyperchloraemic acidosis** is worst with **ureterosigmoidostomy** and colonic pouches; $NH4^+$ also enters via K^+ channels. Mucosal villous atrophy occurs, but transporters persist, so the risk persists.
- Alkali options are sodium bicarbonate, citrate solutions, or potassium citrate when K is needed. **Chlorpromazine 25–50 mg tds** or **nicotinic acid 400 mg 3–4×/day** inhibit cyclic AMP; they limit, but do not correct, acidosis.
- **Hypokalaemia** is commoner with colon. **Trap**: correcting acidosis alone causes profound hypokalaemia and flaccid paralysis, so give K with bicarbonate. Mg and Ca are also low.
- **Altered sensorium** is usually **ammonia** (liver dysfunction plus ureterosigmoidostomy). Treat by draining the diversion, oral neomycin, protein restriction, IV arginine glutamate and lactulose.
- **Drugs are reabsorbed**, notably **phenytoin, some antibiotics and methotrexate**, so **drain continent pouches during chemotherapy**.
- **Osteomalacia** is commonest with ureterosigmoidostomy, from acidosis, vitamin D resistance and renal Ca loss. Alkaline phosphatase is raised. Correcting a base deficit >2.5 mEq/L prevents bone loss; treat with alkali and calcium, then active vitamin D.
- **Infection**: ~3/4 of conduit urines are bacteriuric; **treat pure Proteus or Pseudomonas** and observe asymptomatic mixed cultures.
- **Stones** (struvite, Ca phosphate) follow acidosis, urease organisms, **staples** and mucus: colon conduit 3–4%, ileal conduit 10–12%, continent caecal reservoir ~20%.

Cancer

SAQ Malignancy in augmented bladder and after urinary diversion

Paper answer · P1 · P4 · merged from 2

Asked as: P4 SAQ "Secondary tumours after urinary diversion" · P1 SAQ "Cancer in augmented bladder"

Definition

- New cancers arising in or next to bowel used for urinary diversion or bladder augmentation, usually after a **long latency** (10–20 years or more; usually decades). Distinct from recurrent urothelial cancer.
- Bladder augmentation (enterocystoplasty) is mainly for neurogenic bladder. **Earliest reported tumour 4 years after surgery**; tumours after augmentation are often **aggressive and advanced at diagnosis**.

Risk by diversion (German registry, Kälble 2011)

Diversion	Risk	Surveillance
Ureterosigmoidostomy	2.6% (6–29% in long-term series); mean latency about 26 years	Regular colonoscopy **from 5 years**
Augmentation cystoplasty	1.6%	Endoscopy, often from 10 years (see surveillance below)
Colonic neobladder	1.3%	—
Ileocaecal pouch	0.14%	If symptomatic
Ileal neobladder	0.05%	No routine
Ileal conduit	0.02%	No routine

- **Pattern:** risk is highest where urothelium meets colon and is bathed in faeces; purely ileal diversions are lowest.

Risk factors

- **Gastrocystoplasty** (highest among augments), **exstrophy**, immunosuppression after transplant, smoking, chronic infection, long latency.
- The underlying disease may itself carry risk: bladder cancer in spina bifida is about 2–4%, mostly stage III or IV, and non-gastric augmentation did not appear to increase that risk.

Pathogenesis

- **Urothelium–colonic junction** exposed to urine and faeces; tumours arise at the ureterocolic anastomosis. Ureterosigmoidostomy is the extreme model: urothelium beside colonic mucosa bathed in faeces.
- **N-nitrosamines (N-nitroso compounds)** from bacterial action on urinary nitrates in stagnant, infected urine (probably enhancers, not sole cause).
- Chronic inflammation at the anastomosis releases growth factors (EGF, epidermal growth factor) that drive proliferation; ornithine decarboxylase; abnormal mucin; perianastomotic urothelium is genetically unstable.
- Removing the urothelium prevents tumours experimentally.
- **Defunctioned anastomoses still form cancer:** when converting a ureterosigmoidostomy, excise the old anastomosis.

Pathology

- Mostly **adenocarcinoma**, usually at the anastomosis (ureterocolic or **enterovesical**); also adenomatous polyps, urothelial carcinoma (aggressive, often metastatic), signet-ring and other types, sarcoma.
- **Nephrogenic adenoma:** benign; probably the commonest tumour after cystoplasty.

Clinical features

- **Gross haematuria**, rectal bleeding, change in bowel habit, ureteric obstruction (hydronephrosis), recurrent UTI (urinary tract infection), pelvic pain, new incontinence, mass.
- Neurogenic patients may notice symptoms late.

Surveillance and prevention

- **Ureterosigmoidostomy:** regular colonoscopy from year 5.

- **Augmentation:** chapters differ: Campbell bladder-surgery chapter advises **endoscopy yearly from 10 years**; paediatric chapter: no consensus, not shown cost-effective (Mayo group: unnecessary). Higher-risk groups benefit most.
- Investigate any haematuria promptly: cystoscopy with biopsy, cytology, CT (computed tomography) urography; biopsy and staging.
- **Prevention:** prefer ileal segments; counsel patients; regular clean intermittent catheterisation, daily bladder irrigation (less stasis, mucus and stones), treat urea-splitting infection, stop smoking, lifelong (transitional) follow-up.

Management

- **Ureterosigmoidostomy:** resection of the segment with the anastomosis; conversion to conduit or continent diversion.
- **Augmentation:** partial or total (radical) cystectomy with **excision of the augment**, lymphadenectomy and new urinary diversion (conduit or continent pouch).
- Chemotherapy as for histology; prognosis poor because of late stage.
- **Also know:** after cystectomy, recurrent urothelial cancer may arise in the upper tracts or a retained urethra (urethral recurrence about 7–9%); needs separate surveillance. Stones, perforation and metabolic acidosis also occur after augmentation.

Diagrams to draw: ureterosigmoid anastomosis with tumour at the urothelial–colonic junction; clam ileocystoplasty with tumour at the enterovesical suture line.

> **Key points to remember:**
> - Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
> - Mostly adenocarcinoma at the anastomosis; earliest at 4 years after augmentation.
> - Excise defunctioned anastomoses.
> - Gastric segment, exstrophy, immunosuppression: highest risk in augments.
> - Haematuria needs cystoscopy; treatment is cystectomy.

Also, from the condensed book (not in the paper answer):

- After **ureterosigmoidostomy**, cancer develops in 6–29% after **10–20 yr**. Defunctioned anastomoses still develop cancer, so **excise them, do not ligate**.

10 · Neuromechanical aspects

- Detubularising bowel into a sphere disrupts coordinated peristalsis and, by Laplace's law, increases volume and lowers pressure; this is the basis of reservoirs.

11 · Landmark trials

Trial	Design	Result	Take-home
Alvimopan (Lee 2014)	Randomised double-blind vs placebo, radical cystectomy	Bowel recovery 5.5 vs 6.8 days; stay 7.4 vs 10.1 days; less POI	Use with opioid-based analgesia

134 Continent Urinary Diversion

1 · Definition and basic principles

LAQ Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic

Paper answer · P2 · P4 · merged from 2

Asked as: P4 LAQ "Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion" · P2 LAQ "Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction."

Read as: principles of continence in continent diversion, orthotopic neobladder versus non-orthotopic (heterotopic) urinary diversion after radical cystectomy, and the Mitrofanoff principle in urinary and GI (gastrointestinal, bowel) reconstruction, i.e. antegrade continence enema channels.

Introduction, definition and types

- A continent diversion stores urine in a bowel reservoir and empties it **at will**: by voiding through the native urethra (**orthotopic neobladder**) or by CIC (clean intermittent catheterisation) of a stoma (**continent cutaneous pouch**).
- **Neobladder definition**: an **orthotopic** reservoir of detubularised bowel (usually ileum) joined to the native urethra after radical cystectomy, allowing **volitional voiding** without a stoma.
- Evolution: ileal conduit (Bricker, 1950), then continent cutaneous (Kock, 1982), then orthotopic (Camey, 1979; Hautmann, 1988; Studer, 1989).
- **Ileal conduit** (Bricker, 1950) remains the commonest diversion worldwide; **ureterosigmoidostomy** is the oldest continent diversion.
- AUA (American Urological Association)/SUO (Society of Urologic Oncology) and EAU (European Association of Urology): counsel every patient on all options. **80–90% of men and ~75% of women** are medically eligible for a neobladder, but far fewer receive one.

Group	Examples	How urine is held and emptied
Orthotopic	Studer, Hautmann W, T pouch, Padua, Mainz III	Native rhabdosphincter; voids by relaxing and straining
Incontinent cutaneous	Ileal conduit; transverse colon, sigmoid or jejunal conduit	Stoma and appliance
Continent cutaneous	Indiana pouch; right colon pouch with appendix (Mitrofanoff) or Yang–Monti channel; Kock (historical)	Self-catheterisation through a valved channel
Rectal	Ureterosigmoidostomy, Mainz II (sigma-rectum pouch)	Anal sphincter; urine mixed with stool

Principles of continence

- **Continence is a pressure balance:** low reservoir pressure must stay below adequate outlet resistance. Failure of either side causes leakage.

A. Low-pressure, adequate-capacity reservoir

- **Detubularisation** (opening bowel on its antimesenteric border) and cross-folding abolish coordinated peristaltic contractions.
- **Spherical shape:** greatest volume for a given surface area. By **Laplace's law** (P = 2T/R; wall tension = pressure × radius), a larger radius gives lower pressure for the same wall tension. Result: a low-pressure reservoir.
- **Capacity ≥500 mL** when mature. Bowel stretches with time (ileum more than colon); 44 cm of double-folded ileum holds <200 mL at first and 500–600 mL within a year.
- Ileum is most compliant; colon has pressure waves.

B. Reliable outlet

- **Orthotopic:** intact **rhabdosphincter** (pudendal nerve), healthy urethra, careful apical dissection, nerve sparing. Voiding is by Valsalva and pelvic floor relaxation. Night leakage (20–50%) reflects loss of the guarding reflex.
- **Continent cutaneous:** a stable efferent channel that is easy to catheterise and has a durable, dependable continence mechanism (see continent cutaneous mechanisms below).

C. Protect the upper tracts and metabolism

- **Unobstructed ureterointestinal anastomosis** to protect the kidneys.
- **Enough renal function** (eGFR, estimated glomerular filtration rate, about ≥35–40 mL/min) to clear the acid load reabsorbed by bowel mucosa.
- Ability to self-catheterise.

Studer as an example of the principles: orthotopic low-pressure ileal reservoir (**Studer, 1989**): detubularised ileal reservoir with a long **isoperistaltic afferent limb**, joined to the urethra. **Detubularisation** stops coordinated contractions, so pressure stays low; a **sphere** (Laplace) gives most volume for surface area. The **isoperistaltic afferent limb** limits reflux (damps reflux during Valsalva voiding) and accepts **short ureters**. No staples in the reservoir. Continence depends on the **rhabdosphincter**.

Choice of bowel segment

Segment	Points
Ileum (default)	Most compliant, lowest pressure; mesentery reaches the urethra; mucosa atrophies, so absorption falls
Ileum + caecum (Mainz)	Near-ileal compliance; loses ileocaecal valve
Colon (right, sigmoid)	Less distensible, pressure waves, so more night leakage; used when ileum is short or diseased
Stomach	Secretes acid: helps in renal impairment; haematuria–dysuria; rarely used

Reflux versus obstruction

- Voiding by Valsalva raises pressure equally throughout, so reflux into a low-pressure pouch is benign, while **every antireflux valve can obstruct** (Le Duc tunnel stricture 7–29%; late Kock nipple stenosis).
- Direct refluxing anastomosis (Wallace, Leadbetter): stricture about 3–6%.
- **USC (University of Southern California) RCT (randomised controlled trial)**, 484 patients: Studer vs T pouch, no difference in renal function or infection at 3 years; the antireflux T pouch gave no renal benefit and needed more secondary procedures (reoperations).

- Studer: reflux on cystogram, yet kidneys preserved in about 95%.

Continent cutaneous mechanisms

Mechanism	Example	How it works	Weakness
Nipple valve (intussuscepted)	Kock (historical), Mainz I	Reservoir pressure compresses the intussuscepted segment	Same pressure effaces it; failure ≥15%, reoperation about one-third; late nipple stenosis
Flap valve: narrow tube (appendix) tunnelled submucosally or in taenia	**Mitrofanoff** (Penn, right colon pouch), Yang–Monti ileal tube if no appendix, Kropp	Narrow tube in a submucosal tunnel is coapted as pressure rises	No pop-off; stomal stenosis
Imbricated ileocaecal valve + tapered ileum	**Indiana** pouch	Continence zone lies at the plicated valve	Bowel frequency, B12 loss
- Used when continence is wanted but the urethra is unusable; umbilical stoma preferred.			
- **Also know** (native outlet): Kropp, Pippi Salle, slings, AUS (artificial urinary sphincter), bulking, bladder neck closure.			

The Mitrofanoff principle

- **Mitrofanoff (1980):** any supple, narrow tube implanted in a **submucosal tunnel** of a low-pressure reservoir, with its other end at skin, gives a **continent catheterisable channel**.
- As the reservoir fills, pressure transmitted through the mucosa flattens the tunnelled segment against the muscle backing: a **flap valve**. Continence lies in the tunnel, not in the conduit.
- Appendix: only a **2 cm tunnel** suffices because of its small calibre.

Conduits

Conduit	Points
Appendix (first choice)	Natural tube, minimal harvest morbidity; needs length ≥5–6 cm and ≥12 Fr; caecal cuff or **Mitchell caecal extension** adds length
Yang–Monti	2–3 cm ileum opened on the antimesenteric border and retubularised transversely over 12 Fr; double or spiral Monti for obese; twice the revisions
Tapered ileum	Long enough; kinks if redundant
Ureter, fallopian tube	More stenosis
Casale bladder flap	No bowel; consumes bladder capacity

Technique (appendicovesicostomy)

1. Mobilise caecum; detach appendix with a caecal cuff, preserving mesoappendix.

2. Spatulate tip; tunnel submucosally (posterolateral or anterior bladder wall), intra- or extravesically.
3. Bring the base to the umbilicus or right iliac fossa with skin flaps, without tension or twist; keep it short and straight; fix the bladder to the abdominal wall.
4. Catheterise after each step. Channel catheter 2–4 weeks; then CIC every 4 hours by day.

Mitrofanoff applications in urinary reconstruction

- **Neurogenic bladder** (spina bifida, spinal injury), often with augmentation, when urethral CIC is difficult (wheelchair-bound, sensate urethra).
- After bladder neck reconstruction or closure; exstrophy; valve bladder.
- **Continent cutaneous diversion** after cystectomy: Penn and right colon pouch (appendix in taenia; up to 100% continence); Yang–Monti if no appendix.
- Unusable urethra: complex stricture, urethrectomy.

Mitrofanoff applications in GI reconstruction: MACE

- **MACE (Malone antegrade continence enema; Malone, 1990):** appendix brought to skin as an appendicocaecostomy with an antireflux tunnel (flap valve) so stool does not leak.
- Antegrade washouts empty the colon, giving **faecal continence** in neurogenic bowel (spina bifida), anorectal malformations, cloaca and refractory constipation.
- Options: Monti-MACE; **split appendix** for both MACE and urinary channel; caecostomy button. Yang–Monti for the urinary channel when the appendix is absent or used for MACE.

Results and complications of continent catheterisable channels

- **Continence >95%** for flap valves; nipple valves about 85%.
- **Stomal stenosis is the commonest problem**; also false passage, channel polyps or granulation, pouch stones, revision (suprafascial or subfascial).
- Missed CIC: no pop-off, so **reservoir rupture** or upper tract damage.

Patient selection

	Orthotopic only	All continent diversions
Absolute contraindications	Tumour at the urethral margin (frozen section); significant urethral stricture	Severe renal (eGFR <40 mL/min) or hepatic dysfunction (liver failure); physical or mental inability to self-catheterise
Relative	Poor external sphincter; bladder-neck or anterior vaginal wall tumour (women); prostatic stromal invasion (men); irradiated sphincter (irradiated pelvis)	Frailty, poor cognition, social problems, prior radiation or bowel resection

- **Renal function**: creatinine <1.7–2.2 mg/dL or eGFR **>35–40 mL/min** (below about 35–40 mL/min = severe renal impairment; another source: <40 mL/min), because the kidney must excrete acid reabsorbed from stored urine. Formal tests: urine pH ≤5.8 after ammonium chloride, osmolality ≥600 mOsm/kg after water deprivation.
- **Urethral recurrence**: in men the strongest predictor is prostatic stromal invasion (prostatic urethral tumour raises urethral recurrence risk); in women, bladder-neck or anterior vaginal wall tumour. **Intraoperative frozen section** of the urethral margin decides. CIS (carcinoma in situ) or multifocality alone do not exclude a neobladder.

- **Age and obesity alone are not contraindications**; comorbidity, cognition and support decide. A conduit suits the frail or unsupported patient.
- Continent cutaneous pouch: continence wanted but urethra unusable.
- Mark a conduit stoma site in everyone: up to ~10% change plan intraoperatively.

Orthotopic versus non-orthotopic: comparison

Feature	Orthotopic neobladder	Ileal conduit	Continent cutaneous	Ureterosigmoidostomy
Appliance	None	Bag	None; small stoma	None
Body image	Best	Worst	Good	Good
Complexity	High	Lowest	Highest	Moderate
Continence	Day 80–90%; night incontinence 20–50%	Not applicable	High (Indiana, Mitrofanoff)	Depends on anal sphincter
Catheterisation	10–20% of men, ~30% of women	None	Always	None
Bowel used	Long ileal segment (Studer 54–56 cm)	10–15 cm ileum	Right colon + terminal ileum	None isolated
Typical late problems	Night leakage, retention, acidosis	Stomal stenosis (20–24%), parastomal hernia, renal deterioration	Stomal stenosis, stones, difficult catheterisation	Severe acidosis, pyelonephritis, cancer at the anastomosis

Neobladder types

Design	Segment	Ureters
Studer	40–44 cm ileum + afferent limb	Refluxing, isoperistaltic limb
Hautmann W	About 70 cm ileum	Refluxing; largest capacity
T pouch	44 cm + T limb	Antireflux flap valve
Camey II; Padua	Ileum	Antireflux (Le Duc)
Mainz III; sigmoid	Ileocaecal; colon	Tunnelled

	Studer	Hautmann
Ileum	54–56 cm, U + afferent limb	~70 cm folded in a W
Initial capacity	Smaller early capacity	Larger: earlier continence
Late	Less late retention	More retention, reabsorption

Studer technique (operative)

- Brief: isolate ileum 15–25 cm from the ileocaecal valve; open the distal part on the antimesenteric border, sew into a U, fold into a sphere; end-to-side ureteric anastomoses to the afferent limb; anastomose the dependent point to the urethra. 1. **Isolate 54–56 cm of terminal ileum**, ending **15–25 cm** above the ICV (ileocaecal valve) (another source: **15–20**

cm proximal to the ICV); restore bowel continuity. 2. **Reservoir: distal 40–44 cm** (two 20–22-cm limbs) opened on the antimesenteric border; sew the two limbs (medial edges) into a **U** (back wall). 3. **Afferent limb:** leave the proximal **10–15 cm** tubular as the afferent limb (another source: **about 15 cm**; original 20 cm; 10–12 cm in later figures); close its end. 4. **Fold the U in the opposite direction** (across) and close the anterior wall into a **sphere**. 5. **Ureters:** spatulated **end-to-side, refluxing** ureteroileal anastomoses to the afferent limb, interrupted 4-0, **stented**. 6. **Urethral anastomosis** at the most dependent point with pre-placed sutures; catheter and suprapubic tube.

- Capacity rises from <200 to **500–600 mL** in a year.
- **Intracorporeal robotic Studer** follows the same steps.

Postoperative care, voiding training and follow-up

- Mucus irrigation; oral bicarbonate; cystogram before catheter removal.
- **Voiding training**: void **sitting**, relax the pelvic floor, strain gently; **timed voiding** every 2 hours, extended to 3–4 hours as capacity grows (under 200 mL to 500–600 mL in a year). **Night alarm.**
- Lifelong follow-up: annual metabolic panel, B12, upper-tract imaging; urethral wash cytology if CIS (carcinoma in situ).

Current status of orthotopic neobladder

1. Preferred where suitable: closest to a native bladder. AUA/SUO and EAU: counsel every cystectomy patient on all options and refer to an experienced surgeon.

2. Under-used. 80–90% of men and about 75% of women are eligible, yet only about 9% of US patients receive any continent diversion, and this is falling; continent diversion in RAZOR 21%, iROC 11%. Causes: comorbid patients, surgeon training, intracorporeal robotics.

3. Designs in use. Studer (40–44 cm U plus isoperistaltic afferent limb) and **Hautmann W** (about 70 cm; refluxing since 1997) dominate; T pouch or serous-lined tunnel when antireflux is wanted.

4. Oncological safety.

- Men: urethral recurrence about 7% at 5 years; main risk is prostatic stromal involvement. The neobladder seems protective (meta-analysis HR (hazard ratio) 0.44).
- Women: risk lies with bladder-neck tumour; no skip lesions, so preserve the urethra. **Frozen section** of the urethral margin decides in both sexes.
- Local recurrence <10%, rarely disabling the pouch; nodal disease is not a bar. Selected **post-radiation** patients do well.

5. Function (outcomes)

Outcome	Result
Day continence	80–90% (most by 3–6 months); another source (Studer): 90–98%; about 90%
Night incontinence	20–50% (up to half leak): no sphincter (guarding) reflex, mucosal water secretion, nocturnal diuresis
Retention needing CIC	Men 4–10%; **women 20–30%** (pouch angulation): men leak, women retain
Self-catheterisation	**10–20% of men, ~30% of women**

6. Complications. Early morbidity equals conduit. Late: hyperchloraemic metabolic acidosis, vitamin B12 deficiency, mucus, ureteroileal stricture, stones, pouch rupture, pouch–vaginal fistula (5–10%), hernia.

7. Robotic intracorporeal neobladder: comparable to open; long-term data lacking.

8. Quality of life: no diversion proven superior; body image favours neobladder.

- **Also know:** nerve sparing and uterine preservation aid continence.

Metabolic complications (bowel in the urinary tract)

Segment	Disturbance
Ileum, colon	**Hyperchloraemic metabolic acidosis** (ammonium chloride absorbed); hypokalaemia, worse with colon
Jejunum	Hyponatraemic, hypochloraemic, hyperkalaemic acidosis
Stomach	Hypochloraemic, hypokalaemic alkalosis

- Acidosis is severe in most ureterosigmoidostomies; most colonic (Mainz) pouches need alkali. Treat with sodium bicarbonate or citrate, and replace potassium as well.
- **Also know:** osteomalacia; vitamin B12 deficiency and bile-salt diarrhoea after ileal loss (liver B12 stores last 3–5 years); stones (ileal conduit 10–12%, continent caecal reservoirs ~20%); bacteriuria; drug reabsorption (methotrexate, phenytoin); ammoniagenic encephalopathy; growth delay in children.
- **Secondary cancer**: ureterosigmoidostomy 6–29% after 10–20 years (yearly endoscopy from 5 years); ileal conduit and ileal neobladder lowest.

Surgical complications

- **Early (≤90 days)**: ileus, urine leak (long pouch suture lines; drain percutaneously), infection, mucus retention; not more than with a conduit.
- **Late**: ureteroileal stricture ~3–6% (Wallace or direct end-to-side) (Studer/neobladder: **<3%**); neobladder retention, urethral stricture, rare pouch rupture; ventral and parastomal hernia; **hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, stones, pyelonephritis (commonest late, Studer), urethral recurrence.
- Continent cutaneous channels: stomal stenosis commonest (see results above).

Quality of life

- No randomised trial. Body image favours the neobladder, urinary function the conduit; no diversion is superior on HRQoL (health-related quality of life).

Recent advances

- RARC (robot-assisted radical cystectomy): **RAZOR (2018)** non-inferior 2-year progression-free survival (open diversion in both arms); **iROC**: intracorporeal diversion gave more days alive and out of hospital at 90 days.
- Intracorporeal neobladders and nerve- or organ-sparing cystectomy are increasing.

Diagrams to draw:

- Studer neobladder and ileal conduit side by side; Studer U-pouch with afferent limb, ureters and urethral anastomosis; ileal segment marked (afferent limb plus 40–44-cm U), then the folded spherical pouch with afferent limb.
- Flap valve (tunnelled appendix collapsing as pressure rises) beside a nipple valve.

Key points to remember:

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL; 500–600 mL by 1 year) against a reliable outlet.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.
- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel; Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Flap-valve continence >95%; stomal stenosis is the commonest complication.
- No diversion is proven best for quality of life; counsel all options.

Also, from the condensed book (not in the paper answer):

- Orthotopic is now preferred (natural voiding, no appliance, fewer revisions); cutaneous suits pre-existing incontinence or urethral disease.

KEY POINT

Continence is a pressure balance: low reservoir pressure (from detubularisation) against adequate outlet resistance (sphincter or efferent valve). If either fails, the patient leaks. Camey I (tubular ileum) failed because its pressure was too high.

2 · Patient selection

Selection means excluding a bad urethra, kidneys or hands.

*See the paper answer **Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic** (Section 12, Ch 134).*

Also, from the condensed book (not in the paper answer):

- **Locally advanced or node-positive (N+) disease found at surgery is not a contraindication**: pelvic recurrence is ~7% (13% if extravesical or N+) and rarely disables the neobladder. The exception is a grossly positive margin at pubis, rectum or side wall.
- eGFR 40–59 is safe, as decline is no faster than with a conduit.
- A short or fatty mesentery is the commonest technical reason to abandon orthotopic diversion.

ALGORITHM

1. Estimated glomerular filtration rate (eGFR) >40.
2. Intact external sphincter and adequate bowel.
3. Not frail; no severe liver, cardiac or psychological disease.
4. Men: no distal prostatic urethral cancer. Women: no palpable anterior vaginal tumour or cervical invasion.
5. If all are met and the patient will self-catheterise, offer an orthotopic neobladder.

- Mark a conduit site in everyone, through mid-rectus and away from creases, umbilicus and scars.

Urethral recurrence

Feature	Men	Women
Dominant risk	Prostatic urethral involvement; worst with **stromal invasion** (usually N+, so give neoadjuvant chemotherapy)	**Bladder-neck tumour**; anterior vaginal wall invasion
Pre-op	Transurethral resection (TUR) biopsy at 5 and 7 o'clock beside the verumontanum — optional	Examination under anaesthesia (EUA); bladder-neck biopsy
Definitive test	Intraoperative **frozen section of urethral margin** (more predictive than TUR biopsy; the American Urological Association (AUA) requires a negative margin)	
Bladder carcinoma in situ (CIS)	Not predictive	
Skip lesions	—	None: a negative bladder neck means a negative urethra

- Risk rises with depth: none < CIS/mucosal/ductal < stromal. **The neobladder itself is protective** (hazard ratio (HR) 0.44).
- Prostatic involvement needs urethral surveillance (wash cytology, urethroscopy).
- In women, keep the urethra if bladder-neck biopsy and proximal urethral frozen section are negative.

Patient-related factors

- **Renal function** is the key contraindication. Proceed if creatinine <**1.7–2.2 mg/dL** or eGFR >**35–40 mL/min**, with hydration and oral bicarbonate and no nephrotoxins.
- **Self-catheterisation** is needed by 10–20% of men and ~30% of women with a neobladder, and by all with cutaneous pouches. The need may appear years later, so all must be willing; multiple sclerosis, quadriplegia or frailty favour a conduit.
- **Sphincter weakness** is managed with an artificial urinary sphincter (AUS) in men, and Burch or sling in women (who will likely catheterise).
- **Prior radiation** is acceptable without incontinence, stricture, fistula or severe bowel damage; the final decision is made intraoperatively.
- After **prior prostatectomy**, pre-op continence predicts outcome. After **prior bowel resection** with short bowel, use a sigmoid neobladder.

3 · Preserving continence at cystectomy

The **rhabdosphincter** (pudendal nerve) is critical. It lies **anterior and lateral** to the urethra in both sexes.

Men

- Never dissect posteriorly beyond the apex. Incise endopelvic fascia and puboprostatic ligaments minimally; control the dorsal venous complex (DVC) with **absorbable** suture, as clips or staples erode into the anastomosis.

ALGORITHM

1. Open the endopelvic fascia; incise the puboprostatic ligaments slightly.
2. Ligate the DVC close to the apex.
3. Incise the anterior urethra and place sutures.
4. Place posterior sutures through the rectourethralis, then divide the posterior urethra.
5. Tag the sutures until the ureteroenteric anastomosis is done.

- Place sutures **before** full division, as the male urethra retracts. Apical frozen section with tumour means urethrectomy and cutaneous diversion or conduit; atypia or focal CIS means a second distal margin.
- Nerve sparing has a controversial continence benefit; it may help night continence by keeping urethral sensation.

Women

- Sparing uterus and anterior vagina in selected low-stage disease avoids fistula, improves sexual function and reduces retention. A palpable bladder-neck or vaginal tumour needs en bloc urethrectomy and cutaneous diversion.

ALGORITHM

1. Divide the posterior pedicles and develop the plane off the anterior vagina.
2. Use the Foley balloon to mark the bladder–urethra junction.
3. Free the bladder neck and the first 0.5 cm of urethra.
4. Clamp the bladder neck, divide just distal to it, and send a frozen section.

- Leave endopelvic fascia untouched; the bladder neck is not needed for female continence.
- Always interpose an **omental flap** between vagina and pouch, and keep vaginal and urethral suture lines apart.

4 · Choice of bowel and ureteric reflux

Ileum gives lowest pressure; obstruction, not reflux, loses kidneys.

- **Ileum is the default**, being most distensible; its mucosa atrophies, lowering mucus and absorption. Colon has pressure waves; stomach has poor compliance and haematuria–dysuria. Close with absorbable suture, as staples cause stones.
- **Antireflux is unnecessary**: Valsalva raises pressure equally throughout, and every valve can obstruct. A direct (Wallace or Leadbetter) anastomosis strictures in **3–6%**, against 7–29% for Le Duc. **Obstruction, not reflux, destroys kidneys.** Kock nipple failures are late (>10 yr) and silent.

5 · Orthotopic neobladder designs

*See the paper answer **Intestinal segments in urology and orthotopic neobladder** (Section 12, Ch 133).*

Also, from the condensed book (not in the paper answer):

- Hautmann became refluxing in 1997, cutting stenosis from 9.5% to 1%. Studer is the most popular design; the sigmoid neobladder has the worst night continence.

ALGORITHM

1. **Studer:**
2. Isolate ileum ending 15–20 cm proximal to the ileocaecal valve; keep a 40–44 cm reservoir (two 20–22 cm limbs) and an afferent limb.
3. Open the reservoir, oversew the back wall, and fold it the opposite way.
4. Make stented end-to-side ureteroileal anastomoses.
5. Join the dependent point to the urethra.

KEY POINT

Hautmann vs Studer is a capacity trade-off: a bigger initial pouch is continent sooner but empties worse and absorbs more later.

6 · Continent cutaneous reservoirs

*See the paper answer **Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic** (Section 12, Ch 134).*

Also, from the condensed book (not in the paper answer):

- **Indiana** uses the right colon and ~10 cm of terminal ileum, with a double-imbricated ileocaecal valve and tapered ileum; the whole colon is opened to abolish caecal peristalsis.
- An **umbilical stoma** has less stenosis and is easiest for paraplegics; the appendix takes only 14–16 Fr catheters.

7 · Perioperative care

- **Enhanced recovery after surgery (ERAS)** means prehabilitation, carbohydrate loading, restrictive fluids, no nasogastric (NG) tube, early diet and opioid-sparing analgesia. **Alvimopan** cut ileus from ~23% to ~7%; ileus is the commonest gastrointestinal complication.
- **Drainage**: single-J stents and a 24 Fr urethral catheter irrigated for mucus; remove the catheter at **3 weeks**. Then use timed voids, lengthened progressively, with pelvic floor exercises.

8 · Results and complications

Men tend to leak, women to retain.

Early (<90 days) morbidity is no higher than with an ileal conduit.

Complication	Key facts	Management
Urine leak	Commoner (long suture lines)	Catheter, stents; if undrained, percutaneous drain with or without nephrostomies; **avoid early open repair**
Bacteriuria	Very common	Treat only if symptomatic or Proteus/Pseudomonas; in febrile urinary tract infection (UTI), look for obstruction, stones, residual; if recurrent, nitrofurantoin or methenamine
Ureteroileal stricture	Can appear after 10 yr, often silent	Suspect on CT or ultrasound (US); confirm by pouchogram or antegrade study; endoscopy (retreatment >50%) inferior to open
Pouch stones	Kock and stapled pouches	Endoscopic
Pouch perforation	Rare; radiation, chronic retention	Cystogram; catheter with or without drains if stable
Pouch–vaginal fistula	5–10%; vaginal excision, radiation	Lateral cystogram before catheter removal; beyond the first weeks it won't heal, so repair transvaginally or transabdominally, or convert to cutaneous
Stomal stenosis (cutaneous)	Up to ~20%; after 90 days, radiation	Usually open revision
Efferent limb necrosis	Rare, ischaemic	Endoscopy for level; explore if below fascia
Incisional hernia	Common; impairs emptying	Repair

Continence

- Day continence comes over **3–6 months**, ultimately **80–90%**. Night continence lags into year 2, and nocturnal incontinence persists in **20–50%**. Risks are age >65, colon, diabetes, no nerve sparing and hysterectomy.
- Night leakage reflects loss of the sphincter–detrusor reflex; advise voiding at bedtime and an alarm 2–3 times a night.
- Evaluate persistent leakage after the 12–18-month plateau, excluding fistula and using urodynamics.

Stress incontinence	Men	Women
Bulking agent	Marginal	Mild only; <50%, poorly durable
Definitive	AUS or male sling	Pubovaginal sling — most then catheterise
Safety rule		No blind retropubic needle passage; use bone anchors or the prepubic route

Retention

- It presents as infection, overflow, palpable pouch or new hydronephrosis. It affects men in 4–10% (~21% by 10 yr) and **women in 20–30%**.
- In women the cause is **mechanical**: the pouch falls back and angulates the pouch–urethral junction. Prevent with omentum behind the pouch, vaginal suspension, sacrocolpopexy and uterine preservation.

- Treat with clean intermittent self-catheterisation (CISC), as drugs are useless; exclude stricture or recurrence.

> **KEY POINT**
>
> Men fail by **leaking** (sphincter without reflexes), so augment the sphincter. Women fail by **not emptying** (unsupported pouch kinks), so support the pouch.

9 · Follow-up

- Stricture occurs in the first 4 months, recurrence from 4 months to 3 yr, and stones, late obstruction and urethral or upper-tract tumour after 3 yr.
- Review 4-monthly in year 1, 6-monthly to 3 yr, then annually (examination, metabolic panel, blood count), with CT and chest imaging to 5 yr.
- Check B12 yearly. Do urethral wash cytology in men, and bimanual examination and voided cytology in women. After 5 yr continue lifelong upper-tract US, metabolic panel and blood count.
- No diversion has proven superior quality of life; the neobladder gives better body image and the conduit better urinary function.

10 · Landmark trials

Trial	Design	Result	Take-home
USC Studer vs T pouch	Randomised, 484 radical cystectomy (RC) patients	No difference in renal function or infection at 3 yr; T pouch more secondary procedures	Antireflux adds cost without benefit
RAZOR	Open vs robotic RC, extracorporeal diversion	Continent diversion only 21%	Low uptake even with expert surgeons
iROC	Open vs robotic RC, intracorporeal diversion	Continent diversion 11%	Intracorporeal approach further reduces continent diversion

135 Minimally Invasive Urinary Diversion

1 · Context

- Robot-assisted radical cystectomy (RARC) equals open oncologically in trials (RAZOR), but with **extracorporeal** diversion (ECUD); intracorporeal diversion (ICUD) is untested.
- In the International Robotic Cystectomy Consortium (IRCC), ICUD rose ≈**11%/year**, driven by the **ileal conduit**, the commonest robotic diversion.

2 · Patient selection

A bar to one step is not a bar to all.

Obesity, age, advanced disease and prior surgery no longer exclude.

Decision	Contraindication
Minimally invasive route	Poor cardiopulmonary tolerance of steep Trendelenburg (relative; eased by Xi and AirSeal)
Use of ileum	Short bowel; severe inflammatory bowel disease; radiation enteritis
Neobladder (absolute)	Urethral involvement distal to prostate; creatinine **>2 mg/dL**; poor hepatic function (as for open)

ALGORITHM

1. Pasadena Consensus learning pathway:
2. RARC with extracorporeal diversion.
3. Intracorporeal conduit (≈30–100 cases).
4. Continent diversion (≈25–50 cases). Target time: **≤4 h conduit, ≤5 h neobladder**.

3 · Perioperative bundle

Enhanced recovery (ERAS) uses alvimopan, goal-directed fluids, immediate oral diet and opioid-sparing analgesia, giving less ileus and a shorter stay. Caution: a stay **<5 days** about doubles 90-day major complications.

Element	Recommendation
Fasting	Solids to 6 h, clear fluids to 2 h
Bowel preparation	**None**, and no oral antibiotics (electrolyte upset, liquid spillage, no benefit)
Thromboprophylaxis	Pneumatic compression and stockings; low-molecular-weight heparin before and up to **4 weeks** after
Antibiotics	Broad-spectrum intravenous within **1 h**

4 · Position and ports

- Use lithotomy and steep Trendelenburg, with Hasson entry after prior surgery.
- **Six to seven ports sit more cephalad** to reach over small bowel and take nodes to the aortic bifurcation.
- A neobladder adds a left stapling port; a suprapubic 12-mm port serves bowel reanastomosis and extends for specimen extraction.

5 · Nerve sparing

One plexus serves sexual function and neobladder continence.

	Male	Female
Danger points	Presacral and medial common iliac; high pedicle ligation; seminal vesicles and Denonvilliers fascia; prostatic pedicles and apex	Lateral vaginal wall and bladder base (inferior hypogastric plexus)
Manoeuvre	Athermal, intrafascial, traction-free pedicle control	Preserve tissue around proximal urethra and anterior vaginal wall; extract via Pfannenstiel, not transvaginally
Sphincter	One-third overlapped by dorsal venous complex; dorsal penile nerve branches within **3 mm** of apex	Mainly mid–distal third, so continence survives loss of bladder neck

6 · Intracorporeal ileal conduit

ALGORITHM

1. Pass the left ureter under the sigmoid mesentery to the right.
2. Choose 12 cm ileum, 15–20 cm from the ileocaecal valve; place the marionette stitch.
3. Isolate with a 45-mm stapler.
4. Do ureteroileal anastomoses (left first) and stent.
5. Reanastomose bowel, then form the stoma.

- The **marionette stitch** (1-0 silk through the stoma site and conduit end) is held **untied**; it replaces the missing hand and positions the conduit.
- Avoid a redundant segment (See Ch 133), and tag cut ends to prevent malrotation.
- The **enterotomy** is an athermal cold-scissors buttonhole, because thermal slits gave more strictures.
- Single-J stents are used, and the **guidewire comes out only after anchoring**. Use a Bricker (separate end-to-side) or Wallace (Y, one anastomosis) technique.

7 · Intracorporeal neobladder

The key change is **early neobladder–urethral anastomosis** (Karolinska): dependent ileum is anchored to the urethra first, then the pouch is built in place. Design principles: See Ch 134.

Design	Ileum	Signature
Karolinska (modified Studer)	50 cm	10-cm afferent limb; Penrose sling
USC (Studer)	65 cm (longest)	44+16 chimney+5 discarded
UCLH Pyramid	50 cm	Koie-derived cross-fold
Saint-Augustin modified Y	40 cm (shortest)	Refluxing, uncrossed ureters
Padua	42 cm	Only stapled pouch; split-nipple antireflux
FloRIN	50 cm	Asymmetrical U; L-shaped neo-trigone; antireflux
Roswell Park (Hautmann W)	45–50 cm	One trough at a time; chimneys

ALGORITHM

1. Urethral anastomosis (Van Velthoven barbed suture).
2. Detubularise distal 40 cm (cold scissors); form the posterior plate.
3. Join ureters to afferent limb or chimney, stented.
4. Close the anterior wall once the 22-Fr catheter is confirmed.

- A **tension-free urethral anastomosis** lacks tactile feedback. It is helped by urethral length, less Trendelenburg, perineal pressure and Penrose traction. Incising mesenteric peritoneum risks pouch ischaemia.

Intracorporeal Indiana pouch

ALGORITHM

1. Mobilise the right colon.
2. Divide ileum 15 cm from the ileocaecal junction and transverse colon proximal to the middle colic; restore continuity.
3. Detubularise along the anterior taenia and fold.
4. Bring ureters through the posterior plate on double-J; place a caecostomy.
5. Taper ileum over 14 Fr, imbricate the valve, and form an umbilical stoma.

> **KEY POINT**
>
> Capacity comes from detubularised right colon; continence from tapered ileum and the imbricated ileocaecal valve.

8 · Postoperative care

Avoid overhydration; there is no routine nasogastric tube (Cochrane). Irrigate the neobladder **8-hourly** for mucus.

Tube	Removal
Urethral Foley as pelvic drain (conduit)	Day 1
Neobladder catheter	**3–4 weeks** , after leak-free pouchogram
Externalised stents / double-J	≈1 week / cystoscopically at 3–4 weeks
Feeding-tube stents tied to Foley	With the catheter

9 · Outcomes

Most problems come from the diversion, and many appear late.

- **Operative time** depends most on diversion type (conduit ≈70 min shorter); ICUD and ECUD take similar time. Proficiency needs ≥**66 RARCs**.
- **Complications** are mostly diversion-related, infection commonest. Two-thirds of reoperations occur after 3 months, so 90-day reporting underestimates them.
- **Parastomal hernia** reaches ≈one-third by 3 years (risks: long operation, fascial defect ≥30 mm, low eGFR).
- ICUD versus ECUD (IRCC) is neutral for high-grade complications and readmission after adjustment, with less transfusion.

- **Reoperation** reaches ≈half by 5 years; predictors are neoadjuvant chemotherapy and any high-grade complication.
- **Uretero-enteric stricture** rises over 5 years; robotic revision equals open.

> **KEY POINT**
>
> Night continence is worse because the urethrosphincteric guarding reflex is inactive in deep sleep, so continence depends on reservoir capacity. Hence detubularisation, nerve and sphincter preservation, a night alarm and limiting night output. Assess at 6–12 months.

- Good ECOG status and retained filling sensation predict continence. Up to 25% of men need clean intermittent catheterisation (CIC) after open neobladder.
- Intracorporeal series report day continence 68–100%, night 55–85%, and CIC up to 11%.

10 · Landmark trials

Trial	Design	Result	Take-home
RAZOR	RCT RARC vs open, extracorporeal diversion	Oncological equivalence	Says nothing about ICUD
iROC	Multicentre RCT, intracorporeal vs open	Open: worse QoL at 5 weeks, more disability at 5 and 12 weeks; no difference after 12 weeks; 11% continent	ICUD gains are early and recovery-related

124 Bladder Surgery for Benign Disease

1 · Surgical anatomy essentials

- The bladder is an extraperitoneal organ (mucosa, lamina propria, detrusor, perivesical fat) derived from the endodermal urogenital sinus.
- **Anteriorly** lies the space of Retzius, the usual access. **Posteriorly**, peritoneum reflects near the prostatorectal junction (men) or vesicouterine space (women), and Denonvilliers fascia separates bladder and prostate from rectum. **Laterally** lie the pelvic sidewall, obturator space, and vas or round ligament.
- The **medial umbilical ligaments** are the obliterated umbilical arteries; the **median umbilical ligament** is the urachus, which can persist as a patent urachus, sinus or cyst.
- The **uterine artery** crosses over the ureter near the internal os, and the **vas** crosses it near the seminal vesicle tip. The distal ureter runs medial to the medial umbilical ligament.
- Superior and inferior vesical arteries form the posterolateral pedicles; lymph drains mainly to external iliac nodes. Nerves are sympathetic T10–L2 and parasympathetic S2–S4; the pudendal nerve (Onuf's nucleus) supplies the sphincter.

> **KEY POINT**
>
> Three landmarks organise every benign bladder operation: Retzius for access, the peritoneal reflection for the posterior bladder, and the distal ureter for safety. They decide whether surgery stays extraperitoneal.

> **KEY POINT**
>
> Common aftercare: watertight **two-layer 2-0 polyglactin** closure (mucosa, then seromuscular), leak test, large-bore Foley for about **1 week**, drain out at about **48 h** if output is low, and a **cystogram before catheter removal**. With high drain output, drain creatinine above serum means a urine leak.

2 · Bladder diverticula

A sac without muscle cannot empty: stasis, infection, stones, cancer.

SAQ Bladder diverticulum.

Paper answer · P2

Type: SAQ (10 marks)
Source: 2016-Summer Paper 2, Q3c

Definition

- Herniation of bladder mucosa through the **muscularis propria**, forming a sac with a neck. With no functioning muscle it **cannot empty** (stasis).
- Male:female about 9:1.

Classification

Feature	Congenital (primary)	Acquired (secondary)
Age	Children (<10 years)	Men >60 years

Feature	Congenital (primary)	Acquired (secondary)
Cause	Weak detrusor at the ureteric hiatus	**BOO (bladder outlet obstruction)**: BPH (benign prostatic hyperplasia) in about 70%, stricture; neurogenic bladder
Number, site	Solitary; **paraureteral (Hutch)**	Often multiple, anywhere
Bladder wall	Smooth	Trabeculated
Associations	VUR (vesicoureteral reflux); Ehlers–Danlos, Williams, Menkes syndromes	Stones, cancer
- **Also know:** iatrogenic (after cystotomy or reimplantation); cellule, saccule and diverticulum form a continuum.		

Clinical features

- Usually incidental while investigating LUTS (lower urinary tract symptoms), haematuria or UTI (urinary tract infection).
- **Double voiding**, recurrent UTI, stones, retention, ureteric obstruction.
- **Malignancy** in 0.8–10% (urothelial 70–80%, squamous 20–25%). With no muscle layer, tumours invade the fat early (beyond lamina propria = **T3a**).

Investigations

- **VCUG (voiding cystourethrogram)**: size, VUR, emptying, **paradoxical filling** during voiding.
- **Cystoscopy**: enter and inspect the sac (flexible scope); cytology; biopsy carefully (perforation risk).
- **CT (computed tomography) urography**: tumour, stones, ureteric deviation, hydronephrosis.
- **Urodynamics**: BOO or neurogenic dysfunction (urine "sinks" into the sac, so contractility looks falsely low).

Management

- **Treat the cause first** (BOO); a small diverticulum may then empty and need only surveillance.
- **Observation** (cystoscopy, cytology) if asymptomatic; **CIC (clean intermittent catheterisation)** if unfit with poor emptying.
- **Endoscopic**: neck incision or resection ± fulguration in frail patients.
- **Diverticulectomy indications**: refractory symptoms, recurrent UTI, stones, upper-tract damage or VUR, premalignant change.
- **Tumour inside**: partial or radical cystectomy.

Diverticulectomy technique

- Open, laparoscopic or robotic; endoscopic prostate surgery first, same session.
- Steps: ureteric catheter if near the ureter → cystotomy → evert or dissect the sac from its pseudocapsule → close in two layers → catheter about 1 week, cystogram before removal.

- Complications: ureteric injury, leak.

Diagram to draw: paraureteral (Hutch) diverticulum with narrow neck and ureter.

> **Key points to remember:**
> - No muscle → stasis, UTI, stones, cancer.
> - Acquired type: treat the BOO first.
> - Tumour invading beyond lamina propria is T3a.

Also, from the condensed book (not in the paper answer):

- Voiding cystourethrogram (VCUG) shows hidden diverticula and the outlet obstruction, and decides between a preoperative stent and simultaneous reimplantation.
- A fibrous **pseudocapsule** gives the surgical plane. Most acquired diverticula are asymptomatic and incidental.
- CT urogram shows hydronephrosis, ureteric displacement and stones. Send cytology in all, add flow rate or urodynamics, and obtain a **negative urine culture** before surgery.
- At cystoscopy, enter every sac and note neck size and distance from the orifice; treat stricture or bladder neck contracture before or with the diverticulectomy.

Situation	Management
Small, asymptomatic, uncomplicated	Observe, or endoscopic fulguration/resection
Large with residual urine, recurrent urinary tract infection (UTI), stones, pain	Diverticulectomy
VUR or ureteric obstruction	Diverticulectomy with or without reimplantation
Cancer in diverticulum	Partial or radical cystectomy, not diverticulectomy

Operative principles

- Treat the cause at the same sitting: transurethral prostate surgery, or simple prostatectomy for glands **>80 cc**, and augmentation for neurogenic bladder. Do **endoscopic outlet surgery first**, because later irrigation disrupts fresh suture lines.
- Localise the sac by selective filling or flexible cystoscopic transillumination, and place a ureteral catheter if it lies near the ureter.
- Open extraperitoneal surgery remains the gold standard, even for posterior sacs; laparoscopic and robotic surgery is reasonable in experienced hands.

ALGORITHM

1. **Open intravesical diverticulectomy**
2. Fill the bladder with about 200 mL via an 18 Fr Foley.
3. Lower midline incision; split recti; incise transversalis fascia.
4. Sweep peritoneum off the dome and open a vertical cystotomy.
5. Identify the orifices (ureteral catheter if concerned) and excise the sac.
6. Place a 20–22 Fr Foley, close in two layers, leak-test, and drain.

Step	Intravesical	Combined intra-/extravesical
Delivery	Sac everted or dissected from the pseudocapsule from inside; mucosal incision about **0.5 cm** round the neck	A finger in the sac pushes it out; the neck is incised from outside
Hazard	Ureteric orifice injury	**Excess bladder wall resection** if the inner and outer incisions do not meet
Closure	Two layers 2-0 polyglactin with a secure seromuscular layer, which prevents recurrence	

Laparoscopic / robotic

- Use steep Trendelenburg and cystoscopy first, with pneumoperitoneum at **15 mm Hg**. Place the assistant port opposite the sac, circumscribe the neck, and excise and bag the sac with the ureter in view; staplers risk stones forming on the staples.
- If the sac sits at the ureteral hiatus, amputate the neck, make a new hiatus and reimplant over a double-J. Extraperitoneal, single-site and single-port variants exist.

Complications

- **Urinary extravasation is commonest**. It shows as high drain output, infected urinoma, or ileus and urinary ascites after transperitoneal surgery. Check drain creatinine and do CT cystography.

ALGORITHM

1. **Managing a leak**
2. If asymptomatic, keep Foley and drain.
3. If a symptomatic collection forms, drain it percutaneously.
4. Do a cystogram before removal; if the leak persists, choose nephrostomies or exploration and closure.

- Other complications are UTI, fistula, bladder spasm, wound infection and bowel injury.
- Unrecognised ureteric injury presents with acute kidney injury or flank pain; try a retrograde stent, and if it fails, place a nephrostomy and repair later.

> **KEY POINT**
>
> Diverticulectomy is a dissection problem: find the sac, stay in the sac–pseudocapsule plane, keep the ureter in view, close the seromuscular defect securely, and fix the outlet first.

3 · Bladder stones, foreign bodies and fistulae

A bladder stone signals stasis, so treat the cause too.

- The cause is **stasis**: bladder outlet obstruction, neurogenic bladder, or augmented and substituted bladders. Foreign bodies are mostly iatrogenic (mesh, migrated intrauterine device).
- **Non-contrast CT** is best for stone burden, prostate volume and hydronephrosis. Cystoscopy is needed only for haematuria, suspected cancer or prostate assessment.
- Treat the stone **and** its cause; the concomitant operation often picks the route (transvesical prostatectomy, diverticulectomy or enterocystoplasty).

Approach	Best for	Advantage	Limitation
Transurethral cystolitholapaxy	Most stones	No incision	Prolonged instrumentation risks urethral stricture
Percutaneous cystolithotripsy	Large stones	Nephroscope, short stay, no stricture	Contraindicated after prior surgery, diversion or augmentation (bowel injury)
Open cystolithotomy	Very large burden; with prostatectomy, diverticulectomy, augmentation	Historic gold standard	Incision, longer catheter
Laparoscopic/robotic	Large burden with other procedures; foreign bodies, eroded mesh	Minimally invasive	Technically demanding

- **Percutaneous** access two fingerbreadths above the symphysis is dilated to a **30 Fr** sheath for a 24–26 Fr nephroscope. Stones under 1 cm are grasped whole and larger ones fragmented; the suprapubic catheter is clamped at 24–48 h. After prior surgery, image first to exclude interposed bowel.
- **Transvesical laparoscopy** uses a pneumovesicum of **8–12 mm Hg**, avoids the peritoneum and suits eroded mesh. Transperitoneal laparoscopic or robotic cystolithotomy allows simultaneous prostatectomy or diverticulectomy.
- **Open** cystolithotomy opens the bladder vertically, removes the stones, inspects for tumour and intravesical lobe, and closes in two layers with a drain.
- Refractory fistulae (colovesical, vesicovaginal) may need partial cystectomy, and a refractory urinary fistula may justify simple cystectomy.

4 · Partial cystectomy for benign disease and urachal surgery

- Indications are leiomyoma, paraganglioma, endometriosis, urachal anomalies and fistulae; most benign tumours need only transurethral resection (TUR).
- **Avoid it with small capacity or severe dysfunctional voiding.** Exclude malignancy first, because cancer needs wide margins and lymphadenectomy. Lesions near the orifice need reimplanting.

Lesion	Clue	Investigation	Treatment / trap
Leiomyoma	Women of childbearing age; endovesical, intramural or extravesical	Transvaginal ultrasound (US), MRI	Small endovesical: TUR; large or intramural: wide local resection
Paraganglioma	Episodic hypertension, hypotension or syncope on **micturition or sex**; or painless haematuria	Plasma metanephrines; MIBG scan; yellow raised mucosa	**Alpha-blockade before any instrumentation**; partial cystectomy preferred over TUR (recurrence)
Endometriosis	Cyclical haematuria with menses	US, MRI (bright on T1); cystoscopy limited	Partial cystectomy (disease is extravesical)
Urachal anomaly	Umbilical discharge, mucinuria, infected mass; malignant risk	US/CT; VCUG or sinogram; cystoscopy	Culture and antibiotics, with or without drainage; if recurrent, excise urachus with a dome cuff

- Dome and anterior lesions are approached extraperitoneally, posterior ones transperitoneally, with no node dissection. **Aspirate before opening to avoid spillage**.
- Excise full thickness with about a **2-cm margin** if capacity allows, with frozen-section margins, and reimplant if the orifice is involved.
- **Urachal excision** takes the whole tract with peritoneal wings and a small dome cuff, with **no wide margin or frozen section**; the umbilicus may be kept if benign. Open excision uses a periumbilical midline incision down to a transmural dome cuff.
- Complications are extravasation (commonest), ureteric injury and reduced capacity.

> **KEY POINT**
>
> Leiomyoma, paraganglioma and endometriosis infiltrate and recur if incompletely excised, so benign partial cystectomy borrows cancer discipline (exclude cancer, no spillage, a margin). Urachal surgery is the exception: the whole tract matters, not a wide margin.

5 · Simple (supratrigonal) cystectomy

A last resort that helps only if the bladder is the source.

- It treats end-stage refractory dysfunction: radiation cystitis, neurogenic bladder, refractory incontinence, fistula, bladder neck contracture, refractory anastomotic stricture, and interstitial cystitis/bladder pain syndrome (IC/BPS).
- Prove the bladder is the source, since non-vesical pain persists. In IC/BPS, success is predicted by a **small fibrotic bladder, small capacity under anaesthesia and Hunner lesions**.
- Diversion alone risks pyocystis, bleeding, pain and secondary cancer.

Feature	Simple (benign)	Radical (cancer)
Prostate / anterior vagina	Preserved unless indicated	Removed
Lymphadenectomy	None	Performed
Bladder opened	Often, to find the trigone and bladder neck	Avoided
Excision	Bivalve and amputate near the trigone; stump oversewn and residual mucosa cauterised	En bloc

Steps are to divide the ureters, bivalve the bladder, divide the pedicles, amputate each half and cover the stump with posterior peritoneum. An ileal conduit usually follows, and robotic surgery may reduce complications. Drains stay 5–7 days, until drain creatinine equals serum, and complications are common.

6 · Ureteral reimplantation in adults

Adults need reimplantation mainly for injury and stricture.

- Indications are distal ureteral trauma, iatrogenic injury, **stricture** and late-presenting injury, and reimplantation alongside diverticulectomy or partial cystectomy. Primary VUR is rare.
- In thermal or gunshot injury, debride even normal-looking ureter; assess partial injury with on-table retrograde pyelogram or methylene blue.
- Strictures follow impacted stone, radiation or chronic inflammation. An atrophic kidney needs decompression, then a renogram.

- Measure the gap with antegrade and retrograde imaging and exclude malignancy. A **poorly functioning kidney needs nephrectomy, not reimplantation**. **Persistent extravasation despite adequate stenting** is an indication.

> **KEY POINT**
>
> Refluxing reimplantation is acceptable in adults. Refluxing anastomoses show no faster renal decline, more stenosis or more infection than tunnelled ones, while needing less ureter and less bladder mobilisation.

Feature	Nonrefluxing (tunnelled)	Refluxing (direct)
Tunnel	**3–5 × ureteric diameter** (5:1 robotically)	None
Ureter length / mobilisation	More	Less; reaches the posterior dome
Use	High-grade reflux with renal risk	Generally sufficient in adults

Technique	Access	Neo-hiatus	Signature step
Modified **Lich-Gregoir** (nonrefluxing)	Extravesical	Posterior wall, superior-medial to native orifice	Detrusor trough over intact mucosa; detrusor closed over ureter
Modified **Politano-Leadbetter** (nonrefluxing)	Transvesical	Cephalad-medial to native hiatus	Submucosal tunnel; right-angle clamp draws ureter through
Refluxing extravesical	Extravesical	Posterior dome, lateralised	Direct spatulated anastomosis
Refluxing transvesical	Transvesical	Posterior dome over a Kelly clamp	Clamp pulls ureter in

- Divide the ureter proximal to disease, with a frozen-section margin in stricture. Mobilise the bladder off peritoneum, dividing the ipsilateral medial umbilical ligament. Test reach on the **decompressed** dome.
- **Lich-Gregoir**: a small cystotomy forms the neo-orifice, detrusor is incised toward the bladder neck sparing mucosa, a spatulated 4-0 anastomosis is stented, and detrusor is closed over the ureter.
- **Politano-Leadbetter**: through an anterior cystotomy, a neo-orifice about 1.5 cm superior-medial to the native one leads into a submucosal tunnel.
- For a poorly vascularised ureter, use a non-transecting side-to-side refluxing anastomosis.
- Preserve periureteric adventitia, spatulate, and keep knots outside the lumen. The stent is a **6 Fr × 26–28 cm double-J**.
- Primary reimplantation suits defects up to **4 cm**; longer gaps need a psoas hitch or Boari flap, and renal descent adds up to **5 cm**. Further reach comes from dividing the contralateral medial umbilical ligament and pedicle.
- Enter **superior-medially, toward the bladder neck**, because lateral sites kink on filling.
- Open surgery is preferred for trauma; laparoscopic and robotic surgery give less blood loss and shorter stay with equal patency.
- Aftercare is a Foley for 1 week with a cystogram and a stent for **≈6 weeks**. A **leak** is managed by draining the urinoma and keeping the Foley and stent; if refractory, place a nephrostomy.

- **Late obstruction** usually follows stent removal (ischaemia, tension, narrow neo-orifice, poor spatulation, radiation, lateral site); short strictures are treated endoscopically and high-grade ones by redo.

> **KEY POINT**
>
> Reimplantation fails from ischaemia or tension. Every rule (preserve adventitia, spatulate, enter superior-medial toward the bladder neck, bridge rather than stretch) prevents one or the other.

7 · Bridging distal ureteral defects

*See the paper answer **Boari flap** (Section 11, Ch 93).*

Also, from the condensed book (not in the paper answer):

- Nonrefluxing Boari length is the defect plus about 3 cm; longer flaps need a wider base.
- Beyond reach, options are transureteroureterostomy, ileal ureter and autotransplantation.

Option	Reach	Rule
Primary reimplant	≤4 cm	Toward bladder neck
Psoas hitch	Gains about 5 cm	Sutures superficial, parallel to the fibres; reimplant before tying
Boari flap	Up to **15 cm**	Base ≥**4 cm** , apex ≥**3 cm** , length:base ≤**3:1**

- Hitch plus flap shortens the flap and preserves capacity. The hitch sutures go into the psoas minor tendon, avoiding the **genitofemoral** (surface) and **femoral** (deep) nerves; reimplant on the superolateral dome.
- The Boari flap is based on the diseased side, anchored to psoas and tubularised in two layers.
- A bladder advancement flap gives comparable reach and is simpler. A **small-capacity, poorly compliant bladder** contraindicates both hitch and flap.
- Complications are femoral or genitofemoral neurapraxia, iliac vessel injury, ischaemic stricture of a narrow-based flap, and reduced capacity.

> **KEY POINT**
>
> The bladder is the donor tissue: hitch first, flap second. Every flap rule (base width, 3:1, measure empty) protects the flap's blood supply and bladder capacity.

8 · Enterocystoplasty (augmentation)

Augmentation protects the kidneys from high bladder pressure.

- Indication: neurogenic bladder with poor compliance **despite** clean intermittent catheterisation (CIC), anticholinergics, botulinum toxin and sacral neuromodulation, especially with upper tract deterioration or incontinence with skin breakdown.
- Bowel lowers pressure only if **detubularised**, which abolishes peristalsis.
- Work-up includes video-urodynamics, serum chemistry, B12 and bowel history; **confirm the patient can catheterise**. Metabolic effects: See Ch 133.

- Add a sling, artificial urinary sphincter, bladder neck closure, catheterisable stoma or nonrefluxing reimplant as needed.

Segment	Note
Ileum (commonest)	25 cm taken **≈20 cm** proximal to the ileocaecal valve (spares B12 and bile-salt absorption); acidosis risk
Ileocaecal	When a catheterisable channel is wanted
Sigmoid	Colonoscopy first
Stomach	Possible; needs a higher incision

ALGORITHM

1. Isolate ileum with a broad mesenteric base and check reach to the dome.
2. Restore bowel side-to-side; close the window.
3. **Detubularise** on the antimesenteric border and fold into a U.
4. **Bivalve the bladder widely** and make a wide-mouthed anastomosis.
5. Suprapubic tube with or without Foley; leak-test; drain.

- A narrow mouth creates a poorly emptying pseudodiverticulum.
- Early complications are extravasation (preserve the arcade), UTI, bowel injury, fistula and intra-abdominal infection.
- **Perforation** follows missed catheterisation and is often masked by sensory loss. Diagnose with CT cystogram and proceed to **emergency exploration**.
- Stones are the commonest long-term complication. Neoplasia risk means surveillance endoscopy from **10 years**, with routine metabolic and vitamin bloods.
- If the urethra is not catheterisable, do a continent ileocaecocystoplasty; if the patient cannot catheterise, an incontinent ileovesicostomy.

> **KEY POINT**
>
> Augmentation exchanges a pressure problem for a lifelong bowel-in-bladder problem. Detubularisation and a wide-mouthed anastomosis win the pressure battle; catheterisation, irrigation, bloods and endoscopic surveillance are the price.

9 · Continent catheterisable channels

*See the paper answer **Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic** (Section 12, Ch 134).*

Also, from the condensed book (not in the paper answer):

- Channels suit adequate-capacity bladders after failed urethral CIC; augment if compliance is poor.
- The bladder is drained for 2–4 weeks before channel catheterisation starts. Always have a fallback if the appendix is absent or short.
- A Monti uses 2–4 cm of ileum; a double Monti risks channel diverticula; vas, colon and stomach channels do poorly.

ALGORITHM

1. **Minimally invasive appendicovesicostomy**
2. Detach the appendix with a caecal cuff and spatulate it.
3. Make a ≈4-cm submucosal detrusor tunnel and anastomose.
4. Bring the stoma to the right lower quadrant or umbilicus.
5. Drain the bladder for **2–4 weeks**. Continence is **>95%**.

KEY POINT

Use the appendix when present and long enough; reconfigure ileum when not. Continence comes from tunnel length through detrusor, and missed catheterisation endangers the bladder, not just the channel.

10 · Vesicovaginal fistula (VVF) †

P3 P2 *VVF: management options, early vs delayed repair — causes, double-dye test, CT urography, catheter trial, timing, vaginal (Latzko, flap-split) vs abdominal (O'Conor) routes, interposition flaps*

- Leakage usually starts 1–3 weeks after hysterectomy, and a coexisting ureteric injury is present in about 10%.
- A ureterovaginal fistula is treated with a ureteric stent first, and by ureteroneocystostomy if that fails.

★ Putting it together

A · NMIBC: risk and treatment

AUA/SUO risk is assigned after complete TURBT.

Risk	Features	Adjuvant	Cystoscopy
Low	Low-grade (LG) solitary Ta ≤3 cm; PUNLMP	Single post-op instillation	3 mo, then yearly
Intermediate	LG Ta recurring <1 yr/>3 cm/multifocal; LG T1; high-grade (HG) Ta ≤3 cm	Induction MMC/gemcitabine or BCG; maintain 1 yr	3–6-monthly for 1–2 yr
High	HG T1; HG Ta >3 cm/multifocal/recurrent; CIS; variants; LVI; HG prostatic urethra	Re-TURBT (T1); BCG with **maintenance 3 yr** (SWOG); early RC if highest-risk	3–4-monthly for 2 yr, 6-monthly for 2 yr, then yearly; upper tracts yearly for 2 yr
BCG-unresponsive	HG despite adequate BCG (≥5/6 induction + ≥2/3 maintenance)	RC; if unfit: pembrolizumab (CIS), nadofaragene, gem–docetaxel	—

B · MIBC: cystectomy vs preservation See Ch 131, 132

	NAC + RC + pelvic node dissection	Trimodality therapy
Who	Default if fit; NAC (gem–cis or ddMVAC) if cisplatin-eligible	Solitary T2, complete TURBT, **no CIS/hydronephrosis**, good function; or unfit
Cisplatin-ineligible	GFR <60, ECOG ≥2, grade ≥2 hearing loss or neuropathy, or NYHA III–IV: local therapy	
After	Adjuvant nivolumab (CheckMate 274): ≥pT3/N+ after NAC, or no NAC	Lifelong cystoscopy; salvage RC 10–30%

C · Choosing the diversion See Ch 133–135

	Ileal conduit	Orthotopic neobladder	Continent cutaneous
Best for	Frail, poor renal function, cannot catheterise	Default continent option, either sex	Unusable urethra or sphincter
Must have	Usable (non-irradiated) ileum	eGFR **>35–40** ; intact sphincter; negative urethral frozen section; will catheterise	eGFR >35–40; dexterity; efferent limb
Main problems	Stoma problems, ureteroileal stricture	Night leakage; late retention (women); acidosis	Stomal stenosis, catheterisation difficulty, stones

D · Metabolic complications by segment

Segment	Disturbance	Treatment
Ileum / colon	**Hyperchloraemic metabolic acidosis**	Bicarbonate, potassium citrate

Segment	Disturbance	Treatment
Jejunum	**Hyponatraemic hypochloraemic hyperkalaemic acidosis**	Sodium chloride; avoid jejunum
Stomach	**Hypochloraemic hypokalaemic alkalosis**; haematuria–dysuria syndrome	H2 blocker/PPI

Losing terminal ileum lowers B12 and causes bile-salt diarrhoea; buffering acidosis in bone causes osteomalacia. See Ch 133

E · Metastatic urothelial carcinoma

ALGORITHM

1. **Fitness**: cisplatin-, platinum-only or platinum-ineligible.
2. **First line**: enfortumab vedotin (EV) with pembrolizumab (EV-302); otherwise gem–cis (gem–carbo if cisplatin-ineligible).
3. **If no progression on platinum**, give avelumab maintenance (JAVELIN Bladder 100).
4. **Later lines**: pembrolizumab; erdafitinib (FGFR-altered); EV; sacituzumab govitecan. See Ch 131

Other paper questions in this section

Asked in the papers but not tied to a specific place in the book above.

LAQ Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman

Paper answer · P2

Asked as:

- Describe in detail the management of T1G3 TCC of urinary bladder? — *Summer 2019 Paper II Q1 (LAQ)*
- Discuss management of T1G3 bladder tumor — *Practice Set 7 Paper II Q3a (SAQ)*
- Discuss treatment of 25 yr female with bladder TCC T1NoMo and C13 — *Practice Set 4 Paper II Q1 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 7 Paper II Q3a: also answered in 2019-Summer Paper 2, Q1) (Practice Set 4 Paper II Q1: T1G3 management also answered in 2019-Summer Paper 2, Q1)

Practice Set 4 Paper II Q1: "C13" is read as CIS (carcinoma in situ): T1 N0 M0 urothelial carcinoma with concurrent CIS in a 25-year-old woman.

Introduction and risk category

- T1G3 is high-grade (HG) urothelial carcinoma (the current name for TCC, transitional cell carcinoma) that invades (invading) the lamina propria but not the muscularis propria (not muscle). TCC is now urothelial carcinoma. CIS is flat and always high grade.
- T1 is 20–30% of NMIBC (non-muscle-invasive bladder cancer) and its most dangerous form: recurrence >80%, progression to muscle invasion 30–50%.
- Grade, not stage, drives progression. So every HG T1 is **high risk** (AUA [American Urological Association]/SUO [Society of Urologic Oncology] and EAU [European Association of Urology]).
- **T1 with CIS is high-risk NMIBC** (AUA and EAU). CIS is the second strongest progression factor after grade. Untreated CIS progresses to muscle invasion in 40–83%.
- **EAU 2021 very high risk:** T1 high grade (HG) with CIS plus at least one of age >70, multiple papillary tumours, or diameter ≥3 cm. In this woman (25-year-old), age does not count, so number and size decide. LVI (lymphovascular invasion) and variant histology also make it very high risk.
- **Young woman:** long life expectancy, fertility, pregnancy and sexual function shape every choice.

Risk factors for progression

- **Adverse factors:** concurrent CIS (carcinoma in situ), LVI (lymphovascular invasion), variant histology, size >3 cm, multifocality, prostatic urethral involvement, residual T1 at re-resection.

Factor	Significance
Concurrent CIS (carcinoma in situ)	Second most important factor after grade
Depth of invasion (substaging)	Invasion deep to the muscularis mucosae is worse; T1a/T1b proposed, no agreed method

Factor	Significance
LVI (lymphovascular invasion)	Suggests occult spread
Variant histology	Micropapillary (any amount), plasmacytoid, sarcomatoid, nested
Size >3 cm, multifocal, recurrent	More residual tumour and progression
Prostatic urethra, diverticulum	Poor BCG contact; diverticulum lacks muscle
Residual T1 at re-TURBT	Progression up to 80%
HG T1 at 3 months after BCG	Progression about 82%
- **Also know:** EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables; hydronephrosis suggests muscle invasion.	

Evaluation

- Cystoscopy, urine cytology (usually positive in CIS), CT (computed tomography) urography (upper tract tumour <5%, more with CIS), bimanual examination under anaesthesia (including vagina).
- Smoking and occupational history.
- **Pregnancy test** before CT and BCG (bacille Calmette-Guérin); expert pathology review.

Step 1: Complete, good-quality TURBT (transurethral resection of bladder tumour)

- Resect all visible tumour **with detrusor muscle** in the specimen; send the base separately.
- Blue-light cystoscopy or NBI (narrow-band imaging) finds more CIS (**blue-light cystoscopy** finds about one-third more CIS). **En bloc resection** (where possible) keeps orientation for substaging.
- Biopsy the prostatic urethra if tumour is at the bladder neck or trigone, or with CIS.
- Biopsy red velvety areas and the **bladder neck** (relevant to a later neobladder).

Step 2: Re-TURBT (restaging) in all T1

- Within **2–6 weeks** (within 6 weeks), before BCG (bacille Calmette-Guérin).
- **Why:** residual tumour in ≥40%; upstaging to muscle invasion in 25–30% (about half if no muscle in the first specimen). **Residual T1: progression up to 80%.** Re-TURBT improves recurrence- and progression-free survival: Divrik's RCT (randomised controlled trial) showed better recurrence-free and progression-free survival.

```
HG T1 on TURBT → re-TURBT at 2–6 weeks ├─ ≥T2 → neoadjuvant chemotherapy + cystectomy ├─ residual T1 / very high risk → offer early radical cystectomy └─ T0, Ta or CIS → BCG induction + 3-year maintenance
```

- Practice Set 7 version: HG T1 → re-TURBT; T0, Ta, CIS, T1 → BCG (bacille Calmette-Guérin); very high risk → offer early radical cystectomy.
- Practice Set 4 version: T1 + CIS → re-TURBT within 6 weeks; T0/Ta/CIS → BCG induction + 3-year maintenance.

Step 3a: BCG – standard for most patients (bladder preservation)

- **Mechanism:** BCG attaches to (binds) **fibronectin**, triggers cytokines (IFN-γ [interferon-gamma], IL [interleukin]-2) and a **Th1 (T-helper 1)** immune (cytokine) response that kills

tumour. It is the **only intravesical agent that reduces progression**.

- **Schedule:** start 2–4 weeks after TURBT, never immediately (intravasation and sepsis); induction weekly ×6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** 3 weekly doses at 3 and 6 months, then every 6 months (6-monthly) to **3 years**. Median recurrence-free survival 76.8 vs 35.7 months. **EORTC 30962:** full dose for 3 years in high risk.
- **CIS:** tumour-free up to 84%, about half durable; persistent CIS at 3 months: second course gives a further 30–50% complete response.
- BCG is live: avoid pregnancy and breastfeeding during treatment.
- **Contraindications (avoid with):** gross haematuria, traumatic catheterisation, symptomatic UTI (urinary tract infection), previous BCG sepsis.
- **Toxicity:** cystitis common; BCG sepsis needs isoniazid, rifampicin, ethambutol and steroids (antituberculous drugs and steroids).

Step 3b: Early (immediate) radical cystectomy

- **Rationale:** clinical T1 is understaged in one-third to one-half. Patients who progress under surveillance do worse than those presenting with muscle invasion (progressors do worse than primary muscle-invasive disease). Immediate cystectomy gives about **92% 10-year disease-free survival**.
- **AUA:** offer to fit patients with persistent HG T1 on re-resection, or T1 with CIS, LVI or variant histology.
- **EAU very-high-risk group (2021 model; risk factors: age >70, multiple papillary tumours, diameter ≥3 cm):** T1 HG with CIS plus ≥1 risk factor; T1 HG without CIS but with all three; also LVI, variant histology, CIS of the prostatic urethra (prostatic urethral CIS), persistent T1 at re-TURBT. Offer immediate cystectomy; BCG if declined or unfit.
- **Offer for (Practice Set 4):** very high risk, residual HG T1, LVI, variants, diffuse CIS, or patient preference.
- Also: diverticular tumour, tumour not controllable endoscopically.

Step 3c: Cystectomy tailored to a young woman

- Standard female operation is anterior exenteration (bladder, urethra, anterior vagina, uterus, cervix, ovaries).
- **Organ-sparing** (≤cT2, bladder neck uninvolved): keep the **uterus and ovaries** (fertility, hormones, avoids surgical menopause; ovaries also make testosterone). Keep the **vagina and neurovascular tissue** (vaginal dissection no more dorsal than 2 and 10 o'clock) for sexual function and continence. Oncological safety of sparing is not fully known.
- **Orthotopic neobladder** suits her: needs a negative bladder-neck biopsy and **urethral frozen section** (negative bladder neck means negative urethra; CIS does not predict urethral involvement). Day continence 80–90%; **retention 20–30% in women**, so she must accept CIC (clean intermittent self-catheterisation). See Practice Set 4 Paper 2, Q3e.
- **Fertility:** if uterus or ovaries must go, offer oocyte or embryo cryopreservation first. Pregnancy after a neobladder is possible; caesarean uses a high midline incision to avoid the pouch.

Step 4: BCG failure (BCG-unresponsive disease)

- **Adequate BCG for T1:** ≥5 of 6 induction doses.

- **BCG-unresponsive:** HG T1 at the 3-month check after induction; persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Standard:** timely radical cystectomy (EAU, AUA).
- **Unfit or refusing (declined):** pembrolizumab (KEYNOTE-057: CIS complete response 41%); nadofaragene firadenovec (53%); N-803 with BCG; gemcitabine–docetaxel; TAR-200; trials.
- **Also know:** BCG-intolerant; delayed relapse may respond to BCG rechallenge.

Follow-up (AUA/SUO high risk)

- Bladder preserved: cystoscopy and cytology every 3–4 months (3-monthly) for 2 years, 6-monthly for 2 years, then yearly for life.
- Upper-tract imaging yearly for 2 years (regular upper-tract imaging).
- Positive cytology with normal cystoscopy: check prostatic urethra and upper tracts.
- After neobladder: B12, metabolic panel, upper tracts.

Recent advances

- Checkpoint inhibitors with BCG in BCG-naive disease (trials); urine markers predicting BCG response; TAR-200.

Diagram to draw: bladder wall layers showing CIS, Ta, T1a/T1b (above and below the muscularis mucosae) and T2.

> **Key points to remember:**
> - Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
> - BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
> - T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
> - Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
> - BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
> - Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

SAQ Optical coherence tomography in carcinoma bladder

Paper answer · P2 · P4 · merged from 2

Asked as: P4 SAQ "Optical coherence tomography in carcinoma bladder" · P2 SAQ "What is optical coherence tomography. What is its potential role in management of CA bladder?"

Introduction

- **WLC (white-light cystoscopy)** is the gold standard but subjective: many biopsies are negative and small lesions are missed. WLC misses flat lesions and cannot show depth.
- **OCT (optical coherence tomography)** is an **"optical biopsy"** technique listed among enhanced cystoscopic methods for NMIBC (non-muscle-invasive bladder cancer): real-time cross-sectional imaging of the bladder wall. It works like ultrasound but uses light.

What is OCT: principle

- OCT is the **optical analogue of B-mode ultrasound**. **Near-infrared light** (about 1300 nm) is shone on tissue, and the echo delay of back-scattered light is measured instead of sound, using **low-coherence interferometry** (back-scattered light measured against a reference beam).
- The delay of reflected light gives depth. Scanning across builds a cross-sectional image.
- **Resolution about 10–20 µm** (near histology). **Depth about 2–3 mm** (P2: only **1–2 mm**): urothelium, lamina propria and superficial muscle.
- First described in 1991 (ophthalmology).

Technique

- A fibre-optic probe passes through the cystoscope's working channel.
- WLC or blue light finds the lesion; the probe is placed **in contact** with it and read at once (real time).

Image interpretation

Tissue	OCT appearance
Normal	Three layers: thin, dark, low-scatter urothelium; bright lamina propria with clear boundary; darker muscularis visible below
Inflammation	Layers preserved but thickened
Papillary tumour (Ta)	Thickened urothelium, papillary pattern; lamina propria boundary intact
Invasive (T1)	Layers lost; **loss (break) of the urothelium–lamina propria boundary** = invasion

Potential role in carcinoma bladder

1. **Diagnosis:** tell tumour from inflammation at suspicious areas; separate benign from malignant flat lesions in real time; fewer unnecessary biopsies.
2. **Staging aid: Ta vs T1** (lamina propria invasion) during TURBT (transurethral resection of bladder tumour); helps in diverticula and thin walls.
3. **Resection margins:** checks margins and completeness of TURBT.
4. **Surveillance** of NMIBC, including after BCG (bacille Calmette-Guérin).
5. **Upper tract:** through the ureteroscope, OCT correlates well with final grade and stage of UTUC (upper tract urothelial carcinoma).

Evidence and limitations

- Small studies show **high sensitivity but lower (moderate) specificity**. Inflammation, scar, recent TURBT and post-BCG change mimic tumour (false positives). Adding OCT to fluorescence (blue-light) cystoscopy improves specificity.
- Tiny field of view: needs WLC or PDD (photodynamic diagnosis) to guide it.
- Too shallow to judge (deep) muscle invasion; cannot grade (no cellular detail); cost; learning curve.
- Still **investigational**; not recommended routinely by guidelines.

Other enhanced cystoscopy techniques

Technique	Principle
PDD (blue light, HAL, hexaminolevulinate)	Tumour porphyrins fluoresce red; detects about one-third more CIS (carcinoma in situ)
NBI (narrow-band imaging)	415/540 nm light absorbed by haemoglobin; shows vessels; many false positives
CLE (confocal laser endomicroscopy)	Cell-level surface images after fluorescein
- **Also know:** Raman spectroscopy.	

Diagram to draw: OCT image showing the three normal bladder layers beside a tumour with loss of layering.

> **Key points to remember:**
> - Near-infrared interferometry ("ultrasound with light"); 10–20 µm resolution; 2–3 mm depth (P2: 1–2 mm).
> - Loss of layering suggests tumour; boundary loss suggests T1.
> - Role: Ta vs T1; cannot assess deep muscle.
> - High sensitivity, moderate specificity, false positives from inflammation; investigational.

SAQ Intravesical gene therapy.

Paper answer · P2

Type: SAQ (10 marks)
Source: 2022-Summer Paper 2, Q3a

Concept

- A gene is instilled into the bladder so that urothelial and tumour cells make an anticancer product, or the vector itself kills them.
- **Why the bladder suits it:** direct contact, little systemic exposure, response checked by cystoscopy and cytology.
- **Barrier:** the glycosaminoglycan layer resists vectors → **transduction enhancers** (Syn3; DDM, n-dodecyl-β-D-maltoside).
- **Indication:** high-risk **BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer)** in patients unfit for or declining radical cystectomy. BCG-unresponsive: persistent or recurrent CIS (carcinoma in situ) or high-grade Ta/T1 after adequate BCG, or relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.

Agents

Agent	Vector, mechanism	Key data
Nadofaragene firadenovec	**Replication-incompetent adenovirus type 5** carrying the **interferon-α2b gene**, with **Syn3** surfactant. Urothelium keeps secreting interferon → apoptosis (TRAIL, TNF-related apoptosis-inducing ligand) and immunity, overcoming the short dwell of interferon protein	75 mL every **3 months**, 1-hour dwell. CIS: **complete response 53% at 3 months, 24% at 12 months**; papillary high-grade RFS (recurrence-free survival) 44% at 12 months; serious events 2%. **FDA (Food and Drug Administration) approved December 2022**, first bladder gene therapy
Cretostimogene grenadenorepvec (CG0070)	**Replication-competent oncolytic adenovirus**: E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity	Two 6-week inductions + maintenance; single-arm phase III BOND-003 in BCG-unresponsive CIS: high CR (complete response) rate; combination with pembrolizumab under study
Inodiftagene vixteplasmid (BC-819)	Plasmid: **diphtheria toxin A** gene under tumour-active **H19** promoter	Investigational

Administration and toxicity

- Catheter instillation, patient turned during dwell; biosafety precautions.
- Mostly grade 1–2: **bladder spasm, urgency, discharge around catheter, fatigue**.

Place in practice

- Alternatives: radical cystectomy (standard), **pembrolizumab** (KEYNOTE-057), gemcitabine–docetaxel, **N-803** (interleukin-15 superagonist) + BCG.
- Limits: responses wane (median CR duration about 10 months); single-arm trials; cost.

Also know: non-viral IL-12 (interleukin-12) plasmid (EG-70); early adenoviral attempts failed through poor transduction.

Diagram to draw: adenovirus transducing a urothelial cell → interferon secretion → tumour apoptosis.

> **Key points to remember:**
> - Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
> - CIS CR 53% at 3 months, 24% at 12 months.
> - Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

SAQ Von Bruns Nest

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 10 Paper 2, Q3e

Read as: von Brunn nests of the urothelium.

Definition

- **Von Brunn nests** are solid nests of benign urothelial cells lying in the **lamina propria**. They form when the basal layer of the surface urothelium buds downward and the buds pinch off.
- A common normal or reactive finding in most adult bladders, especially at the **trigone and bladder neck**; also in the ureter and renal pelvis.

Pathogenesis: the proliferative cystitis spectrum

`Chronic irritation (infection, stones, catheter, outlet obstruction) → urothelial buds → von Brunn nests (solid) → central degeneration and fluid → cystitis cystica (cysts) → glandular (columnar / goblet cell) metaplasia → cystitis glandularis`

Lesion	Histology	Malignant potential
Von Brunn nests	Solid urothelial nests, lamina propria	None
Cystitis cystica	Nests with central cyst, urothelial lining	None
Cystitis glandularis (typical)	Cuboidal or columnar lining	Low
Cystitis glandularis, **intestinal type**	Goblet cells, mucin	Possible link to **adenocarcinoma**

Clinical features

- Usually incidental. When florid: irritative symptoms, haematuria, infection; cystoscopy shows submucosal nodules or cysts that can mimic tumour.
- **Associations:** chronic inflammation, outlet obstruction, **pelvic lipomatosis** (florid cystitis glandularis), bladder exstrophy.

Differential diagnosis (pathologist's trap)

- **Nested variant of urothelial carcinoma:** deceptively bland but aggressive (male:female 6:1). Clues: deep infiltrative nests reaching muscle, nuclear atypia.
- **Inverted papilloma:** endophytic nests with trabeculae, usually at the trigone.
- Nephrogenic adenoma; urothelial carcinoma with glandular differentiation.

Management

- None for isolated nests; treat the cause (infection, stone, catheter, obstruction).
- **TURBT (transurethral resection) or biopsy** if a mass is present, to exclude carcinoma.
- **Cystitis glandularis:** remove the irritant and arrange **endoscopic surveillance**, because rare change to adenocarcinoma is reported (especially the intestinal type and in exstrophy).
- Upper tract: benign, but coexisting upper-tract urothelial cancer cannot be excluded, so follow up.

Diagram to draw: urothelium budding into lamina propria, forming a solid nest, then a cyst, then glandular metaplasia.

Key points to remember:

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

SAQ Mitomycin

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 11 Paper 2, Q3b

Pharmacology

- MMC (mitomycin C) is an antitumour antibiotic from *Streptomyces caespitosus*.
- **Mechanism:** after reductive activation in the cell it acts as an **alkylating agent**, cross-linking DNA and stopping DNA synthesis; cell-cycle non-specific.
- Molecular weight 334 Da: **little systemic absorption** from the bladder.

Uses in NMIBC (non-muscle-invasive bladder cancer)

1. **Single immediate post-TURBT (transurethral resection of bladder tumour) instillation:** 40 mg. Given **within 6 hours** it reduces recurrence; at 24 hours it does not, because it works by preventing implantation of floating cells. Best for a solitary low-grade papillary tumour. Avoid if perforation or extensive resection is suspected.
2. **Adjuvant induction:** 40 mg weekly for 6 weeks (± maintenance) in low- and intermediate-risk disease. Reduces recurrence but **not progression**; BCG (bacille Calmette–Guérin) with maintenance is superior in high risk.
3. **Optimised delivery (Au, 2001):** empty bladder, overnight fluid restriction, sodium bicarbonate (less drug breakdown), 40 mg in 20 mL; retain 1–2 hours.
4. **Device-assisted:** EMDA (electromotive drug administration) increases tissue uptake; chemohyperthermia at 42 °C (RITE [radiofrequency-induced thermochemotherapeutic effect], HIVEC [hyperthermic intravesical chemotherapy]).
5. **Sequential gemcitabine then MMC** for BCG-exposed disease.

Other urological uses

- **Upper tract:** single bladder dose after nephroureterectomy (ODMIT-C: bladder recurrence 28% to 16% at 1 year). **UGN-101 (mitomycin gel)**, FDA (Food and Drug Administration) approved 2020 for low-grade UTUC (upper tract urothelial carcinoma): complete response 59%, ureteric stricture about 44%. UGN-102 for low-grade intermediate-risk bladder tumours.
- **Bladder-sparing chemoradiation:** 5-FU (5-fluorouracil) + MMC (BC2001: better locoregional control than radiation alone).
- **Antifibrotic:** injected after incision of recurrent bladder neck contracture or urethral stricture.

Side effects

- Chemical cystitis (30–40%); **skin rash, palmar or genital dermatitis** (signature toxicity; wash after voiding); contracted bladder (about 5%); encrustation cystitis; myelosuppression

rare.
- Extravasation through a perforation can cause pelvic necrosis.

Diagram to draw: timeline: TURBT, single MMC within 6 hours, then weekly induction ×6.

> **Key points to remember:**
> - Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
> - Reduces recurrence, not progression.
> - Signature toxicity: skin rash; contracted bladder about 5%.

SAQ Mullerionosis of UB

Paper answer · P2

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 2, Q3b

"Mullerionosis of UB" is read as müllerianosis of the urinary bladder.

Definition

- Benign lesions of **müllerian-type epithelium** in the bladder wall, in three forms:

Lesion	Tissue
Endometriosis (commonest)	Endometrial glands **and stroma**
Endocervicosis	Endocervical mucinous glands
Endosalpingiosis	Ciliated tubal epithelium

- **Müllerianosis** strictly means two or more of these together.
- Women of reproductive age, often after **caesarean section**. Usually **posterior wall or dome** (vesicouterine pouch).
- Bladder is the commonest urinary tract site of endometriosis, ahead of the ureter.

Pathogenesis

- **Implantation:** retrograde menstruation (Sampson) into the vesicouterine pouch.
- **Iatrogenic implantation** at caesarean or pelvic surgery (primary vs iatrogenic forms).
- **Metaplasia** of coelomic or subperitoneal müllerian remnants (the "secondary müllerian system").
- Lymphatic or vascular spread; direct extension of anterior uterine adenomyosis.

Clinical features

- **Cyclical haematuria** or irritative LUTS (lower urinary tract symptoms) with menses; suprapubic pain, dysmenorrhoea, dyspareunia.
- Endocervicosis: non-cyclical mass or LUTS.
- **Also know:** menouria differential includes vesicouterine fistula (Youssef syndrome).

Evaluation

- **Ultrasound** (transabdominal plus transvaginal): size and depth of a hypoechoic nodule.
- **MRI (magnetic resonance imaging):** extent; haemorrhagic foci **bright on T1**, fibrotic thickening dark on T2.
- **Cystoscopy:** bluish submucosal nodule with neovascularity, best seen during menses. Of **limited value**, because disease is mainly **extravesical** (from serosa inwards).

- **Biopsy:** excludes adenocarcinoma, especially with endocervicosis. Immunohistochemistry: ER/PR (oestrogen/progesterone receptor) positive; CD10 marks endometrial stroma.
- Ultrasound of upper tracts for ureteric involvement.
- **Differential:** adenocarcinoma, cystitis glandularis, nephrogenic adenoma, urachal remnant.

Management

- **Hormonal suppression** (combined pill, progestins, GnRH [gonadotropin-releasing hormone] agonists): symptom relief; recurrence after stopping.
- **TUR (transurethral resection):** incomplete for transmural disease; perforation risk; recurs.
- **Partial cystectomy** (laparoscopic or robotic) with a margin: **primary treatment**. Ureteric stent or reimplantation if the orifice is involved.
- Excise other pelvic endometriosis with the gynaecologist.
- Malignant change (endometrioid or clear cell adenocarcinoma) is rare.

Diagram to draw: sagittal pelvis with a dome nodule growing from the vesicouterine pouch into the detrusor.

> **Key points to remember:**
> - Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
> - Cyclical haematuria; MRI bright on T1.
> - Disease is extravesical: cystoscopy limited, partial cystectomy preferred.

SAQ Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder.

Paper answer · P3

Type: SAQ (10 marks)
Source: 2023-Winter Paper 3, Q3e

Definition

- **CIS (carcinoma in situ, Tis):** flat, non-invasive, **high-grade** urothelial carcinoma. It is not "premalignant".
- Histology: severe atypia, loss of polarity, non-cohesive pleomorphic cells, **loss of umbrella cells** (dysplasia keeps them).
- **Types:** primary (isolated, about 10%), secondary or concurrent (with papillary tumour, about 90%).

Pathogenesis

- **Flat pathway:** dysplasia → CIS → high-grade invasive cancer.
- Driven by **TP53, RB and PTEN** loss; **no FGFR3** mutation (unlike low-grade papillary tumours).
- **Field change:** multifocal; spreads to prostatic urethra, distal ureters and upper tract.

Clinical importance

- Untreated, **40–83% progress** to muscle invasion. Primary CIS progresses in >50%.
- "CIS alone" hides invasion at cystectomy in up to 20%.

- **Second strongest prognostic factor after grade**; any CIS makes NMIBC (non-muscle-invasive bladder cancer) **high risk**.
- About 80% have **irritative symptoms**; it mimics cystitis.

Diagnosis

- **Urine cytology:** most useful in CIS, because the cells are poorly cohesive. Positive cytology with normal cystoscopy needs mapping.
- Cystoscopy: red velvety patch or normal-looking mucosa; **blue-light cystoscopy** (hexaminolevulinate) detects about one-third more CIS; NBI (narrow-band imaging).
- **Biopsies:** cold-cup from suspicious and random sites; prostatic urethra; upper tract imaging.

Management

- **BCG (bacillus Calmette–Guérin) is first line:**
- Induction: 6 weekly instillations.
- **Maintenance** (SWOG [Southwest Oncology Group] 2000: 3 weekly doses at 3 and 6 months, then 6-monthly for 3 years): median recurrence-free survival 76.8 vs 35.7 months.
- Complete response up to 84%; durable in about half.
- **Persistent CIS at 3 months:** second BCG course; 30–50% more respond.
- **BCG-unresponsive** (persistent or recurrent CIS within 12 months of adequate BCG): **radical cystectomy**.
- Unfit or unwilling: **pembrolizumab** (KEYNOTE-057: complete response 41%), nadofaragene firadenovec, gemcitabine–docetaxel.
- Prostatic urethral CIS: TURP plus BCG; ductal involvement needs cystoprostatectomy.

Follow-up

- Cystoscopy and cytology 3-monthly for 2 years, then 6-monthly; upper-tract imaging.

Diagram to draw: flat versus papillary pathways of bladder cancer.

> **Key points to remember:**
> - Flat, high-grade; TP53/RB; no FGFR3.
> - Untreated 40–83% progress; always high risk.
> - BCG with maintenance; cystectomy if BCG-unresponsive.

SAQ Virtual cystoscopy

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3a

Definition and principle

- **Virtual cystoscopy (VC)** is a computer-generated, three-dimensional (3D) endoluminal view of the bladder. It is reconstructed from thin-section **CT (computed tomography)** or **MRI (magnetic resonance imaging)** data of a distended bladder.
- Software "flies through" the lumen using **surface or volume rendering**; multiplanar images also show wall thickness and extravesical spread.

Technique

Step	CT virtual cystoscopy	MR virtual cystoscopy
Distension	Air or CO2 through a catheter (negative contrast), or excretory-phase contrast (positive contrast, no catheter)	Urine-filled bladder; urine bright on T2
Acquisition	Thin slices, supine and prone (moves dependent urine off the posterior wall)	3D T2 or T1 contrast sequences
Drawback	Radiation; catheter for gas	Cost, time, motion artefact

Advantages

- Non-invasive; no anaesthesia.
- Sees where a scope cannot: through a **tight urethral stricture**, inside **diverticula**, beyond a large median lobe.
- With **CTU (CT urography)**, upper tracts and staging are imaged in one study (Chapter 5: high accuracy for bladder tumours if well distended).

Limitations

- Detects polypoid tumours ≥5 mm well; misses **small and flat lesions, including CIS (carcinoma in situ)**.
- No colour or mucosal detail, **no biopsy and no treatment**.
- False positives: trabeculation, clot, debris, folds of an underdistended bladder, prostatic impression.

Indications (selective)

- Unfit for, or refusing, cystoscopy; urethral stricture; diverticulum; bladder evaluation during CTU for haematuria.
- **Guideline position:** EAU (European Association of Urology) and AUA (American Urological Association) do not recommend VC in place of cystoscopy. Flexible cystoscopy remains the standard.

Related advances

- **Also know:** MRI **VI-RADS (Vesical Imaging–Reporting and Data System)** scores muscle invasion (Chapter 6); blue-light and narrow-band cystoscopy improve CIS detection (Chapter 8); virtual ureteroscopy uses the same principle.

Diagram to draw: thin-slice CT of a gas-distended bladder, then a 3D endoluminal fly-through showing a polypoid tumour.

> **Key points to remember:**
> - 3D endoluminal reconstruction from thin-section CT or MRI.
> - Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
> - No biopsy; does not replace cystoscopy.
> - Useful beyond strictures and inside diverticula.

SAQ Bladder transplantation

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 4, Q3f

Rationale

- After cystectomy, or when the bladder cannot be reconstructed, the usual substitute is **bowel**. Bowel in the urinary tract causes mucus, metabolic acidosis, stones, infection and malignancy risk.
- Chapter 135: two frontiers aim to **remove bowel from urinary reconstruction**: tissue engineering and bladder transplantation.

Tissue engineering (why it has not succeeded)

- **Atala et al. (2006):** first seeded biodegradable construct for neurogenic augmentation. A later multi-institution trial showed **no significant gain in compliance or capacity** (Chapter 48).
- Engineered conduit trial: **half developed stenosis or stricture**, because of stiff material, poor cell coverage and **no blood supply** (Chapter 135).
- Engineered bladder: storage only (Chapter 111). See 2022-Summer Paper 4, Q2.

Bladder transplantation: development (Chapter 135)

- A **VCA (vascularised composite allograft)**: the donor bladder is transplanted with its own blood vessels.
- Stepwise preclinical work at USC (University of Southern California): pigs, perfused cadavers, then brain-dead human donors.

Technique (robotic)

1. **Robotic retrieval** of the bladder on its vesical vascular pedicles from the donor.
2. **Back-table vascular reconstruction**.
3. **Robotic implantation**: vessels to recipient iliac vessels; ureters joined to the graft.
4. **ICG (indocyanine green) fluorescence** confirms perfusion. Vascularity and mucosal integrity held for over 12 hours.

First clinical case

- Reported in **2025** from Los Angeles: a combined kidney and bladder transplant from a deceased donor, in a dialysis patient who had lost most of his bladder.

Problems

- **Lifelong immunosuppression**: infection, malignancy, drug nephrotoxicity. This is the major limitation (Chapter 135), especially after cancer.
- **Denervated graft**: stores urine; voiding may need straining or CIC (clean intermittent catheterisation).
- Rejection monitoring, donor supply, unknown long-term function.

Likely candidates

- Already immunosuppressed patients (kidney transplant recipients) with an absent or non-functioning bladder.

Diagram to draw: donor bladder on its vascular pedicles anastomosed to recipient iliac vessels, ureters entering the graft.

Key points to remember:

- Goal: avoid bowel in the urinary tract.
- Tissue-engineered bladders failed on blood supply and function.
- Robotic VCA; first human case 2025.
- Limits: immunosuppression and denervation.

SAQ Artificial urinary bladder

Paper answer · P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3d

Read as a man-made bladder substitute (prosthetic, engineered or transplanted). Bowel neobladder: see Practice Set 4 Paper 2, Q3e.

Need

- Bowel substitutes absorb urine and secrete mucus, causing acidosis, mucus, stones, infection, perforation and cancer. The aim is a substitute that **is bladder-like, not bowel**.

Requirements of an ideal substitute

- Low-pressure storage of adequate volume; continence.
- Complete, voluntary emptying; no reflux or upper-tract damage.
- Biocompatible: no encrustation, infection, rejection or cancer risk; durable.

Approaches

Approach	Principle	Status
Alloplastic (prosthetic) bladder	Synthetic reservoir (silicone, polymers) joined to ureters and urethra	Failed: encrustation, infection, leak at the tissue–prosthesis junction, migration, fibrosis
Tissue-engineered bladder	Autologous urothelial and muscle cells seeded on a biodegradable scaffold, omental wrap	Atala (2006): 7 patients improved; phase II trial showed no significant gain in capacity or compliance; experimental
Engineered conduit	Seeded scaffold as urinary conduit	Neo-Urinary Conduit trial: half developed stenosis or stricture; stiff, no blood supply
Bladder transplantation	Robotic retrieval and transplant of a vascularised composite bladder allograft	Feasible in models and brain-dead donors; first human case reported in 2025; needs immunosuppression

Why progress is slow

- **Vascularity:** thick constructs die centrally.
- **Innervation and contraction:** an engineered bladder stores urine but cannot yet empty voluntarily.
- **Urine–material interface:** foreign material encrusts and becomes infected.
- Fibrosis and contracture.

Alternatives in clinical use

- Augmentation (bowel, ureterocystoplasty, autoaugmentation); continent or orthotopic diversion; ileal conduit.

- **Also know:** neuroprosthetic emptying (sacral anterior root stimulator), artificial urinary sphincter, intraurethral valve-pump devices for underactive bladders.

Future

- Bioprinted vascularised constructs, stem cells (urine-derived, iPSCs, induced pluripotent stem cells), smart scaffolds.

Diagram to draw: biopsy → cell culture → seeding on bladder-shaped scaffold → omental wrap → anastomosis to bladder base.

> **Key points to remember:**
> - Prosthetic bladders failed (encrustation, infection, leak).
> - Atala 2006 promising; later trial no gain; still experimental.
> - Barriers: vascularity, innervation, urine–material interface.

Rapid Revision Sheet · Section 12

CA bladder, etiology, newer investigations and treatment. (LAQ)

- Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
- Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
- High-risk NMIBC: BCG with 3-year maintenance.
- MIBC: cisplatin NAC + cystectomy; trimodality in selected.

Re-look (second-look) TUR bladder tumour (SAQ)

- Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
- Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
- Residual T1 → progression (progresses) in up to 80%.

Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation (LAQ)

- Single instillation within 6 h (at most 24 h); never BCG immediately (SWOG S0337 gemcitabine).
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Withhold if perforation; best for low-risk solitary tumours.
- Only maintenance BCG reduces progression; SWOG 3-year schedule.
- NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab (41% CR), nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

BCG therapy and BCG failure (SAQ)

- Only BCG reduces progression; maintenance is essential.
- Never immediately after TURBT; start at 2–4 weeks.
- Adequate BCG = 5/6 induction + 2/3 maintenance.
- Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
- Timely cystectomy is standard; salvage CR rates halve by 12 months.

Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy (LAQ)

- NAC: cisplatin combinations only; ABC 5% absolute 5-year survival gain; SWOG 8710 pT0 38% vs 15%, P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab (adds durvalumab) improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.

- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

Bladder-preserving treatment of TCC (LAQ)

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
- TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
- BC2001: MMC/5-FU (mitomycin C/5-FU) + RT, 2-year locoregional DFS 67% vs 48%.
- Salvage cystectomy 10–30%; follow up the bladder for 10 years.

Intestinal segments in urology and orthotopic neobladder (LAQ)

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
- Wallace has the lowest stricture rate; detubularisation gives low pressure.
- Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
- Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
- Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

Methods of preoperative bowel preparation. (SAQ)

- Mechanical lowers amount; antibiotics lower concentration.
- Nichols–Condon: neomycin + erythromycin.
- Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

Malignancy in augmented bladder and after urinary diversion (SAQ)

- Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
- Mostly adenocarcinoma at the anastomosis; earliest at 4 years after augmentation.
- Excise defunctioned anastomoses.
- Gastric segment, exstrophy, immunosuppression: highest risk.
- Haematuria needs cystoscopy; treatment is cystectomy.

Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic (LAQ)

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL) against a reliable outlet; 500–600 mL by 1 year.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR (estimated glomerular filtration rate) >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.

- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; refluxing ureteric anastomosis is acceptable (USC (University of Southern California) trial).
- Obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel.
- Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Flap-valve continence >95%; stomal stenosis is the commonest complication.
- No diversion is proven best for quality of life; counsel all options.

Bladder diverticulum. (SAQ)

- No muscle → stasis, UTI, stones, cancer.
- Acquired type: treat the BOO first.
- Tumour invading beyond lamina propria is T3a.

Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman (LAQ)

- Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
- BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
- T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
- Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
- BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
- Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

Optical coherence tomography in carcinoma bladder (SAQ)

- Near-infrared interferometry ("ultrasound with light"); 10–20 µm resolution; 2–3 mm depth (P2: 1–2 mm).
- Loss of layering suggests tumour; boundary loss suggests T1.
- Role: Ta vs T1; cannot assess deep muscle.
- High sensitivity, moderate specificity, false positives from inflammation; investigational.

Intravesical gene therapy. (SAQ)

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

Von Bruns Nest (SAQ)

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

Mitomycin (SAQ)

- Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.

- Reduces recurrence, not progression.
- Signature toxicity: skin rash; contracted bladder about 5%.

Mullerionosis of UB (SAQ)

- Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
- Cyclical haematuria; MRI bright on T1.
- Disease is extravesical: cystoscopy limited, partial cystectomy preferred. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder. (SAQ)

- Flat, high-grade; TP53/RB; no FGFR3.
- Untreated 40–83% progress; always high risk.
- BCG with maintenance; cystectomy if BCG-unresponsive. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Virtual cystoscopy (SAQ)

- 3D endoluminal reconstruction from thin-section CT or MRI.
- Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
- No biopsy; does not replace cystoscopy.
- Useful beyond strictures and inside diverticula.

Bladder transplantation (SAQ)

- Goal: avoid bowel in the urinary tract.
- Tissue-engineered bladders failed on blood supply and function.
- Robotic VCA; first human case 2025.
- Limits: immunosuppression and denervation.

Artificial urinary bladder (SAQ)

- Prosthetic bladders failed (encrustation, infection, leak).
- Atala 2006 promising; later trial no gain; still experimental.
- Barriers: vascularity, innervation, urine–material interface. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*